FOC Sys Baseline Design Definition Doc_Part 2: Physical Architecture

Written by	**Responsibility**
Bernd Guth	Signed as AUTHOR on 04/06/2019 17:23
Verified By	
Alessandro Postinghel	Signed as CHECKER on 04/06/2019 17:31
Tallarida Domenico	Signed as CONFIGURATION MANAGER on 05/06/2019 09:41
Massussi Fabio	Signed as PRODUCT ASSURANCE on 05/06/2019 09:00
Gianmarco Massaro	Signed as PROJECT CONTROL on 04/06/2019 17:35
Marinelli Mauro	Signed as SYSTEM ENGINEER on 05/06/2019 16:12
Approved By	
Pighetti Luca	Signed as PROGRAM MANAGER on 05/06/2019 17:11
Released By	
Gianni Carella	Signed as DOCUMENTATION MANAGER on 06/06/2019 10:22

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GALILEO FOC – SYSTEM SUPPORT

System Baseline Design Definition Document Part 2: Physical Architecture

Written by	*Responsibility-Signature*
B. Guth	Galileo System Architect
Verified by	
A. Postinghel	System Engineering Manager
F. Massussi	Product Assurance
D. Tallarida	Configuration Control
G. Massaro	Project Control
M. Marinelli	System Manager
Approved by	
L. Pighetti	Project Manager
ESA Approval	

Approval evidence is kept within the documentation management system.

CHANGE RECORDS

ISSUE	DATE	§ CHANGE RECORDS
1.0	29/10/2010	First version of the document for the FOC contract delivered as part of formal SQR1A-KP2 Milestone. This version is an evolution of former IOV DDF2 (ref. GAL-RPT-ALS-SYST-R-0064 issue 13) in line with FOC TRB v1.1.
1.1	14/03/2011	Version updated for FOC System REKP in line with FOC TRB v1.2.
2.0	31/07/2015	Version for System DCDR internal red team review.
2.1draft	08/10/2015	Intermediate issue for SdCDR Readiness check-point.
2.1	20/11/2015	Update for System DCDR kick-off.
3.0	09/06/2016	• RID 297 Updating of section 7.2.5.3.1, to include in GSRP the generation of an alarm and backup EOP in case valid EOP parameters from IERS are not available. Updating dynamic scenario #SFC2.3.5 in section 7.2.5.4.5. • Refinement of dynamic scenario #SFC2.3.2 SLR Measurements in section 7.2.5.4.2 and updating of section 7.2.5.3.7) • Updating of dynamic scenario #SFC2.3-3 (Support Data Provision to GCS and GMS, including removal of MSF product validation (section 7.2.5.4.3) • Updating of dynamic scenario #SFC3.1.1 in section 7.3.1.6.1 to remove frequency adjustment and OSPF. • Updating of section 7.2.4.3 and refinement dynamic scenario #SFC2.1-1bis (section 7.2.4.4.2) to reflect WA-154. • Updating dynamic scenarios #SFC2.2.1 (section 7.2.4.4.1), SFC2.2.1 bis 1 (section 7.2.4.4.2 and SFC2.2.3 1 (sections 7.2.4.4.1), to add a step to configure the GGTO in the PTF. • Section 6.3.1.2.5 updated to clarify MUCF-SCAT capability to support long-term characterisation of hot back up clocks, to provide feedback on technology development (as per SDCDR Action P1_NW-199 related to RID System-199) • Table 86, section Section 6.5.3.3.2.4:update to remove #SFC5.3.2-4 (Off line Mission Monitoring) scenario, not

		implemented in MGF
4.0	23/09/2016	• Update for SDCDR action P1_NW_259: o Section 6.2.3.3.2 : Implementation of GST(MC) and PTF-GSS redundancy at PTF deployed at GCC site. o Section 6.2.3.3.3 : Clarification that the implementation of a combined TWSTFT/CV time transfer blended has been disregarded o Section 6.2.3.3.4 : Implementation of PTF-GSS redundancy at PTF deployed at GCC site: o Section 6.2.3.3.5 : Editorial o Sections 6.2.3.3.6 and 7.2.3.3.9 :Clarification that TW is used to provide GST to UTC steering for the master GST(MC) at master PTF. o Sections 6.2.3.3.13 : Updating of PTF's monitoring and control through GACF o Sections 6.2.3.3.14 :Updating of OSPF processing of PTF-GSS measurements and GST(MC)/PTF-GSS switch. o Sections 6.2.3.4: Updating of Dynamic Scenarios SFC 2.1-1, SFC 2.1-2 and SFC2.1-3 • Update of §4 to reflect system boundaries (support, service facilities), "Galileo Global Component" replaced by "Galileo Core Infrastructure"
5.0	28/05/2019	• §5.2/5.3 Integration Chain added • §5.3.2.11 3rd Maser added with internal redundancy concept • LEOP from GCC introduced with new LEOP TT&C stations replacing the ESCC • §4.2.1.1, §6.5.2 SFC5.2 updated for LEOP TT&C, figures changed • GCS context diagram changed §5.2.1 / SSeg boundaries updated • GMS context diagram corrected §5.3.1 / MKMF context corrected • §5.6.2.1 TTC sites updated • §5.6.2.3 GSS sites updated • New § 5.3.1.3 for GMS virtualization • §5.3.2.12 SPF front end concept added • §4 Figure 1 System architecture updated, REEF added • §6.2.3 updated with Dual TSP (in line with Fig. 1)

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1. INTRODUCTION

1.1 Scope of the Document

The scope of the System Design Definition File Physical (eq. Organical) Architecture volume (SYSBSLDEF part 2 referred as DDF2) is to describe the physical architecture of the Galileo System as specified by System and as implemented at segment level.
As such, the DDF2 content is directly traced to segment reference documents (See Reference Documents list).

The System DDF2 contains dedicated sections to:

- Describe the physical architecture and constraints of the Galileo System,
- Map segment functions defined in the system functional architecture as described in SYSBSLDEF part 1 onto physical elements of the system,
- Describe the physical implementation of the system functional chains (as identified in SYSBSLDEF part 1 [SYS-RD.01]). This description includes static and dynamic views.

Note: The organic architecture should be limited to the level of segments. However, in order to give a full understanding of the system architecture, the segments design are also summarized making reference to the needed segment documentation.

The mapping between System Design Definition File and STRB is provided below:

- **DDF1** (Design Definition File – Logical architecture) includes design changes up to STRB2.0 to describe the system concepts from the logical perspective, except for CR18/19 changes.

- **DDF2** (Design Definition File – Physical architecture- **This document**) includes design changes up to STRB1.5 to describe the associated physical implementation. It does not contain the physical description of STRB1.5.1 and 2.0 since it is not yet contracted at segment level (thus no segment physical implementation available).

- **Change request 18/19:** On top of MRD7.1 (AD.47), change request 18/19 are processed through dedicated TN (SYS-RD.89, SYS-RD.90, SYS-RD.91 and SYS-RD.92). Associated DCN are listed in the RJF (SYS-RD.70) in section 3.1.2 and anticipation of expected logical architecture evolution captured directly in these above mentioned TNOs.

1.2 Scope of Application

This document is generated as part of Galileo Project Phase CDE1 and applicable to all teams in charge of designing, and validating the Galileo System. It is used during all the phases of design, development up to the acceptance and hand-over of the system to the customer.

1.3 Document Structure

This document is composed of following main sections:

- **Section 3: Galileo Architecture Summary:** as an introduction, this section recalls the main physical characteristics of the Galileo System including a high level presentation of the main components of the Galileo System and Interfaces with External Entities.
- **Section 4: Galileo Core System Perimeter**: this section intends to gives a physical description, functional mapping and organical context for all the external entities interacting with the Galileo Core System. These external entities are needed as support to provide the Galileo services.
- **Section 5: Galileo Core System Physical Description**: this section gives a physical description of the physical components of the Galileo Core system. It includes:
    - The physical description of the 4 Galileo segments, namely, the Ground Mission Segment, the Ground Control Segment, the Space Segment and the Test User Segment.
    - The detail physical description of all the elements (N-2 level) composing each segment including a physical description, a functional mapping and an organical context for each one of them.
    - The description of the ground site locations (Ground Control Center and Remote stations, TTC, GSS and ULSs), and the ground networks.
- **Section 6: Galileo System Functional Chains Physical Description:** this section includes the detailed physical description of all the System Functional Chains needed to provide the different Galileo Services. The main goal of this section is to provide the physical mapping of the SFC functional description included in SYSBSLDEF part 1 [SYS-RD.01]. This section is organized per SFC with the following structure:
    - SFC overall physical description including a physical static diagram showing the physical elements involved in the SFC and the exchanges between them. This physical static diagram is the physical realization of the functional static block diagram of the SFC described in SYSBSLDEF part 1 [SYS-RD.01].
    - Detailed physical description of the processes within the SFC, including a detail role of each physical element involved.
    - Physical Dynamic Scenarios related to the SFC. **Dynamic scenarios are identified with the hashtag #SFC.**
- **Section 7: Dual GCC Management:** brief section recalling the dual GCC concept within the Galileo System. Refer to dedicated TN on GCC split concept (SYS-RD.76) for further details.

Remark: System Physical Configurations for the different system milestones are described:

- for what concerns the intermediate system configurations (ATL, IATO): in SYS-RD.74.
- for what concern the targeted final FOC2 architecture : in AD.48.

2. DOCUMENTS

2.1 Applicable Documents

AD#	Title	Document No.	Issue
AD.00	FOC STRB 2.0.x	GAL-LI-ESA-SYST-X-3849	2.0
AD.01	GSRD	GAL-REQ-ESA-SYST-X-0011	7.0
AD.02	TUSREQ	GAL-REQ-ESA-REC-X-0025	2.8
AD.03	GMSREQ	GAL-REQ-GLI-SYST-A-0165	15.1
AD.04	FOC SSREQ	GAL-REQ-ESA-SYST-X-0116	2.1
AD.05	GCSREQ	GAL-REQ-GLI-SYST-A-0019	17.3
AD.06	FOC SISICD	GAL-ICD-ESA-SYST-X-0027	2.0
AD.07	GMS SS ICD	GAL-ICD-GLI-SYST-A/0014	16.4
AD.08	GCS GMS ICD	GAL-ICD-GLI-SYST-A-0166	17.0
AD.09	FOC GCS SS ICD	GAL-ICD-ESA-SYST-X-0030	2.6
AD.10	SS Ext IRD	GAL-IRD-ESA-SYST-X-00314	2.0
AD.11	GMS Ext IRD	GAL-IRD-ESA-SYST-X-0031	2.0
AD.12	GCS Ext IRD	GAL-IRD-GLI-SYST-A-0371	9.2
AD.13	GMS to TSP ICD	GAL-ICD-GLI-SYST-A/0677	9.0
AD.14	GMS to GRSP ICD	GAL-ICD-GLI-SYST-A/0679	9.0
AD.15	GMS to GGTO ICD	GAL-ICD-GLI-SYST-A/0678	9.0
AD.16	GMS to ERIS ICD	GAL-ICD-GLI-SYST-A/0685	--
AD.17	GMS to SC ICD	GAL-ICD-GLI-SYST-A/0686	10.0
AD.18	SS to ERIS ICD	GAL-ICD-GLI-SYST-A-0725	--
AD.19	SAR-Beacon to SSeg to SAR-MEOLUT ICD (COSSARGALICD Vol 2)	GAL-ICD-ESA-SYST-X-0089	1.1
AD.20	GMS to SAR RLSP ICD (COSSARGAL ICD, Vol.3)	GAL-ICD-GLI-GMS-A/0375	13.0
AD.21	SS to SLR ICD	GAL-ICD-GLI-SYST-I-0726	3.0
AD.22	Deleted		
AD.23	GCS SATMAN ICD	GAL-ICD-GLI-SYST-A-0717	4.6
AD.24.	GMS to CSP ICD	GAL-ICD-GLI-SYST-A/1028	5.0
AD.25	GSS Site IRD (GSA responsibility)		
AD.26	ULS Site IRD (GSA responsibility)		
AD.27	TTC Site IRD (GSA responsibility)		
AD.28	REDU Site IRD	ESA-DTEN-NG-REQ/02788	
AD.29	FOC System Verification Requirements	GAL-REQ-ESA-SYST-X/0017	2.1
AD.30	LEOP Requirement Document (removed from STRB 2.0)	GAL-REQ-ESA-OPS-X/0003	2.1
AD.31	GCS to IOT ICD	ESA-DTEN-NG-ICD/03147	1.5
AD.32	IOT Services Requirements Document	GAL-REQ-ESA-OPS-X/0025	1.0
AD.33	GCC Site IRD (GSA responsibility)		
AD.34	GMS to GSMC ICD	GAL-ICD-GLI-SYST-A/1029	TBD
AD.35	Operations Service Requirement Document (OSRD)	GAL-REQ-GLI-SYST-A-0023	3.2
AD.36	Galileo FOC Launch Services Technical Requirements	GAL-REQ-ESA-LNS-X/0005	1.2
AD.37	Galileo Ground Segment Integration Standards	ESA-DTEN-NG-SPEC-03338	4.0
AD.39	IOT Security Service Requirements Document	GAL-REQ-ASTD-IOT-R-0019	2.2
AD.40	GDDN ERD	GAL-REQ-GLI-SYST-A/10021	2.6

AD.41	Requirements Engineering Plan and Specifications (DOORS)	GAL-PL-ALS-SYST-A-0606	--
AD.42	BBKME ICD	GAL-ICD-GLI-SYST-A/0376-x	6.1
AD.43	Galileo Segments to GDDN ICD	ESA-DTEN-NG-ICD/03764	2.8
AD.45	Galileo Constellation Geometry Requirements	GAL-REQ-ESA-SYST-X-1241	1.0
AD.46	SAR GS REQ	GAL-REQ-ESA-SYST-X-0538	1.3
AD.47	Mission Requirement Document	MRD Issue 7.2,	7.2
AD.48	ESA FOC2 roadmap	GAL-TN-ESA-SYST-X-2486	3.0
AD.49	SSREQ	GAL-REQ-ESA-SYST-X-0116	1.9
AD.50	Ground Control Segment to External Network Operations Centre ICD (GCS ENOC ICD)	GAL-ICD-ESA-SYST-X-4053	1.0

Table 1: List of Applicable Documents

2.2 Reference Documents

SYS-RD#	Title	Document No.	Issue
SYS-RD.1	FOC System Design Definition File part I Functional architecture	GAL-RPT-TASI-SYST-X-0005	6.0
SYS-RD.2	Deleted		
SYS-RD.3	FOC System Design IOV Justification File	GAL-RPT-ALS-SYST-R-0005	15
SYS-RD.4	System Performance Budget File	GAL-TN-GLI-SYST-I-0006	23.0
SYS-RD.6	FOC MOCD	GAL-MGT-ESA-OPS-X/0026	2.3
SYS-RD.7	Classified Annex to Baseline System Design Definition Document - Part 1: Logical Architecture	GAL-DDF-CIMC-SEC-X/5185x	2.1
SYS-RD.8	Classified Annex to Baseline System Design Definition Document - Part 2: Physical Architecture	GAL-DDF-CIMC-SEC-X/5186x	2.1
SYS-RD.10	SIS Flags usage	GAL-TN-ESA-SYST-X-2069	1.3
SYS-RD.12	Annex D (Commercial Service) of Cryptographic Algorithms Definition and Requirements	GAL-REQ-GLI-SYST-A/0599	1.10
SYS-RD.14	Classified Annex of FOC MOCD	GAL-MGT-ESA-SEC-X-0629x	1.0
SYS-RD.16	Navigation Message Definition and Justification	GAL-TN-ALS-SYST-I-0659	2.0
SYS-RD.23	System States and Modes and System Start-Up	GAL-TN-ALS-SYST-A-0653	2.0
SYS-RD.29	Operations concept for PL IOT and SC commissioning using GCS	GAL-TN-ALS-SYST-R-0658	2.0
SYS-RD.30	LEOP Services DDVP	GAL-PL-CNES-LEOP-A/0007	1.3
SYS-RD.38	IOT DDJF	GAL-JF-INM-IOT-R/0001	3.1
SYS-RD.47	SISICD Justification File	GAL-TN-ASTG-SYST-A-0027	1.3
SYS-RD.54	TTCF, GSS and ULS Sites List	GAL-TN-GLI-SYST-I-1223	1.0
SYS-RD.69	System Start-Up	GAL-TN-ALS-SYST-A-0693	3.0
SYS-RD.70	Requirement Justification File	GAL-DJF-TASI-SYST-X-0002	2.1
SYS-RD.71	GST-UTC Offset Interface Options	GAL-TN-ESA-SYST-X/267	1.3
SYS-RD.72	GGTO Implementation and Operations	GAL-TN-ESA-SYST-X/2494	3.0
SYS-RD.73	Galileo Single Frequency Ionosheric Model for User Receivers	Issue 1.1, June 2015. From European Union 2015	1.1
SYS-RD.74	Galileo System Architecture Configuration	GAL-TN-ESA-SYST-X-1075	5.1
SYS-RD.76	GCC split concept configuration	GAL-TN-ESA-SYST-X-2662	1.0
SYS-RD.77	Galileo FOC SAR Ground Segment Input to Element Requirement Documents (SGSERD)	GAL-REQ-ESA-SYST-X-0572	1.7
SYS-RD.78	SGS DDJF	GAL-DJF-CAP-SGS-X-0036	1.4
SYS-RD 79	Early Services Search and Rescue Service Operational Interfaces	GAL-ICD-ESA-OPS-X/0734	1.0
SYS-RD.80	Galileo FOC – System Support IOV SAR Link Budget	GAL-TN-TASI-SYST-X-0116	1.0
SYS-RD.81	SGS Performance Budget File	GAL-PBF-CAP-SGS-X/0034	1.3
SYS-RD.82	SAR Ground Segment SGS Definition file (DDF)	GAL-DDF-CAP-SGS-X/0012	1.5
SYS-RD.83	SAR/Galileo MEOSAR Reference Beacon Requirements	GAL-REQ-ESA-SYST-X/1532	2.0
SYS-RD.84	SAR/Galileo SGSKPI Collection Platform Requirements	GAL-REQ-ESA-SYST-X/1954	2.0

SYS-RD.85	Trade-off on FOC Return Link Service Provision	GAL-TN-TASI-SYST-X-0240	1.0
SYS-RD.86	SAR Return Link System Description Document	GAL-TN-ESA-SYST-X/2686	1.0
SYS-RD.87	Constellation Deployment and Maintenance plan	GAL-PL-ESA-SYST-X-0478	4.0
SYS-RD.88	GDDN Design Documentation	GAL-DDF-BT-OPS5-X-0001	1.9
SYS-RD.89	Use of External Data Broadcast Service for OS navigation message authentication Technical assessment	GAL-TN-TASI-SYST-X-0222	3.0
SYS-RD.90	TN#02: Availability of Galileo Commercial Service Data Dissemination as specified in EC-GAL-CR-18 on GMRD	GAL-TN-ASTG-SYST-X-13782	1.0
SYS-RD.91	CR 19 I/NAV Optimization Technical Assessment	GAL-TN-ASTG-SYST-X-13754	1.1
SYS-RD.92	Implementation of Separated E6 Pilot and E6 data encryption	GAL-TN-ASTG-SYST-X-13767	1.0
SYS-RD.93	RAMS contribution to System Availability	GAL-TN-TASI-SYST-X-0060	6.0
SYS-RD.94	OD analysis to support DCN0763 to GCSREQ-26588	ESA-DTEN-NG-MEMO-0014362 dated 9/03/2011	
SYS-RD.95	TTCF Network Sizing	GAL-TN-TASI-SYST-X-0185	1.2
SYS-RD.96	System Performance Budget File Classified	GAL-TN-ESA-SYST-X-0396-x	4.0
SYS-RD.97	SAR Annex to PBF/DJF	GAL-TN-ASTG-SYST-X-12126	3.0

Table 2: List of System Reference documents

2.2.1 Space Segment Reference Documents

2.2.1.1 IOV Space Segment Reference Documents

SS-RD#	Title	Document No.	Issue
IOVSS-RD.1	Space Segment Design Description and Design Justification File	GAL-DD-ASTD-SS-R-0001	4.1
IOVSS-RD.2	Satellite Design Description and Design Justification File	GAL-DDJF-ASTD-SA-R-0007	2.1
IOVSS-RD.3	Payload DDJF	GAL-DDJF-ASTD-PL-R-0017	3.0
IOVSS-RD.4	Avionics DDJF	GAL-DDJF-ASTD-AV-R-0004	5.0
IOVSS-RD.5	TT&C Subsystem Requirement Document	GAL-REQ-ASTD-TTC-R-0013	3.2
IOVSS-RD.6	Satellite Security Units DDJF (Annex 9 to SAT)	GAL-DDJF-ASTD-SU-R-0016	1.0
IOVSS-RD.7	Galileo IOV PUS Definition	GAL-ICD-ASTD-SS-A-0028	1.0
IOVSS-RD.8	CMCU Requirements specification	GAL-REQ-ASTU-PLC-R-0630	3.3
IOVSS-RD.9	Avionics Subsystem Requirements Specification	GAL-REQ-ASTD-AV-R-0010	3.0

IOVSS-RD.10	NSGU Element Requirements Definition	GAL-RQS-SES-PLF-R-0211	2
IOVSS-RD.14	Space Segment and Satellite Mission Analysis	GAL-AN-ASTD-SS-R-0001	4.0
IOVSS-RD.15	SSUM Volume 1: Satellite System Summary	GAL-MAN-ASTD-SA-R-0001	1.0
IOVSS-RD.16	SATREQ - Satellite Requirements Documents	GAL-REQ-ASTD-SA-R-0001	6.1
IOVSS-RD.18	Space Segment and Satellite Operations concept	GAL-DD-ASTD-SS-R-0004	3.0

Table 3: List of IOV Space Segment Reference documents

2.2.1.2 FOC Space Segment Reference Documents

SS-RD#	Title	Document No.	Issue
FOCSS-RD.01	Satellite Design Description and Justification File	GS-SAT-OHB-TN-0001	7.0
FOCSS-RD.02	Galileo FOC Payload Design Description and Justification File	GS-PL-SSTL-RP-0018	8.0
FOCSS-RD.03	FDIR Analyses	GS-SAT-OHB-RP-0026	6.0
FOCSS-RD.05	General Design and Interface requirement	GS-SAT-OHB-SP-0002	3.0
FOCSS-RD.06	Satellite Mission Analysis	GS-SAT-OHB-RP-0006	2.0
FOCSS-RD.07	AOCS DDJF	GS-AOCS-OHB-TN-0092	1.0
FOCSS-RD.08	Satellite User Manual Vol 2b OBHB	GS-SYS-OHB-MA-0015	2.0
FOCSS-RD.09	OHB architectural Design Document	GS-OSW-OHB-ADD-0002	3.0

Table 4: List of FOC Space Segment Reference documents

2.2.2 GMS Reference Documents

GMS-RD#	Title	Document No.	Issue
GMS-RD.1	GMS Design Definition File (DDF) Part I: Logical Architecture	GAL-DD-TAS-GMS-X-500002	3.0
GMS-RD.2	GMS Design Definition File (DDF) Part II: Physical Architecture	GAL-DD-TAS-GMS-X-500003	5.0
GMS-RD.3	GMS Design Justification File (DJF)	GAL-JF-TAS-GMS-X-500003	2.0
GMS-RD.4	MUCF Design Definition File (DDF) Part I: Functional Analysis	GAL-DD-ASP-MUCF-R/100006	4.F
GMS-RD.5	MSF Design Definition File (DDF)	GAL-DD-DMS-MSF-R/130144	2.E
GMS-RD.9	GACF Element Requirement Document (ERD)	GAL-REQ-ASP-GACF-R/00052	11.0
GMS-RD.10	MUCF Element Requirement Document (ERD)	GAL-REQ-ASP-MUCF-R/0053	10.0
GMS-RD.11	MUCF Interface Control Document (ICD)	GAL-ID-ASP-MUCF-I/010529	6.0
GMS-RD.12	GACF Interface Control Document (ICD)	GAL-ID-AAS-GACF-I/10144	6.0
GMS-RD.16	SPF Interface Control Document (ICD)	GAL-ID-ASP-SPF-I/10087	6.0
GMS-RD.18	OSPF Element Requirement Document (ERD)	GAL-REQ-ASP-OSPF-R/0050	11.0

GMS-RD.19	OSPF ICD	GAL-ID-TAS-OSPF-400435	3.0
GMS-RD.24	PTF ERD	GAL-REQ-ASP-PTF-R/00054	8.0
GMS-RD.25	MSF ERD	GAL-REQ-ASP-MSF-R/00060	12.0
GMS-RD.26	MDDN ERD	GAL-REQ-ASP-MDDN-R/0059	10.0
GMS-RD.27	MGF ICD	GAL-ID-AAS-MGF-I/010097	6.0
GMS-RD.30	PTF DDF	GAL-DD-TASI-PTF-400592	3
GMS-RD.31	MSF ICD	GAL-ID-AAS-MSF-I/010158	6.0
GMS-RD.32	GSS ICD	GAL-ID-TAS-GSS-400429	3.0
GMS-RD.34	PTF ICD	GAL-ID-AAS-PTF-I/10146	6.0
GMS-RD.36	SPF ERD	GAL-REQ-AAS-SPF-R/0055	9.0
GMS-RD.37	GSS DDF	GAL-KSX-DD-GSS-R/80039	4.B
GMS-RD.38	MGF ERD	GAL-REQ-ASP-MGF-A/0056	8.0
GMS-RD.39	MKMF ERD	GAL-REQ-TAS-MKMF-400396	1.4
GMS-RD.41	GMS Input to Operation Concept	GAL-TN-TAS-GMS-400860	7.0
GMS-RD.44	GSS ERD	GAL-REQ-ASP-GSS-R/0045	9.0
GMS-RD.46	MKMF ICD	GAL-ID-TAS-MKMF-400432	3.0
GMS-RD.49	ULS ERD	GAL-REQ-ASP-ULS-A/0047	8.0
GMS-RD.55	ALG_NAV Algorithms Detailed Processing Model Volume 2: OD&TS	GAL-DPM-GMV-OSPF-430075	1.G
GMS-RD.57	PRS KMF Requirements Document (ERD)	GAL-REQ-TAS-MKMF-400396	1.2
GMS-RD.58	PTF design definition file	GAL-DD-CTT-PTF-A/145053	6.C
GMS-RD.59	GMS Upgrade from V2.0.1 to V2.1	GAL-TN-TAS-GMS-401120	3.0
GMS-RD.60	GAL-DD-TAS-ISC-X-500001-X	ISC Design Definition & Justification File	

Table 5: List of GMS Reference documents

2.2.3 GCS Reference Documents

GCS-RD#	Title	Document No.	Issue
GCS-RD.1	GCS Design Definition File (DDF)	GALF-DDF-ASTU-0006	4.0
GCS-RD.2	GCS Design Justification File (DJF)	GALF-DJF-ASTU-0009	3.2
GCS-RD.3	TTCF Detailed Design description	GAL-TN-IE-TTCF-R/0072	3.C
GCS-RD.5	FDF Element Requirement Document (ERD)	GALF-REQ-ASTU-FDF/00053	3.1
GCS-RD.8	SCCF Software Design Document	GAL-ADD-LOG-SCCF-R/0002	7A
GCS-RD.9	SCCF Design Definition File (DDF)	GAL-DDF-LOG-SCCF-A/0015	8D
GCS-RD.11	SCCF Requirement Specification (ERD)	GALF-REQ-ASTU-SCCF-0052	6.1
GCS-RD.17	SCPF Element Requirement Document (ERD)	GALF-REQ-ASTU-GCCS-0049	2.5
GCS-RD.21	SCCF to TTCF Interface Control Document (ICD)	GAL-ICD-ASTU-C-I/0536	5.2
GCS-RD.28	GCS DDF secure annex	GALF-DDF-ASTU-0005X	

Table 6: List of GCS Reference documents

2.2.4 TUS Reference Documents

TUS-RD#	Title	Document No.	Issue
TUS-RD.1	System Segment Design Document	GAL-SPE-THA-TUS-R/00066	5.1
TUS-RD.2	TUS System Design Justification File	GAL-JF-THA-TUS-R-00004	5.3
TUS-RD.3	Annex to TUS System Design Justification File -RF DESIGN TRADE-OFF AND PERFORMANCES ALLOCATIONS	Annex to GAL-JF-THA-TUS-R-00004 issue 5.3, TUS-DJF-01	5.0
TUS-RD.4	TUS SYSTEM SEGMENT SPECIFICATION DOCUMENT	GAL-SPE-THA-TUS-R/00065	3.1
TUS-RD.5	Acquisition performances	Annex to GAL-JF-THA-TUS-R/00004 issue 5.3. TUS-DJF-06	3.3

Table 7: List of TUS Reference documents

2.2.5 SAR Reference Documents

SAR-RD#	Title	Document No.	Issue
SAR-RD.1	SPECIFICATION FOR COSPAS-SARSAT 406 MHz DISTRESS BEACONS	CS-T.001	Is.3 rev15
SAR-RD.2	Trade-off on FOC Return Link Service Provision	GAL-TN-TASI-SYST-X-0240	Is.1
SAR-RD.3	SAR Return Link System Description Document	GAL-TN-ESA-SYST-X/2686	Is.1 rev.0

Table 8: List of SAR Reference documents

3. GALILEO ARCHITECTURE SUMMARY

3.1 Galileo Core System Physical Description

The Galileo Core System is composed of four segments:

- **Space Segment**: The Galileo space segment includes the constellation satellites (excluding the launch services, LEOP and IOT activities considered as external to the core system). The Galileo constellation in FOC will comprise thirty satellites in medium-Earth orbit (MEO) deployed in a Walker 24/3/1 plus 6 in-orbit spares. Each satellite will broadcast four ranging signals carrying clock synchronization, ephemeris and other data, depending on the particular signal. A user equipped with a suitable receiver will be able to determine his position to within a few meters when receiving signals from visible Galileo satellites.

  Note: currently the constellation status doesn't not perfectly reflect the baseline target as, the 4 IOV satellites have been placed in orbital slots according to IOV baseline (27/3/1), and the satellites relevant to L3 (Sat 5 / GSAT0201 and Sat 6 / GSAT0202) were wrongly left in an elliptical orbit caused by launcher injection anomaly and after recovery measure are now place into a safe orbit which anyway is not nominal. Their positions now mirror each other, meaning they overfly the same location on the ground every 20 days. This compares with a standard Galileo repeat pattern of 10 days, helping to synchronize their ground tracks with the rest of the constellation.

- **Ground Segment**: The Galileo Ground Segment is composed of two distinct segments physically collocated in a Ground Control Center (GCC) and a corresponding redundant one, and in remote sites world wide spread (TTC, GSS and ULS Stations). The GCCs shall be located in European territories.
  - **Ground Control Segment (GCS)**: responsible for monitoring and controlling the whole Galileo constellation (including maintenance). The GCS includes the TTC stations allowing S-Band TT&C communication with the Galileo satellites.
  - **Ground Mission Segment (GMS):** responsible for:
    - monitoring the constellation and the satellite health from a mission point of view.
    - generating and up-load mission data needed for the user to perform the different Galileo mission services (OS and PRS). The key data such as clock synchronization, ephemeris will be calculated from measurements made by a network of Galileo Sensor Stations (GSS) distributed world wide receiving the Galileo SIS, and uplinked through a network of Uplink Stations (ULS) connected to the Galileo constellation.
    - forwarding properly the external mission data received for external services (CS and SAR Return Link).
  - **Communication Network**: responsible for the communication connections between all ground elements.

- **User Segment**: As part of Galileo Core System, a Test User Segment is developed to demonstrate the service functionalities and associated performances. The user segment relies only on the Galileo SIS.

Figure 1 gives an overview of the targeted **FOC2** Galileo Core System architecture (AD.48) (corresponding to system build 2.0):

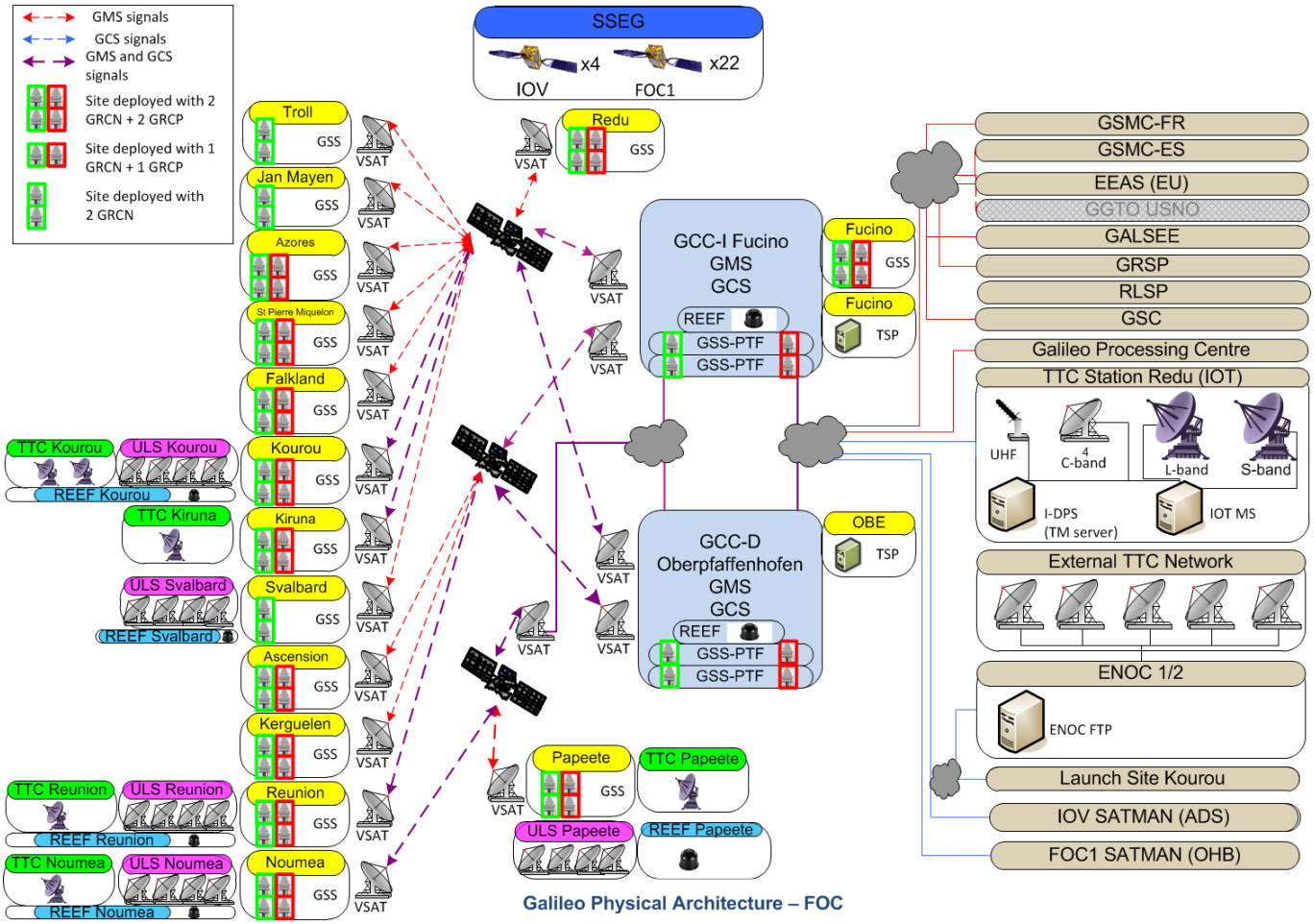


Figure 1: Galileo System FOC2 Architecture Overview

Remark: for detail description of network element and inter GCC communication, please refer to dedicated section 5.6.3.

4. GALILEO PERIMETER

This section briefly describes the entities and systems within the contour of the Galileo System Global Component as specified in the external ICDs.

4.1 Galileo System Boundary overview

The final perimeter of the Galileo System is illustrated in the figure below. It provides the perimeter of the Galileo core infrastructure, as well as interaction with service and support facilities and external entities.

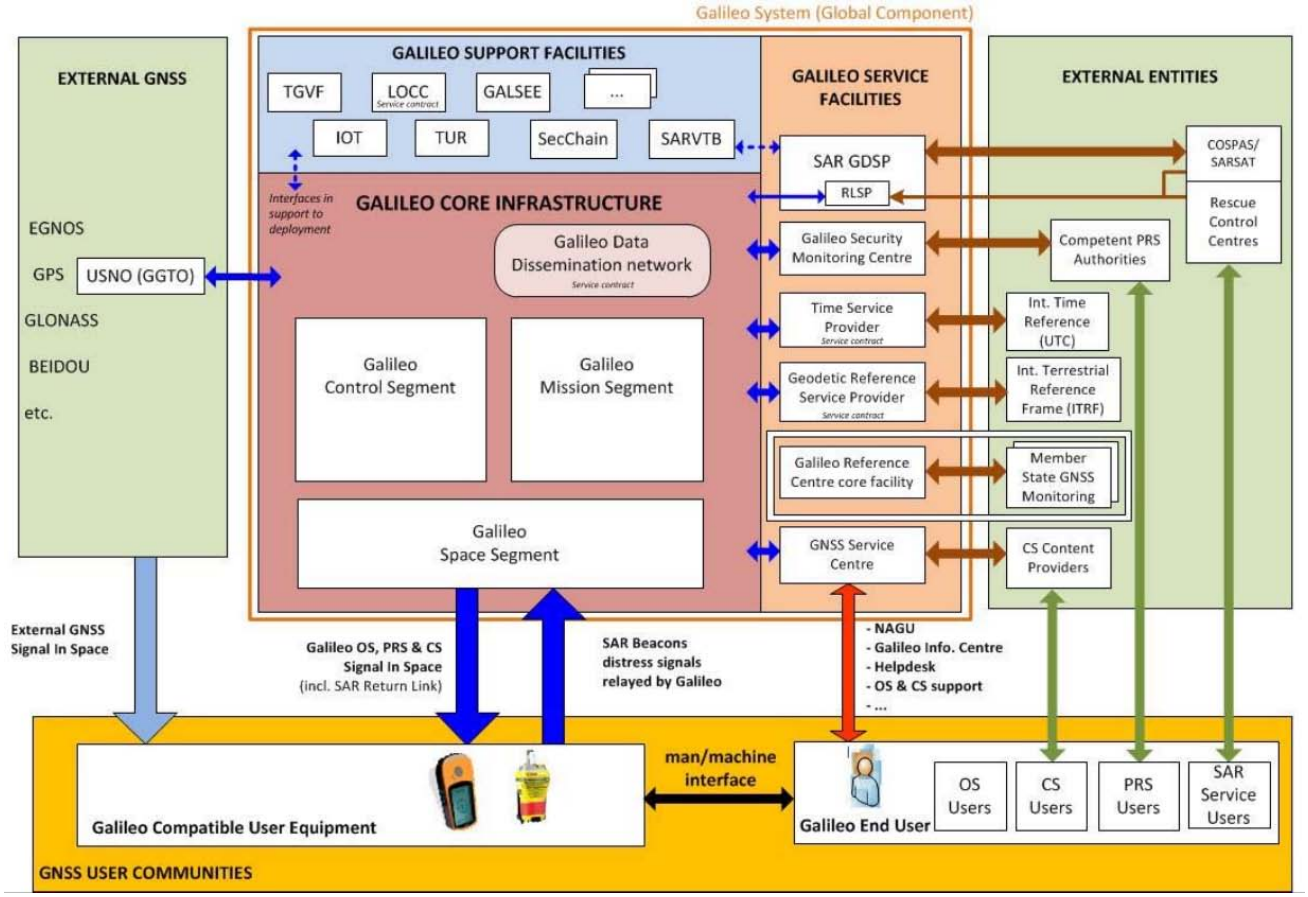


Figure 2: Galileo System Boundaries

Support facilities:

- LEOP Operation Control Centres LOCChave been replaced by the LEOP TT&C stations, used during LEOP activities under control of the GCC and interconnected via the External Network Operation Centre (ENOC)
- Launch Service Provider (LS)
- Galileo System Evaluation Equipments (GALSEE)
- In-Orbit-Test Provider (IOT)
- Test User Receiver segment (TUR)
- Time and Geodetic Validation Facitlity (TGVF)
- Search and Rescue Validation test bed (SARVTB), corresponding to RLSP prototype (future service facility)

- Security Chain Validation Test Bed (SECVTB or SecChain)
- Satellite and Payload Manufacturers (SATMAN)

Service facilities:

- Geodetic Reference Service Provider (GRSP)
- Time Service Provider (TSP)
- Galileo Security Monitoring Center (GSMC)
- Galileo Reference Center (GRC)
- Galileo Service Center (GSC)
- SAR GDSP (providing link to COSPAS/SARSAT)
- USNO

Other external entities connected to Galileo core infrastructure:

- International Laser Ranging Service (ILRS) providing SLR data

4.2 Support & Service Facilities Description

This section gives a detailed description of all the External Entities to the Galileo Core System. For each entity, a dedicated section includes:

- a high level description
- a functional mapping: list of the functions identified in the logical architecture [SYS-RD.01] mapped to the entity
- an entity organical context: to show the physical interface of the external entity (with which Galileo element the entity interface with, and which signals are exchanged).
- A list of the main design drivers

4.2.1 Support facilities

4.2.1.1 External LEOP TT&C stations

4.2.1.1.1 High Level Description

The External LEOP TT&C stations are connected to the GCC via External Network Operation Centres (ENOC) during Launch and Early Orbit Phase operations. Those additional external TT&C stations are controlled by the GCS, in order to be able to perform those operations directly from the GCC for all new satellites to be deployed as part of Galileo constellation, from launch (after separation of the satellite from the Launcher dispenser) to their operational position.
The internal Galileo TT&C stations are nominally already used for constellation maintenance and cannot be allocated for the LEOP phase operations on top of the constellation maintenance work.

The GCS infrastructure will allow for proper LEOP operations training and Operation products validation, as well as for all necessary System compatibility test campaigns.
The GCS shall be capable of supporting interfaces to two ENOCs in parallel. The interfaces provide the following data:

- Ranging and Meteorological data, in Galileo formats, from external TT&C stations.
- SLE services to external TT&C stations with ENOC as the service provider.
- Orbit Data from the GCS for external TT&C tracking.

The main functions executed during LEOP are:

- Monitoring and control of satellite for mode transitions from launch mode through init mode, sun acquisition, earth acquisition, to normal mode.
- Initial orbit determination
- Correction of injection errors (out-of-plane)
- Station acquisition manoeuvres
- Platform commissioning
- Potential additional TT&C coverage in support of contingency operations.

4.2.1.1.2 External LEOP TT&C stations Organical Context

The LEOP TT&C stations shall implement an interface only with the GCS during the LEOP process, as shown in the following the LEOP TT&C context diagram:

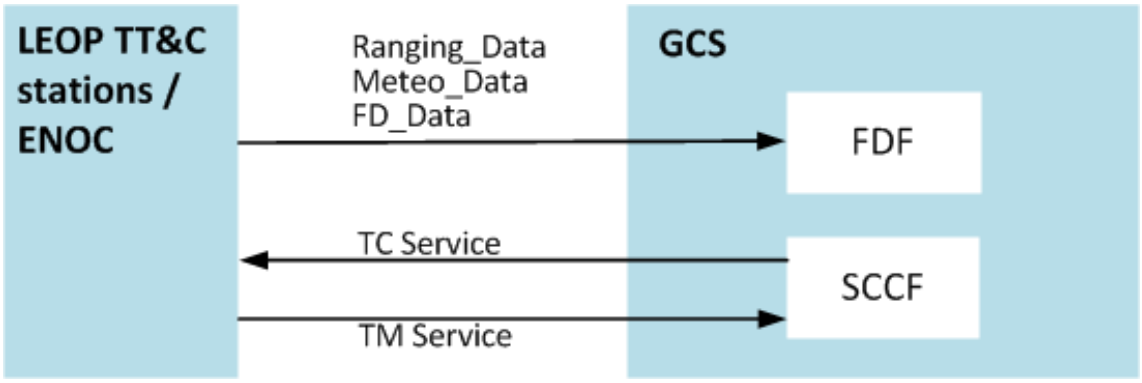


Figure 3: LEOP TT&C stations Organical context

4.2.1.1.3 Main Design Drivers

4.2.1.1.3.1 System Design Drivers

LEOP needs to be performed before IOT in a short time (35 days).

The ENOC must be designed to interface to the GCC and to provide a back-up to ENOC in case of problems.

Satellite manoeuvres must be accurately controlled to meet the intial bias conditions with respect to the Walker constellation parameters.

4.2.1.1.3.2 Segment Design Drivers

Interface to Satellites:
Communication to the satellite will be handled through a network of S-Band TT&C stations providing 100% access to the satellite during the early phases.
The TTC stations will be compatible with ESA Authentication and use standard mode for communications and ranging.

Interface to GCS – Content:
The GCS-ENOC interface will be established during LEOP operations for all spacecraft in order to transfer all required data between the LEOP TT&C stations/ENOC and the GCS, including:

- Reports related to planning of drift stop and fine positioning manoeuvres.
- Full transcripts of LEOP Telemetry, Telecommands and Meteorological and Ranging Measurements.
- The Launch Service Provider's Initial Orbit Vector.
- Orbit Vectors and historical (not real-time) ranging

Interface to GCS - Security
ENOC must provide the means to interface securely with the GCS. This includes the provision of non time-critical VPN-Protected equipment aligned with the system security requirements.

Link Security Modes
Due to LEOP execution from within the GCC, Galileo Key Management Facilities will be available. As a result communications with the spacecraft can and will be performed in ENCRYPTED mode. This applies to both TT&C and mission related interfaces.

TT&C Interface Compatibility for LEOP
In order to maximise interoperability with external TT&C stations, the TT&C mode during LEOP will nominally be Standard PM mode with Tone ranging.

4.2.1.2 Launch Service Provider (LS)

4.2.1.2.1 Launch Service Description

Two types of launchers are considered in the constellation deployment strategy analyses and equivalent performances in the renewal phase:
- Soyuz 2.1b (medium launcher): with launching capacity of 2 satellites per launch from Kourou or from Baikonour
- Ariane 5-ECA with a Fregat upper stage (heavy launcher): with launching capacity up to 4 satellites per launch from Kourou

4.2.1.2.2 Launch Service Functions

The Launcher Service shall perform the following functions:
- Launch and direct injection of a number of Galileo Satellites mounted on a dispenser, into the required orbit.
- Disposal of the Launcher upper stage to graveyard orbit

Function Allocation
This covers the following LS functions defined in the system DDF1 (SYS-RD.01).

Function ID	Function name
LSPF1	Perform Launch

Table 9: LS Function List

4.2.1.2.3 Launch Service context

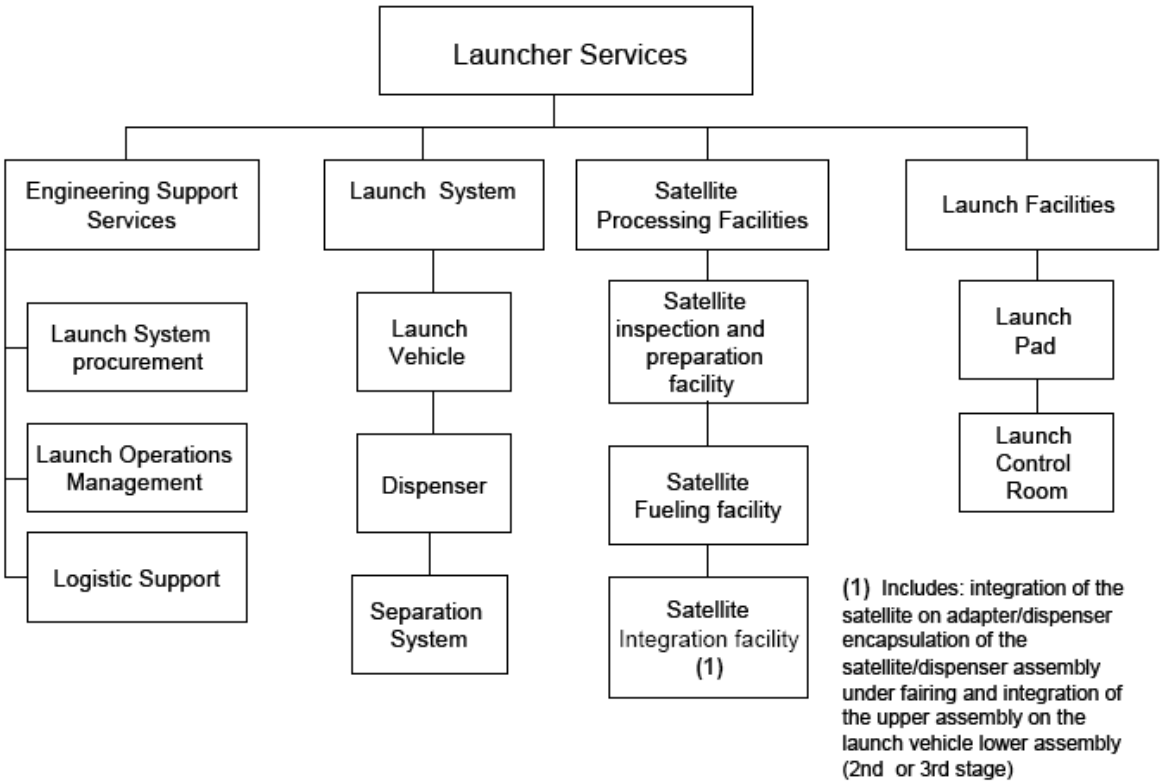


Figure 4: Launch Service Organical context

4.2.1.2.4 Main Design Drivers

4.2.1.2.4.1 System Design Constraints

To reduce both system cost and time to deploy multiple satellite launches are required.
Orbit parameters and tolerances are a major driver for the launch service.
to ensure constellation availability, the launcher upper stage must be disposed of into a graveyard orbit and the satellite deployment sequence must ensure collision risk is minimised.

4.2.1.2.4.2 Segment Design Constraints

The launcher environment (shock, vibration, EMC, thermal, pressure) has a major impact to the unit design in terms of qualification levels. In particular the shock levels at the dispenser/satellite interface must be minimised to avoid requalification of existing designs or restrictions on type of components that can be used (relays, etc)
The Launcher Payload is the only driver for the satellite mass requirement. Dual launch on Soyuz is the driving case.
The launcher payload volume is the main driver for overall satellite volume requirements.
The launch profile is a major driver for the thermal design an battery sizing requirements.

4.2.1.3 Galileo System Evaluation Equipments (GALSEE)

4.2.1.3.1 High Level Description

The GALSEE system validation platform, already supporting the In Orbit Validation (IOV) test campaign in 2013, will carry on system performance monitoring activities during the Initial Services validation campaign and later on during the Initial Services provision phase
The “Galileo System Evaluation Equipments” (GALSEE) facility originates from the necessity to support, with a more detailed investigation, all the verification activities since the GALILEO IOV Phase with an independent and reliable system tool. It has been upgraded in order to provide external monitoring of GALILEO System KPIs during nominal operation.
The GALSEE major characteristics with respect to the existing tools at different levels are to be an independent and flexible platform, to be capable to support the verification activities at system and sub-system levels and to manage real data. The high level GALSEE architecture is shown in the next figure. A main difference of GALSEE compared to TGVF is the availability in GALSEE of a direct interface to the Galileo Mission Segment, allowing the retrieval of system internal data, such as the navigation message at the output of Orbit Determination and Time Synchronization processing facility of the Galileo Control Center and before uplink. Specifically, GALSEE is connected to one GCC directly and receives GSS data received by that GCC (via GMS-SPF), and it shall receive also the data received by the other GCC thanks to the inter-GCC link.

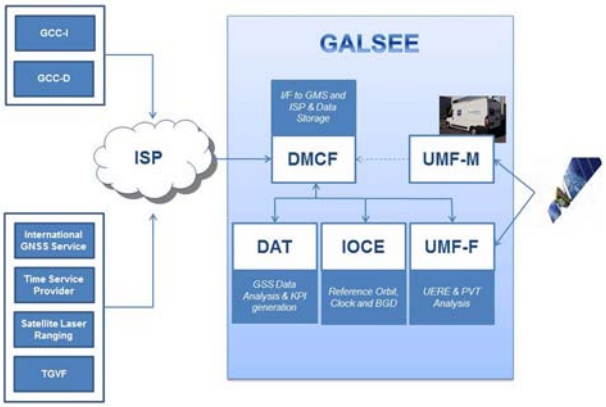


Figure 5 : GALSEE High Level Architecture

The GALSEE main objective is to post-process raw data and GMS products as received from Galileo, compare them with a reference product generated inside GALSEE and generate corresponding Key Performance Indicator evaluation and statistic to support Performance Verification and Monitoring. One of the key features of GALSEE is the capability to generate reference products (orbits, clocks, biases…) using the Galileo GSS data andSIS data as input, elaborating them with different algorithms with respect to those used in the Galileo operational system. GALSEE is also equipped with a facility equipped with User Receiver to support Ranging Error and PVT assessment in the field. The facility is designed to process Galileo signals (Open signals, SoL and PRS signals, the last ones limited to de-crypted and unclassified parts) at carrier frequencies L1, E5a, E5b and E6 transmitted by each Galileo Spacecraft (SV). This capability promotes the GALSEE system as a qualified facility able to give

an effective contribution to the verification activities relevant to Performance Verification and Monitoring Campaign.

GALSEE generates system performance figures of merit directly related to the end user's navigation performance such as:

- the broadcast navigation message data accuracy in terms of Signal In Space Ranging Error (SISRE), both for single and dual frequency users and as a function of the navigation message Age of Data (AOD);
- Global and per satellite availability of healthy SIS;
- Galileo System Time (GST) - UTC offset;
- UTC and Galileo to GPS Time Offset (GGTO) time and frequency dissemination accuracy;
- Position, Velocity and Time (PVT) computations
- User Equivalent Ranging Error (UERE) budget assessment at a number of fixed and mobile stations;
- On board atomic clock characterization
- Single frequency user IONO model characterization.

4.2.1.3.2 Functions Allocation

The following table shows the functions identified in the system DDF1 (SYS-RD.01) mapped to GALSEE:

Function ID	Function name
GALSEEF1	Data Management and Control

Table 10: GALSEE Function List

4.2.1.3.3 Entity Organical Context

The following figure shows the GALSEE context:

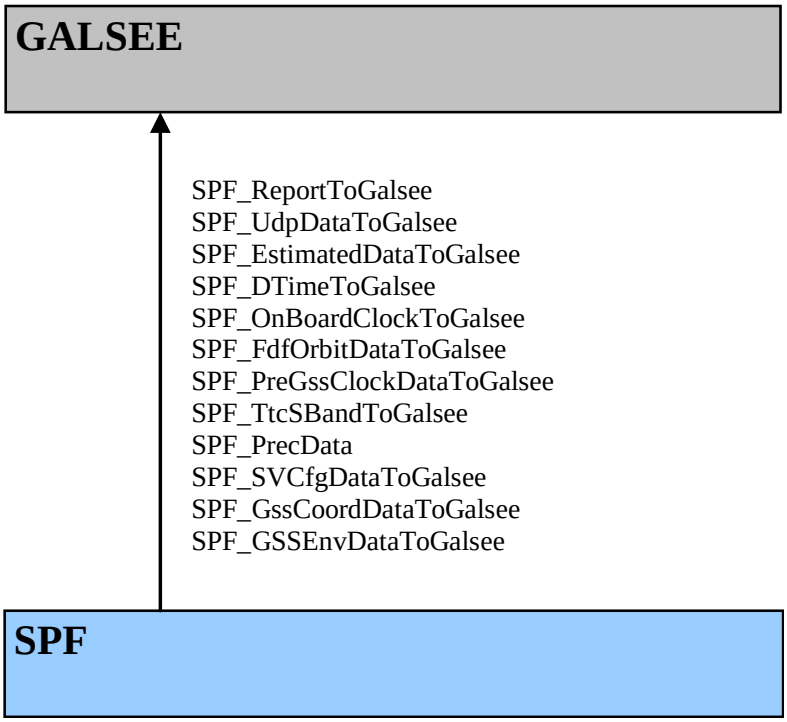


Figure 6: GALSEE Organical context

4.2.1.3.4 Entity Main Design Drivers

The “Galileo System Evaluation Equipment” (GALSEE) facility major characteristics are to be an independent and flexible platform, to be capable to support the verification activities at system and sub-system levels and to manage real data.
The GALSEE main objective is to generate reference products (orbits, clocks, biases…) using the Galileo GSS data and SIS data as input, elaborating them with different algorithms with respect to those used in the Galileo operational system.
The facility is designed to process Galileo signals (both Open signals and PRS signals the last ones limited to de-crypted and un-classified parts) at carrier frequencies L1, Ea, Eb and E6 transmitted by each Galileo Spacecraft (SV).
Relevantly to the verification activities, the GALSEE will process and analyse only the decrypted not-classified data, comparing them with the reference not-classified products from GMS. The GALSEE will not generate any PRS observable and will not include any decryption/encryption function.

4.2.1.3.4.1 Main system design drivers

- GALSEE Independence

The GALSEE shall be developed as an independent facility which is able to reproduce the Galileo environment, using as much as possible independent instruments and analysis methods w.r.t. the operational ones, in order to guarantee the maximum support to all the Galileo analysis and verification activities related to the system/segment design and development.

o GALSEE Flexible Environment

The GALSEE shall ensure a flexible environment, organized in modular components (both hardware and software) so as to be suitable to changes or evolutions in the Galileo mission operational scenario. This means that the facility shall be able to add, to adapt, to upgrade or to remove elements from the entire system.

o GALSEE Main Elements

The GALSEE elements which will be used in the Initial Services validation campaign are:
Data Analysis Tool (DAT), the high level performance reporting tool generating a set of performance indicators to be continuously monitored in the ES validation campaign (this is the equivalent of TGVF DPAF).
UERE Monitoring Facility (UMF), supporting the test activities associated to the verification of the performance for the selected user typologies and expressed in terms of User Equivalent Ranging Errors (UERE) budgets. Multiple fixed and mobile UMF units, will be used to collect measurements for static analyses and dynamic UERE tests in field.
Independent Orbit and Clocks Estimation (IOCE). This element supports all the other GALSEE elements in terms of reference satellite orbit and clock estimates, it provides the truth reference data specifically associated to the observed satellites.
Data Management and Control Facility (DMCF), including Data Configuration and Interface Network Management. This element supports all the other GALSEE elements by providing the required high data storage capability, data configuration environment, element interfacing capability, access to external world and monitoring and control for the overall system.

4.2.1.4 In-Orbit-Test Service Provider (IOT)

4.2.1.4.1 IOT Service Description

The IOT Service provides a means to test the satellite functions and performance after launch and separation. The aim is to check that all functions are available and that performance has not degraded since ground acceptance tests.

The IOT is split into 4 phases:
- Platform commissioning (performed by GCS)
- Payload commissioning (performed by GCS)
- Non-Secure Payload IOT (performed by IOT service provider with support of GCS TTC for satellite M&C)
- Secure IOT (performed with GMS elements with support of GCS for satellite M&C)

This covers the following IOT functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
IOTF1	Perform IOT Measurements

Table 11: IOT Function List

The following figure gives an overview of the IOT system for both non-secure and secure IOT

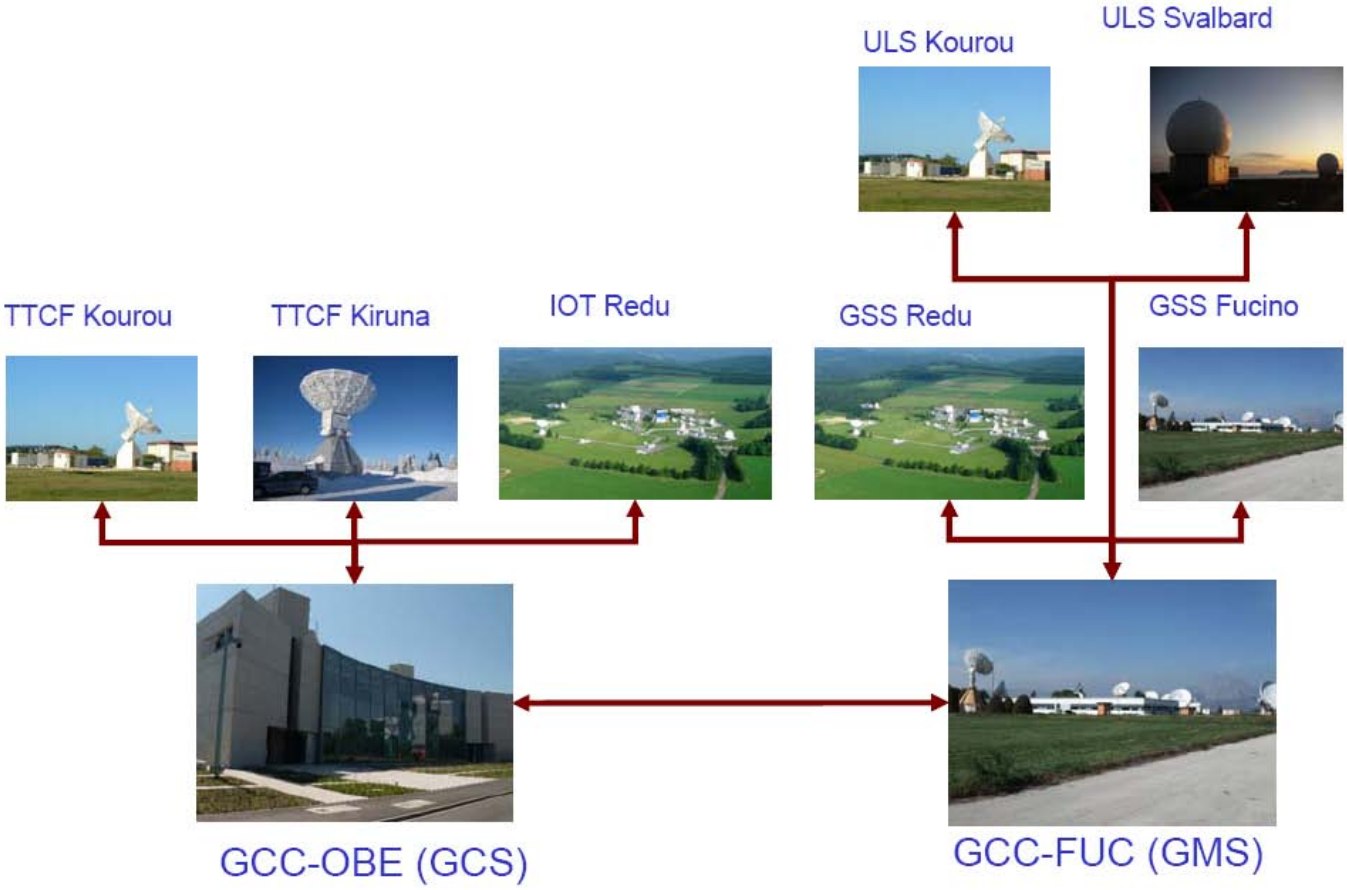


Figure 7: IOT Organical context

4.2.1.4.2 Non-secure IOT infrasctructure description

The non-secure payload IOT campaign relies on :

- GCS TT&C stations for satellite M&C (e.g. payload activation, initial satellite synchronisation…)
- Selfstanding IOT station in REDU to perform End-to-End payload testing. To do so, the IOT site in Redu includes the following facilities:
  - An L-band reception station to receive and demodulate both the navigation services and the search & rescue messages and to assess RF performance etc
  - Navigation receivers, mainly TURs, to process and handle the service data on the various channels
  - Measurement System MS (hardware & software) to verify the characteristics of the received signals and derive the achieved performance
  - A C-band transmission station to allow the content and performance of the individual navigation services to be tested;
  - UHF transmitters (up to 5) to uplink a known distress messages to test the SAR payload;

The following figure gives an overview of the IOT system and interaction with GCS in the frame of non-secured IOT:

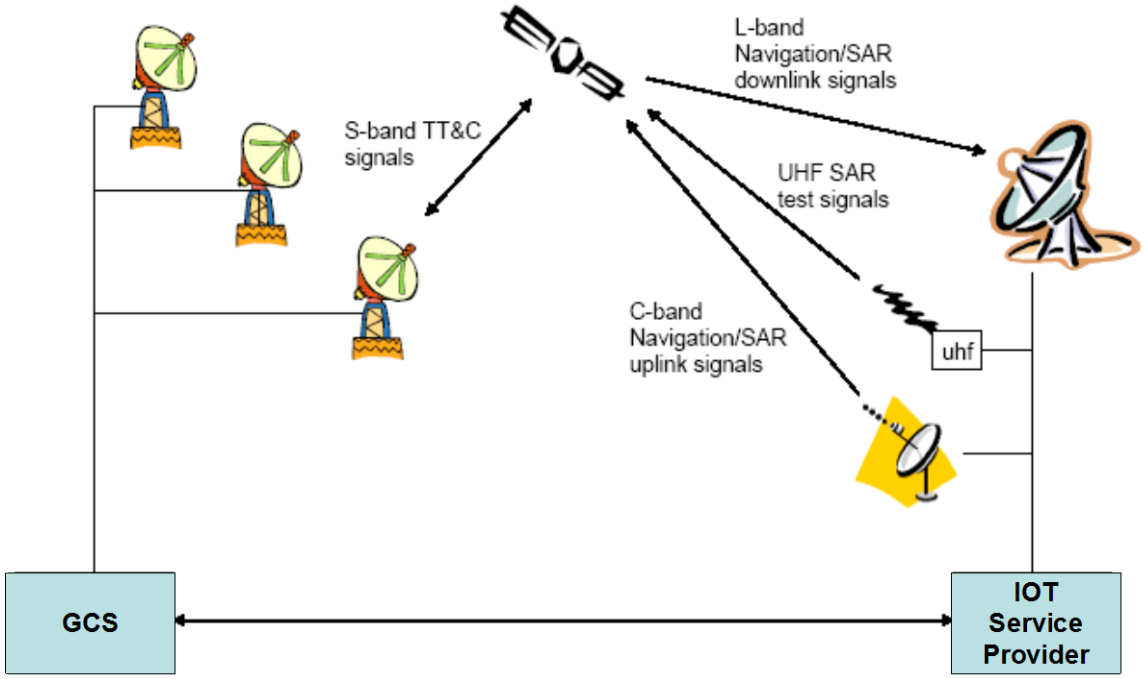


Figure 8: Non secure IOT Organical context

4.2.1.4.3 Secure IOT infrastructure description

The secured IOT includes three phases:

- Early Secure IOT-1 to initialize the PFSU/CSU and configure into E1 and E6 Encrypted protected mode
- Early Secure IOT-2 to initialize PLSU/CSU and configure into “Test Mode”
- Secure IOT in which the on board security Units PFSU/ PLSU on IOV satellites, CSU on FOC satellites) are configured into protected mode and End to end data dissemination test is performed during PLSU/CSU PRS mode management.

The Secured payload IOT campaign relies on:

- GCS TT&C stations for satellite M&C (e.g. PFSU/CSU encryption mode control…)
- GMS elements for PLSU/CSU M&C and PRS End-to-End test

The following GCS and GMS elements will then be used during Secure IOT:

- During all Secure IOT Phases:
  - SCCF for S/C commanding, STC/STM encapsulation & degraded data U/L.
  - GCS TTCFs for TC/STC & TM/STM delivery and S band degraded navigation message uplink. Kiruna and Kourou are the preferred TTCFs for IOT operations since they provide a significant coverage overlap with the IOT antenna at Redu, allowing IOT procedure TC to be sent to the payload from the SSCF to be executed step by step, interleaved with payload signal feedbacks received at the Redu IOT facility.
- During early Secure IOT (S-Band Protection):
  - GCSKMF for STC/STM management.
- During Secure IOT (C-Band and L-Band PRS Protection):
  - MKMF for STC/STM management & C-Band protection.

- PKMF for STC/STM management and PRS L-Band protection.
- GSS Fucino and Redu:
  - GSS antenna and GRCN for confirmation of C-Band flow transition in S/C, Commercial Service key encryption test on E6, uncla L-Band testing (PRS PN Test code) and GRCP for PRS decryption and end to end check.
  - PRS TUS-1 & TUS-2 for PRS Message Analysis.

  Note:
  i. REDU GSS is upgraded with TUR-P in FOC to support S-IOT in addition to the Fucino GSS
  ii. PRN PN "test code" is only used for L1A/E6A RF measurements during IOT uncla tests. No use of TUR or GRCx is foreseen for these tests.
- ULS Kourou/Svalbard (providing also good coverage with the Redu antenna) for:
  - C-Band Authentication
  - Uplink of Navigation Message in protected mode
- GMS elements for End-to-End dissemination:
  - MGF for message multiplexing and routing.
  - MNE/CNE for enabling communications with ULS/GSS & GCS respectively.
  - OSPF in emulation mode for dummy message generation.
  - MUCF for ULS schedule.
  - GACF for assets M&C.

The following figure shows the configuration of the GSSs and TUR at Fucino and Redu to support secure IOT:

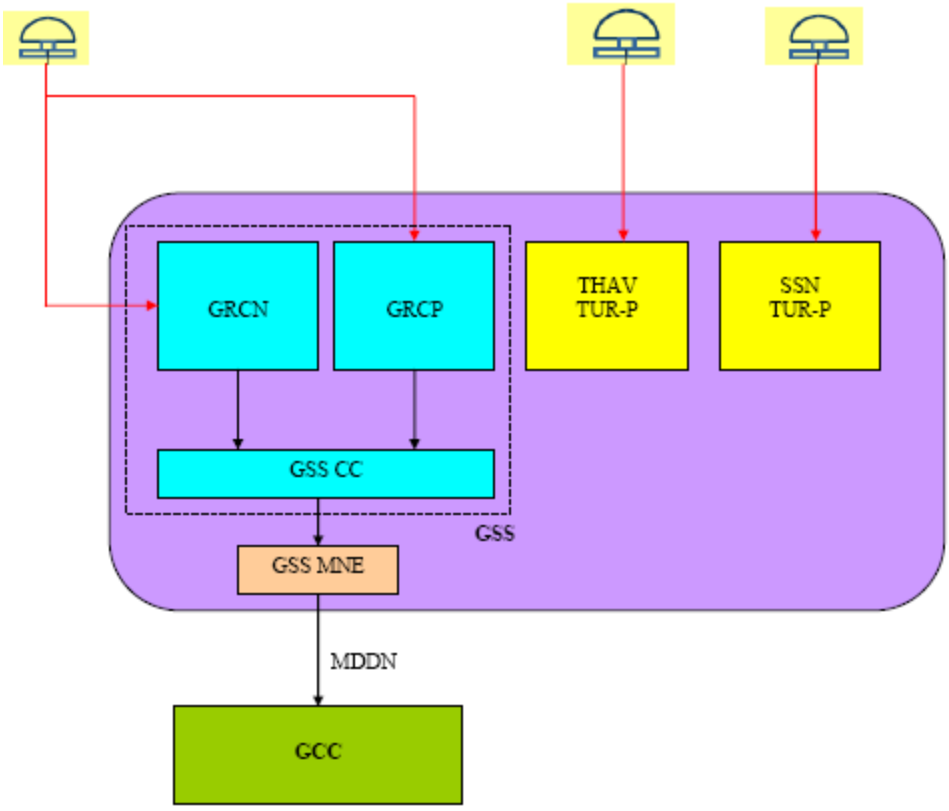


Figure 9: Secure IOT reception chain

4.2.1.4.4 IOT Service Design Drivers

The main system driver is that the IOT should be completed as soon as possible from separation from launch vehicle to entering service. This duration depends on the IOT tasks to be performed as well as availability of OPS personnel.

4.2.1.5 Test User Receiver (TUR)

4.2.1.5.1 TUS Overview

The Test User Segment (TUS) main responsibilities are to demonstrate that the mission service performances can be achieved when relying on the ranging and data provided in the Galileo SIS. They are used during SCTC and IOT tests.

The test of mass-market receivers is not addressed in the frame of TUS contract. It is subject of specific activites perforemdn in the frame of Early Services.

The TUS contributes to the following system functional chains:
- SFC4 - Mission Data Provision
  - o Verification of OS positioning, timing, SOL Integrity, and PRS positioning, timing , integrity solution performances
- SEC-SFC14 – Security Protection of SoL
  - o Check of the integrity table authentication
- SEC-SFC15 – Security Protection of PRS

4.2.1.5.2 TUS Organical Description

Two manufacturers are producing the TUS Receviers, namely :
- Thales Avionics THAV
- Septentrio SSN

The following figure shows the overall architecture and context of the TUS:

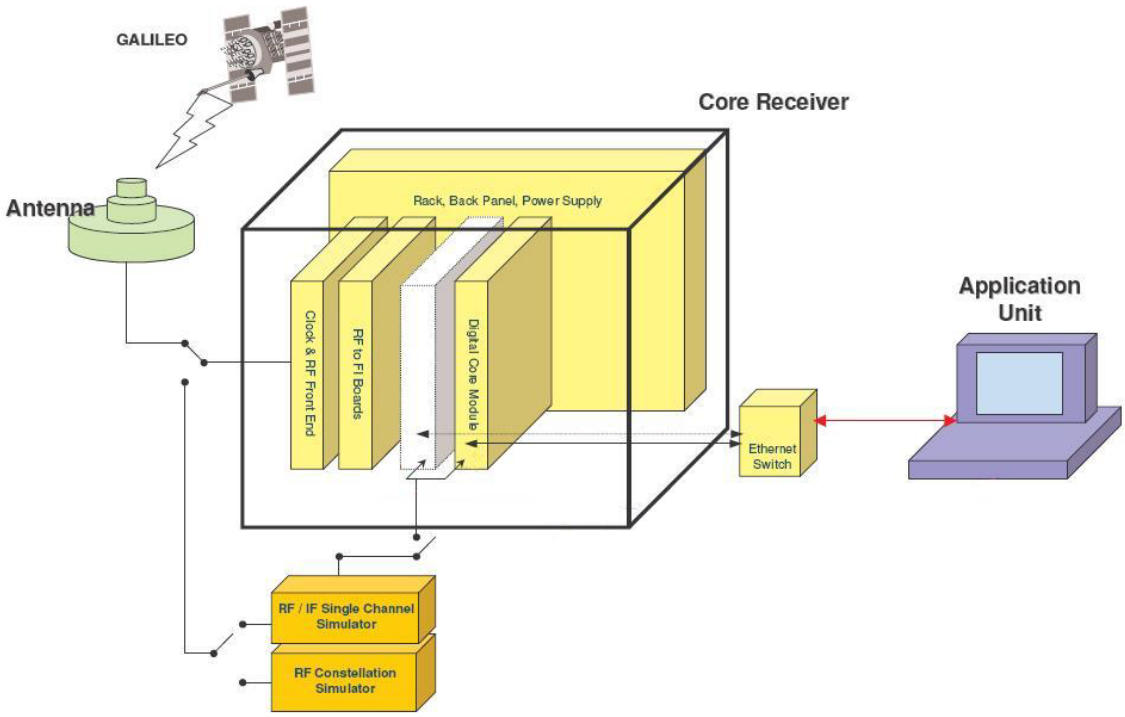


Figure 10: TUS Organical context

4.2.1.5.3 TUS Functions Physical Mapping

The following table lists the system functions allocated to the TUS and the mapping to TUS elements:

Function ID	Function name	Mapping to elements
TUSF1	PseudoRange and Phase Measurement on SIS	Antenna + Core receiver
TUSF2	Navigation Message Recovery	Core Receiver
TUSF4	Position Determination	Application Unit
TUSF5	Corrected Measurements Determination	Application Unit
TUSF5.1	Clock Correction	
TUSF5.2	Relativistic Effect Computation	
TUSF5.3	Troposphere Compensation Computation	
TUSF5.4	Sagnac Correction Determination	
TUSF5.5	Group Delay Determination	
TUSF5.6	Iono Free PR Measurements Determination	
TUSF6	Measurement Pre-Processing	Application Unit
TUSF6.1	Code Carrier Phase Filter	
TUSF6.4	Valid Measurement Determination	
TUSF7	Timing Determination Algorithm	Application Unit
TUSF7.1	GST Determination Algorithm	
TUSF7.2	GST-UTC Conversion Algorithm	
TUSF8	Analyses Tool	Application Unit
TUSF9	Autonomous Integrity Determination	Application Unit

TUSF10	Search and Rescue	Application Unit
TUSF11	Differential Data	Application Unit
TUSF11.1	Differential Data Generation	
TUSF11.2	Differential Data Reception	

Table 12: TUS Function List

4.2.1.5.4 TUS Elements Description

The TUS is composed of the following elements:
- RF components
- Core Receiver
- Application Unit

4.2.1.5.4.1 RF components

The TUS RF components include the following sub-assemblies:

- the antenna (HWCI) responsible for receive all Galileo signals on L1, E5 & E6 frequencies ,
- the preamplifier (HWCI) for the Galileo signals,
- the cable (COTS) between the preamplifier and the TUS Receiver.

4.2.1.5.4.2 Core receiver

Inside the Core Receiver, the Core Modules are interfaced with the needed RF/IF boards: E5a/E5b or L1/E6.

The clock synchronization signal is distributed from the RF front end to each core module. This signal is generated from 10 MHz internal clock (OCXO or VCO) or 10 MHz/5 MHz external clocks.

The following figure gives an overview of the Core Receiver architecture.

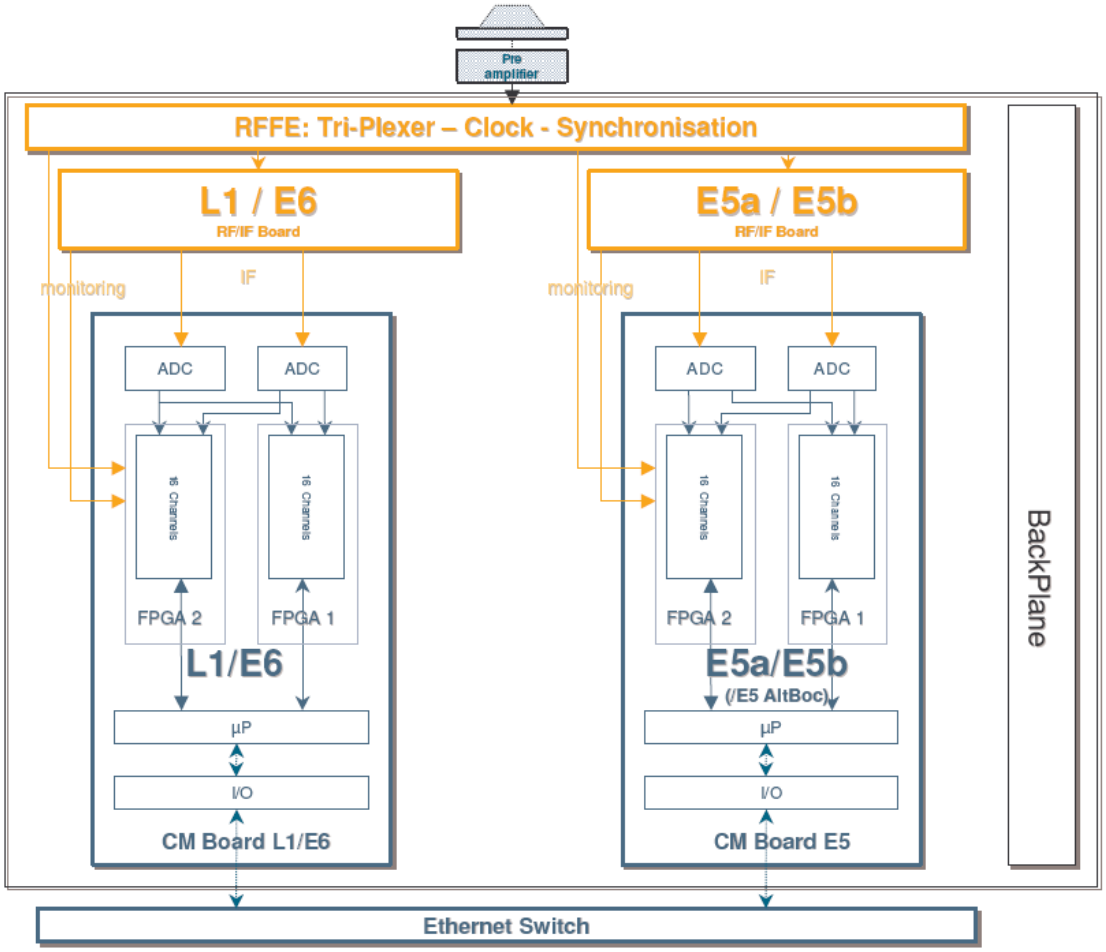


Figure 11: TUS Core Receiver architecture

Each core module performs the digital signal processing and is mainly in charge of performing pseudo-range measurements on the SIS ranging codes and pseudo-phase measurements on the SIS carrier. 16 parallel channels allow processing all the satellite in view. The main functions performed are (TUSF1):

- Carrier Tracking
- Carrier phase measurement
- Ranging code tracking
- Ranging code measurement
- Multipath mitigation
- Interference mitigation
- Signal monitoring

The Core module is also in charge of recovering of the navigation data transmitted by the Navigation Signals (satellite parameters, SISA, etc.). To carry out this function the following tasks are needed (TUSF2):

- Bit synchronisation
- Frame synchronisation
- Frame Demultiplexing

- Symbol recovery
- CRC check
- Viterbi decoding
- De-interleaving
- Bit recovery

4.2.1.5.4.3 Application Unit

The Application Software of the Application Unit is decomposed in:

The **Navigation PC software** which processes the following main functions: overall Test Receiver control (MMI and configuration files), Navigation Data Recovery, Raw Measurement pre-processing, Navigation Solution Determination, Autonomous Integrity Monitoring and HMI related Computations.

The **Performances Analysis Software** (**PAS**) which analyses, stores and displays the data provided by the Core Modules, the Navigation PC Software. The PAS can control the RF signal and constellation simulators, recover the "truth data" from the simulator and compare with the Receiver outputs. The PAS can also send commands and control the operating parameters of the different subsystems in order to enable automated tests.

4.2.1.6 Time and Geodetic Validation Facility (TGVF)

Corresponding to GRSP and TSP prototypes. GRSP and TSP are described in section 4.2.2 (Service facility).

4.2.1.7 Security Chain Validation Test Bed (SecChain or SECVTB)

The SECVTB is composed of 3 test platforms representative of the independent security chains of the Galileo security architecture (S-Band, C-Band and PRS Chains) that allows for verifying System security functionalities.

The SECVTB consists of a set of Security Items (xKMF, PxSU, CSU, GRCP CB, TUR-P CB) and Test Tools developed for both IOV and FOC system configurations. All Security Items are interconnected by Test Tools emulating the unclassified elements of a given chain (SCCF, MGF, NSGU, etc.), these Test Tools are able to intercept, analyse and modify data flows between the Security Items. Due to its flexibility the SECVTB offers a unique platform that allows for negative testing that cannot be performed anywhere else.

The SECVTB hosts and integrates security equipment from various segments (GMS, GCS, SSEG and TUS), this allows for inter-segment verification. Due to this, the Sec Chain activity is considered a System level verification activity.

SECVTB provides an important contribution to the risk mitigation of System integration activities: evolutions of security equipment are typically first deployed on the SECVTB. This allows for an 'early verification' of the upgraded security functional chain before relevant upgraded security equipment will be deployed on VAL/OPE Chain. Note that this also allows for identifying System issues in advance of System testing on VAL/OPE Chain.

Thus the SECVTB provides a platform for (non-regression) testing of the (functional) Security Chains. The main driver for the non-regression testing on the SECVTB will be to demonstrate that:

- The FOC KMF is able to manage both Type#1 and Type#2 Satellites in parallel
- The GRCP and TUR-P (Test Tool equivalent) are compatible with both Type#1 and Type#2 Satellites (NAVSEC SF)
- The System NCRs are corrected (e.g. confirming patch corrects issue before the execution of the System test)

These non-regression testing activities are typically performed with the support of the OPS team. Note that this allows for training and preparing the OPS team (familiarization with new functionalities/MMI etc) for System tests such as SCTC. In addition the SECVTB allows for verification of Security Operational Procedures (SOP) by OPS teams.

4.2.1.8 SAR Validation Test Bed (SAR VTB)

Support facility corresponding to the prototype of SAR RLSP used during the IOV phase. This prototype is not part of the Galileo infrastructure anymore (contract ended).

4.2.1.9 Satellite and Payload Manufacturers (SATMAN)

4.2.1.9.1 High Level Description

The Satellite and Payload Manufacturers (SATMAN) define a single point of contact (from both a communications and operations perspective) for all spacecraft platform and payload manufacturer entities. Currently, two SATMAN are identified, depending on the type of satellite in the Galileo constellation:

- AST-D for the 4 IOV satellites
- OHB for the 14 FOC1 satelltes

Each SATMAN will supply the GCS with extracts and references from the Satellite Reference Database (SRDB) for each satellite including in particular the TC/TM database (classified and unclassified), operational procedures (classified and unclassified) and satellite On-Board Software (OBSW) images and patches.

SATMAN also provides specialist support to spacecraft anomaly investigations and other contingency operations. To this end, the GCS will supply SATMAN with the information necessary for it to support these activities. This includes the provision of OBSW dumps, operational history reports and transcripts of TC, TM and Event packets sent to and received from the satellite.

The functions performed by SATMAN are expected to be required throughout all Galileo mission phases. The interface will be operated on an irregular basis, both during and outside of normal operations throughout all Galileo mission phases.

It is to be expected that any contact via this interface will be preceded by some form of off-line communication between the involved parties, in order to plan the data exchange.

4.2.1.9.2 Functions Allocation

Function ID	Function name
SATMANF1	Satellite Manufacturer

Table 13: SATMAN Function List

4.2.1.9.3 Entity Organical Context

The following figure shows the SATMAN context:

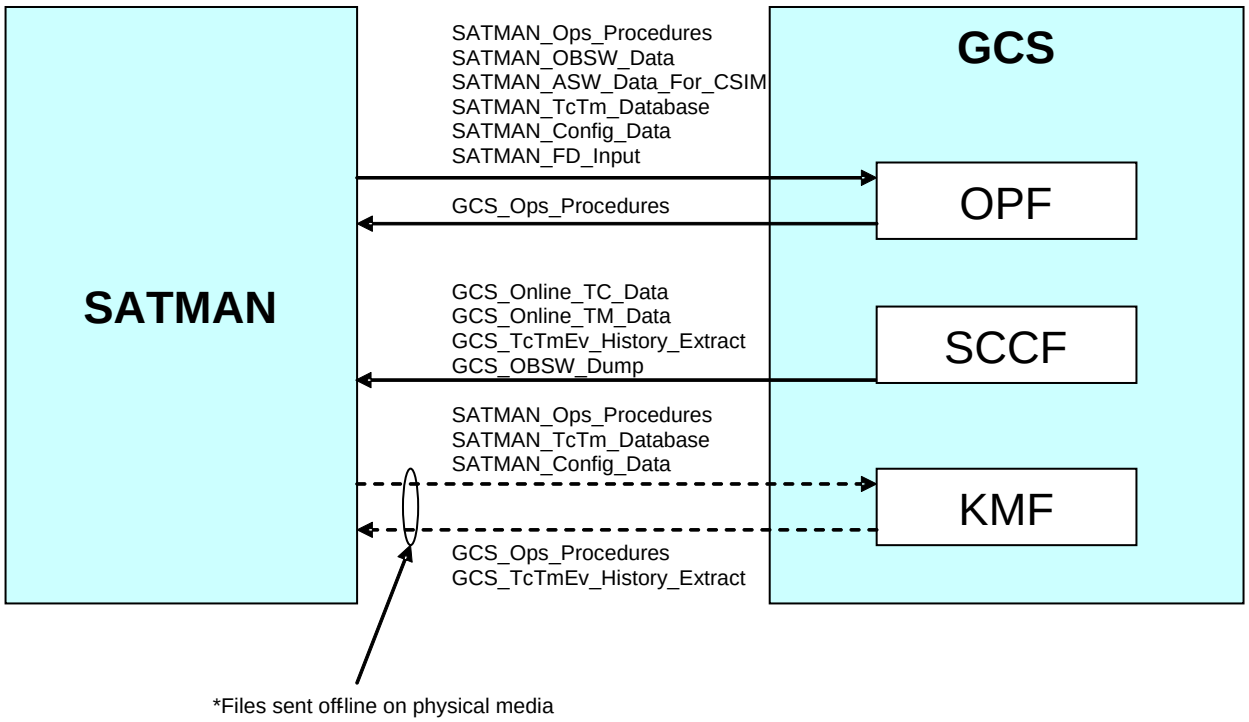


Figure 12: SATMAN Organical context

4.2.1.9.4 Entity Main Design Drivers

Interface to GCS – Security:
SATMAN must provide the means to interface securely with the GCS. This includes the provision of VPN equipment aligned with the system security requirements. The interface will be non-time critical.

Interface to GCS – Data Formats:
SATMAN is required to support the following key data formats for compatibility with the Galileo ground segment:

- Satellite databases (TC/TM and STC/STM) in SCOS-2000 ASCII MIB database format
- OBSW patch and dump in SCOS-2000 OBSM image format

- Operational procedures in MOIS V5 format
- Avionics Software images in S-REC format for input to the Constellation Simulator
- Flight dynamics data for input to the GCS flight dynamics facility in XML format. Content definition and associated schemas depends on the type of Satellite/Manufacturer (IOV-AST-D / FOC1-OHB)
- Support for classified and unclassified deliveries of satellite databases and operational procedures (SATMAN_Config_Data) in XML format.

Satellite Reference Database (SRDB):
SATMAN is required to maintain a reference database to maintain under strict configuration control all data required to characterize each satellite in the Galileo constellation during each step of the system lifecycle. This will act as the central repository for the following data items:

- TC/TM data base (including packet and format definitions, derived parameters)
- Flight dynamics data
- Software parameters
- Sensor/actuator alignment
- Calibration and performance data
- Configuration control data
- Verification data
- Mechanical, thermal, electrical and P/L data
- Flight operations procedures, on-board control procedures

Secure SRDB:
SATMAN is also required to maintain a classified or secure SRDB to manage all classified spacecraft data including in particular:

- Secure TC/TM (STC/STM) database
- Classified operational procedures

Interface from GCS – Data Formats:
SATMAN is required to accept the following key data formats for compatibility with the Galileo ground segment:

- Operational procedures in MOIS V5 format
- Online TM and TC in SCOS-2000 RAPID format.
- Export of TC, TM and Event packet data for an specific satellite over a time period (a PARC export of RAPID files format in SCOS-2000 format)
- OBSW patch and dump in SCOS-2000 OBSM image format
- Operational databases (TC/TM) files in SCOS-2000 ASCII MIB database format

4.2.2 Service facilities

4.2.2.1 Geodetic Reference Service Provider (GRSP)

4.2.2.1.1 High Level Description

GRSP (Geodetic Reference Service Provider) is an entity external responsible for generating the GTRF (Galileo Terrestrial Reference Frame) and assist GMS with its maintenance through the enabling of SLR measurements/campaigns and the provision of the SLR data including the GTRF reports.

Pending the availability of the external entities, the necessary GRSP products are provided via the TGVF (support facility).

Remark: It is important to note that the MRD specification of the geodetic service performance and availability are conditioned to the availability and quality of the GRSP products. Therefore, it is essential to ensure that the GRSP product quality and availability is covered by appropriate GRSP Design and Performance requirements in-line with the MRD geodetic availability and accuracy targets. Please refer to the description of SFC2.3.

4.2.2.1.2 Functions Allocation

GRSP provides:

- the interface with scientific entities for periodical updates of scientific products such as:
  - from NGDC (National Geophysical Data Center):, Earth rotation parameters, Earth model
  - From CSSI (Center for Space Standards and Innovation): Norad Two-Line Elements data
  - From NSSDC (National Space Science Data Center): Ionospheric IRI (International Reference Ionosphere) model/data
  - NGDC: National Geophysical Data Center: Solar flux data
- the interface between Galileo and ILRS for SLR campaigns. Satellite Laser Ranging is a high-precision two-way ranging technique allowing measurement of the distance between an SLR ground station and a spacecraft down to millimeter accuracy. Its products are used by ILRS to feed the Scientific Community involved in geodetic studies, as well as the Geodetic Reference Service Provider.
- an independent validation to the Galileo generated orbits, clocks and ERPs

This covers directly the following GRSP functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GRSPF1	Generate GTRF and Provide related/not Related Products

Table 14: GRSP Function List

4.2.2.1.3 Entity Organical Context

The following figure shows the GRSP context as defined by the GMS-GRSP-ICD and GMSEIREQ

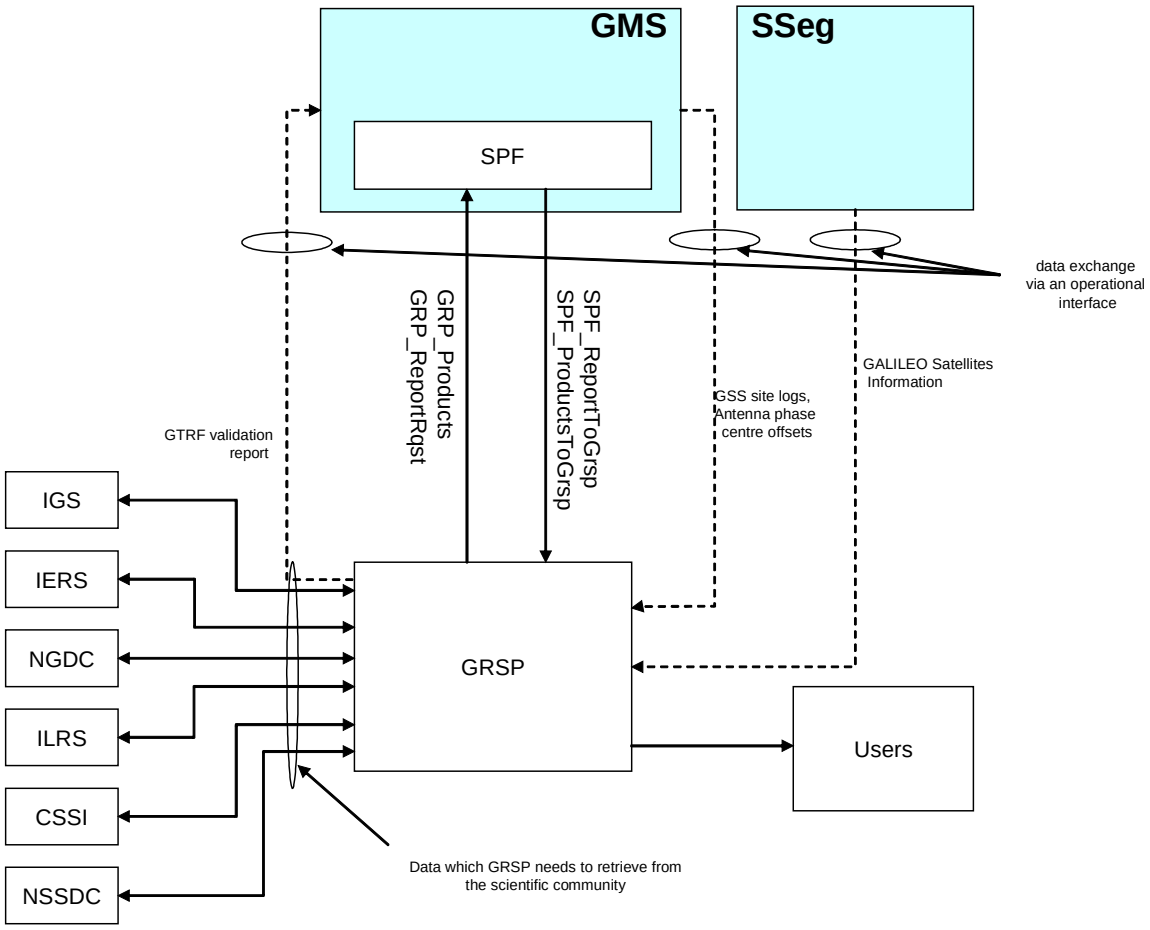


Figure 13: GRSP Organical context

Where:

IGS: International GNSS Service, IGS has been realizing a GPS only ITRF to assure consistency of IGS products by avoiding possible distortions caused by system differences, inaccurate local ties and inaccurate velocity estimation of GPS sites due to short time series because un-modelled non-linear effects may affect the velocity significantly, and so on. Considering the important role of the GPS technology in realizing the ITRF, the IGS realized ITRF and ERP series will be also used by the GRSP for validation. GRSP will retrieve GPS data from the IGS database for realizing the GTRF.

IERS: International Earth Rotation Service, GTRF station coordinates shall be identical to the ITRF station coordinates within 3 cm (2 sigma) for all station markers common for both GTRF and ITRF. Thus, GGSP will rely on IERS, the producer of ITRF and the corresponding EOP series.

NGDC: National Geophysical Data Center, GMS will use solar flux data (10.7 cm) to generate ionosphere correction information for the broadcast message. GRSP is responsible to collect solar flux data and pass them to GMS. Solar flux data can be retrieved from NOAA's NGDC.

ILRS: International Laser Ranging Service, GRSP is responsible for contacting ILRS to coordinate SLR tracking of the Galileo satellites. GRSP forwards orbit prediction and manoeuvre information to ILRS from GMS. GRSP forwards SLR data from ILRS to GMS, including both historical and new. SLR data will be used by GRSP mainly for validation, the feedback information will be sent to ILRS by GMS through GRSP to improve the SLR tracking of the Galileo satellites.

CSSI: Center for Space Standards and Innovation, GRSP will retrieve from the Celestrack, part of the CCSI the Norad Two-Line Elements data which will be provided to GMS for a further provision to GCS.

NSSDC: National Space Science Data Center, IRI: GRSP will retrieve the IRI (International Reference Ionosphere) data which will be provided to GMS for a further provision to GCS.

4.2.2.1.4 Entity Main Design Drivers

GTRF as provided by the GRSP shall be calibrated to International Terrestrial Reference Frame (ITRF) and its updates to within 3 cm (2-sigma).

Note the current realization of the ITRF (ITRF2005) will be considered throughout the lifetime of the Galileo. In case there is a new realization for which the 3cm accuracy can not be met the GTRF will provide also the transformation matrix between the GTRF and the latest ITRF.

Regarding the GTRF maintenance GRSP shall deliver alarms for larger deviations at sites experiencing problems, e.g. antenna problems, earth quakes, etc. These alarms shall be delivered as soon as the problem is detected.

The required GTRF performance with the respect to ITRF can be met by imbedding the GSS's into a larger network of IGS stations and solving their station coordinates simultaneously with all the stations of the whole network. A subsequent network alignment will automatically align the GSSs to the ITRF.

4.2.2.2 Time Service Provider (TSP)

4.2.2.2.1 High Level Description

Galileo establishes its own Reference System Time: GST or Galileo System Time. For specific timing application, this GST needs to be steered to other international time scales such as UTC or TAI. The TSP is an external entity in charge of providing any relevant information to allow GST steering to TAI.

Pending the availability of the external entities, the necessary TSP products are provided via the TGVF (support facility).

Remark: It is important to note that the MRD specification (MRD-260/885) of the timing service performance and availability are conditioned to the availability and quality of the TSP products. Therefore, it is essential to ensure that the TSP product quality and availability is covered by appropriate TSP Design and Performance requirements in-line

with the MRD timing availability and accuracy targets. Please refer to the description of SFC2.1.

4.2.2.2.2 Functions Allocation

The TSP is responsible for:

- measuring the GST to UTC(k) offset, by means of TWSTFT techniques and Common View as backup.
- Computing the GST-TAI offset prediction from these measurements
- Provide GST steering parameter to Galileo system
- Interface with BIPM

Where:

- UTC(k): UTC(k) laboratories provide the TSP with individual clock data and time-transfer data (measured using either GPS CV or TWSTFT or both) on a daily basis
- BIPM: The Bureau International des Poids et Mesures (BIPM) provides the GTSPF with its Circular T on a monthly basis (including the offset between UTC & TAI). The TSP will provide the BIPM with the GMS PTF clock and time-transfer data for possible inclusion in the Circular T.

Galileo is in charge of the generation of GST and its steering to TAI applying the UTC steering correction computed and provided by the TSP.

The following figure presents the main functions associated to the GST generation and the boundaries between Galileo and the external TSP.

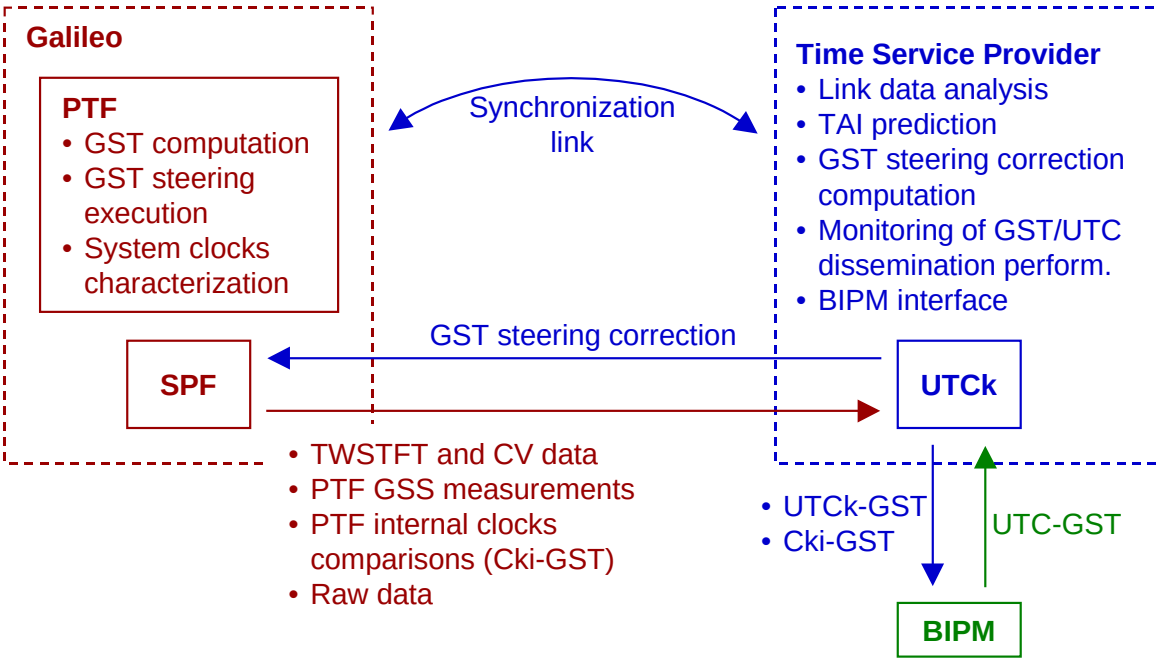


Figure 14: Boundaries between Galileo and the external TSP

This covers directly the following TSP functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
TSPF1	Metrological Timekeeping of GST

TSPF1.1	Time Transfer
TSPF1.2	GST Steering Generation

Table 15: TSP Function List

4.2.2.2.3 Entity Organical Context

The following figure shows the TSP context as defined by the GMS-TSP-ICD and GMSEIREQ.

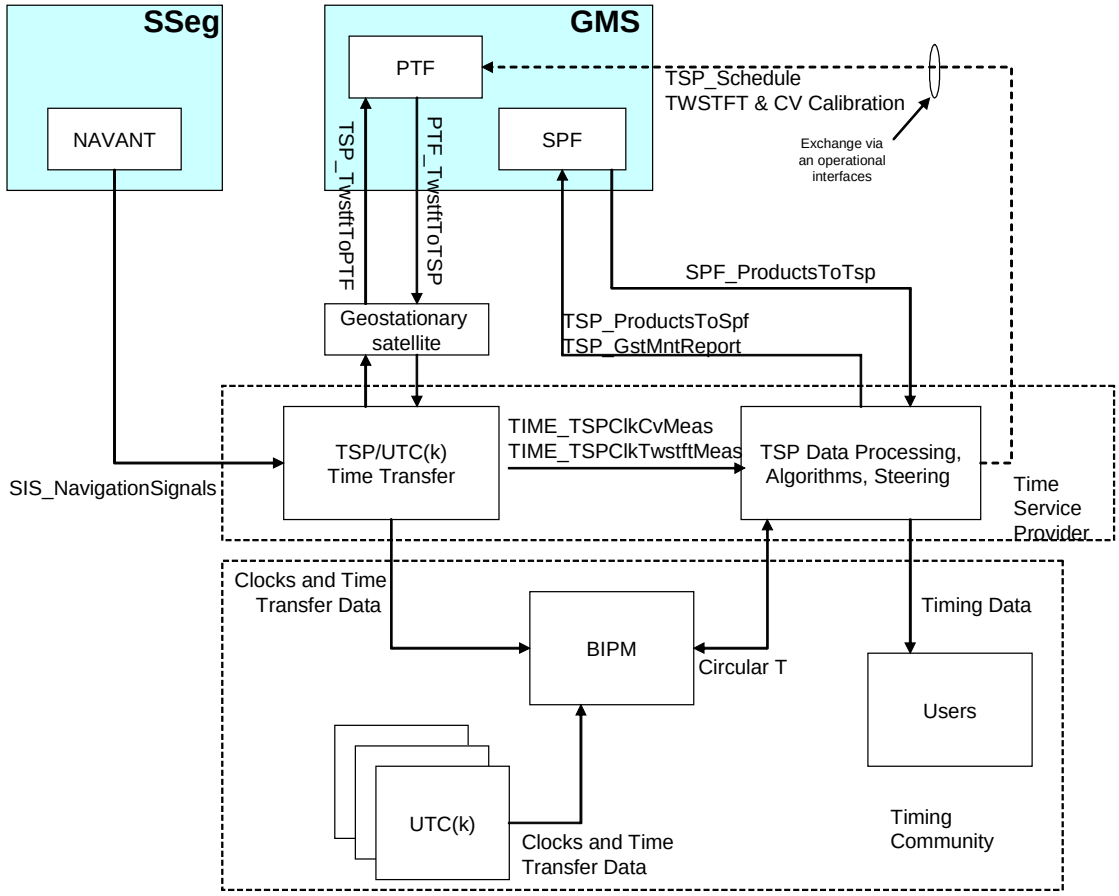


Figure 15: TSP Organical context

Where:

- BIPM: The Bureau International des Poids et Mesures (BIPM) provides the GTSPF with its Circular T on a monthly basis (including the offset between UTC & TAI). The TSP will provide the BIPM with the GMS PTF clock and time-transfer data for possible inclusion in the Circular T.
- UTC(k): UTC(k) laboratories provide the TSP with individual clock data and time-transfer data (measured using either GPS CV or TWSTFT or both) on a daily basis.
- Users: TSP provides a web site including performance reports on GST (MC) to TAI steering for interested parties (eg, ESA, GSA etc).

4.2.2.2.4 Entity Main Design Drivers

TSP main design drivers can be split into two groups:

- Technical implementation drivers (correct realization and the deployment of all interfaces)
- Performance drivers (delivery of the products meeting the specification)

4.2.2.2.4.1 Technical Implementation Drivers

The overview of the deployed interfaces between the TSP and Galileo core system is provided in the following figure:

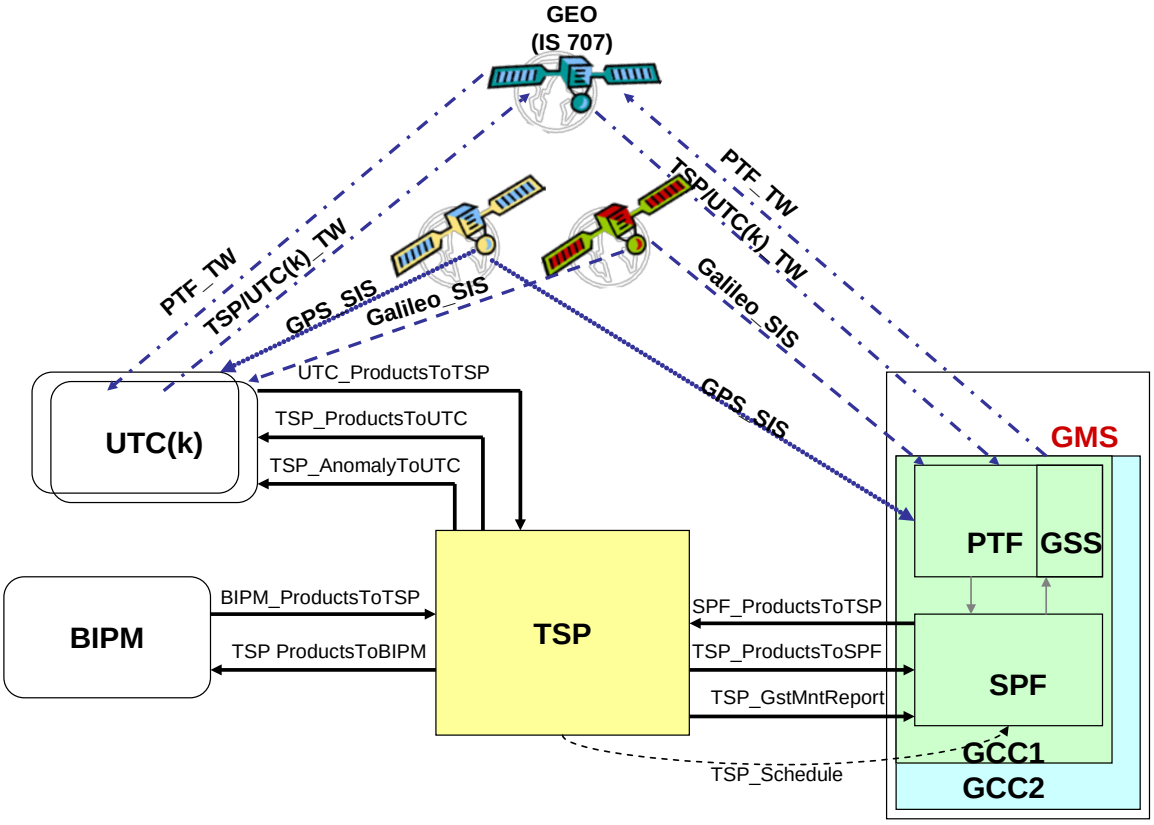


Figure 16: TSP-GCC interface architecture overview

Since TSP does not possess any time transfer hardware a number of UTC(k) laboratories contractually bounded with the TSP will assist TSP to deliver its steering products. The term:" TSP/UTC(k) laboratory" is used to distinguish the UTC(k) laboratories contractually bounded with TSP to play role in GST steering process from the UTC(k) laboratories contributing to TAI but which are outside the scope of the TSP contract.

TWSTFT method for GST-TAI steering:
To enable the TWSTFT operations it is necessary to select the appropriate Geo-stationary satellite allowing for the measurements to be carried within EU and between EU and US. The provision of the required services is subject to the SLA with the satellite operating company. Further more the TWSTFT operations need to comply with the telecommunication statutory directives of the country hosting the PTF.

Notes:
1. with the use of IS-3R satellite, the polarization needs to be changed when switching from the EU to US (transatlantic) sessions. Consequently both PTFs will have to modify their current TWSTFT hardware base, for instance by including the configurable OMT (otherwise TWSTFT measurements between the EU and US will not be possible).
2. TSP/UTC(k) laboratories were required to incorporate into their hardware configurable OMT within the frame of their contribution/role in TAI determination (TAI links).
3. the usage of the IS-3R is only temporary as Intelsat is planning to relocate the TWSTFT operations onto a different satellite which has not been launched yet. It is not known whether the configurable OMT will be required once the service is handed over to a new satellite.

BIPM usage:
TSP will rely also on the support from the BIPM and the GMS provision of the Galileo timing data as specified in GMSTSP ICD.

4.2.2.2.4.2 Performance drivers

GST shall be a continuous time scale and it shall be steered towards TAI with a maximum offset of 50 ns for 95% of the time during any yearly time interval and the difference shall be predictable with an uncertainty of 28 ns (95% confidence).

Note the uncertainty between GST and TAI stems mainly on the fact that TAI is made available only in deferred time once per month. Real-time operations require GST steering to be made towards a prediction of TAI, TAIp, and the maximum prediction interval can be up to six weeks.

Given the superior time transfer performance achievable with the TWSTFT, the successful implementation and operation of the TWSTFT interfaces between GMS and TSP is representing a necessary prerequisite, for the above GST-TAI requirements to be met.

4.2.2.3 Galileo Security Monitoring Center (GSMC)

The Galileo Security Monitoring Centre is an external facility to the Galileo system that acts as a European focal point for the monitoring and oversight, at EU level, of Galileo security-related functions. The GSMC shall be operated by an agency external to the Galileo core system, with the authority to make policy decisions relating to PRS security operations on behalf of the participating member states of the Galileo project.
The primary responsibility of the GSMC is the implementation, with regard to Galileo, of EU security policy, and providing direction to the implementation of the PRS.
For detailed description, see Classified Annex.

4.2.2.4 Galileo Reference Center (GRC)

The primary mission of the Galileo Reference Centre (GRC) is to perform an independent monitoring of the Galileo Services - Open Service (OS), Public Regulated Service (PRS) Navigation and Commercial Service (CS) and report it to the relevant stakeholders.

The GRC interfaces with the other elements of the Galileo System are under consolidation at the date of writing of this document. It is foreseen that the GRC will interface:

- With the Space Segment, to receive the Signals in Space (OS, PRS, CS)
- With the TSP and GRSP, to retrieve offline the timing and geodetic products required to support the performance analysis task (details TBD)
- With the GSC (details TBD)
- With the GSMC, to provide PRS navigation performance monitoring assessment and expert support on navigation (details TBD)
- With designated Member States facilities, in order to exchange data and products (details TBD)

4.2.2.5 Service Centre (GSC)

4.2.2.5.1 High Level Description

The GSC is the external entity dealing with all the administrative, legal and commercial issues related to the provision of Galileo services. This is the single interface between the Galileo core system and the other service centres. It implements its own Service Centre to deal with various operational issues and manages requests from other Service Centres.

The GSC is only available as of System Build 1.5.1.

Functions Allocation
The main functionalities of the GSC are:

- Managing Contracts, Litigation, Operations and Marketing, Insurance and Liability, Certification, Direct Local Service Operation etc.
- Centralizing requests to Galileo: resolving possibly conflicting requests, prioritising requests from different Service Centres
- Supporting Commercial Service: sorting priorities of Commercial Data requests for transmission via the CS, billing, controlling key management requests.

This covers directly the following GSC functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SCF1	CS Data Upload
SCF2	CS Data Response Process
SCF3	Key Management
SCF4	CS Resource Management

Table 16: GSC Function List

4.2.2.5.2 Entity Organical Context

The following figure shows the GSC context:

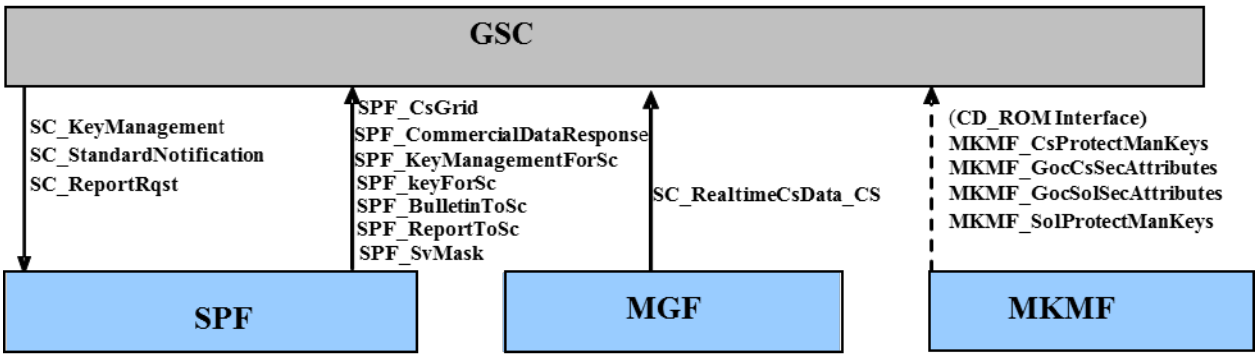


Figure 17: GSC Organical context

4.2.2.5.3 Entity Main Design Drivers

- **Single Interface with GMS:**

The Galileo Service Centre (GSC) is the only Service Centre that interfaces directly with the Ground Mission Segment of the Galileo Core Infrastructure. GSC manages requests from other Service Centres: centralizing requests to GMS, resolving possibly conflicting requests, prioritizing requests from different Service Centres, sorting priorities of Commercial Data requests for transmission via the CS, billing, controlling key management requests, etc.
Service providers are understood as the entities providing the users with specific Galileo services. The term "Service Provider" refers to any facility implemented to offer navigation, timing, other navigation-related data or revenue-generating services to the users. Identified facilities can be called "Service Centres" (SCs).

- **Commercial Service Support:**

One of the most important design drivers for GSC is the real time commercial service capacity to be used, defined by the CS_RealtimeData. This message will include the Commercial Service contents required for uploading the message format according to the following table:

User Level	**CS_RealTimeData Message**											
	< 25 kB											
SIS Level	**CS_Data Satellite 1**			**CS_Data Satellite 2**			**…**	**CS_Data Satellite 36**		
	PT 14	**CD** 448 bits		**PT** 14	**CD** 448 bits		…	**PT** 14	**CD** 448 bits	
Uplink Level / GSC Level	**CS_Data Packet (1/2)**		**CS_Data Packet (2/2)**	**CS_Data Packet (1/2)**		**CS_Data Packet (2/2)**	**…**	**CS_Data Packet (1/2)**		**CS_Data Packet (2/2)**
	PT 14	**CD** 247	**CD** 201	**PT** 14	**CD** 247	**CD** 201	…	**PT** 14	**CD** 247	**CD** 201

Table 17: CS_RealtimeData Message Format (without headers)

The following acronyms are used in

User Level	CS_RealTimeData Message											
	< 25 kB											
SIS Level	CS_Data Satellite 1			CS_Data Satellite 2			…	CS_Data Satellite 36				
	PT	CD		PT	CD		…	PT	CD			
	14	448 bits		14	448 bits			14	448 bits			
Uplink Level / GSC Level	CS_Data Packet (1/2)		CS_Data Packet (2/2)	CS_Data Packet (1/2)		CS_Data Packet (2/2)	…	CS_Data Packet (1/2)		CS_Data Packet (2/2)		
	PT	CD	CD	PT	CD	CD	…	PT	CD	CD		
	14	247	201	14	247	201		14	247	201		

Table 17: PT for Page Type and CD for Commercial Data.: The Page Type (14 bits) content is defined by Commercial Service Provider/GSC to identify service data for Commercial Data (448 bits) in the same page.
In order to fit GMS uplink format, the GSC shall provide CS data in grouped packets of 261 bits (14 bits for Page Type +247 bits for Commercial Data) and 201 bits for Commercial Data), plus an identifying header (used for routing on the ground, but not for uplink). The CS Data may be encrypted using the CS service keys, and may include key management messages for this purpose. The content of the CS Data is not relevant to the GMS, which will simply strip the CS Data out and insert it into the uplink.
Note that CS data packet (1/2) and CS Data Packet (2/2) shall be considered as part of the same single page (contains a total of 14 bits for Page Type description and 448 bits of Commercial Data), this means that both packets are merged in space segment and included as a single page (disseminated in 1 second) in the C/NAV.

4.2.2.6 Search And Rescue Service Provider (SAR)

4.2.2.6.1 High Level Description

COSPAS-SARSAT is a satellite system designed to provide distress alert and location data to assist Search And Rescue (SAR) operations. Spacecraft and ground facilities are used to detect and locate the signals of distress beacons operating at 406 MHz.

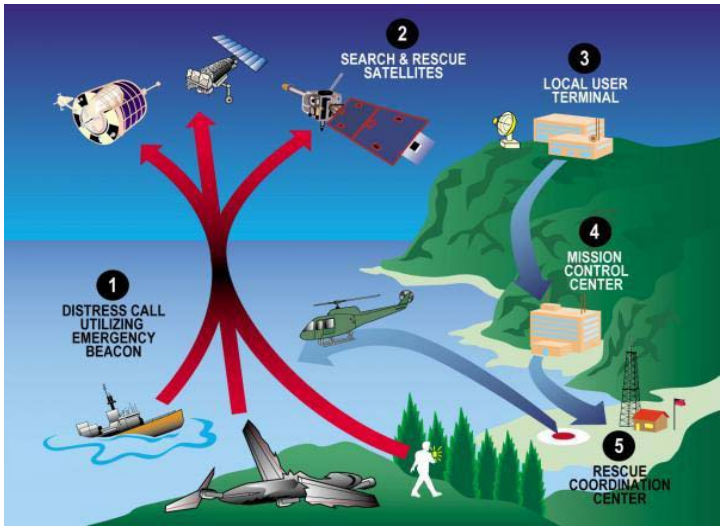


Figure 18: SAR System Operational Overview

The Galileo system will provide SAR/Galileo Search and Rescue service in support to the COSPAS-SARSAT Medium Earth Orbit (MEOSAR) program by:

- relaying, detecting and localizing distress signals from Search and Rescue (SAR) beacons operating according to COSPAS-SARSAT standards : Forward Link Service
- relaying responses to these distress signals back to those SAR beacons equipped by Galileo receivers: Return Link Service

The figure below shows the overall Galileo/MEOSAR system, where the Galileo satellites the MEOLUTs and the RLSP are part of the Galileo Core system.

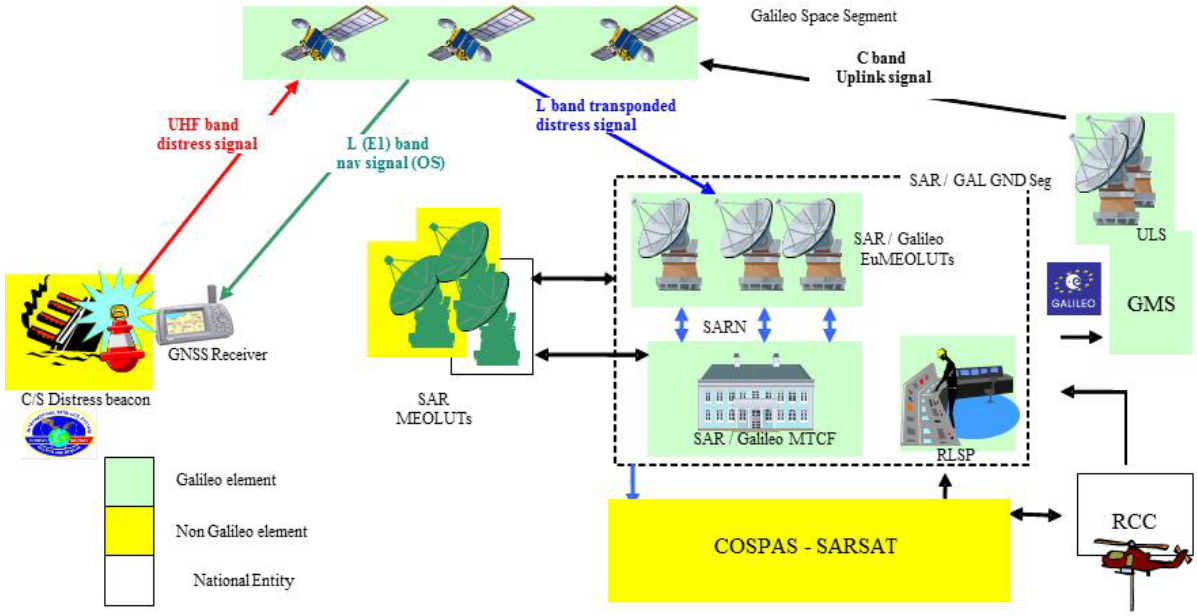


Figure 19: SAR Architecture Overview

The SAR Galileo system includes the following components:

- SAR Galileo Space segment including:
  - SAR/Galileo Payload on Galileo satellites, responsible for relaying the 406MHz distress signals from user beacons to the Local User Terminal (MEOLUT): support to forward link.
  - Galileo satellite Navigation payload, responsible to relay the Return Link Messages

- The SAR/Galileo Forward Link Ground Segment infrastructure including:
  - Three SAR/Galileo European MEOLUTs allowing the reception of distress messages from the SAR payloads in their visibility and the processing of the messages for distress localisation and message detection. The number of SAR/Galileo MEOLUTs, their capacity (in terms of the number of tracked satellites) and their locations, are such that the forward link service coverage requirements as specified in GSRD-641 are met.
  - One MEOLUT Satellite Tracking Coordination Facility (MTCF) coordinating the tracking schedule the European MEOLUTs and the External MEOLUTs in order to maximize the SAR forward link coverage capability. MTCF is also involved in the centralized functions such as the Monitoring and Control functions, Archive Data Management functions and KPI collection, computation and reporting
  - RLS emergency beacons (REBs): 10 RLS emergency beacons for test purposes for Demonstration and Evaluation test phase
  - SAR/Galileo Reference Beacons: 5 Reference beacons with stable characteristics placed at known locations that allow the monitoring of SAR/Galileo system performance parameters at the MEOLUT sites and the collection of SAR Key Performance Indicators (KPI)
- The SAR/Galileo Return Link Ground Segment infrastructure including:
  - One Return Link Service Provider (RLSP) that interfaces with Cospas/Sarsat, concentrates all inbound distress messages and generate return link messages to be transmitted to the distress beacons

The SAR/Galileo MEOLUTs, RLSP, MTCF and Reference Beacons are hosted on sites located within the European territories of EU or ESA member states.

4.2.2.6.2 SAR Forward and Return Link Components

This section provides a short description of the elements involved in the SAR service provision, either internal or external to the SAR/Galileo perimeter

4.2.2.6.2.1 SAR Beacons

The Cospas Sarsat System provides alerting services for Emergency Beacons operating at 406MHz these are one of the following types:
- Emergency Locator Transmitters (ELTs) for aviation use
- Emergency Position Indicating Radiobeacons (EPIRBs) for maritime use
- Personal Locator beacons (PLBs) for applications which are neither aviation or maritime

Some SAR Beacons are equipped with GNSS receiver and have the capability to receive the Galileo downlink navigation signal which contains the SAR Return Link Message (RLM) as a feedback from the distress signal

Up-Link Signal RF Characteristics*:*

The SAR beacon transmits two types of digital message, short and long messages, with a transmission time of 440ms and 520ms respectively.

Current Cospas-Sarsat emergency beacons within the 406.0-406.1 MHz band according to the Cospas-Sarsat channel Assignment (SAR-RD.1).

The transmitter power output is 5 W +/- 2 dB (35 to 39 dBm). Power output rise time is less than 5 ms (measured between the 10% and 90% power points).

The following antenna characteristics are defined for elevation angles greater than 5 degrees and less than 60 degrees:

- - Pattern: hemispherical
- - Polarisation: circular (RHCP) or linear
- - Gain: between -3 dBi and 4 dBi over 90% of the above region
- - Antenna VSWR: not greater than 1.5:1

4.2.2.6.2.2 SAR MEOLUT

The MEOLUT is a ground receiving station responsible for the detection and localization of emergency beacon signals relayed by Galileo satellites and of forwarding the extracted distress information to its associated MCC.

Functions Allocation

The main functions of the MEOLUT are:

- Galileo/not Galileo satellite tracking according to their own tracking plan (Stand Alone mode) or to the tracking plan received by the MTCF (Networked Mode)
- SAR burst detection and processing, including message demodulation
- TDoA/FDoA parameter estimation
- SAR beacon localization through DOA method using the TOA/FOA estimated locally and /or exchanged with other MEOLUTs or exploiting the position data that may be embedded within the beacon distress messages
- Generation of alert data (i.e. location data, beacon ID, alert message)
- Interface with MTCF (reception of tracking plan and transmission of the related telemetry)
- Interface with MCC associated to the MEOLUT (transmission of alert data)
- Interface with other MEOLUTs (exchange of TDoA / FDoA measurements)

Performance requirements for SAR/Galileo European MEOLUTs *:*

The SAR MEOLUT is capable to receive Galileo/SAR signals from Galileo,GPS and Glonass satellites and the following characteristics are assumed:

- The ratio of effective antenna gain to overall system noise temperature (G/T) of the MEOLUT receiving sub-system is selected so that the target BER (< 5 10-5 for elevation angles higher than 5 degrees) can be achieved. This corresponds to G/T higher than 4 dB/K
- The MEOLUT receiver antenna is also capable of tracking satellite transmissions in LHCP.

The MEOLUT receiver meets the following specifications:

- Time reference accuracy: within +/- 50 ns of UTCProcessing threshold: C/N0 > 34.8 dB-Hz
- Time tagging accuracy: < 7 µsec (1 sigma), measured at MEOLUT processing threshold using a calibrated input signal fed directly into the MEOLUT

- Frequency measurement accuracy: < 0.1 Hz (1 sigma), measured at MEOLUT processing threshold using a calibrated input signal fed directly into the MEOLUT

The SAR/Galileo MEOLUTs are hosted on sites located within the European territories of EU or ESA member states.

Country	Location	Element
Spain	Maspalomas	Operational MEOLUT
Norway	Svalbard	Operational MEOLUT
Cyprus	Larnaca	Operational MEOLUT

Table 18: European MEOLUT locations

4.2.2.6.2.3 MTCF (MEOLUT Tracking Control Facility)

The MTCF is mainly in charge of coordinating the European MEOLUT network to optimize the use of the available resources (satellites and tracking antennas) and to exchange TDoA / FDoA data among MEOLUTs.

Functions Allocation

The main functions of the MTCF are:
- Tracking plan definition for Galileo MEOLUTs
- SAR Forward Link Monitoring and Control
- SAR data archiving
- KPI collection, computation and reporting

MTCF is located in Toulouse at CNES premises

4.2.2.6.2.4 MCCs (Mission Control Centres)

MCC is in charge of the routing of the distress information received by the MEOLUT/MTCF.

Depending on the beacon location, the MCC can:

- Forward the data to another MCC
- Contact appropriate SPOC (RCC)

Data Distribution Regions: A DDR comprises two or more MCC service areas. Cospas-Sarsat alert data and System information are exchanged between DDRs through a single MCC which acts as the point of contact for that DDR. This MCC is identified as the nodal MCC of the DDR. However, bilateral arrangements can be implemented between adjacent MCC service areas included in different DDRs to facilitate the exchange of alert data in overlapping service areas or adjacent search and rescue regions.

4.2.2.6.2.5 SAR RLSP (Return Link Service Provider)

The purpose of the Return Link Service is to relay SAR Return Link Messages, e.g. messages carrying information from the RLSP to the beacons in distress, or rescue services, using the Galileo system
The Return Link Service Provider acts as the single interface between Galileo Ground Segment and the Cospas-Sarsat system. Cospas/Sarsat processes distress signals from active beacons and interacts with the SAR community.

The purpose of the RLSP is to relay SAR Return Link Messages carrying information to the beacons in distress. RLSP transmit the RLM request to the GMS that then uplinks the Return Link Message via two Galileo satellites, based on support information provided by the RLSP.

The RLSP provides, within each RLM request, a list of S/Cs visible by the user at the RLM release, calculated based on FW link and ephemeris computations considering the beacon location. Updates of this list are sent to GMS if the visibility of satellites as seen by the user in distress changes.

The RLSP receives information from the GMS regarding the status of the messages to be disseminated/cancelled, the current GMS SAR status and the current/future status of the constellation.

Functions allocations:
The RLSP carries out the following functions:

- Interface to Cospas Sarsat
- Resource management
- Dissemination management

4.2.2.6.2.6 Forward and Return Link Physical Context

The distribution of Cospas-Sarsat alert data throughout the world is summarized as follows:

1. the MEOLUTs receive the beacon signals relayed by the satellites
2. the relayed signals are processed and alert data is sent to the associated MCC for distribution.
3. Each MCC distributes alert data to any country within its service area which has agreed to accept such services. Alert data is provided to Service Point of Contacts (recognized national points of contact) that will use the data to enable fast and effective rescue of persons in distress.
4. Some alert messages, processed by Cospas Sarsat, result in a request for RLM sent to the RLSP.

The following figures give an overall view of the SAR/Galileo Forward and Return Link physical context:

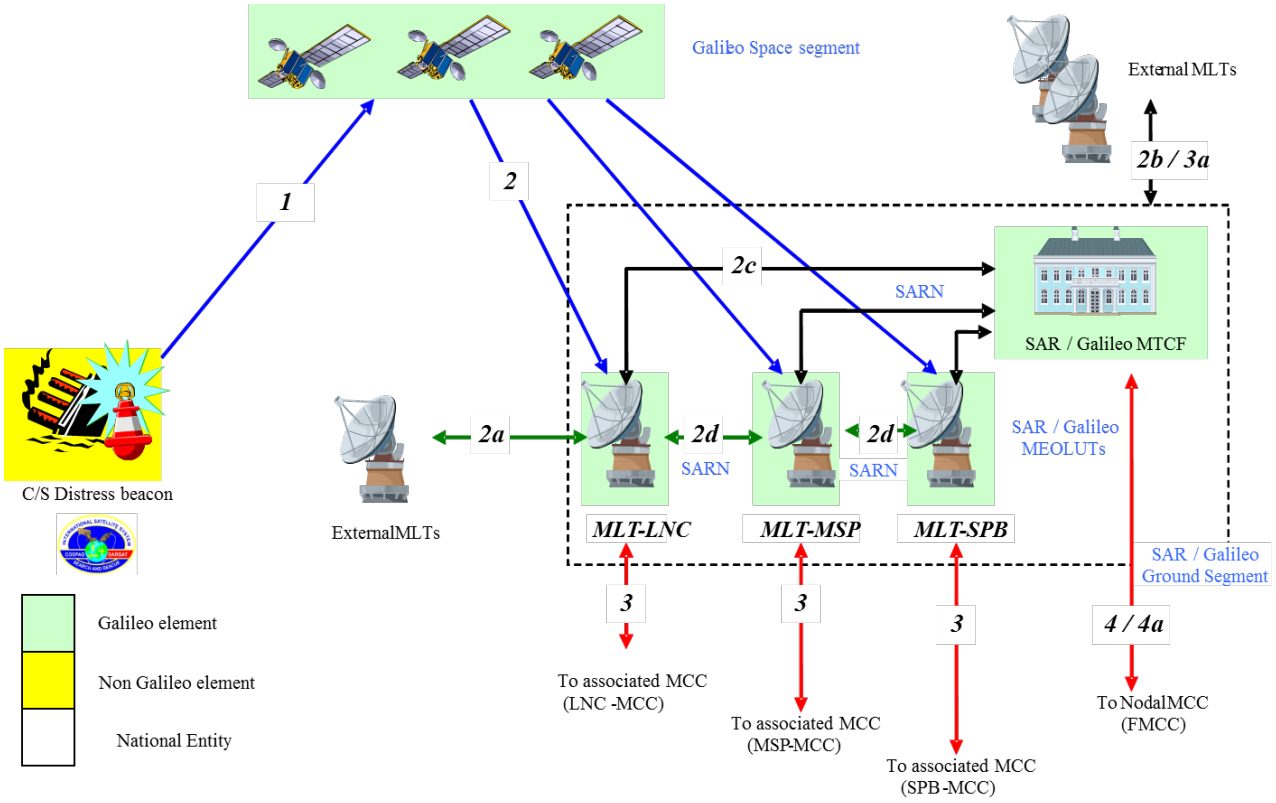


Figure 20: SAR Forward Link context

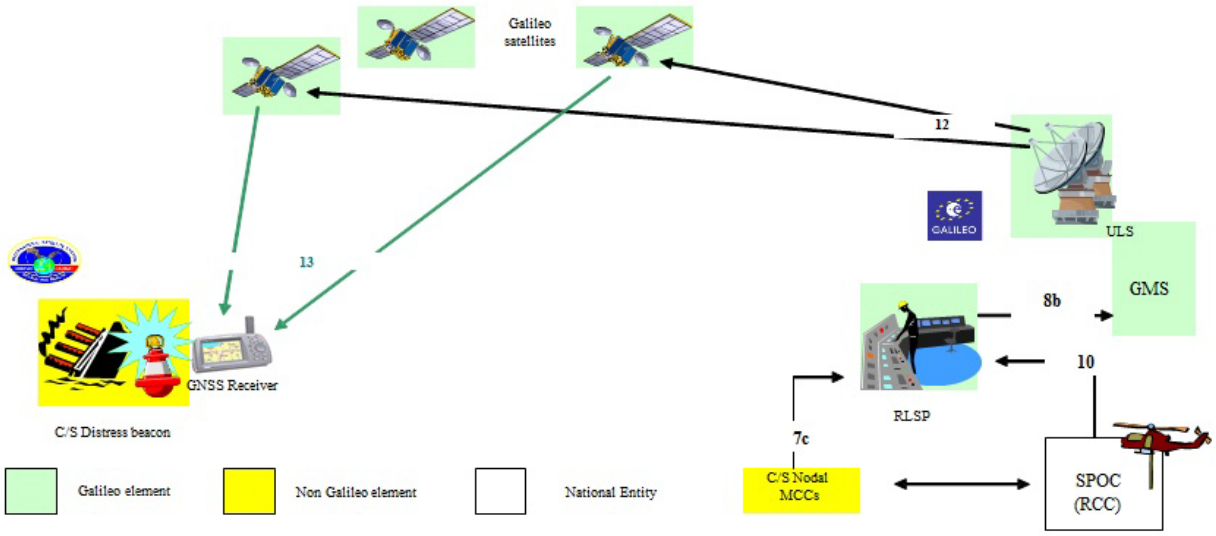


Figure 21: SAR Return Link context

Remark: The RLSP will not be available in the SB 1.5.0 but is to be connected in a future System Build

4.2.2.7 USNO

4.2.2.7.1 High Level Description

In the frame of Galileo interoperability with GPS Galileo provides in the navigation message the offset between the GPS and the Galileo timescales (Galileo to GPS Time Offset). This allows users to determine their position through reception of both Galileo and GPS SIS.
The computation of the GGTO parameters is performed in coordination with GPS/USNO (United States Naval Observatory), to which a dedicated interface has been identified.

The following figure shows the interface between Galileo and GPS/USNO:

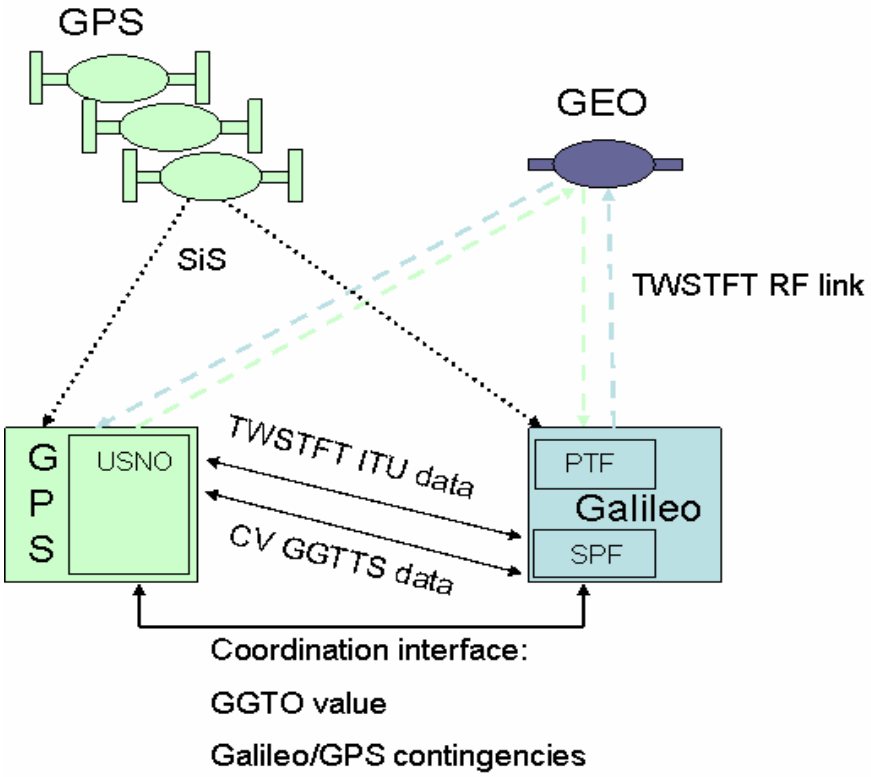


Figure 22: GGTO interfaces between Galileo and GPS

4.2.2.7.2 Functions Allocation

The GPS/USNO provides within the scope of the GGTO interface the following functionality:

- Access to the GPS system time by the means of TWSTFT and GPS-CV time transfer links.
- Ccalculation/determination of GGTO using combined GPS/Galileo receiver (FOC baseline, needing at least 14 operational satellites) and TWSTFT/CV (IOV baseline)
- GGTO result coordination and error debugging/handling
- exchange of GGTO interface related contingency scenarios

This covers directly the following GPS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name

GPSF1	Provide GPS System Time (USNO)
GPSF2	Generate GPS SIS (CV Time Transfer)

Table 19: GPS/USNO Function List

4.2.2.7.3 Entity Organical Context

The following figure shows the GPS context as defined by the GMS-GRSP-ICD and GMSEIREQ.:

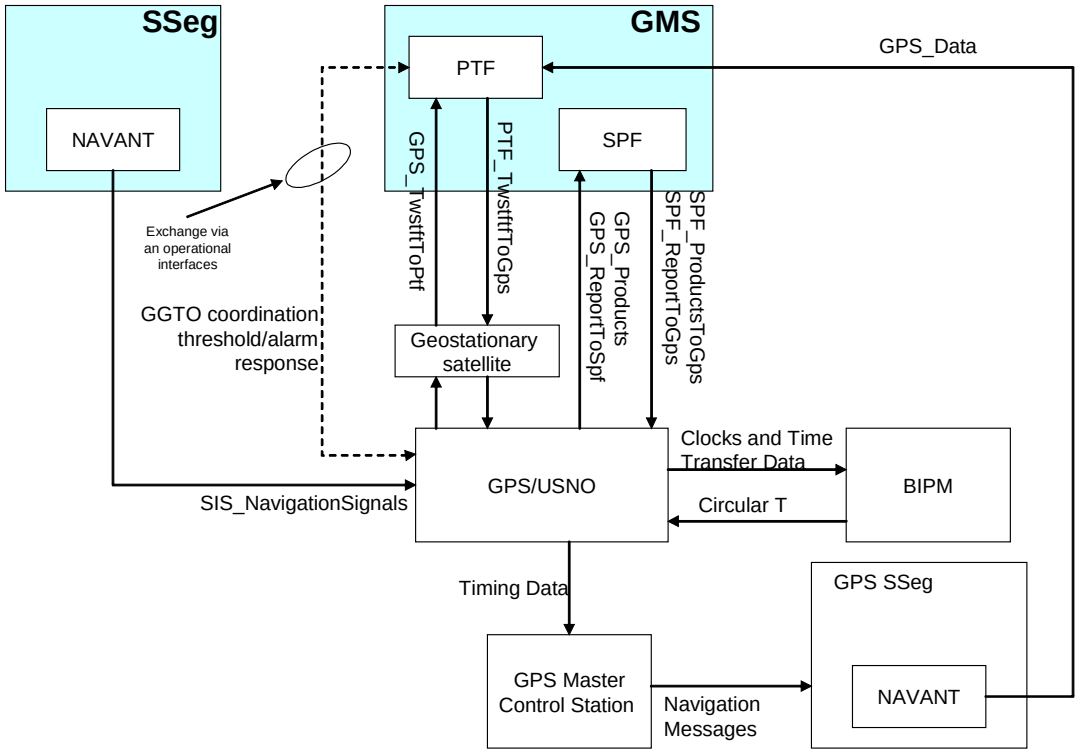


Figure 23: GPS/USNO Organical context

4.2.2.7.4 Entity Main Design Drivers

The accuracy of the offset between GST and GPS Time (modulo 1 s) shall be less than 5 ns with 2-sigma confidence level over any 24 hours.

The stability of the GGTO, expressed as an Allan deviation, shall be better than 2e-14 over any one day.
“Note: The GGTO instability refers to the GST(MC)-UTC(USNO) offset as measured via TWSTFT link between Galileo PTF and USNO”.

The validity period of the GGTO shall be minimum 24 consecutive hours.

GPS/USNO needs to ensure that their contribution to the GGTO overall performance is in line with the above specification.

4.2.3 Other external entities connected to Galileo core infrastructure

4.2.3.1 Satellite Laser Ranging (SLR)

4.2.3.1.1 High Level Description

Satellite Laser Ranging is a high-precision two-way ranging technique allowing measurement of the distance between an SLR ground station and a spacecraft. Its products are used by ILRS to feed the Scientific Community involved in geodetic studies, as well as the Geodetic Reference Service Provider.

There is no direct interface between the Galileo Core Infrastructure and the ILRS (apart from the satellite beam between SLR stations and the Galileo satellites), all data exchange is performed between GMS and the GRSP.

GRSP based on the data received from GMS coordinates with ILRS the SLR campaigns and is also in charge of the SLR data retrieval .

The figure below provides the outline of the Galileo satellites SLR tracking campaign with the emphasis on the interfaces which needs to be in place in order to enable the SLR tracking of the Galileo satellites.

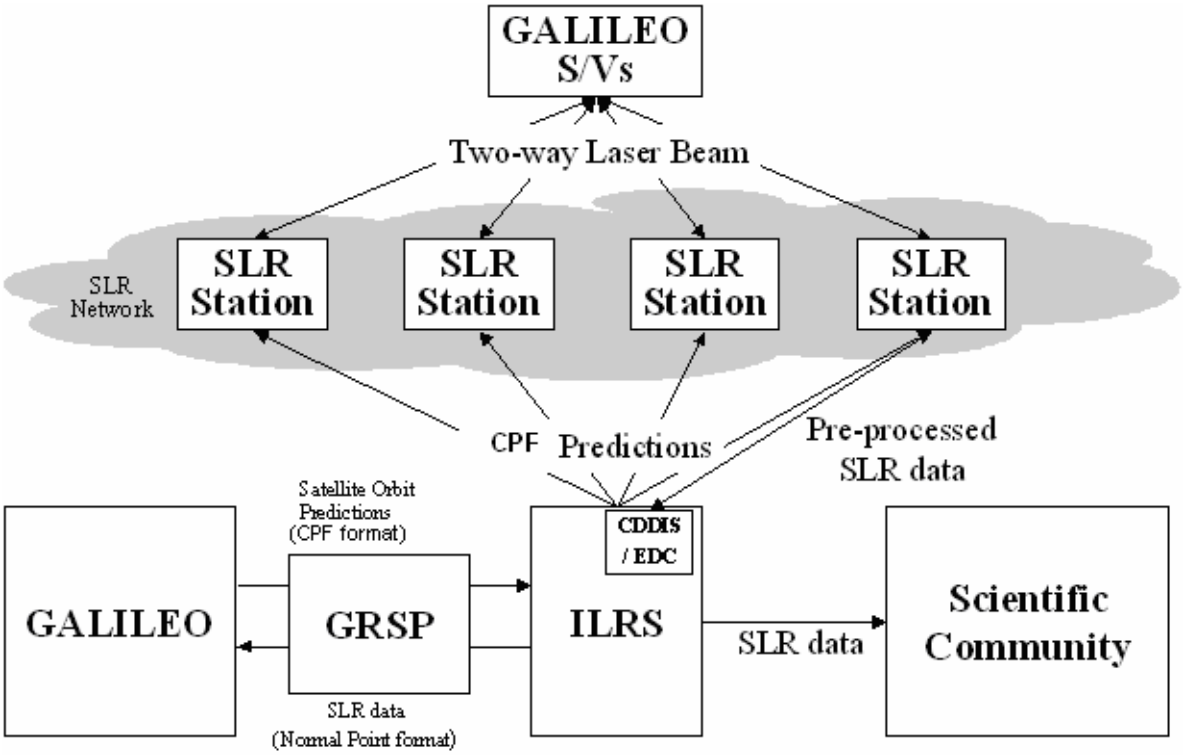


Figure 24: Galileo satellites SLR tracking campaign principle

ILRS includes a network of SLR stations. Based on orbit predictions for spacecraft provided with Laser Retro-Reflectors, and a table of priorities and/or special arrangements. SLR stations can autonomously identify visibility passes by performing numerical propagation of the prediction state vectors according to predefined models.

4.2.3.1.2 Functions Allocation

SLR contributes to the GRSP function for GTRF generation:

Function ID	Function name
GRSPF1	Generate GTRF and Provide related/not Related Products

Table 20: SLR Function List

4.2.3.1.3 Entity Organical Context

The following figure shows the SLR context as defined by the GMS-GRSP-ICD and GMSEIREQ

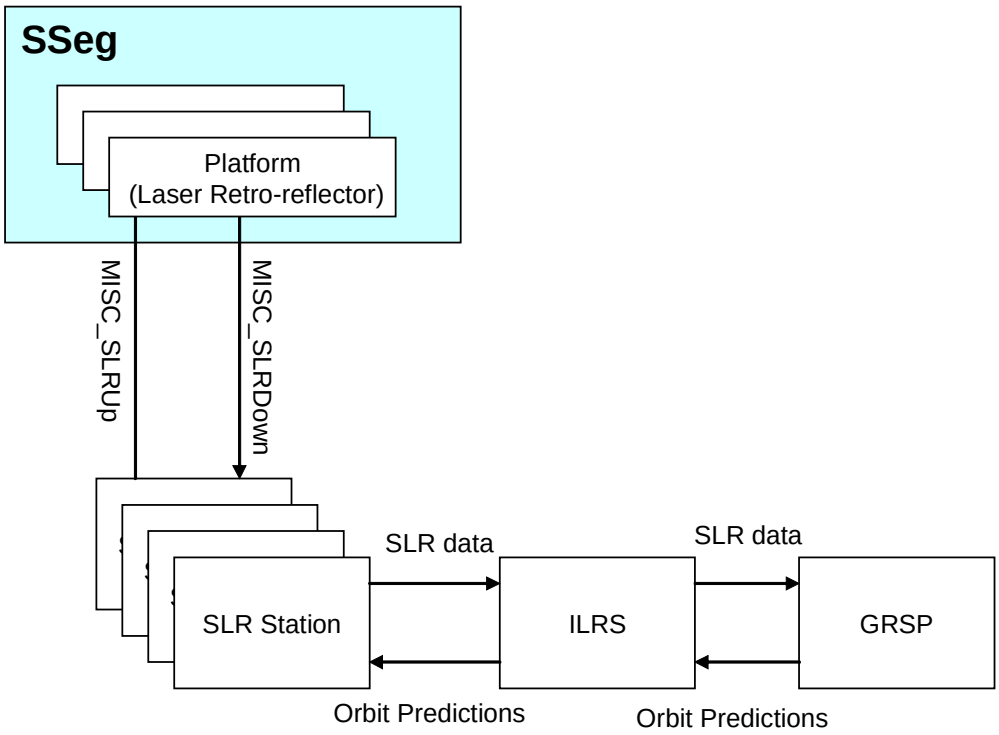


Figure 25: SLR Organical context

Where:

- ILRS: International Laser Ranging Service, ILRS mission is to support, through satellite and lunar laser tracking data and related products, geodetic and geophysical research activities.
- SLR station: Satellite Laser Ranging Station

4.2.3.1.4 Entity Main Design Drivers

The success of the SLR campaign (accuracy of SLR measurements shall be better than 2 cm RMS) depends on the three aspects:

- Limited availability of the SLR stations: the SLR campaigns require careful planning as ILRS is engaged in many scientific projects. (In practice GMS will have to request the SLR campaigns well in advance).
- Technical restrictions: the elevation of the Galileo satellites (23000 km) limits the number of SLR stations able to track Galileo satellites. (In practice only the stations equipped with the laser transmitters powerful enough will be able perform the tracking). The technical characteristics of the SLR stations are provided below.
- Quality of provided orbit predictions: The GRSP shall obtain 120 hours of orbit predictions in the CPF format with an accuracy (1sigma) to be, during the first 48 hours of the predictions, better than or equal to:
    - 9 m radial
    - 165 m along-track
    - 15 m cross-track

Note the quality of the SLR measurements is directly linked with the quality of the GCS generated orbit predictions. The better the predictions are the more successful SLR measurement points are collected, resulting into the measurement noise to be lower.

The main characteristics of the SLR stations, which currently and operationally provide laser-ranging measurements on the LEO-MEO satellites, are:

- Laser wavelength: 532 nm
- Energy per pulse: 300 – 50mJ (50mJ is considered here as the dimension value)
- Pulse frequency: typically 10 Hz
- The satellite is observed for different elevation angles, in the range (90° - 15° above the horizon): 15° is considered here as the dimension value.
- Emission/reception telescope: 100 cm diameter (typically)
- Required number of return photoelectrons: >=1 per pulse

5. GALILEO CORE SYSTEM ORGANICAL BREAKDOWN

5.1 INTRODUCTION: PHYSICAL MAPPING OF FUNCTIONAL ARCHITECTURE

The following section intends to describe the mapping of the functions identified in the Logical part of the DDF (DDF part 1) to physical elements part of the FOC Galileo Core System. The system physical DDF presents architecture details down to element level. The objective is not to substitute segment documentation but highlight element architecture constraints useful to understand system issues. Anyhow, for each constraint described, it is indicated if it comes from system constraints (and linked to related system requirements in DOORS) or from segment design.

In addition, any intermediate configuration and associated limitation during each phase of development and deployment up to FOC configuration are detailed in SYS-RD.74 and AD.48.

For each element within the Galileo segments a dedicated section includes:

- a high level description
- a functional mapping: list of the functions identified in the logical architecture [SYS-RD.01] mapped to the element
- an entity organical context: to show the physical interface of the element (with which other components the element interfaces with, and which signals are exchanged).
- A list of the main design drivers

The following figure shows the context diagram of the Galileo core system.

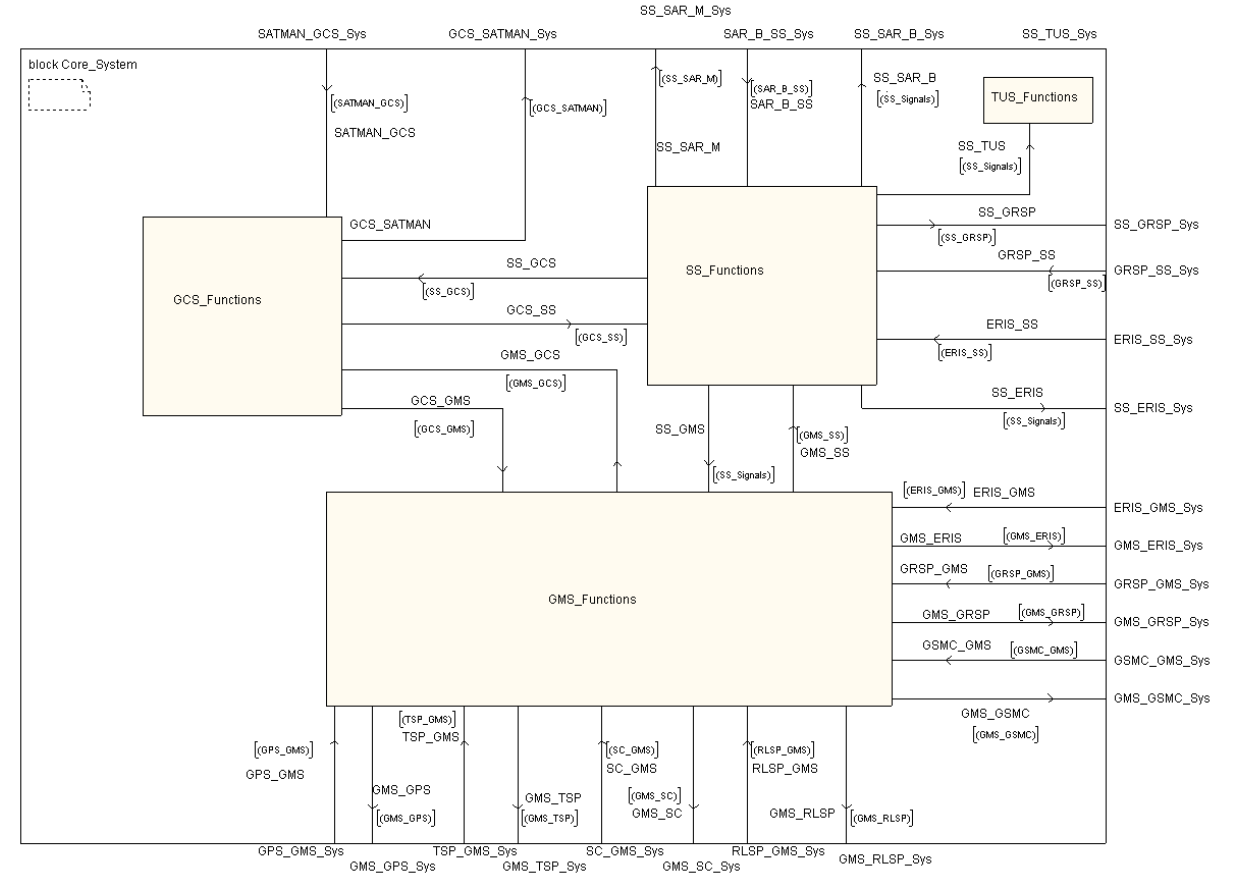


Figure 26: Galileo Core System Organical context

5.2 GCS SEGMENT

5.2.1 GCS OVERVIEW

The Ground Control Segment (GCS) is mainly responsible for the monitoring and control of the Galileo constellation.

The GCS contributes to the following system functional chains:

- SFC3 - Maintain Overall System Synchronization
  - Maintain the satellite payload time synchronization to the GST (by sending synchronization TCs to the satellites)
  - Maintain the GCS elements synchronized to the GST
- SFC4 - Mission Data Provision
  - Transmission of "degraded navigation data" and PRS data received from the GMS and uplink in S-Band (via TTCs)
- SFC5 - Monitor and Control
  - Constellation Monitoring and Control (reception of TM and generation of TCs)
  - Perform ground asset monitoring and control for the GCS (in coordination with GMS)
  - Manage TTC Links
  - Management of the L-Band TM data relayed by the GMS (for constellation monitoring and control)
  - GCS planning activities (short and mid term planning in coordination with GMS)
- SFC6 - Archiving
  - Manage the GCS local and central archive
  - Databases Management (spacecraft, operational procedures, anomaly)
- SEC-SFC12 Security Protection Of Satellite Monitoring And Control Security management:
  - Key management and data protection management for the Secure TCs/TMs

This involves the following GCS functions as shown in DDF Part 1 (SYS-RD.01):

- GCSF1 - GCS Short Term Planning (SFC5)
- GCSF2 - Manage TTC Link (SFC5, SFC6)
- GCSF3 - Archiving (SFC5, SFC6)
- GCSF4 - Constellation Monitoring and Control (SFC3, SFC5, SFC6, SFC7)
- GCSF5 - GCS Monitoring and Control (SFC5, SFC6)
- GCSF6 - Provide Training
- GCSF7 - Perform Simulation and Test
- GCSF8 - Manage Databases
- GCSF9 - Security

5.2.1.1 GCS ORGANICAL DESCRIPTION

The following figure shows the overall architecture and context of the GCS:

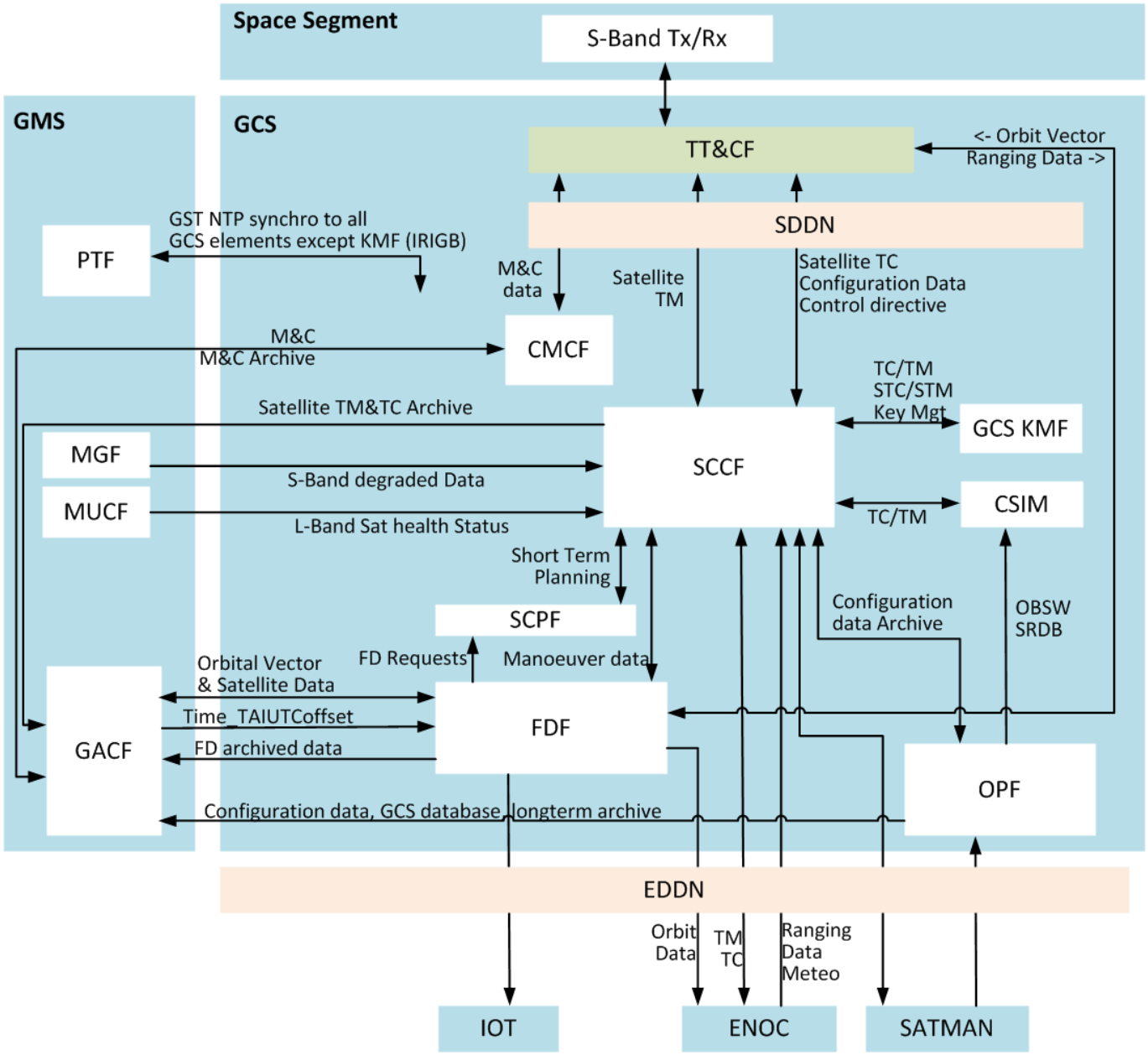


Figure 27: GCS Organical context

Notes:
1) In order to keep it readable, this context diagram does not represent all the links but shows only the most relevant ones. For detail interface between elements, refer to internal ICDs
2) For details on time distribution from PTF to GCS elements, see SFC3.3 physical diagram

The GCS is composed of the following element:
- CMCF: Central Monitoring and Control Facility:
- CSIM: Constellation Simulator
- FDF: Flight Dynamics Facility

- GCS KMF: GCS Key Management Facility
- OPF: Operations Preparation Facility
- SCCF: Spacecraft & Constellation Control Facility
- SCPF: Spacecraft & Constellation Planning Facility
- Network: SDDN (Satellite Data Distribution Network) and S-EDDN (External Data Distribution Network)
- TTCF: Telemetry Tracking and Control Facilities (TT&C Stations)

Further details on GCS elements are provided in the following sections.

5.2.1.2 GCS FUNCTIONS PHYSICAL MAPPING

The following table lists the system functions allocated to the GCS and the mapping to GCS elements:

Function ID	Function name	Mapping to Element
GCSF1	GCS Short Term Planning	SCPF/FDF/SCCF
GCSF2	Manage TTC Link	TTC/SCCF/SDDN
GCSF2.1	TTC Link Modes	TTC/SCCF/SDDN/KMF
GCSF2.2	Two Way Ranging	TTC/SDDN
GCSF3	Archiving	OPF/CMCF/FDF/SCCF
GCSF3.1	Central Archiving	OPF/CMCF/FDF/SCCF
GCSF3.2	Local Archiving	Elements
GCSF4	Constellation Monitoring and Control	SCCF/FDF/KMF
GCSF4.1	Execute Schedule and Special Operation	SCCF/TTC/FDF/KMF
GCSF4.2	Constellation Monitoring	SCCF/FDF/KMF
GCSF4.3	Manage OBSW	SCCF
GCSF4.4	Flight Dynamics	FDF
GCSF5	GCS Monitoring and Control	CMCF
GCSF5.1	GCS Element Monitoring and Control	CMCF/TTC
GCSF5.2	GCS Central Monitoring and Control	CMCF
GCSF5.3	GCS Network Monitoring and Control	SDDN/CMCF
GCSF6	Provide Training	CISM
GCSF7	Perform Simulation and Test	CSIM
GCSF8	Manage DataBase	OPF
GCSF8.1	SC DataBase	OPF/KMF
GCSF8.2	Operational Procedure DataBase	OPF/KMF
GCSF8.3	Anomaly Database	OPF
GCSF8.4	Document Database	OPF
GCSF9	Security	KMF
GCSF9.1	Security protection of S-Band	KMF
GCSF9.2	Security protection of ground infrastructure	KMF

Table 21: GCS Function List

Note: the mapping of functions per element is detailed in each element description.

5.2.1.3 GCS OPERATIONS, VALIDATION AND SW UPGRADE CAPABILITIES

The physical implementation of the GCS allows supporting simultaneously live space constellation operations in parallel to ground and validation testing activities, allocating the relevant activities on redundant but physically separated hardware.
In particular the GCS foresees the deployment in each GCC of three functionally identical "chains" of classified and unclassified equipment, namely:

- Operations Chain (OPE) that is used to support operations with the space constellation. The specific characteristics of this set of equipment are:
  - Connectivity via Site Data Handling Set (SDHS) to remote TT&C ground stations for the exchange via SLE protocol of TM and TC data with spacecraft in orbit.
  - Frozen configuration of operational products (Databases, procedure etc), which are not modifiable using nominal procedures on the OPE chain.
- Validation Chain (VAL), which is a fully representative replica of the GCS HW deployed on the OPE chain, intended for Operations product preparation, validation and for operators training in an environment that is as close as possible to the real operational context. The specific characteristics of this equipment set are:
  - The connection of the SCCF to the Constellation Simulator (CSIM), replacing the spacecraft and the connection to TT&C stations for the exchange of TM and TC data
  - The OPF instance on the VAL chain can connect to the external Satellite Manufacturer to receive spacecraft data (SW patch data, Ops procedures, databases) and is able to send data back TC/TM data from the archive ;
  - The capability to be configured with same operative conditions applied on the OPE chain.
  - Provision of the support to prepare and validate of all operational products (Databases, procedures, etc).
    Once fully validated using the CSIM, Operational products can be transferred to the OPE chain via the OPF (see sect. 5.2.2.5).
  - The OPF is the GCS element where all the operational products are kept under configuration control and where all GCS Configuration Items (CI) required by other GCS elements are stored. The OPF is provided with all necessary isolation and security measures to establish a unidirectional connection from the VAL to the OPE chain to transfer products that are validated on the OPE chain.
  - The support different types of Tests to be performed in conditions have to be similar to real operational conditions, e.g.:
    - o System Validation Testing (SVT) with new spacecraft on ground. This allows reserving the GCS resources of the OPE chain to the support of M&C functions for satellites already in orbit, while fulfilling security constraints related to the separation of Operational and Test contexts (e.g. use of separate encryption keys). For this purpose the replica SDHS part of the VAL chain are used to establish SLE connections with remote spacecraft EGSE installed at the spacecraft integration site. Anomalies Troubleshooting: for this purpose the capability is provided to configure the operational context as for the OPE chain and playback real TM data transferred from the archive
    - o Testing of GCS SW upgrades. The VAL chain allows testing the upgraded version of GCS element SW that have already passed preliminary tests in an operational-like context to get the final approval for the transfer on the OPE chain.

  - Testing of on board SW patches received from the SATMAN using the CSIM, before upload on board.
  - Testing of manoeuvre commands before actual use on real satellites.
  - System testing that benefit of the connection between the GCS VAL and GMS VAL to validate exchanges over the GCSGMS Interface.
- Preparation Chain (PREP). For FOC an additional instance of the OPF is provided separately from other GCS elements. This installation is connected on the "administration LAN" at the GCC, with no direct interfaces to other GCS elements. This facility was provided as support to the Operations team for Operational products preparation on top of the facilities provided on the VAL chain. The PRE chain:
  - is provided with a connection to the VAL via an dedicated EDDN (External Data Distribution Network) in order to meet security requirements. Data can be also transferred between PREP and VAL chain using offline media;
  - supports Real time synchronisation of PREP OPF instances across GCCs via a suitable network link (outside GCS, on the administrative network);
  - allows the GCC Ops team to access to the OSM (OPF Supervisor MMI) and to MOIS from their office computers by using remote desktop technology (provided that the PREP chain is integrated with the GCC administration LAN;
- Integration Chain (INT). This is a only a functional replica of the GCS elements, with less HW resources that those deployed on the OPE and VAL chain. The INT chain supports preliminary local integration testing of element evolutions and in particular for GCS SW upgrade test.

The equipment replica on the VAL and INT chains are not expected to provide the same RAMS and performance requirements of the OPE chain. Separate LANs are foreseen to interconnect the equipment of each GCS chain.

- OPE, VAL and INT LANs are physically and logically independent in both GCC
- The OPE and VAL LANs will be interconnected across GCCs and further connected to the GMS NRT LAN
- Independent local INT LAN will exist in each GCC for the only purpose of connecting the GCS elements of the INT chain (no connection to GMS LANs and other external connections).

5.2.1.4 GCS HW CONFIGURATION IN THE TWO GCCS

The GCS dual GCC deployment for FOC implied the procurement of new additional facilities to be installed at GCC-I, to complement the GCS IOV installation at GCC-D.
During the procurement of the new HW, an issue had to be tackled related to the evolution of the Operating System (OS) running on the new generation of HP server families used in the previous GCS installation.

The new HP machines (e.g. HP DL380 G7 servers), based on an evolution of the OS (SUSE Linux enterprise v11), did not support any more the previous OS version (SUSE v9) supporting key SW applications like SCOS, used by the SCCF.

To overcome this limitation, a solution was identified to run the GCS SW in a virtualised environment. This allows running, through the so called a hypervisor layer, the previous SUSE v9 OS without additional changes. Further advantages were identified when combining

virtualisation with the increased processing power offered by the new machines, like the capability to host several GCC elements application in a single platform and the capability to share CPU and Hard disk capacity, with an overall resource optimisation. Specific care was taken to aggregate elements on a single platform to guarantee the necessary redundancy level in every aspect of the solution, including networking and storage to avoid any single point of failure. See GCS-RD.1 for further details.

In the present build 1.5.0 two different GCS configurations have been implemented:

- at GCC-D, where the first IOV operational installation was deployed, the GCS is based on IOV HP platforms supporting the SUSE v9 OS with no virtualisation. The HW configuration is capable of supporting a space constellation made of up to 18 spacecraft on both OPE and VAL chain (8 spacecraft only on the INT chain used for on-site testing) using up to 5 concurrent TT&C links.

- at GCC-I, taking advantage of the newer and more powerful HP machines evolution, the GCS is based on virtualised environment, capable of aggregating on single server several virtual elements with apportioned resources in terms of CPU, RAMS, and disk capacity. The original element SW (as used in GCC-D) and relevant OS is installed on a virtual machine, which runs exactly on the same specifications as the physical hardware in GCC-D. Separation of HW is foreseen for the different GCS chains. The virtualised environment at GCC-I is dimensioned to support a space constellation of up to 26 satellites on both the OPE and VAL chain (8 satellite on the INT chain) with up to 6 concurrent TT&C links (however only the 5 TTCF of Kiruna, Kourou, Noumea and Reunion are integrated : the 6th TTCF of Papeete is being procured but not yet integrated)

As consequence of the above, the GCS capability to support the GCC handover operations is limited by the GCS HW resources available at GCC-D, able to cope with a constellation comprising 18 satellites and 5 TTCF at most.

The next step for Build 1.5.1 [AD.48] foresees the installation of the GCS virtualised environment capable of supporting 26 spacecraft and 6 TTCF also at GCC-D, with the 6th TTCF of Papeete fully integrated and capable of being operated within the GCS.

5.2.2 GCS ELEMENTS DESCRIPTION

5.2.2.1 CENTRAL MONITORING AND CONTROL FACILITY (CMCF)

5.2.2.1.1 HIGH LEVEL DESCRIPTION

The CMCF is responsible of the Monitoring and Control functions of the Ground facilities of the GCS (excluding satellite M&C managed by the SCCF).
The CMCF also supports the high-level M&C interface with the GMS Ground Asset Control Facility (GACF).

5.2.2.1.2 FUNCTIONS ALLOCATION

According to GCS DDF [GCS-RD.1] the CMCF is responsible for the following main functions:
- Monitor and Control of the GCS Hardware and Software within the local GCC
- Monitor and Control of TT&C stations
- Interface with SCCF to provide TT&C station availability status
- M&C interface to the GACF (local and remote)
- Maintenance of local GCS elements and TT&C stations
- Receipt, storage and processing of summary status information on the local GCS networks from the local GMS
- Central archive of all the GCS M&C data
- Security event centralisation

This covers directly the following GCS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GCSF3.1	Central Archiving
GCSF3.2	Local Archiving
GCSF5.1	GCS Element Monitoring and Control
GCSF5.2	GCS Central Monitoring and Control
GCSF5.3	GCS Network Monitoring and Control

Table 22: CMCF Function List

5.2.2.1.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the CMCF context according to the GCS DDF (GCS-RD.1), GCS-GMS-ICD (AD.8):

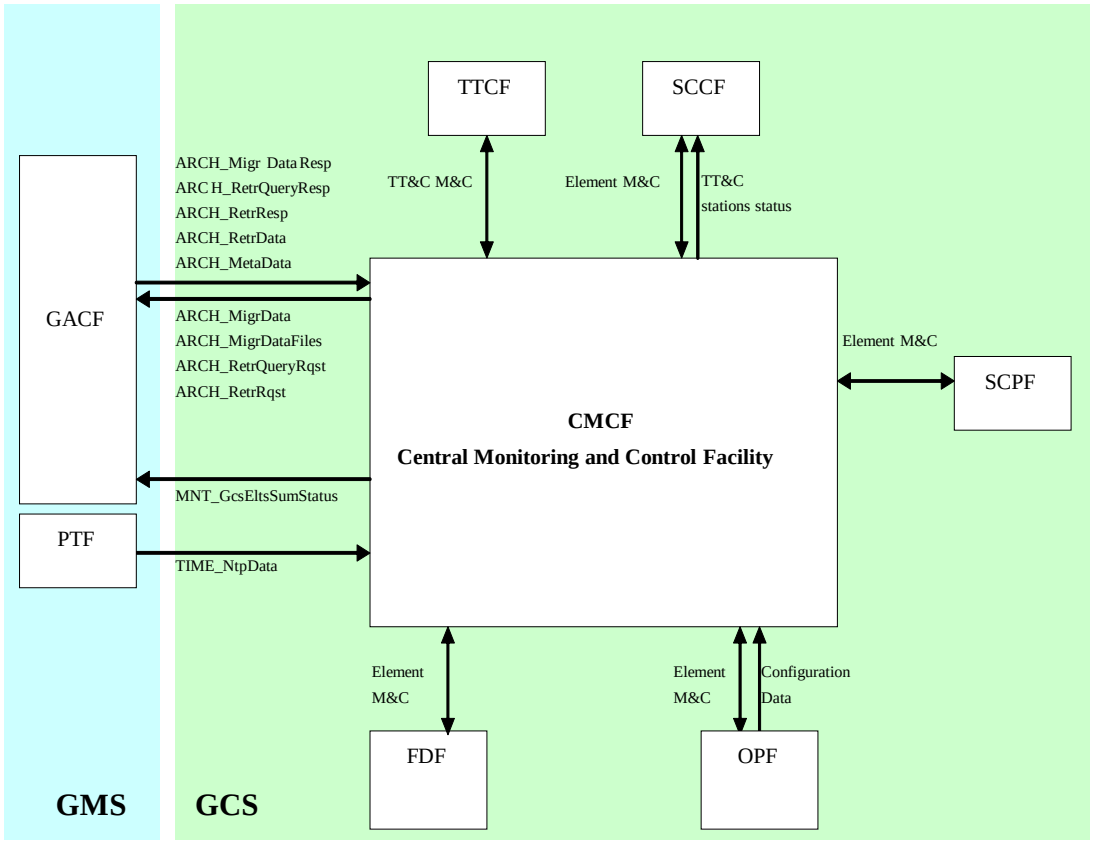


Figure 28: CMCF Organical context

5.2.2.1.4 ELEMENT MAIN DESIGN DRIVERS

5.2.2.1.4.1 SYSTEM LEVEL CONSTRAINTS

Sharing of M&C Responsibility with GMS

In the GCS:

- the CMCF, with its associated MMI, provides the level 0 and Level 1 M&C,
- the GACF provides the level 2 M&C.

The concept of three M&C operational levels in each GCC is the following:

- Level 0: level for GCS Element specific maintenance, via local dedicated Element MMIs available at each Element and at the CMCF.
- Level 1: level for both GCS mid-level management and remote element maintenance through a GCS specific central MMI within the GCC.
- Level 2: level to monitor and control all ground segment Elements (both GCS and GMS) during normal operations via an MMI located within each GCC. For the GCS, Level 2 focuses on high level M&C operations.

CMCF Archiving Capacity

- The CMCF have sufficient storage to record all M&C data changes that have occurred in the past 3 years.
- The CMCF retrieve any Element M&C archived data (up to 3 years old) within 1 minute for off line analysis, excluding time to uncompress the data.

Security event centralisation
The CMCF is responsible for the collection, and processing of security related events within the GCS and the forwarding of Security Events to the GMS. This is performed within the Security Event Centralisation function of the CMCF (cf DDF CLA).
Remark: for what concerns the S-KMF, S-KMF security events are directly sent to GMS P-KMF without CMCF involvement.

5.2.2.1.4.2 MAJOR SEGMENT DESIGN CHOICES

Operability:
- CMCF is normally unmanned and is used by operators to carry out GCS specific maintenance tasks and when the GACF in the GMS is unavailable.
- CMCF archived data are automatically transferred to the Galileo Global Archive in the GACF. Transfer frequencies are configurable by operator.

Redundancy Management:
Each CMCF operates independently of the other CMCF in both GCC and therefore there is no synchronization of data between them.

5.2.2.2 CONSTELLATION SIMULATOR (CSIM)

5.2.2.2.1 HIGH LEVEL DESCRIPTION

The Constellation Simulator (CSIM) supports a number of pre-launch and operational maintenance activities for the GCS. It provides a high fidelity simulation of the entire Galileo constellation of spacecraft and all of the operational ground stations.
The CSIM provides the following capabilities:
- It models the behaviour of the constellation and the reception and transmission of Telemetry (TM) & Telecommand (TC) data by modelling the satellites in the constellation and the TT&C stations.
- It also provides a real time interface to the SCCF so the overall mission control system can be validated before the ground and space networks are fully deployed
- It provides training of operations engineers and procedure validation throughout the Galileo constellation lifetime corresponding to GCS “GCSF6 - Perform Training.
- It allows for verification of procedures throughout the constellation lifetime corresponding to GCS functions “GCSF7 - Perform Simulation and Test ”
To meet these needs, the CSIM can be used in a single spacecraft environment according to the testing and training required or in a constellation environment where all satellites in the actual constellation are modelled.

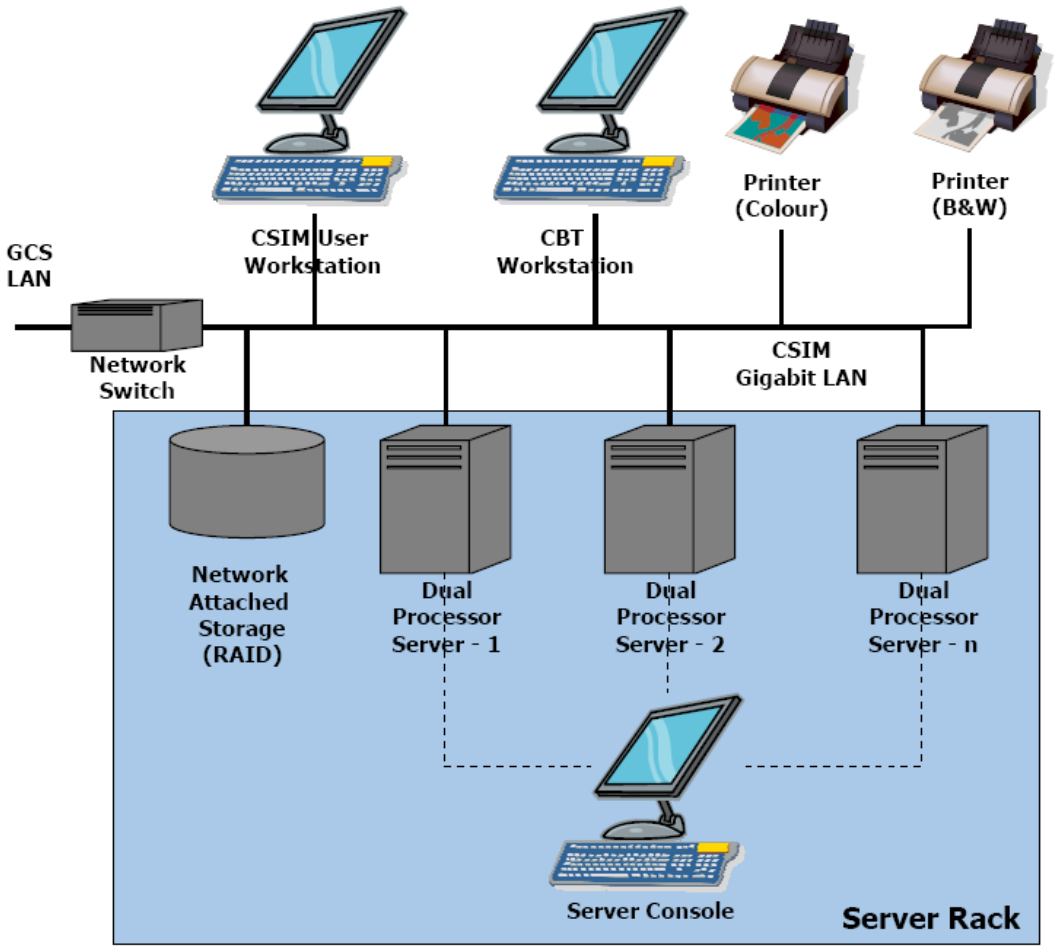


Figure 29: CSIM Overall Architecture

5.2.2.2.2 FUNCTIONS ALLOCATION

The functional scope of the CSIM includes the following major functions:

- Initial validation and commissioning of the Spacecraft Constellation Control Facility (SCCF)
- Training Flight Control Staff
- Validating flight control procedures for both LEOP and nominal operations
- Validating flight control procedures for use in anomaly situations
- Validating on-board software patch procedures prior to upload to the spacecraft

The CSIM supports the following functions identified in the System DDF1 (SYS-RD1):

1) The system functional chain SFC5.2 Satellite Monitoring and Control ;
2) The following GCS functions:

Function ID	Function name
GCSF6	**Provide Training**
GCSF7	**Perform Simulation and Test**

Table 23: CSIM Function List

Several CSIM instances are deployed, namely:

- None in GCS OPE chain.
- 4 IOV and 1 FOC in GCS-D VAL chain
- CLA CSIM deployed in GCC-I VAL chain
- only one in GCC-D INT chain (1 IOV)
- one at the LEOP control centre

Remark: In GCC-I, Classified CSIMs are deployed for simulation of IOV and FOC spacecraft.
The FOC version of the Classified CSIM includes an instance of the on-board Central Security Unit (CSU) for the first series of FOC satellites and interfacing hardware. This allows the simulation of protected mode operations for a single FOC satellite. The IOV Classified CSIM is integrated within the instances of the on-board PFSU and PLSU. Both CLA CSIM are installed in GCC-I but are accessible from GCC-D.

The CSIM instance present on the VAL chain of each GCC supports Operations activity, in particular for Operations Products validation and for operator training purposes.
The CSIM instance present on the INT chain of each GCC supports SW testing at GCS level before any new SW configuration is used on VAL or OPE chain.
The CSIM instance deployed at the LEOP control centre is used primarily to support Operations run from the LEOP Control Centre, however this can be also used to fulfil some testing activity (e.g. LEOP control centre GCC interface).

The following use cases are addressed by the CSIM:

- GCS functional Tests

- Ground system interface tests,
- Spacecraft behaviour familiarisation and training,
- Operations products validation, including satellite databases, procedures, on board SW patches and manoeuvre commands.
- On board anomalies investigation

Each CSIM chain comprises a separate physical subsystem, with its own internal LAN segment, servers, network addressable storage, workstations. A network switch is also provided, which can be configured to combine the functionality offered by the CSIM elements of each chain.

A central workstation provides the simulator user interface and centralised control capability whilst a number of servers host the simulation instances.
A separate network addressable RAID storage device is included as a centralised storage source. Printers are provided for hard copy purposes.
All CSIM Instances will have a workstation and two servers. When viewed as a whole, this gives a great deal of flexibility. This could allow several different simulations to be run in parallel using one or more servers. For example a LEOP demonstration simulation (6 emulated spacecraft) could be run with 3 servers, while an IOV constellation simulation (2+2) could be run with 2 servers and procedure validation could be run using a single server.

A separate Computer Based Training (CBT) Tools Workstation is also provided as part of the CSIM delivery. This is a computer based training PC equipped with all training information provided by the various GCS element manufacturer, designed to work standalone.

5.2.2.3 FLIGHT DYNAMICS FACILITY (FDF)

5.2.2.3.1 HIGH LEVEL DESCRIPTION

The Flight Dynamics Facility (FDF) supports orbit determination & prediction, manoeuvre planning and attitude monitoring for both individual satellites and overall constellation management. The FDF contains both real time and non-real time tools used for the calculation of orbits, orbital events, manoeuvres and supporting calculations. Despite of being used off-line for most of the time, while S/C are not contact with ground, the FDF can also be used during special operations to monitor in near real time orbit manoeuvres and S/C attitude.

5.2.2.3.2 FUNCTIONS ALLOCATION

The aims of the FDF are as follows:

- Orbit Determination: Calculates spacecraft orbits based on: precise ephemeris data from GMS, or; ranging, ranging rates and tracking data received from TT&C stations, or; an initial orbit vector provided by the LEOP provider, or; historical ranging data from the LEOP provider.
- Orbit Propagation: Propagates an orbit state vector and generates orbit ephemerides.
- Manoeuvre Planning: Analyses manoeuvres required by each spacecraft to maintain correct orbit within the constellation. Determines the thruster firing times, orientation and duration of the needed manoeuvres. Attitude manoeuvres can be calculated if need depending on the space segment design.
- Calculate fuel usage for satellite mass and end of life predictions.
- Attitude Propagation: Propagates the attitude using attitude law provided by the space segment.
- Attitude Determination: Monitors the current spacecraft attitude based on spacecraft telemetry. It can determine attitude sensor bias. In case during in-orbit operation some de-pointing is observed a compensation of Earth sensors can be commanded via TC in both Pitch and Roll directions after estimating the offset on ground.
- Receive satellite orbit vectors from the GMS.
- Distribute orbit vectors to TTC facilities and GMS.
- Provide manoeuvre parameters to SCCF.
- Manage TT&C station ranging campaigns and receive the output data, including related Doppler measurements, antenna pointing angles, and meteo data.
- Manage archiving and retrieval of Flight Dynamics data and synchronisation of the FDF facilities across the two GCC sites.
- Import relevant databases, configuration files and other data needed to support Flight Dynamics operations.
- Provide an MMI for operator access to management and control of FDF functions, including a printing capability.

This covers directly the following GCS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GCSF1	**GCS Short Term Planning**
GCSF3.1	**Central Archiving**
GCSF3.2	**Local Archiving**
GCSF4.1	**Execute Schedule and Special Operation**
GCSF4.2	**Constellation Monitoring**

GCSF4.4	Flight Dynamics

Table 24: FDF Function List

5.2.2.3.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the FDF context according to the GCS DDF (GCS-RD.1), GCS-GMS-ICD (AD.8), GCSExt IRD (AD.12):

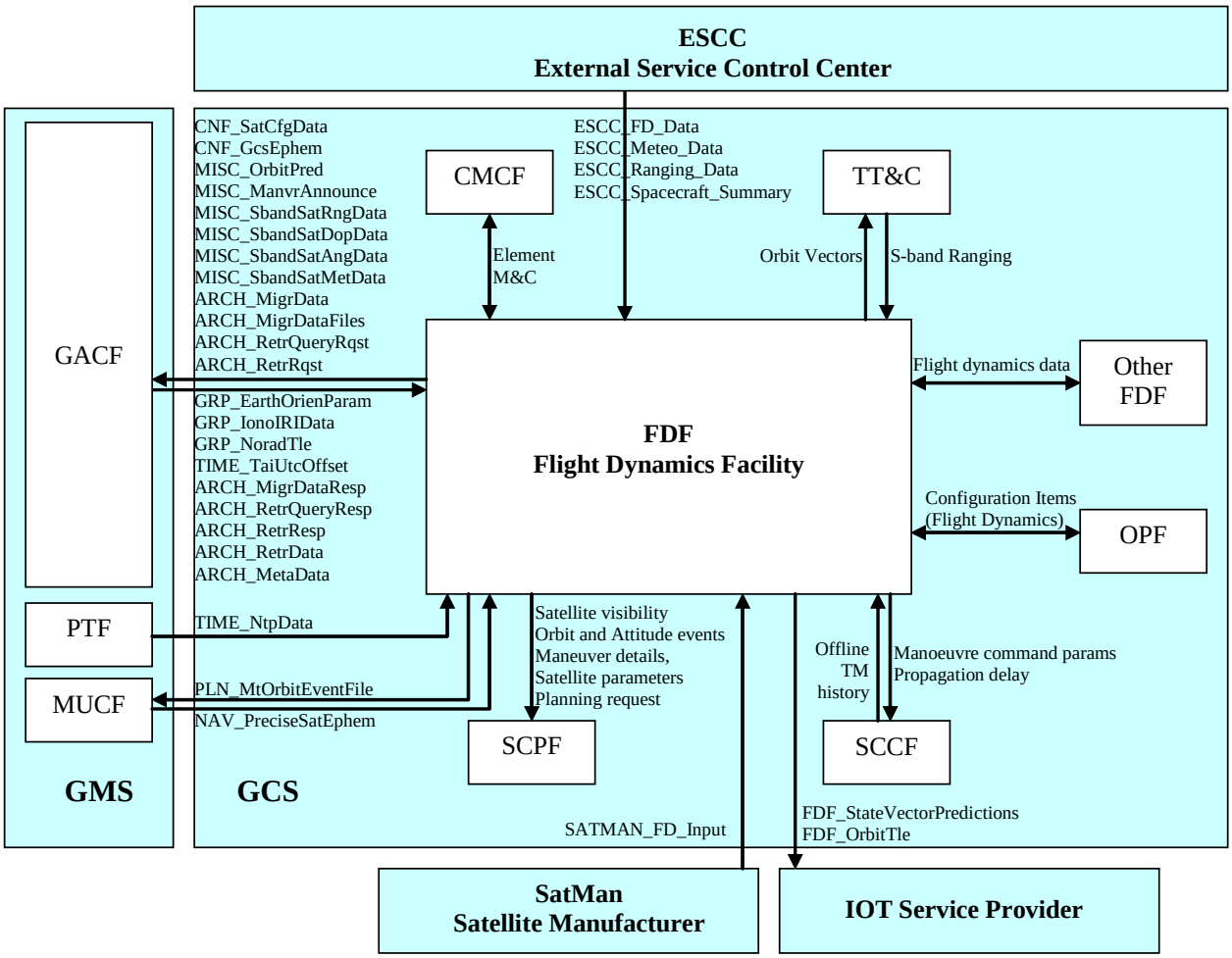


Figure 30: FDF Organical context

5.2.2.3.4 ELEMENT MAIN DESIGN DRIVERS

5.2.2.3.4.1 SYSTEM LEVEL CONSTRAINTS

Performance Constraints:

In mission phases where no GMS navigation solution is available for a given satellite, assuming that an adequate TTCF contacts schedule is feasible, the Ground Control Segment shall be able to determine the orbital position of any operational Galileo satellite based on S-band

ranging, Doppler, and tracking measurements using a minimum of two Galileo GCS TTCF stations with the following accuracy (3 sigma), within no more than 4 orbits:

- Semi Major axis <= 2m
- Eccentricity < = 2.36 E-7
- Inclination < = 0.000020 degrees
- RAAN < = 0.000025 degrees
- Phasing angle < +/- 0.000125 degrees.

Based on the best available orbit knowledge, the FDF shall be able to generate 48 hours of propagated orbit position with an accuracy (1-sigma) better than or equal to:

- 9 m radial.
- 165 m along-track.
- 15 m cross-track.

Such orbit predictions shall be provided via GMS (in accordance with AD.14) to the GRSP to support Laser ranging measurements via external Laser stations.

The FDF supports the maintenance of absolute satellite positioning in the constellation with the following constraints:

- +/-1 degrees for the out-of-plane position w.r.t. the nominal RAAN;
- +/-2 degrees for the out-of-plane-position of any S/C w.r.t the nominal orbit plane inclination;
- +/- 2 degrees (nominal) for the absolute along track positioning relative to any other S/C on the same plane.

The FDF is also in charge of:

- computing the orbit manoeuvres to transfer satellites to a disposal orbit located at least 300km above the operational MEO orbit
- performing collision avoidance manoeuvre for the Galileo satellites based on the analysis of debris trajectories retrieved on regular basis from the NORAD command centre.

5.2.2.3.4.2 MAJOR SEGMENT DESIGN CHOICES

Dual GCC Management:

The Flight dynamics Facility is expected to work mainly off line. Each FDF instance present in the two GCCs can perform the same functions and can support the whole constellation.

The following operating Modes are defined for the FDF:

MASTER: this FDF instance receives inputs from all other elements, processes them under operator control and delivers output products to any other element.

STANDBY: this FDF instance receives input from all other elements but does not distributes any output. It is kept synchronised with all input data to be ready to take the Master role in case of need. All FDF functions are in any case locally available to an operator.

The Role switching among the two FDF instances has to be manually coordinated to ensure that at any time only one instance takes to Master role.

Operability:

The equipment will be connected to the GCS LAN. All operational data is redundantly stored in the prime and redundant servers and all processes are run on top of this data and also in the servers. The hot back-up between servers is carried automatically by the COTS used for the implementation of the central data store (MySQL).

The elements involved in the operational physical architecture implementation are:

- LAN: Through the appropriate network cards, all equipment will be connected to the LANs.
- Two redundant servers (OPE Chain): This will ensure that the data is permanently available after failure of one of the servers. The computational power of these machines will be sufficient to guarantee that the performance requirements are fulfilled.
- Three workstations (OPE Chain): they can be connected to any of the two servers.

The FDF of both GCCs shall be capable of receiving the same input data.

Note that It shall be possible for a single FDF to support routine operations for the whole constellation.

The operator of the particular instance of the FDF in Master Mode has responsibility for overall GCS flight dynamics / orbit determination, including operational inputs and outputs of data.

5.2.2.4 GCS SATELLITE KEY MANAGEMENT FACILITY (GCS KMF)

5.2.2.4.1 HIGH LEVEL DESCRIPTION

The GCS Key Management Facility is responsible for all TM & TC related security key management, both on the ground and on the spacecraft including the ground based encryption / decryption of TM & TC with cryptographic devices. It also provides M&C for the spacecraft security units.

5.2.2.4.2 FUNCTIONS ALLOCATION

The functional allocation made by GCS covers the following GCS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GCSF2	**GCSF2_Manage_TTC_Link**
GCSF2.1	**GCSF2.1_TTC_Link_Modes**
GCSF3.2	**GCSF3.2_Local_Archiving**
GCSF4	**GCSF4_Constellation_Monitoring_and_Control**
GCSF4.1	**Execute Schedule and Special Operation**
GCSF4.2	**Constellation Monitoring**
GCSF8.1	**SC DataBase (classified)**
GCSF8.2	**Operational Procedure DataBase (classified)**
GCSF9	**GCSF9_Security**
GCSF9.1	**Security protection of S-Band**
GCSF9.2	**Security protection of ground infrastructure**

Table 25: GCS KMF Function List

5.2.2.4.3 ELEMENT ORGANICAL CONTEXT

The following diagram shows the GCS KMF context:

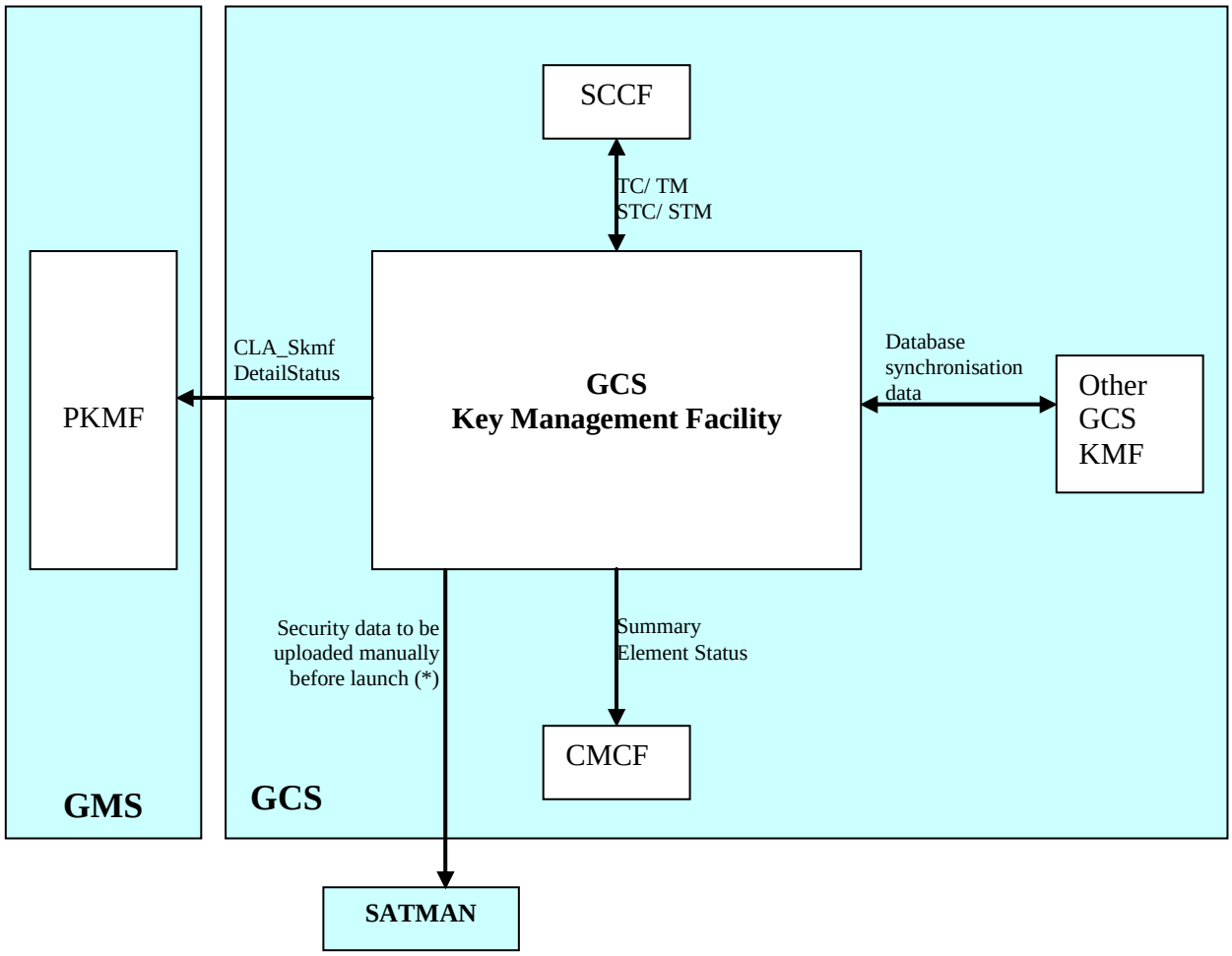


Figure 31: GCS KMF Organical context

(*) signals from GCS KMF to SATMAN:
CLA_SATMAN_Ops_Procedures,
CLA_SATMAN_TcTm_Database,
CLA_SATMAN_Config_Data,
CLA_SATMAN_FOM
CLA_GCS_Ops_Reports,
CLA_GCS_Ops_Procedures,
CLA_GCS_TcTmEv_History_Extract

5.2.2.4.4 ELEMENT MAIN DESIGN DRIVERS

5.2.2.4.4.1 SYSTEM LEVEL CONSTRAINTS

The GCS KMF shall allow the processing of classified information with a classification level up to Confidential. This has an impact (as specified in the applicable security certification/evaluation/approval criteria and standards) on the whole life cycle of the element including:

- the development process and environment;
- the delivery procedures of the element components;

- the security functions provided by the element;
- the architectural and low level design choices;
- the SW and HW COTS that can be used;
- the security features of the site where the GCS KMF shall be installed;
- the operational procedures (including personnel clearance requirements).

The applicable security certification/evaluation/approval criteria and standards are defined in the classified Annex to this document.

5.2.2.4.4.2 MAJOR SEGMENT DESIGN CHOICES

Information flows between the GCS KMF and other elements at a different classification level have been avoided or controlled by means of approved equipments or procedures Therefore it must be assured that there is separation for the following domains:

- Unencrypted traffic;
- Encrypted traffic;
- Unencrypted key material;
- Encrypted key material;
- Control and Monitoring;
- TEMPEST red power;
- TEMPEST black power.

The GCS KMF manages its own archiving separately from other GCS elements. This separation is due to the higher security classification of KMF data, the need to maintain security boundaries and the differing data retention periods. The GCS KMF facilities include an online time archive to support GCS KMF performance evaluation and reporting functions, and an offline secure archive that is shared with the GMS.

5.2.2.5 OPERATION PREPARATION FACILITY (OPF)

5.2.2.5.1 HIGH LEVEL DESCRIPTION

The OPF is responsible of the configuration data and databases of the GCS including:
- databases used by GCS elements (supports the storage and modification of the data on a non-real time basis),
- satellite databases
- operations data (operations procedures)
- on-board software maintenance

5.2.2.5.2 FUNCTIONS ALLOCATION

According to GCS DDF (GCS-RD.1) the OPF is responsible for the following main functions:
- Spacecraft Database Editors - Allows the appropriate operator to modify the spacecraft database used by the spacecraft constellation control facility.
- Spacecraft Procedure Development - Tools to allow the editing of the operations procedure database containing the routine and contingency procedures (Automated Flight and Ground Control procedures) for each satellite. Also provides document generation tools for production of the manual Flight and Ground Control procedures.
- Avionic Software Maintenance Ingestion - The tools required to import new platform software, as delivered by the manufacturer, and to provide the software image and patch data to the SCCF for upload to the satellites in the constellation.
- Import/Export - To import the Satellite database and Operational Procedures from the Spacecraft Authority.
- Consistency Checker - To ensure consistency between separate data and to be able toidentify differences between data of the same type.
- Configuration Control - To provide configuration control of all data within the GCS holding a record of each current item and to hold previous versions in a local archive.

This covers directly the following GCS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GCSF3.1	Central Archiving
GCSF3.2	Local Archiving
GCSF8	Manage Data Base
GCSF8.1	SC Data Base
GCSF8.2	Operational Procedure Data Base
GCSF8.3	Anomaly Data Base
GCSF8.4	Document Database

Table 26: OPF Function List

5.2.2.5.3 ELEMENT ORGANICAL CONTEXT

The following diagram shows the OPF context:

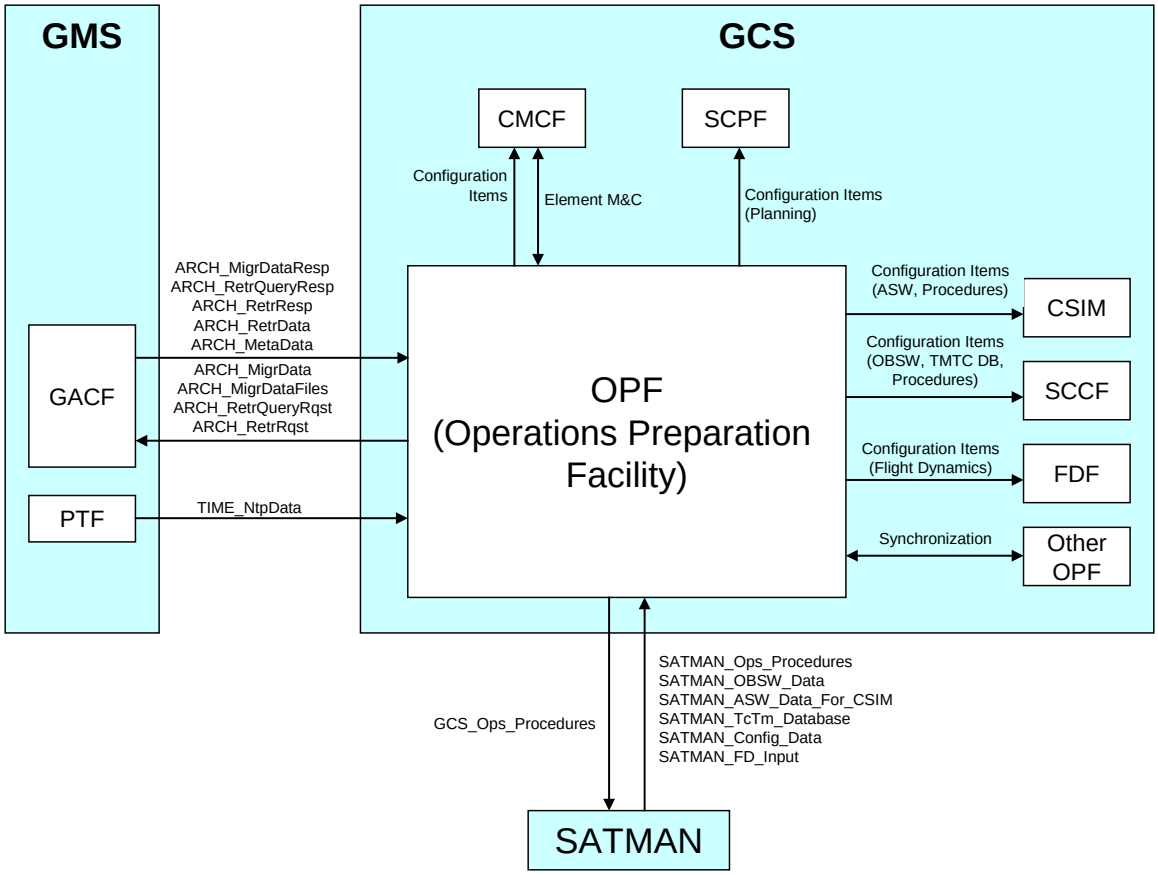


Figure 32: OPF Organical context

5.2.2.5.4 ELEMENT MAIN DESIGN DRIVERS

5.2.2.5.4.1 SYSTEM LEVEL CONSTRAINTS

Database import:
The OPF is able to to import the satellite/EGSE database in no longer than 5 minutes.
Consistency checking:
The OPF is able to perform a consistency check on a single spacecraft database in less than 5 minutes.
The OPF is able to perform a consistency check on every confirmed data modification in less than 10 seconds.
Difference Listing:
The OPF is able to perform difference listing, between one spacecraft database and another, in less than 2 minutes.
Database export:
The OPF is able to derive and export a complete single spacecraft consistency checked database in less than 5 minutes.

5.2.2.5.4.2 MAJOR SEGMENT DESIGN CHOICES

Operability:
The OPF will support 2 workstation positions on the OPE chain and 3 on the VAL chain.

Redundancy:

- The OPF instances on the OPE chain at each GCC site will be interconnected, one acting as MASTER, the other as STANDBY.
- The OPF instances on the VAL chain at each GCC site will be similarly interconnected and also operate as a MASTER/STANDBY pair.

COTS:

OPF will make use of the following COTS products:

- the MOIS product for procedure management
- the SCOS FARC product for configuration control of all file types

Operating Systems:

- The OPF servers and workstations will operate in a Linux environment
- The MOIS product will operate in a Windows environment

5.2.2.6 SPACECRAFT CONSTELLATION CONTROL FACILITY (SCCF)

5.2.2.6.1 HIGH LEVEL DESCRIPTION

The Spacecraft Constellation Control Facility (SCCF) is the core element within the GCS that supports real-time operations for both routine and special satellite operations. It provides TM and TC processing status displays, manual commanding facilities together with operations automation at both schedule and procedure levels.

5.2.2.6.2 FUNCTIONS ALLOCATION

According to GCS DDF (GCS-RD.1) the SCCF is responsible for the following main functions:

- **Real-time Functionalities:**
  - Preparation and transmission of telecommands (including pre-validation), verification of the command execution, as well as on-board command queue management (ie mission time line).
  - Telemetry processing, including reception, interpretation, calibration and out of limits checking.
  - Spacecraft automation, encompassing automatic reception and processing of telemetry, automatic upload of TCs into on-board command schedule, automatic schedule execution (ground-based execution of satellite operations, including establishing TM/TC session for each contact and performing ranging operations).
  - Operations Progress Monitoring providing a real-time display of the status of the planned operations for each spacecraft in the constellation. This includes the status of operations on ground and on-board the spacecraft compared to the active plan.
  - On-board Software Maintenance (OBSM) including the provision of the tools required to upload new payload and platform software and to manage the software distribution to all spacecraft in the constellation.
- **Off-line Functions:**
  - The Medium Term Operations History (MTOH) that provides storage of all telemetry received and telecommands sent, as well as event logs and reports. This is used extensively by non-real time analysis tools.
  - Performance Evaluation tools that allow analyses to be performed on received telemetry, both on a spacecraft or constellation basis, to show parameter trends.

This covers directly the following GCS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GCSF1	**GCS Short Term Planning**
GCSF2	**Manage TTC Link**
GCSF2.1	TTC Link Modes
GCSF3.1	Central Archiving
GCSF3.2	Local Archiving
GCSF4	**Constellation Monitoring and Control**
GCSF4.1	Execute Schedule and Special Operation
GCSF4.2	Constellation Monitoring
GCSF4.3	Manage OBSW

Table 27: SCCF Function List

5.2.2.6.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the SCCF context:

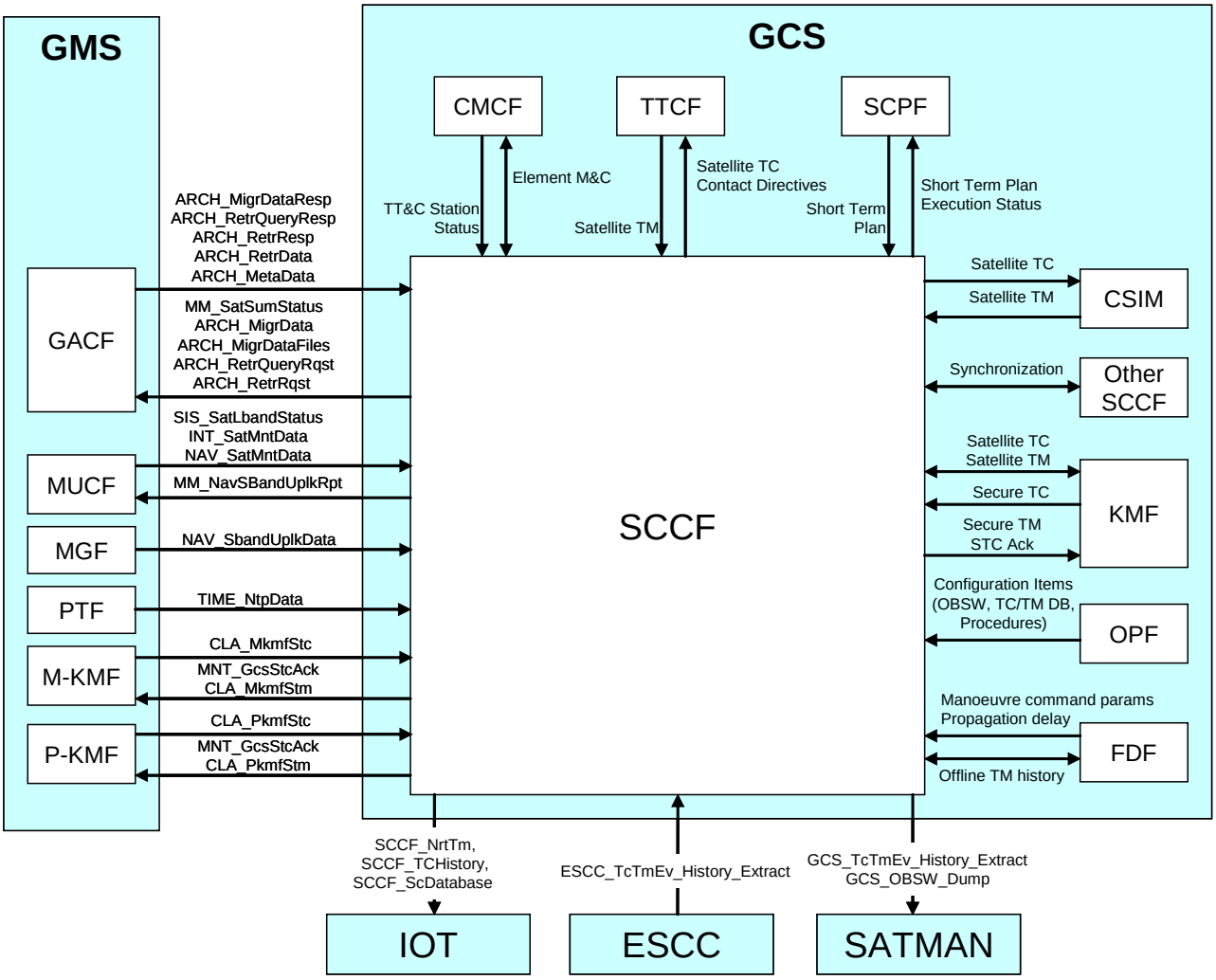


Figure 33: SCCF Organical context

5.2.2.6.4 ELEMENT MAIN DESIGN DRIVERS

The GCS SCCF element is a spacecraft constellation control system based on the SCOS 2000 SW developed by ESA/ESOC. As this was developed as single satellite control system, the SCCF had to be customised to allow for multiple spacecraft control, taking also into account of the following Galileo specific features:

- the number of in orbit Satellites to be controlled (up to a max of 41 satellites including operational, in-orbit spares and replacements satellites);
- the maximum number of satellites simultaneously in control at any instant via TTCF (6 in parallel, as the number of TTCF, see sect. 5.2.2.9);

- automation of operations: individual satellite operations are planned and scheduled in advance over single S band passes of 60-90 minutes each, once per orbit;
- Allocation of satellite passes supported by individual TTCF to be shared among two GCCs;
- Execution of special operations in separate contexts to minimize interference with routine operations, up to a max of 2 special operations contexts per GCC
- Capability to receive a continuous flow of health status data from the entire constellation via L band data received by the GMS

5.2.2.6.4.1 SYSTEM LEVEL CONSTRAINTS

The following system level constraints apply to the design of the SCCF element.

Constellation M&C – Worst Case Performance Scenario:

- The SCCF will provide the capability to:
  - manage a constellation of up to 41 satellites (including operational, in-orbit spare and replacement satellites);
  - command in parallel at least 6 satellites in orbit, plus 4 satellites on ground (2 simulated, 2 during LIT or SVT);
  - concurrently receive and process real-time telemetry from 6 satellites in orbit, plus 4 satellites on ground (2 simulated, 2 during LIT or SVT);
  - perform orbit and attitude control manœuvres for at least 4 satellites in parallel;
  - support ongoing special operations with up to 4 satellites in parallel.

Display Performance:

- The capability to update the information of the Constellation Status Display in not more than 5 second after reception of the pertinent data at the GCC,
- The capability to process and display to the operator in the GCC real time telemetry less than 5 seconds after reception,

Archive Retrieval:

- The capability to retrieve and display of historical TM & TC data (from at least the past 7 days) within 3 seconds,
- The capability to retrieve and display of historical TM & TC data (from at least the past 3 years) within 10 seconds,
- The capability to retrieve TM & TC data older than 3 years within 5 minutes,
- The capability to generate the executable schedules from the STP in not more than 5 minutes,
- The capability to update the historical TM & TC data such that the most recent data is no more than 5 hours old.

Consistency Checking:

- The capability to check the consistency of an updated schedule in not more than 1 minute,

Back-up Contact Activation:

- The activation of backup contact in no longer than 10 minutes

Time Correlation:

- The correlation of the on-board time with the Galileo System Time with an accuracy of +/- 5 msec

5.2.2.6.4.2 MAJOR SEGMENT DESIGN CHOICES

Redundancy:

Fault tolerance is provided by the inclusion of redundant components in the SCCF design. Each active satellite domain is mirrored by a fully redundant hot-standby using the SCOS-2000 server family (Prime/Backup) configuration. In the event of a non-recoverable failure within a critical component on one server the user may immediately switch to the redundant server and continue operations with no degradation in service. The only impact on active operations will be the temporary loss of the commanding chain to the satellite while re-configuration takes place.

Operability:

The number of 'SCCF SCOS workstations' for FOC is six, with one workstation for the shift leader and five for the spacons. No spare will be required on-line but spares will be kept in storage.

Note that these workstations carry TMTC (SCOS), AUTO and authorisation displays.

Load sharing:

The SCCF will support load sharing across all servers by making use of the SCOS multi domain concept which allows prime and backup domains to be distributed across the available servers.

Data synchronisation between GCCs

The SCCF has two SCOS systems one with access to live data from the satellite and one with access to the archived data from the past 3 years:

- Live system

  SCCF will synchronise (in near real time) the TC and Event history between sites. The status display will also be synchronised in real time.

- Archive system

  This system archive contains all received telemetry for the past 3 years. It is mirrored between sites and will be almost as up-to-date as the real time system (with about a 1 minute lag).

Use of ESA COTS:

The SCCF technical solution is built around the ESA spacecraft control kernel SCOS-2000, recognised throughout Europe as the state-of the-art in terms of generic spacecraft control systems. The SCCF architecture based on the latest version of SCOS 2000, Release 5, which incorporates much of the necessary functionality without modification.

5.2.2.7 SPACECRAFT CONSTELLATION PLANNING FACILITY (SCPF)

5.2.2.7.1 HIGH LEVEL DESCRIPTION

The Spacecraft Constellation Planning Facility (SCPF) is responsible for generating the short term planning for spacecraft operations for the Galileo constellation, and therefore the SCPF plans for the resources involved in these operations (e.g. S-Band TTC stations and GCS spacecraft control facilities).

The SCPF planning function requires coordination with the Ground Mission Segment (GMS) that is responsible for managing the overall provision of the Galileo mission services and generation of the Mission Plan, which is the primary input to the GCS mission planning function.

The SCPF generates a Short-Term Plan (STP) which contains:
- TT&C contact plan,
- on-board satellite operations,
- ground-based satellite operations plan.

5.2.2.7.2 FUNCTIONS ALLOCATION

According to GCS DDF (GCS-RD.1) the SCPF is responsible for the following main functions:
- Planning exchange and coordination with GMS and SCCF (Reception of GMS Mid Term Plan as input for the GCS Short term plan elaboration, support to Mid term Plan elaboration)
- Coordinate Spacecraft Operations Planning Requests.
- Generate TT&C Contact Scheduling
- Provide a Spacecraft Operations Planning Database

This covers directly the following GCS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GCSF3.1	Central Archiving
GCSF3.2	Local Archiving
GCSF1	GCS Short Term Planning

Table 28: SCPF Function List

5.2.2.7.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the SCPF context according to the GCS DDF (GCS-RD.1), GCS-GMS-ICD (AD.8):

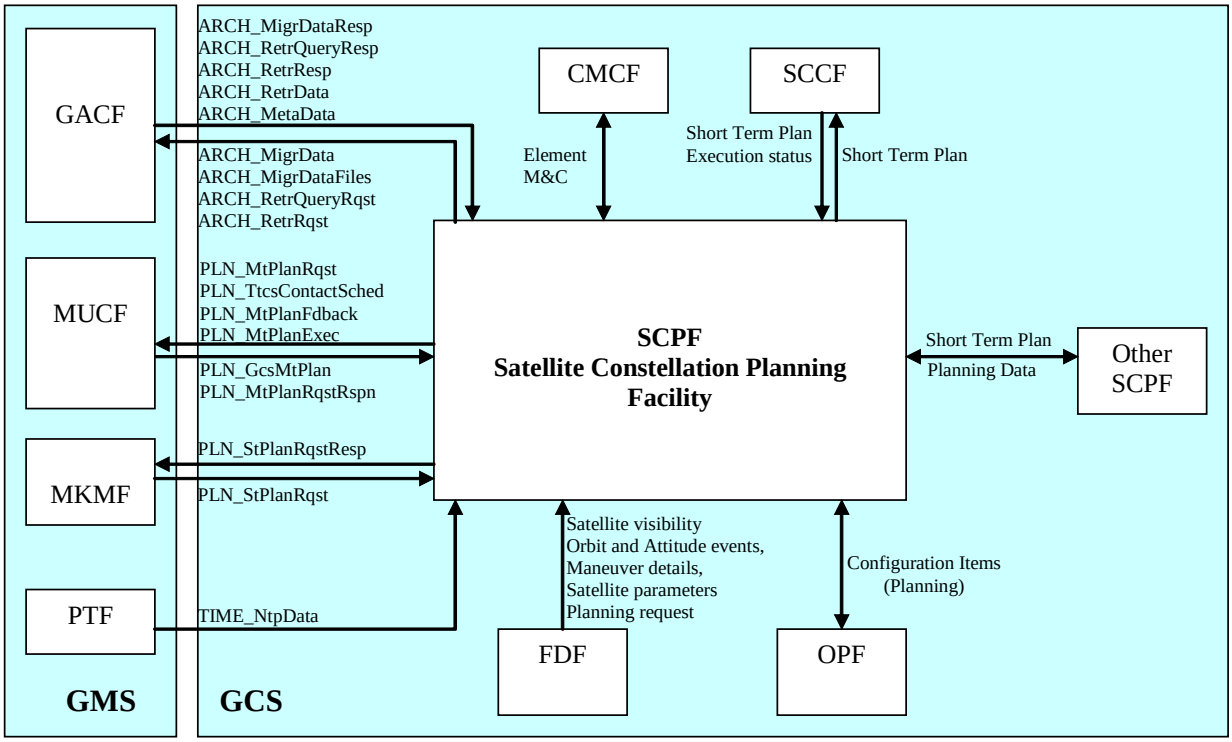


Figure 34: SCPF Organical context

5.2.2.7.4 ELEMENT MAIN DESIGN DRIVERS

5.2.2.7.4.1 SYSTEM LEVEL CONSTRAINTS

Planning Constraints:

- The SCPF is in charge to generate a detailed and conflict-free Short Term Plan, within the framework provided by the Medium Term Plan received from the MUCF. This Short Term Plan includes at least:
  - TT&C Contact Plan (including a backup contact plan)
  - GCS Ground-based operation timeline (Automatic/Manual)
  - On-board operations timeline
  - Spacecraft Availability Data
- The GCS need also to process additional requests when generating the STP:
  - GCS Ground Engineer Request,
  - non-routine operation requests deriving from analysis,
  - constraints arising from near-real-time resource availability changes,
  - Addition of scheduling constraints from satellite operations,
  - other security related requests.

Performance Constraints:

- The duration of the SCPF generated short-term plan is configurable, between one day and four weeks.
- The SCPF is able to generate a Short Term Plan in less than 10 minutes and modify it in less than 5 minutes. For replanning purpose, the SCPF is able to display all unallocated contacts to the constellation from all TT&C stations.

5.2.2.7.4.2 MAJOR SEGMENT DESIGN CHOICES

Operability:

The submission of planning requests can occur at any time, and the requests can arrive from defined sources. Most requests are submitted electronically to the SCPF but an operator can submit a request manually.

External Ground Facilities may request hand-over of spacecraft operations to the GCC. This is anticipated to be a co-ordination interface that will be managed procedurally, using telephone and e-mail.

Redundancy Management:

- The SCPF hosted at geographically separate GCC sites operate in Master/Stand-by mode. Planning data are replicated by the Master to the standby SCPF. The stand-by SCPF can be used for experiment purpose, generating STP that will not be used or distributed.
- The master SCPF send within one minute the STP to the stand-by SCPF.

Planning cycle:

The basic planning cycle consists of 4 major processes that occur in sequence:

- Check and accept planning requests
- Determine contact requirements and select satellite contacts using FDF data
- Place the requested tasks onto the Constellation Plan timeline
- Generate the Short Term Plan for the required period

SCPF Archiving Capacity:

Historical Plans are retained within the SCPF for 12 months. In order to ensure history is retained over a period exceeding the planned operational lifetime, the GMS maintains a global long-term archive.

5.2.2.8 SATELLITE DATA DISSEMINATION NETWORK (SDDN)

5.2.2.8.1 HIGH LEVEL DESCRIPTION

The SDDN is the logical network responsible for providing all the needed functionalities to ensure communication between GCS elements within the GCC and:

- the GCS remote sites, namely TTCFs, via the GDDN WAN: role of the SDDN network equipments.
- The GCS External Entities: role of the S-EDDN network equipments, permitting non real-time products exchange between GCCs and external entities (satellite payload & platform manufacturers, external satellite control centers).

In addition, as part of SDDN, the GCS is responsible for procuring the GCC-CNEs, responsible for communication between the two GCC sites.

Finally, as part of SDDN, the GCS is responsible for procuring the GCS nRT LAN within the GCC and the TTCF.

5.2.2.8.2 FUNCTIONS ALLOCATION

According to GCS DDF (GCS-RD.1) the SDDN is responsible for the following main functions:

- Data routing and redundancy management: end-to-end IP networking services between the GCC centres and the TT&C stations with priority assignment between the TTCF flows through ToS (Type of Service marking). At least four different priorities are expected, to be shared by SDDN, GSS sub-network and ULS sub-network traffic profiles.
- Redundancy management with automatic failover: a redundant IP Routing over two paths, with automatic failover is achieved using a dynamic routing protocol between SNEs at two communicating sites.
- Firewall and data communication protection: every communication link over GDDN and over EDDN will be protected by strong VPN enforcement with packet filtering at layers 3/4.
- IP management: message fragmentation and reassembly with configurable MTU (Maximum Transfer Unit) in order to allow that maximum priority messages are transmitted first
- Performance monitoring: this is performed through two software:
  - o A packet capture probe function is present at each SNE. The probes provide per-packet observations to the GNMF..
  - o A network monitoring mediator gateway function. It consists of the distributed component in charge of local M&C of the network elements and of the proper transmission of this information to the GNMF at both GCC centres, in a controlled way.

This covers directly the following GCS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GCSF2	**Manage TTC Link**
GCSF2.1	**TTC Link Mode**
GCSF2.2	**Two Way Ranging**
GCSF3.2	**Local Archiving**
GCSF5.3	**GCS Networks Monitoring and Control**

Table 29: SDDN Function List

The SDDN is involved in the following system functional chains:
- SFC1.2: Maintain constellation
- SFC3: Maintain overall system synchronization
- SFC5.1: Ground Asset Monitoring and control
- SFC5.2: Satellite Monitoring and Control
- SFC5.4.2: Short term planning
- SFC6: Archiving

5.2.2.8.3 ELEMENT ORGANICAL CONTEXT

- **SDDN Context Diagram**

The following figure shows the SDDN context:

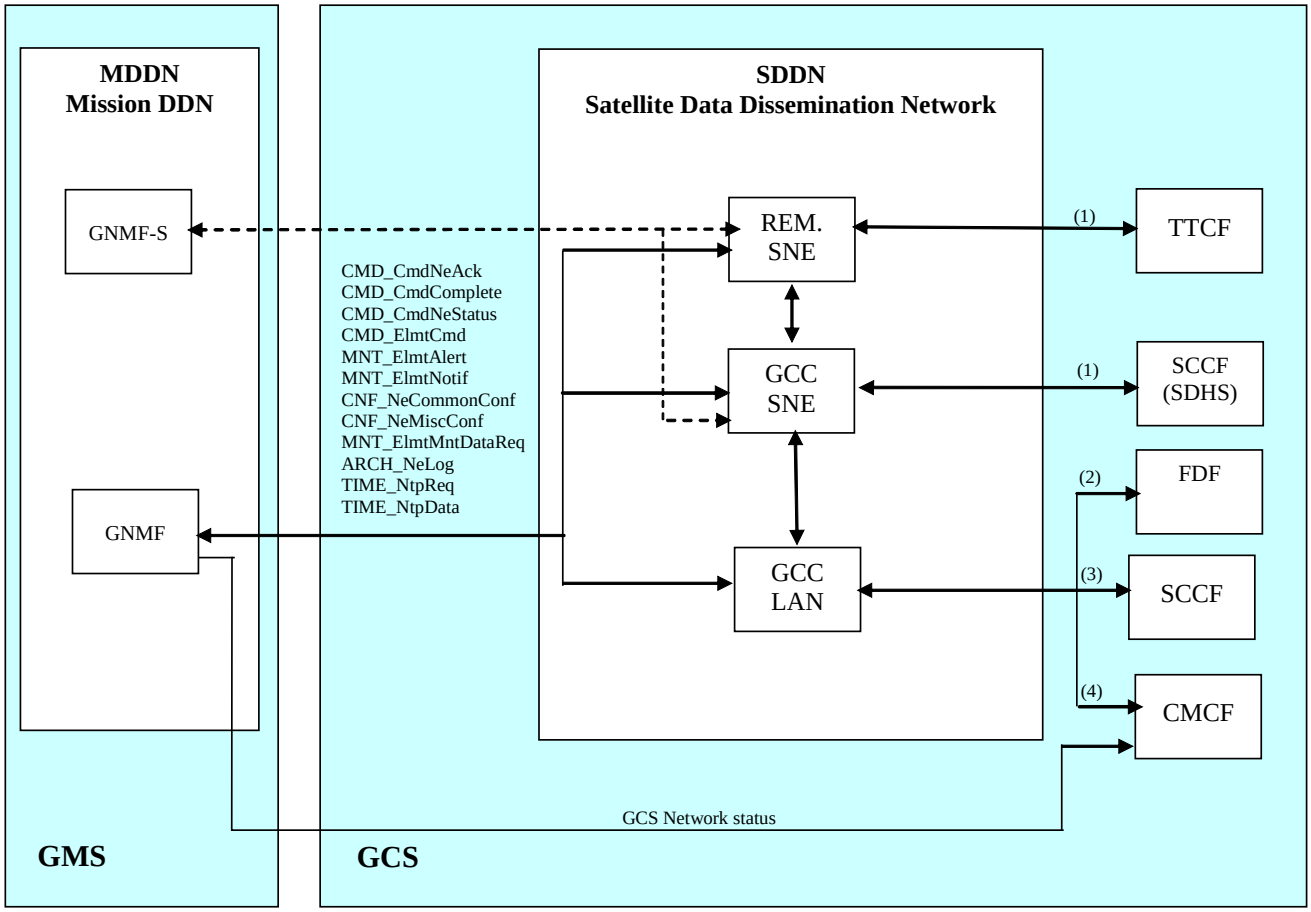


Figure 35: SDDN Organical context

where:

(1)	TM/TC
(2)	Orbit Vectors, Ranging Data, Meteo Data

(3)	Control Directives
(4)	Element History

Note that the traffic between GCS and GMS flows are ensured over the Common Network Elements (CNEs), one CNE per GCC. The CNEs are not included in the diagram, for clarity.

- **S-EDDN Context Diagram**

The following figure shows the S-EDDN context:

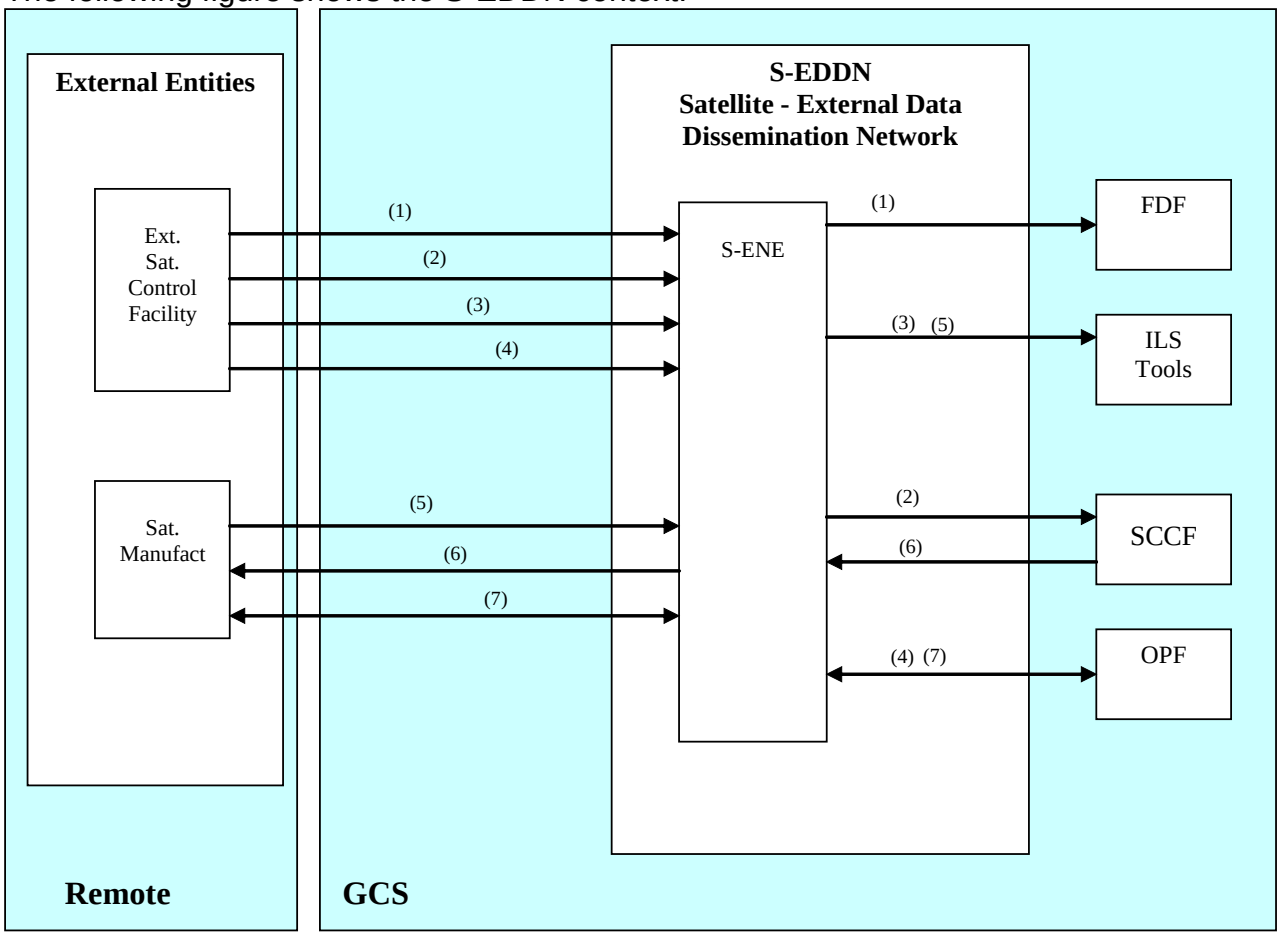


Figure 36: S-EDDN Organical context

where:

(1)	LEOP, Ranging Data, Meteo Data
(2)	LEOP TC/TM History
(3)	S/C Summary and IOT Reports
(4)	LEOP FD Data
(5)	FOM & OPS Reports
(6)	TC/TM History and OBSW Dumps
(7)	ASW, OBSW, Procedures, Satman Config, FDF Input, TC/TM DB, GCS OPS Reports

5.2.2.8.4 ELEMENT MAIN DESIGN DRIVERS

5.2.2.8.4.1 SYSTEM LEVEL CONSTRAINTS

The SDDN is procured by GCS but monitored and controlled by GMS, and more specially the Galileo Network Management Facility (GNMF). GCS network management is based on performance monitoring agents located in the SNEs, and consists of the following performance assessment (note that GCS has NO continuity requirement):

- SDDN network latency
- SDDN network availability

All SDDN network devices are managed using the SNMP v3 protocol, and use Internet Protocol (IP) as their network protocol.

5.2.2.8.4.2 MAJOR SEGMENT DESIGN CHOICES

During normal operation conditions, network management is performed by the GNMF. Therefore, the controls and commanding from GNMF directly access the individual management interfaces provided by the SNE network devices. The GNMF also directly accesses the S-EDDN SNE infrastructure components.

However, local (level 0) network management capability is provided, for AIV and as standby. This will provide minimal ability of configuring and of monitoring the SDDN SNE and S-EDDN ENE equipments in the absence of GNMF functions.
Note that the same traffic monitoring probe software and the same network monitoring mediator gateway software will be deployed on auxiliary computers within every SNE. They will present the same interface towards the common GNMF as their counterparts in the MNEs.

The SDDN, together with S-EDDN, is composed of the following equipment:

- TTC-SNE, interfacing the TTC LAN and the GDDN, The TT&C SNE offers the following functions:
  - o routing the nRT traffic,
  - o distributing observations / measurement of GDDN traffic flows, and
  - o monitoring the network elements.

  The core of the TTC-SNE architecture is made by the redundant pairs of chains, each consisting of a switch-router and a VPN gateway: one of the TTC-SNE computers runs the active mediator gateway and packet capture probe software. The other computer is in cold-standby. The failover of these functions is performed automatically, and requires therefore intervention from the GNMF.

- GCC-SNE, interfacing the GCC LAN and the GDDN, covers the same functions as the TTC-SNEs. It also provides stateful firewall capabilities.

- SDDN ENE, interfacing the GCC LAN and the External Entities provided CSP WAN. The S-EDDN ENE is mostly a secure access infrastructure providing strong latch between the EDDN WAN services and their connected VPN-protected External Entities, on one hand, and the GCS LAN at GCC on the other hand. It provides access control and intrusion detection.

TTC Network equipment architecture:
The TTC-SNE, located in each remote site supporting a TT&C station, connects the TT&C station to the GDDN. The TTC-SNE architecture consists of two redundant identical lanes, as derived by RAMS analyses, on the basis of availability constraints for FOC.
Each lane is equipped with

- a switch-router,
- a Monitoring and Control (M&C) computer,
- IP Security (IPSec) VPN gateway used to protect the traffic, through IPSec encapsulation, and of the site, through layer 3/4 packet filtering,
- an Ethernet 10/100 Base-TX interface that is converted to a fibre optic interface through an media converter, allowing each GSS-MNE instance to interface with the GDDN via either a copper or a fibre port.

Each lane connects to a separate line (or path) on the GDDN.
It should be noted that the SIV site also connects to the GCC via a TTC-SNE.

GCC- GCS-SDDN Network equipment architecture:
At a GCC, a single, redundant subsystem, the GCC-SNE, interfaces to all remote TT&C stations.

The GCC-SNE, located in the GCC, connect GCS elements in the GCC to the WAN network (GDDN). The Telemetry & Tele-commanding subsystems connect directly, and redundantly, to the two lanes of the GCC-SNE, whilst other subsystems connect via the appropriate GCC LANs. The GCC-SNE presents the same basic architecture as that of the TTC-SNEs, i.e., each lane is equipped with

- a switch-router,
- a Monitoring and Control (M&C) computer,
- IP Security (IPSec) VPN gateway used to protect the traffic, through IPSec encapsulation, and of the site, through layer 3/4 packet filtering,
- an Ethernet 10/100 Base-TX interface that is converted to a fibre optic interface through an media converter, allowing each GSS-MNE instance to interface with the GDDN via either a copper or a fibre port.

Lane aggregation is performed in a similar way to that on the GMS GSS Subnet, by the GDDN delivering all the TTC-SNE lane 1 traffic to lane 1 of the GCC-SNE, and all the TTC-SNE lane 2 traffic to lane 2 of the GCC-SNE.

GCC- SDDN-CNE Network equipment architecture:
The CNE consist of two lanes, only one of which is connected to the GDDN at any one time.

During FOC, when both GCCs contain GMS and GMS, the CNE lanes consist of:

- a computer devoted to RT flow handling,
- a Monitoring and Control (M&C) computer,
- a switch-router, used to connect the site components to the gateway

- IP Security (IPSec) VPN gateway used to protect the traffic, through IPSec encapsulation, and of the site, through layer 3/4 packet filtering.

In addition, a firewall cluster is distributed across both lanes, used to provide secure connectivity to GMS RT and nRT LANs, GCS Operational (OPE) and Validation (VAL) LANs, the GNMF and GNMF-s.

During IOV, the basic architecture of the CNEs is the same, but the connectivity is different, depending on the location: the GCS-GCC-CNE connects to the GCS Operational (OPE) VLAN and the GNMF, whilst the GMS-GCC-CNE connects to the GMS nRT LAN, the GNMF, and the GNMF-s.

5.2.2.9 TTC STATIONS (TTCF)

5.2.2.9.1 HIGH LEVEL DESCRIPTION

The TTC stations are mainly responsible for satellite tracking, TC reception from SCCF and up-linking towards the satellites, telemetry reception from satellite and forwarding to SCCF.
As per [SYS-RD.95], six TTCF are foreseen to support the system FOC configuration. Only 5 TTCF however are presently deployed in System Build 1.5.0 [AD.48], since the TTCF of Papeete is not yet integrated with the system.TTC stations are co-located with ULS at Kourou, Noumea, Reunion and Papeete. The other two sites for TTCF are Kiruna and Redu, which is collocated with the Galileo IOT facility.
All TTCF are linked to the GCC through the GCS networks, called SDDN, via redundant links supporting the well-established Multi-Protocol Label Switching (MPLS) Virtual Private Network protocol (see sect. 5.2.2.8 and 5.6.3 for more info on the GDDN).
Each TT&C station provides the S-band Uplink and Downlink transmission and reception facilities for the satellite control and management of the Galileo Constellation:

- for telecommands transmission, it processes by modulation, up conversion and amplification to the satellites
- for telemetry reception from the satellites, it processes by amplification, down conversion, demodulation and transmission decoding.
- For relaying the degraded navigation data and PRS data received from the GMS and uplinked to the satellite in backup mode
- For tracking and ranging

The TT&C facility interfaces with the:

- Galileo satellites through the RF links for telecommand transmission, telemetry reception
- Spacecraft Constellation Control Facility (SCCF), for the transmission and reception of TC's, TM and acceptance of ranging requests including station configuration data / Contact plan provision
- Flight Dynamics Facility (FDF) to receive the orbit vectors for antenna pointing and provide the S-Band ranging and tracking data as well as meteorological data (on request)
- Central M&C Facility (CMCF) for the TT&C remote M&C. Each TTC also provides equipment diagnostic functions.
- Site infrastructure

With regard to the TTCF at Redu, is must be remarked that, unlike the other TTCFs, specifically procured and installed for Galileo by the GCS, this is an existing ESA TT&C station. Due to the re-use of local M&C equipment, the remote M&C function from the GCCs presents some limitations, in particular for what concerns the possibility to automate the station configuration in support to routine and contingency operations.

5.2.2.9.2 FUNCTIONS ALLOCATION

The following table lists the segment functions allocated to the TT&C Stations:

Function Id	Function Name
GCSF2	**Manage TTC Link**
GCSF2.1	TTC Link modes
GCSF2.2	Two way ranging

GCSF3.2	Local archiving
GCSF4.1	Execute schedule and special operations
GCSF5.1	GCS element Monitoring and Control

Table 30: TTCF Function List

The TTCF is involved in the following system functional chains:
- SFC1.2: Maintain constellation
- SFC5.1: Ground Asset Monitoring and control
- SFC5.2: Satellite Monitoring and Control
- SFC6: Archiving

5.2.2.9.3 ELEMENT ORGANICAL CONTEXT

The following context diagram shows the TTCF location inside the Galileo core system:

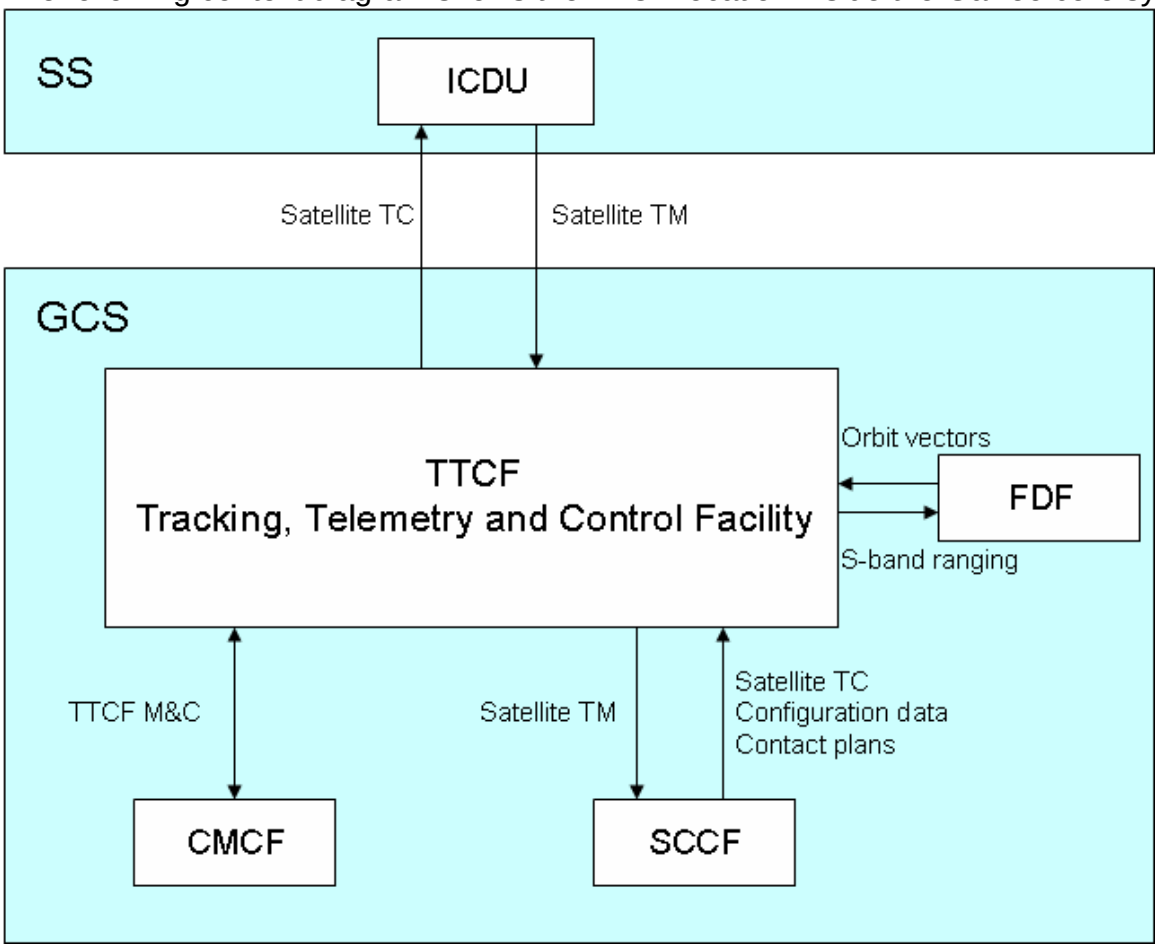


Figure 37: TTCF Organical context

5.2.2.9.4 ELEMENT MAIN DESIGN DRIVERS

5.2.2.9.4.1 SYSTEM LEVEL CONSTRAINTS

Telecommands are handled by the TT&C facilities in the form of CLTUs, the format of which are defined in the ESA standard (PSS-04-107).
The TT&C facility receives CLTUs from the SCCF in a pre-protected format, and they are modulated onto the IF carrier, up converted to the S-band frequency and then transmitted to the spacecraft.
Telemetry is handled in the form of Transfer Frames, the format of which is defined in the ESA standard (PSS-04-106). The telemetry frames received from the spacecraft at the TT&C facilities, are firstly demodulated from the RF carrier, the frame data is then recovered and then each is subject to CV and R-S decoding. The resulting frames are forwarded to the SCCF.
The TTCF supports both Spread Spectrum and Standard PM modes, for receive and transmit, spread spectrum being the nominal mode for TTC.
The TTC stations are dimensioned according to the satellite control needs only. They are also used for the up-link of degraded navigation data in S-Band embedded in the TC frame in the case ULS mission uplinks in C-band are interrupted.

The following main constraints are imposed to the TTC station:
- The diameter of the TTC S-band antennas on ground is 13m.
- The TTC is able to track the satellite, either using the ephemeris downloaded from the FDF or using the satellite Auto-track mode.

5.2.2.9.4.2 MAJOR SEGMENT DESIGN CHOICES

The TTCF element is designed according to the following segment main design choices:
- **Monitoring and control**:
  Each TT&C station includes it own local M&C subsystem, which provides M&C of all the equipments of the TT&C station. This local M&C sub-system interfaces with the CMCF to allow remote M&C of the TT&C station from the GCC.
  During the FOC phase, there will be two CMCFs, one at each GCC. In this case the token exchange function will allow only one CMCF to be in control of any one TT&C station at a time. This token mechanism is extended to local operator, who would need access during periods of maintenance or testing.

- **Contact plan**:
  The TTCF contact plans are passed from SCCF to TTCF element through local M&C sub-system, which is in charge to execute it locally.

- **Interface with FDF:**
  The TTCF interfaces with the FDF for the following data transfer:
  - Spacecraft Orbit Vectors: the FDF generates orbit vectors for each spacecraft of the Galileo constellation and send them to the TTCF. These orbit vectors are stored in the Antenna Control Unit and used to create a spacecraft tracking file that allows the TT&C station antenna to track individual spacecraft.
  - Ranging Files: these files are sent only on request to the FDF.

- **Telecommands and Telemetry**

The TTCF interfaces with the SCCF (SDHS) based upon the SLE service standards, for the transfer of telecommands and telemetry. For telecommands, the F-CLTU service is adopted, and for telemetry the Return All Frames (RAFs) service is used.

- **Hosting sites:**
  The TTCF interfaces with Hosting Sites for infrastructure and site security monitoring.

The TTCF element is composed of the following subsystems:
- Local M&C: in charge of the local monitor and control of all equipment of the TT&C station and transfer of these data to / from the remote CMCF located at the GCC. This M&C is also in charge of receiving the contact plans from SCCF.
- Communications sub-system: which comprises the equipment necessary to interface the TT&C station to the SDDN
- Timing subsystem
- Baseband subsystem and switching, which comprises the base-band units arranged as one real time and one stand-by, with the associated switching equipment that allows the switchover from one unit to the other.
- RF Sub-system and Switching, which provides the RF conditioning between the IF sub-system and the TT&C antenna.
- Antenna Sub-system, including infrastructure (13 meter diameter Cassegrain design antenna), waveguide runs and switching, antenna pedestal or building and its foundations, antenna wave-guide dehydration system and antenna servo and tracking equipment
- Test and calibration equipment, included for the purposes of remote control and monitoring (via the local M&C), of the quality and integrity of the data being processed within the TTCF.

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5.3 GMS SEGMENT

5.3.1 GMS OVERVIEW

The Ground Mission Segment (GMS) main responsibilities are to generate Galileo main services mission data (OS and PRS), to manage their proper dissemination to the users via the space segment and to monitor and control the overall ground segment (i.e. GCS and GMS). In addition, the GMS provides a support to the following external entities:

- Service Centres: Support to dissemination of CS data via Galileo satellites and service monitoring, plus generation of reports on Galileo services and status.
- SAR Service: Support to dissemination of RLM messages and service monitoring

The GMS contributes to the following system functional chains:

- SFC2 - Generate System Time and Geodetic Reference
  - Provision and maintenance of Galileo System Time
  - Galileo/GPS Time Offset Determination
  - Contribute to geodetic reference determination (in particular by relaying SLR-related data from GCS to GRSP)
- SFC3 - Maintain Overall System Synchronization
  - Maintain the satellite payload time synchronization to the GST (by sending synchronization requests to the GCS)
  - Maintain the GMS elements synchronized to the GMS
  - Provide timing reference to the GCS
- SFC4 - Mission Data Provision
  - Provision (generation and dissemination) of navigation data for Open Service and PRS services to Galileo satellites
  - Provision of "degraded navigation data" to the GCS for uplink in S-Band (via TTCs)
  - Provision of service mode information to be uplinked to the satellites.
- SFC5 - Monitor and Control
  - Perform ground asset monitoring and control for the GMS and the GCS
  - Provision of L-Band TM data to GCS (for constellation monitoring and control)
  - Perform mission monitoring
  - Support to GCS planning activities (including GCS planning at mid-term)
- SFC6 - Archiving
  - Manage the GMS local and central archive
  - Manage the system global archive (including also GCS archive long-term storage).
- SFC7 - Support CS Services
  - Support to commercial services: dissemination of commercial data and commercial ranging code encryption management
  - Provision of Galileo products (service and general information reports) to "outside world"
- SFC8 - Provide SAR Service
  - Support COSPAS-SARSAT service: dissemination of Return Link messages
- SFC9 - Support EDBS Service
- SFC11 – Control Access to Ground Infrastructure
- SFC13 – Security Protection for Mission Data
  - Management of the C-Band uplink authentication
- SFC14 – Security Protection of SoL
  - Management of the integrity table authentication

- SFC15 – Security Protection of PRS

This involves the following GMS functions as shown in DDF Part 1 (SYS-RD.01):
- GMSF1 - Generate SIS Observables (SFC3, SFC4, SFC5)
- GMSF3 - Schedule and Perform Message Dissemination (SFC4, SFC7, SFC8, SFC9)
- GMSF4 - Generate Time and Geodetic Reference Data (SFC2, SFC4, SFC5)
- GMSF5 - Determine Navigation (SFC2, SFC4, SFC5, SFC6)
- GMSF6 - Manage System Archives (SFC3, SFC5, SFC6)
- GMSF7 - System Monitoring and Control (SFC2, SFC3, SFC5, SFC6, SFC8, SFC9)
- GMSF8 - Security Management (SFC4, SFC7, SFC9)
- GMSF9 - Perform Mission Planning (SFC3, SFC5, SFC7, SFC9)
- GMSF10 - Provide Training
- GMSF11 - Perform Simulation and Test
- GMSF12 - Collect and Route External Entities Data (SFC4, SFC5, SFC7, SFC9)
- GMSF13 - Manage Databases

5.3.1.1 GMS ORGANICAL DESCRIPTION

The following figure shows the overall architecture of the GMS with the main data flows exchanged between elements:

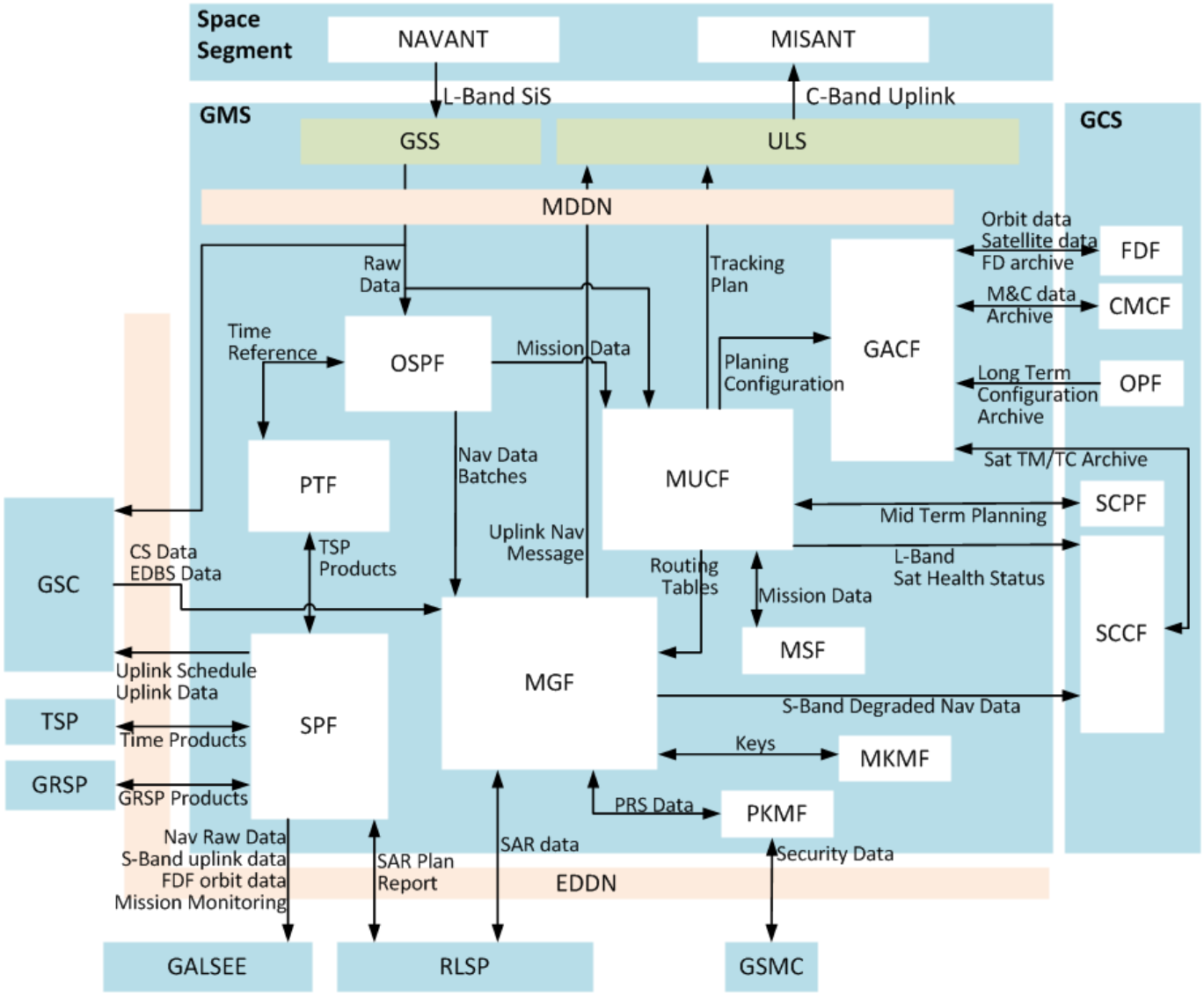


Figure 38: GMS Organical context

Note: In order to keep it readable, this context diagram does not represent all the links but shows only the most relevant ones.

In particular the following links are not drawn on the figure:

- *PTF to all elements (GMS and GCS) for GST Timing synchronisaiton*
- *GACF to all elements (GMS and a few in GCS) for M&C, archive retreival and configuration data distribution*

For detail interface between elements, refer to internal ICDs

The GMS is composed of the following element:

- GACF: Ground Asset Control Facility
- GSS: Galileo Sensor Station
- MDDN: Mission Data Dissemination Network (including GNMF)
- MGF: Message Generation Facility

- MKMF: Mission Key Management Facility
- MSF: Mission Support Facility
- MTPF: Maintenance and Training Platform
- MUCF: Mission and Uplink Control Facility
- OSPF: Orbit and Synchronisation Processing Facility
- PKMF: PRS Key Management Facility
- PTF: Precise Time Facility
- SPF: Service Products Facility
- ULS: Up-Link Station

The GMS includes a standalone RF Environment Estimating Facility (REEF), in charge of detecting and localizing basic Radio-Frequency (RF) interferences (jamming) targeting GSS, TTC and VSAT on GCC and GSS sites.
Further details on GMS elements are provided in the sections after.

The GMS interfaces with the following entities:
- SS: Galileo Space Segment
- GCS: Galileo Ground Control Segment
- GSC: Galileo Operating Company (interfacing with the Service Centers)
- GSMC: Galileo Security Monitoring Center
- GRSP: Geodetic Reference Service Provider (including interface with SLR Service Provider)
- TSP: Time Service Provider
- RLSP: Return Link Service Provider (COSPAS-SARSAT service)
- GPS system (GPS SIS for synchronisation purpose during IOV and GPS Ground Segment for GGTO)
- CSP: Communication Service Provider
- GALSEE: GALileo System Evaluation Equipments

5.3.1.2 GMS FUNCTIONS PHYSICAL MAPPING

The following table lists the system functions allocated to the GMS and the mapping to GMS elements:

Function ID	Function name	Mapping to Element
GMSF1	Generate SiS Observables	GSS
GMSF3	Schedule and Perform Message Dissemination	xxx
GMSF3.1	Format and Route Messages	MGF/MDDN/ULS
GMSF3.2	Up Link C Band Messages	ULS
GMSF3.3	Select Navigation Message	MGF/MDDN
GMSF3.5	Generate Uplink Schedule	MUCF
GMSF4	Generate Time and Geodetic Reference Data	xxx
GMSF4.1	Generate Coordinates Reference	GACF/GSS/MSF
GMSF4.2	Determine GGTO	PTF/MUCF/GACF/MSF
GMSF4.3	Generate System Time	PTF/GACF/MSF
GMSF4.4	Time Transfer	PTF
GMSF5	Determine Navigation	OSPF
GMSF5.1	Determine Orbit	OSPF

GMSF5.2	Determine BGD	OSPF
GMSF5.3	Determine SISA	OSPF/GACF/MSF
GMSF5.4	Determine Satellite Clock Parameters	OSPF
GMSF5.5	Determine Single Frequency Ionospheric Model	OSPF
GMSF5.6	Generate Navigation Batches	OSPF
GMSF6	Archiving	GACF
GMSF6.1	Global Archiving	GACF
GMSF6.2	Central Archiving	GACF
GMSF6.3	Local Archiving	All elements
GMSF7	Monitoring and Control	xxx
GMSF7.1	Integrated Mission Monitoring	GACF/MUCF
GMSF7.2	Monitor Mission Off Line	MUCF/MSF
GMSF7.3	Monitor Mission On Line	MUCF
GMSF7.4	Ground Asset Monitoring and Control	GACF
GMSF7.4.1	GMS Elements Monitoring and Control	All elements
GMSF7.4.2	GMS Central Monitoring and Control	GACF
GMSF7.4.3	Overall Monitoring and Control	GACF
GMSF7.4.4	Network Monitoring and Control	GACF/MDDN
GMSF8	Security	xxx
GMSF8.1	Security Protection of PRS	PKMF
GMSF8.2	Security Protection of SOL	MKMF, SPF
GMSF8.3	Security Protection for Mission Data	MKMF, SPF
GMSF8.4	Security Protection of CS	MKMF, SPF
GMSF9	Perform Mission Planning	MUCF
GMSF9.1	Long/Mid Term Planning	MUCF
GMSF9.2	Short Term Planning	MUCF
GMSF10	Provide Training	MTPF
GMSF11	Perform Simulation and Test	MTPF/MSF/MUCF
GMSF12	Collect and Route External Entities Data	SPF/MGF/MUCF
GMSF13	Manage Database	GACF/MTPF

Table 31: GMS Function List

Note: the mapping of functions per element is detailed in each element description.

5.3.1.3 GMS ELEMENT VIRTUALISATION

GMS non-Real Time elements are running on virtual servers which are provided by the Infrastructure Service Capability (ISC). This common infrastructure hosts the GACF, MUCF (including GRF), GNMF, MSF, SPF, part of VTP, and unclassified U-MKMF. The ISC is composed of a HW and a SW layer offering services (virtual machines) to the different hosted applications.

The HW layer incorporates:

- a GMS main cluster running the hosted applications
- dedicated machines for the SPF Back End (BE) and Front End (FE) execution

- Operation Positions (Working Stations) for operator interactions with the different applications/elements
- a dedicated HW for ISC administration (Management cluster), and
- network components for connectivity.

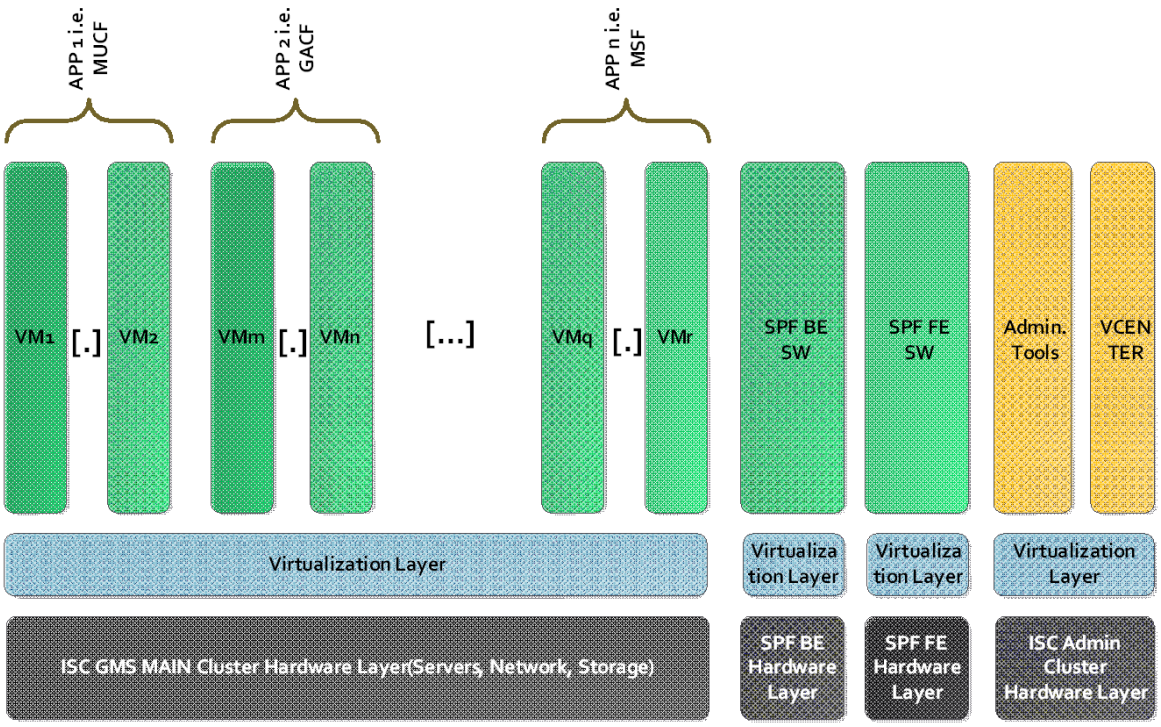


Figure 39: ISC infrastructure

The SW layer is spread over the servers of the GMS main cluster and is based on virtualization. In addition to the hypervisor present in each ISC server, the GMS SW elements are composed of one or more Virtual Machines (VM) running in the GMS Main Cluster.
To comply with security requirements, the SPF BE and FE run on dedicated hardware, but are based on VM as well to take advantage of the administration tools (i.e. deployment) offered by the virtualisation.

For the same rationale as for the SPF (mainly security aspects), a dedicated ISC administration cluster hosts the VCENTER for the virtualization administration and the other administration tools. Each administration tool is hosted by a VM, as for Vcenter. The identified tools are:

- vRealize Loginsight: ESXi – virtualization layer logs retrieval and collection,
- vrealize operations: Dashboard for infrastructure monitoring
- Sanitization/Built in Test: for VM isolation and built in tests run
- ISC User database for ISC User Management (based on Active Directory)
- ISC deployment VM: for automated deployment of the different tools on the different servers

The ISC architecture relies on virtualization to offer VM redundancies and therefore helps to improve global robustness. On top of this, the ISC has been sized to cope with several types of failures (see [RD.46 for more information):

- management cluster is based on two standalone independent servers and only one server is needed to host all the applications
- GMS main cluster relies on 16 servers whereas only 12 (+1 in case of maintenance activities) are needed to support GMS activities
- SPF BE and SPF FE rely on 2 servers each whereas only 1 server at each SPF is needed.

5.3.1.4 GMS OPERATIONS, VALIDATION AND SW UPGRADE CAPABILITIES

The physical implementation of the GMS allows supporting simultaneously live mission operations in parallel to validation testing activities allocating the relevant activities on redundant GMS elements of the mission chain, responsible for the real time processing of GSS raw data.

In particular the GMS is deployed in each GCC with 2 functionally “chains” of classified and unclassified equipment, namely:

- Operations Chain (OPE) that is used to support operations for the mission provision. The specific characteristics of this set of equipments are:
  - It is composed of 2 mission chains (GSS, OSPF, MGF, MNE, KMF) allowing to perform software reconfiguration activities on one of the 2 chains while the other one remains in operational mode. The OPE chain can only be used to support special operations tests, such as IOT, specific SFT (System Functional Test) where dissemination to satellite is requested.
  - The management of the chain can be achieved by configuring the relevant redundant elements in either “Operational” or “Running“ mode. No uplink to the satellite is executed when the chain is in “Running” mode.
  - GACF and MUCF allow monitoring and control of both “Operational” and “Running” chains.
  - Once a new SW/HW configuration is deemed validated in VAL chain, the deployment onto the operational facilities will follow a staggered approach whereby whenever possible, the new SW/HW configuration is initially activated in only one of the redundant elements of the mission chain to verify absence of anomalies before extending the change over the other chains.

- Validation Chain (VAL) is under responsability of GSOP and intended for Operations product preparation, validation, operators training, as well as other system tests (if not feasible on INT), The specific characteristics of this equipment set are:
  - The VAL chain receives the flux of OPE chain GSS data via a network diode . Only 1 side connection is possible (OPE towards VAL) while the opposite side traffic is blocked. This allows the VAL chain to replay in real-time the system behavior with possible different software for test purposes. ULS can be connected to VAL chain for test purposes but no real uplinks to satellites can be executed.
  - Another testing mode is possible by replaying past recorded data, using the MTPF.

- Integration Chain (INT), intended for system integration and testing deployed at GCC-I only. The specific characteristics of this chain are:
  - Based on a replica of the GMS with the exception of the PTF Hydrogen Maser, so that the corresponding clock signal is provided by the VAL chain.
  - Several Real Time data streams are provided by OPE chain, such as GSS raw data/NAV data, GSS and SV clock estimations from OSPF, CS and EDBS from the GSC
  - Non Real Time data (ephemeris, GRSP, TSP,…)
  - Connected via an optical diode and firewall to the OPE chain.

5.3.2 GMS ELEMENTS DESCRIPTION

5.3.2.1 GROUND ASSET CONTROL FACILITY (GACF)

5.3.2.1.1 HIGH LEVEL DESCRIPTION

The GACF is responsible for:

- the overall Monitoring and Control of the Ground segments (supervising GMS and GCS elements) including the maintenance management.
- the GMS central archive
- the Global Galileo archive for the whole life system including GCS long term archiving.

5.3.2.1.2 FUNCTIONS ALLOCATION

The GACF is responsible for the following main functions:

- High level control and overall monitoring, including GCS element monitoring report
- GMS Ground facilities monitoring and control
- Configuration control of GMS Element (software and hardware), including the mission configuration parameter dissemination (from MSF to OSPF, PTF)
- Maintenance activity management (commands, on-line/off-line reconfiguration, SW upgrades)
- Galileo System Archiving (Maintenance of the GMS Central and global Archive)
- Network data flow capture and storage

This covers directly the following GCS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF4.1	Generate Coordinates Reference
GMSF4.2	Determine GGTO
GMSF4.3	Generate System Time
GMSF5.3	Determine SISA
GMSF6	Archiving
GMSF6.1	Global Archiving
GMSF6.2	Central Archiving
GMSF6.3	Local Archiving
GMSF7.1	Integrated Mission Monitoring
GMSF7.4	Ground Asset Monitoring and Control
GMSF7.4.1	GMS Elements Monitoring and Control
GMSF7.4.2	Central Monitoring and Control
GMSF7.4.3	Overall Monitoring and Control
GMSF7.4.4	Network Monitoring and Control
GMSF13	Manage Database

Table 32: GACF Function List

Note: The local monitoring and control of the element is performed by each element

5.3.2.1.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the GACF context according to the GMS DDF2 (GMS-RD.2), GCS-GMS-ICD (AD.8):

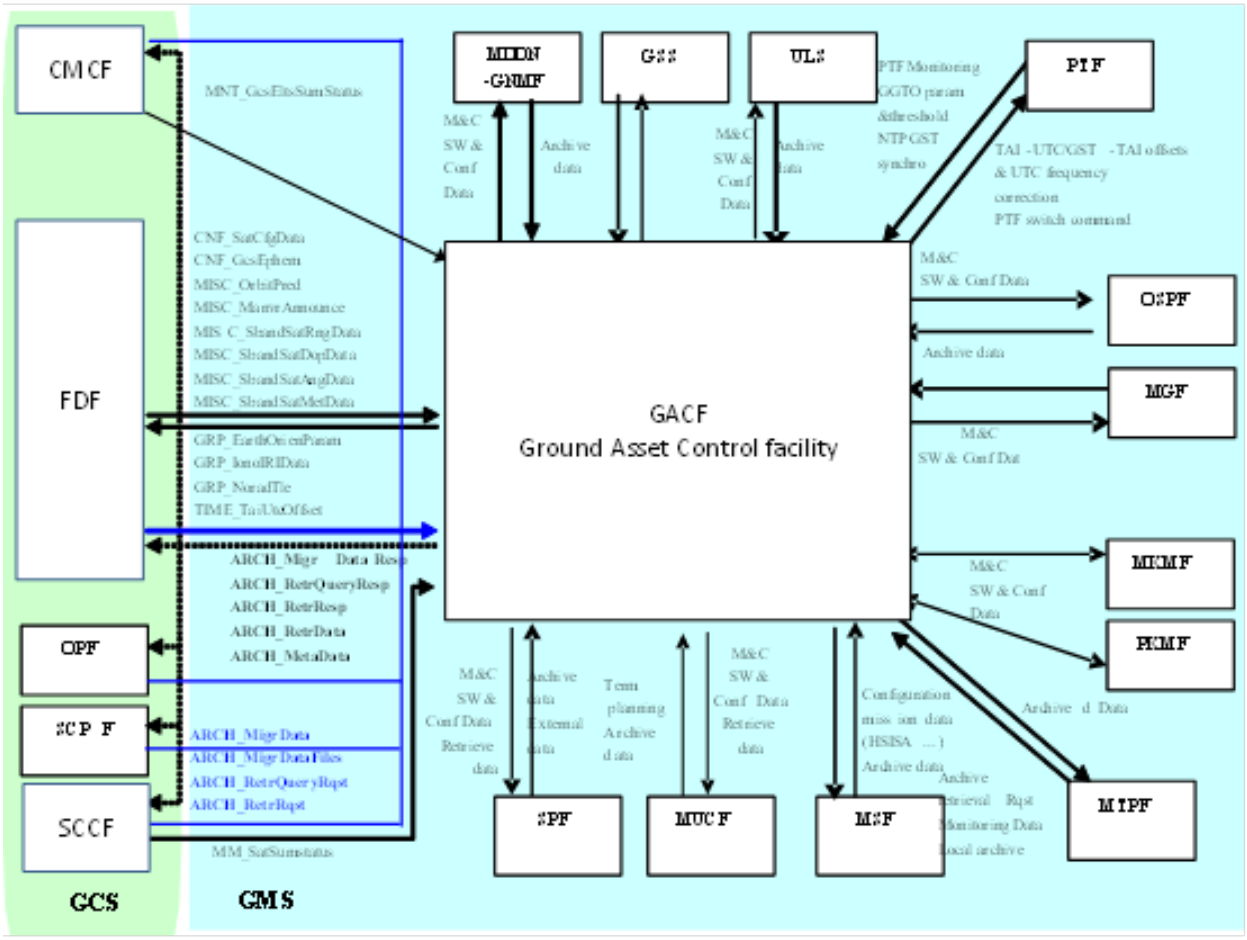


Figure 40: GACF Organical context

5.3.2.1.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.1.4.1 SYSTEM LEVEL CONSTRAINTS

Performances:

- Storage capability from GCS to GACF Global Archive is limited to 1 GB / day
- Archive retrieval capability from GACF to any GCS element is limited to 10 GB

Protocols:

- Files exchanged between the GACF and the GCS use a format based on the XML language. The archive and retrieval requests and responses also use the XML language.
- The SNMPv3 protocol is used by the GACF for technical monitoring.

5.3.2.1.4.2 MAJOR SEGMENT DESIGN CHOICES

Redundancy Management:

- M&C: One GACF instance is deployed in each GCC. On a per domain basis, each GCC is either active (with the full monitoring and control capability) or passive (with monitoring

capability and limited control capability). A unique technical M&C working position is authorised to supervise the assignment of domains to GACFs..

- Archive: The GACF ensure the consistency between the GACF Global Archive in the two GCCs, this is managed by an automated process to merge the data.
- Handover: The GCC handover regarding the GACF M&C operations can be a seamless transition thanks to the:
  - Synchronisation between archives and monitoring data between both GCCs
  - Mirroring of the maintenance configuration database

Operability:

GACF is permanently manned by staff and have different categories of users:

- Observer: Only interested in the overall status of the GMS;
- GACF Operator: Responsible for manning the GACF
- GMS Element Responsible: In charge of in-deep analysis & troubleshooting
- GMS Operations Supervisor: on occasion make use of the GACF to oversee the nominal operations as well as to perform troubleshooting or unplanned emergency reconfiguration activity

The GACF does not transit in operational State, whenever occurs a failure in one of the following functions:

- Technical monitoring
- Commanding
- Maintenance Support
- GMS Central Archive
- Galileo Global Archive
-

Performances constraints:

The Global storage capability of the GACF is:

- 2000 GB for the Central archive
- 100 GB / day for 20 years for the Global Archive

Number of terminals:

The numbers of terminals dedicated for the GACF are:

- 3 for the GACF on-line activity,
- 4 for the GACF off-line activity

5.3.2.2 GALILEO SENSOR STATION (GSS)

5.3.2.2.1 HIGH LEVEL DESCRIPTION

The GSS is responsible for collecting the SIS observables for Navigation and Integrity and providing those data to the navigation and integrity processing chains.

5.3.2.2.2 FUNCTIONS ALLOCATION

According to GSS DDF (GMS-RD.37) the GSS is responsible for the following main functions:

- Reception and demodulation of the four carrier frequencies (L1, E5a, E5b & E6)
- Provision of Raw Data measurements and observables to the GCC

- Generate a local realization of the GST
- Management of the CS and PRS keys
- Local monitoring and control

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF1	**Generate SiS Observables**
GMSF4.1	**Generate Coordinates Reference**
GMSF6.3	**Local Archiving**
GMSF7.4.1	**GMS Element Monitoring and Control**

Table 33: GSS Function List

The GSS are involved in the following system functional chains:
SFC4.1 Open Services (OS) provision
SFC4.3 Public Regulated Services (PRS) provision
SFC5.1 Ground Asset Monitoring and Control
SFC6 Archiving

5.3.2.2.3 ELEMENT ORGANICAL CONTEXT

The following context diagram shows the GSS location inside the Galileo core system:

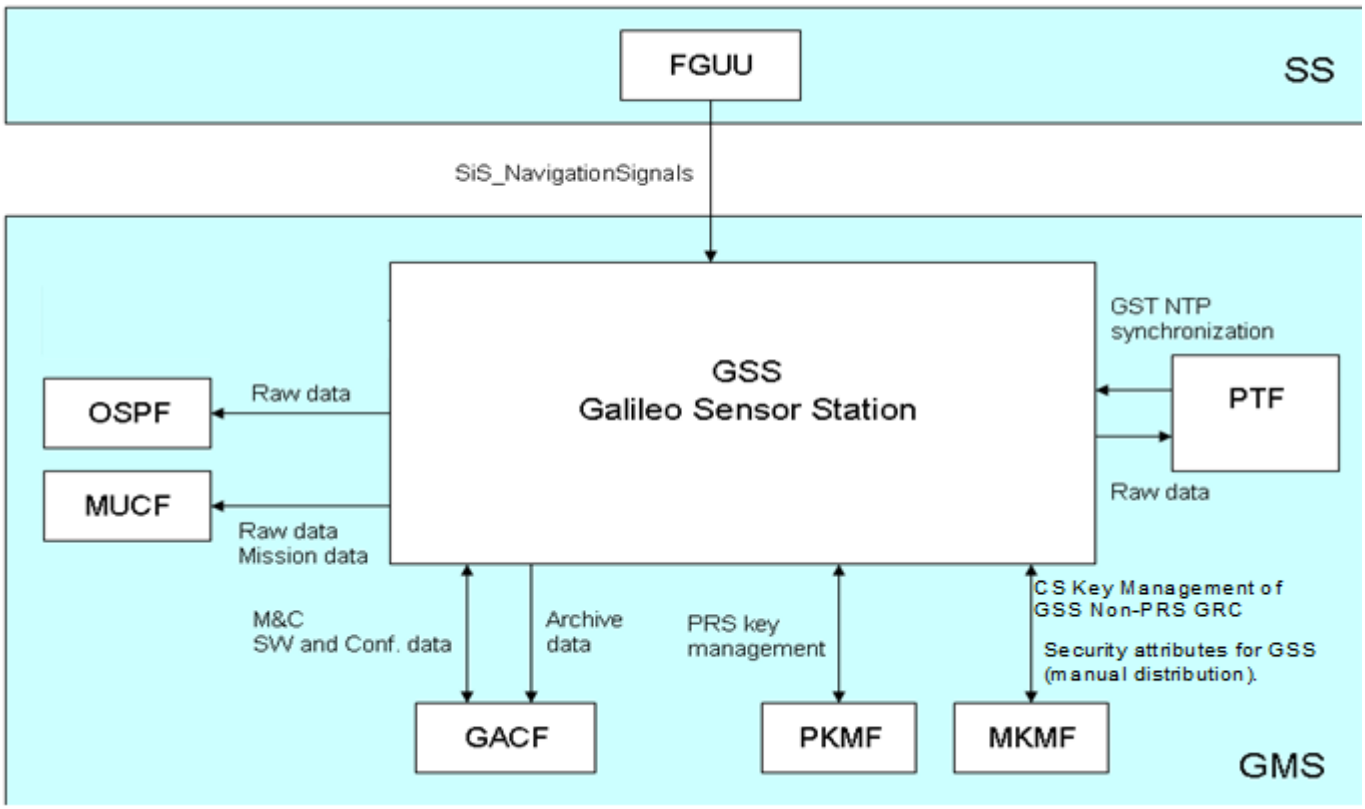


Figure 41: GSS Organical context

5.3.2.2.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.2.4.1 SYSTEM LEVEL CONSTRAINTS

- GSS Station constraints

  The GSS sites are constrained by the requirements specified in GSSSIRD [AD.25].

  The GSS are capable of processing simultaneously all the signal types related to each of the Galileo services:
  - $E5a_I$, $E5a_Q$, $L1_B$ and $L1_C$ for F/NAV
  - $E5b_I$, $E5b_Q$, $L1_B$ and $L1_C$ for I/NAV
  - E5 AltBOC
  - $E6_B$, $E6_C$, $L1_B$ and $L1_C$ for C/NAV
  - $L1_A$ and $E6_A$ for PRS

  Upon reception of the satellite signals, the GSS performs the following measurements on each carrier:
  - Code phase measurement
  - Carrier phase measurement (accumulated Doppler satellite acquisition)
  - Signal C/N0 ratio
  - Code carrier phase coherency

  Each station provides the navigation observables and measurements for every satellite frequencies n view with a 1-second sampling rate.

  The GSS measurements are time stamped with the local realization of the GST (Galileo System Time), derived from the atomic clock in the GSS.

5.3.2.2.4.2 MAJOR SEGMENT DESIGN CHOICES

The GSS are composed of the following sub-elements:
- Galileo receivers, including an antenna and a preamplifier
- GSS core computer
- An atomic clock and its associated distribution unit
- A PPS buffer board

All these sub-elements are housed in a single rack.

The following scheme shows the GSS architecture:

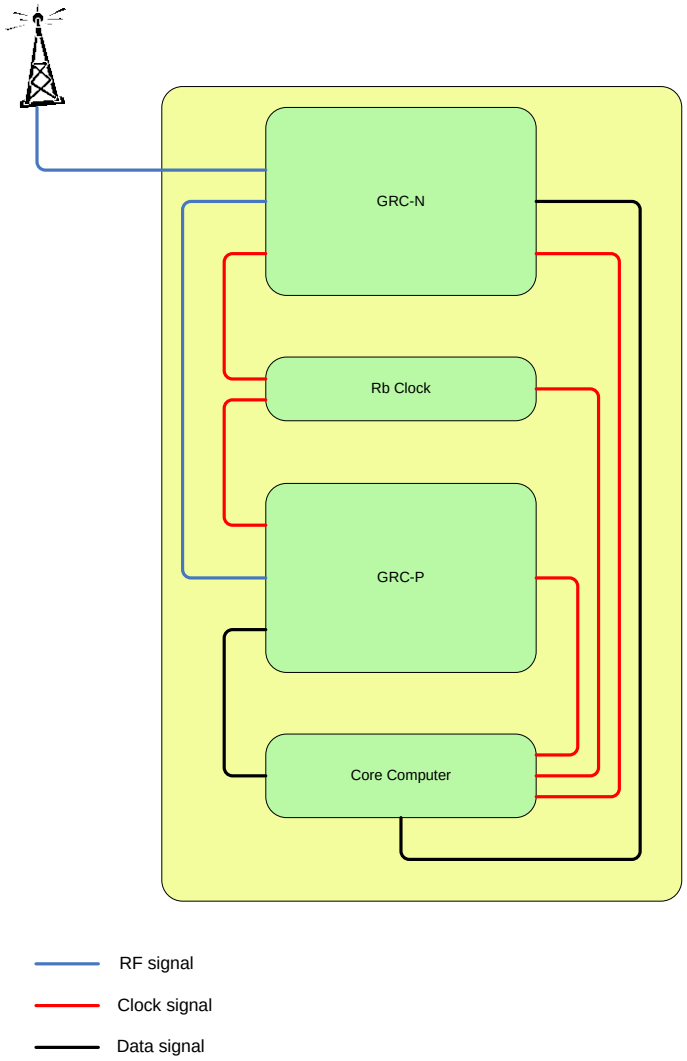


Figure 42: GSS architecture

The GSS are mainly composed of Galileo receivers, whose purpose is to measure and provide the SIS observables to the Core Computer. The GSS receivers provide have the following main functions:

- Down conversion, acquisition, tracking and detection
- Observables
- Signal anomalies (including interference, multipath or cycle slip) / distortion detection and mitigation if possible
- Inter-frequency bias compensation
- Internal calibration
- Synchronization to GST and provide pulse to core at each start of time epoch
- M & C interface with Core Computer
- interface with Core Computer

- **Non PRS receiver**

The receiver providing the non PRS observables is called non PRS Receiver (GRC-N). This receiver is responsible for Open Service (OS) and Commercial Service (CS).

The GRCN is composed of the following major items:

- Antenna, LNA (low noise amplifier) and RF filter which interfaces with Galileo SIS and GPS SIS.
- GSR (GALILEO Station Receiver) with the following modules:
  - Triplexer, performing the distribution of the RF signals from the antenna to the GRC-N internal RF board and to the GRC-P
  - Radio Frequency Front End
  - Galileo RF/IF/CM to process the Galileo signals
  - IO-master to interface for data and M&C
  - GPS Receiver to provides GPS data and M&C
- Rubidium clock / frequency standard for output of 10 MHz reference clock signal to both GRC-N and GRC-P.

The following diagram shows the GRC-N architecture:

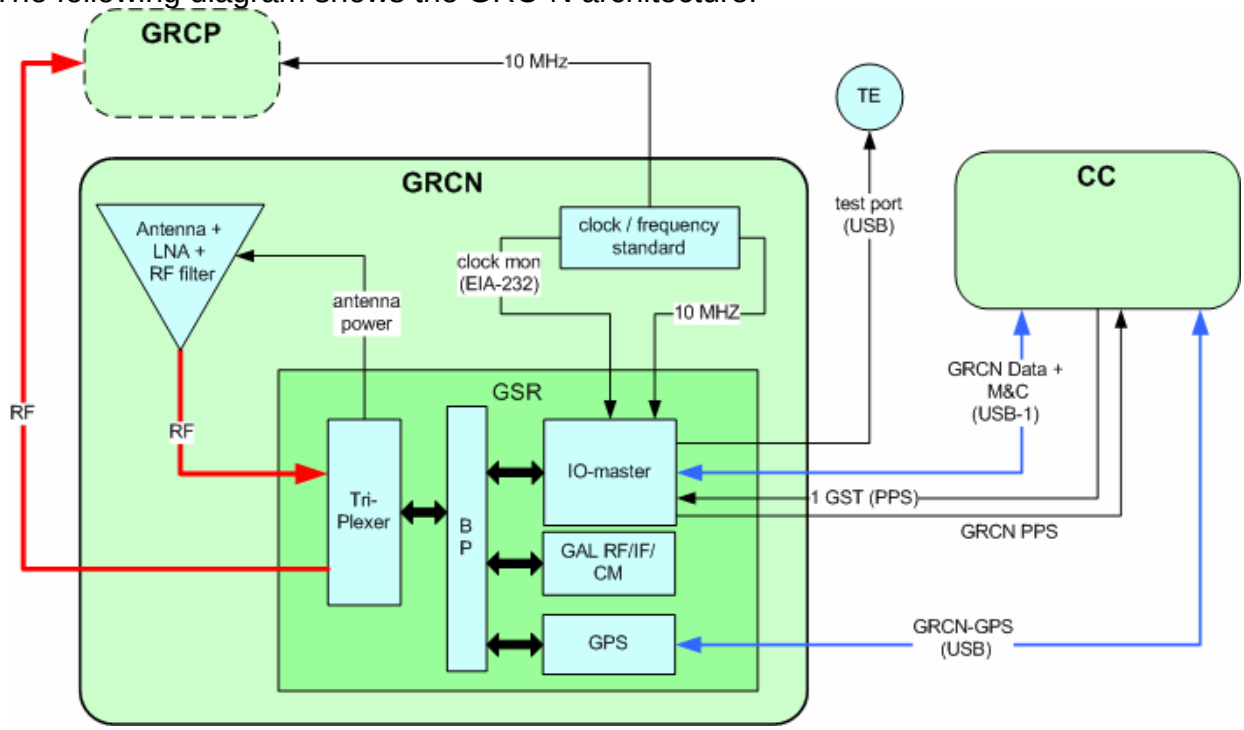


Figure 43: GRC-N architecture

- **PRS receiver**

The receiver providing the PRS observables is called PRS Receiver (GRC-P).
This receiver shares antenna and clock with the non PRS receiver. The PRS Receiver will contain a crypto board for security functions, and a TEMPEST rack module.
The GRCP is composed of the following major items:

- PRS Receiver which has the following modules:
  - A Radio Frequency Front End
  - A Galileo RF/IF to process the GALILEO signals
  - A Core Module to interface for data and M&C
  - A Crypto Module for GRC-P security functions
- GPS Receiver which provides GPS data and M&C

The following diagram shows the GRC-P architecture:

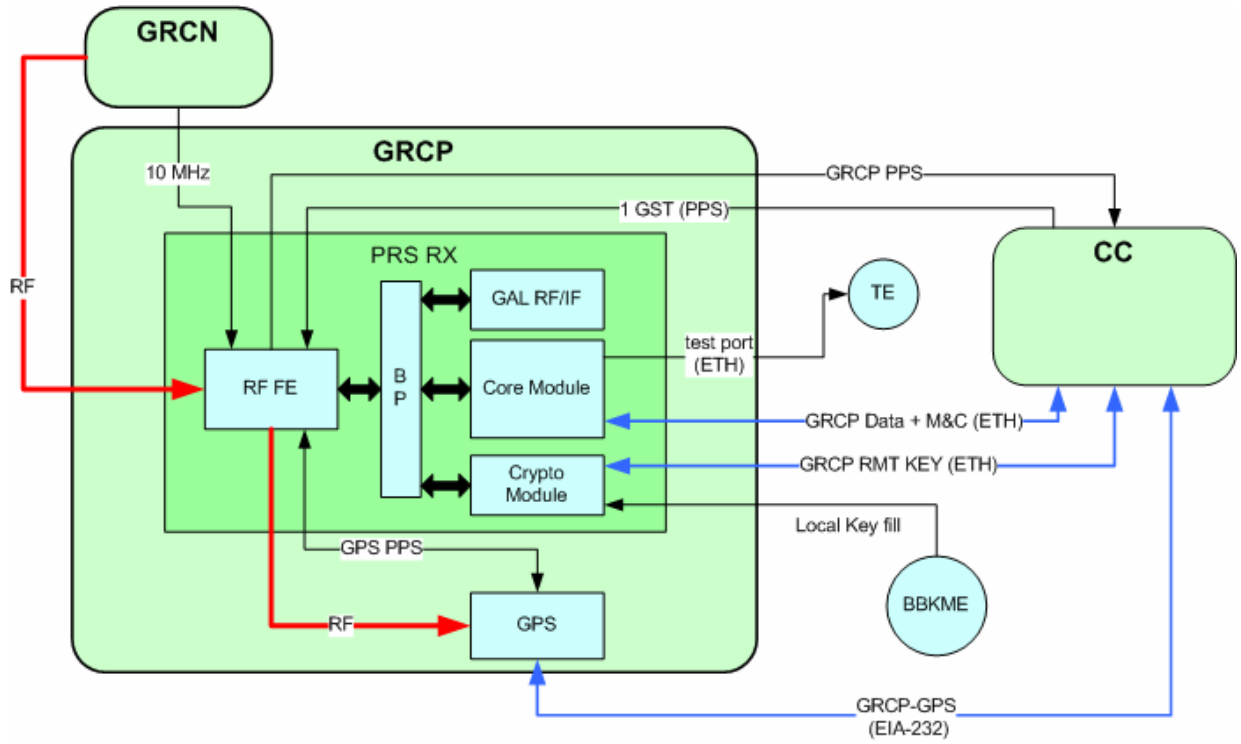


Figure 44: GRC-P architecture

Note: Both GRC-P and GRC-N will contain a COTS GPS receiver. For GRC-P, the GPS is delivered as a separate rack module, whereas it is integrated into the GRC-N rack module.
The GPS receivers will be kept for both the IOV and FOC phases. During the IOV phase the GPS receiver is necessary for time synchronization of GSS. The GPS receiver is a COTS product which is capable of processing GPS L1 and L2 signals. The GPS receivers share antenna and clock with Galileo receivers.

- **GSS Core computer**

The main purpose of this component is to receive observables from the GSS receivers, site sensors and to send those observables to the processing and archiving elements at Control Center (GCC). The transmission is made via MDDN. In addition, the GSS Core is performing the monitoring and control of the GSS. As such, it provides an interface between the GACF at the GCC and each component at the GSS.
The core consist of two main sub-components, the GSS Monitoring & Control board, which acts as the local M&C Agent and local archive and the GSS SIS Collector that receives raw measurements and navigation data from the receivers, and transfers this data to the GCC through the MDDN via the MNE co-located at the GSS site.

- **Time synchronization of the GSS**

The GSS receivers shall reconstruct the GST from the Galileo observables based on a PVT solution, when synchronized to its external clock.
When this is not possible (during the IOV phase for example), the GSS shall use the GPS constellation or as a back-up the disseminated GST provided by the network (NTP).

- **GSS PTF**

Two redundant GSSs will also be installed in each GCC, co-located with the PTF. In those cases, the configuration of the GSS is slightly different:

- The GSS has no clock but is connected to the PTF, which is used to synchronize the receivers
- Each GSS is fit with a set of calibration tools for the PTF to GSS and GSS receiver to GSS antenna path calibration

5.3.2.3 MISSION DATA DISSEMINATION NETWORK (MDDN-GNMF)

5.3.2.3.1 HIGH LEVEL DESCRIPTION

The MDDN is responsible for providing all the needed functionalities to ensure communication between GMS elements within the GCC and:

- the GMS remote sites (namely ULSs and GSSs), via the GDDN WAN: ensured by the MDDN network equipments.
- The GMS External Entities: ensured by the M-EDDN network equipments

In addition, the MDDN includes a dedicated facility, the GNMF, responsible for the network performance M&C. This includes:

- Network HW component monitoring and control (for any network equipment, i.e. MNEs and SDDN including the GCC LAN). This task consists in:
  - Fault Management, i.e. management of the status of the network equipment (gathered from receiving SNMP traps (alarms) and performing regular polling of relevant equipment status information). A summarizeed fault management information is forwarded to GACF.
  - Network M&C, i.e. network administration, discovery & topology, inventory management, fault management, status monitoring, troubleshooting, event console, etc …
- Network performance monitoring

Finally, as part of MDDN, the GMS is responsible for procuring the GMS RT and nRT LANs within the GCC and the GSSs/ULSs.

5.3.2.3.2 FUNCTIONS ALLOCATION

According to GMS DDF2 (GMS-RD.2) the MDDN is responsible for the following main functions:

- Transfer of the raw navigation acquired by the GSSs to the GCCs
- Transfer of the GCC processed data to the ULSs
- Transfer of data between GCC's
- Interface with External Entities
- Management of the GCS network
- Management and assessment of the overall Galileo network performance, including, the MDDN itself and the GCS SDDN. This network monitoring and control function also applies for the GDDN which the WAN part of the two above networks. Management of the overall Galileo networks is critical for the mission monitoring but also for the assessment of the fulfillment of GGDN CSP service level agreement.

The MDDN functionally interfaces with:

- the SDDN for SDDN management purpose
- the GACF for network overall monitoring and control
- the CMCF give information about the GCS networks status

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name

GMSF3.1	**Format and Route Messages**
GMSF3.3	**Select Navigation Message**
GMSF6.3	**Local Archiving**
GMSF7.4.1	**GMS Elements Monitoring and Control**
GMSF7.4.4	**Network Monitoring and Control**

Table 34: MDDN-GNMF Function List

5.3.2.3.3 ELEMENT ORGANICAL CONTEXT

- **MDDN Context Diagram:**

The following figure shows the MDDN context:

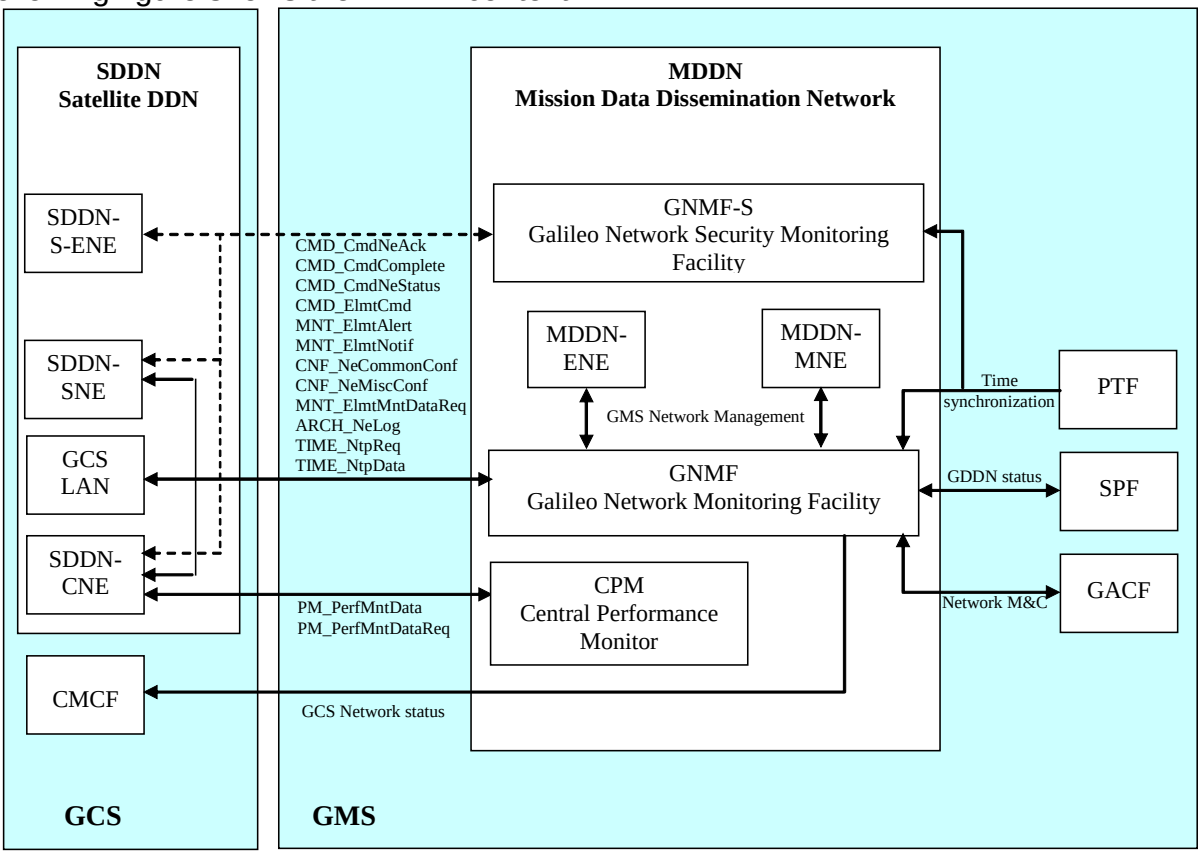


Figure 45: MDDN-GNMF Organical context

- **M-EDDN Context Diagram:**

The following figure shows the M-EDDN context:

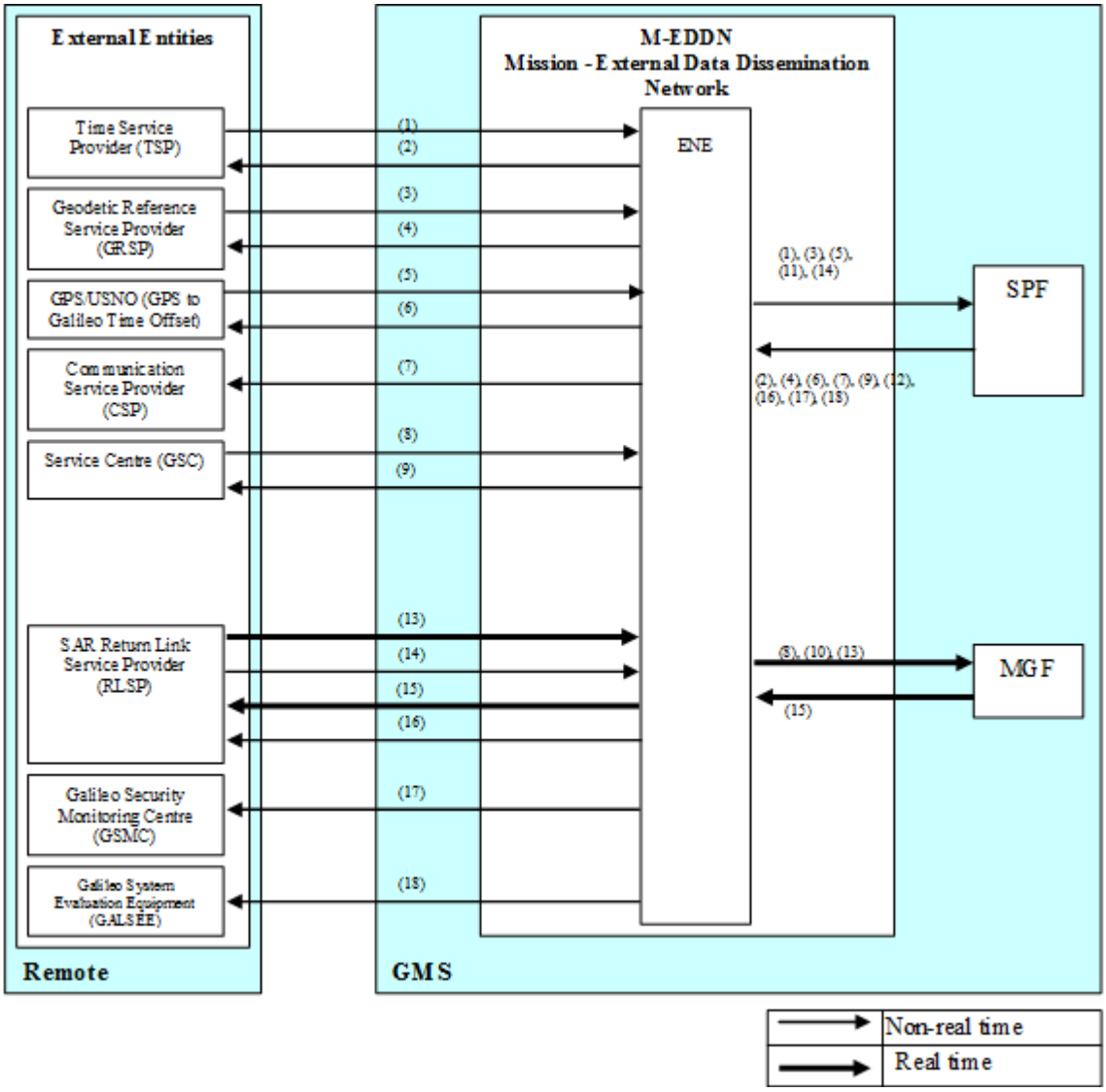


Figure 46: M-EDDN Organical context

where:

(1)	Timing Related Products to Galileo
(2)	GST &Timing Related Products to TSP
(3)	Data to implement GTRF, Report Request
(4)	Products to GRSP, Pseudo-range and phase measurement Galileo and GPS including navigation message, satellite configuration data.
(5)	Time-related info. provided by GPS for GGTO computation & Information on planned and unplanned events impacting GGTO performance
(6)	Time-related info. provided by Galileo to GPS for GGTO computation & Information on planned and unplanned events impacting GGTO performance
(7)	GDDN Status Report
(8)	Commercial Data for uplink, Report Requests, Key Management Data,
(9)	Commercial Service Grids, Key Management Data, Galileo System

	Performance Reports and bulletins
(13)	SAR Return Link Message Requests
(14)	SAR Report Requests
(15)	SAR RLM flow Control Info
(16)	SAR Status Reports and Bulletins
(17)	Galileo Status report
(18)	Galileo System Information for GALSEE evaluation

5.3.2.3.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.3.4.1 System Level Constraints

The following provides a high level overview of the general constraints on the MDDN, and constraints relevant to its sub-elements.

In the context of a high level description of the MDDN, any reference to GNMF can be considered as being applicable also to the GNMF-S.

Network Performance Constraints

Since the GDDN is provided by an external CSP, it is necessary to monitor the negotiated Service Level Agreement (SLA) in order to verify its fulfillment.
The Performance Monitoring agents at the MNEs and SNEs will collect performance data of all traffic streams going across the GDDN and generate QoS stamps that will be used by the GNMF to derive end-to-end performance data and check the fulfillment of the CSP SLA.
Three types of measurements are made:

- Delay (sometimes called latency):
  - the traffic latency is measured by the MNEs and SNEs, taking into account the time at exit from the sending site with that when entering the receiving site and using synchronization of all Galileo sites to the GST. This latency measurement is of particular importance for operational flows and associated budgets.
  - if the delay is out of limit this leads to the generation of an alarm, which is forwarded at the end of the M&C chain to the GNMF.
- Traffic integrity or packet loss measurement
  - the MNEs and SNEs at each side of the links are regularly comparing the amount of packets of each type. Moreover, by means of a CRC check the received packets are controlled on errors.
  - If packet loss is over the limit, this leads to the generation of an alarm, which is forwarded at the end of the M&C chain to the GNMF.
- Network availability
  - Using the detected packet loss and time-stamps of received packet as described above, actual reception and sequence of each critical packet is checked.
  - As a result, even interruptions that last only a few seconds can be detected, and a highly precise availability can be calculated.
  - This availability control mechanism focuses on the WAN network performances provided by the CSP, since the tagging and evaluation are done right at the WAN endpoints.

Network Interfaces

- GNMF interfaces:

In order to properly monitor and control the networks, the GNMF interfaces with:

- GACF for:
  - transfer of the overall network availability status, monitored from the MNES and SNEs and required as part of the overall monitoring and control
  - reception of the network management commands (redundancy activation, traffic priority management, …)
- MUCF for:
  - transfer of some network performance assessment and network availability status required for integrated performance monitoring
- CMCF for:
  - transfer of the GCS network availability status, since GCS networks is not directly managed by the CMCF

- Interface with GACF

The GACF does not control the GNMF and the networks directly, this is performed locally from GNMF. The GACF can issue commands to GNMF only to modify passwords and to support archive migration.

- Interface with MUCF

The MUCF is in charge of the overall mission performance assessment and takes therefore the network performance issue into account.

For this purpose, the MUCF interacts with the MDDN for the transfer of the GNMF network performance assessment.

- Interface with CMCF

The CMCF is part of the GCS segment, and is responsible for the reception, from the GNMF, of summary status information on the local GCS networks, and the processing and storage of such information.

For this purpose, the CMCF interacts with:

- the MDDN-GNMF in GMS for:
  - GCS network monitoring status reception
  - GCS summary status transfer, including GCS networks summary status received from GACF

Real-Time and Non-Real Time data flows

The MDDN is involved in the GSS Real Time (RT) and non-Real Time (nRT) and ULS RT and nRT system functional chains:

- **GSS RT Functional Chain**

GSS RT Functional Chain data consists of a unidirectional flow of GSS Raw data

At the ground stations, GSS sensors connect to GSS-MNEs via the ground station LAN, which then connects to the GDDN.

At the GCCs, the GDDN connects to the GCC-MNEs, which are part of the MDDN. The MDDN then connects over the GCC GMS RT LAN, which is specified and procured as part of the MDDN, to the OSPF, where the (Navigation) is processed.

- ULS RT Functional Chain
ULS RT Functional Chain data consists of a unidirectional flow of GMS navigation data
At the GCCs, the processed MGF data is sent over the GCC GMS RT LAN, which is specified and procured as part of the MDDN, to the GCC-MNEs which connects to the GDDN.
At the ground stations, the GDDN connects to ULS-MNEs, which then connect, via the ground station LAN to the ULS antenna, where the data is then up-linked to the Galileo satellites.

- GSS nRT Functional Chain
GSS nRT Functional Chain data consists of a bi-directional flow of non-Navigation data, such as M&C data, between GSS components in the ground stations, and the GNMF in the GCCs.
At the ground station, data is transported between the GSS components and the GDDN, over the same physical path as the GSS RT data, to / from a different port on the GSS-MNE.
At the GCCs, the GDDN connects to the same GCC-MNEs used for the RT Chains, which are part of the MDDN. The MDDN then connects over the GCC GMS nRT LAN, which is specified and procured as part of the MDDN, to the GNMF. The GCC GMS nRT LAN is a physically separate infrastructure from the GCC GMS RT LAN.

- ULS nRT Functional Chain
ULS nRT Functional Chain data consists of a bi-directional flow of non-Navigation data, such as M&C data, between ULS components in the ground stations, and the GNMF in the GCCs.
At the ground station, data is transported between the ULS components and the GDDN, over the same physical path as the ULS RT data, to / from a different port on the ULS-MNE.
At the GCCs, the GDDN connects to the same GCC-MNEs used for the RT Chains, which are part of the MDDN. The MDDN then connects over the GCC GMS nRT LAN, which is specified and procured as part of the MDDN, to the GNMF. The GCC GMS nRT LAN is a physically separate infrastructure from the GCC GMS RT LAN.

5.3.2.3.4.2 Major Segment Design Choices

Redundancy:

The MDDN components are drivers for real-time data flows, under stringent performance and continuity requirements,
In order to achieve these performance, the network architecture implement the needed redundancy (double GCC and multiple lines per WAN link) while taking care of carefully mixing non-Real Time (nRT) traffic with the performance sensitive Real Time (RT) flows.

GSS Network equipment architecture:
The GSS-MNE, located in each remote site supporting a GSS station, connects the GSS station to the GDDN. The GSS-MNE architecture consists of 2 identical lanes, as derived by RAMS analyses, on the basis of the availability constraints for FOC.
Each lane is equipped with

- a computer devoted to RT flow handling,
- a Monitoring and Control (M&C) computer,

- a switch-router, used to connect the site components to the gateway
- IP Security (IPSec) VPN gateway used to protect the traffic, through IPSec encapsulation, and of the site, through layer 3/4 packet filtering,
- an Ethernet 10/100 Base-TX interface that is converted to a fibre optic interface through an media converter, allowing each GSS-MNE instance to interface with the GDDN via either a copper or a fibre port.

Each lane connects to a different switch on the GDDN, and each switch on the GDDN connects to a different line (or path) to each of the GCCs.
Note that, in IOV, relaxed availability and continuity requirements render multiple lanes and GDDN lines unnecessary. Therefore all GSS sites are initially equipped with a single-lane GSS-MNE.

ULS Network equipment architecture:
The ULS-MNE, located in each remote site supporting a ULS station, connects the ULS station to the GDDN. The basic ULS-MNE architecture is the as that of the GSS-MNE, however, there are some significant differences, which are identified here.
The standard ULS-MNE architecture consists of two non-identical lanes, as derived by RAMS analyses, on the basis of availability constraints for FOC. Each antenna must connect to a different switch-router. This results in the following possible configurations:
For a standard, two-lane location, the first lane is equipped with

- a computer devoted to RT flow handling,
- two (for a 4 ULS antenna site) or three (for a 5 ULS antenna site) switch-routers,
- a Monitoring and Control (M&C) computer,
- IP Security (IPSec) VPN gateway used to protect the traffic, through IPSec encapsulation, and of the site, through layer 3/4 packet filtering,
- an Ethernet 10/100 Base-TX interface that is converted to a fibre optic interface through an media converter, allowing each GSS-MNE instance to interface with the GDDN via either a copper or a fibre port.

The second lane is equipped the same, except for the number of switch routers: there are two installed.

Each lane connects to a different switch on the GDDN, and each switch on the GDDN connects to a different line (or path) to each of the GCCs.

GCC- GMS-MDDN Network equipment architecture:
At a GCC, a single subsystem, the GCC-MNE, simultaneously interfaces to the GSS Subnet and the ULS Subnet.
The GCC-MNE, located in the GCC, connects (via the GMS-LAN) GMS elements in the GCC to the WAN network (GDDN). The GCC-MNE presents the same basic architecture as that of the GSS- and ULS-MNEs, with some significant differences, which are described below.
The GCC-MNE architecture consists of three non-identical lanes, as derived by RAMS analyses, on the basis of the combined continuity and availability constraints for FOC.
Each lane is equipped with

- Two computers devoted to RT flow handling,
- a switch-router for aggregated GSS flows, (i.e., GCC-MNE lane 1 receives the aggregated lane 1 traffic from all GSS-MNEs, GCC-MNE lane 2 receives the aggregated lane 2 traffic from all GSS-MNEs.

- a Monitoring and Control (M&C) computer,
- IP Security (IPSec) VPN gateway used to protect the traffic, through IPSec encapsulation, and of the site, through layer 3/4 packet filtering.
    - 1 gateway for aggregated GSS flows, (i.e., GCC-MNE lane 1 receives the aggregated lane 1 traffic from all GSS-MNEs, GCC-MNE lane 2 receives the aggregated lane 2 traffic from all GSS-MNEs)
    - 1 gateway per ULS ground station

Additionally, a cluster of two redundant firewalls is placed between the GCC-MNE and the GCC LANs, for the nRT traffic flows, with one firewall cluster member logically part of each of the first two GCC-MNE lanes.
The Ethernet ports of GCC-MNE RT computers on GCC-internal side are directly plugged in the GMS RT LAN (different lanes connected to different switches of the LAN). Note that, since the GCC-MNE and GDDN equipment are closely located to each other, there is no need for a fibre-copper converter in the GCC-MNE.

GMS-GCC-M-EDDN Network equipment architecture:
There are two types of M-ENE installed at the GCC; the Real-Time M-ENE, and the non-Real-Time M-ENE.
The non-redundant RT M-ENE subsystem shares much of the specific RT handling functions with the MNE. The M-ENE contains,

- a dedicated RT computer, based on the common MNE RT software, connecting to the GMS RT LAN
- a firewall
- a switch-router
- taps, for connecting latency measurement and intrusion detection equipment
- a VPN gateway, providing a copper, Ethernet connection to the EDDN

The non-redundant nRT M-ENE subsystem contains,

- two firewalls, each from different manufactures,
- taps, for connecting packet capture and intrusion detection equipment,
- a switch-router,
- a VPN gateway, providing a copper, Ethernet connection to the EDDN.

5.3.2.4 MESSAGE GENERATION FACILITY (MGF)

5.3.2.4.1 HIGH LEVEL DESCRIPTION

The Message Generation Facility (MGF) is the meeting point for all the data streams needed for generating the C-Band Uplink Messages.

The MGF is then responsible for:

- selecting the OSPF data flows (on the basis of QoS information)

- generating messages as specified in the GMSSSICD: by multiplexing the input data and generated data needed to provide the OS, CS, PRS, SAR services.

- interacting with MKMF and PKMF for data protection (authentication and encryption when needed)

- generating a QoS indicator to reflect the presence of PRS data in the produced sub-frames

- performing the sequencing of mission data dissemination to the ULS according to pre-defined routing tables provided by the MUCF.

In addition, the MGF produces every 10 minutes a selected subset of navigation data (F/NAV and G/NAV excluding integrity pages) to be up-linked through TTCS (after transmission to GCS).

5.3.2.4.2 FUNCTIONS ALLOCATION

According to GMS DDF2 (GMS-RD.2) the MGF is responsible for the following main functions:
- Elaborate messages as specified in the GMS to Space ICD by multiplexing the data coming from OSPF, SAR and GSC.
- Multiplexed streams to be forwarded to the ULS according to predefined routing tables provided by the MUCF.
- Transmit messages dedicated to the GCS (Degraded Navigation Service)

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF3.1	Format and Route Messages
GMSF3.3	Select Navigation Message
GMSF6.3	Local Archiving
GMSF7.4.1	GMS Elements Monitoring and Control
GMSF12	Collect and Route External Entities Data

Table 35: MGF Function List

5.3.2.4.3 ELEMENT ORGANICAL CONTEXT

The MGF interacts with the following elements:
- SCCF in the GCS to provide the S-band degraded navigation data

- ULS to provide the uplink navigation navigation messages
- PTF to be synchronized with the Galileo System Time
- MUCF to receive the Galileo mission data and SAR RLM and CS dissemination routing tables
- Direct connection to GSC to receive the commercial data to be uplinked
- Interaction with PKMF to encrypt the PRS data
- Interaction with the MKMF to authenticate the uplink navigation messages
- RLSP provider to receive the SAR data to be uplinked
- RLSP to receive warning information regarding inability to process requests or capacity restrictions on a particular area
- OSPF to receive all navigation data batches to be uplink through the S-band and C-band.

The following figure shows the MGF context according to the GMS DDF2 (GMS-RD.2), GCS-GMS-ICD (AD.8), GMSExtIRD (AD.11):

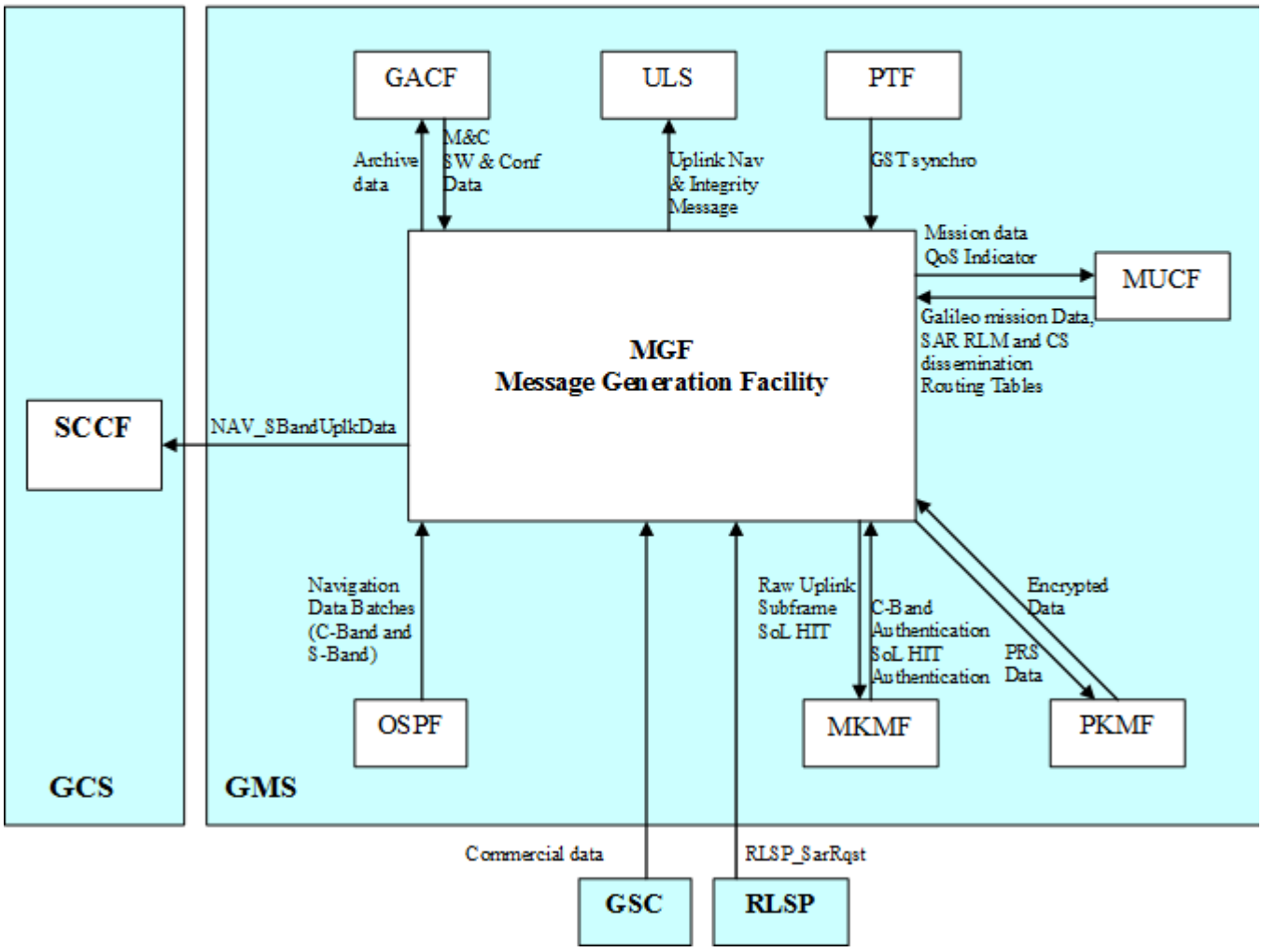


Figure 47: MGF Organical context

5.3.2.4.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.4.4.1 SYSTEM LEVEL CONSTRAINTS

Performances: real-time ummaned element

5.3.2.4.4.2 MAJOR SEGMENT DESIGN CHOICES

Redundancy:

Three instances are working in parallel in each GCC.

Data management flexibility:
Needed wide flexibility in the change of message format and additional new satellites, that without redesign of the code (MGF-ERD-0990)

Operability:
The MGF is a fully automated element (no interaction with any human user for nominal GMS operations, and no operational Man Machine Interface). Operations of MGF by human users are performed through other elements, such as the MUCF (Mission Monitoring) and the GACF (Monitoring and Control).

5.3.2.5 MISSION KEY MANAGEMENT FACILITY (MKMF)

5.3.2.5.1 HIGH LEVEL DESCRIPTION

The MKMF is responsible for the key management and data protection management for the C-band protection, the key management of the CS 3rd carrier encryption as well as the management of the various Security Modules (SMs) related to these issues.

5.3.2.5.2 FUNCTIONS ALLOCATION

The MKMF is responsible to perform the following main functions:

- Key generation for CS 3rd carrier encryption, C-band up-link protection:
  - o generating all the keys needed by the satellite NSGU
  - o generating all the keys needed by the GSS CS
  - o generating all the keys needed by the CS user terminals
  - o generating all the keys needed by the MKMF C-band up-link protection
- Manual and automatic distribution of the keys to the relevant ground and space SMs and SUs (directly or through the relevant service provider)
- Authentication of the C-band messages
- Monitoring and control of all the relevant SMs and SUs.

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF6.3	**Local Archiving**
GMSF7.4.1	**GMS Elements Monitoring and Control**
GMSF8.3	**Security Protection for Mission Data**
GMSF8.4	**Security Protection of CS**

Table 36: MKMF Function List

5.3.2.5.3 ELEMENT ORGANICAL CONTEXT

The MKMF organical context is shown in the following figure:

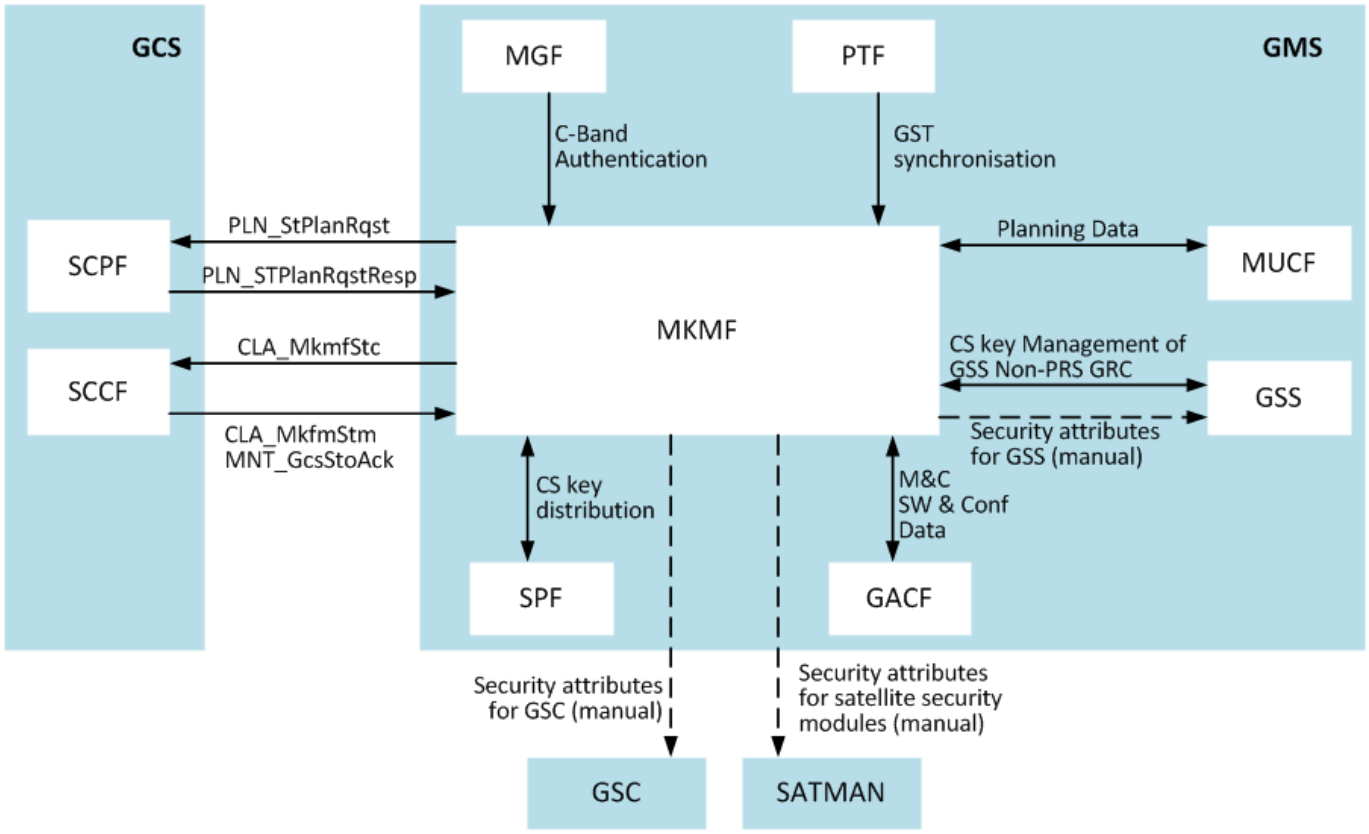


Figure 48: MKMF Organical context

5.3.2.5.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.5.4.1 SYSTEM LEVEL CONSTRAINTS

- **Functional constraints related to C-Band Mission uplink protection:**

- The MKMF is responsible for the C-band Integrity and Message subframe authentication, to authenticate subframes between the GMS and the satellite.
- The MKMF is reponsible to distribute the keys for C-band protection to the satellites through secure telecommands via GCS and to monitor and control the security unit on-board the satellite.

- **Functions related to Commercial Service:**

- The protection of the Commercial Service third carrier ranging code can be (activated or) deactivated by command from GMS on request of the GSC.
- The CS ranging code encryption key can be changed in the satellite by command from the GMS.
- The CS ranging code encryption key can be distributed manually (i.e. using removable media) from GMS to CS Test User Receivers, Ground Segment CS Receivers and the CS Service Provider.
- The CS ranging code encryption key can be distributed remotely (i.e. via network links) from GMS to Ground Segment CS Receivers, the CS Service Provider and to GCS (for distribution to Space assets).

- MKMF has the capability to analyse any failure of key integrity or authenticaton of the CS ranging code encryption key, detected in the satellite and reported via telemetry (standard TM).
- GMS informs the service centres 7 days in advance when the CS ranging code encryption keys are going to be changed (in accordance with GMS to SC ICD). The GOC SC may request GMS to change the CS ranging code encryption key. When requested to change this key, this change will be implemented within 30 days time, in the ground assets and in the satellites. The CS third carrier protection status is verified by GMS, which is capable to access the signals for TCAR on L-Band

- Security constraints

The MKMF shall allow the processing of classified information with a classification level up to Confidential. This has an impact (as specified in the applicable security certification/evaluation/approval criteria and standards) on the whole life cycle of the element including:

- the development process and environment;
- the delivery procedures of the element components;
- the security functions provided by the element;
- the architectural and low level design choices;
- the SW and HW COTS that can be used;
- the security features of the site where the MKMF shall be installed;
- the operational procedures (including personnel clearance requirements).

The applicable security certification/evaluation/approval criteria and standards are defined in the classified annex to the element requirements.

5.3.2.5.4.2 MAJOR SEGMENT DESIGN CHOICES

Architectural choices:

Information flows between the MKMF and other elements at a different classification level have been avoided or controlled by means of approved equipment or performed off-line (as detailed in the classified element documentation).

Failure propagation between pieces of HW/SW having different DAL level are avoided by appropriate barriers and/or mechanisms (as detailed in the classified element documentation).

Hot redundancy has been assured by means of MKMF hot back-up unit (as detailed in the classified element documentation).

The MKMF is connected through dedicated connections to

- the GCC Real-Time (RT) LAN
- the GCC Near Real-time (NRT) LAN
- the PTF time dissemination network (IRIG-B)

Development Assurance Levels:

- SW DAL for C-Band authentication functions

- All MKMF software which may cause an interruption (or an out-of-specification delay) of the C-Band uplink data authentication function has been assigned a DAL Level C or higher.
- Each offline function providing the implementation of barriers has been assigned a DAL level E.
- Each security module function providing a CRC barrier against risk of corruption of integrity data has been assigned a DAL Level C

5.3.2.6 MISSION SUPPORT FACILITY (MSF)

5.3.2.6.1 HIGH LEVEL DESCRIPTION

The MSF is in charge of the off-line analyses enabling to tune the on-ground facilities including simulations, training and maintenance.
In particular, the MSF:

- performs an offline navigation performance evaluation
- provides operational configuration data (mainly for OSPF).

5.3.2.6.2 FUNCTIONS ALLOCATION

According to MSF DDF (GMS-RD.5) the MSF major functions are summarized as follows:
- Provision of calibrated parameters for OSPF (HSISA and SRP)
- Provision of reference data for external entities (GRSP) and MUCF, orbits and resituated clocks for both GSS and satellites,
- Provide configuration parameter (GGTO threshold) via an operational interface to GPS/USNO
- Generation, modification and validation of configuration files for GMS facilities,
- Analysis of SiS digital and RF characteristics
- Archiving.

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF4.1	Generate Coordinates Reference
GMSF4.2	Determine GGTO
GMSF4.3	Generate System Time
GMSF5.3	Determine SISA
GMSF6.3	Local Archiving
GMSF7.2	Monitor Mission Off Line
GMSF7.4.1	GMS Elements Monitoring and Control
GMSF11	Perform Simulation and Test

Table 37: MSF Function List

5.3.2.6.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the MSF context according to the MSF ICD(GMS-RD.31):

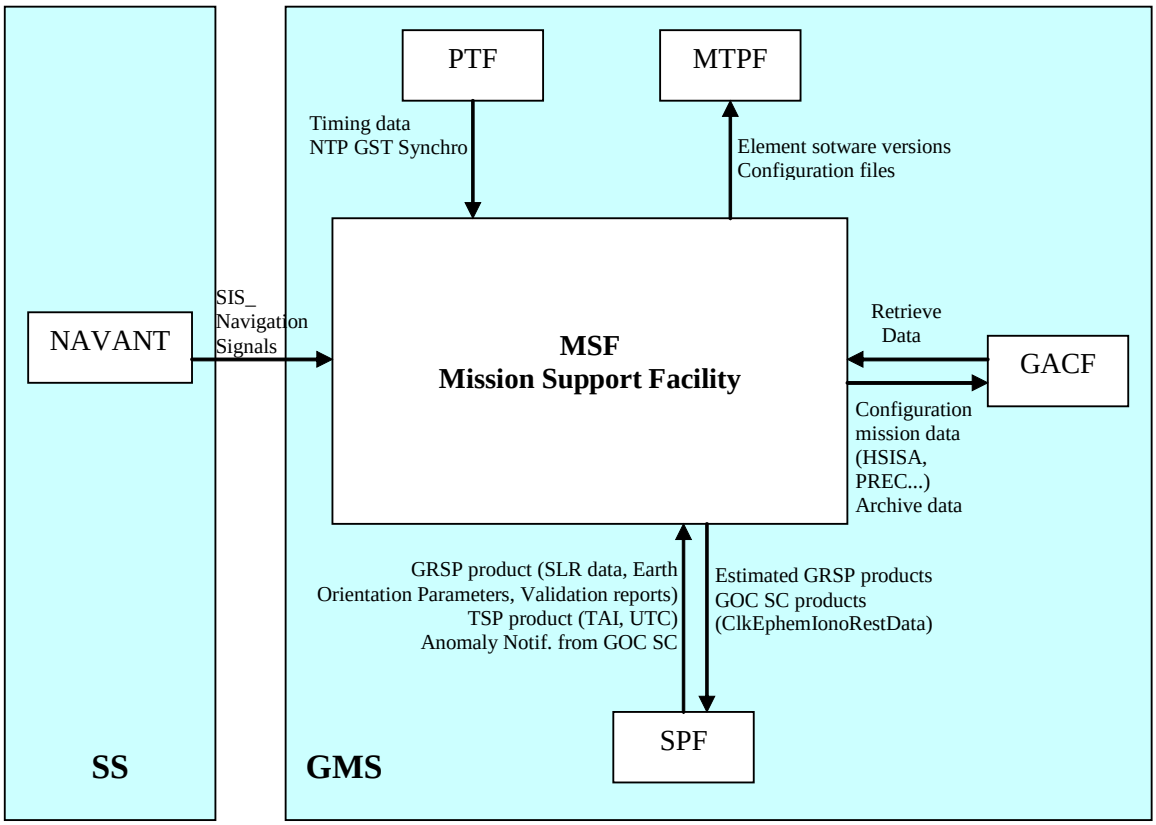


Figure 49: MSF Organical context

5.3.2.6.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.6.4.1 SYSTEM LEVEL CONSTRAINTS

Performance Constraints:

- Reference Orbits and clocks accuracy: 10 cm (1 sigma) for orbits and 0.3 ns (1 sigma) for clocks (GMSREQ-10746).

- SISA (whose major component is the HSISA computed by MSF) shall be lower than 85 cm (GMSREQ-10765 for SOL and GMSREQ-10766 for PRS).

5.3.2.6.4.2 MAJOR SEGMENT DESIGN CHOICES

Operability:

MSF is a manned element. The MSF operator is in charge of changing the element state. When the user requests a change of state, the CSF requests (through GACF) a confirmation before trying to execute anything. If the user confirms the operation, a change of mode procedure is started in the CSF server.

Redundancy:
1 MSF instance is deployed in each GCC, which is managed with a nominal/back-up concept. The handover is based in having a MSF working in NOMINAL mode and another MSF working in BACKUP mode. Both of these MSFs are getting the same input data and are generating the same output data. However, only the MSF in NOMINAL mode provides data to the MSF external entities and the MSF in BACKUP mode doesn't provide any data.

SISA Computation:
SISA is build by adding HSISA values plus the known Message fitting and discretisation errors at OSPF level.
HSISA, computed by MSF, is a post-processed estimate of the minimum 1-sigma which over-bounds the predicted orbit and clocks errors before message formatting.

5.3.2.7 MAINTENANCE AND TRAINING PLATFORM (MTPF)

5.3.2.7.1 HIGH LEVEL DESCRIPTION

MTPF provides the training, trouble shooting and validation facility to the GCC operators. Furthermore the MTPF will exchange operations procedures with the Central Documentation Management System (CDMS) located inside the GCS.

The MTPF is composed of the VAL chain used in offline mode plus add-on tools:
- The MTPF time is adjusted with a time distribution component.
- GMS AIVP to test element interfaces.

Synthetic data (from RDG Raw Data Generator) can be injected or real data collected in OPE chain can be replayed.

5.3.2.7.2 FUNCTIONS ALLOCATION

The MTPF element shall be used for
1. Segment upgrade preparation and trial installations of LRUs/SWRUs
2. Anomaly investigations and trouble shooting
3. Segment operations procedures set-up, analysis, validation, and maintenance
4. Operators hands-on training using prepared and replayed scenarios
5. Operators theoretical training (e.g. operations processes, system design, other background information presentations)
6. Support to performance assessments (depending on closed-loop capabilities and GSS data sets)

All these activities shall be performed without impacting the operational system. The main assumption is made that only one of those activities will be performed at a time (although practically an overlap e.g. between anomaly investigation and trial installations is technically feasible and could be organised).

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF6.3	**Local Archiving**
GMSF7.4.1	**GMS Elements Monitoring and Control**
GMSF10	**Provide Training**
GMSF11	**Perform Simulation and Test**
GMSF13	**Manage Database**

Table 38: MTPF Function List

5.3.2.8 MISSION AND UPLINK CONTROL FACILITY (MUCF)

5.3.2.8.1 HIGH LEVEL DESCRIPTION

The MUCF is responsible for the tasks related to planning, mission data dissemination support and mission monitoring.

5.3.2.8.2 FUNCTIONS ALLOCATION

According to GMS DDF2 (GMS-RD.2) the MUCF is responsible for the following main functions:

- Mission maintenance operations planning:
  - Production of Mid-Term/Long-Term plannings for Galileo system
  - Production of Shirt Term planning for GMS
- Uplink Scheduling:
  - ULS contact plan generation
  - MGF Data dissemination routine table generation for Galileo Mission Data, SAR RLM and CS data.
- Online Mission Monitoring:
  - Ground segment assets monitoring: GMS data determination and dissemination chain, network performance
  - Satellite element health monitoring and control: satellite clock synchronisation and switching management
  - On-line mission service performance monitoring:
    - OS and PRS service performance
    - SAR and CS performance
- Mission Performances Analysis (on-line and off-line mission monitoring):
  - OS and PRS performance off-line analyses (service availability, positioning accuracy)
  - SAR RLM service status
  - CS Service Status

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF3.5	Generate Uplink Schedule
GMSF4.2	Determine GGTO
GMSF6.3	Local Archiving
GMSF7.1	Integrated Mission Monitoring
GMSF7.2	Monitor Mission Off Line
GMSF7.3	Monitor Mission On Line
GMSF7.4.1	GMS Elements Monitoring and Control
GMSF9	Perform Mission Planning
GMSF9.1	Long/Mid Term Planning
GMSF9.2	Short Term Planning
GMSF11	Perform Simulation and Test
GMSF12	Collect and Route External Entities Data

Table 39: MUCF Function List

5.3.2.8.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the MUCF context according to the GMS DDF2 (GMS-RD.2) and GCS-GMS-ICD (AD.8):

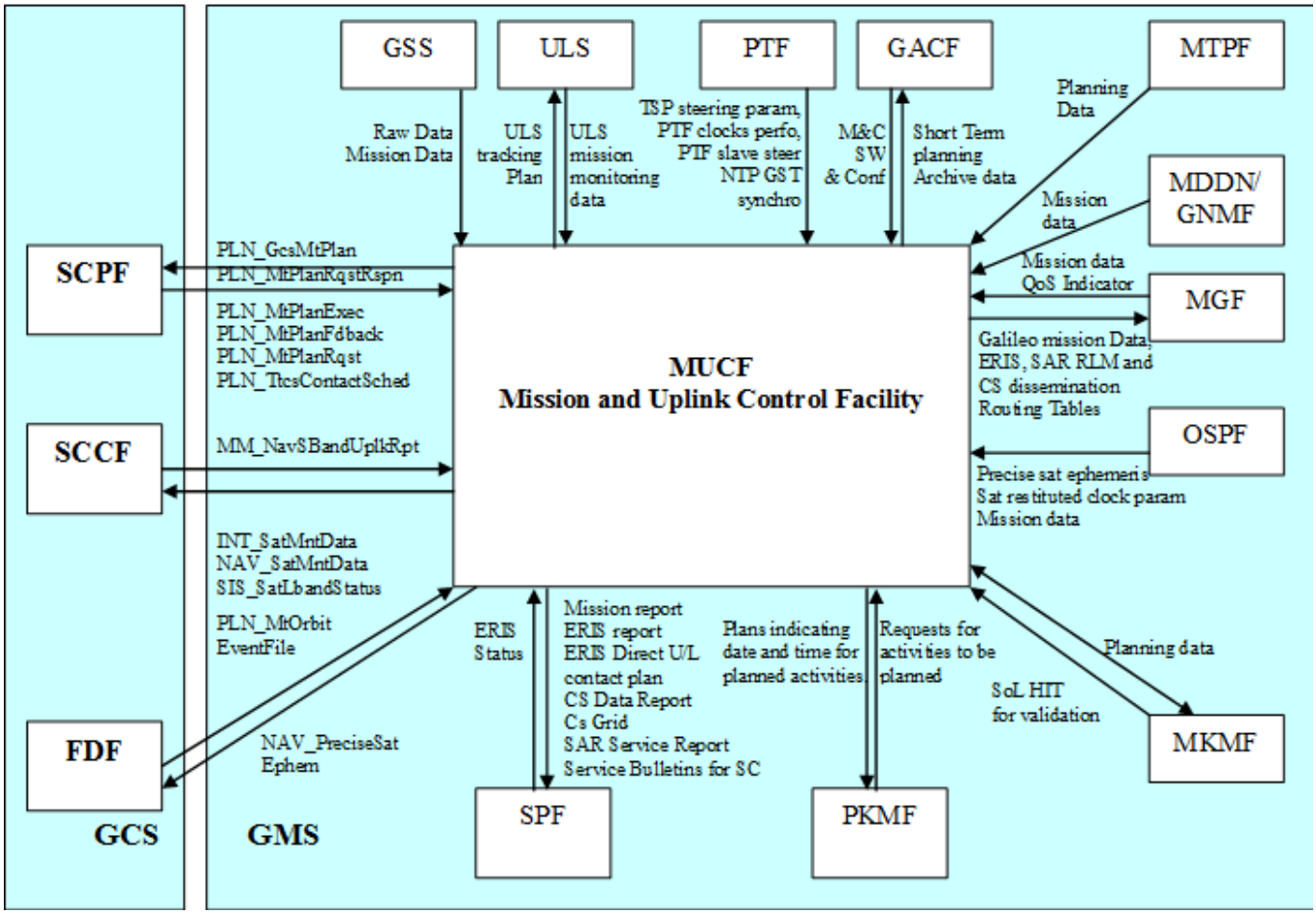


Figure 50: MUCF Organical context

5.3.2.8.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.8.4.1 SYSTEM LEVEL CONSTRAINTS

The MUCF support the following main functions, required to be supported by dedicated Man Machine Interface (MMI) with the operators (see section 5.3.2.8.4.2 on operability):

1. Near real time Mission services monitoring and control, providing the relevant Man Machine Interface with operators. This is allocated as part of the Galileo Mission On-line Supervision (GMOS) component, in charge of providing both real time and predictive parameters supporting Navigation services monitoring. The flexibility to reconfigure displays and MMI

features has being object of different operator driven improvements throughout the system evolution from IOV to the FOC configuration.

2. Off line preparation of the Galileo mission overall Long/Mid Term Plan (MTP), allocated to the Galileo Mission Overall Management (GMOM) component.
3. Off line Galileo Mission Simulation and Performance Assessment (GMPA) based on GSS raw data and prediction capabilities.

The MUCF is also capable of working in in "Analysis" mode, allowing conducting troubleshooting activities by re-processing GSS raw data retrieved from the archive.
However because of:

- the need to run simultaneously the near real time GMOS component and
- the high level of HW resources required,

it is recommended to conduct troubleshooting sessions on the MUCF replica provided as part of the MPTF, with resources that are fully dedicated to processing of retrieved data.

5.3.2.8.4.2 MAJOR SEGMENT DESIGN CHOICES

Operability:

The MUCF on the OPE chain is required to support the following working positions:

- Two working positions for Planning engineers, for access to the GMS Long / Mid Term Plan (MTP) and GMS Short Term Plan (STP functions and uplink scheduling function, one in the MCR to support emergency re-planning operations and one the POAR for nominal planning preparation. Note that either the GMS MTP or the GMS STP can be edited by a single operator at a time, which has to grant exclusive access to the product.
- Two working positions for Mission Engineers (one in the MCR, one in the SOR) providing concurrent access to on-line Mission Performance Monitoring & Prediction, Integrated Mission Monitoring and mission performance assessment (each working position is supported by two workstations to guarantee the concurrent access to separate functions). (**STRB#1.5.0 change: GAL-DCN-TASI-SYST-X-0842 v1: one extra working position required in the SOR**);
- One working position for Mission Engineer providing concurrent access to the Off line mission performance assessment functions and the functions related to the Monitor and Control of the S/C atomic oscillators.

Dual GCC Management:

There is one MUCF within each GCC. To support the dual GCC operations,

- the MUCF monitoring component (GMOS) is fully cross strapped in each GCC. The MUCF GMOS component is therefore capable of monitoring and controlling in separate contexts the mission parameters produced in each GCC.
- The MUCF planning component (GMOM) in each GCC is designed to work according to a Master/Backup split role logic, where the Master GMOM only is able to process and implement the Plan, providing the output data to the Backup MUCF/GMOM component which is kept synchronised and capable of taking up the Master role at any time using the last available Plan.
- With regard to the GMPA component, no specific dual GCC feature is implemented. This is only invoked locally in each GCC by manual operator actions, using data present in the archive, which is kept synchronised in each GCC by the GACF element.

MUCF main components :

According to GMS DDF2 [GMS-RD.2] and MUCF DDF [GMS-RD.4] the MUCF main functionalities have been allocated to the following components:

- GMOS (Galileo Mission On-Line Supervision): responsible for on-line mission monitoring including:
  - The monitoring of the segment assets: monitoring of the Data Determination chain of the GMS, the Data Dissemination chain (C-Band and S-Band), the Space Segment (satellite Health monitoring and satellite clocks).
  - The monitoring of the service performances: monitoring of the Navigation Services OS & PRS, the Dissemination support to services SAR, CS, the OS, PRS, SAR & CS predicted availability.

- GMPA (Galileo Mission Performance Assessment): responsible for Mission Performance analyses including:
  - the Overall analysis Control and Configuration (OCC)
  - a Performance Assessment Tool (PATBox) analysing the following parameters:
    - A posteriori Navigation and Dissemination performance (PATX)
    - Mission Performance Prediction (MPP)
    - Satellite Clock Analysis (SCAT) to support the GMOS

- GMPS (Galileo Mission Planning & Scheduling): responsible for mission planning and uplink schedule generation:
  - Mission Planning generation and validation (GMOM)
  - Uplink Schedule computation (USPT)
  - Uplink Scheduling Safety Monitor [USSM]

- CSF (Common Service Functions): to support the above functions with dedicated technical means (remote monitoring, DB management, local archive, access control..)

5.3.2.9 ORBIT AND SYNCHRONISATION PROCESSING FACILITY (OSPF)

5.3.2.9.1 HIGH LEVEL DESCRIPTION

The OSPF is responsible for Orbit determination and time Synchronization Processing (OD&TS), computing mission navigation data to be transmitted to the user:

This includes the following main parameters :
- Precise ephemeris of all satellites in the Galileo constellation computed in the Galileo Terrestrial Reference Frame (GTRF), based on GSS raw data
- Precise Clock model parameters predicting the offset of satellite on-board clocks to the GST, based on GSS raw data
- Signal In Space Accuracy (SISA) for the signals of all satellites
- Almanac for all of the satellites in the Galileo constellation
- Additional correction parameters for single frequency users such as ionospheric corrections, estimated Broadcast Group Delay.

5.3.2.9.2 FUNCTIONS ALLOCATION

According to GMS DDF2 (GMS-RD.2) the OSPF is responsible for the following main functions:
- GSS data acquisition and pre-processing
- Galileo satellites orbit determination, providing almanacs and precise ephemeris
- Satellites clocks correction data determination
- SISA determination
- Navigation data elaboration
- OSPF Internal Monitoring and Control, allowing to take into account the various commands and monitoring requests

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF5	**Determine Navigation**
GMSF5.1	**Determine Orbit**
GMSF5.2	**Determine BGD**
GMSF5.3	**Determine SISA**
GMSF5.4	**Determine Satellite Clock Parameters**
GMSF5.5	**Determine Single Frequency Ionospheric Model**
GMSF5.6	**Generate Navigation Batches**
GMSF6.3	**Local Archiving**
GMSF7.4.1	**GMS Elements Monitoring and Control**

Table 40: OSPF Function List

The OSPF is involved in the following system functional chains:
- SFC1 Provide and maintain the Galileo constellation
- SFC2.1 Generate Galileo System Time
- SFC2.2 Provide GPS-Galileo time offset
- SFC4.1 Navigation data provision

- SFC4.3 Uplink Scheduling
- SFC6 Archiving

5.3.2.9.3 ELEMENT ORGANICAL CONTEXT

The OSPF interacts with some other elements within the GMS:

- GSS: The GSS stations will provide the OSPF with the various observations on the basis of which the navigation message is computed.
- MGF: to which the OSPF will provide the navigation messages to be uplinked to the Galileo constellation for dissemination to users. Additionally, the OSPF will provide QoS data on the basis of which the MGF will perform the selection of the master OSPF among the 4 on-line redundant units running in parallel.
- GACF: to exchange of configuration and technical monitoring data.
- MSF: Indirectly via the GACF, which will provide the OSPF with some calibration parameters for the navigation algorithm or the Galileo satellites SRP model
- MUCF: to which the OSPF will provide mission monitoring information.
- PTF: to which the OSPF will provide with clock synchronisation data.

The following figure shows the OSPF context according to the GMS DDF2 (GMS-RD.2):

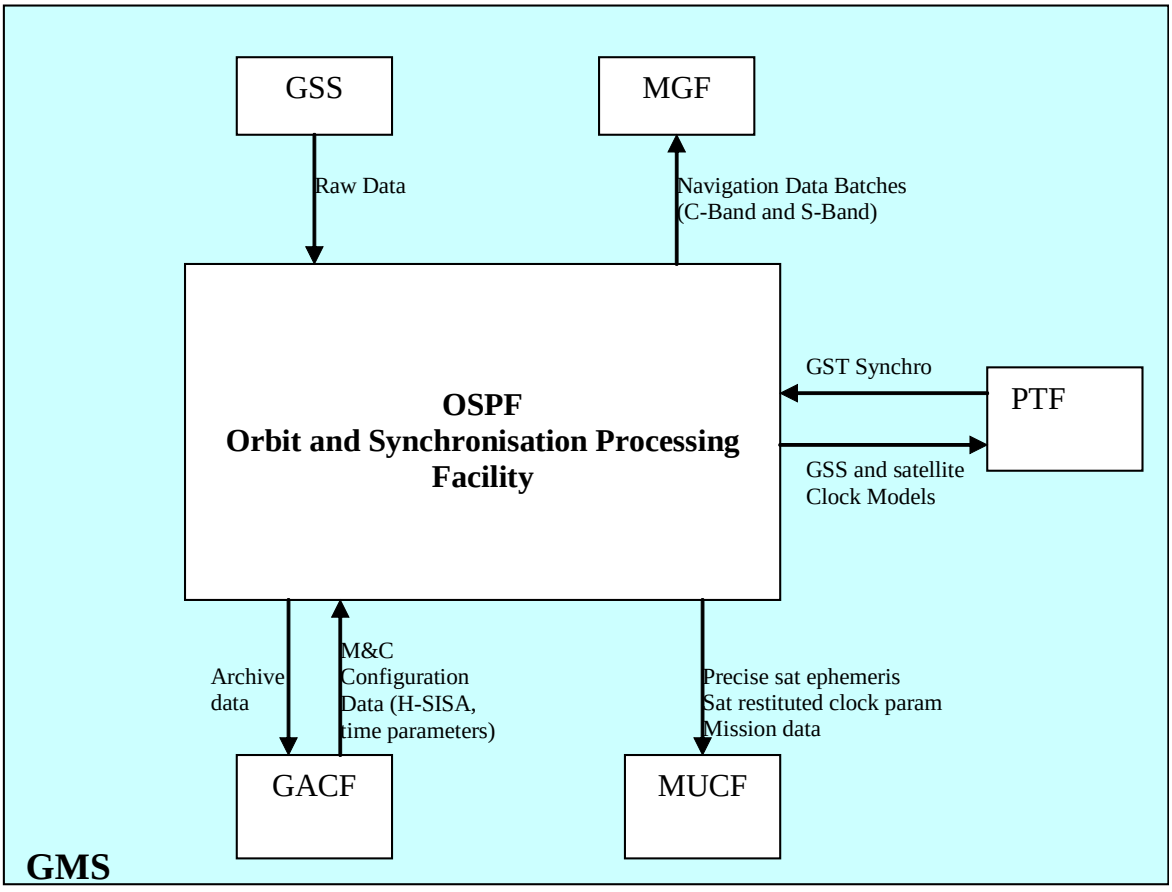


Figure 51: OSPF Organical context

5.3.2.9.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.9.4.1 SYSTEM LEVEL CONSTRAINTS

The OSPF is in charge of Orbit determination and time Synchronization Processing (OD&TS) i.e. Galileo satellites ephemeris and clock correction parameters estimation and prediction as well as Signal In Space Accuracy (SISA) determination for Galileo satellites.

The navigation data generated together with some other data processed in the PTF (GST offsets respect GPS time and UTC) are sent to the MGF to be formatted and sent to the user receivers.

The OSPF shall be capable to provide several batches of ephemeris and satellite clock corrections for each satellite in the constellation once every 10 minutes (GMSREQ-369).

The SISA computed by the OSPF associated to the navigation data generated and clock correction prediction shall be lower than 85 cm (GMSREQ-10765).

5.3.2.9.4.2 MAJOR SEGMENT DESIGN CHOICES

- **Redundancy:**
  To achieve the required availability and continuity requirements in FOC, it has been estimated that each of the 2 GCCs will be endowed with 2 on-line OSPF units. This configuration is deemed to provide the redundancy which is required to meet the corresponding GMS availability and continuity of service requirements.

- **OSPF Processing:**
  The OSPF is a real-time facility that performs a number of real time processes namely:
  - GSS data acquisition and pre-processing.
  - Galileo satellites orbit determination: this included the computation of ephemeredes and almanacs for each satellite in the constellation.
  - Galileo satellites and ground stations time synchronisation against GST.
  - Signal-In-Space (SISA) determination for the signals of all satellites
  - Computation of the Broadcast Group Delay correction parameters.
  - Computation of the ionospheric model for single-frequency users, including the so-called Ionosphere Disturbance Flag (previously referred to as “storm bit”).
  - Navigation data elaboration and product check.
  - Generation of Quality of Service (QoS) for the computed navigation data.
  - OSPF Internal Monitoring and Control, allowing to take into account the various commands and monitoring requests.
  - Generation of technical and mission monitoring data.

5.3.2.10 PRS KEY MANAGEMENT FACILITY (PKMF)

5.3.2.10.1 HIGH LEVEL DESCRIPTION

The PKMF is responsible for the Key management and data protection management for the PRS service, as well as the management of the various SMs and SUs involved in the PRS data flow protection. In addition the PKMF is in charge of regulating the access to the PRS service as requested by the GSMC.

5.3.2.10.2 FUNCTIONS ALLOCATION

The main function of the PRS KMF is to control and manage the access to PRS service, in order to achieve this objective it implements:

- PRS terminals management,
- Security Modules management,
- Key management,
- Monitoring and Control,
- PRS Data protection,
- Accounting of security related events,
- Backup and archive,
- Redundancy management.

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF6.3	**Local Archiving**
GMSF7.4.1	**GMS Elements Monitoring and Control**
GMSF8.1	**Security Protection of PRS**

Table 41: PKMF Function List

5.3.2.10.3 ELEMENT ORGANICAL CONTEXT

The PKMF organical context is shown in the following figure:

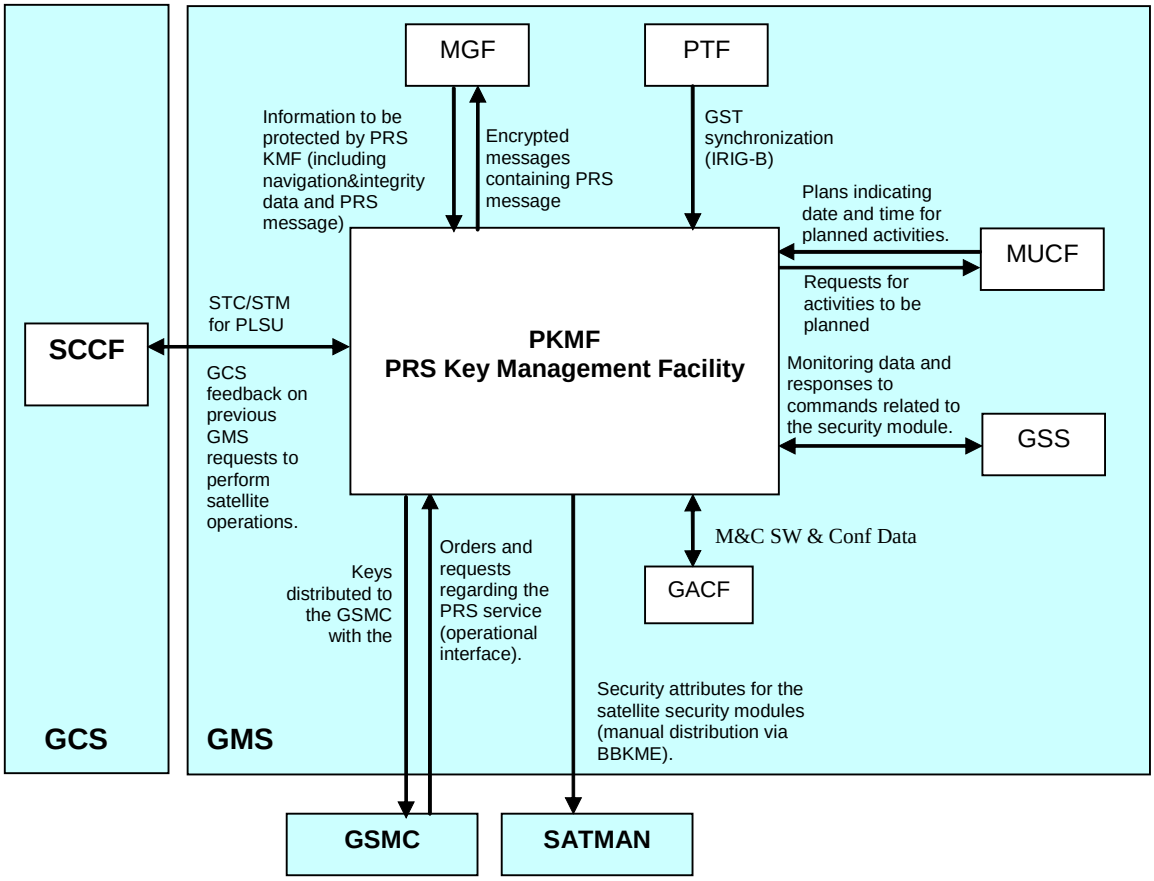


Figure 52: PKMF Organical context

5.3.2.10.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.10.4.1 SYSTEM LEVEL CONSTRAINTS

The PKMF shall allow the processing of classified information with a classification level up to Secret. This has an impact (as specified in the applicable security certification/evaluation/approval criteria and standards) on the whole life cycle of the element including:

- the development process and environment;
- the delivery procedures of the element components;
- the security functions provided by the element;
- the architectural and low level design choices;
- the SW and HW COTS that can be used;
- the security features of the site where the PKMF shall be installed;
- the operational procedures (including personnel clearance requirements).

The applicable security certification/evaluation/approval criteria and standards are defined in the classified Annex to this document.

In addition, the effect of failures of the functions provided by the PKMF has an impact on the required Development Assurance Level of both hardware and SW components of the element. For the PKMF this level can be as high as DAL B and must be estimated in accordance with the severity of the hazardous events caused by the failure.

5.3.2.10.4.2 MAJOR SEGMENT DESIGN CHOICES

- **Architectural choices**
  Information flows between the PKMF and other elements at a different classification level have been avoided, controlled by means of approved equipments or performed off-line (as detailed in the classified Annex to this document).

  Failure propagation between pieces of HW/SW having different DAL level are avoided by appropriate barriers and/or mechanisms (as detailed in the classified Annex to this document).

  Hot redundancy has been assured by means of PKMF hot back-up unit (as detailed in the classified Annex to this document).

  The PKMF is connected through dedicated connections to:
  - o the GCC Real-Time (RT) LAN
  - o the GCC Near Real-time (NRT) LAN
  - o the PTF time dissemination network (IRIG-B)

- **SW DAL choice**
  - o Each PRS KMF SW which may cause an interruption (or an out-of-specification delay) of the PRS authentication function shall be assigned with a DAL (Development Assurance Level) Level C or higher.
  - o OFF line functions providing implementation of barrier shall be assigned with a DAL level E
  - o Security module providing ON line functions shall be assigned with a DAL level B.

5.3.2.11 PRECISE TIME FACILITY (PTF)

5.3.2.11.1 HIGH LEVEL DESCRIPTION

The PTF represents the core of Galileo timing infrastructure and it is inevitably one of the most important elements inside the GMS. The PTF's main task is to provide the GST "navigational" timekeeping, hence to generate the stable timescale referred to as GST and distribute its physical realization (GST(MC1/2)) within the Galileo Core System via NTP, IRIG-B and locally also via 10MHz RF and 1pps interface. Thus the PTF uses atomic references, basis for the Galileo timescale generating the Galileo System Time (GST), the time and frequency distribution systems, processing, measurement, steering and time transfer facilities all connected to each other. The communication and distribution of the data and reference signals within the PTF will be catered for with the LAN, RF and 1pps interfaces. Being a basically un-manned Element, the PTF implements a very high level of automation. This automation minimizes operator intervention during operation and also allows maximal element robustness. Recovery from internal failure is automated through rapid detection, internal repair mechanisms and switching management among internal redundancies.

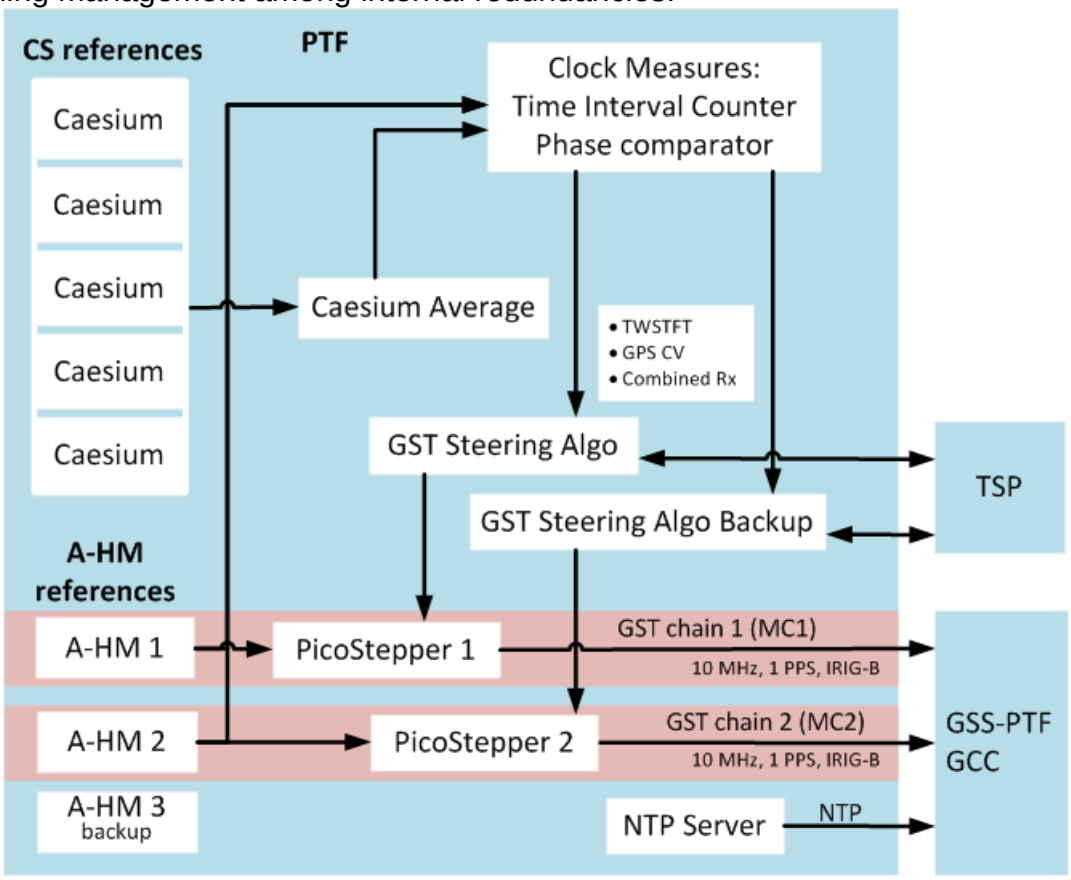


Figure 53: PTF architecture

5.3.2.11.2 FUNCTIONS ALLOCATION

According to GMS DDF2 (GMS-RD.2) and the PTF-ERD (GMS-RD.24) the PTF is responsible for the following main functions:

- Generate the GST time scale, on the basis of measurements performed on a clock ensemble;
- Generate a physical realization of the GST time scale
- Distribute the GST externally and within the GCC,
- Contribute to GST steering to TAI by providing relevant information to the TSP
- Compute the Time offset between GST and GPS time scale,
- Ensure Steering of the 2 GST time scales from the 2 GCCs.
- Support the GMS for Timing mission monitoring

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF4.2	**Determine GGTO**
GMSF4.3	**Generate System Time**
GMSF4.4	**Time Transfer**
GMSF6.3	**Local Archiving**
GMSF7.4.1	**GMS Elements Monitoring and Control**

Table 42: PTF Function List

The PTF is involved in the following system functional chains:

- SFC2.1 - Generate Galileo System Time
- SFC2.2 - Provide Galileo GPS Time-Offset (GGTO)

5.3.2.11.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the PTF context according to the GMS DDF2 (GMS-RD.2), GCS-GMS-ICD (AD.8), GMSExtIRD (AD.11):

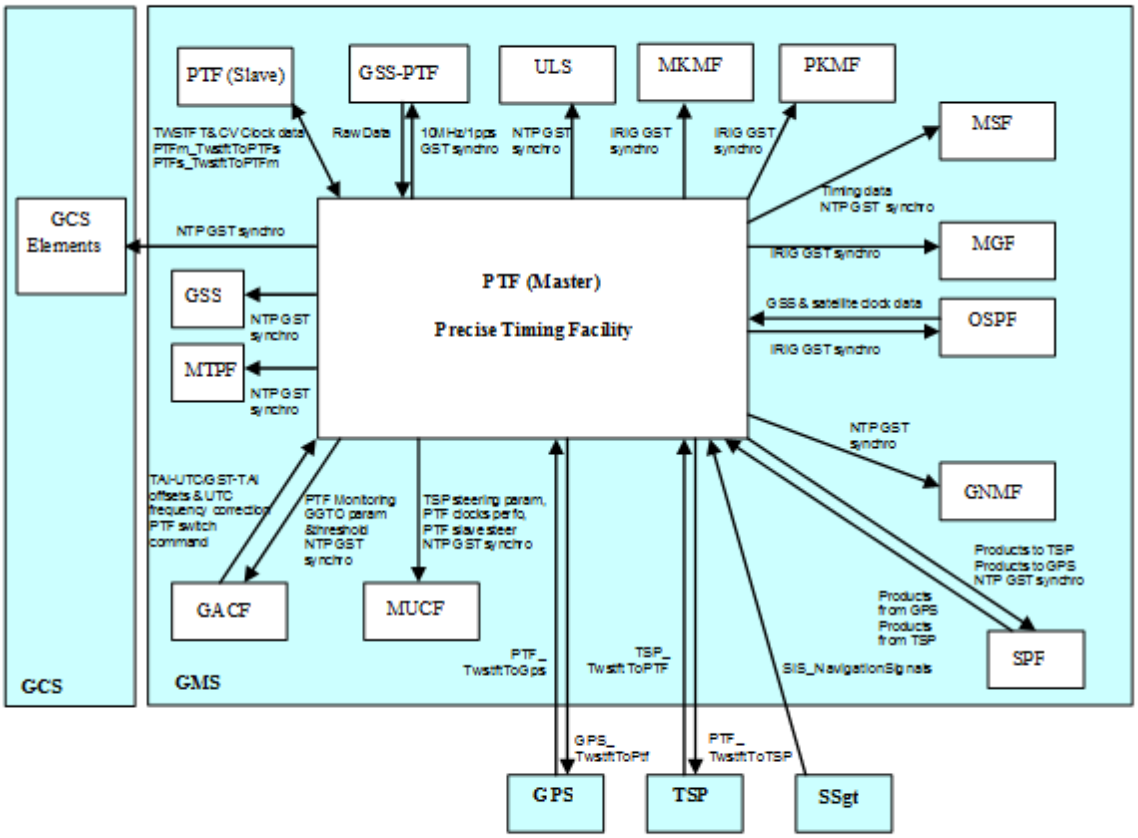


Figure 54: PTF Organical context

5.3.2.11.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.11.4.1 SYSTEM LEVEL CONSTRAINTS

- **PTF External Redundancy**

  The PTF is one of the few elements for which the redundancy is imposed at system level. Indeed GMSREQ (AD.3) specifies that the GMS includes two PTF (one in each GCC).

  As mentioned above, due the importance of the GST generation task, thus the PTF element, there will be two PTFs (present baseline, the number of PTFs might change) working in parallel in the Master-Slave concept. In the case of the Master PTF failure the Slave PTF will become the Master correspondingly the respective GST realization will be used within the Galileo System.

  In the nominal operation the master PTF is steered to TAI using TSP generated steering corrections. Note although TSP generates two sets of the GST-TAI steering corrections as if

both PTFs were the Master, the steering corrections are applied only by the Master PTF (provided that the navigation functionality is not deteriorated by doing so).

The slave PTF will disregard the TSP steering corrections and will steer itself to the Master to ensure that their respective timescales are kept within the specified time and frequency offset.
Note: Slave PTF is not smoothly steered to GST. However, seemless transition is possible due to the transition law implemented in OSPF (see dedicated section 6.2.3.3.14 in SFC2 description).The PTF switch will be commanded by GACF based on the results of the PTF monitoring performed by PTF, GACF &MUCF. Note the event of PTF switching represents the worse case scenario following the PTF critical failure, this is also reflected in the PTF design. The functional blocks inside the PTFs will be multiple redundant with most of the critical hardware working in the “hot” backup mode, to minimize the number of instances when the actual switching of the PTF needs to be performed.
In addition to critical failure cases, PTF switching can be also initialized based on scheduled maintenance.

The PTF’s are connected to each other by the means of TWSTFT, GPS& Galileo CV links, see below, all of which will be periodically calibrated in order to know the difference between the GST(MC) realisations within the required uncertainty boundaries.

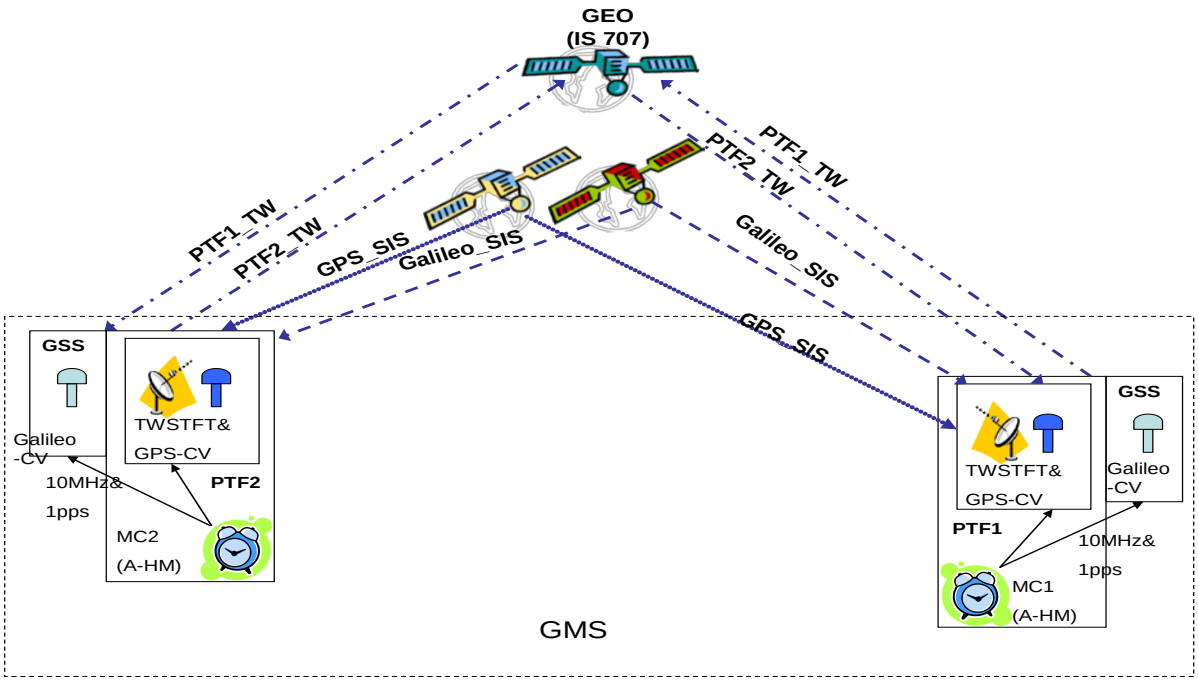


Figure 55: PTF calibration concept

Note: It is under GMS design responsibility to implement a proper steering process which enables seamless transition in case of master PTF switch. In case of PTF failure, the navigation processing shall be informed.

Note: PTF will use the deployed time transfer hardware also for the detemination of the Galileo-GPS Time Offset (GGTO).

- **PTF Internal Redundancy**

  Additional internal PTF redundancy is ensured by the fully independent (PTF v3) generation of 2 Master Clocks (MC1/2) per PTF. The PTF has 2 independent GST generation chains, each based on one A-HM (with a potential redundancy by the third A-HM) and adjusted via a picostepper. Every picostepper is controlled by a chain-dedicated GST Steering algorithm and stabilised by the stable mid/long-term Caesium atomic references.
  A third A-HM has been introduced in PTF v3 as an offline hot spare, permanently monitored, thus allowing better anomaly detection via 3 corner comparison. In case of maintenance of one of the A-HM, still 2 A-HM are operational.

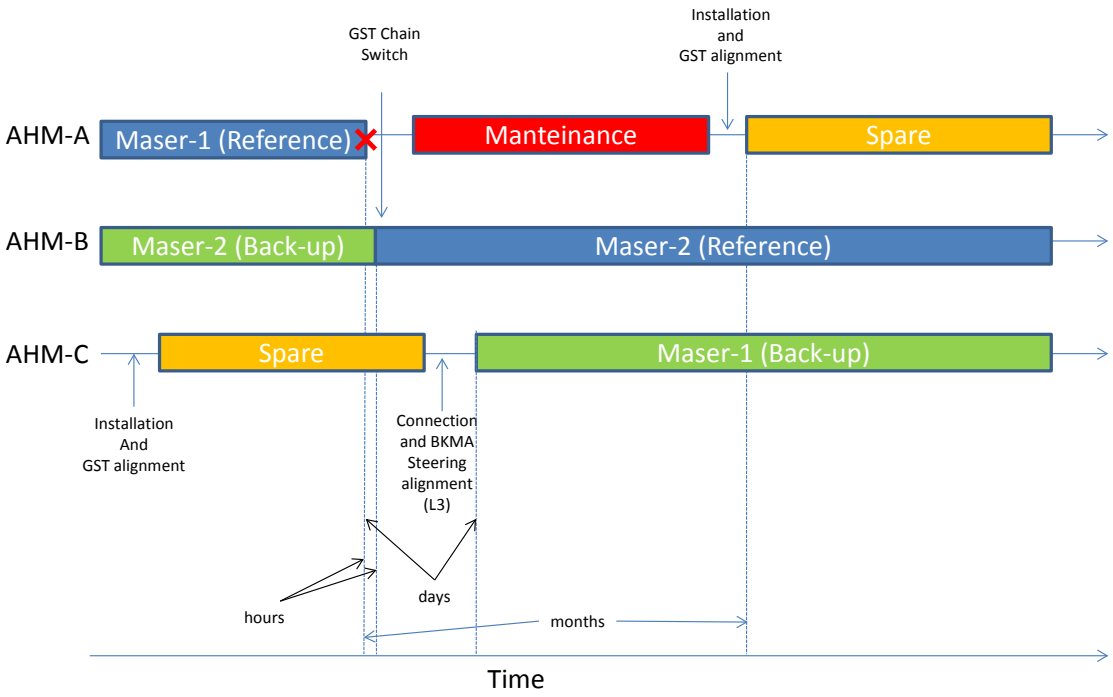


Figure 56: PTF 3 A-HM operational concept

- **GSS-PTF Co-located Galileo Receiver**

  Two GSSs (inlcuding GRC-P and GRC-N) are co-located with each PTF in each GCC. This is a GMS implementation choice to comply with some availability performance (according to GAL-RFD-TAS-GMS-A/224)
  The GSSs collocated with the PTF are not equipped with the RAFS as the remaining GSSs, but instead use directly GST as a reference. GRC inside each GSS collects measurements that will be processed by OSPF to restitute the drift of the satellite clocks for the computation of navigation data.

The GRC raw measurements are processed also by the PTF. The resulting Galileo Common View data (Galileo-CV) is used in the PTF to PTF synchronization and GST to TAI steering process.

Note: PTF task is to calibrate the signal paths (1PPS/10MHz signals) between the PTF GST output and the GRC (Galileo Receive Chain) input inside the GSS-PTF. Note the calibration information is applied by the GSS-PTF when outputting the raw measurement to OSPF. In that way, the raw measurements are tagged by the true GST.
The GSS raw measurements are processed with respect to the GST physical realisation time reference provided at GSS input. GST reference is represented with the GST epoch by means of a one-pulse-per-second (PPS) signal with calibrated phase stabilised cables and the GST frequency by means of a precise 10 MHz signal.
The measurements which are calibrated with the respect to GST(MC) can be used for instance by OSPF to restitute the GST (common view process at OSPF with the other GSS stations.

- **PTF Atomic References**
  The atomic reference ensemble at each PTF consists of 5 high performance Caesium (Cs) clocks and 3 Active Hydrogen Masers (A-HM). Each of the 2 GST chains provides stable clock references at 10MHz and 1 PPS signal.

  Caesium references are the atomic references which have outstanding mid/long-term performances. Caesium references are available as COTS and they are by far the most common type of atomic references used. The required performance of the Caesium references is given in the table below:

Noise	Experimental Allan Variance
White frequency	$\sigma_y^2(\tau) = (8.5\text{x}10^{-12}\text{x}\tau^{-1/2})^2$
Flicker frequency	$\sigma_y^2(\tau) = (2\text{x}10^{-14}\text{x}\tau^{0})^2$

Table 43: Caesium Clock Performances

Masers are atomic references which have outstanding short-term stability. They are also COTS items, however due to complexity of their production, only a several type of them are made per year. Hydrogen clocks are compared to caesium ones much bigger and their price is normally by factor of 3 more expensive the caesium clocks. This is also one of the reasons why only a primary timing laboratories and a few very specialized companies have the masers. The required performance of the Active Hydrogen Maser clocks is given in the table below:

Noise	Experimental Allan Variance
White phase 1 HzBW	$\sigma_y^2(\tau) = (2.0 \times 10^{-13} \times \tau^{-1})^2$
White frequency	$\sigma_y^2(\tau) = (4 \times 10^{-14} \times \tau^{-1/2})^2$
Flicker frequency	$\sigma_y^2(\tau) = (2 \times 10^{-15} \times \tau^{0})^2$
Random walk of frequency	$\sigma_y^2(\tau) = [(2 \times 10^{-15})^2/\text{day}] \times \tau$

Table 44: Maser Clock Performances

The A-Maser Frequency Standards shall provide the generation of a stable clock reference at 5 and 10 MHz. The A-Maser Frequency standard shall be capable of receiving and executing a clock drift adjustment command.

5.3.2.11.4.2 MAJOR SEGMENT DESIGN CHOICES

- **PTF Switching:** As explained in GMS DDF2 (GMS-RD.2), the PTF switching process is controlled by the OSPF as part of its ODTS (Orbit Determination and Time Synchronisation) duties. In order to minimise the impact of switching to the Galileo navigation and UTC dissemination services, the OSPF will apply during the transition period a transition law to compensate for the time and frequency offset between the PTFs.

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5.3.2.12 SERVICE PRODUCT FACILITY (SPF)

5.3.2.12.1 HIGH LEVEL DESCRIPTION

The SPF provides the interface between the GMS elements located in the GCC and the external world. The SPF is responsible for forwarding data from authenticated external entities (GSC, TSP, RLSP, GRSP, GGTO, GSMC) to the required internal entity, and to export product data from internal entity to external entities (GSC, TSP, RLSP, GRSP, GGTO, CSP, GALSEE, GSMC). SPF also converts raw data from GSS to RINEX files with the appropriate filtering to be provided to different external entities. The SPF is also of providing feedback of the properly uplinked commercial service data (commercial service data is directly provided to MGF using a real-time data flow between GCS and MGF).

Remark: At the time of System builds 1.5.0 and 1.5.01 the available interfaces and dataflows are commensurate with the external entities that are in place. As the external entities evolve towards System Build 2.0, then the SPF will need to be updated accordingly.

5.3.2.12.2 FUNCTIONS ALLOCATION

According to GMS DDF2 (GMS-RD.2) the SPF is responsible for the following main functions:

- Maintain a Galileo Web Site and provide Web Services (software executable which can be accessed using web protocols) and FTP services.
- Provide scheduled Event Service
- Physical interface with the following External Entities (retrieving information and providing reports):
  - RLSP service: to retrieve the SAR dissemination plan
  - GSC: Provide reports and service keys
  - GRSP: to exchange the GRSP products
  - TSP: to exchange TSP products (except Twstft data for which a direct interface between TSP and PTF is identified)
  - GPS/USNO: to exchange GPS products (except Twstft data for which a direct interface between TSP and PTF is identified)
  - GSMC: to provide Galileo overall status
  - GALSEE: interface to support SI&V activities, GALSEE receives the agreed products from GMS as per GMS-SC ICD, and has no on-line possibility to request new ones.
  - CSP: provide GDDN status

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
GMSF6.3	**Local Archiving**
GMSF7.4.1	**GMS Elements Monitoring and Control**
GMSF8.2	**Security Protection of SOL**
GMSF8.3	**Security Protection for Mission Data**
GMSF8.4	**Security Protection of CS**
GMSF12	**Collect and Route External Entities Data**

Table 45: SPF Function List

5.3.2.12.3 ELEMENT ORGANICAL CONTEXT

The following figure shows the SPF context according to the GMS DDF2 (GMS-RD.2), GMSExtIRD (AD.11), SPF ICD, and External Entities ICDs:

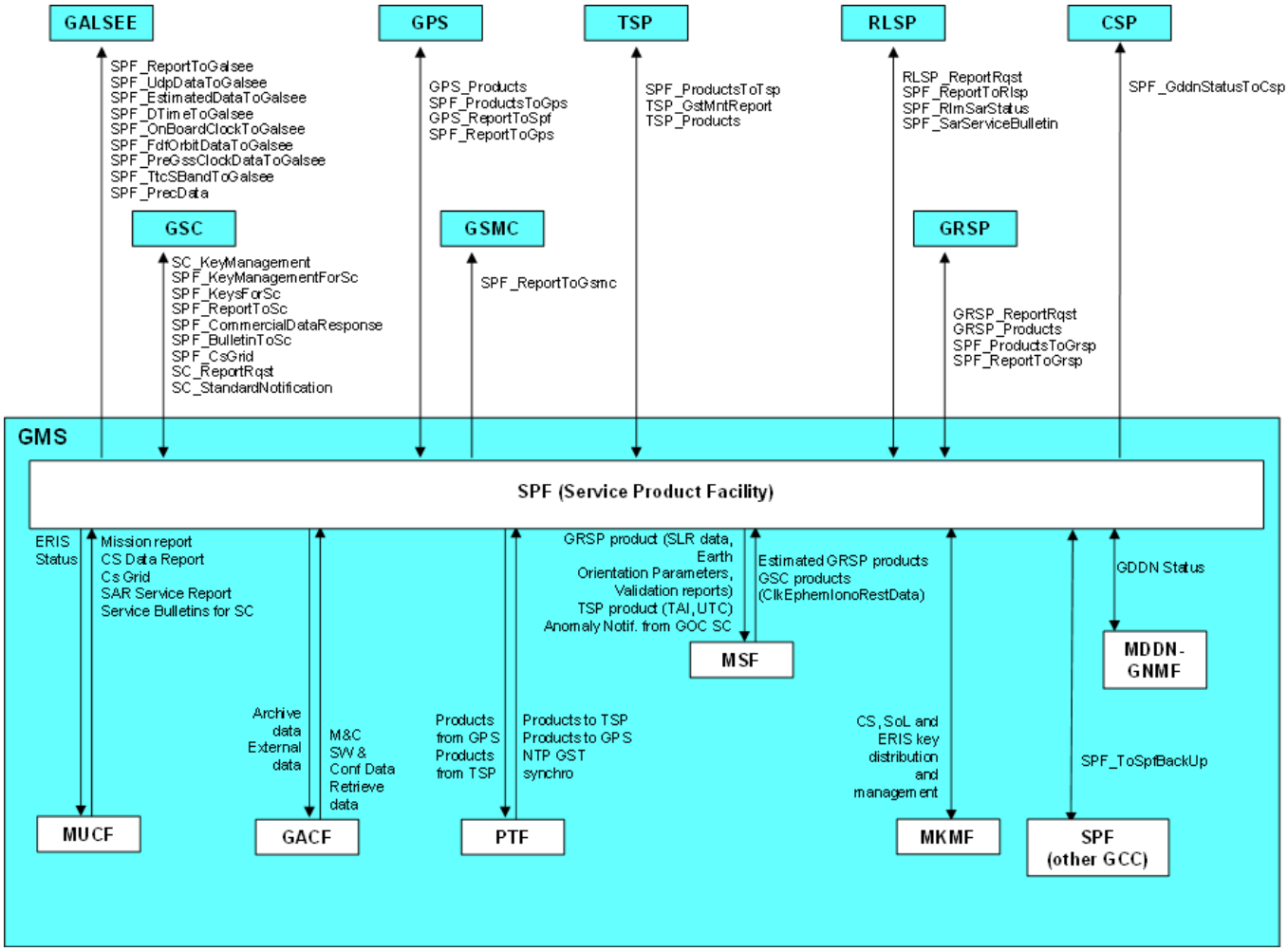


Figure 57: SPF Organical context

5.3.2.12.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.12.4.1 SYSTEM LEVEL CONSTRAINTS

The SPF Element shall be expandable to receive and process required data and services concerning up to 36 Galileo satellites (SPF-ERD-1548).

The SPF Element shall be easily configurable to allow report message content reflecting up to 50 GSS sites. (SPF-ERD-1549)

The SPF should be able to provide the data to external entities by FTP and should be able to receive request through HTTPS web portal.

The SPF must never initiate connections to elements within the gcc

As un-trusted element within the GCC, the SPF shall never initiate any transaction with other GCC elements, when SPF need to transmit any information towards other GCC Elements, it

should wait for a request (poll) from the GCC Element in order to open an specific channel to transmit this information.

5.3.2.12.4.2 MAJOR SEGMENT DESIGN CHOICES

- **Main Drivers identified by GMS**

The following main SPF design drivers have been identified by the GMS:

- Automatic conversion of GSS binary data to RINEX files (version 3) (SPF-ERD-2080)
- SPF software developed in accordance to level E requirements (see SPF-ERD-1150)
- Functionalities provided by following generic methods: web site, web service, and scheduled event service (SPF-ERD-1569)
- XML format conversion (SPF-ERD-2070)
- Number of external connections (SPF-ERD-0850)

  An SPF shall be capable to interface simultaneously with up to:

  - 1 Galileo Service Centre (with throughput for up to 50 external service centres)
  - 1 TSP
  - 1 GRSP
  - 1 RLSP
  - 1 GSMC
  - 1 GPS Ground Segment
  - 1 GALSEE
  - 1 CSP
- Local cache with capacity to hold GMS data for 7 days (SPF-ERD-1990). The regular reports sent by the SPF to the external entities are stored in the SPF cache for 7 days (10 days in some cases). After this period, the reports are only available in the GMS central archive (GACF).

- **SPF Security domains**

  One SPF instance is composed of two different security domains: front-end security domain and back-end security domain. Each domain has 2 servers, one active and one backup.
  The active front-end server acts as an application-level proxy with no data stored, to ensure that no connection is directly established between the outer EDDN and the core of the SPF, where more critical data is stored.
  The back-end (BE) security domain hosts the operational part of the SPF where the element's main logic is housed. The active BE Server stores all the files received from the external entities and the GMS elements.

- **SPF Redundancy**

  One SPF instance is deployed in each GCC. The Master SPF (in the nominal GCC) receives the incoming data products and distributes them as appropriate. It also replicates them to the Backup SPF (in the back-up GCC).
  SPFs are planned to be deployed in a Master/Slave configuration. In order to offer a unique access point for external Service Centers (web server type access), this requires the GNMF to manage the redirection of the Service Centers (external) requests to the Master SPF.

- **Maximum interruption during an SPF switch-over**

Upon GACF command to switch an SPF from Back-Up to Master, the Back-Up SPF shall change mode to Master and respond within 60 seconds. Furthermore, the entire process of switching SPF, from the initial drop of the SPF service to its resumption, shall take no longer than 5 minutes (SPF-ERD-0860).

- **SPF MTBF**
  The Mean Time Between Failure (MTBF) shall be better than 5000 hours (SPF-ERD-1790).

5.3.2.13 UP-LINK STATION (ULS)

5.3.2.13.1 HIGH LEVEL DESCRIPTION

The ULS is responsible for uplinking messages to the satellites.

5.3.2.13.2 FUNCTIONS ALLOCATION

According to GMS DDF2 (GMS-RD.2) the ULS is responsible for the following main functions:

- Reception of the multiplexed sub-frames produced by the MGF, construction and coding of the up-link frames
- Provide pointing to Galileo spacecrafts from each ULS 3,5-m antenna according to the tracking plans produced by the MUCF
- Ensure compatibility between the received tracking plan and the up-link message to be transmitted to the satellites
- Provide baseband modulation of the carrier
- Transmit C-Band RF signals
- Local monitoring and control

This covers directly the following GMS functions defined in the system DDF1 (SYS-RD.01):

Function Id	Function Name
GMSF3.1	Format and route messages
GMSF3.2	Up-link C-band messages
GMSF6.3	Local archiving
GCSF7.4.1	GMS element Monitoring and Control

Table 46: ULS Function List

The ULS is involved in the following system functional chains:
- SFC4: Mission service provision
- SFC5.1: Ground Asset Monitoring and control
- SFC6: Archiving
- SFC7.1: CS data dissemination
- SFC8.2.2: SAR return link dissemination

5.3.2.13.3 ELEMENT ORGANICAL CONTEXT

The ULS stations interface with the:

- Galileo spacecraft through the RF links for C-Band messages transmission
- Message Generation Facility (MGF): the MGF sends the message and integrity sub-frames to the ULS for up-link
- Mission and Uplink Control Facility (MUCF): the MUCF sends the up-link schedules to the ULS in order to allow satellite tracking. ULS sends to MUCF some relevant information for mission monitoring data
- Ground Assets Control Facility (GACF) for the ULS remote M&C.
- Site infrastructure

The following context diagram shows the ULS location inside the Galileo core system:

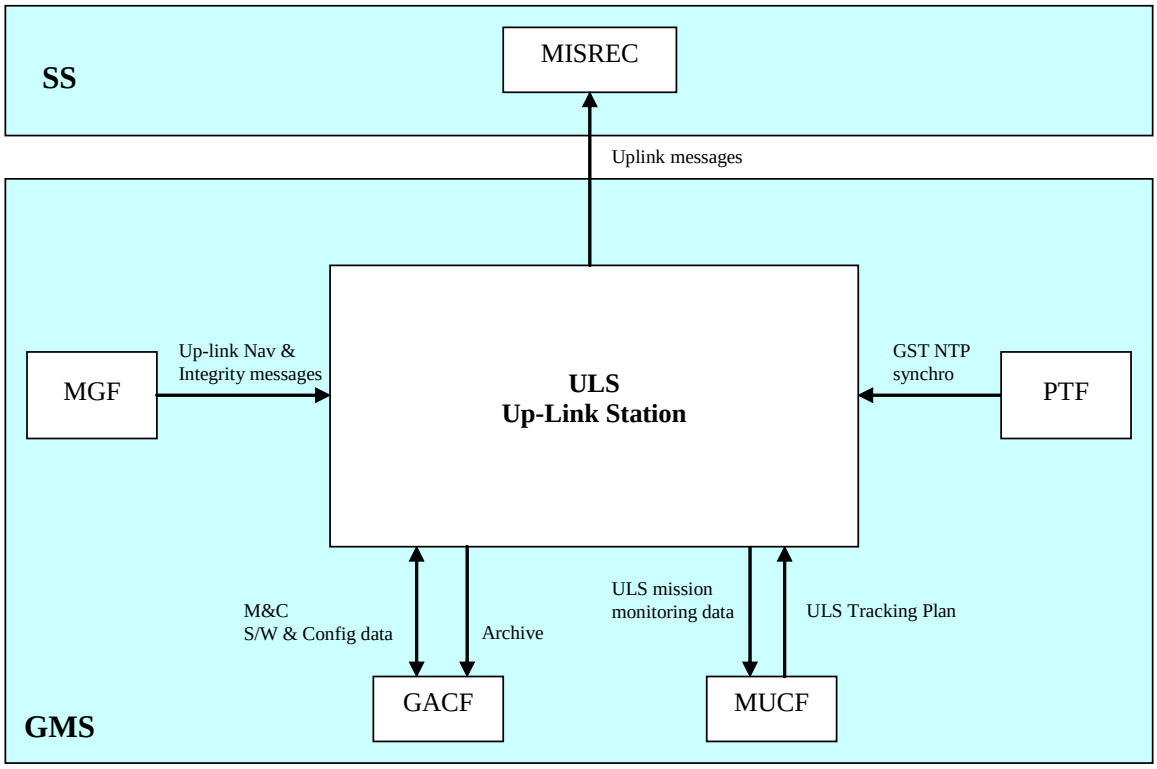


Figure 58: ULS Organical context

5.3.2.13.4 ELEMENT MAIN DESIGN DRIVERS

5.3.2.13.4.1 SYSTEM LEVEL CONSTRAINTS

- **C-Band Uplink Signal Characteristics**
  The ULS Station physical architecture is driven by the link budgets and the data dissemination strategy. Thus the main characteristics of the uplink signal and the related impact on ULS architecture are presented in the table below:

Mission Up-link Frequency	5005 MHz (Allocated 5000-5010MHz)
Up-link symbol-rate	4800 sps
Chip rate	5 Mchips/sec
Polarisation	RHCP
EIRP	57dBW

Table 47: C-band up-link signal main characteristics of the signal

5.3.2.13.4.2 MAJOR SEGMENT DESIGN CHOICES

- ULS Stations Characteristics
  The ULS Station includes:
  - An IF generator and modulator: receives the frames to be up-linked, perform differential encoding and modulates it onto an IF carrier, to which spreading is applied.
  - An Up-converter translating the IF signal to the carrier frequency.

- A High-Power amplifier: according to current link budgets, this will deliver a minimum transmit power of 30W over 5000-5010MHz to the antenna dish feed point.
- A coupler feeding back the transmitted signal linked to a complete down-link chain for monitoring purposes.
- The antenna dish and pedestal with motor drive and optional ancillary equipment (dehydrator, de-icing, radome) depending on location. This is a 3.5m diameter Cassegrain class, low-cost, full-motion unit. This relatively small diameter allows programmed pointing as opposed to tracking, simplifying the equipment and further optimising cost. Each ULS site will include 4 operational 3.5m antennas
- A time and frequency subsystem
- A Local Monitoring and Control system, divided into an up-link chain M&C sub-system and a ULS main M&C sub-system, in charge to perform monitoring and control of the ULS and managing the up-link schedule as well as Doppler compensation
- A Mission Processing system, in charge of receiving the up-link sub-frames (message and integrity sub-frames), format the up-link frames (construction, randomization and inner coding), check coherency with the implemented tracking plan and transmit them to the baseband modem

The ULS block diagram is shown here-after (for two up-link chains):

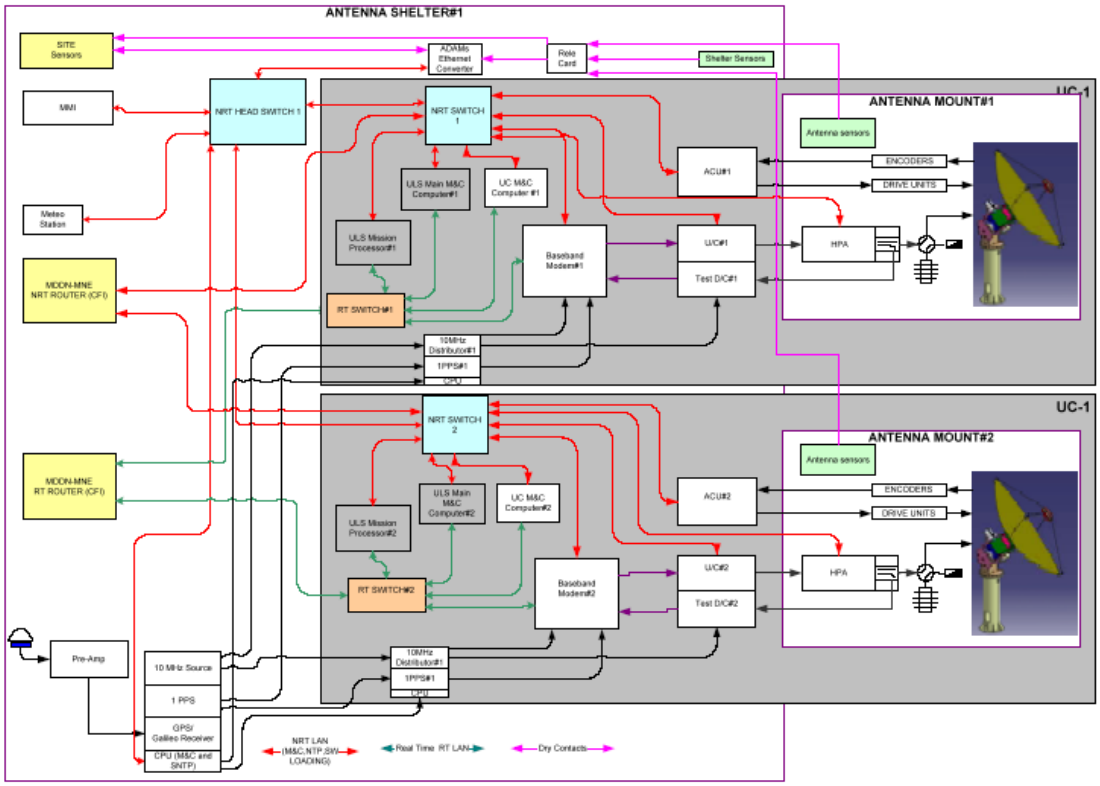


Figure 59: ULS block diagram

5.4 SPACE SEGMENT

5.4.1 SPACE SEGMENT OVERVIEW

The Space Segment (SSgt) is mainly responsible for the provision of the satellites composing the Galileo constellation.

The provision of a nominal constellation relies on the following tasks:

- Under Space Segment (IOV AST or FOC OHB) responsibility:
  - The Galileo Satellites: specified by SSREQ [AD.4]
- Under System responsibility:
  - The Constellation deployment and maintenance strategy definition covered by dedicated TN on constellation deployment strategy (SYS-RD.87) and specified in OSRD [AD.35]
  - LEOP operations for newly launched Galileo satellites will be run from a security accredited GCC, using an external LEOP TT&C network to extend the TT&C station coverage required during LEOP. External LEOP TT&C are operated via External Network Operations Centres (ENOCs), GCS-ENOC ICD [AD.50]
  - The Galileo Launcher Services provision (ESA CFI): Ariane or Soyouz based on LSREQ [AD.36] and dedicated interface between SSgt and Launcher service
- Under OPS contract responsibility (GSA/Space Opal):
  - LEOP: performed by GCC on the basis of OSRD [AD.35] and GCS-ENOC ICD [AD.50]
  - IOT activities for all satellite launches: non-secure IOT performed by IOT service provider (Inmarsat), and secure IOT based on GMS elements on the basis of IOT Services Requirements document IOTREQ [AD.32], and GCSIOT ICD [AD.31])

The space segment context is described in the following figure:

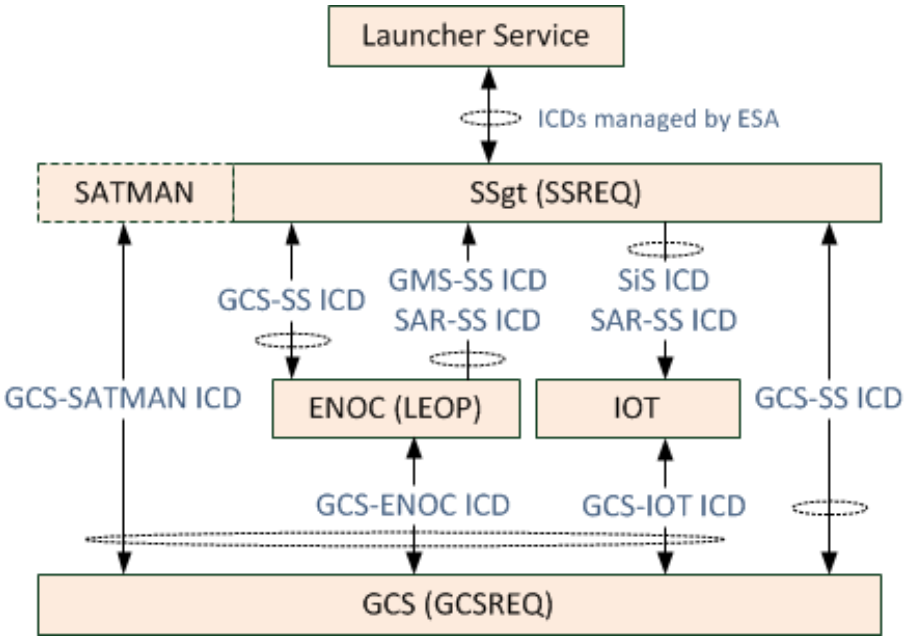


Figure 60: Space Segment Boundaries

Note: the space segment organical breakdown presented in the following sections includes the constellation and satellite descriptions.
The 3 main functions for Launch Service, LEOP and IOT are covered by External Entities (respectively Launchers, GCC and IOT Service provider) as described in dedicated section on External Entity description.

5.4.2 CONSTELLATION DESCRIPTION

5.4.2.1 Orbit Characteristics

The Galileo constellation is composed by 30 satellites in a Walker 24/3/1 configuration plus 6 in-orbit spares. This means that 24 satellites are distributed uniformly in 3 different orbit planes, plus 2 spares per orbital plane:

- All the satellites have circular orbits completing 17 orbits in 10 days.
- Each satellite in a given plane is separated from his neighbour satellites by 45°
- The ascending nodes of the orbital planes is uniformly distributed around the equator, i.e. at intervals of 120°
- The minimum phase separation between the satellite of two different planes is 1°
- The 6 spare satellites are distributed in the 3 different orbit planes and are located between two operating satellites, i.e. 22.5° apart.

The nominal orbital parameters are as follows:

- Semi Major Axis: 29,600.300 km
- Inclination: 56 degrees

The nominal orbital parameters are optimised (i.e. initial biases) in such a way to minimise the number of manoeuvres needed to keep the relative geometry of the satellites to a level sufficient to provide the navigation services with the required performances: see Galileo Constellation Deployment Strategy (SYS-RD.87).

Satellites that have failed or reached the end of their life are repositioned into a graveyard orbit to allow a new satellite to be placed in the operational slot.

Spare satellites are repositioned to the position, where the operational satellite had failed. The repositioning duration will depend on the distance to travel tot he orbital slot from the spare position. On average this will be less than 8.9 days.

The satellites have a 12-year lifetime. It is expected that satellites will start be replenished after 10 years to ensure constellation availability.

5.4.2.2 Satellite types

Several contracts are covering the provision of Galileo satellites:

- 4 IOV satellites by Astrium-GmBH
- 14 FOC1 satellites by OHB corresponding to Work-order 1

- 8 FOC1 satellites by OHB corresponding to Work-order 2

The ground segments GCS and GMS have the capability to handle both types of satellites (from a M&C point of view for GCS, and from a mission point of view for GMS).

5.4.3 SATELLITE OVERVIEW

Two types of satellites are provided by Astrium and OHB contractors. This section intends to describe the architecture of both satellite types. It has to be noted that most of the architecture and components are similar, except for some specific equipments (security unit, platform control unit). Common architecture will be described and specific elements and differences will be highlighted.

For each satellite type, the following table indicates the elements composing the different subsystems (for detail description of each element and associated functionality, refer to dedicated section):

	Subsytem	AST IOV element	OHB FOC1 element
Navigation Payload	C-Band Reception	- 1 C-Band Mission Antenna (MISANT) - 2 redundant C-Band mission Receiver (MISREC)	- 1 C-Band mission Antenna (MISANT) - 2 redundant C-Band mission Receiver (MISREC)
	Timing	- Atomic clocks: - RAFS1 (nominal master) - RAFS2 (cold backup) - PHM1 (nominal hot backup) - PHM2 (cold backup) - 2 redundant Clock Monitoring and Control Unit (CMCU)	- Atomic clocks: - PHM1 (nominal master) - PHM2 (cold backup) - RAFS1 (nominal hot backup) - RAFS2 (cold backup) - 2 redundant Clock Monitoring and Control Unit (CMCU)
	Payload Security Unit	2 redundant Payload Security Unit (PLSU)	Covered by platform CSU (see platform elements)
	Navigation Signal Generation	- 2 redundant Navigation Signal Generator Unit (NSGU) - 2 redundant Frequency Generation & Upconversion Unit (FGUU)	- 2 redundant Navigation Signal Generator Unit (NSGU) - 2 redundant Frequency Generation & Upconversion Unit (FGUU)
	Navigation output section	- Switch matrix - TBD nominal nav HPA (+ TBD redundant) - 2 OPF - 1 OMUX - 1 L-Band Navigation Antenna	- Switch matrix - 3 nominal nav TWTA (+ 3 redundant) - 1 OPF - 1 OMUX - 1 L-Band Navigation Antenna
	Remote Terminal Unit (RTU)	2 redundant RTU	2 redundant RTU
	Payload Thermal control	- Payload Heaters - Payload Thermistors	- Payload Heaters - Payload Thermistors
	Laser Retro Reflector (LRR)	Laser Retro Reflector	Laser Retroreflector Array (LRA)
	Other	-	Environmental Monitoring Unit (EMU)
SAR Payload	SAR Transponder Assembly	Only on FM3 and FM4: 1 SAR Transponder (including HPA, input filter, output filter)	1 SAR Transponder (including HPA, input filter, output filter)
	SAR Antennae	Only on FM3 and FM4:	- 406 MHz Receive UHF antenna

		- 406 MHz Receive UHF antenna - 1544 MHz Transmit LBand antenna	- 1544 MHz Transmit LBand antenna
Payload /Platform I/F	Communication Link	MIL BUS1553	MIL BUS 1553
	Payload power distribution		
Platform	Avionics Subsystem	- 2 redundant Integrated Control and Data handling unit (ICDU) - Attitude and Orbit Control System (AOCS) sensors and actuators: - 2 Fine Sun Sensors (FSS) - 2 Coarse Sun Sensor (CSS) - 2 Earth Sensors (ES) - 2 Gyroscopes (GYRO) - 2 Magnetic Torque Rods (MTR) - 4 Reaction wheels (RWL)	- 2 redundant Control and Data handling (CDU) including On Board Data Handling Software (OBDH) - Attitude and Orbit Control System (AOCS) sensors and actuators: - 2 Fine Sun Sensors (FSS) - 2 Coarse Sun Sensor (CSS) - 2 Earth Sensors (ES) - 2 Gyroscopes (GYRO) - 2 Magnetic Torque Rods (MTR) - 4 Reaction wheels (RWL)
	TT&C (Telemetry, Tracking and Command) Subsystem	- 2 S-Band antenna: - TTCANTZ (+Z) - TTCANTN (-Z) - 2 redundant TTC transponders (TX/RX)	- 2 S-Band antenna: - TTCANT+Z - TTCANT-Z - 2 redundant TTC transponders (TX/RX)
	Platform Security Unit	2 redundant Platform Security Unit (PFSU)	2 redundant Common Security Unit (CSU) Note: CSU includes also functionalities to manage payload security (as per Astrium CSU)
	Thermal Control Subsystem (TCS)	- Platform Heaters - Platform Thermistors	- Platform Heaters - Platform Thermistors
	Electrical Power Subsystem (EPS)	- 2 Solar Arrays (SA-Y, SA+Y) - 2 Solar Array Drive Mechanism (SADM-Y, SADM+Y) - Battery (BATT) - 2 redundant Power Conditioning and Distribution Unit (PCDU)	- 2 Solar Arrays (SA-Y, SA+Y) - 2 Solar Array Drive Mechanism (SADM-Y, SADM+Y) - Li-Ion Battery (BATT) - 2 redundant Power Conditioning and Distribution Unit (PCDU)
	Propulsion Subsystem	- 2 Fill and Drain Valves (F&DV) - 1 Hydrazine Tank - 3 Pressure Transducers (PT) - 1 Filter - 2 Latching Valves (LV) - 8 Thrusters	- 2 Fill and Drain Valves (F&DV) - 1 Hydrazine Tank - 3 Pressure Transducers (PT) - 1 Filter - 2 Latching Valves (LV) - 8 Thrusters
	Harness		
	Structure Subsystem		

Table 48: IOV and FOC1 satellite subsystem configurations

The physical architecture of the IOV satellite is detailed in Figure 61 below:

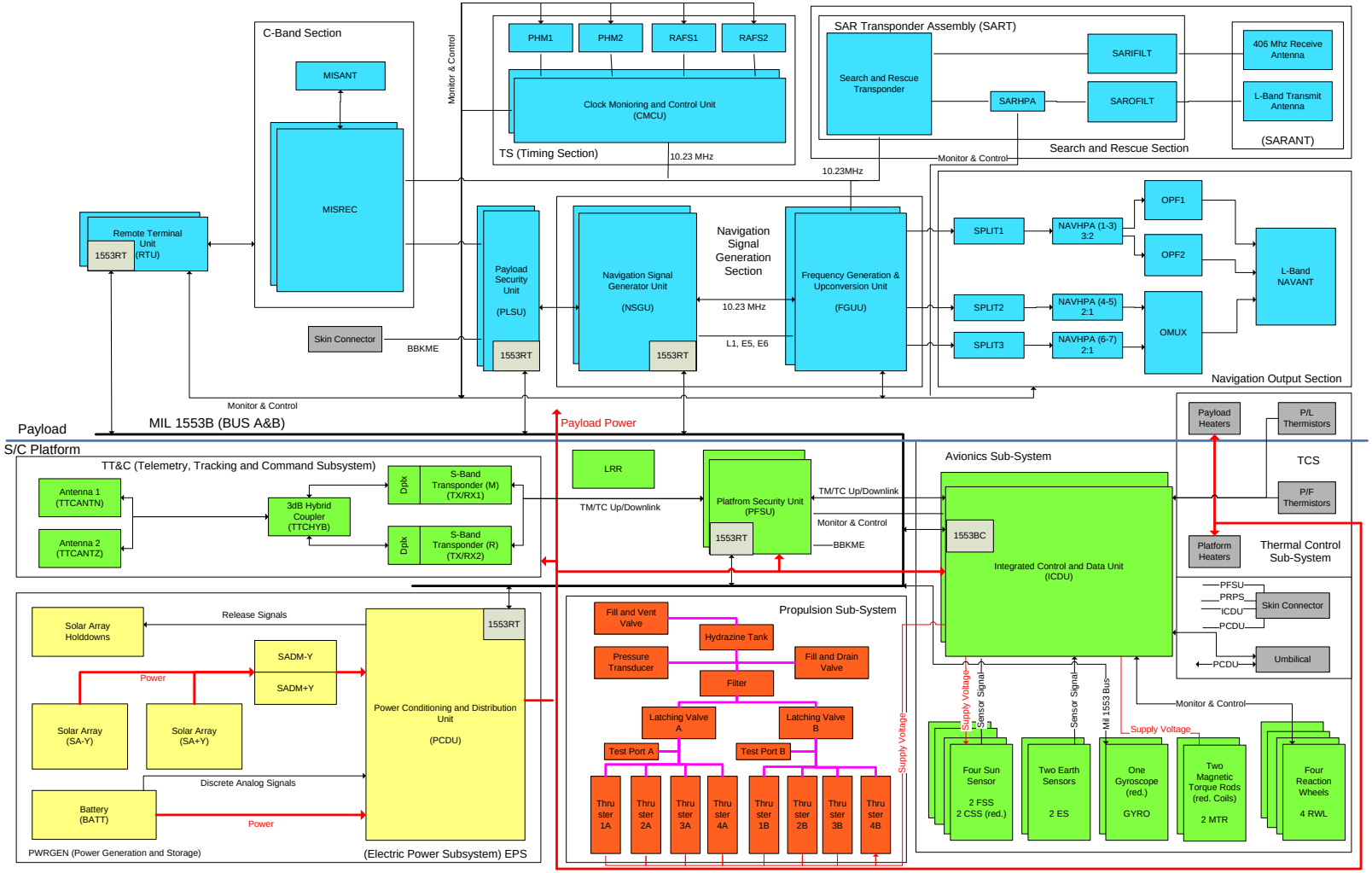


Figure 61: IOV Satellite physical architecture overview

The physical architecture of the FOC1 satellite is detailed in Figure 62 below:

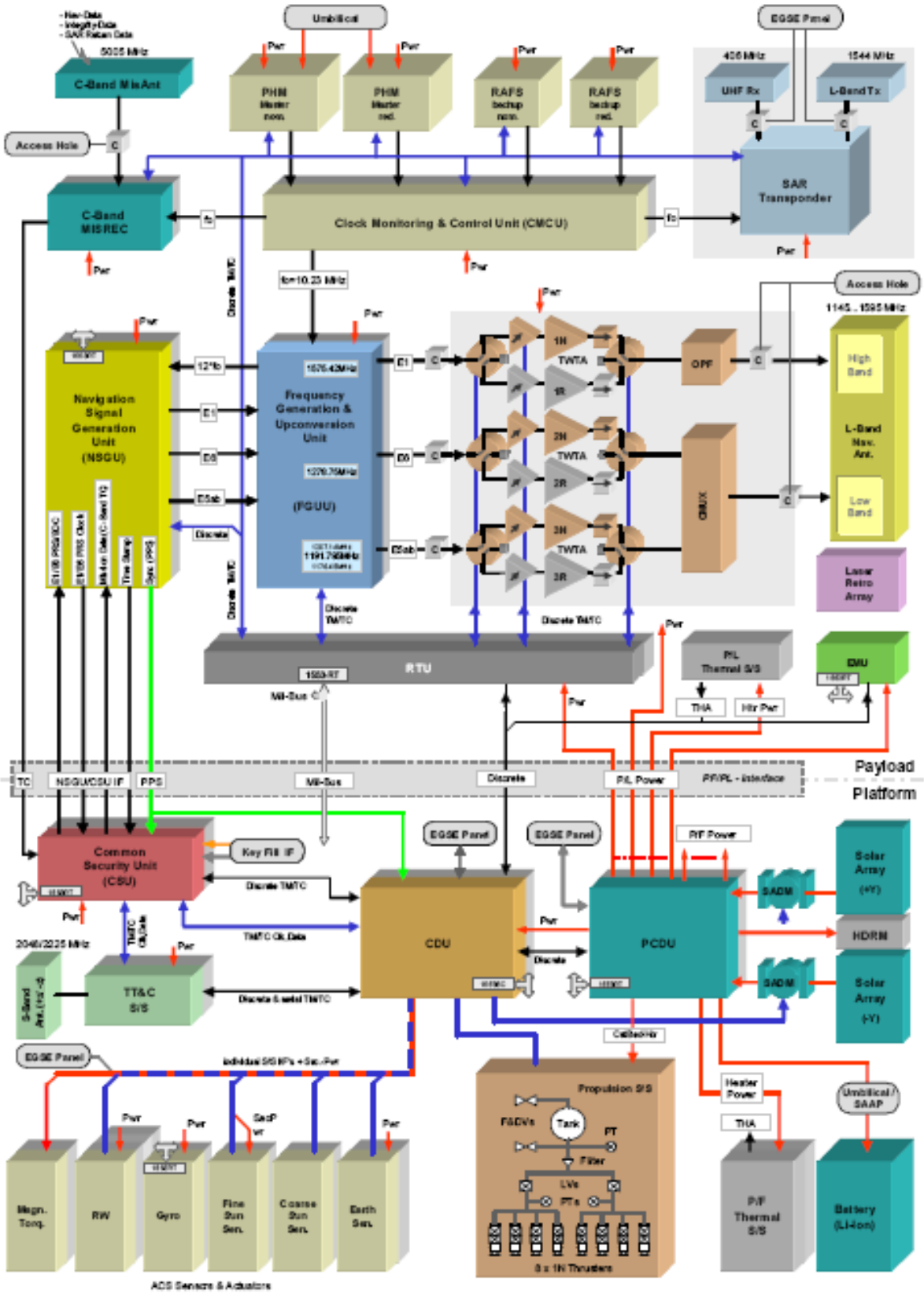


Figure 62: FOC1 Satellite physical architecture overview

The following table lists the system functions allocated to the Satellite and the mapping to Satellite elements:

Function ID	Function name	Mapping to IOV Element	Mapping to FOC1 Element
SSF1	Receive Mission Data	xxx	xxx
SSF1.1	Receive Uplink Signals	MISREC	MISREC
SSF1.2	Data Authentication	PLSU	CSU
SSF1.3	Distribute Data	NSGU/PLSU	NSGU/CSU
SSF2	Generate Ranging Code	xxx	xxx
SSF2.1	Generate Navigation Ranging Code	NSGU	NSGU
SSF2.2	Generate PRS Ranging Code	PLSU	CSU
SSF2.3	Generate CS Ranging Code	NSGU	NSGU
SSF3	Prepare Message	xxx	xxx
SSF3.1	Generate On-Board Data	NSGU/AVS	NSGU/AVS
SSF3.2	Compose Navigation Message	NSGU	NSGU
SSF3.3	Modulate Navigation Messages On Ranging Code	NSGU	NSGU
SSF4	Transmit Mission Data	xxx	xxx
SSF4.1	Generate Carriers	FGUU	FGUU
SSF4.2	Modulate Messages	NSGU/FGUU	NSGU/FGUU
SSF4.3	Amplify and Transmit Signals	LTX	LTX
SSF5	Provide Time Reference	xxx	xxx
SSF5.1	Generate OnBoardClock	CMCU	CMCU
SSF5.2	Generate OnBoardTime	NSGU	NSGU
SSF6	Manage Platform	AVS	AVS
SSF6.1	FDIR	AVS	AVS
SSF6.2	Health Status	AVS	AVS
SSF6.3	Satellite Mode	AVS/RTU	AVS/RTU
SSF6.4	AOCS	AVS	AVS
SSF6.5	On-Board Data Management	AVS/RTU	AVS/RTU
SSF6.7	Autonomous Action and Control	AVS	AVS
SSF7	Manage Payload	AVS/RTU	AVS/RTU
SSF8	SAR Payload	SAR	SAR
SSF9	Provide TTC Communication	AVS/TTC	AVS/TTC
SSF9.1	Receive and Control TC	AVS/TTC/RTU/PFSU	AVS/TTC/RTU/CSU
SSF9.2	Transmit and Manage TM	AVS/TTC/RTU/PFSU	AVS/TTC/RTU/CSU
SSF9.3	Two Way Ranging	AVS/TTC/RTU	AVS/TTC/RTU
SSF10	Optical Ranging	LRR	LRR

Table 49: Space Segment Function List

5.4.3.1 PAYLOAD OVERVIEW

The Galileo Payload is composed of a navigation payload and a SAR payload.

The main functions of the navigation payload are:
- Provision of on-board timing signals
- Receipt and management of up-linked navigation message data
- Receipt and management of up-linked external data (EDBS)
- Generation of downlink navigation messages (multiplexing data) including error correction coding, channel encoding, interleaving
- Generation of ranging codes
- Encryption of ranging codes if required
- Generation and modulation of L-Band carrier signals
- Broadcast of navigation signals

The main function of the SAR payload is to provide forward links for the SAR service (the Return Link being implemented in the L1b mission link).

5.4.3.1.1 PAYLOAD DESCRIPTION

The navigation payload is composed of the following element:
- On-board clocks: Ultra Stable Oscillators (Passive Hydrogen Maser (PHM), Rubidium Atomic Frequency Standards (RAFS))
- CMCU: Clock Monitoring and Control Unit
- FGUU: Frequency Generator & Up Converter Unit
- MISANT: C-Band Rx Mission Antenna
- MISREC: C-Band Mission Receiver
- NAVANT: L-Band Tx navigation antenna, used as well for SAR Return Link Message transmission
- NAVHPA: Navigation High Power signal Amplifiers (SSPA) and switches
- NAVOFILT and NAVOMUX: Navigation Output Filter and Output Multiplexer (OMUX)
- NSGU: Navigation Signal Generator Unit
- Payload Security Unit :
  - PLSU within payload in AST IOV satellite
  - Common Security unit (CSU) within platform in OHB FOC1 satellite
- RTU: Remote Terminal Unit
- Laser Retro-Reflector (LRR)

These elements are briefly detailed in the sections after.

The SAR Payload is composed of the following elements:
- SARANT: Combined UHF/L-Band Search and Rescue Antenna for forward link
- SART: Search and Rescue (SAR) Transponder

5.4.3.1.2 PAYLOAD FUNCTIONS

The following table lists the system functions allocated to the Payload and the mapping to Payload elements

Function ID	Function name	Mapping to IOV Element	Mapping to FOC1 Element
SSF1	Receive Mission Data		
SSF1.1	Receive Uplink Signals	MISREC	MISREC
SSF1.2	Data Authentication	PLSU	CSU
SSF1.3	Distribute Data	NSGU/PLSU	NSGU/CSU
SSF2	Generate Ranging Code		
SSF2.1	Generate Navigation Ranging Code	NSGU	NSGU
SSF2.2	Generate PRS Ranging Code	PLSU	CSU
SSF2.3	Generate CS Ranging Code	NSGU	NSGU
SSF3	Prepare Message		
SSF3.1	Generate On-Board Data	NSGU	NSGU
SSF3.2	Compose Navigation Message	NSGU	NSGU
SSF3.3	Modulate Navigation Messages On Ranging Code	NSGU	NSGU
SSF4	Transmit Mission Data		
SSF4.1	Generate Carriers	FGUU	FGUU
SSF4.2	Modulate Messages	NSGU/FGUU	NSGU/FGUU
SSF4.3	Amplify and Transmit Signals	LTX	LTX
SSF5	Provide Time Reference		
SSF5.1	Generate On-Board Clock	CMCU	CMCU
SSF5.2	Generate On-Board Time	NSGU	NSGU
SSF6.3	Satellite Mode	RTU	RTU
SSF6.5	On-Board Data Management	RTU	RTU
SSF7	Manage Payload	RTU	RTU
SSF8	SAR Payload	SAR	SAR
SSF9	Provide TTC Communication	RTU	RTU
SSF9.1	Receive and Control TC	RTU	RTU
SSF9.2	Transmit and Manage TM	RTU	RTU
SSF10	Optical Ranging	LRR	LRR

Table 50: Payload Function List

5.4.3.1.3 PAYLOAD ELEMENT DESCRIPTION

5.4.3.1.3.1 ATOMIC CLOCKS (RAFS, PHM)

5.4.3.1.3.1.1 High Level Description

The atomic clocks are responsible for generating a very stable frequency reference as input to the CMCU.

5.4.3.1.3.1.2 Functions Allocation

According to Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2), the Atomic clocks are responsible for generating and delivering a very high stable frequency reference of 10 MHz to be used by the CMCU to generate the 10.23MHz satellite frequency reference.

This covers directly the following SS functions defined in the system DDF1 [SYS-RD.01]

Function ID	Function name
SSF5.1	Generate On-Board Clock

Table 51: Atomic Clock Function List

5.4.3.1.3.1.3 Element Organical Context

The following figure shows the Atomic clock context:

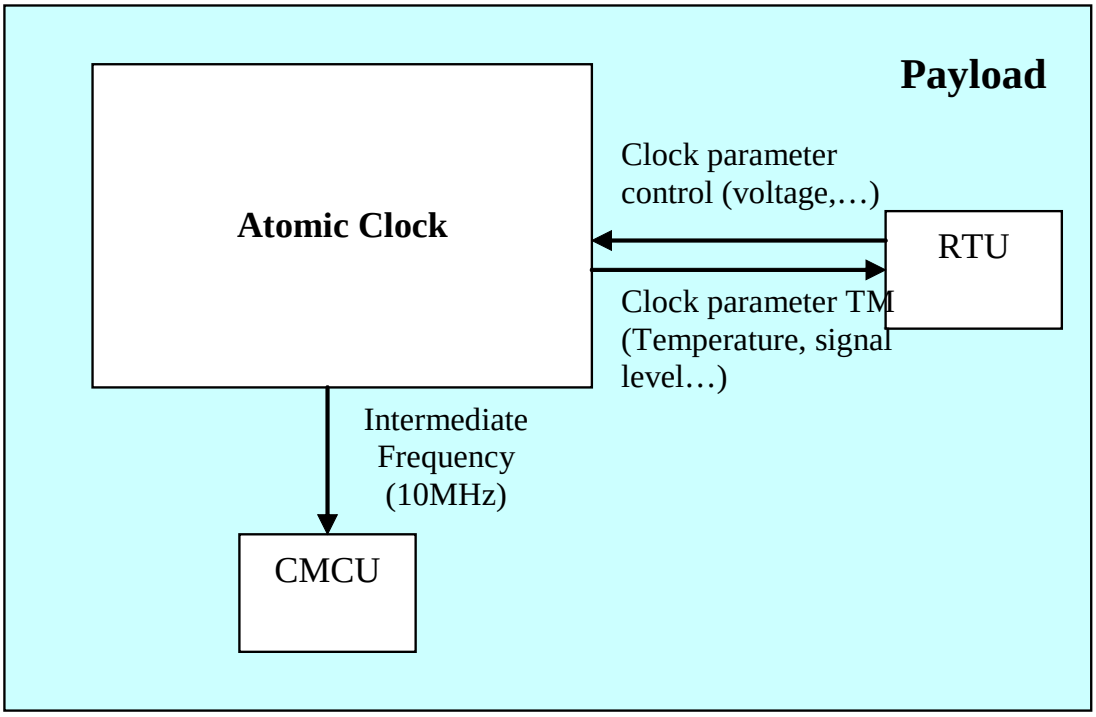


Figure 63: Atomic Clocks Organical context

5.4.3.1.3.1.4 System Level Constraints

As the atomic clocks are furnished to the Space Segment, only system constraints apply to them.

- Redundancy

Each Galileo satellite payload includes the following atomic clock types, as required in GSRD-1158:

- 2 Space qualified Rubidium Atomic Frequency Standards (S-RAFS), in double-redundant configuration (1 master on and one backup off)
- 2 Space qualified Passive H-Masers (S-PHM), in double-redundant configuration (1 master on and one backup off)

Note: in the specification, “oscillator” and “clock” are commonly used to refer to the same unit. In the truest sense, a clock is a device that generates periodic, accurately spaced signals. It is composed of the oscillator on one hand, and a device that counts the oscillations and converts them to units of time interval. Here, the atomic clock refers clearly to the atomic oscillator.

The oscillator is composed of atomic resonator (delivering an atomic frequency) combined with a quartz oscillator in lock loop to deliver a stable standard frequency (5 or 10MHz).

The purpose of each oscillator is to deliver a high stable resonance frequency. Each oscillator uses the quantized energy levels (discrete values) in atoms or molecules as the source of its resonance. An electromagnetic field at a particular frequency can boost an atom from one energy level to a higher one. Or, an atom at a high energy level can drop to a lower level by emitting energy. The resonance frequency (f) of an atomic oscillator is the difference between the two energy levels divided by Planck’s constant (h):

$$f = \frac{E_2 - E_1}{h}$$

The Resonance Width is the range of possible frequencies where a resonator can resonate.

Since all atoms are identical in an atomic oscillator, they should produce exactly the same frequency when they absorb or release energy. In theory, the atom is a perfect “pendulum”. Concretely, this frequency drifts in time according to external environment, aging of the clock, resonance width…

- Hydrogen Maser: “MASER” is the acronym of “microwave amplification by stimulated emission of radiation”. Masers operate at the resonance frequency of the hydrogen atom 1,420,405,752 Hz. In the Maser, an hydrogen gas is sent through a magnetic gate allowing atoms in certain energy states to pass through. The selected atoms enter a storage bulb surrounded by a tuned, resonant cavity. In this cavity, some atoms drop to a lower energy level, releasing photons of microwave frequency. These photons stimulate other atoms to drop their energy level, and they in turn release additional photons. In this manner, a self-sustaining microwave field builds up in the bulb. The tuned cavity around the bulb helps to redirect photons back into the system to keep the oscillation going. The result is a microwave signal that is locked to the resonance frequency of the hydrogen atom and that is

continually emitted as long as new atoms are fed into the system. This signal keeps a quartz oscillator in step with the resonance frequency of hydrogen to deliver a stable 10MHz frequency, as shown in the following figure:

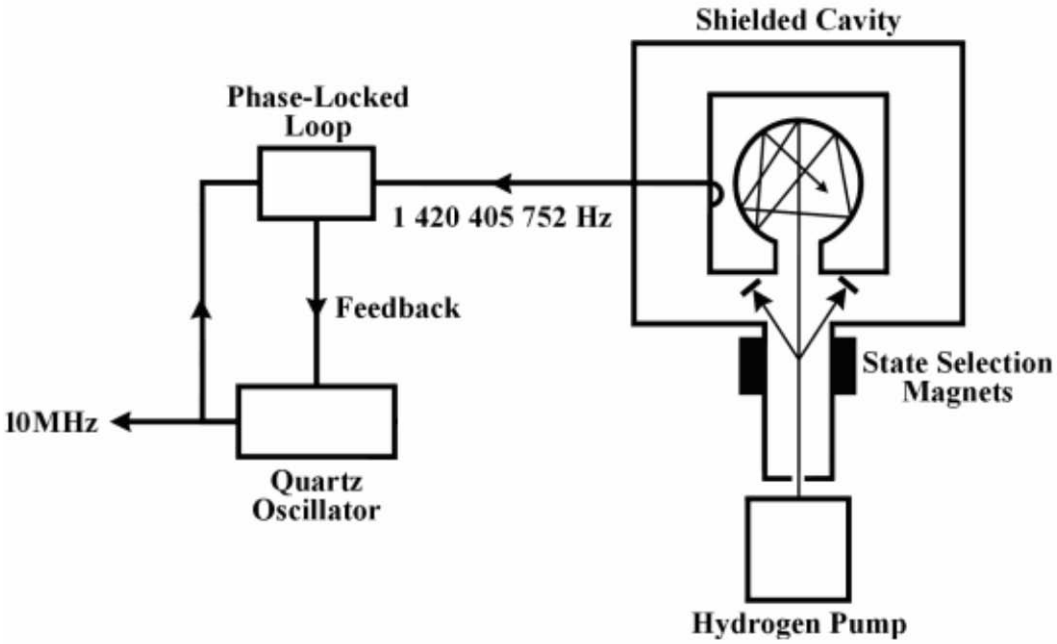


Figure 64: PHM physical architecture

- Rubidium clock: Rubidium oscillators operate at 6,834,682,608 Hz, the resonance frequency of the rubidium atom (87Rb). The optical beam from the rubidium lamp pumps the 87Rb buffer gas atoms into a particular energy state. Microwaves from the frequency synthesizer induce transitions to a different energy state. This increases the absorption of the optical beam by the 87Rb buffer gas. A photo cell detector measures how much of the beam is absorbed and its output is used to tune a quartz oscillator to a frequency that maximizes the amount of light absorption. The quartz oscillator is then locked to the resonance frequency of rubidium, and the stable 10MHz frequency is derived from the quartz oscillator as shown in the figure below:

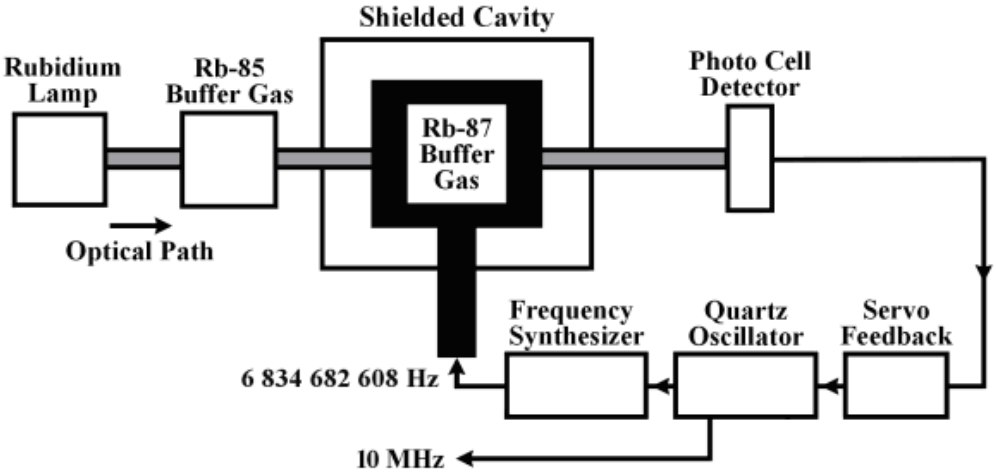


Figure 65: RAFS Physical Architecture

- Atomic clock performance.
  The performance of an atomic clock is driven is characterised by 2 parameters, namely its stability, and its accuracy.

  o **Stability**: the stability characterizes how well the oscillator can produce the same frequency over a given time interval (basically, this doesn't indicate whether the frequency is right or wrong, but only whether it stays the same).
  The stability of an oscillator is:
    - characterize by a **Quality Factor, Q**: defined as the resonance frequency divided by the resonance width. A high resonance frequency and a narrow resonance width increase the Q parameter value. Generally speaking, the higher the Q, the more stable the oscillator (a high Q means that an oscillator will stay close to its natural resonance frequency).
    - specified by a statistic such as the **Allan deviation** that estimates the frequency fluctuations of the device over a given time interval.

  The following table gives the main characteristics of each type of atomic oscillator used in the Galileo System

	Hydrogen Maser	**Rubidium**
Resonance Frequency (Hz)	1 420 405 752	6 834 682 608
Quality Factor, Q (approximate value)	10^9	10^7

Table 52: Atomic Clock Main Characteristics

=> *Allan deviation***:** At system level, the stability of the standard frequency at the output of the clock is not specified. Indeed, the performance key driver is the stability of the

10.23MHz frequency reference derived from the atomic clock. In that sense the Allan deviation specification apply to this 10.23MHz

- **Accuracy**: the accuracy refers to the frequency offset of the oscillator. Frequency offset is the difference between a measured frequency and an ideal frequency with zero uncertainty. This difference is evolving in time and is called “drift”. As for the stability performance, the performance of the drift of the clock itself is not specified at system level. The key driver is the accuracy of the 10.23MHz frequency reference that is specified

- **Operability**

Two S-PHM and two S-RAFS are implemented on board the satellites. Nominally, only one RAFS out of two is switch ON, and one S-PHM out of two (the other one is switch OFF in cold redundancy due to power consumption and emission limitation). The atomic clocks will operate in a hot redundant, parallel configuration.
In nominal configuration, the S-PHM is the primary reference clock operating at switch on of the satellite with the RAFS in hot redundancy. In that case, the navigation signal is generated based on the S-PHM (PHM1). The RAFS (RAFS1) stands in hot redundancy.

Remark: it was originally foreseen that IOV satellites would run on RAFS for the primary clock. Due to the good demonstration of S-PHM performance during the IOV experimentation phase, the selected primary clock is now S-PHM.

The redundancy of the clocks is managed in the CMCU by telecommand.

5.4.3.1.3.2 CLOCK MONITORING AND CONTROL UNIT (CMCU)

5.4.3.1.3.2.1 High Level Description

The CMCU is responsible for generating and maintaining a highly stable frequency reference to the satellite (10.23MHz) to be used for satellite time and carrier frequency generation.

5.4.3.1.3.2.2 Functions Allocation

According to Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2), the CMCU is responsible for the following main functions:

- Generate high stable frequency of 10.23MHz derived from on-board atomic clocks
- Perform clock switch based on TCs
- Perform clock steering
- Provide clock phase comparison measurements

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF5.1	SSF5.1_Generate_OnBoardClock

Table 53: CMCU Function List

5.4.3.1.3.2.3 Element Organical Context

The following figure shows the CMCU context:

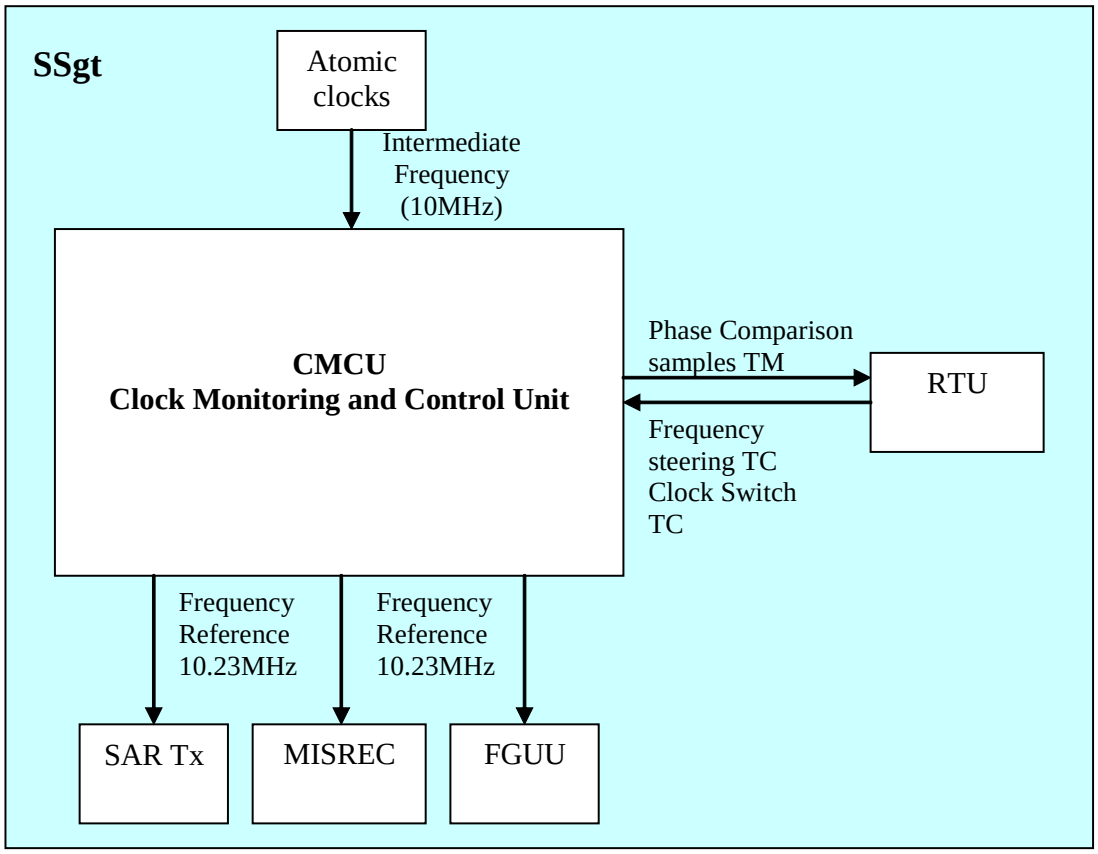


Figure 66: CMCU Organical context

5.4.3.1.3.2.4 Element Main Design Drivers

5.4.3.1.3.2.4.1 System Level Constraints

At system level are specified the performances of the frequency reference delivered by the CMCU, and the way to monitor and control it. Indeed, this frequency reference is directly used to generate the satellite times and downlink signals.

- ***CMCU frequency reference performance:***
  The performance of frequency reference produced by the CMCU is characterized by its stability and its accuracy (drift).

  o **Frequency reference stability:**
  At system level, the stability of this frequency is specified through its Allan Deviation: The Allan deviation is specified for two configurations, whether the frequency reference is derived from the S-RAFS atomic clock or if it is derived from the S-PHM clock:
    - S-RAFS: The S-RAFS flicker floor is better than 3.5x10-14, and the Allan deviation of the S-RAFS output signal does not exceed the values given in the following table:

Integration Time	Allan Deviation
1	5.1×10^{-12}
10	1.64×10^{-12}
100	5.1×10^{-13}
1000	1.64×10^{-13}
10000	5.1×10^{-14}

Table 54: S-RAFS Allan Deviation

Concretely this leads to an autonomy of the S-RAFS of 100 minutes wrt the performance specifications, as specified at system level: *SSREQ-3996: Frequency Drift Stability: The RAFS drift stability shall not affect the clock Allan deviation over 100 min.*

- S-PHM: The Allan deviation of the S-PHM output signal does not exceed the values given in the following table:

Integration Time	Allan Deviation
1	1.1×10^{-12}
10	3.5×10^{-13}
100	1.1×10^{-13}
1000	3.5×10^{-14}
10000	1.1×10^{-14}

Table 55: S-PHM Allan Deviation

Concretely this leads to an autonomy of the S-PHM of 8.3 hours wrt the performance specifications, as specified at system level: *SSREQ-1259: Frequency drift stability: The PHM drift stability shall not affect the clock Allan deviation over 8 hours.*

Note: *The Signal clock Allan deviation characteristics for both S-RAFS and S-PHM, is specified as observed (for any carrier and any service carrier combination) at the transmitting antenna phase centre. This is to take into account any potential degradation introduced by the satellite RF circuitry.*

- **Frequency reference accuracy (drift):**
  At system level, the stability of this frequency is specified through its Long term drift
  - S-RAFS performance*: SSREQ-1191: Long term frequency drift: The output long term frequency drift per year including ageing (12 years) and drift, over the full operating temperature range shall be lower than or equal to 3.65E-10.*
  - S-PHM performance*: SSREQ-1255: Long term frequency drift: The output long term frequency drift per year including aging (12 years) and drift, over the full operating temperature range shall be lower or equal 3.0 x 10-12.*

- ***Clock monitoring and control description***
  The performance of the clocks is continuously monitored and controlled in order to ensure that there are no undetected failures or degradation.

As shown in the physical description in the figure above, the controller equipment in the CMCU is responsible for:

- Monitoring the phase comparison of the 10.23MHz signal at the output of the two synthesizer fed by the primary and back-up active clocks. The phase comparison samples are then transmitted within the telemetry (TM link) to GCS (then forwarded to GMS). Note that the sampling rate is configurable in order to cope with potential ambiguity when initial offsets between clocks is not know,

SSREQ-1319: Phase comparison samples collection: The Clock Control and Monitoring shall provide the collection of phase comparison samples between primary and back-up clock to be made available in the nominal telemetry sent to the ground, according to the following:

- *The sampling rate shall be programmable between 0.1 second and 1000 seconds with a resolution of 0.1 seconds or less and shall be selectable by telecommand*
- *88 millidegrees resolution for a 360 degrees range*

The phase comparison enables to monitor off-line the back up that is not directly observable in the SIS. It enables to maintain the back up in hot redundancy.

- Applying the following commands received from the ground via the TC link:
  - Primary to hot redundant clock switch
  - Hot to cold backup clock switch
  - Frequency adjustment to the DDS Synthetiser

5.4.3.1.3.2.4.2 Major Segment Design Choices

The CMCU architecture and redundancy shall fulfill the required atomic clock redundancy.

The following figure shows the CMCU physical architecture:

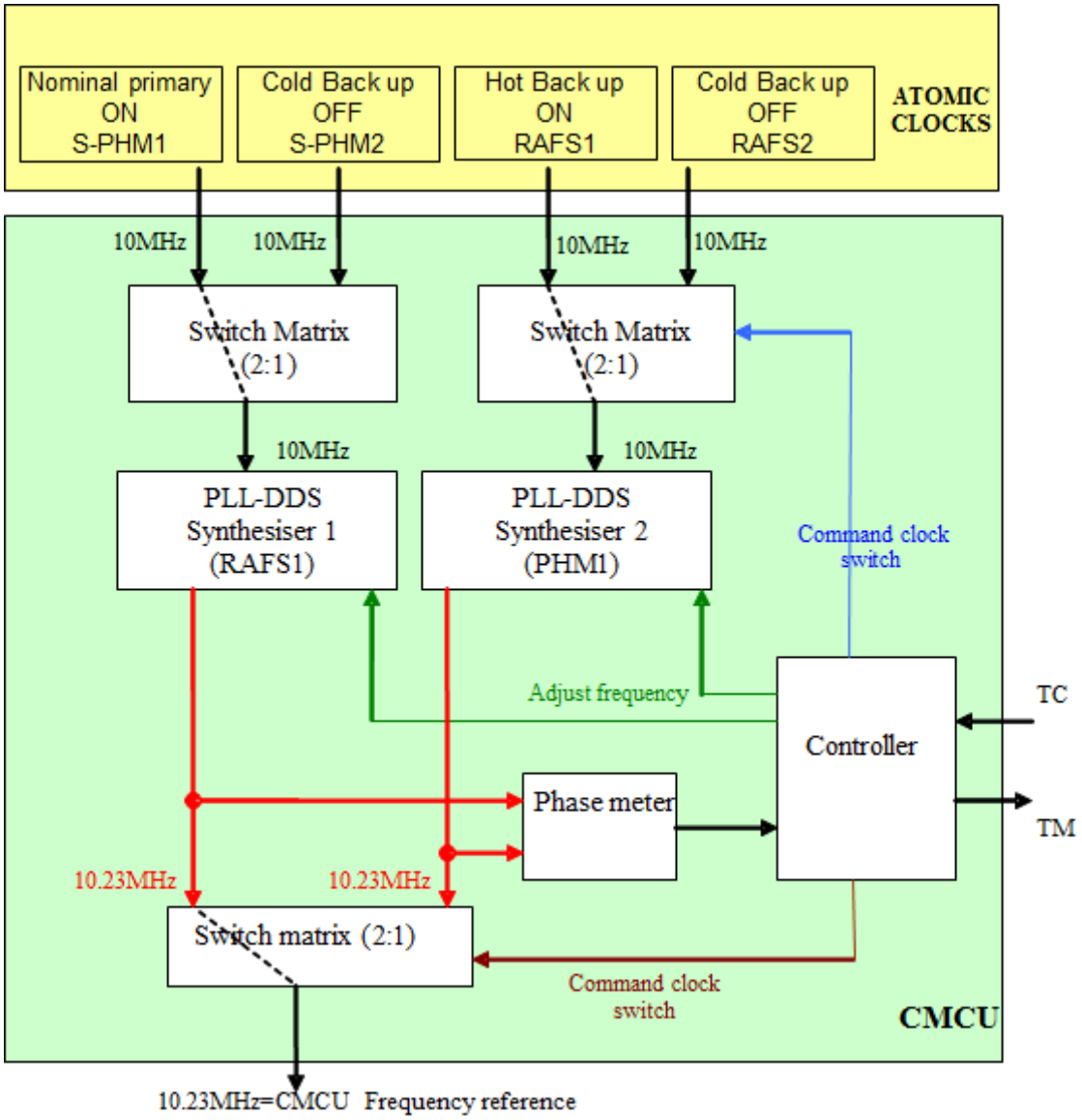


Figure 67: CMCU Physical Architecture

5.4.3.1.3.3 FREQUENCY GENERATION & UPCONVERSION UNIT (FGUU)

5.4.3.1.3.3.1 High Level Description

The FGUU is responsible for performing frequency up conversion to provide different frequencies for the NSGU or the output RF chain.

5.4.3.1.3.3.2 Functions Allocation

According to Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2) the FGUU is responsible for the following main functions:

- Multiplication of the CMCU frequency reference to 122.76MHz (=10.23x12) to be provided to the NSGU for timing generation
- Upconversion of signals provided by the NSGU to be downlinked in L-Band.
- EIRP adjustment by varying the signal level into the SSPA.

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF4.1	Generate Carriers
SSF4.2	Modulate Messages

Table 56: FGUU Function List

5.4.3.1.3.3.3 Element Organical Context

The following figure shows the FGUU context:

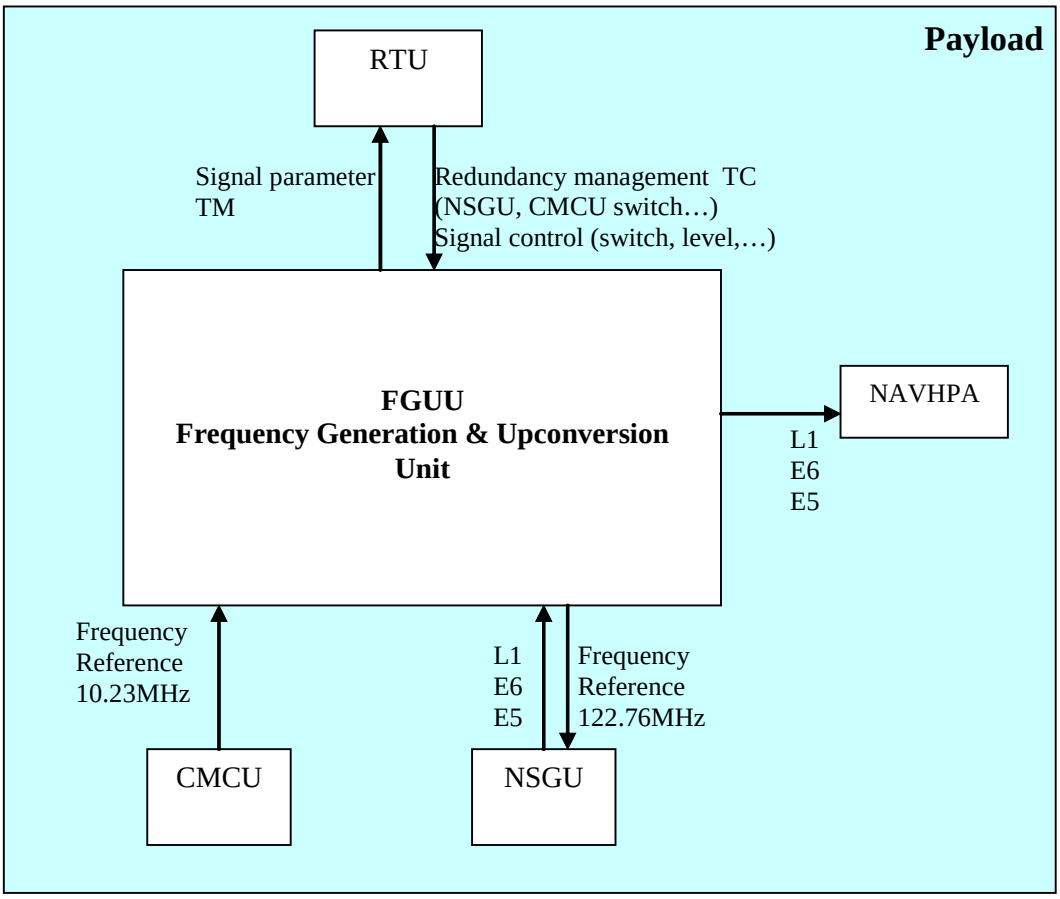


Figure 68: FGUU Organical context

5.4.3.1.3.3.4 Element Main Design Drivers

5.4.3.1.3.3.4.1 System Level Constraints

As the FGUU is responsible for converting of the navigation signals received from the NSGU and up-converting them to L-band, the system constraints applicable to the FGUU are the downlink signal characteristics as recalled in the following table:

Signal	Carrier	Carrier-Frequency	Polarisation	Transmitted bandwidth
E5	fE5	1191.795 MHz	Right-hand circular	92.07 MHz
E6	fE6	1278.750 MHz	Right-hand circular	40.92 MHz
L1	fL1	1575.420 MHz	Right-hand circular	40.92 MHz

Table 57: Signal-Carrier RF Characteristics

5.4.3.1.3.3.4.2 Major Segment Design Choices

In the on-board timing management, the FGUU is responsible for converting the frequency reference 10.23MHz delivered by the CMCU up to 122.76 MHz (12 x 10.23 MHz), to be delivered by the NSGU (related to the digital sampling rate used by the NSGU).

5.4.3.1.3.4 C-BAND MISSION RECEIVER (MISREC)

5.4.3.1.3.4.1 High Level Description

The C-Band mission Receiver is responsible for processing the RF signal received from the ground (6 separate CDMA signals at the same operating frequency)

5.4.3.1.3.4.2 Functions Allocation

According to Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2) the MISREC is responsible for the following main functions:

- Signal demodulation
- Forward Error Correction (FEC) and decoding
- Provide serial output data stream to the payload

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF1.1	Receive Uplink Signals

Table 58: MISREC Function List

5.4.3.1.3.4.3 Element Organical Context

The following figure shows the MISREC context:

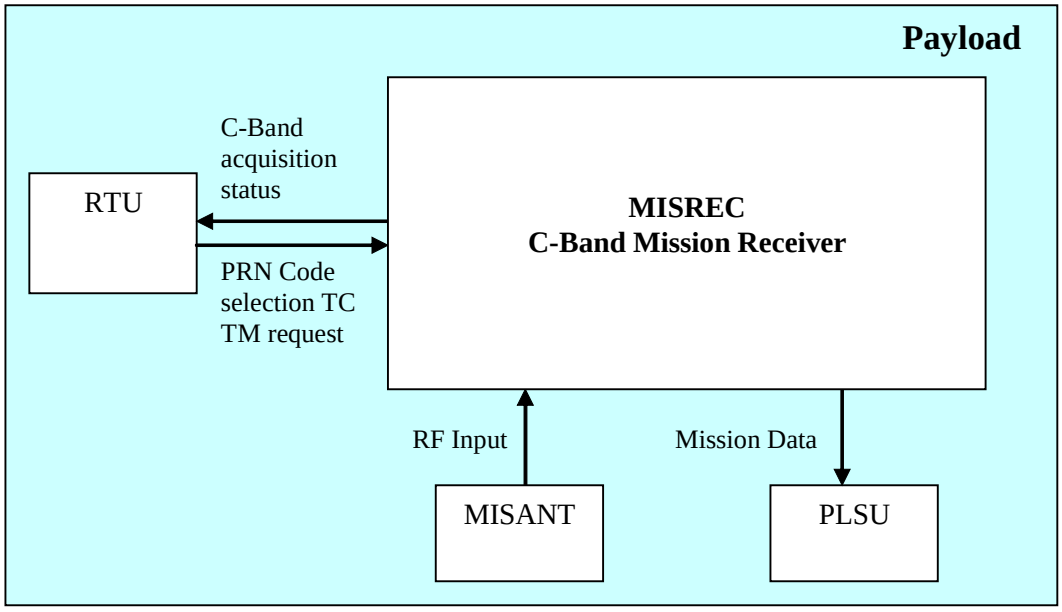


Figure 69: MISREC Organical context

5.4.3.1.3.4.4 Element Main Design Drivers

5.4.3.1.3.4.4.1 System Level Constraints

Physical

The constraints defined at system level are:

1) Limitation to the G/T range of variation due the MISANT antenna pattern (including antenna beam profile error and ripple within beam) and system noise temperature. This is shown in the following table

Antenna half-beam width	For Earth coverage	12°		
S/C G/T	Within beam width	-19 dB/K	-20.5 dB/K	
S/C G/T spread	Within beam width Includes: antenna beam profile error, ripple within beam; system noise temperature variations		0 dB	+3.9dB

Table 59: MISANT Antenna pattern drivers

2) The maximum latency introduced by MISREC to output INT message bits is constrained. The overall allocation to the payload is 152ms. The allocation made by the payload manufacturer to the MISREC is 40ms for the INT message and 120ms for the NAV message

Operability

Although allocation of spreading codes to the 6 channels of each satellite is fixed, flexibility is required and implemented in terms of possibility to change this allocation by telecommand.
In addition each channel can be enabled/disabled by telecommand

5.4.3.1.3.4.4.2 Major Segment Design Choices

Redundancy

The redundancy concept has been selected in order to meet the following requirements:

- No single point failures: this implies to have at least two redundant branches, except for the common elements at the input (which cannot be avoided).
- For the failure of a single branch it has been shown that the requirement defined at P/L level is met with comfortable margins, so in order to keep the design simple, no crosstrapping is added..

In order to meet the reliability requirements of the equipment the Mission Receiver has been designed applying cold redundancy at the main electrical chain. This means that two identical end-to-end chains work in cold redundancy (only one of them is working simultaneously).
If part of one receiver unit fails, all the operating receiver unit is switched off, enabling the other one.
A single RF input coming from the antenna is provided to the equipment. Then, this input signal is divided in order to drive the nominal and redundant units. The circuit used is a passive power splitter with a very low failure probability.

The mission receiver is fed with two possible reference clocks: nominal or redundant. By means of two internal passive cross-strapping circuits with very low failure probability, these two signals can be selected at nominal or redundant sides.
The provided external DC primary power lines and TC/TM lines are doubled.
Finally, each Receiver unit produces dual redundant outputs in order to feed the nominal or redundant PLSU/CSU and NSGU. The acknowledgement protocol implemented between MISREC and PLSU/CSU and NGSU requires that the acknowledgement signal will be cross-strapped at MISREC level.

5.4.3.1.3.5 L-BAND AMPLIFICATION, FILTER AND NAVIGATION ANTENNA (NAVHPA, NAVOFILT, NAVOMUX, NAVANT)

5.4.3.1.3.5.1 High Level Description

The RF output chain is responsible for amplifying, filtering and downlinking the L-Band signal. It is composed of the following elements:

- Power Amplifiers:
  - On IOV satellites: NAVHPA: Navigation High Power signal Amplifiers (SSPA) and switches
  - On FOC1 satellites: TWTA
- NAVOFILT and NAVOMUX: Navigation Output Filter and Output Multiplexer (OMUX)
- NAVANT: L-Band Tx navigation antenna, used as well for SAR Return Link Message transmission

5.4.3.1.3.5.2 Functions Allocation

According to Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2) the Output RF Chain is responsible for the following main functions:

- Amplification of the modulated navigation signals, E5, E6 and L1 (SSPA)
- Filter High band channels (Hi Band Filters)
- Combine low band channel signals E5 and E6 (OMUX Diplexer)
- Satellite to the Air interface (L-Band Antenna)

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF4.3	Amplify and Transmit Signals

Table 60: RF Output Chain Function List

5.4.3.1.3.5.3 Element Organical Context

5.4.3.1.3.5.4 Element Main Design Drivers

5.4.3.1.3.5.4.1 System Level Constraints

The following system constraints are considered for the design of the RF output chain:

- The minimum power levels received on the ground by a receiver equipped with a 0 dBi right-hand circularly-polarised antenna at all elevation angles down to 10° masking angle, excluding shadowing or other propagation effects were originally specified by ESA (see for example GSRD v2 section 13.4.1). The EIRP specification for each service was derived from these figures under the specified conditions.
- Multiple launch specification: restricts available envelope for the antenna, limiting the gain and pattern shape
- Out of band spurious: impacts SSPA linearity and (more importantly) the OMUX rejection characteristics

5.4.3.1.3.5.4.2 Major Segment Design Choices

- RF output chain trade-offs:
  - o Choice of amplifier technology and sizing:
    The space segment selected SSPA technology for the HPA after a trade-off that considered mass, power consumption, heritage, and impact to signal performance.
    For L1 the total power is achieved with 2x55W SSPAs in parallel. The power of each is fed to separate parts of the antenna and combines in space. Such a scheme was chosen to limit impact of combination losses from a passive combiner. A spare SSPA can be selected if either of the two prime SSPAs fail.
    For E5 and E6 single 65 and 60 W SSPAs are used in each chain, with a cold redundant spare for each.
    It has to be mentioned that ESA strongly pushed during the early stages of the Galileo program for using 60W class solid state amplifiers (as requested in GSRD v1, section 7.2.2.12.11).
  - o Choice of OMUX order:
    For the outputs filters and OMUX, the filter order selected is a compromise between out-of-band spurious and in-band performances that effect the signal distortion (amplitude and phase variation) and navigation performance (group delay stability). An eigth order elliptic filter was selected in order to achieve the stringent radio astronomy and COSPAS SARSAT band requirements.

  - o Choice of antenna type:
    An active phased array antenna is selected to allow the beam to be optimised to give an isoflus pattern on the ground. Due to limitations on the size of the antenna this is only partially achieved with a maximum EIRP achieved 8 degs from boresight.

5.4.3.1.3.6 NAVIGATION SIGNAL GENARATOR UNIT (NSGU)

5.4.3.1.3.6.1 High Level Description

The NSGU is responsible for managing the on-board navigation data dissemination including the formatting, encoding and modulation of the downlink L-Band message. In addition the NSGU is responsible for generating the Satellite Time and distributing it as reference to Payload elements.

5.4.3.1.3.6.2 Functions Allocation

According to Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2) the NSGU is responsible for the following main functions:

- Reception and storage of up-linked navigation data from MISREC via the PLSU/CSU
- Storage of the navigation data
- Error correction encoder (CRC)
- Generation of Navigation ranging codes
- Generation of Commercial Service (CS) ranging code (encrypted or not)
- CS key management
- Formatting of Navigation messages
- Encoding (multiplexing of navigation data message with ranging codes)
- Modulation of the data and schedule the provision to the FGUU (note: for E6 and L1 signals, direct digital modulation is used/ for E5 signal, baseband signals are generated)
- Receive TCs from ICDU
- Provide TM to RTU
- Generate Satellite system Time (LGST)

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF1.3	Distribute Data
SSF2	Generate Ranging Code
SSF2.1	Generate Navigation Ranging Code
SSF2.3	Generate CS Ranging Code
SSF3	Prepare Message
SSF3.1	Generate On-Board Data
SSF3.2	Compose Navigation Message
SSF3.3	Modulate Navigation Messages On Ranging Code
SSF4.2	Modulate Messages
SSF5.2	Generate On-Board Time

Table 61: NSGU Function List

5.4.3.1.3.6.3 Element Organical Context

The following figure shows the NSGU context:

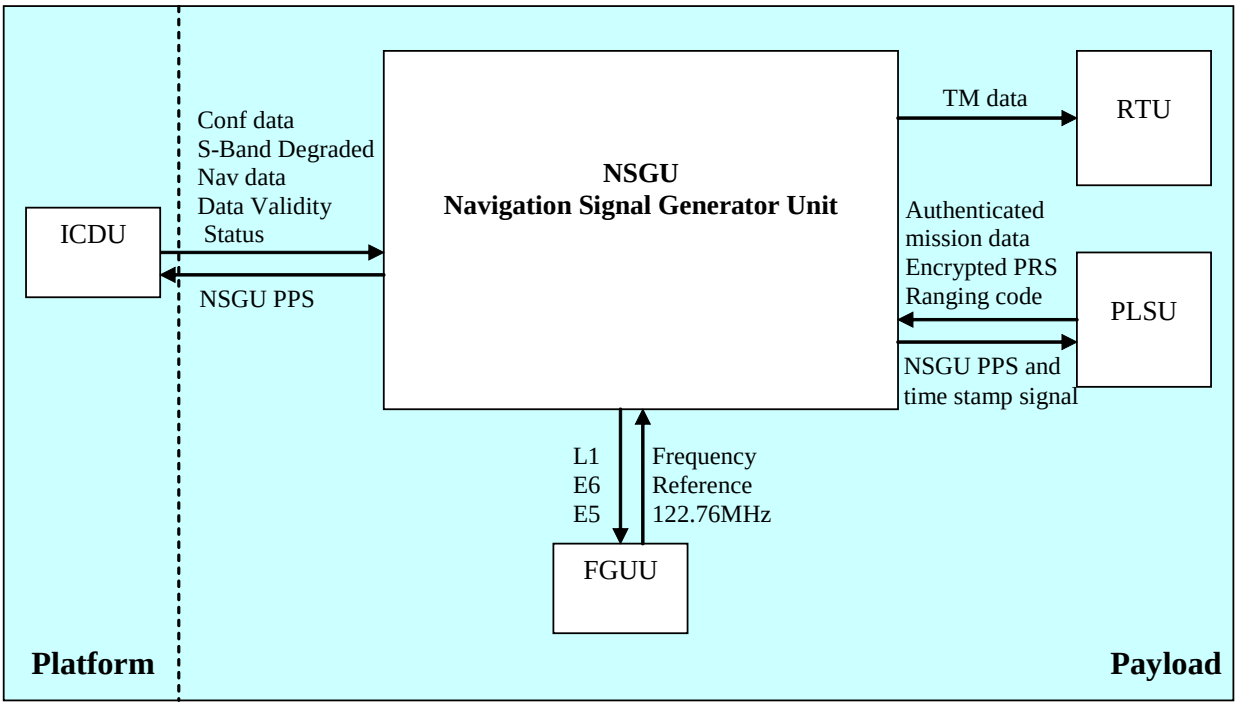


Figure 70: NSGU Organical context

5.4.3.1.3.6.4 Element Main Design Drivers

5.4.3.1.3.6.4.1 System Level Constraints

- **Message handling System constraints:**

The NSGU shall store on board several batches of navigation data generated on ground and disseminate them according to the Time-of-Applicability associated to them.
However the maximum time to process all the navigation data, once it is completely received is limited to 300ms for the overall payload.
On the other hand shall be able to disseminate on real time all the integrity data received from ground. The time allocated to the overall integrity data processing in the payload is set to 152ms. The allocation made by the payload manufacturer to the NSGU is 80ms.
The NSGU gets from the PLSU/CSU is in charge of providing different BOC signals for E6-A and L1-A signal components generation (PRS signal flexibility).

- **CS COMSEC and NAVSEC module location:**

The NSGU shall include the NAVSEC and COMSEC protection modules for the Commercial Service

5.4.3.1.3.6.4.2 Major Segment Design Choices

- **NSGU tasks constraints:**

The following tasks are detailed at segment level:

- Code Generation: The Code Generator provides the spreading codes for all signal components except the classified codes used for the G/Nav E6-A and L1-A components.

Where applicable these spreading codes use a tiered construction including the use of primary and secondary codes.

- o Spectrum Spreading: This function spreads the navigation message data or pilot signal components with the relevant spreading codes from the code generator. For the G/Nav signal components the classified spreading codes are provided to the NSGU by the PLSU/CSU.
- o Signal Generation: There are three signal generators generating navigation signals E5, E6, and L1. They use the output data of the spreader and generate the signals for the FGUU. The signal generation process includes signal shaping, modulation, multiplexing and digital-to-analogue conversion. The shaping filter coefficients are programmable by telecommand.
- o Navigation Message Handing: This function performs the receiving, updating and formatting of either the C-band navigation data received via the PLSU/CSU or the S-band navigation data received from the ICDU.

- **Redundancy:**
  The NSGU shall be cold redundant with physical separation between the prime and redundant channels and/or Units.

5.4.3.1.3.7 PAYLOAD SECURITY UNIT (PLSU OR CSU)

5.4.3.1.3.7.1 High Level Description

The Payload Security Unit is responsible for all the security aspects of the payload. This function is ensured by:

- Dedicated PLSU element on AST IOV satellite
- Embedded in CSU element on OHB FOC1 satellite

5.4.3.1.3.7.2 Functions Allocation

According to Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2), the PLSU/CSU is responsible for the following main functions:

- C-Band uplink signal check: authentication and anti-replay verification of Mission Messages received through C-Band, and transmission to the NSGU
- Generation of encrypted ranging codes for the PRS.
- Payload Secured Telecommands (STC) handling: decryption, authentication and anti-replay checking of all the Secure Telecommands (STC) pertaining to the Payload.
- Payload Secure Telemetries (STM) handling: encryption, authentication/integrity tagging, and anti-replay protection of all Secure Telemetry (STM) at the output of the Payload
- Monitoring and control (including self-test): to generate audit records of security events (key changes, attacks, authentication failures etc).
- Payload Key management: all keys required by the PLSU/CSU will be manually loaded into the unit before launch from the BBKME (Black Box Key Management Equipment).

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF1.2	**Data Authentication**
SSF1.3	**Distribute Data**
SSF2.2	**Generate PRS Ranging Code**

Table 62: PSLU Function List

5.4.3.1.3.7.3 Element Organical Context

The following figure shows the PLSU context on IOV Payload:

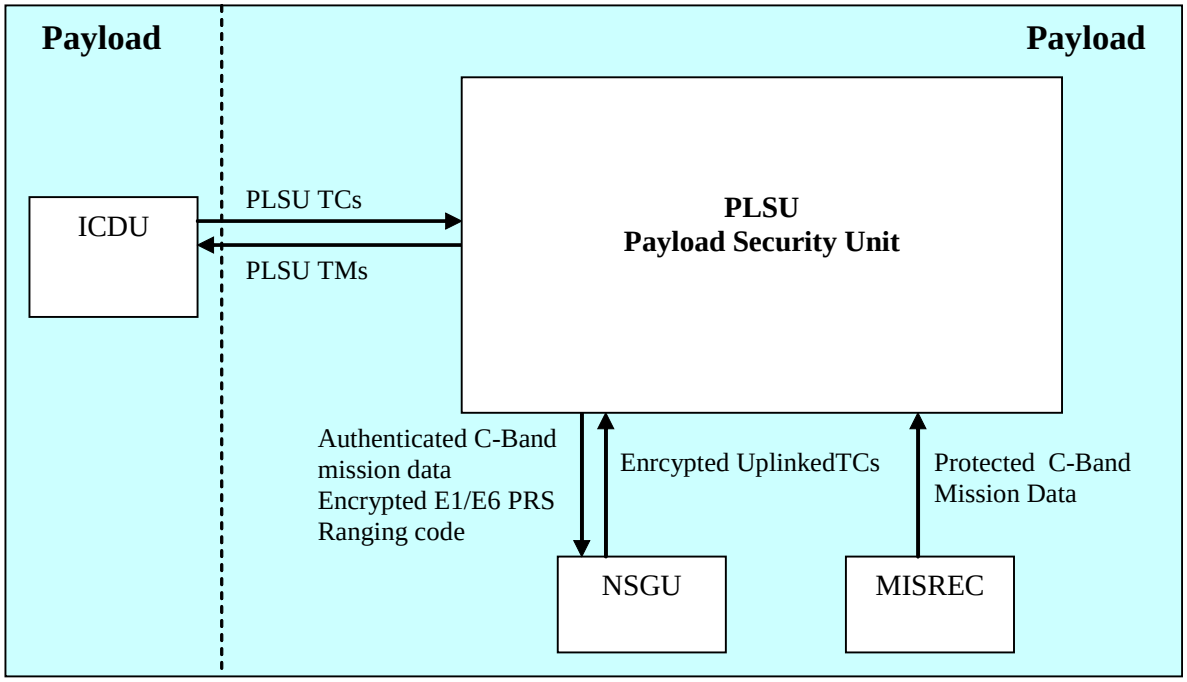


Figure 71: PLSU Organical context on IOV Payload

The following figure shows the CSU context on FOC1 Payload:

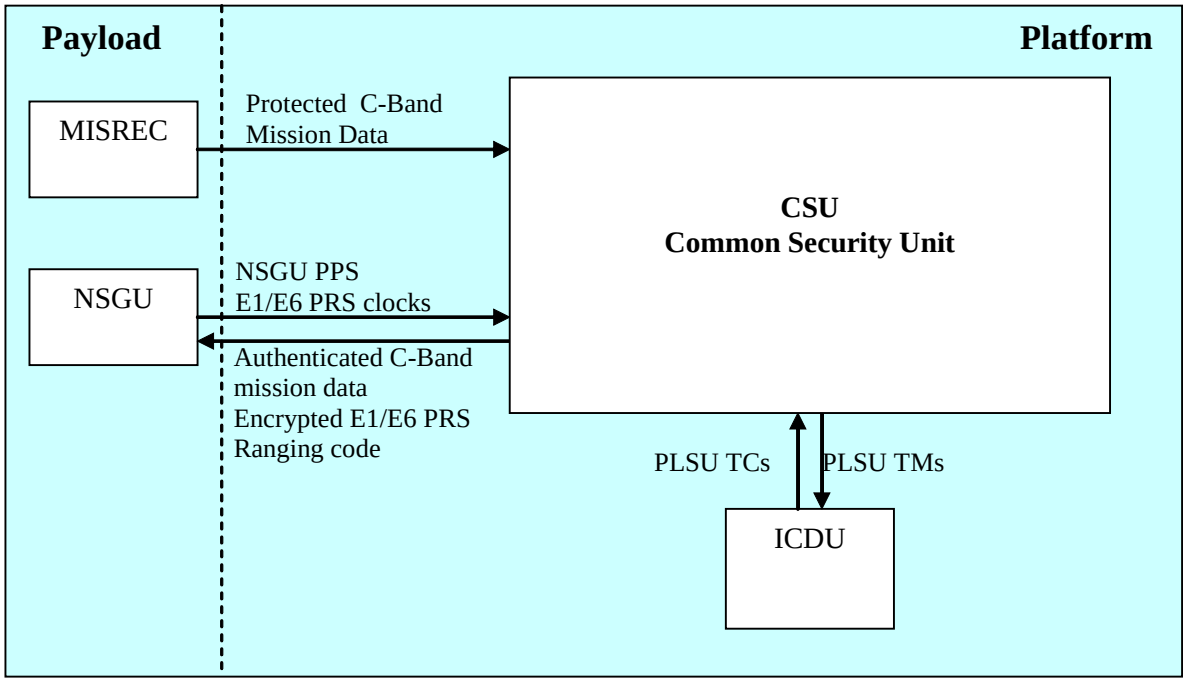


Figure 72: CSU Organical context on FOC1 Payload

5.4.3.1.3.7.4 Element Main Design Drivers

5.4.3.1.3.7.4.1 System Level Constraints

The constraints imposed by the System on the PLSU/CSU are security related: As the PLSU/CSU will process classified data, it must be subjected to a security evaluation, certification, and accreditation in order to demonstrate its suitability and compliance with the relevant security requirements.
The maximum latency introduced by the PLSU/CSU on the INT message bits is constrained. The overall allocation to the payload is 152ms. The allocation made by the payload manufacturer to the PLSU/CSU is 30ms.
The PLSU/CSU has to be able to provide different BOC signals to the NSGU for E6-A and L1-A signal components generation (PRS signal flexibility).

5.4.3.1.3.7.4.2 Major Segment Design Choices

A significant proportion of the functional and non-functional requirements of the PLSU/CSU are constrained to exact parameters (e.g. it must produce mission output data exactly as defined in applicable ICDs). Areas in which design choice have been executed include:

- The PLSU/CSU shall consist of two identical and fully independent functional branches, one main branch and one redundant branch. These branches shall operate in a cold redundant configuration, and a fault in one half of the unit shall not affect the operation of the other half.
- All flight software and firmware shall be designed to DAL B, unless otherwise agreed and justified.
- The selected ASIC technology was based on risk minimization. A more innovative technology such as 90nm would have allowed the implementation of more gates improving the overall performances of the ASIC.

5.4.3.1.3.8 REMOTE TERMINAL UNIT (RTU)

5.4.3.1.3.8.1 High Level Description

The RTU is a unit responsible for relaying the TM/TC information between the platform (ICDU/CDU) and the payload (Units)

5.4.3.1.3.8.2 Functions Allocation

According to Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2) the RTU is responsible for:

- Receive the discrete/serial TMs from the payload element and transfers them to on-board data bus for the platform (ICDU) to be included in the S-Band downlink.
- Convert the TCs from the platform into discrete/serial TCs for the payload to command the payload elements (except for NSGU and PLSU/CSU)

This includes:

- General TMs for all the payload elements (MISANT, MISREC, PLSU/CSU, NSGU, FGUU, NAVHPA, CMCU, Atomic clocks, NAVOFILT and NAVOMUX, NAVANT, SAR): e.g. temperature, current…
- General TCs for some payload elements (e.g. FGUU redundancy management, Atomic clock voltage control…)
- Specific TM/TCs:
  - CMCU: frequency reference steering and clock redundancy management (switches...) / Phase comparison samples.
  - MISREC: PRN Code selection and C-Band Acquisition status TM

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF1.3	Distribute Data
SSF6.3	Satellite Mode
SSF6.5	On-Board Data Management
SSF7	Manage Payload
SSF9	Provide TTC Communication
SSF9.1	Receive and Control TC
SSF9.2	Transmit and Manage TM

Table 63: RTU Function List

5.4.3.1.3.8.3 Element Organical Context

The following figure shows the RTU context:

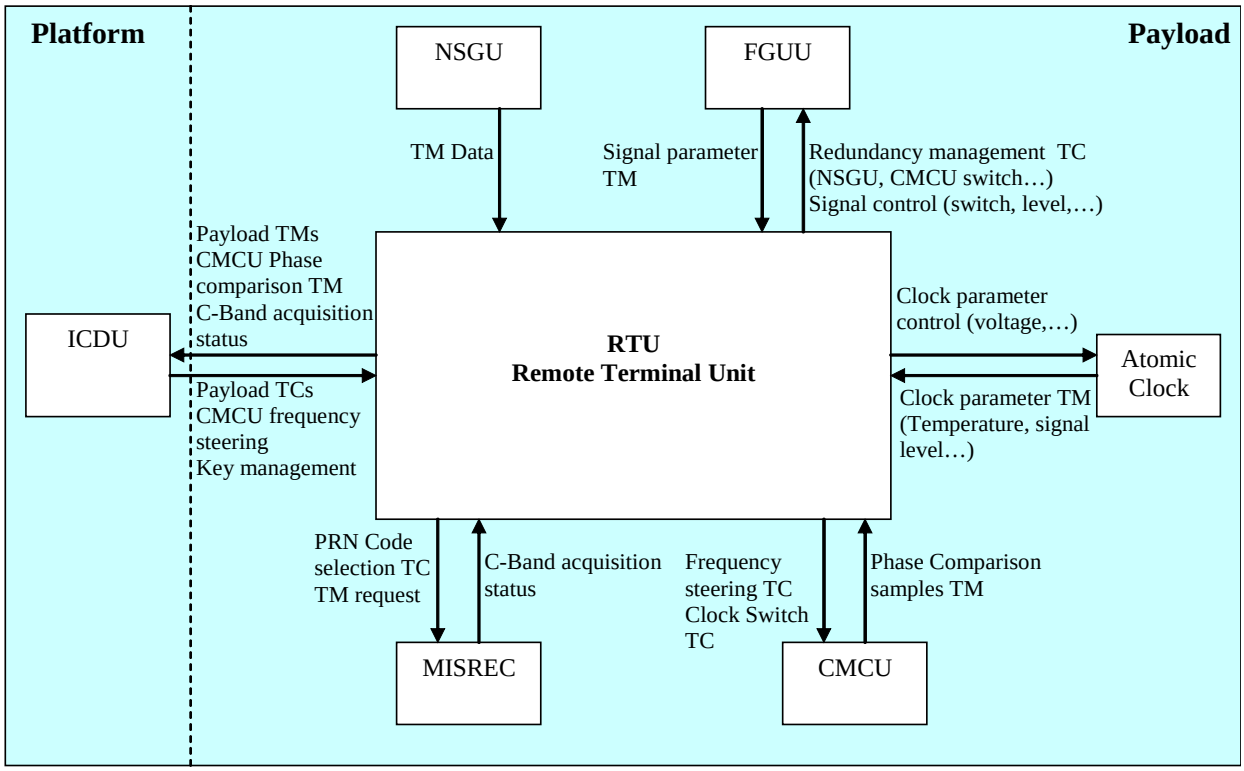


Figure 73: RTU Organical context

5.4.3.1.3.8.4 Element Main Design Drivers

5.4.3.1.3.8.4.1 System Level Constraints

The RTU design shall take into account GSTB-V2 experience showing an issue on the redundancy and cross strapping design. Considering the cross-strapping of redundant halves, a switch over from Side-A to B of RTU could be needed because of a failure of payload unit Side A: this should be rectified for IOV Payload .

5.4.3.1.3.8.4.2 Major Segment Design Choices

The physical constraints apply on the link interface and the connection to the serial on-board bus MIL-1553.

Redundancy:
The RTU consists of two redundant sections, RTU A (nominal) and RTU B (redundant), mounted inside a common housing.

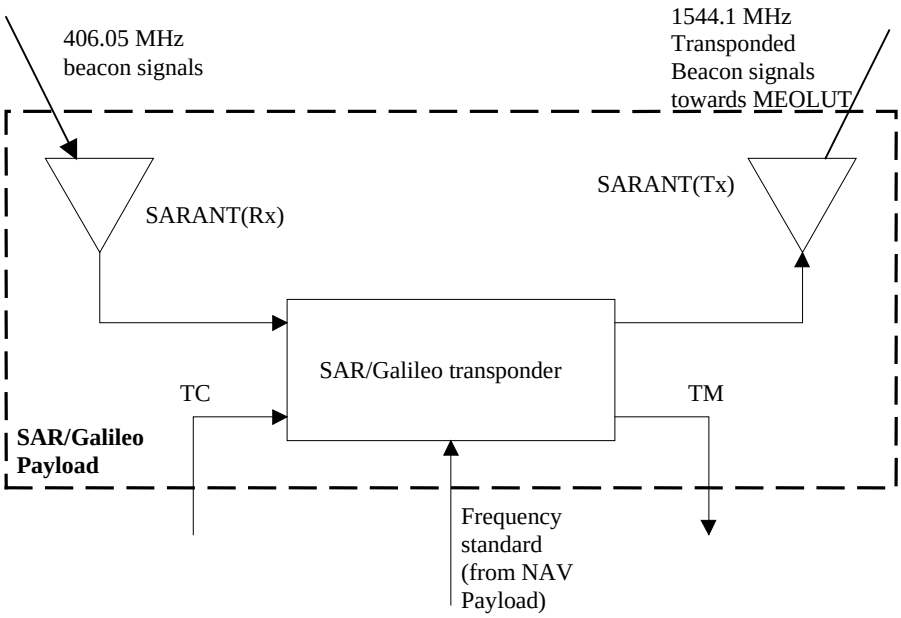
Operability:
Each side of the RTU contains three modules:
- Control Module (CTRL)
- TM Acquisition Module (TMAQ)
- DC/DC Converter Module (DCDC)

5.4.3.1.3.9 SAR PAYLOAD

5.4.3.1.3.9.1 High Level Description

The SAR payload consists of an UHF Receive and L-Band transmit antenna and a SAR transponder.

The SAR Payload (in fact SAR Forward Link Transponder) is responsible for relaying the SAR messages for the forward link (receive 406MHz distress beacon from the user) to the SAR service provider (L-Band downlink). It is a transparent non-inverting transponder and therefore does not do any processing of the received signal.



Legend:

SARANT(Rx)	- Forward Link Receiver Antenna
SARANT(Tx)	- Forward Link Transmitter Antenna
TC	- Telecommand to spacecraft
TM	- Telemetry from spacecraft

Figure 74: SAR Payload Architecture

5.4.3.1.3.9.2 Functions Allocation

The SAR Payload has the following top level functions

- To be able to receive up to 5 simultaneous beacon bursts at different frequencies in the 406MHz COSPAS-SARSAT band.
- To transparently frequency translate the received signal of 406MHz to the downlink frequency of 1544MHz.
- To transmit the signal in L-Band to the Medium Earth Obit Local User Terminals (MEOLUTs)

The following table lists the system functions allocated to the SAR Transponder:

Function ID	Function name
SSF8	SSF8_SAR_Payload

Table 64: SAR Payload Function List

5.4.3.1.3.9.3 Element Organical Context

The following figure shows the SAR Payload context:

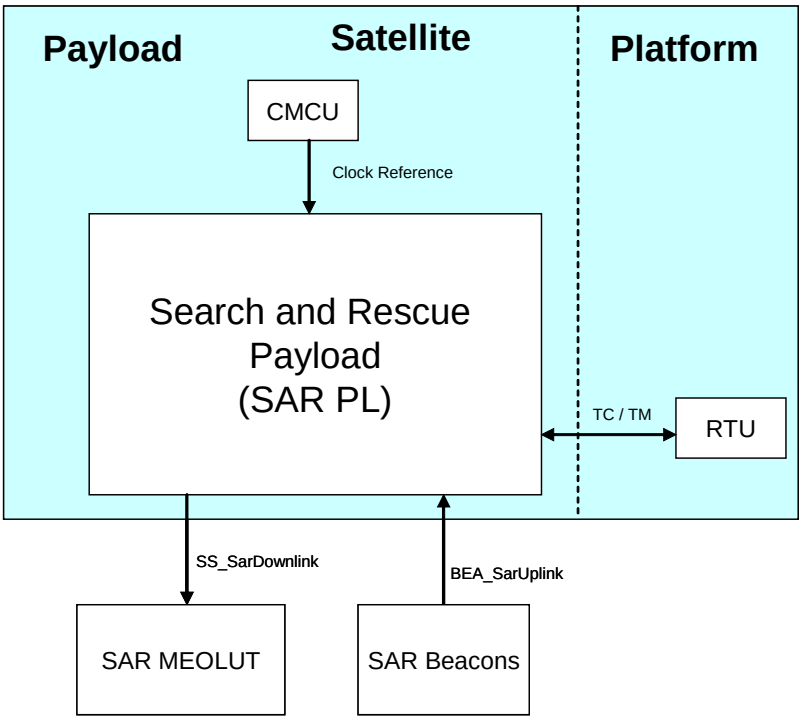


Figure 75: SAR Payload Organical context

5.4.3.1.3.9.4 Element Main Design Drivers

5.4.3.1.3.9.4.1 System Level Constraints

The constraints imposed by the system on the SAR payload are mainly with regards to the quality of the downlink signal, to enable the SAR MEOLUTs to successfully receive the signal with the required quality.

As the SAR payload is the only part of the SAR Forward link service within the Galileo system, all system level requirements on SAR forward link service are in some way associated with the functionality of the SAR payload.
To meet the performance needs of the SAR forward link service the SAR payload shall have the following main constraints as defined in the GSRD

- To receive 5 simultaneous bursts in different frequencies the distress signal from SAR beacons and transmit them to MEOLUT of the international SAR satellite service (GSRD-236, 865, 1195 & 2580)
- To provide coverage of the whole surface of the Earth for both receive and transmit (GSRD-238 & 1193)
- SAR Forward link susceptibility to Out-of-Band interference (GSRD-3131)
- SAR Forward link susceptibility to In-Band interference (GSRD-1129)
- The payload elements providing SAR services shall comprise 406MHz receive and 1544MHz transmit antenna functions and a transparent transponder (GSRD-1197)
- The SAR transponder shall use the reference frequency generated by the Navigation Payload CMCU.(GSRD-1199)

5.4.3.1.3.9.4.2 Major Segment Design Choices

The Search And Rescue (SAR) payload is principally based on a transparent transponder receiving the uplink signal in the 406.0 - 406.1 MHz band and retransmitting it in the 1544.05 - 1544.15 MHz band. The SAR payload includes no redundancy at satellite level.

Redundancy at mission level is obtained via the satellites of the constellation.
The satellite is not considered failed due to a failed SAR payload and will not be replaced. The SAR payload reliability is <1920 FITs.

The SAR antenna performs both the receive function at UHF-band (RHCP polarization) and the transmit function at L-band (LHCP polarization). The baseline solution consists of a ring of 6 quadri-filar helix radiating elements for the UHF receive function, with a central ‘splash plate’ feed acting as the L-band transmit antenna.

The receive signal is routed to the Search and Rescue Transponder (SART). The SART performs low pass filtering of the received signal, amplification via a Low Noise Amplifier (LNA), output filtering and then down-conversion into IF, close-to-band filtering using crystal filtering (switchable in order to provide a wide-band (90 kHz) or narrow band mode (50 kHz)) and up-conversion into L-band.

The signal is then amplified with a nominal RF output power of 5 W, filtered and routed to the SAR antenna transmit port for transmission.

Test couplers are included in order to allow the testing of the SAR payload during the payload and satellite AIT cycle.

To achieve the linearity, whilst meeting power consumption requirements, a linearizer is being included in the Transponder SSPA design.

Major trade-off is to optimize the filter design to meet out-of-band requirements and limit impact on in-band group delay stability and gain variation. Currently the required rejection is not compatible with meeting group delay stability, due to the steep filter roll-off requirements.

To meet the G/T requirements the test coupler may be removed and the LNA repositioned on Z-panel to reduce losses before LNA and operating temperature.

5.4.3.2 PLATFORM OVERVIEW

5.4.3.2.1 PLATFORM DESCRIPTION

According to Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1), the Galileo platform is composed of the following subsystems:

- Avionics Subsystem (including ICDU/CDU, Sensors and Actuators)
- TTC Subsystem, including S-Band Antenna (TTCANT) and TTC Transponder)
- Platform Security Unit :
  - PFSU (Platform Security Unit) on AST IOV satellites
  - CSU (Common Security Unit) on OHB FOC1 satellites
- Electrical Power Subsystem (EPS) (including PCDU: Power Conditioning and Distribution Unit)
- Propulsion Subsystem

These elements are briefly detailed in the sections after.

5.4.3.2.2 PLATFORM FUNCTIONS

The main functions of the navigation platform:

- Manage the satellite TM/TCs and S-Band link
- Perform AOCS

The following table lists the system functions allocated to the Platform and the mapping to Platform elements:

Function ID	Function name	Mapping to Element	Mapping to Element
SSF6	Manage Platform	AVS	AVS
SSF6.1	FDIR	AVS	AVS
SSF6.2	Health Status	AVS	AVS
SSF6.3	Satellite Mode	AVS	AVS
SSF6.4	AOCS	AVS	AVS
SSF6.5	On-Board Data Management	AVS	AVS
SSF6.7	Autonomous Action and Control	AVS	AVS
SSF7	Manage Payload	AVS	AVS
SSF9	Provide TTC Communication	AVS/TTC	AVS/TTC
SSF9.1	Receive and Control TC	AVS/PSFU/TTC	AVS/CSU/TTC
SSF9.2	Transmit and Manage TM	AVS/PSFU/TTC	AVS/CSU/TTC
SSF9.3	Two Way Ranging	TTC	TTC

Table 65: Platform Function List

5.4.3.2.3 PLATFORM ELEMENTS

5.4.3.2.3.1 AVIONICS SUBSYSTEM

5.4.3.2.3.1.1 High Level Description

The Avionics subsystem provides all satellite command and control, data handling and attitude and orbit control functions, required for safe operation of the satellite during all mission phases including non-nominal situations. The GALILEO avionics is an integrated control and data system which consists of an Integrated Control and Data Unit (ICDU) containing the On-Board Computer, the sensors and actuators for attitude and orbit control and the Avionics Software (ASW).
The overall avionics function is the result of close interaction of functions at hardware and software level. The core equipment for implementation of this functionality is the ICDU, which provides the major hardware resources and hosts the avionics software.

The AVS subsystem includes the following components:
- Integrated Control and Data Unit (ICDU) in IOV satellite of Control and Data Unit (CDU) on FOC1 satellites:
- AOCS Softwares
- AOCS Hardwares
  - Sensors to determine the orientation and the movement of the satellite including:
    - Earth Sensors: to determine satellite pitch and roll
    - Sun Sensors: to determine satellite yaw (3rd satellite axis in complement to the Earth sensor measurements), based on measurements of the orientation of the satellite relative to the Sun.
    - Gyros: to determine the movement of the satellite, such as the rate of rotation and the angular change in attitude of the satellite.
  - Actuators to dynamically control the satellite attitude:
    - Reaction Wheels: 4 wheels properly placed on-board enable to control and steer the 3 satellite axis
    - Magnetic torquers: used to correct the orbit perturbation not handled by the wheels: the principle is to generate a magnetic moment in the satellite which interacts with the Earth magnetic field establishing a pure torque on the satellite.

5.4.3.2.3.1.2 Functions Allocation

According to Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1), the avionics subsystem is responsible for the following functionalities:
- Satellite System Control:
  - Mode management
  - TM and TC processing (TC decoding, TM encoding and storage)
  - HK – monitoring and event management
  - Overall FDIR functionality
  - Onboard data processing and distribution: in particular, the avionics are responsible for the generation of Satellite status and Data Validity Status (DVS) flags

- Attitude and Orbit Control:
  - o Satellite attitude maintenance
  - o Orbit control capability
  - o Attitude acquisition, safe mode and de-orbiting capabilities
  - o AOC FDIR

- Satellite BUS Control
  - o Electrical power subsystem management
  - o Thermal control
  - o TTC subsystem management
  - o Propulsion Control
  - o Release and Pyrotechnic activation
  - o Platform Security Unit management
  - o BUS FDIR

- Payload Control: The ICDU within the Avionics subsystem is the only platform unit interfacing with the payload, handling the following functionalities:
  - o Payload management
  - o Payload Security Unit management
  - o Payload TM/TC handling
  - o Payload FDIR

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF3.1	**Generate On-Board Data**
SSF6	**Manage Platform**
SSF6.1	**FDIR**
SSF6.2	**Health Status**
SSF6.3	**Satellite Mode**
SSF6.4	**AOCS**
SSF6.5	**On-Board Data Management**
SSF6.7	**Autonomous Action and Control**
SSF7	**Manage Payload**
SSF9	**Provide TTC Communication**
SSF9.1	**Receive and Control TC**
SSF9.2	**Transmit and Manage TM**
SSF9.3	**Two Way Ranging**

Table 66: Avionic Subsystem Function List

5.4.3.2.3.1.3 Element Organical Context

The following figure shows the architecture of the Avionics subsystem with its main interfaces:

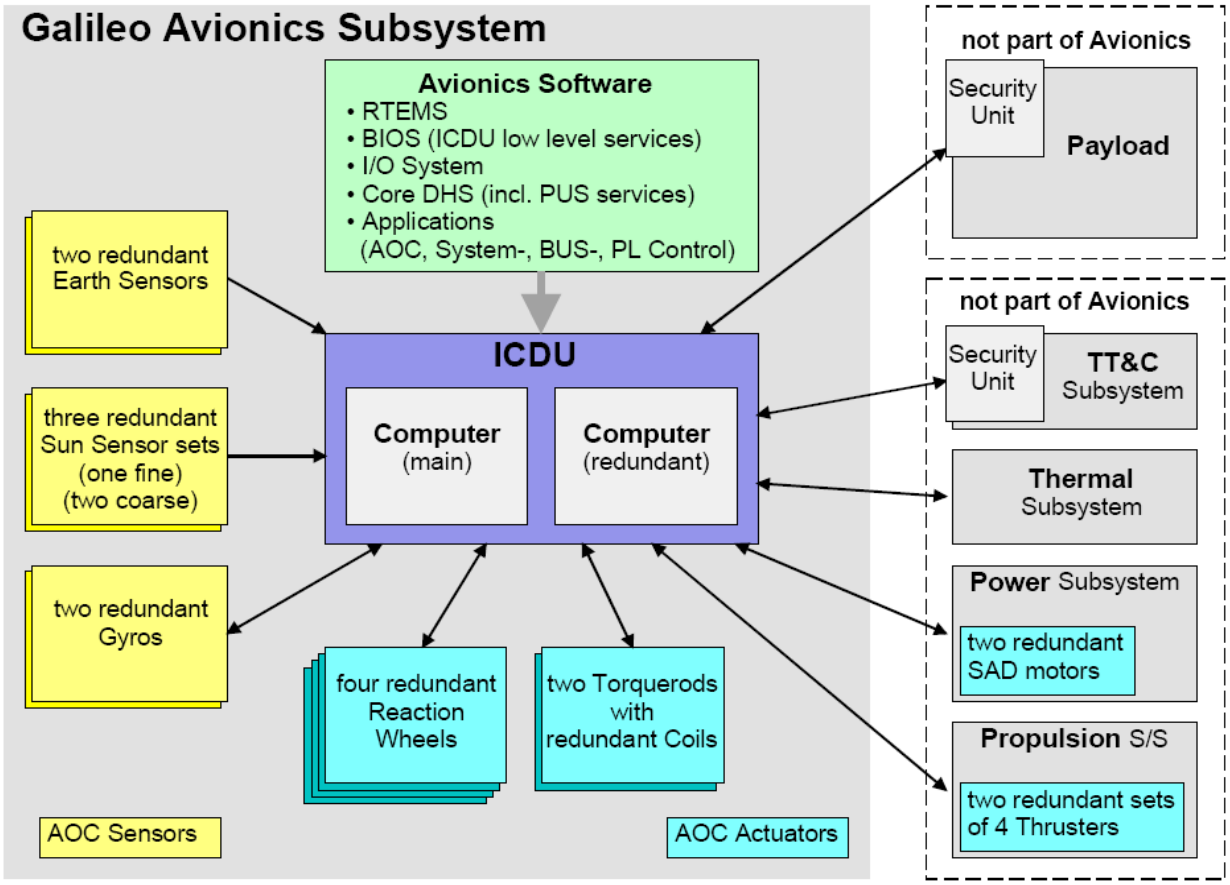


Figure 76: Avionic Subsystem Organical context

5.4.3.2.3.1.4 Element Main Design Drivers

The following parameters are to be considered for the design of the Avionics subsystem:

- Satellite control: commandability and observability
- Satellite Autonomy
  - Automatic mode transitions
  - Satellite performance versus duration without ground support
- Satellite availability - Satellite FDIR
- Authentication approach (ESA standard)
- Minimisation of the pointing errors

5.4.3.2.3.1.4.1 System Level Constraints

The main drivers are:

- The high reliability and life time: Subsystem reliability is very high, this has a significant impact on redundancy schemes and hardware selection.
  - Avionics: 0.932 (AVREQ-4410)
  - ICDU : 0,947

- A very high availability: Satellite Availability requirements imply a maximum of 2 transitions to Safe Mode or Intermediate Safe Mode over 12 years – figures assume mission recovery from ISM within 13 hours and within 3 days from Safe Mode. This drives the FDIR HW and SW architecture, requiring the capability to identify and isolate at least at subsystem level most of the possible failures and to provide the mechanisms to recover them using redundancy without degrading the mission performances. The availability have to be considered in particular for the periods, when the satellite is in nominal operation. Consequently, the availability can be relaxed during other phases (only driven by the reliability requirements).
- Telecommand functions are mainly driven by stringent operational security requirements leading to the use of different protection mechanisms and checks.
- Low ground communication bandwidth induced by the narrow communication window (1 hour for telemetry download of 2 orbits with a 20 kbit/s data rate, AVREQ- 2192), leading to synthetic TM generation and selective retrieval mechanisms to make available the relevant parameters for expertise
- Severe EMC radiated requirements to protect the UHF receive band, induced by the SAR Payload, drive the ICDU design. A notch between 406 and 406.1 MHz of – 10dBµV/m is required.
- Severe Radiation requirements in MEO

In particular, The satellite should provide an Attitude and Orbit Control Sub-system (AOCS) that include all on-board hardware and software to provide during all the missions phases an autonomous capability to determine and to maintain the required satellite attitude pointing, rate and acceleration.

5.4.3.2.3.1.4.2 Major Segment Design Choices

5.4.3.2.3.1.4.2.1 ICDU or CDU

In accordance to the Avionic specification requirement, the ICDU/CDU shall offer the following functions and performances:

- Telecommand interface, capable to receive and process telecommands at a 1 kbit/s rate and to execute Direct Commands (DTC). Telecommand receiver has to be hot redundant.
- FDIR functions that should include :
    - o A Reconfiguration Module, capable to perform the processing unit monitoring and reset and reconfiguration to the redundant processing unit, to acquire some system alarms (namely : Battery discharge – Low voltage, Earth or Sun Presence, Thruster On Time …). This function is necessary to autonomously cope with ASW or processor failures.
    - o A Safe Guard Memory, which shall save the necessary ASW states to be able to continue after a processor reconfiguration in the required mission mode. The SGM has to be protected against SEU, power drop and S/W induced data corruption.
    - o A HPC (High Power Command generator) that shall be directly connected to the Telecommand interface and the Reconfiguration Module to enable the execution of hardware reconfiguration and the execution of DTC.
- OBT clock that shall maintain and distribute the on-board time with at least 10-5 stability (required for platform accuracy).

- Telemetry functions that shall provide a 20 kbit/s data rate, provide at least 3 Virtual Channels, and provide a minimum 2.5 Gbits storage capability in a dedicated Telemetry memory.
- Processing module that shall provide at least a 14 Mips processing resources, 6 Mbytes SRAM and a 2 Mbytes EEPROM.
- A MIL-STD1553B bus coupler that shall interface with:
  - On AST IOV satellite: the payload RTU, NSGU and PLSU units and with the platform PCDU and PFSU units.
  - On OHB IOV satellite: the payload RTU and NSGU units and with the platform PCDU and CSU units.
- Interfaces shall be provided with the different satellite subsystems

The following figure shows the ICDU context on IOV satellite:

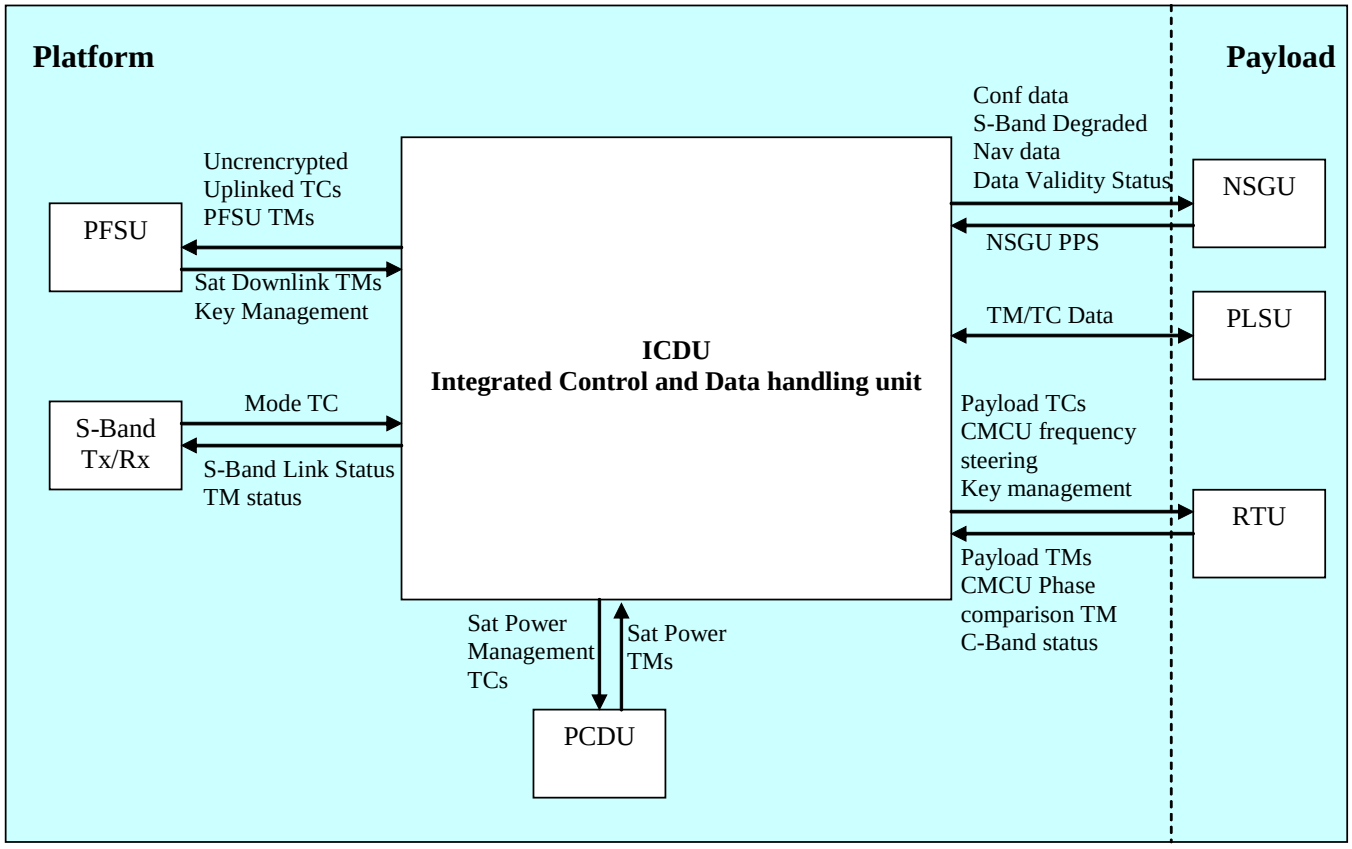


Figure 77: ICDU Organical context on IOV satellite

The following figure shows the CDU context on FOC1 satellite:

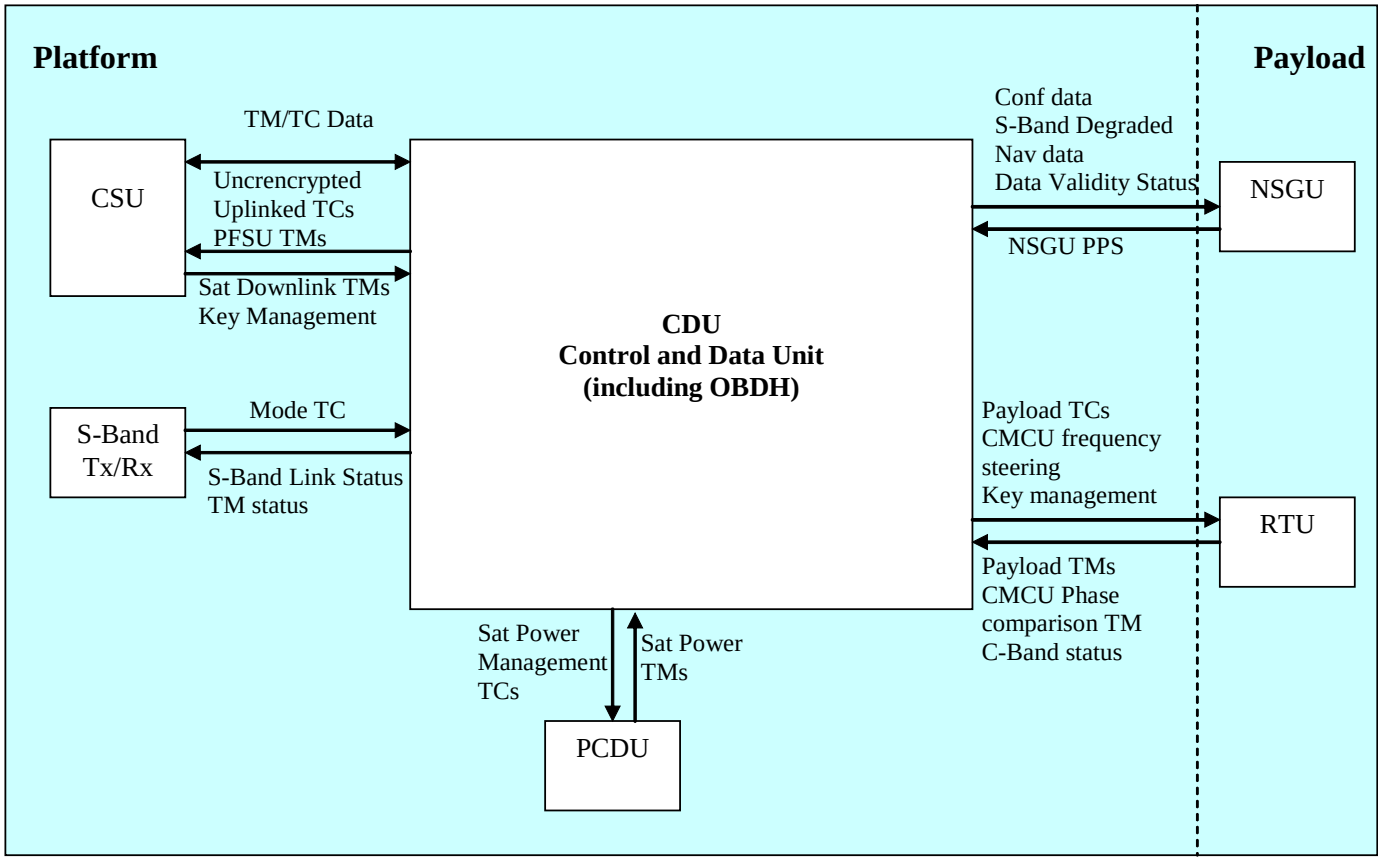


Figure 78: CDU Organical context on FOC1 satellite

5.4.3.2.3.1.4.2.2 Avionics Softwares

The Avionics Softwares manage AOCS, data handling, command and control of subsystems. They are specified as DALC, although developed to DALB without independent software verification.

The ASW is broken down in 5 main blocks in a bottom up approach

- RTEMS: the operating system.
- BIOS: it is a library of low level services, without any task, for access to the HW interfaces of the ICDU. It is provided by the ICDU supplier in order to simplify the HW/SW integration on the ICDU.
- I/O System: this element implements all the Input/Output operations that are necessary at application level. It is dedicated to the operations that take place in the ICDU and on the busses and links that are connected to it. The detailed operations of equipment units that are connected to these links are implemented by the relevant application.
- Core DHS: implements all the packets services (PUS) and supports the applications.
- Applications: these elements implement the functional processing associated to the management of the spacecraft: Attitude and Orbit Control, Bus Control, Payload Management and System Control. They rely on the Core DHS for the PUS services and on the I/O System for Input/Output operations. The coupling between the different applications is kept as low as possible to ease parallel development.

5.4.3.2.3.1.4.2.3 Avionics Hardwares

AOCS Hardwares- Sensors:

- Three 2-axes Sun sensors providing Sun vector measurements with their optical axes arranged in the spacecraft x/z-plane providing full coverage around the S/C y-axis
    - o One fine Sun sensor with its optical axis aligned with S/C - –X-axis, FOV: +/- 64deg
    - o Two coarse Sun sensors with their optical axis tilted away from the plus- and minus S/C z-axis by 25 degree towards the S/C +x-axis, FOV: 75deg cone angle around optical axis
- One 2-axes Earth sensor pointing into spacecraft +z – direction,
- One 1-axis RIG arranged along a space diagonal of the S/C coordinate system

AOCS Hardwares- Actuators:

- 4 Reaction Wheels in a pyramid arrangement,
- Torque Rods arranged perpendicular to each other, with redundant coils.
- 8 Mono – prop. Thrusters slightly tilted and arranged in two 4 –thrusters sets at –x- side
- Single Axis Solar Drive Mechanisms rotating the solar array around the S/C-y – axis,

The need to control dynamically the satellite attitude is due to the following constraints:

- Over the orbit period, the satellite rotates by 360 ° around the normal of the orbital plane, to ensure that the antennas always point towards the Earth (at the same time, the mechanical disturbances must be cancelled by the torquers)
- In order to limit the error in the yaw determination, the angle between spacecraft-Sun and spacecraft-Earth must be maintained sufficiently large (co-linearity problems of an Earth-/ Sun sensor measurement system limited by a relatively large constant angular momentum perpendicular to the orbital plane).
- In order to optimise the solar power, the yaw attitude must be steered to keep the solar arrays perpendicular to the Sun.

Redundancy Management:

In order to eliminate critical failure modes and single point failures that will adversely affect the satellite mission performance and to provide the required reliability, redundancy is implemented at different levels:

- Interface Circuit Level
- Equipment/Component Level
- Functional Level

Interface Circuit Level

Each of the two redundant ICDU parts includes similar AOC interface boards accommodating the sensor and actuator interface circuits. Each of the sensors and actuators is connected to the main and redundant interface circuit and thus can be controlled either by the main ICDU or the redundant ICDU part.

- Each of the Sun Sensors, Earth Sensors and Gyros is connected via redundant interface circuits.
- Each of the Torque Rods provides redundant coils for connection to the main and redundant interface board.

- Each of the Reaction Wheels is connected via redundant interface circuits.
- The thruster valve is based on standard design of a single valve with a single solenoid coil. Two standard valves are combined in a serial configuration in order to achieve a dual coil, dual seat configuration, one coil operating one seat and the second coil operates the second seat. This valve design allows for valve seat redundancy. The single coils are connected via the harness to the main and redundant interface board.
- Each cat bed heater consists of 4 elements arranged in an internal redundant configuration. The complete cat bed heater is connected via the harness to the PCDU
- and commanded from the ICDU.
- Each of the Solar Array Drive Motors provides redundant windings, which are connected to the main and redundant interface boards.

Redundancy at Equipment/Component Level
The equipment (sensors and actuators) of the defined AOC configuration provides the following redundancy at equipment/component level:
- 2 Earth Sensors, 1 active, 1 cold redundant
- 2 Fine Sun Sensors, 1 active, 1 cold redundant
- 2 Coarse Sun Sensors, 2 active, internally redundant
- 1 Gyro, 1 active internally redundant
- 4 Reaction Wheels, 4 active, hot redundant
- 8 Thrusters, a set of 4 active, a set of 4 cold redundant

This means one of two failure tolerant redundancy for each sensor type, one of four failure tolerant Hot Redundancy of Reaction Wheels and one of two failure tolerant Redundancy of chemical Thrusters Sets.

Redundancy at Functional Level
The AOC configuration provides the following functional redundancy:
- SUN and EARTH ACQUISITION with or without GYRO use
- SUN and EARTH ACQUISITION with Wheels or Thrusters for attitude control
- NORMAL MODE with or without SUN SENSOR
- Wheel unloading by magnetic torques or solar sailing

In addition the function of direct actuation commands to the Propulsion Subsystem thruster valves, cat bed heaters and the Solar Array Drive is provided.

Operability:
The operability, composed by commandability and observability, is defined in the section 5.4.3.2 of SSREQ (AD.4).
The main driver for the Galileo satellite operation, in particular for avionics is:
The satellite shall, with the exception of well identified scenarios, be capable of autonomous operation, including the defined performance related to routine operations, for at least two orbital periods in nominal or contingency cases.

In case of any single failure, the avionics (as other subsystem) contents a set of functions which allows automatic reconfiguration mechanism based mainly on FDIR.
This concept is defined considering several type of level:
FDIR - Level 0
Equipment single failure should be autonomously detected and corrected internally by the equipment.

FDIR - Level 1
Equipment single failure should be autonomously detected and corrected by switching to a redundant equipment.
FDIR - Level 2
Anomaly of a vital function, where it can be autonomously known that a function can not be fulfilled by surveillance of parameters that go out of range without knowing that an equipment has failed or a software error has occurred, should be recovered autonomously by switching to a redundant configuration.
FDIR - Level 3
System alarms should rely on hardwired signals to the computer or on software processes. System alarms should lead to Safe mode.

In addition to the automatism, the avionics subsystem allow special and contingency to be performed via ground command as the unit control (On/Off), the attitude and orbit manoeuvres, the reconfiguration and parameter updating.

5.4.3.2.3.2 TTC SUBSYSTEM

5.4.3.2.3.2.1 High Level Description

The TT&C subsystem is responsible for S-Band two way transfer of information between the satellite and the GCS on ground. It includes:

- S-Band Antennae: the Satellite to Air RF interface in S-Band
- TTC transponder

5.4.3.2.3.2.2 Functions Allocation

According to Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1), the TTCF subsystem is responsible for the following functionalities:

- telecommand reception and control for distribution to PFSU/CSU or ICDU/CDU
- telemetry transmission when received from PFSU/CSU
- two way ranging handling: for accurate distance measurement between the satellite and the ground station with S-Band signal.

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):

Function ID	Function name
SSF9.1	**Receive and Control TC**
SSF9.2	**Transmit and Manage TM**
SSF9.3	**Two Way Ranging**
SSF6.2	**Health Status**

Table 67: TTC Subsystem Function List

5.4.3.2.3.2.3 Element Organical Context

The following figure shows the organical context of TTC subsystem

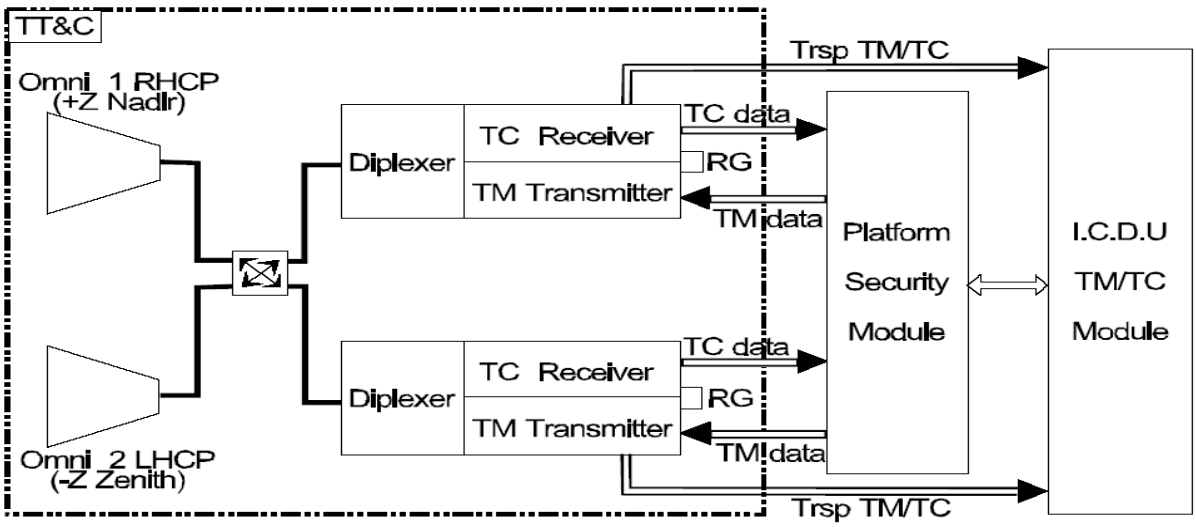


Figure 79: TTC Subsystem Organical context

5.4.3.2.3.2.4 Element Main Design Drivers

5.4.3.2.3.2.4.1 System Level Constraints

The TTC subsystem shall comply with the RF and message specification as specified in GCSSS ICD [AD.9]

5.4.3.2.3.2.4.2 Major Segment Design Choices

TTC Architecture:

The TT&C subsystem is composed of:

- two S-Band transponders in charge of receiving TC and transmitting TM simultaneously,
- two omni-antennas ensuring a global coverage around the spacecraft,
- one 3 dB hybrid coupler allowing the up and downlinks through both antennas,
- a RF harness realizing connections on one hand between transponders and hybrid and on the other hand between hybrid and antennas.

Redundancy:

The S-Band transponders are operated in hot redundancy for the receive part and in cold redundancy for the transmit part.

TTC Operability:

The TT&C subsystem is able to operate in three different modes:

- standard mode with ESA standard code ranging,
- standard mode with tone ranging mode,
- spread spectrum mode.

The functional TC/TM/RNG modes availability vs operational mode is given in the following table.

TT&C Mode	Function	Operational Modes		
		LEOP	Special	Routine
STD Mode with ESA Ranging	TC alone	X	X	
	Ranging alone	X	X	
	TC + Ranging	X	X	
	TM alone	X	X	
	TM + Ranging	X	X	
	TC + TM + Ranging	X	X	
STD Mode with Tone Ranging	TC alone	X		
	Ranging alone	X		
	TC + Ranging			
	TM alone	X		
	TM + Ranging	X		
	TC + TM + Ranging			
Spread Spectrum Mode	TC alone			X
	Ranging alone			X
	TC + Ranging			X
	TM alone			X
	TM + Ranging			X
	TC + TM + Ranging			X

Table 68: TC/TM/RNG functional modes vs operational modes

5.4.3.2.3.3 PLATFORM SECURITY UNIT (PFSU OR CSU)

5.4.3.2.3.3.1 High Level Description

The Platform Security Unit (PFSU/CSU) is responsible for protecting the S-band Telecommand (TC) and Telemetry (TM) links. In nominal operating mode all TC to and TM from the satellite will be encrypted and authenticated.
This function is ensured by:

- Dedicated PFSU element on AST IOV satellite
- Embedded in CSU element on OHB FOC1 satellite

5.4.3.2.3.3.2 Functions Allocation

According to Security Units DDJF (IOVSS-RD.6) and satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1), the PFSU/CSU is responsible for the following main functions:

- Apply cryptographic algorithms for decryption & authentication check on any TC sent to the spacecraft on the S-band link.
- Acquisition and processing of encrypted TC dedicated to the PFSU/CSU
- Apply cryptographic algorithms for encryption & authentication on any TM sent from the spacecraft to ground on the S-band link.
- Generation and encryption of specific PFSU/CSU internal TM (secure telemetry - STM)
- Reception and management of PFSU/CSU keys internal keys for the various cryptographic functions
- Management of PFSU/CSU operational modes, on-board data bus interface, etc
- Generation of internal supply voltages

This covers directly the following SS functions defined in the system DDF1 (SYS-RD.01):
Function ID Function name

Function ID	Function name
SSF9.1	Receive and Control TC
SSF9.2	Transmit and Manage TM

Table 69: PFSU/CSU Function List

5.4.3.2.3.3.3 Element Organical Context

The following figure shows the PFSU/CSU context:

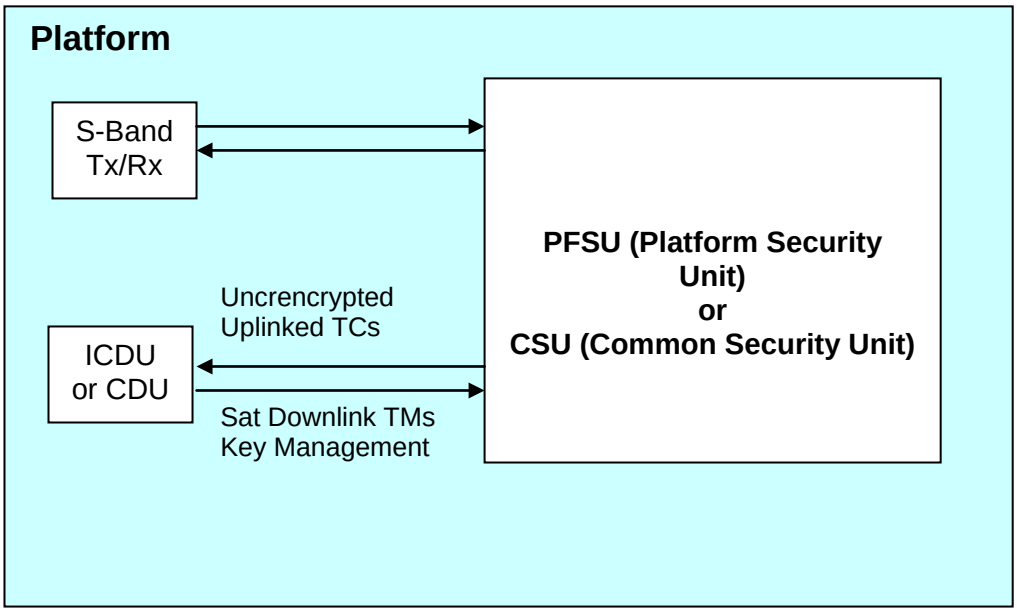


Figure 80: PFSU/CSU Organical context

5.4.3.2.3.3.4 Element Main Design Drivers

5.4.3.2.3.3.4.1 System Level Constraints

The constraints imposed by the System on the PFSU/CSU are security related: As the PFSU/CSU will process classified data, it must be subjected to a security evaluation, certification, and accreditation in order to demonstrate its suitability and compliance with the relevant security requirements.

5.4.3.2.3.3.4.2 Major Segment Design Choices

A significant proportion of the functional and non-functional requirements of the PFSU/CSU are constrained to exact parameters (e.g. conformance to Telemetry Packet standard). Areas in which design choice have been executed include:

- The PFSU/CSU shall consist of two identical and fully independent functional branches, one main branch and one redundant branch. These branches shall operate in a hot redundant configuration. This hot redundant TC reception chain shall ensure that no single failure shall lead to the loss of satellite commanding capability. Each chain will be connected to one of the redundant outputs of the TTC transponder.
- It shall be possible to disable and enable by STC any autonomous function of the PFSU/CSU.
- During platform/satellite level integration, it shall be possible to substitute the classified data (algorithms, keys and any other classified element) of the PFSU/CSU in order to allow the operation of the satellite and the PFSU/CSU in nominal mode as well as in clear mode without imposing security constraints on the AIT environment and the handling of the satellite.

- The design of the security unit shall be such, that exchange and installation, respectively of the security relevant items containing classified information or technologies shall be possible as the last step of launch preparation.
- All Software and Firmware shall be designed to DAL C or better unless justified and agreed with the customer.
- It shall be possible to allow a memory load of any part of the application part of the onboard SW and at any time on ground request.

5.4.3.3 OTHER SUBSYSTEM OVERVIEW

The other satellite subsystems are not directly specified by System. The following section just gives an overview of these subsystems with potential constraints from System when relevant.

The following subsystems are described in dedicated section as transverse to payload and platform:

- Thermal Control Subsystem (TCS)
- Structure Subsystem
- Electrical Power Subsystem (EPS) composed of:
  - Solar Array (SA)
  - Solar Array Drive Mechanism (SADM)
  - Battery
  - Power Conditioning and Distribution Unit (PCDU)
- Harness
- Propulsion Subsystem

5.4.3.3.1 THERMAL CONTROL SUBSYSTEM (TCS)

See Space Segment Documentation [IOVSS-RD.1] and [FOCSS-RD.1]

5.4.3.3.2 STRUCTURE SUBSYSTEM

See Space Segment Documentation [IOVSS-RD.1] and [FOCSS-RD.1]

5.4.3.3.3 ELECTRICAL POWER SUBSYSTEM (EPS)

See Space Segment Documentation [IOVSS-RD.1] and [FOCSS-RD.1]

5.4.3.3.4 HARNESS

See Space Segment Documentation [IOVSS-RD.1] and [FOCSS-RD.1]

5.4.3.3.5 PROPULSION SUBSYSTEM

5.4.3.3.5.1 PROPULSION SUBSYSTEM DESCRIPTION

The propulsion subsystem uses a monopropellant system to perform manoeuvres. The Subsystem comprises:

- Tank to store fuel
- latching valves
- Prime and redundant 4x 1N thruster branches
- pipework
- Pressure transducer
- Propellant Filter
- Fill and drain valves for propellant and pressurant
- Test Ports

5.4.3.3.5.2 PROPULSION FUNCTIONS AND CONTEXT

Manoeuvres are performed by a thruster valve being commanded open, which causes pressurized fuel to flow from the tank to the thruster. In the thruster there is a reaction and gas is expelled.

The manoeuvre is controlled by the avionics in a closed loop. the attitude is monitored by the AOCS sensors and the thrusters modulated on/off to ensure the correct attitude is maintained.
The manoeuvre is monitored by satellite FDIR functions and terminated if safety thresholds are reached.
The propulsion system is used as a back-up mode for unloading momentum stored in the AOCS wheels.

5.4.3.3.5.3 MAIN DRIVERS

The main driver to determine the propellant quantity is the amount of deltaV the propulsion system must give the satellite.
The budget is composed of:

- launch injection errors and initial orbit positioning
- mid-term station-keeping
- repositioning of spare

The tank can store up to 78kg. 60 kg is required for a dry mass of 630 kg.

5.5 Inter-Segment Interfaces

The Intersegment interfaces are:

- GMS to GCS Interface
- GCS to SS Interface
- GMS to SS Interface
- SIS Interface

The physical chacteristics of each interface are described in the dedicated ICDs.

5.6 HOSTING SITES AND NETWORKS PHYSICAL DESCRIPTION

5.6.1 GCC SITES

The Physical constraints of the GCC site (under Hosting Entity Responsibility) are defined in a dedicated document: [AD.37], GCC Infrastructure Requirements Document, ref. ESA-APPNG-REQ/01179.

The GCC is the place where all Galileo operations will be conducted from. If the remote sites are supposed to be unmanned, this is not the case for the GCC, where operators, support engineers and staff will be in charge of performing all the operations to maintain the Galileo constellation and ensuring the mission of the system.

Two GCC are installed in Fucino (Italy) and Oberpfaffenhoffen (Germany).The GCC will host almost all the GMS and GCS elements, including a GSS station as well as the necessary GDDN equipment (VSAT antennas and associated equipment). However, no ULS or TTCF element will be installed on those sites.

5.6.2 REMOTE SITES

The remote sites are the places where the TTCF, ULS and GSS elements are installed. Those remote sites are located worldwide on territories under European control, ensuring a global coverage of the world
The sites are owned by various Hosting Entities, which are in charge of upgrading the sites for Galileo ground elements hosting and offering a hosting service to Galileo (including security and guarding, maintenance of the infrastructure hosting the Galileo elements, first level of maintenance for the elements to be installed …).
Dedicated applicable documents, the site ICDs, show clearly the boundary betwen site (under Hosting Entity Responabiltiy) and element (ULS, GSS and TTC) deployed by Galileo segments. Refer to the following subsection for detail references.
The remote site list is given in dedicated TN TTCF, ULS and GSS site list [SYS-RD.54], for both IOV and FOC configurations.

On each site, the Galileo area is completely independent from other facilities, if any.
Furthermore, each Galileo element on site is as far as possible independent from other elements, allowing to divide the site into independent domains, one for each element. Each domain has its own power supply, hosting building...
4 kinds of sites are foreseen for Galileo:

- TTCF only site
- common TTCF, ULS and GSS sites
- common ULS and GSS sites
- GSS only sites

5.6.2.1 TTCF SITES

TTCF site constraints and boundaries are defined in document [AD.27] - TTCF SIRD.

These sites host the following elements:

- one TTCF element deployed by GCS
- one associated SDDN element, called TTC-SNE deployed by GCS
- GDDN equipment, shared with other elements in case of combined site provided by the Communication Service Provider

In FOC, 7 S-band TT&C stations will be deployed over the world. These sites will be located in:

1. Kourou (French Guyana) with 2 TT&C stations
2. Kiruna (Sweden)
3. Reunion (Reunion Island)
4. Noumea (New Caledonia)
5. Papeete (French Polynesia)
6. Redu (Belgium)

For further details, refer to SYS-RD.74.

5.6.2.2 ULS SITES

ULS site constraints and boundaries are defined in document [AD.26] - ULS SIRD. As required in [AD.26], these sites shall not have any obstacle above 5 deg elevation, in order to allow proper transmission of the integrity messages to the satellites.

These sites host the following elements:

- 4 antennas in the FOC2 phase deployed by GMS (System Build 2.0)
- up to 2 associated MDDN element, called ULS-MNEs deployed by GMS
- GDDN equipment, including up to 2 VSAT terminals, shared with other Galileo elements provided by the Communication Service Provider

Note: The ULS sites also host GSS elements.

ULS sites are the following (further detailed in SYS-RD.74):

1. Svalbard (Norway)
2. Kourou (French Guyana)
3. Reunion (Reunion Island)
4. Noumea (New Caledonia)
5. Papeete (Fench Polynesia)

5.6.2.3 GSS SITES

GSS site constraints and boundaries are defined in document [AD.25] - GSS SIRD.

As required in [AD.25], the environment of the station is assumed to not exceed the specified level of interference and multipath. Furthermore, these sites shall not have any obstacle above 5deg elevation.

These sites host the following elements:

- one GSS element in IOV and up to 3 GSS in FOC phase deployed by GMS
- up to 2 associated MDDN element, called GSS-MNEs deployed by GMS
- GDDN equipment, including up to 3 VSAT terminals, shared with other Galileo elements provided by the Communication Service Provider

The final number of stations that will compose the GSS world wide network is constrained by a list of sites specified at system level; however the network dimensioning is under GMS network design responsibility.

GSS sites are the following (further detailed in SYS-RD.74):

1. Fucino (Italy)
2. Papeete (French Polynesia)
3. Noumea (New Caledonia)
4. Reunion (Reunion Island)
5. Kourou (French Guyana)
6. Troll (Antarctica)
7. Redu (Belgium)
8. Svalbard (Norway)
9. St.Pierre & Miquelon (France)
10. Falklands
11. Kerguelen (France)
12. Terre Adelie
13. Jan Mayen
14. Wallis (France) (deployment under negotiation)
15. Kiruna (Sweden)
16. Azores

5.6.3 NETWORKS

5.6.3.1 Network context overview

The Galileo Ground System Communication Network is illustrated in logical form in the following figures.

For details on network physical architecture, refer to SYS-RD.74.

This network is partitioned into several logical networks:

- **Communications subnetwork between GMS elements of the GCC and the ULS/GSS remote sites** (see Figure 81). This network supports the dissemination of GMS mission and M&C data. This subnetwork relies on the following physical elements:
  - GMS MDDN including the GMS MNEs, the GMS RT and nRT LAN, and GNMF
  - GDDN-WAN

- **Communications subnetwork between GCS elements of the GCC and the TTC remote sites** (see Figure 81). This network supports the dissemination of Satellite TM/TC and M&C data. This subnetwork relies on the following physical elements:
  - GCS SDDN including the GCS SNEs and GCS nRT LAN
  - GDDN-WAN

- **Communications subnetwork between GCS and GMS elements of the 2 GCC sites** (see Figure 82). This subnetwork relies on the following physical elements:
  - GCS SDDN including CNEs and GCS LAN
  - MDDN GMS LANs
  - GDDN-WAN

- **Communications subnetwork between GMS elements of the GCC and the GMS External Entities** (see Figure 83). This subnetwork relies on the following physical elements:
  - GMS M-EDDN including M-ENEs
  - GMS RT and nRT LAN
  - EDDN-WAN

- **Communications subnetwork between GCS elements of the GCC and the GCS External Entities** (see Figure 83). This subnetwork relies on the following physical elements:
  - GCS S-EDDN including S-ENEs
  - GCS nRT LAN
  - EDDN-WAN

The following figures sum up the network context with associated procurement policy:

- **Network context for communication between GCC sites and Remote sites (ULS, GSS, TTC):**

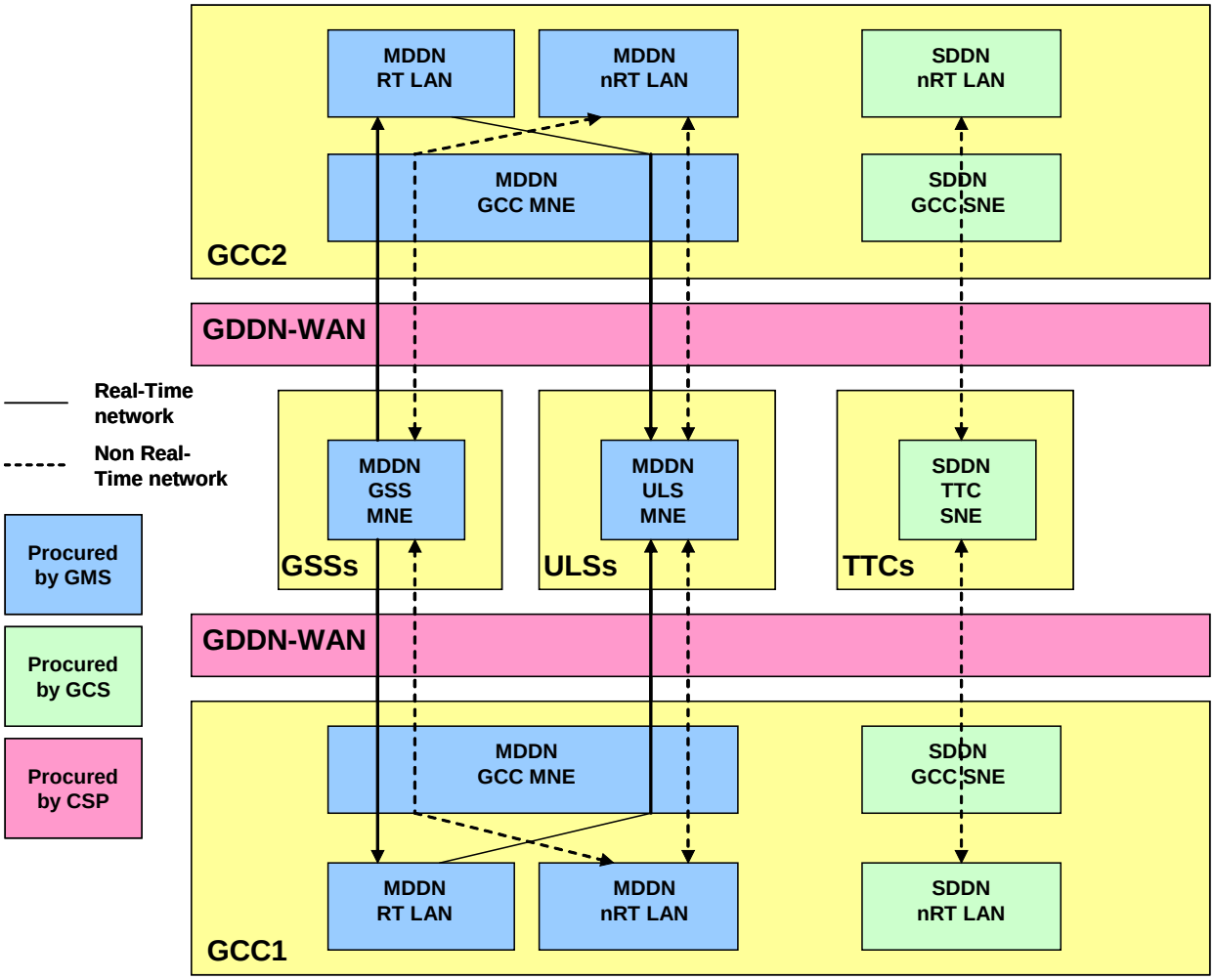


Figure 81: Network architecture between GCC site and remote sites

- **Network context for communication between 2 GCC sites:**

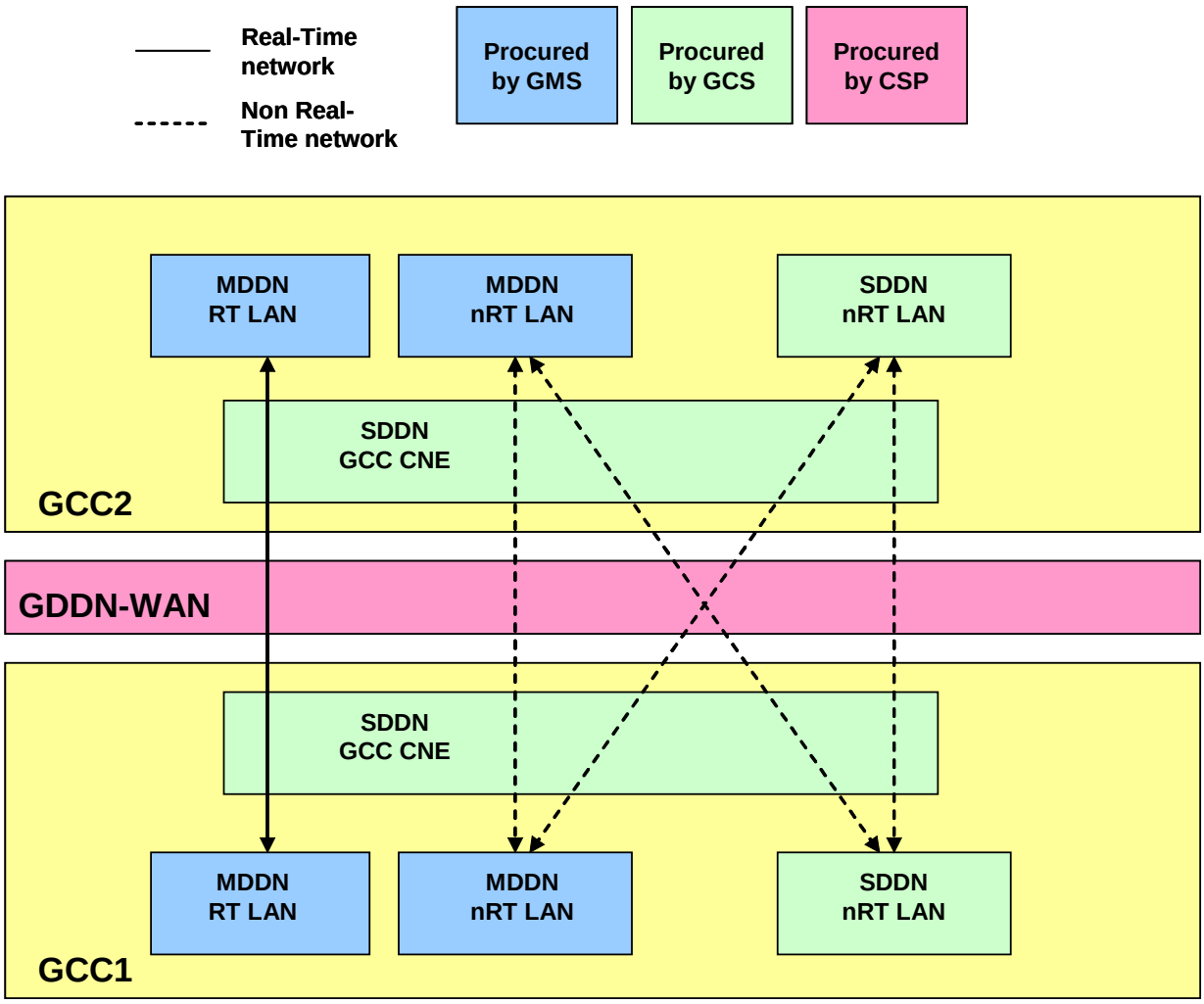


Figure 82: Network architecture between GCC sites

- **Network context for communication between GCC sites and External Entities:**

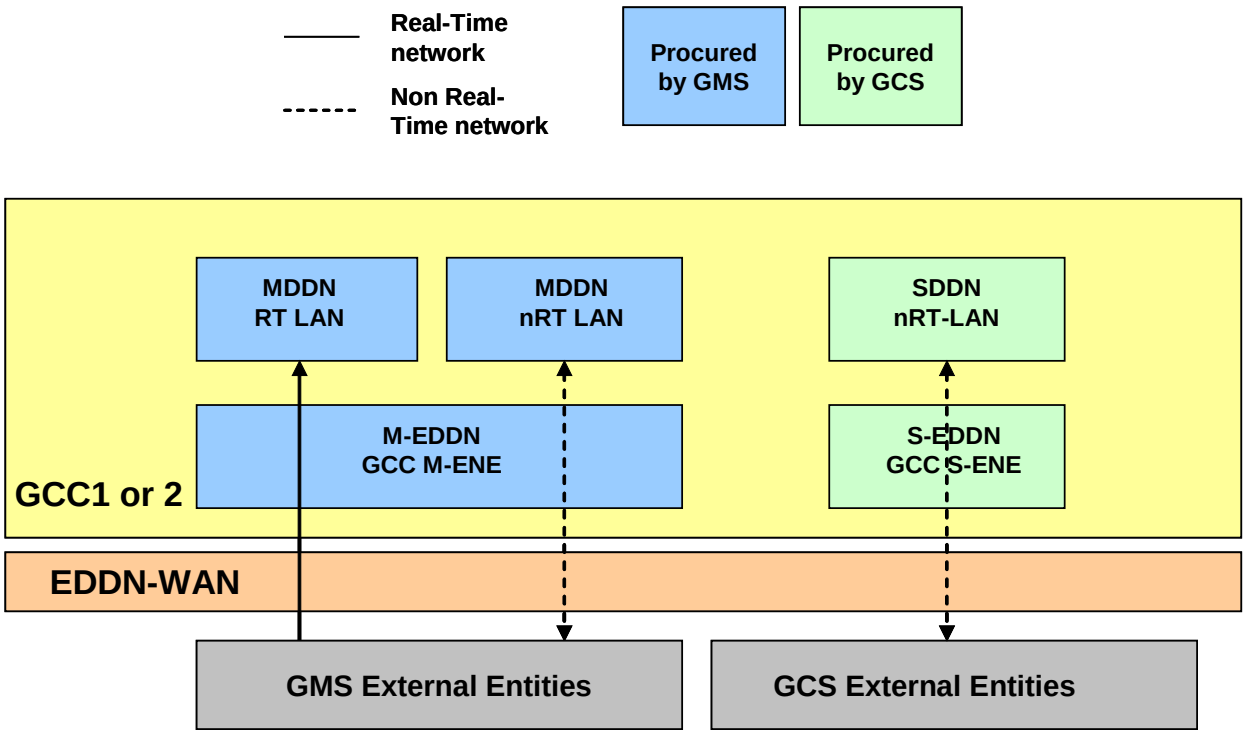


Figure 83: Network architecture between GCC sites and External Entities

5.6.3.2 GMS NETWORKS: MDDN AND M-EDDN

GMS procures and uses two Logical Networks to deliver its service; the Mission Data Dissemination Network (MDDN) and the Mission-External Data Dissemination Network (M-EDDN).

5.6.3.2.1 MDDN

The GMS-MDDN is responsible for providing the physical elements needed to:
- Link the GCC-GMS elements to all the GMS remote sites, namely ULSs and GSSs
- Perform network performance Monitoring and Control.

The MDDN is composed of the following elements:
- The MNEs (Mission Network Equipment) in each GMS element site (GCC, ULS and GSS). The tasks of the MNEs are:
  - o to allow access and perform the data routing from the GMS element LAN (within GCCs, ULSs and GSSs) to the GDDN WAN network (and vice versa). These data are mission data.
  - o to measure, via integrated traffic probes, the network performance according to the following parameters: data integrity, data latency and lines availability

    Three versions of the MNE are used by GMS in the MDDN: GSS-MNE, ULS-MNE, and GCC-MNE. There are two MNEs per element on site, configured in redundant mode
- The GNMF responsible for network management. The GNMF is located inside the GCC. The Performance Monitoring agents at the MNES and SNEs will collect performance data of all

traffic streams going across the GDDN and generate QoS stamps that will be used by the GNMF to derive end-to-end performance data
- The GMS nRT and RT LANs within the GCC and the GMS remote sites (GSSs and ULSs) see section 5.6.3.4.2 for detail architecture description.

MDDN Chains:
- **GSS RT Functional Chain: GSS Raw data**
GSS-MNE → GDDN-WAN → GCC-MNEs → GCC GMS RT LAN → OSPF

- ULS RT Functional Chain: GMS navigation data
MGF → GCC GMS RT LAN → GCC-MNEs → GDDN-WAN → ULS-MNE

- GSS nRT Functional Chain: M&C data
GSS-MNE → GDDN-WAN → GCC-MNEs → GCC GMS nRT LAN → MUCF, GACF, PKMF, MKMF, PTF, GNMF (and vice-versa)

- ULS nRT Functional Chain: M&C data
MUCF, GACF, PTF, GNMF → GCC GMS nRT LAN → GCC-MNEs → GDDN-WAN → ULS-MNE (and vice-versa)

For details refer to MDDN physical description section: see section 5.3.2.3

5.6.3.2.2 M-EDDN

The GMS-M-EDDN is responsible for ensuring the physical interface connection between the GMS elements within the GCC and the GMS external entities:
- 1 External Time Service Provider
- 1 Geodetic Reference Service Provider,
- 1 Galileo Service Centre
- 1 GALSEE
- 1 Search and Rescue - Return Link Service Provider
- 1 Galileo Security Monitoring Centre
- 1 GPS/Galileo Time Offset provider

The M-EDDN is composed of the following elements:
- the M-ENEs (MDDN-External Network Equipment), located in the GCC, connecting GMS elements in the GCC (via the GMS-LAN) to the WAN (External Data Dissemination Network (EDDN)). Since the M-ENE is used to connect to an External Entity, security plays an important role in the functionality of this element. GMS is then responsible for defining the VPN equipment (security level and configuration) to be deployed at the External Entity locations. The External Entity is responsible for procuring the VPN equipment. There are two types of M-ENE installed at the GCC; the Real-Time M-ENE, and the non-Real-Time M-ENE.

5.6.3.3 GCS NETWORKS: SDDN AND S-EDDN

GCS procures and uses two Logical Networks to deliver its service; the Satellite Data Distribution Network (SDDN) and the Satellite-External Data Distribution Network (S-EDDN).

As part of SDDN are also procured the GCC CNEs responsible for communication between two GCC sites.

5.6.3.3.1 SDDN

5.6.3.3.1.1 GCC to TTCF chain

The GCS-SDDN is responsible for providing the physical elements needed to link the GCC-GCS elements to all the GCS remote sites, namely TTCFs

The SDDN is composed of the following elements:

- The SNE (Satellite Network Equipment) in each GCS element site (GCC and TTCFs). The tasks of the SNEs are to allow access and perform the non real time data routing from the GCS element LAN (within GCCs and TTCFs) to the GDDN WAN network (and vice versa). Two versions of the SNE used by GCS in the SDDN: TTC-SNE and GCC-SNE. The SNEs are based on the same architecture as the GMS MNEs (they are produced by the same manufacturer).
- The GCS nRT LAN within the GCC and the GCS remote sites (TTCFs): see section 5.6.3.4.1 for detail architecture description.

SDDN Chains:
- TTCF nRT Functional Chain: TM/TC data
TTC-SNE → GDDN-WAN → GCC-SNEs → GCC GCS LAN → GCS Elements

For details refer to SDDN physical description section: see section 5.2.2.8.

5.6.3.3.1.2 GCC SDDN-CNE

As part of SDDN, the GCS is responsible for procuring the Common Network Equipment (CNEs). These equipments are responsible for the connectivity between two different GCCs.

Although procured by GCS, the inter-GCC link is monitored by the GMS GNMF.

For details, see SDDN physical description section.

5.6.3.3.2 S-EDDN

The GCS-S-EDDN is responsible for ensuring the physical interface connection between the GCS elements within the GCC and the GCS external entities:

- the satellite payload & platform manufacturers
- the external satellite control centres.

The S-EDDN is composed of the following elements:

- S-ENEs (Satellite External Network Equipment) located in the GCC, connecting GCS elements in the GCC (via the GCS-LANs) to the WAN (External Data Dissemination Network (EDDN)). Since the S-ENE is used to connect to an External Entity, security plays an important role in the functionality of this element. GCS is then responsible for defining the

VPN equipment (security level and configuration) to be deployed at the External Entity locations. The External Entity is responsible for procuring the VPN equipment.

5.6.3.4 GCC LAN

The GCC LANs can be broken down into several parts: GMS procured (for intra-GMS data exchange) as part of MDDN, GCS procured (for intra-GCS data exchange) as part of SDDN, and GCC Hosting Service Provider procured Admin LAN and Ops Support LAN, based on ESA-provided functional requirements .

- GCS, GMS LANs: The GCS & GMS LANS active elements are provided by their respective segments at the GCC where they are deployed in IOV. In FOC the two ground segments will both install their respective LAN infrastructures at each GCC.
- The GCC Admin LAN is provided by the hosting entity at each GCC. This is not used for any GCS or GMS operational systems, but provides such services as Internet access, and desktop tools such as word processing, spreadsheets, etc.
- The Ops Support LAN connects systems such as the Video Data Distribution System and the Voice Distribution System.
- Both the GCS and GMS LANs are split into a number of network domains, implemented on different switches.

5.6.3.4.1 GCS LAN Architecture

The GCS LAN must have high availability and therefore will be based on a two layer architecture. The design will be based upon a core/distribution layer and an access layer, as shown the following figure:

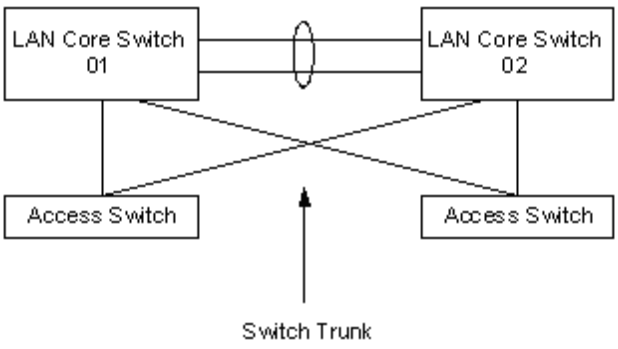


Figure 84: GCS GCC LAN Switch Architecture

The GCS domains or VLANs then connect to the Access Switches. Examples of such connectivity are given in Figure 85, which shows how the SDDN interfaces to the GCS GCCLAN, and Figure 86, which shows how the OPF's on the Operational LAN and the Validation LAN connect to the GCS GCC LAN

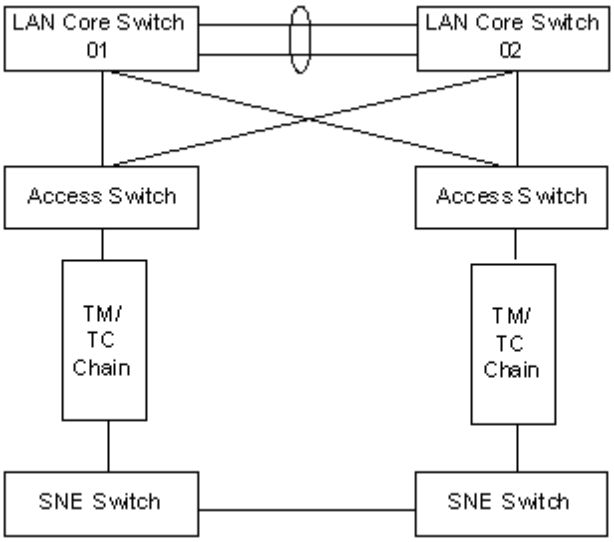


Figure 85: SDDN - GCS GCC LAN connectivity.

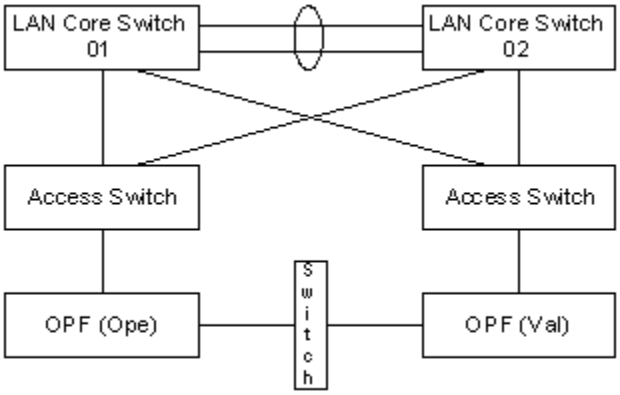


Figure 86: GCS GCC LAN connectivity.

5.6.3.4.2 GMS LAN Architecture

The GMS LAN architecture is similar in structure to the GCS architecture, with the design based upon a core/distribution layer and an access layer, as shown in Figure 84, above.

IN GMS the operational elements are connected via either the Real Time (RT) or non-Real Time (nRT) LANs, which are connected to separate Access Switches. There is no connection between the RT and nRT LANs..

In addition, GMS contains a Maintenance & Training Platform Facility (MTPF), which is a copy of the GMS elements, including the LANs, used for training purposes and preparing for, and testing, hardware or software changes to the GMS environment.

5.6.3.5 GDDN-WAN

5.6.3.5.1 High Level Description

The CSP (Communication Service Provider) is the entity that provides GDDN connectivity services to Galileo system. In order to provide this connectivity with the agreed level of service, the Galileo system (GMS) provides feedback with information on the GDDN link and performance status periodically to CSP.

Data communications between GMS and CSP are ensured by a Wide Area (the EDDN) Network connecting the CSP with both GALILEO GCCs. All data flow types between CSP and GMS determine the implementation of the interface as non time-critical. These flows are interconnected with the GMS through a VPN infrastructure.

5.6.3.5.2 Functions Allocation

The following table shows the functions identified in the system DDF1 (SYS-RD.01) mapped to CSP:

Function ID	Function name
CSPF1	Provide GDDN Connectivity

Table 70: CSP Function List

5.6.3.5.3 Entity Organical Context

The following figure shows the CSP context:

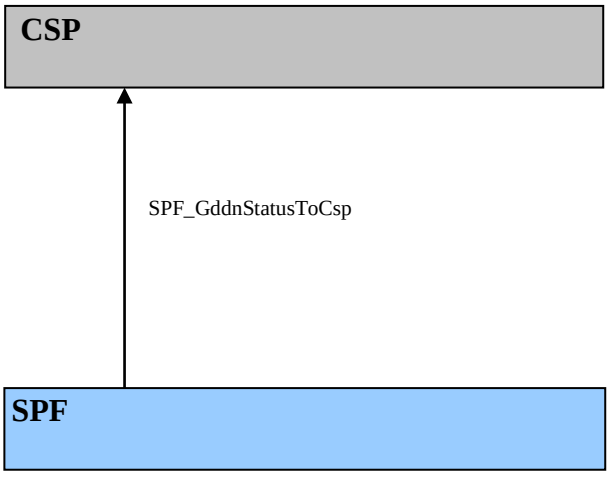


Figure 87: CSP Organical context

5.6.3.5.4 Entity Main Design Drivers

The CSP is responsible for providing connectivity service up to the interfaces with the MNE/SNE, which are under responsibility of GMS/GCS. The CSP can only obtain performance information from its own equipment. Figure 2 shows, schematically, the areas of responsibility and visibility, using GMS connectivity, as an example.

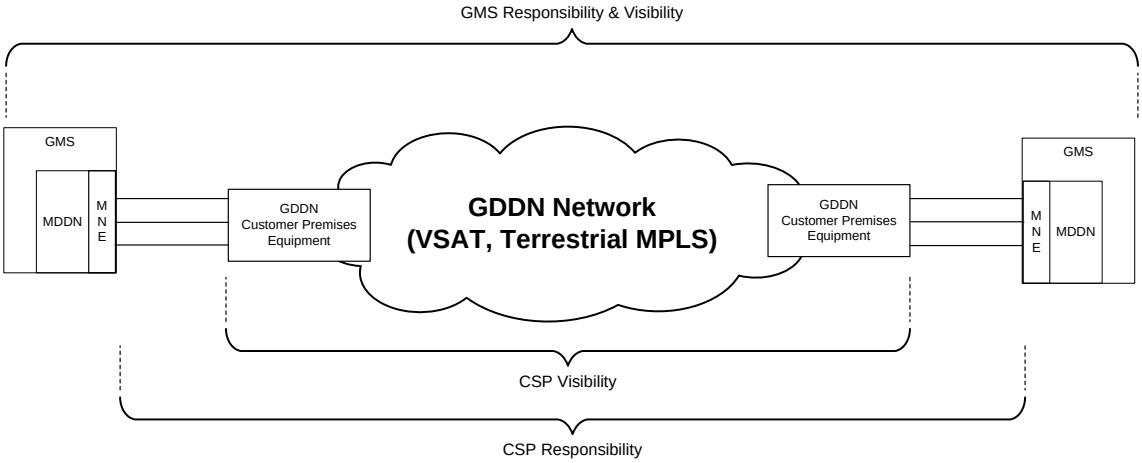


Figure 88: CSP-GMS Interface

The traffic within GDDN and in the lines connecting the GDDN CPEs to the MNEs or SNEs are monitored by the MNEs and SNEs and forwarded to the Galileo Network Management Facility (GNMF). This facility is then responsible for monitoring the performance of the MNEs and the SNEs, and over the GDDN. At regular intervals the GNMF forwards this performance information for each line connected to an NME or SNE, to the CSP (via the SPF and the EDDN).

The main drivers for the design of the GDDN are the performance requirements of GMS, and to a lesser extent, GCS. In addition, as per [AD.48], full resilience to link failure is expected in the GDDN architecture for the FOC configuration.

To meet the requirements, the CSP solution is based on various COTS technologies (C-Band VSAT, high-performance terrestrial point-to-point lines, and terrestrial MPLS). Different combinations of these technologies are deployed, depending on ground station type and location, and the associated performance requirements.

Basically, the solutions are:

- For combined sites (GMS + GCS), with the most stringent performance requirements being driven by GMS, ground stations in the central Pacific Ocean without visibility to AOR or IOR VSATs, use Pacific Ocean Region (POR) use a resilient direct/indirect VSAT combination: VSATs to connect to VSAT hubs located on the US West Coast or Asian Pacific coast, then redundant, high-performance terrestrial circuits ("backhaul" circuits) from the hubs to/from the GCCs.
- For combined sites (GMS + GCS), with the most stringent performance requirements being driven by GMS, ground stations with visibility to Atlantic Ocean Region (AOR) or

Indian Ocean Region (IOR) VSATs, use VSAT point-to-point, ground station to/from GCC.

- TT&C-only site (less stringent performance requirements) in Kiruna foresees in FOC a dual VSAT, as per Kourou (dual VSAT will exist) or will use a combination of VSAT plus the MPLS connecting to GCC-D (as per IOV),
- TT&C part of Kourou, in FOC foresees a connection vi dual VSAT TTC (same links as GSS/ULS), replacing the single MPLS connection to GCC-D.
- For GMS-only (most stringent performance requirements) site located in Svalbard (with poor visibility to AOR/IOR VSATs), use resilient direct/indirect VSAT combination.

5.6.3.5.5 Physical Implementation

The GDDN is a managed service, provided by the Communication Service Provider (CSP), who is responsible for

- the design of the GDDN, based on Galileo performance requirements,
- deployment of the GDDN components in the ground stations and GCCs,
- operation and management and GDDN, to meet the performance requirements, as defined in the Service Level Agreement,
- service level assessment, in co-operation with GMS, who are responsible for measuring the mission-level performance, i.e., performance measured between MNEs, over the GDDN.

This architecture is provided in the figure below. For more details please refer to SYS-RD.88:

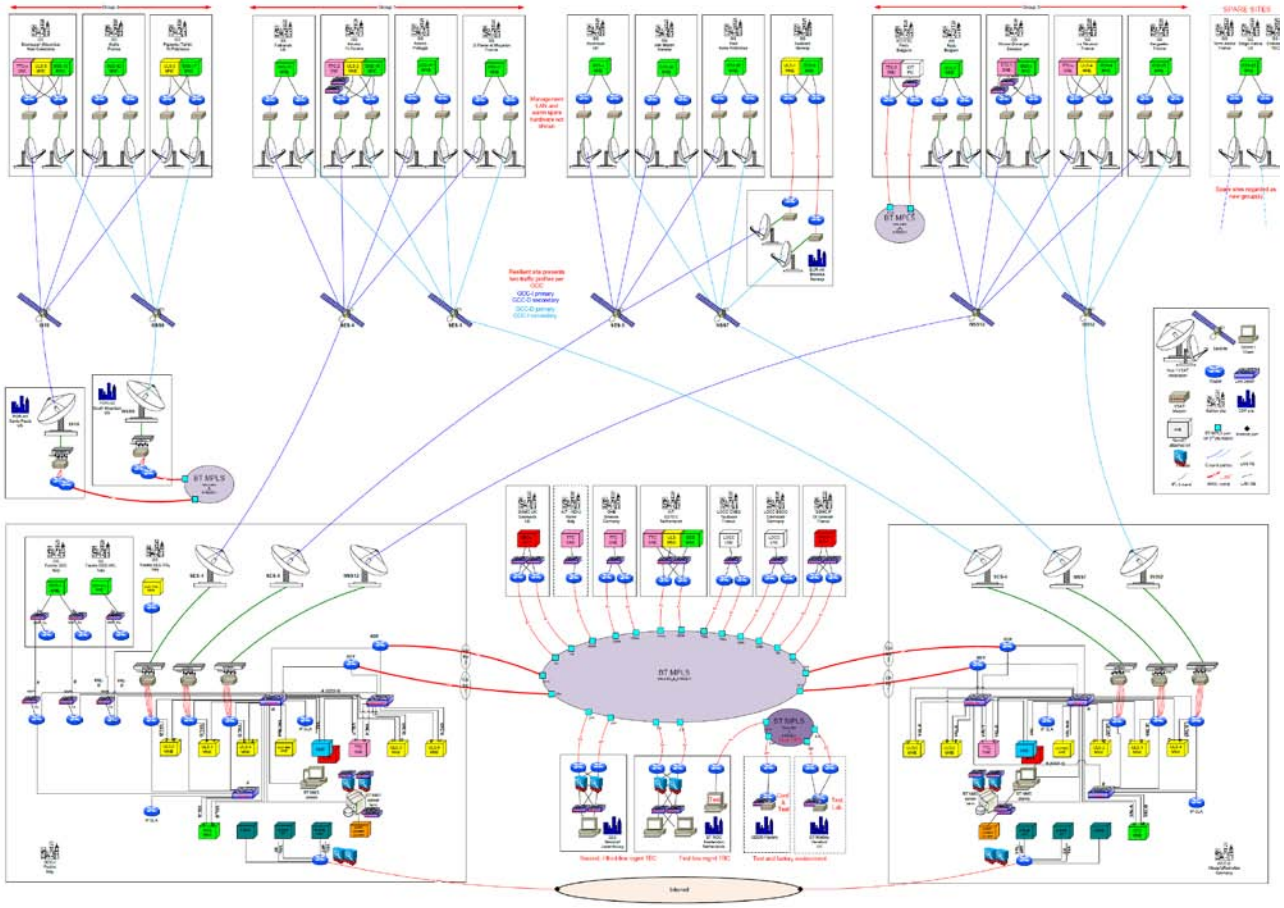


Figure 89 : GDDN FOC1 high-level design

Redundant GSS and ULS links (2 diverse-routed links between GCC-I and GCC-D and all remote GSS and ULS sites) are present, capable of achieving an availability of 99.9%.

STRB#2.0 change: GAL-DCN-ASTG-SYST-X-13820 v1 is increasing the GDDN specified availability from 99.6 to 99.9 to take advantage of the GDDN margin at system level.

The following aspects related to links resilience are highlighted:

GCC – TTC links

- Kourou will be connected via dual VSAT TTC (same links as GSS/ULS). The design notes below relating to ULS will apply.
- The two separate TTCF streams will be kept apart end-to-end by separated GDDN infrastructure.
- Kiruna will be connected via either a dual VSAT, as per Kourou (dual VSAT will exist) or will use a combination of VSAT and MPLS.
- The Redu TTCF site is handled as a terrestrial connection using dual MPLS coupled with a GSS-only site from the VSAT perspective.
- Reunion will operate over dual VSAT, as per Kourou.

- Noumea will operate over dual VSAT to two Pacific hubs, separately linked into MPLS to complete the connection back to the GCCs.

Note that where GSS traffic is supported in combination with a TTCF, resilience is provided via VSAT.

GCC - ULS links:

1. Sites located in the Atlantic/Indian Ocean regions (Kourou, Svalbard, Reunion) use a resilient direct/indirect VSAT combination:
   - Each remote site will have two VSAT terminals aligned to separate satellites.
   - Each satellite is hubbed at just one GCC –each remote site connect to one hub at each GCC, ensuring good separation of European hubs.
   - Each VSAT link will carry a stream of traffic destined for the GCC supporting the link, along with a second stream of traffic destined for the other GCC. This latter stream will be onward-routed to its destination via inter-GCC terrestrial MPLS (separate from the Inter GCC link). Each GCC will then receive (from each remote site):
     i. a primary stream of traffic directly;
     ii. a secondary stream of traffic (backup), received via MPLS through the remote GCC.
   - Each ULS is handled by a unique pair of satellites (POR excepted) .
   - The two ULS streams will be kept apart end-to-end by separated GDDN infrastructure.
   - Both streams for each GCC should operate in parallel.
   - GSS and ULS will share the same paths.
2. Sites located in the Pacific Ocean region (Noumea, Papeete) use a resilient dual-indirect combination. This is unavoidable due to the location of the remote sites out of sight of any GCC-visible satellites (the term “dual-indirect” is defined as two lines, each indirect, with each VSAT path not delivering the data directly into/out of the final destination but instead into/out of a hub at some intermediate location (here, the Pacific hubs). Redundantly routed terrestrial services complete the paths between the GCCs and the hubs.
   - Each remote site will have two VSAT terminals aligned to separate satellites.
   - Each satellite is hubbed at a dedicated Pacific hub.
   - Each VSAT link will carry two streams of traffic, one destined for each GCC. All streams will be onward routed to GCC destinations via terrestrial MPLS. So each GCC will receive two streams of traffic indirectly via MPLS.
   - Both streams for each GCC must operate in parallel.
   - GSS and ULS will share the same paths.

GCC - GSS links:

GSS stations co-located at TTC or ULS sites are, as stated above, integrated into the TTC and/or ULS paths, so they benefit from the same features as described above. A similar design model to GSS-only sites has been implemented, as per ULS and TTC.

The sites are allocated to satellites/hubs in such a way that each set of non- GCC infrastructure (which, as stated, supports just one ULS station, or two in POR) supports no more than four GSS sites in total.

Inter GCCs link:

Inter-GCC connectivity is achieved using two separate terrestrial MPLS accesses per GCC.

These operate in a primary-and-backup arrangement for resilience

GCC - GSMC link

The GSMCs will also be connected using dual, separated MPLS in a similar way to the GCCs.

5.6.3.6 EDDN-WAN

The EDDN is built on a combination of COTS WAN platforms, the platform for each connection matched to the situation and requirements in order to deliver compliant service. The platforms are:

- Terrestrial Internet connectivity
  - For always-available service, three EDDN Fast Ethernet presentations in each GCC, provide access to a 2Mb Internet connection in each case, complete with a small assignment of public IP addresses,

- Basic Rate ISDN
  - For on-demand service, 2 x ISDN BRI 128k at Oberpfaffenhofen only (since this connectivity is an extension of S-EDDN).

The WAN platforms are specified with COTS IP routers, the infrastructure being managed from BT's COTS Network Management System (BT NMS).

6. SYSTEM FUNCTIONAL CHAINS PHYSICAL DESCRIPTION

The objective of this section is:

- Physical mapping of the SFC functional description included in SYSBSLDEF part 1 [SYS-RD.01]: to describe the physical architecture related to each SFC
- Show consistency between design and OPS, RAM: identifying physical scenarios for each SFC including a description of the dynamic data flow exchanges. Within these scenarios described in sequential steps, a link is included:
  - To OPS scenarios (MOCD) in case of OPS involvement.
  - To RAMS FMECA in case of contingency case at some steps.

The section is organized per SFC. The structure of each SFC physical description section is organized as follow:

- SFC overall physical description including a physical static diagram showing the physical elements involved in the SFC and the exchanges between them. This physical static diagram is the physical realization of the functional static block diagram of the SFC described in SYSBSLDEF part 1 [SYS-RD.01].
- Detailed physical description of the processes within the SFC, including a detail role of each physical element involved.
- Physical Dynamic Scenarios related to the SFC

The list of the SFCs is recalled in the table below:

SFC ID	System Functional Chain
SFC1	**Provide and Maintain the Galileo Constellation**
SFC1.1	Provide Nominal Constellation
SFC1.2	Maintain Nominal Constellation
SFC2	**Generate System Time and Geodetic Reference**
SFC2.1	Generate Galileo System Time
SFC2.2	Provide GPS-Galileo Time-Offset
SFC2.3	Contribute to Geodetic Reference Determination
SFC3	**Maintain Overall System Timing Synchronization**
SFC3.1	Satellite Times Synchronization
SFC3.1.1	Payload Frequency Reference Steering
SFC3.1.2	Satellite LGST Steering
SFC3.1.3	Satellite Clocks Monitoring and Switching
SFC3.1.4	Platform Timing Parameters Provision
SFC3.2	GMS Times Synchronization
SFC3.3	GCS Times Synchronization
SFC4	**Provide Mission Services (OS, PRS)**
SFC4.1	Open Services (OS) Provision
SFC4.1.1	OS Positioning Service Provision
SFC4.1.2	OS Timing Service Provision
SFC4.1.3	OS Signal Quality Monitoring Provision
SFC4.1.3.1	OS Signal-In-Space Accuracy (SISA) Provision
SFC4.1.3.2	Health Status Parameters Provision (on OS Signals)
SFC4.1.4	OS Degraded Navigation Data Provision
SFC4.3	Public Regulated Services (PRS) Provision
SFC4.3.1	PRS Positioning Service Provision
SFC4.3.2	PRS Timing Service Provision
SFC4.3.3	PRS Signal Quality Monitoring Provision

SFC4.3.3.1	PRS Signal-In-Space (SISA) Provision
SFC4.3.3.2	Health Status Parameters Provision (on PRS Signals)
SFC4.3.4	PRS Degraded Navigation Data Provision
SFC4.3.6	PRS Security Service Provision (Uncla)
SFC5	**Monitoring and Control**
SFC5.1	Ground Asset Monitoring and Control
SFC5.1.1	Ground Elements Monitoring and Control
SFC5.1.2	Ground Network Management
SFC5.2	Satellite Monitoring and Control
SFC5.2.1	On Line Satellite Monitoring and Control (incl. uncla security req.)
SFC5.2.2	Off Line Satellite Monitoring and Control
SFC5.3	Mission Performance Monitoring
SFC5.3.1	Mission Performance On Line Monitoring
SFC5.3.2	Mission Performance Off Line Monitoring
SFC5.4	Mission Planning Provision
SFC6	**Archiving**
SFC6.1	Archive Migration
SFC6.2	Archive Retrieval
SFC7	**Support Commercial Service**
SFC7.1	CS Data Dissemination
SFC7.2	Provide CS Ranging Code
SFC7.3	Provide Information to GOC Service Centre, GSMC, and GALSEE
SFC8	**Provide SAR Service**
SFC8.1	Provide SAR Forward Link Service
SFC8.2	Provide SAR Return Link Service
SFC8.2.1	SAR Return Link Resource Management
SFC8.2.2	SAR Return Link Message Dissemination
SFC8.2.3	SAR Return Link Interface to Cospas/Sarsat
SFC9	**Support EDBS Service**
SFC9.1	EDBS Management
SFC9.2	EDBS Dissemination
SFC9.2.1	EDBS Uplink
SFC9.2.2	ERIS Direct Uplink Inhibition
SFC10	**Deleted**
SEC-SFC11 (Cla)	**Control Access to the Ground Infrastructures**
SEC-SFC11.1	Security protection of the Galileo networks
SEC-SFC11.1.1	Protection of Intersite Internal Connection (Real-time and Non-Real-time)
SEC-SFC11.1.2	Protection of External Real Time Connection
SEC-SFC11.1.3	Protection of External Non Real Time Connection
SEC-SFC11.1.4	Network Security Management
SEC-SFC11.1.5	Network Security Device Key Management
SEC-SFC11.1.6	Protection of Inter GCC Connection
SEC-SFC11.1.7	Network Security Device M&C Dataflow Protection and Communication
SEC-SFC11.1.8	Intra-Site Network Protection
SEC-SFC11.2	Security protection of ground infrastructure
SEC-SFC11.2.1	Site Alarming and Physical Intrusion Detection System
SEC-SFC11.2.2	Physical Access Control
SEC-SFC11.3	Access control
SEC-SFC11.3.1	User Access control
SEC-SFC11.3.2	Asset Access control
SEC-SFC11.4	Information dataflow protection

SEC-SFC11.4.1	Anti-malware management
SEC-SFC12 (Cla)	**Security Protection Of Satellite Monitoring And Control**
SEC-SFC12.1	TT&C Data Flow Protection
SEC-SFC12.3	Satellite Key Management
SEC-SFC13 (Cla)	**Security Protection For Mission Data**
SEC-SFC13.1	C-Band Data Flow Protection
SEC-SFC13.3	Key management
SEC-SFC14 (Cla)	**Security Protection Of SoL**
SEC-SFC14.1	SoL Data Flow Protection
SEC-SFC14.2	Remote SM Management
SEC-SFC14.3	Key Management for SoL
SEC-SFC15 (Cla)	**Security Protection Of PRS**
SEC-SFC15.1	PRS Data Flow Protection
SFC15.1.1	NAVSEC PRS Protection
SFC15.1.2	COMSEC PRS MSG Protection
SFC15.1.3	COMSEC PRS OTAR Protection
SEC-SFC15.3	Key Management for PRS
SEC-SFC15.4	GSF Security Data Protection
SEC-SFC15.5	PRS Security Redundancy
SEC-SFC16 (Cla)	**PRS Access Management**
SEC-SFC16.1	PRS Users Management
SEC-SFC16.2	PRS Receivers Management (including secondary channel)
SEC-SFC16.3	PRS Orders Management (including secondary channel)
SEC SFC16.4	UNCLA PRS data Management through Secondary Channel
SEC-SFC17 (Cla)	**System and PRS Service Monitoring**
SEC SFC 17.1	PRS Service Performance Monitoring
SEC-SFC18 (Cla)	**Galileo SECURITY Monitoring and Management**
SEC-SFC18.1	Galileo Security Operational Intelligence Facility Monitoring and Management
SEC-SFC 18.1.1	Critical Assets Auditing and Monitoring
SEC-SFC 18.1.2	Cyber Security Processing
SEC-SFC 18.1.3	Cyber Security Monitoring, Control & Reporting
SEC-SFC 18.1.4	Forensics - Storage & analysis (offline)
SEC-SFC18.2	Security Events Monitoring and Management
SEC-SFC 18.2.1	UNCLA Security Events Centralization
SFC 5.1.1	Ground elements monitoring and control
SFC 5.1.2	Ground Network Management
SFC 5.2.1	On Line Satellite Monitoring and Control
SFC 6.3	Event Log collection and forwarding to GSMC
SEC-SFC 18.2.2	CLA Security Events Centralization
SEC-SFC 18.2.2.1	Satellite Monitoring and Control Remote SU Management
SEC-SFC 18.2.2.2	Mission Data Remote SU Management
SEC-SFC 18.2.2.3	PRS Data Remote SU Management
SEC-SFC 18.2.2.4	CLA Sec Event PKMF Centralization
SEC-SFC	UNCLA and CLA Sec Events Distribution to GSMC

18.2.3	
SEC-SFC 18.2.4	Security Events Analysis and Management in GSMC (T&V Tool)
SEC-SFC18.3	Jamming Detection and Management

Table 71: SFC List

Note: the classified SFCs are detailed in DDF2 classified annex [SYS-RD.8].

6.1 SFC1 PROVIDE AND MAINTAIN THE GALILEO CONSTELLATION

6.1.1 Introduction

This SFC is responsible for the establishment of the Galileo satellite constellation and for the maintenance of this constellation over the programme lifetime. The functionality provided via this SFC can be broken down into the following sub-SFCs:

- SFC1.1 Provide Nominal Constellation: this SFC provides all the functionality to launch and commission each of the Galileo satellites into the required position in the constellation
- SFC1.2 Maintain Nominal Constellation: this SFC provides all the functionality to maintain the nominal Galileo constellation

The following sections describe the physical implementation of the SFC. The information provided here is based on the following documentation:

- System documents:
  - System Logical Architecture (SYS-RD.1)
  - System operational concept: MOCD (SYS-RD.6)
  - Operations concept for PL IOT and SC commissioning using GCS (SYS-RD.29)
  - LEOP Services DDVP (SYS-RD.30)
  - IOT DDJF (SYS-RD.38)
- GCS documents:
  - GCS DDF (GCS-RD.1)
  - TTCF DDF (GCS-RD.3)
  - SCCF DDF (GCS-RD.9)
  - SCCF ERD (GCS-RD.11)
  - SCCF to TTCF ICD (GCS-RD.21)
- SSeg documents:
  - SSeg DDJF (IOVSS-RD.1)
  - Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1)
  - Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2)
  - Avionics DDJF (IOVSS-RD.4 and FOCSS-RD.07)
  - TT&C DDJF (IOVSS-RD.5)
  - SSeg IOV PUS Definition (IOVSS-RD.7)
  - Avionics Subsystem Requirements Specification (IOVSS-RD.9)

6.1.2 SFC1.1 Provide Nominal Constellation

The following physical entities are involved in this SFC:

- Launch site
- External Network Control Centre (ENOC)
- LEOP-TT&C stations
- Ground Control Segment - Telemetry, Tracking and Control Facility(GCS-TTCF)
- Ground Control Segment - Operation Preparation Facility (GCS-OPF)
- Ground Control Segment - Spacecraft and Constellation Control Facility (GCS-SCCF)
- Ground Control Segment - Flight Dynamic Facility (GCS-FDF)
- Ground Control Segment - Spacecraft and Constellation Planning Facility (GCS-SCPF)
- IOT site
- GMS (for Secured IOT)

In the context of SFC1.1, the GCS facilities are involved in the same way as SFC5.2 (Satellite M&C): the GCS-related sections of the SFC1.1 physical description will therefore simply point to the relevant sections of SFC5.2.

6.1.2.1 Physical Static View

An overview of the physical implementation of SFC1.1 is provided in the series of figures below:

- the physical context of the Launcher Services.
- the physical context of the LEOP Services.
- the detailed network architecture of the LEOP Ground Segment.
- the physical context of the IOT System.
- the physical context of the GCS-IOT interface.

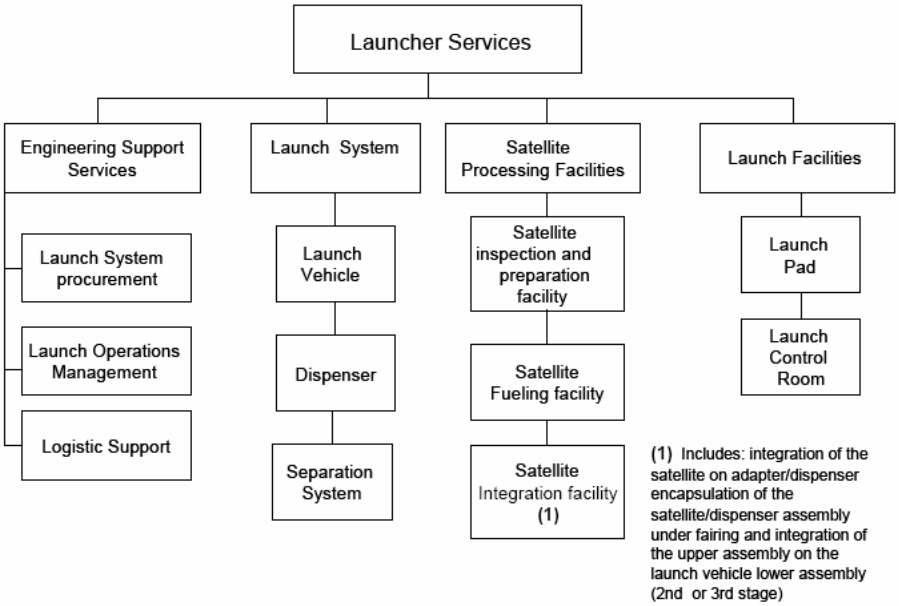


Figure 90 : Launcher Services physical static view

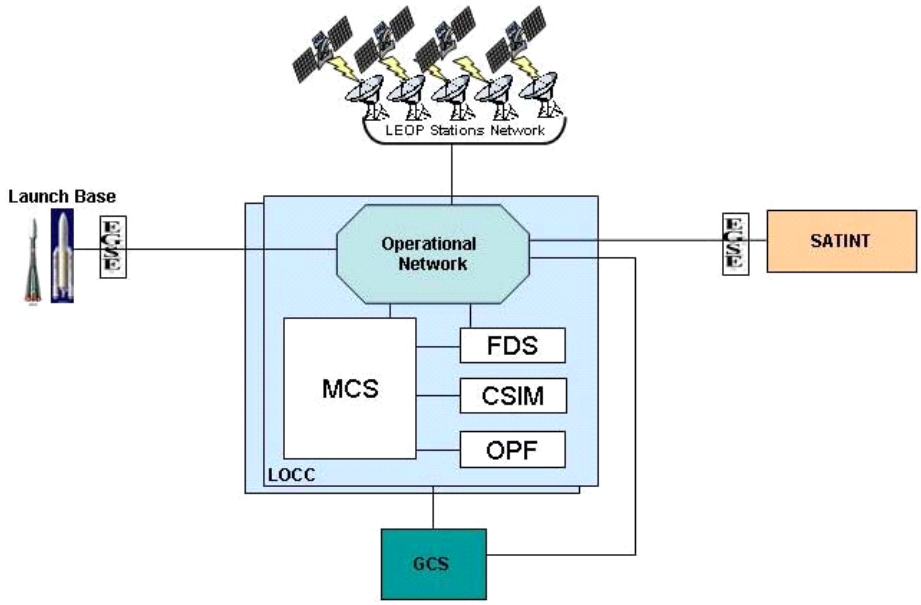


Figure 91 : LEOP Services physical context

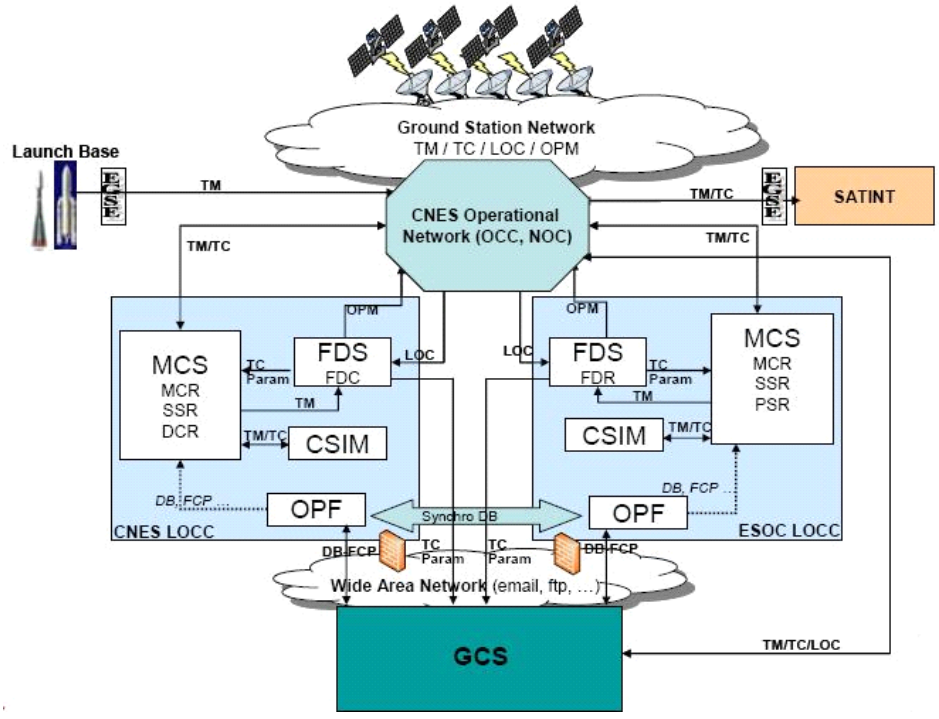


Figure 92 : LEOP Ground Segment Network Architecture

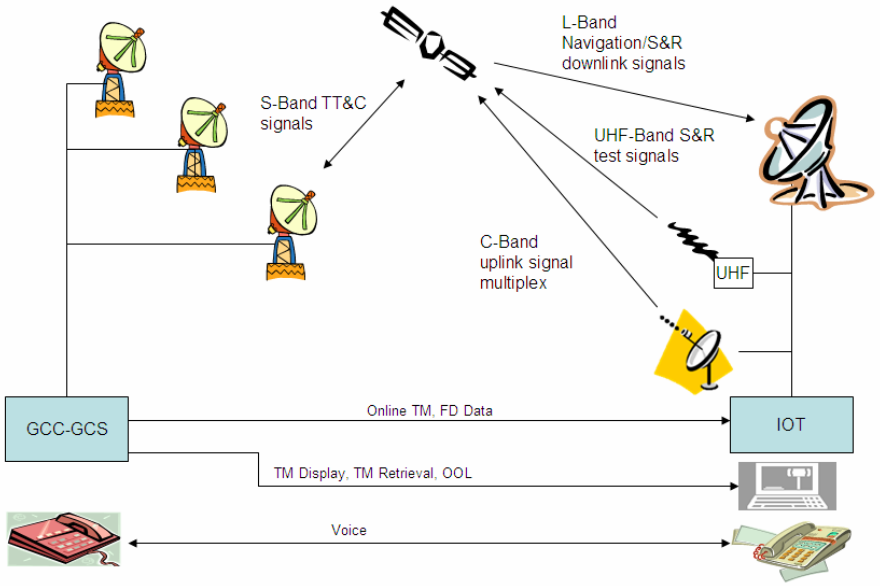


Figure 93 : IOT System physical context

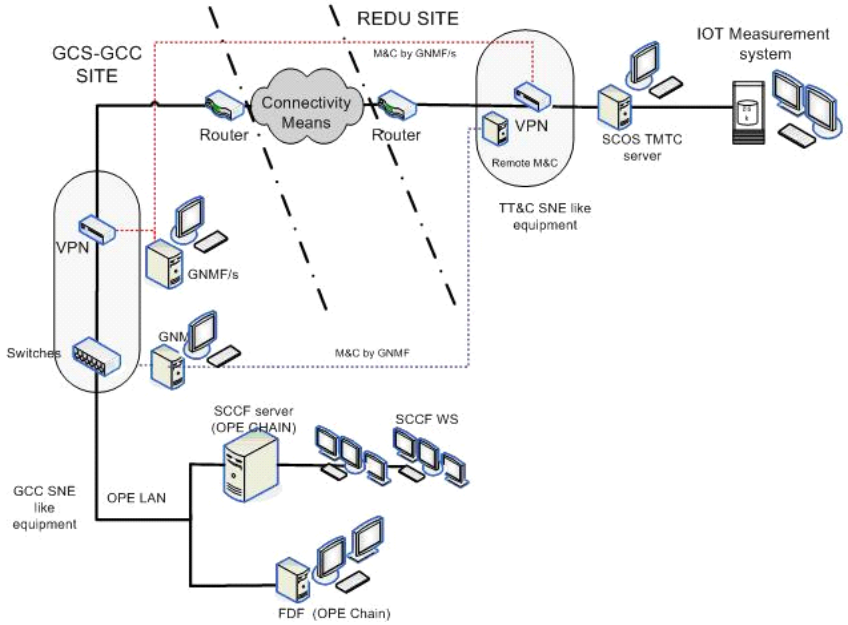


Figure 94 : Physical description of the GCS-IOT interface.

6.1.2.1.1 GCS Role Description

6.1.2.1.1.1 Executable schedule generation: SCCF

See corresponding section of SFC5.2

6.1.2.1.1.2 Satellite TC / TM, STC / STM and Health Status Processing: SCCF

See corresponding section of SFC5.2

6.1.2.1.1.3 Operations Progress Monitoring: SCCF

See corresponding section of SFC5.2

6.1.2.1.1.4 Constellation status: SCCF

See corresponding section of SFC5.2

6.1.2.1.1.5 Ground S-Band TT&C Uplink and Downlink: TTCF

See corresponding section of SFC5.2

6.1.2.1.1.6 Satellite S-band Uplink and Downlink: TTC

See corresponding section of SFC5.2

6.1.2.1.1.7 Ground COMSEC TC/TM: S-KMF

See corresponding section of SFC5.2

6.1.2.1.1.8 Satellite COMSEC TC/TM: PFSU

See corresponding section of SFC5.2

6.1.2.1.1.9 On-board Automation: ICDU and Avionics Software

See corresponding section of SFC5.2

6.1.2.1.1.10 On-board Observability: ICDU and Avionics Software

See corresponding section of SFC5.2

6.1.2.1.1.11 Satellite Health Status Summary: ICDU and Avionics Software

See corresponding section of SFC5.2

6.1.2.1.1.12 Satellite Health Status Summary: NSGU

See corresponding section of SFC5.2

6.1.2.1.1.13 Providing PRS Message Data to MUCF: GSS

See corresponding section of SFC5.2

6.1.2.1.1.14 Extraction of Satellite TM for Delivery to GCS: MUCF

See corresponding section of SFC5.2

6.1.2.1.2 Launcher Services facilities

Note: no information is available for Launcher Service as no dedicated DDJF has been provided as reference document.

6.1.2.1.3 LEOP Services facilities

The LEOP services will be performed using 2 LEOP Operation Control Centres (LOCC) within CNES and ESOC premises, linked to several external entities.
These two geographically separate LOCC's will be implemented and validated during the IOV LEOP Service Provision.

This is necessary as a preparation for the deployment phase where both the schedule density of the launches and the potential of launches of 6 spacecrafts on one launcher would exceed the capacity of a single LOCC.
Each of the LOCC's will require interfaces to other facilities within the overall Galileo ground segment.

The LOCC's will be located at CNES (Toulouse) and ESOC (Darmstadt).
In both cases the LOCC will be implemented by making extensive use of the existing institutional infrastructure at the two sites. For certain key elements identical facilities (SCCF, OPF, and CSIM) will be deployed on both sites to ensure transferability of operational procedures and knowledge, as well as subsystem validation status, between the two centres.
In the IOV phase each LOCC will support one of the two planned dual spacecraft launches.
Current planning allocates the first IOV launch to be performed from the CNES LOCC, the second from the ESOC LOCC.
The proposed architecture allows ESOC LOCC to be a cold back-up of CNES one for the first IOV launch, and CNES LOCC to be a cold back-up of ESOC one for the second IOV launch.

The Ground Station Network (GSN) composed of five TT&C stations will receive telemetry data from the satellite, transmit telecommands to the satellite and collect localization data.
Dedicated leased lines, between the ground stations and the LOCC are used for nominal data transfer and voice communication.
Each leased line is redundant (by an ISDN line). The routing difference between nominal and redundant lines is based on the difference of lines technology.
LOCC is in charge of the acquisition, processing and archiving of the satellite telemetry, of the satellite health monitoring, of the elaboration and sending of telecommands to the satellite, of the satellite attitude control, of the orbit determination using localization data from the different ground stations, of the different manoeuvres computation.

The LEOP Ground Segment will interface with the following entities:

- Satellite Integration and Test Site (SATINT), in order to test the flight procedures and exchange telemetry and telecommand data during System Validation Tests (SVT)
- Launch Base (LB), in order to receive telemetry data, receive the countdown, exchange the go/no go criteria, receive the video stream during the launch, receive ephemeris and covariance data at satellite separation, and provide bi-directional voice communication to support the launch campaign;
- GCC-GCS, in order to:
  - transfer telemetry on line and data archive (TM and TC) from LOCC to GCS at satellite handover
  - send LEOP report after successful LEOP of a specific satellite
  - provide satellite status, flight procedures, tracking/ranging data and initial orbit Vector requested by the GCS
  - retrieve the flight procedures, satellite reference database (SRDB) on which the operational database will be built
  - retrieve a copy of the image of the OBSW and provide a copy of the downloaded OBSW
  - transfer orbit determination, several manoeuvre parameters from LOCC Flight Dynamics Service to GCS in order to place the S/C in its final position

6.1.2.1.4 IOT Site

The IOT physical context has been illustrated in two figure previously.

6.1.2.1.5 GMS

The GACF inside the GMS is in charge of maintaining the constellation status.

6.1.2.1.5.1 GACF: Constellation Status Configuration Control

The constellation status is a configuration file (CNF_SVMask) which contains major information to allow the Galileo missions to work. It includes the parameters of the available Galileo satellites, providing information on:

- Which are the available SVID
- Which are the PRN allocated to the SVID (both C-band and L-band)
- What are the almanacs for available SVID
- CCSDS numbers (it includes SCID number)

The Constellation status configuration file is manually modified by a mission engineer each time one of the following activities is planned:

- Satellite decommissioning
- New satellite deployment
- SVID number change
- PRN code number change
- CCSDS number (SCID) change

The last three bullets (SVID, PRN and SCID changes) are managed through the MTP (Mid-Term Plan) and imply a decommissioning of a satellite with the old numbers and the deployment of the same satellite with the new numbers.

The GALILEO satellites are identified in different ways, for different purposes:

- SCID, used for spacecraft M&C, assigned by the CCSDS
- SVID, used to identify a satellite from a mission perspective, used in the navigation messages
- PRN codes, used to identify the signal-in-space in each of the bands (C-band, S-band and L-band) for each satellite

The figure below shows the physical links (C-band, S-band and L-band PRN codes) used to identify the signals-in-space and protocol links (SCID and SVID) between the Galileo segments used to identify the satellites.

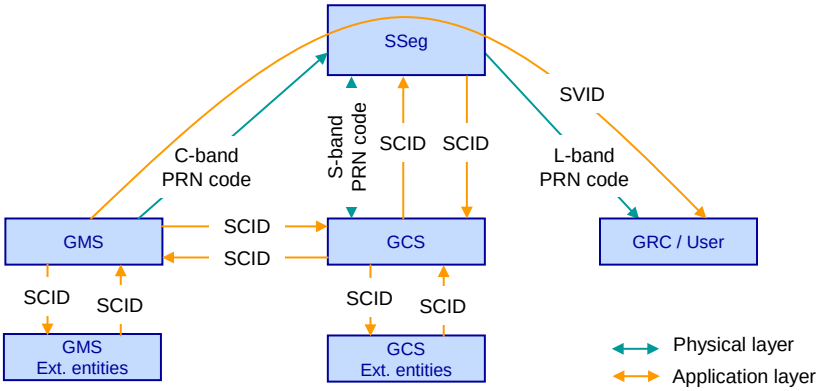


Figure 95: Use of SCID, SVID and PRN codes.

6.1.2.2 Dynamic Scenarios

The objective of the SFC1.1 Dynamic Scenarios is to demonstrate that the System Design allows the System to perform all the functions necessary to provide the service specified by the Customer.
The scenarios presented here are consistent with the MOCD and with the relevant operational documentation (Mission Analysis, Preliminary Constellation Deployment Strategy, LEOP Ops Concept, IOT Ops Concept for Payload IOT).
Due to the fact that different types of launchers can be used and each one can accommodate a different number of satellites, it is necessary to organize the scenarios in such a way that the common aspects to all deployment strategies are not repeated for every scenario.
To this effect a generic scenario has been defined, which describes in detail all the activities pertaining to a single launch, without specifying the launcher type and number of satellites deployed: in facts, all the steps in this scenario all valid, regardless of the launcher type and number of satellites.
This scenario will be then referenced in higher-level scenarios that will illustrate the different deployment strategies, which do not need to repeat the lower-level details common to every deployment strategy.
All the deployment scenarios assume that the 4 IOV satellites are deployed in two orbital planes (i.e. no alternatives are considered).
The constellation slots will be identified in the following way:

- Plane A: A01 to A08
- Plane B: B01 to B08
- Plane C: C01 to C08

The IOV satellites are B05, B06, C04, C05.

6.1.2.2.1 #SFC1.1-1 Deploy Satellites (N,L)

This scenario describes the generic sequence of activities performed in order to deploy N satellites using launcher L.

It will be used as a building block to construct the higher-level scenarios that illustrate the various constellation deployment strategies (see #SFC1.1-2 to #SFC1.1-7).

1. System calculates reference orbits and biases (from Station-Keeping strategy) and passes the values to the Launcher Provider and LEOP Provider

2. Launch (Launch Service Provider)

3. Direct injection into target orbit (Launch Service Provider)

4. Compute separation information (Launch Service Provider)

5. Transmit separation information file (Launch Service Provider)

6. LEOP operations (via ENOC)
   - One Out-of-Plane Manoeuvre to correct the inclination and RAAN due to Launcher dispersion and N In-Plane Manoeuvres to correct the phase and phase rate)
   - Spacecraft mode transitions
   - Start drift manoeuvre

7. Perform Platform Commissioning (GCS): the GCS will perform spacecraft M&C using the LEOP TTC stations.
   Alternative 1:the GCS will perform spacecraft M&C using its own TTC stations network

8. Perform fine positioning Manoeuvres (GCS): the GCS will perform spacecraft M&C using th LEOP TTC stations.
   Alternative 1: the GCS will perform spacecraft M&C using its own TTC stations network. Up to 3 In-Plane Manoeuvres will be performed for drift stop and coarse positioning, followed by up to 3 In-Plane Manoeuvres for fine positioning. These operations may be interleaved with payload IOT operations, if the latter cannot be completed during the free drift phase. The phasing angles of all the satellites are determined by the Station-Keeping strategy, taking into account. the constraints dictated by the constellation geometry (e.g. the phasing angle of the 2nd satellite will be offset by 40 degrees w.r.t. to the phasing angle of the first satellite)

9. Perform Clear Payload IOT (GCS) the GCS can perform spacecraft M&C using the LEOP TTC stations or the GCS TTC stations. The IOT will include Functional Tests, RF Tests and Base-band Tests. At the end of the Clear IOT, the satellite will be initialized in the following configuration:
   - Normal Mode
   - All equipment on the Main side
   - Clocks stabilized
   - PxSU in Clear Mode

10. Perform Secure Payload IOT (GCS + GMS) the GCS will perform spacecraft M&C using the GCS TTC stations, as the satellites have now reached their final orbital position. At the end of the Secure IOT, the satellite will be initialized in the following configuration:
    - Normal Mode
    - All equipment on the Main side

- Clocks stabilized
- PxSU in Protected Mode

11. Satellite and GMS configuration for service provision (see #SFC1.1-8 Dynamic Scenario). At the end of this step, the system is ready to provide a service using the newly launched satellite.

Deployment strategies & logic:

The following figure shows the possible path (development logic) from current to final generic configurations.

Those configurations achieving the performance milestones sooner are highlighted in green. These
are pursued and the paths are in solid black lines. Dashed black lines show not preferred paths segments, either because of performance preferences (section 3.1.5.2) or/and freedom in plane selection (Ωtotal). Dashed grey lines are not desired paths, to be avoided if another alternative exists.
Each path segment shows the number of planes (Ωi) to which the next launch would achieve the
desired configuration. As higher as better in order to have the chance to decide the plane until the
very last moment.

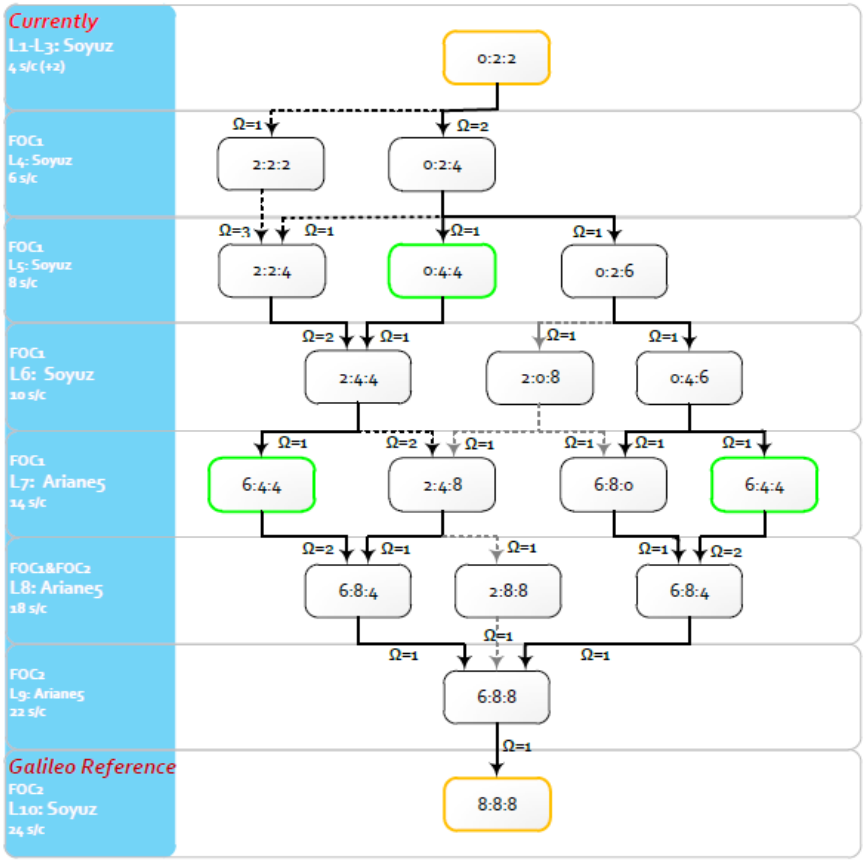


Figure 96 : Deployment strategy

6.1.2.2.2 #SFC1.1-2 Deploy Constellation (Case 1: plane by plane, only Soyuz)

Not used

6.1.2.2.3 #SFC1.1-3 Deploy Constellation (Case 2: plane by plane, only Ariane)

Not used

6.1.2.2.4 #SFC1.1-4 Deploy Constellation (Case 3: plane by plane, Soyuz + Ariane)

Not used

6.1.2.2.5 #SFC1.1-5 Deploy Constellation (Case 4: gradual plane filling, only Soyuz)

Not used

6.1.2.2.6 #SFC1.1-6 Deploy Constellation (Case 5: gradual plane filling, only Ariane)

Not used

6.1.2.2.7 #SFC1.1-7 Deploy Constellation (Case 6: gradual plane filling, Soyuz + Ariane)

Not used

6.1.2.2.8 #SFC1.1-8 GMS re-configuration for service new satellite (evolution for STRB 2.0)

This dynamic scenario describes the activities necessary to configure the GMS in order to add the newly launched satellite to the constellation, from a service provisioning point of view.

1. SVID for new SV (and therefore PRN codes) is defined before launch to allow execution of key loading tests.

2. The GCS will know the orbit based on a specific ranging campaign run at completion of orbit manoeuvring phase and will send the relevant ephemeris file to the GMS.

3. A GMS MTP will be generated to cover the following operations (see figure below) :
   a. The GSS network (including GSS-PTF) re-configuration to track new launched satellites and provide related pseudo-ranges to be processed by OSPF.

   b. Execution of the SCAT analysis from MUCF to synchronize the NSGU time reference with GST after sufficient raw data are collected by the PTF GSS during a SV common visibility with GCS TTCF.

   c. Generation of a Test Navigation Message for a Test satellite; segregation between Test satellites and operational satellites shall be ensured:

      i. The operational navigation messages shall be generated independently from the Test navigation message in order to avoid possible contamination between the Testing and Operational chains. However Test navigation messages can be generated using both Test and operational satellites data.

      ii. The operational satellites shall not disseminate data from Test satellites. However, function of segment implementation, it is possible that Test satellites contain data from operational satellites (related to satellite almanacs).

   d. A manual contact to support the S-IOT for the new launched SVs in parallel to the nominal MTP/STP, using a dedicated ULS. The SHS for the satellites undergoing the manual contact will be automatically set to “InTest”.

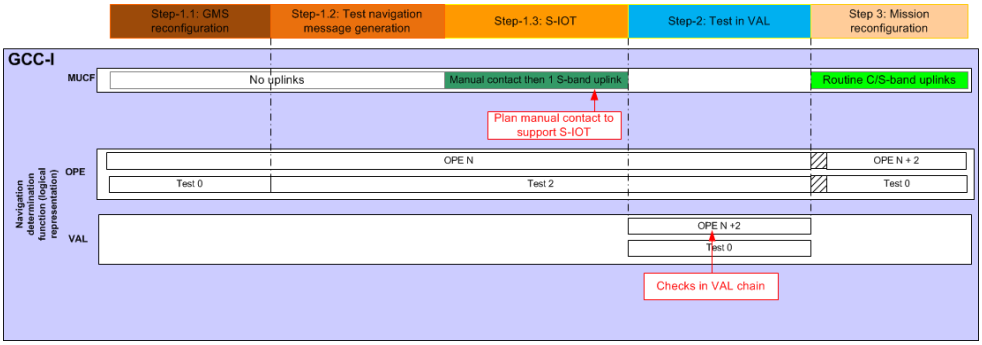
   e. A window for functional and performance checks in order to confirm the correct of the OSPF (which could be performed in Val chain or on another specific OSPF).

   f. Online Reconfiguration of the main OSPF to support the new and previous satellites

   g. A new operation contact to support routine operation for the updated constellation. This will enable nominal message dissemination to all SVs.

4. The GMS will send the MTP with the corresponding task description to the GCS that will generate the relevant STP and command schedule. At the same time GMS will also translate the MTP in STP and GMS command schedule.

5. At the contact time, the GCS uploads the TC generated in accordance to the plan, including the STCs previously transferred to the SCCF from the x-KMF



It is noticed that what above reported is still TBC as no input from the Segments are currently available since the requirement introduced by STRB 2.0 are not yet applicable to them

6.1.3 SFC1.2 Maintain Nominal Constellation

6.1.3.1 Physical Static View

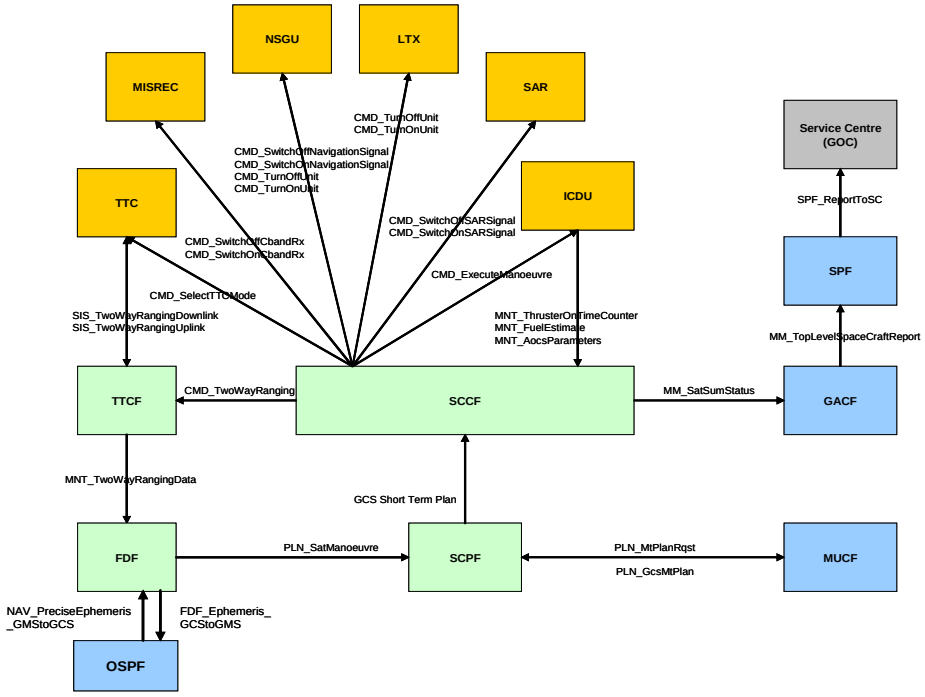


Figure 97 : SFC1.2 physical static view

6.1.3.1.1 Overall physical description

The following physical entities are involved in this SFC:

- Ground control Segment
  - SCCF (Spacecraft and Constellation Control Facility)
  - SCPF (Spacecraft and Constellation Planning Facility)
  - FDF (Flight Dynamic Facility)
  - TTCF (Telemetry, Tracking and Control Facility)
- Ground Mission Segment
  - OSPF (Orbit Determination and Clock Synchronization Processing Facility)
  - MUCF (Mission Uplink Control Facility)
  - GACF (Ground Asset Control Facility)
  - SPF (Service Product Facility)

6.1.3.2 Dynamic Scenarios

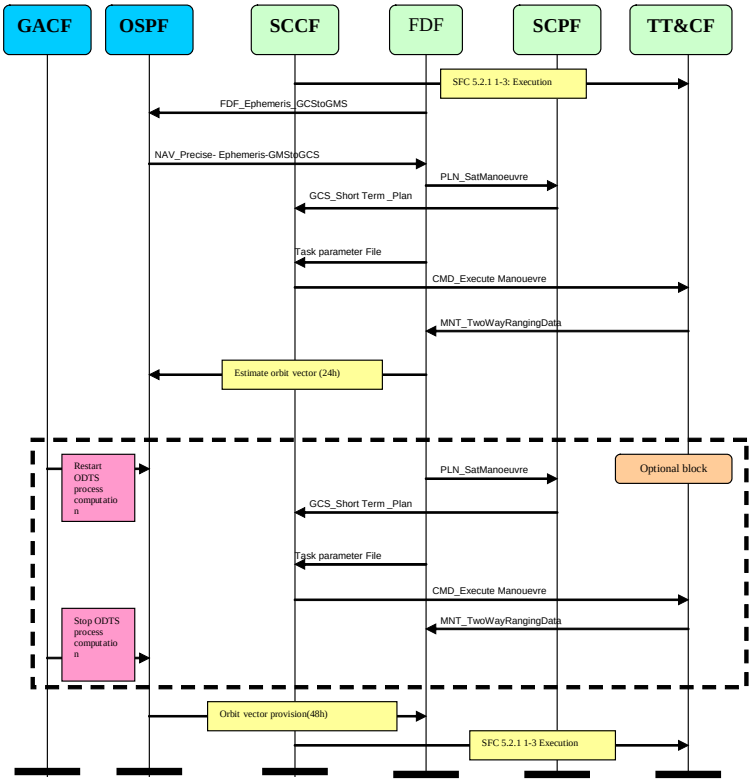
6.1.3.2.1 #SFC1.2-1 Station keeping manoeuvre

6.1.3.2.1.1 General Description

This dynamic scenario provides details on the data flow exchanged to perform the station keeping manoeuvre. Before the manoeuvre, the satellite is excluded from services, as described in the step 5a) of the SFC 5.2.1. Mission Services support by this satellite will be resumed after the manoeuvre completion.
The operations that are strictly necessary to carry out an orbit keeping manoeuvre are the following ones:

- All necessary satellite mode changes and payload reconfiguration takes no longer than 2.5 hours
- Reorientation of the S/C to the target attitude (max. rotation angle 180 deg), with a maximum duration of 34 min.
- The orbit correction manoeuvre itself, which is basically aimed at applying a correction to the velocity of the satellite, with a maximum duration in the order of 10 seconds.
- Reorientation of the S/C to the yaw steering attitude, with a maximum duration of 34 min. This will comprise the transition to Normal pointing mode
- Payload reconfiguration will take no longer than 2 hours (TBC), however, the P/L may take between one day (PHM as master clock) and 4 weeks (RAFS as master clock) to reach specified clock performances.

6.1.3.2.1.2 Flow Chart



6.1.3.2.1.3 Scenario sequence description

The strategy to maximise service availability during a station keeping manoeuvre can be described through the following steps:

1. GCS FDF starts planning the manoeuvre based on the Orbit vector provided regularly by the OSPF, as part of the standard mission services. It is expected that information on planned manoeuvre should be available at least 15 days in advance of the operation.
2. The satellite is taken out from service shortly before the start of the manoeuvre setting the DVS flag to “Working without guarantee” Soon after it is expected that the satellite will stop providing downlink signal to the Users and GMS.
3. The manoeuvre operations are executed comprising payload reconfiguration, attitude reorientation, burn, attitude re-orientation and payload reconfiguration to make the L band downlink available again.
4. As soon as the manoeuvre operations are completed the following operations are run as far as possible in parallel:
    a. A Ranging campaign is started using dedicated S band TT&C passes of about 20 minutes each from all available TT&C stations, planned in advance and distributed as far as possible uniformly over the following 48 hours. It is foreseen however (GCSREQ-23985) that an orbit can be already determined by the FDF after 24 hours, with semi-major axis accuracy of 5 meter (1-sigma).
    b. If needed, the satellite clock frequency will be corrected via S band TC, based in historical information collected on ground. After verifying the correct execution of the clock correction commands transmitted at the earliest S band contact, the ODTS process is restarted to speed up convergence based on the FDF predicted orbit vector (or even the orbit vector available before the manoeuvre) and a predefined HSISA value (this will be consolidated in an experimental phase). At

this moment, the satellite is not yet available for the users as the DVS flag is still set to “Working without guarantee”.

5. At the earliest opportunity the SCAT analysis is run from the MUCF to verify the synchronisation between on board clock and PTF clock (GST). The prerequisite to run this functions are:
    a. The availability of the Orbit ephemerides for the satellite, which can be selected among the predicted orbit vector provided by the GCS, or the new estimate that may be already available from the GCS FDF 24 hours after the manoeuvre. Note that the SCAT analysis is expected to have an uncertainty of 10 ms while a slant range error of 30 meters affects the time estimate by 1 microsecond only, so the initial orbit knowledge is not required to be extremely accurate.
    b. Some hours of satellite raw data collected from the PTF.GSS This may therefore require waiting until the satellite becomes visible from the PTF GSS.

In nominal cases the SCAT analysis should confirm that good synchronisation is achieved and no need for further commands is needed.

6. Approximately 48 hours after the manoeuvre, in nominal condition it is expected that:
    a. the GCS FDF will provide an orbit vector accurate enough to asses the manoeuvre performance (e.g., semi-major axis accuracy of 2 meters three sigma)
    b. The ODTS has achieved convergence and provides satellite ephemeredes and clock parameters compliant with the accuracy requirements.

In nominal case this would complete the manoeuvre operations, the DVS flag will be reset to zero and a new navigation message would be broadcasted to the users. This will allow Open Services to be resumed, while for SoL and PRS services some performance degradation is expected until full convergence of HSISA is reached, which may take a few days more.
The following non nominal events may however occur, expected to be detected before resuming nominal mission service operations that will require keeping the satellite out of service for longer time:

- The GCS FDF precise orbit determination shows that a further correction manoeuvre is necessary
- The SCAT analysis shows that the on board clock requires a further correction.

The ODTS process will be stopped and restarted only after completing the additional necessary operations involving commanding the satellite in S band in some special contact slot. The completion of the operations would be longer in this case, especially in case a new manoeuvre is to be performed.
It is expected however that a good chance exists that a single burn will be sufficient to complete the station keeping manoeuvre.
In such a case the satellite unavailability is essentially dominated by the 48 hours required by the ODTS algorithms to start the initial converge.

6.1.3.2.2 #SFC1.2-2 Re-phasing of Satellite Orbit

The dynamic scenario provided #SFC1.2-1 is suitable also for the re-phasing satellite orbit manoeuvres. The main difference between this scenario and the re-phasing satellite orbit scenario is focused on the ODTS algorithm, which computes data inside the OSPF unit; it restarts after the completion of the following steps requested for the orbit re-phasing:

- satellite transferring into the drift orbit

- ranging campaign to assess the drift orbit
- satellite transferring into the operative orbit
- new ranging campaign to assess the final operative orbit

6.1.3.2.3 #SFC1.2-3 Constellation Replenishment

This Dynamic Scenario contains a summary description of the sequence of activities that will be carried out to replenish the constellation. It does not assume any particular replenishment strategy and it makes use of the same system functionality needed for the deployment scenarios.

1. The Operations Team and the Operating Company jointly identify the need to replace old satellites in the constellation with new satellites

2. One or more satellites stored on ground (guaranteed to be no more than 5 years old) will be selected

3. SVT will be performed

4. A normal Launch Campaign will be conducted

5. The satellite(s) will be launched and commissioned following the same sequence of steps identified in the deployment scenario, #SFC1.1-1 Deploy Satellites (N,L). In this case, the SVID/PRN codes will be chosen to be in the non-operational range (i.e. the PRN codes family number will be greater than 36 so that the user receivers will ignore the signal). The new satellite(s) take the role of spares
   Contingency Scenario #SFC1.2-3/1: if the replenishment launch fails and it carried 1 satellite, the in-orbit spare can be relocated to the orbital slot originally foreseen for the replenishment satellite (also a spare)
   Contingency Scenario #SFC1.2-3/2: if the replenishment launch fails and it carried 2 satellites, another launch can be scheduled to take place within 6 months (TBC)

6.1.3.2.4 #SFC1.2-4 Satellite Disposal (to Graveyard Orbit)

This scenario purely operational and is fully described in dedicated scenario SYSOPS-S-PLN-07 (Satellite Disposal) detailed in MOCD [SYS-RD.6].

6.2 SFC2 GENERATE SYSTEM TIME AND GEODETIC REFERENCE

6.2.1 Introduction

The purpose of the SFC2 is to:

1. Generate Galileo System Time (SFC2.1) to support the system synchronization, navigation functionality and UTC dissemination service
2. Provide GPS-Galileo Time Offset (SFC2.2) to further support Galileo-GPS interoperability
3. Contribute to Geodetic Reference Determination (SFC2.3) to support the navigation functionality of the system

The following section describes the physical implementation of the SFC. It is based on the following documentation:

- ☐ GMS documents:
  - o GMS DDF1 (GMS-RD.1)
  - o GMS DDF2 (GMS-RD.2)
  - o GMS DJF (GMS-RD.3)
  - o GACF ERD (GMS-RD.9)
  - o MUCF ERD (GMS-RD.10)
  - o MUCF ICD (GMS-RD.11)
  - o GACF ICD (GMS-RD.12)
  - o SPF ICD (GMS-RD.16)
  - o OSPF ERD (GMS-RD.18)
  - o OSPF ICD (GMS-RD.19)
  - o PTF ERD (GMS-RD.24)
  - o MSF ERD (GMS-RD.25)
  - o MSF ICD (GMS-RD.31)
  - o GSS ICD (GMS-RD.32)
  - o PTF ICD (GMS-RD.34)
  - o SPF ERD (GMS-RD.36)
  - o GMS input to Operation Concept (GMS-RD.41)
  - o GSS ERD (GMS-RD.44)
  - o ALG_NAV Algorithms (GMS-RD.55)

- ☐ GCS documents:
  - o FDF ERD (GCS-RD.5)

6.2.2 SFC2 High Level Description

The Galileo Navigation System, to provide its functionality to the Galileo users, utilizes an internally generated timescale GST (Galileo System Time) and its own terrestrial reference frame GTRF (Galileo Terrestrial Reference Frame). In order to further enhance its interoperability with the GPS, Galileo System determines the time offset to GPS and distributes the respective GGTO (Galileo-GPS Time Offset) value to the users.

6.2.3 SFC2.1 Generate Galileo System Time

6.2.3.1 SFC2.1 High Level Description

The purpose of this SFC is to provide all timing functionalities needed by the Galileo System to support the provision of the mission services. This includes:

1. Navigation time keeping: GST provision for the OS and PRS navigation services
2. Metrological time keeping: In addition to the tasks performed with in the frame of Navigation time keeping , the Metrological time keeping adds GST(MC) to UTC (modulo 1 second) steering and generation of GST to UTC conversion parameters needed to support the timing services for OS and PRS (UTC dissemination service).
3. GST provision for system overall synchronization

Note: Metrological timekeeping can be considered as an add-on to provide an additional functionality on top of the core Navigation timekeeping which is crucial for the functioning of the whole Galileo system in particular the provision of the navigation services. The metrological timekeeping in its principle encompasses the functionality of the Navigation timekeeping and exploits its notion by adding the UTC (modulo 1 second) steering features. The Metrological timekeeping allows expanding Galileo services by adding the timing (UTC) dissemination service.

It is important to note that the MRD specification (MRD-260/885) of the timing service performance and availability are conditioned to the availability and quality of the TSP products. One TSP is deployed in each GCC, configured in Master/Slave warm redundancy scheme. TSP states are not known by the GMS. Both instances are up and running, processing the same input data received from UTC(k) laboratories/BIPM from one side and the GMS from the other side. However, only the Master TSP connects to the SPF to provide the clock steering data to the GMS. Execution of an operational coordination procedure is required in case of a switch over between the 2 TSP.

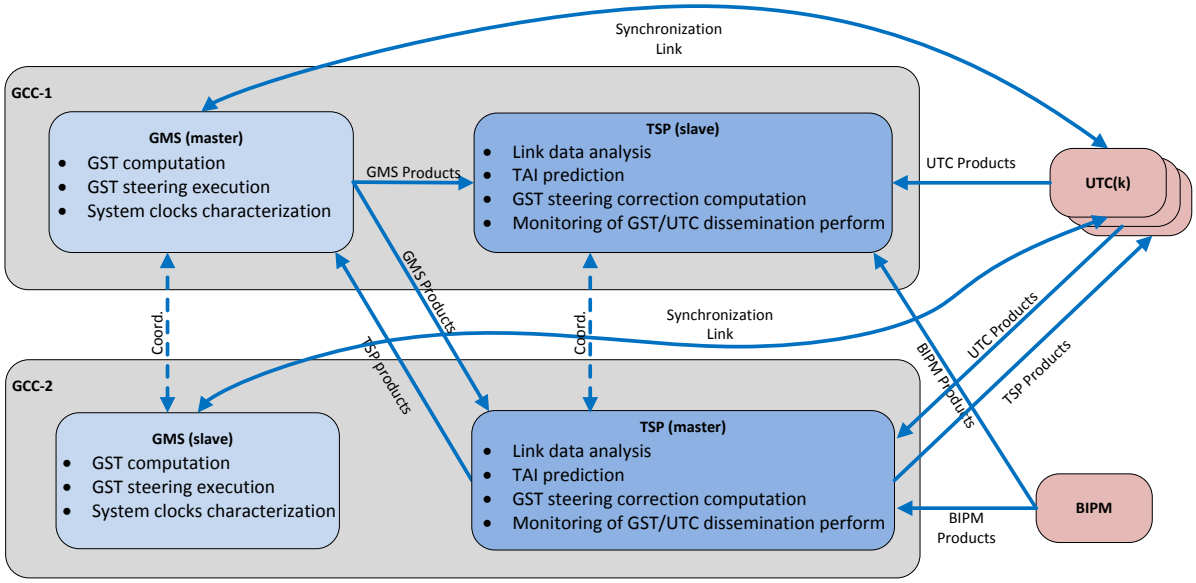


Figure 98: Dual TSP design (in cross-site configuration)

6.2.3.2 Physical Static Block Diagram

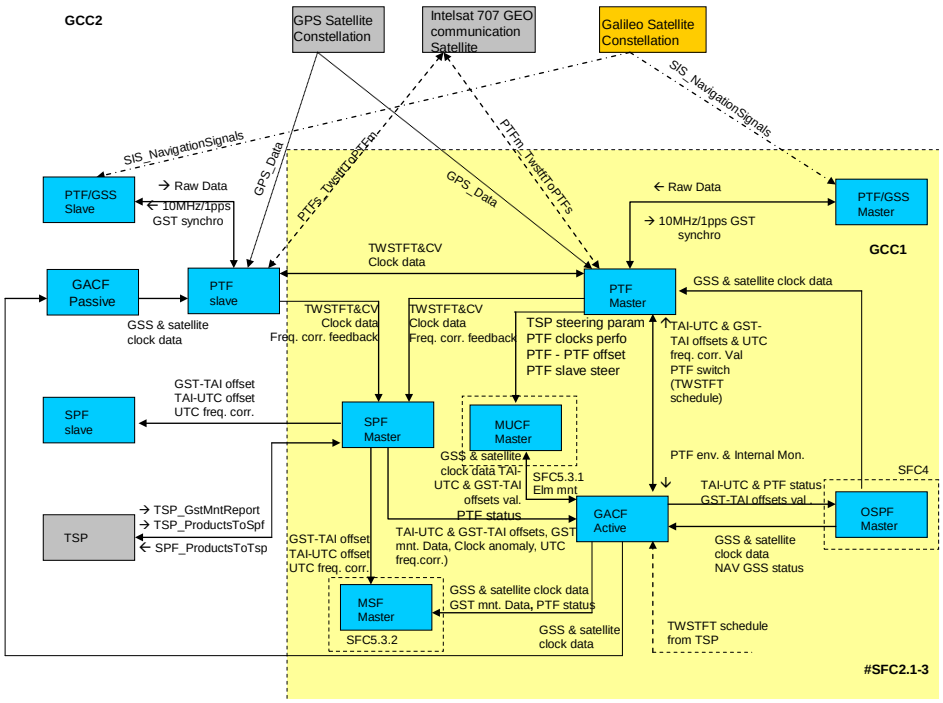


Figure 99: SFC2.1 Physical Static View

The above figure summarizes the overall system configuration, including also the iterations between the two GCCs, required to deliver the SFC2.1 functionality. In order to capture the concept of the metrological and the navigation timekeeping while keeping the clarity of the diagrams by removing the unnecessary elements and the GCC duality the following two static block diagrams are used.

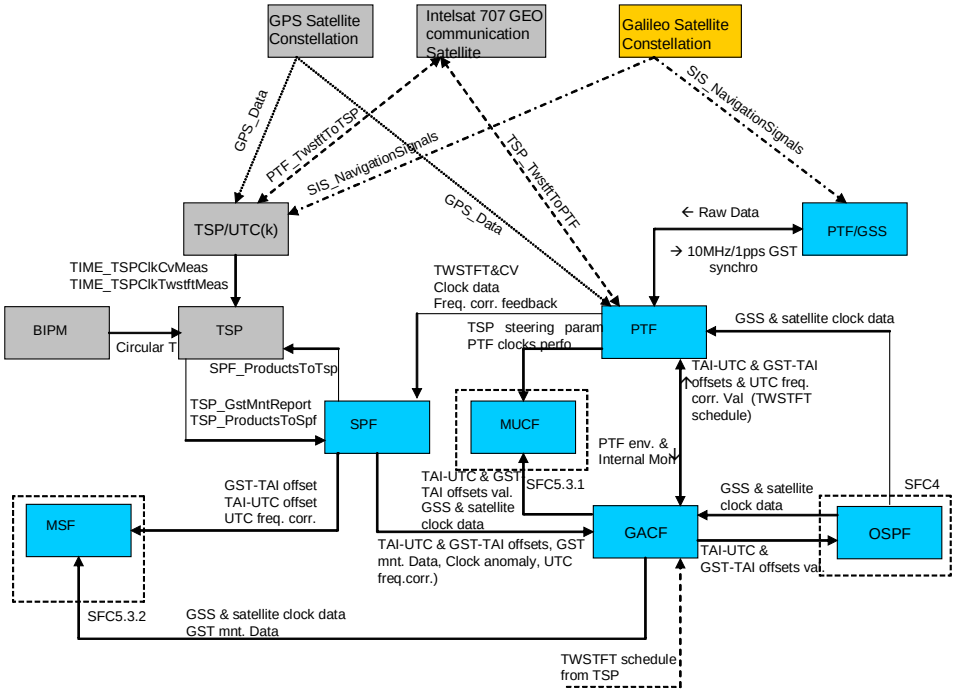


Figure 100 : Metrological time keeping Block Diagram

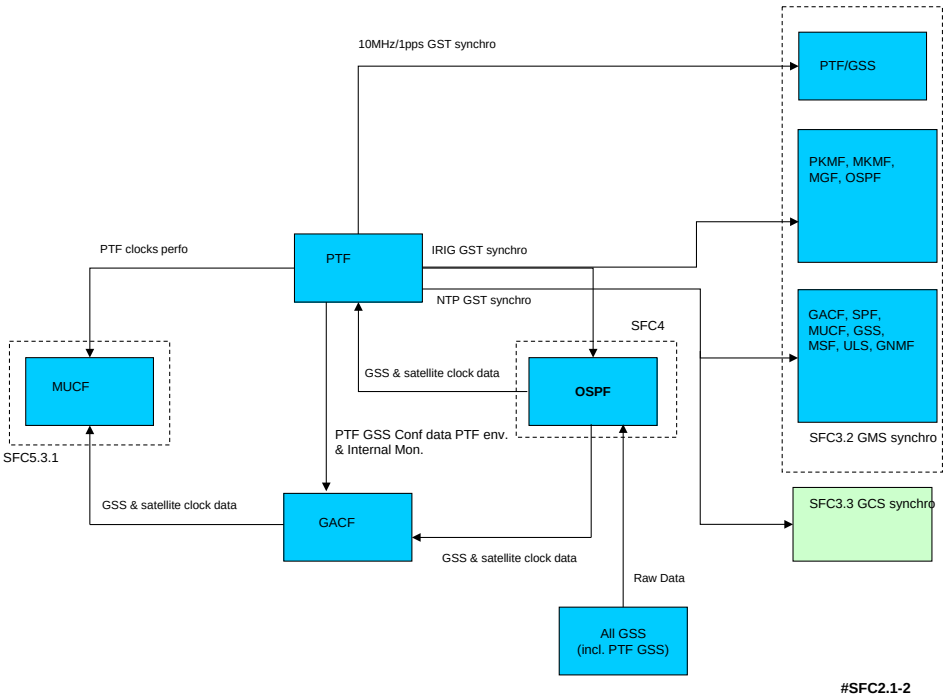


Figure 101 : Navigation time keeping Block Diagram

6.2.3.3 Overall Physical Description

The purpose of this SFC is to:

- Generate and maintain Galileo System Time, based on the Galileo internal timescale realized with the clocks from the PTF clock ensemble and OSPF generated satellite clocks. Galileo system needs to ensure for the GST to be generated autonomously and maintained to be stable in frequency in order to provide the navigation services meeting the required scope and performance.

- Maintain the time offset between the GST and UTC within the required range by steering GST(MC) to UTC modulo 1 second using the steering corrections provided by TSP. Generate based on the GST and TSP timing products the GST to UTC conversion parameters for their inclusion into OS and PRS navigation messages to enable the UTC dissemination services for OS and PRS.

- GST provision for system overall synchronization

Notes:

- The third task (GST provision for the system overall synchronization) is realized by the Navigation timekeeping. Regarding the actual overall system synchronization, only the GST generation/provision is discussed within this document, for more details on system and element synchronization please refer to the dedicated SFC3.2&SFC3.3.
- It is considered that the same set of GST to UTC parameters applicable to OS, and PRS services is broadcast to the Galileo users.
- The GST to UTC parameters are generated by OSPF with the use of the TSP provided parameters which are in the case of Galileo-CV link based on the use of Galileo-GPS combined receiver SIS observables (code phase add carrier measurement s) on E5a, E5b, E5 AllBOC, E6C and E1C and GPS observable on L1 and L2.

6.2.3.3.1 Definitions

- GST reference points:

Following GST versions/realisations are defined within the Galileo System as per segments:

- Ground Mission Segment:
  - GST(MC)
  - GST(GSS)
  - GST(OSPF)
- Space Segment
  - L-GST

The above GST realizations are internal to the Galileo system, thus not visible/accessible to Galileo users. Finally the GST as retrieved by Galileo users from the Galileo signal in space is referred to as GST(USER).

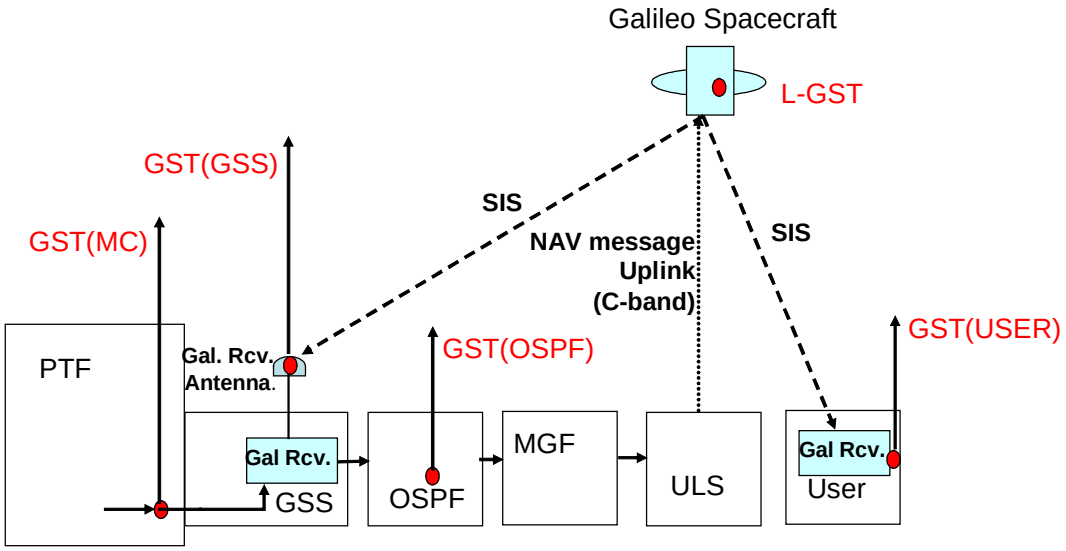


Figure 102 : GST reference points

By the definition, the GST(MC) is defined at a fixed and clearly identified physical point of the electrical path at PTF output of the timing signal used as reference by the GSS Galileo receiver. Note that the GST(MC) at the Master PTF is the reference point where the TSP has to reference his steering products to.

It is possible to reconfigure, through operator command, the GST reference point to any reference point connected via GSS.

- The GST(GSS) is defined at the antenna phase centre of the Galileo receiver located at the GSS collocated with the PTF.
- The GST(OSPF) is the representation of the GST as seen by the OSPF through the processing of Galileo SIS measurements collected by the Galileo receiver at the GSS collocated with the PTF

Note: Different processes often require as an input different GST realisation. For instance GGTO,TSP and GST distribution activities require access to the physical realisation of the GST, hence the GST(MC) is used. On the other hand for the navigation and UTC dissemination services the “paper” realisation of GST referred to as GST or (GST(OSPF)) is employed. It is important that for the difference between these GST realisations it is always accounted for when the final products such as GGTO or GST to UTC conversion parameters are generated.
GST(USER) is the realisation of the GST as obtained by the Galileo user through the reception of the Galileo SIS. The realisation of the GST is essentially based on the GST(OSPF), however additional uncertainty contributions apply due to the user equipment biases, uncompensated SIS propagation effects and uncertainties of the satellite orbit and clock parameters in the Galileo navigation message.
L-GST so called local GST, GST as realised on board of particular Galileo satellite and is derived from the on board satellite clock (P-HM (Passive Hydrogen Maser) or SRAFS (Space qualified Rubidium Frequency Standard). Note: Navigation messages contain the clock model parameters characterising the on board satellite clock (L-GST) with the respect to GST(OSPF)

6.2.3.3.2 GST Generation: PTF

Precision Timing Facility has assigned the primary function of the generation and maintenance of the Galileo System Time (GST).

This task can be further broken down to the following subtasks:

- Generate the GST time scale, on the basis of measurements performed on a clock ensemble;
- Generate a physical realization of the GST time scale (GST(MC))
- Distribute the GST externally and within the GCC,
- Contribute to GST(MC) steering to UTC (modulo 1 second) by providing relevant information to the TSP
- Compute the Time offset between GST and GPS time scale,
- Ensure Steering of the 2 GST time scales from the 2 GCCs.
- Support GMS for Timing mission monitoring.
- Support the generation of GST to UTC conversion parameters by providing the relevant information to the TSP

PTF maintains its own clock ensemble formed by Cesium and A-HM clocks and frequency standards respectively. PTF continuously measures the difference between each of the clocks from the PTF clock ensemble. These measurements are processed and fed including the OSPF generated satellite clock data into the PTF internal GST algorithm. The algorithm produces the correction to be applied to the output of the PTF master clock (A-HM) to provide GST physical realisation.

One PTF is deployed at each GCC site, one being configured as Master PTF and the other as Slaved PTF.
In order to meet GMS availability requirements, the following redundancy scheme is deployed:

- the PTF at each GCC site is able to generate 1+1 redundant GST physical realisations, which are referred to as GST(MC1) and GST(MC2), by using two Active Hydrogen Maser clocks (A-HM1 and A-HM2). On each PTF site, one GST physical realisation is configured as a master GST physical realization and called GST(MC). The hot back-up GST physical realization at each PTF site is slaved to the co-located master GST physical realization GST(MC).
- 1+1 PTF-GSSs are deployed at each GCC site, which are referenced to as PTF-GSS1 and PTF-GSS2. Each PTF-GSSx, x=1 or 2, uses as reference the associated GST physical realisation GST(MCx), x=1 or 2, respectively. Consequently, two GST(MCx) /PTF-GSSx (x=1 or 2) lanes are present at each GCC site.
- Each PTF is equipped with five Cesium frequency standards, to provide robustness against failure of one Cesium.

The inclusion of Satellite and GSS clocks in the generation of GST improves the PTF performance, in terms of GST stability and robustness against clock failure. In addition, each PTF is able to reach the GST performance requirements by using 4 of its internal clock ensemble only.

Notes:

- GST stability is crucial for the navigation functionality of the system will be ensured as mentioned above by combining Cesium clocks and Active Hydrogen Masers frequency standards together. The A-HM with the very good short term stability, however exhibiting rather large linear frequency drift in a long term (for observation times >1 day in general weeks/months) will be steered to the ensemble of the Cesium clocks (exhibit better long term stability). By doing so the A-HM linear frequency drift will be effectively removed, while its short term performance remains preserved.

- In the PTF configured as Master PTF, the master physical realiation GST(MC) is steered to TAI thanks to Two-Way Time Transfer or Common View (GAL-CV or GPS-CV) synchronisation links with the TSP. The slaved GST(MCx), x=1 or 2, at Master PTF is steered to the co-located GST(MC) thanks to the Master PTF algorithms.
  On the PTF site configured as Slave PTF, the master physical realisation GST(MC) is steered to the GST(MC) of Master PTF using PTF-to-PTF links. The slaved GST(MCx) is steered to the co-located GST(MC) thanks to the Slave PTF algorithms.
- The PTF operating mode is controlled by GACF. This includes the selection of Master PTF site and Master GST(MC) at each PTF site.
  In addition, the GACF provides the monitoring and control of the two GST(MC)/PTF-GSS lanes deployed at each PTF site.
- Each GST(MC) physical realization, GST(MCx), x=1 or 2, at each PTF is defined at a fixed and clearly identified physical point of the electrical path used to inject the timing signal (1PPS/10MHz) used as reference by the associated PTF-GSSx, x=1 or 2.
- The Master GST(MC) at the Master PTF is the reference point where the TSP has to reference his steering products to.
- In order to improve PTF robustness, two independent static signal paths (1PPS/10MHz signals) are created for distribution of GST(MCx) to the associated PTF GSSx, whithin each GST(MCx)-PTF-GSSx, x=1 or 2, lane.
- The task of each PTF is to calibrate the 2 independent signal paths (1PPS/10MHz signals), which are defined between each PTF GST(MCx), x=1 or 2, output and the GRC (Galileo Receive Chain) input inside the associated co-located PTF-GSSx. The calibration result is applied by each GSS-PTFx when providing the raw measurement to OSPF.
- For each GST(MCx)/PTF-GSSx lane, the GMS provides automated recurrent calibration function for all GSS PTF receiver chains, such that no control inputs from operators are required in routine operations. In that way, the raw GSS measurements are tagged by the broadcast GST (i.e. the GST(OSPF)).
- The OSPF is notified by the GACF of which PTF site and which GST(MCx)/PTF-GSSx lane within this site is assigned as master. The raw measurements coming from the Master GST(MC)/PTF-GSS lane are used by the OSPF as the reference to compute the offsets of satellite clocks with respect to GST(OSPF), that are part of the navigation data broadcast to the users. In addition, the OSPF determines the offsets of the GST seen by the OSPF using raw measurements from the remaining back-up GST(MCx)/GSS-PTFx lanes, namely:
  - Slaved GST(MC)/PTF-GSS lane at Master PTF site,
  - Master GST(MC)/PTF-GSS lane at Slaved PTF site,
  - Slaved GST(MC)/PTF-GSSx lane at Slaved PTF site,

with respect to the GST(OSPF) seen through the master GST(MC/PTF-GSS lane at master PTF. Such offsets are used to implement a proper the GST transition law in case of PTF-GSS switch.

- Each PTF is equipped with a Galileo-GPS Combined Receiver which generates SIS observables (code phase and carrier measurements) used for both GPS and Galileo Common View to support the determination of GST(MC) to UTC (modulo 1 second) parameters at TSP.

The Galileo SIS observables are collected on E1C, E5a, E5b, while the GPS observable are collected on L1 and L5.
The Galileo/GPS data observables and relevant navigation messages collected at PTF are sent to the TSP, which provides their post processing, jointly with other measurements collected by the European UTC(k) laboratories contributing to the TSP for the comparison of GST(MC) with UTC(k) timescales (in line with the Common View practice).
The Galileo SIS observables collected by the PTF Combined Receiver are used for Common View to support the synchronisation between the two PTF and also for GGTO determination, please refer to SFC2.2.

Note :

- Currently, the Galileo Time and Geodesy Validation Facility takes the TSP function
- L5 signal specified in GMSREQ-11751 will be transmitted IIF block GPS satellites. Currently, the combined receiver output is configured to operate with L1-L2 combination

The task of the PTF is to calibrate the signal paths (1PPS/10MHz signals) between GST(MC) output and the Combined Galileo-GPS Receive input inside PTF.

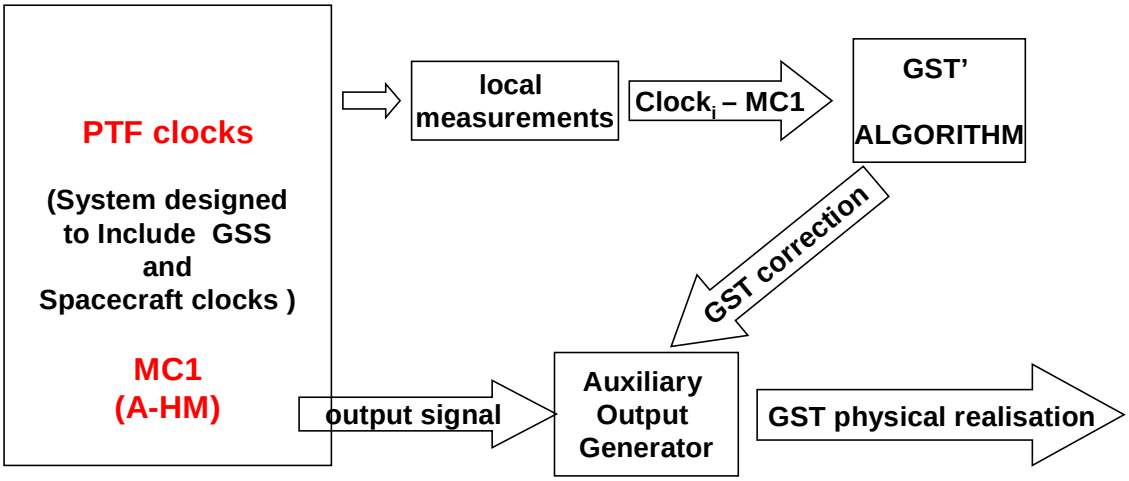


Figure 103 : Overview of the GST generation inside PTF for each GST(MC)-PTF lane (redundancy not shown)

The concept of the Metrological time keeping (GST(MC) to UTC (modulo 1 s) steering) is derived from the Navigation timekeeping described above. The PTF internally generated correction (outcome of the GST algorithm) is combined with the steering correction provided by TSP.

Note that:

- PTF contractors during IOV (CTT&KT) recommended a different steering approach to be implemented in IOV. CTT PTF thus applies either the UTC correction or the internal one. On the other hand, KT PTF proposed to always apply the internal correction and UTC correction is applied separately after the internal one.
- CTT PTF represents the baseline PTF for IOV and FOC. This steering correction is generated based on the TSP knowledge of the GST(MC) versus UTC (modulo 1 s) performance and its evolution. The TSP generates the steering correction with the aim to keep the GST to UTC (modulo 1 s) offset within a specified range as required, under the assumption of the correct implementation of the correction by PTF.

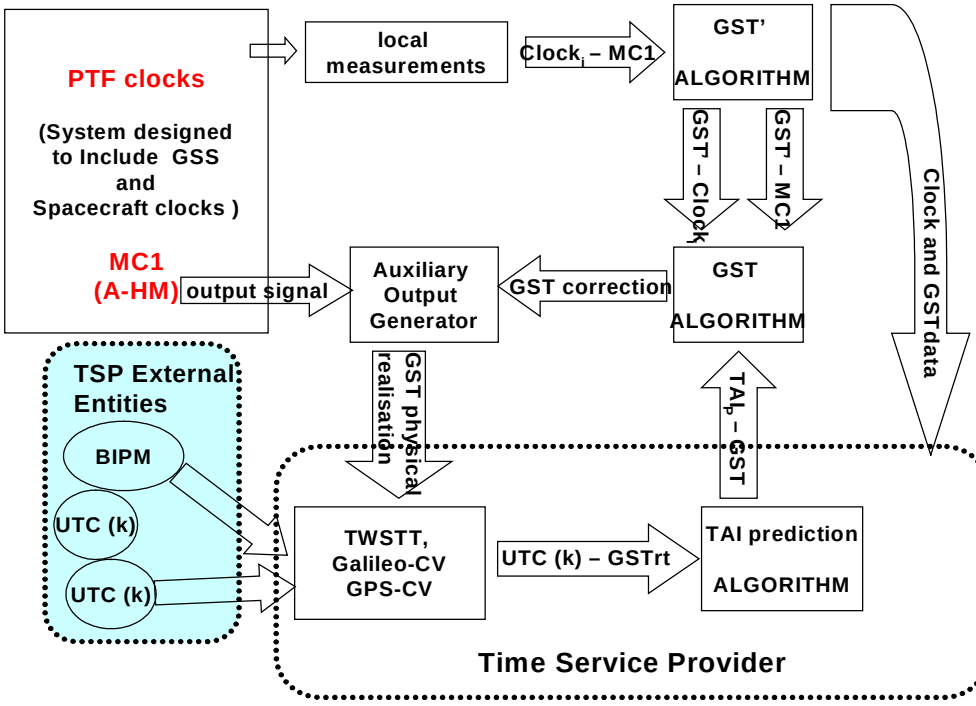


Figure 104 : Overview of the Master GST(MC) to UTC (modulo 1 s) steering concept (steering of Back Up GST (MC) to Master GST(MC) not shown)

6.2.3.3.3 Galileo-CV: Galileo Satellite Constellation (SSeg)

The Galileo OS SIS (E5a E5b and E1C) transmitted by the Galileo constellation are collected by Combined Galileo-GPS Receivers inside the PTF and TSP/UTC(k), to support through Common view the determination of the UTC to GST(MC) offset and PTF-PTF synchronization

The collected pseudo-range measurements and navigation messages are following their post processing (in line with the Common View practice) used for the comparison of the distant clocks.

Notes :

- The Galileo SIS observable collected through the Combined Receiver inside the PTF are also used for GGTO determination with the combined Rx method, please refer to SFC2.2.
- The implementation of a combined TWSTFT/CV time transfer blended initially foreseen in Galileo has been disregarded.

6.2.3.3.4 Galileo-CV: PTF/GSS

Two PTF-GSSs (PTF-GSS1 and PTF-GSS2), 1+1 redundant, each one including a GRC-P and a GRC-N, are deployed at each GCC site.

The PTF-GSS1 and PTF-GSS2 are driven by an associated GST physical realisation, i.e. GST(MC1) and GST(MC2), respectively.

The implementation of 1+1 redundant PTF-GSSs and GST physical realisation is aiming to improve the redundancy level of the GMS. The design of PTF at each GCC site is carried out to allow to connect and to manage up to two co-located GSS in a way to avoid common mode failures that would result in a reduction of the redundancy level provided by the second co-located GSS.

The PTF-GSSs are not equipped with the RAFS as the remaining GSSs, instead they use directly the associated GST(MCx) as a reference.

The GSS raw measurements are processed with respect to the master GST physical realisation time reference (GST(MCx)) provided at PTF-GSSx input. GST(MCx) reference is represented with the GST epoch by means of a one-pulse-per-second (PPS) signal with calibrated phase stabilised cables and the GST(MC) frequency by means of a precise 10 MHz signal.
The PTF-GSSx receiver measurements are automatically calibrated with respect to GST(MC).

The raw measurements coming from the Master GST(MC)/PTF-GSS lane are used by the OSPF as a reference for the computation of the orbit and satellite clock corrections data broadcast in the naviation data.
In addition, raw measurements coming from Master GST(MC)/PTF-GSS lane and the 3 GST(MC)/PTF-GSS lanes are used by the OSPF for determination of the GST transition law in case of PTF-GSS Switch as indicated in section 6.2.3.3.14.

Each PTF-GSS hosts one GRCP and one GRCN receiver. GRCN is in charge of processing the non-PRS signals (i.e. the F/NAV-related signal E5a, the I/NAV-related signals E5b and L1-B&C, and the C/NAV related signals E6-B&C, and the PRS GRC (GRCP) is in charge of the G/NAV-related signals E6-A and L1-A.
The OSPF processes the timing observables from each receiver in the own relevant process (either the PRS or the non-PRS process).

Each PTF-GSSx, x=1 or 2, is equipped with one PTF Calibrator Set in each GRC and with an antenna coupler (with 2 inputs) supplied for RF calibration signals from GRCN and GRCP calibrators. Indeed, the raw measurements correction precision must be warranted, and their correction stability insured: the synchronization requirement of 4.5 ns (1 sigma) implies the need to “cancel” the associated GRC/PTF hardware group delays and to synchronise the satellites signal arrival instant time at antenna level onto the associated GST(MCx), x=1 or 2, time.
A calibration signal is thus generated externally (to reduce the impact on the standard GRC design), synchronized onto the 1 PPS supplied by the associated PTF, and transmitted to the relevant GRC-PTF through the RF channel in a way to share the largest common GRC RF channel with satellites signals received at antenna level. The calibration path is always present and connected in the PTF antenna (it is injected through two separate connectors, one for non-PRS signals and one for PRS signals); a single coupler is used to mix the calibration signals from GRCN and GRCP and the RF signal from the antenna.
The GSS PTF recurrent calibration is provided in automatic way, such that no control inputs from operators are required in routine operations.

The calibrator generates a calibration signal similar to Galileo signals, and a pulse, synchronized on the calibration signal. These signals are repeated every second. A counter is dedicated to the measurement of the time interval between the 1PPS PTF (which is the time reference) and the calibration 1PPS. This allows to know the time at which the calibration signal is emitted toward the antenna at uplink cable input with respect to the 1PPS PTF pulse. Then, with prior laboratory calibration, the uplink cable propagation group delay can be gauged out. Hence the time in associated GST(MCx), x=1 or 2, time at which the calibration signal arrives at antenna level is known exactly. When a measurement reading occurs in GRC-PTF receiver, it is possible to have access to satellite and calibration signals code phases or time difference. Since the time in the associated GST(MCx), x=1 or 2, time at which the calibration code phase measured in the receiver arrived at antenna phase centre is known, and since the downlink HW components are shared by both satellite and calibration signals, it is possible to recover the arrival instant time at antenna phase centre level - in GST(MC) time - of the satellite code phase measured in the receiver. Then, the corrected raw measurements need to be computed as well as the measurement dating expressed in associated GST(MC) time.

Finally, the PTF cables - i.e. the additional cables needed for the GSS/PTF configurations- need to be calibrated with a very high precision, i.e.: the cables from PTF to the Clock and 1PPS splitters (to deliver the very precise clock and 1 PPS hardware signals), the cables from the splitters to the GRC or the Calibration Set, the cables internal to the Calibration Set and the cable from the Calibration Set to the RF Pre-Amplifier (i.e. the cable to uplink the RF Calibrator signal to the Antenna sub-system).

6.2.3.3.5 GPS-CV: GPS Satellite Constellation

The GPS-SiS observables on L1 and L5 are collected by Combined Galileo-GPS receivers inside PTFs and by the GPS receivers inside the TSP/UTC(k)to support through Common view the determination of the UTC to GST(MC) offset at TSP.
The collected GPS pseudo-range measurements and navigation messages are following their post processing (in line with the Common View practice) used for the comparison of the distant clocks.

Note :
The GPS SIS observable collected through the Combined receiver inside the PTF are also used for GGTO determination, please refer to SFC2.2.

6.2.3.3.6 TWSTFT: Geo-stationary Communication Satellite

Geostationary commercial communication satellites are used for the retransmission of RF signals carrying timing information required for the TWSTFT (Two Way Satellite Time and Frequency Transfer) between the TSP/UTC(K) laboratories, GPS/USNO and PTFs.
In particular, TWSTFT is used likewise the GPS/Galileo-CV for the comparison of the distant clocks, for master PTF - slaved PTF synchronization and master GST(MC) to UTC steering at master PTF.
TWSTFT is also used for GGTO determination, please refer to SFC2.2.

6.2.3.3.7 GST(MC) to UTC Steering Generation: TSP/UTC(K)

The TSP/UTC(k) laboratories are the timing laboratories, the ones selected by the TSP to assist TSP with the generation of the GST to UTC (modulo 1 s) steering corrections. TSP has to rely on the external entities such as UTC(k) laboratories as TSP is a processing facility and as such does not possess any time transfer hardware, which is used for comparison of the timescales. The time transfer results obtained from Galileo-CV, GPS-CV and TWSTFT measurements carried out at TSP/UTC(k) and master PTFs are fed by TSP into the GST(MC)-UTC steering algorithm.

6.2.3.3.8 GST(MC) to UTC Steering Generation: BIPM

Bureau international des poids et mesures, the organization responsible for the international standards of measurements. The BIPM time section is responsible for the UTC calculation and dissemination. The TAI/UTC information is available every month via so called Circular-T bulletin.

6.2.3.3.9 GST(MC) to UTC Steering Generation: TSP

The Time Service Provider is the service facility for Galileo in charge of the metrological timekeeping of the GST. This task is fulfilled by supplying from the Master TSP to the master PTF the steering corrections to keep GST to UTC modulo 1 second offset within the specified range. In order to generate the steering correction, TSP needs to process the time transfer and clock data provided by the master PTF, and BIPM /UTC(K) laboratories. Overall TSP provides the following products:

- GST(MC) to UTC modulo 1 second & TAI to UTC offsets (leap seconds), used by MSF, MUCF for the mission monitoring and by OSPF for the generation of the navigation batches
- Clock anomaly data: report on the performance and anomalies of the individual clocks from the PTF ensemble
- UTC frequency correction: frequency correction to be applied by PTF in order to steer GST(MC) towards UTC modulo 1 s

Following text describes the OSPF performed translation of the TSP generated products into the UTC dissemination timing parameters which are included in the OS and PRS navigationmessages:

- TAI-UTC offset (leap seconds):
  GMS in order to fill in the message field for GST-UTC, see [AD.6], SISICD-1423 derives from the TSP products the following information:
  - GMS will compute the GST in WNlsf, DN converted into GST from "nextUpdate" date received in MJD, with:
    - o deltaTls = "currentOffset" in sec,
    - o deltaTlsf = "currentOffset"+"nextChange" in sec

  To perform the MJD to GST correspondence, the following definition shall be applied:
  - The GST start epoch is as defined in [SISICD], SISICD-1481.
  - The Modified Julian Day or MJD is defined as: MJD= JD - 2400000.5
    so that MJD = 0.0 corresponds to November 17.0, 1858. The MJD begins at Greenwich mean midnight and is sometimes used in reference to artificial satellites.

The Julian Day Number, Julian Day, or JD of a particular instant of time is the number of days and fractions of a day since 12 hours Universal Time (Greenwich mean noon) on January 1 of the year -4712, where the year is given in the Julian proleptic calendar.
The conversion algorithm is defined on the following pages:
<<http://www.pdc.ro.nu/jd-code.html> or
<<http://aa.usno.navy.mil/faq/docs/JD_Formula.html>
Example: YEAR = 1970, MONTH = 1, DAY = 1, JD = 2440588.

- GST-UTC modulo 1 second offset:
  The GMS will use the GST-UTC offset data (TIME_GstTaiOffset) supplied by the TSP to fill the user message defined in SISICD 1423 [AD-6]. The mapping between the parameters provided by TSP and parameters defined in SISICD1423 is as follows:
  - SISICD1423 “A0”= “offset” in steering condition = predicted offset between GST and UTC modulo 1 second calculated by the GTSPF .The offset is given in ns.
  - SISICD1423 : “A1”= "offsetfreq" = normalised frequency offset between GST and UTC modulo 1 second, the normalised frequency offset is given in 1E-16 s/s.

  Conversion from the MJD to GST shall be done in the same way as it is done for TAI-UTC offset.

  - at start, the GMS will force values for A0 and A1 to 0, the date being the current GST as known in GMS when it starts.
  - any new received GST-UTC modulo 1 second offset will supersede the forced values,
  - in case no GST-UTC offset products from TSP after the 24 hours and up to 10 days , the GMS will provided the GST-UTC conversion parameters (including leap second information) by using the information included in the last TIME_GstUtcOffset signal received from TSP.
  - in case of lack of GST-UTC offset products from TSP for more than 10 days, the GMS will provides the GST-UTC conversion parameters including valid leap second information, but with Ao and A1 coefficients of GST-UTC offset set to their dummy values (“binary all ones”), to indicate that the unavailability of GST-UTC offset.
    The dummy values of GST-UTC offset coefficients correspond to the following decimal values:
    - Ao=-2^{-30} s=0.931322574615479 ns
    - A1=-2^{-60} =8.88178419700125x10^{-16}

Details on the application of GST-UTC offset at GMS level and dissemination of dummy values to the user are given in SYS-RD.71.

6.2.3.3.10 Galileo Data Provision to TSP and Vice Versa: SPF

Service Product Facility is serving as a secure interface between GMS and the service facilities. Any data (incoming/ outgoing) between GMS and external interfaces, for instance TSP, have to be routed via the SPF. SPF handles also an internal distribution of the data inside GMS. TSP data are routed from SPF to GACF & MSF.

The SPF connects to the TSP (Master or Slave) for GMS Time data products distribution since both TSP are running. On the other hand, only the Master TSP connects to the local SPF to supply the TSP time data products to the SPF.

The SPF provides a configuration file (CNF_SvMask) including a correspondence table to allow TSP the mapping between Galileo satellites SVID and CCSDS Identifier.
The file is sent on event, i.e whenever there is a change in the constellation or in case of test involving a satellite

6.2.3.3.11 Mission Online Monitoring: MUCF

In general, MUCF is in charge of mission Planning (mid/long-term for Galileo and short-term for GMS), C-Band uplink scheduling including distribution of schedules mission On-line Supervision, on-line prediction and a posteriori assessment.

Mission online supervision/monitoring:

- MUCF detects PTF failures that imply a clock switching using PTF and OSPF mission monitoring data
- MUCF is set up to detect an operational GSS failure - that prevents the provision of Galileo raw measurements towards based on the alerts raised by GSS and OSPF.

Mission planning:

- Part of the Mission and Uplink Control Facility (MUCF) is dedicated to control the mission on a high level through generation of plans by amongst of other Generating the maintenance planning of the full system, thus including elements such as PTF

Mission assessment:

- Mission Uplink Control Facility is in charge of the mission online monitoring. MUCF uses the information on the PTF internal clocks and TSP steering parameters with the GST-UTC, TAI-UTC data retrieved from the GACF for the GST and mission online monitoring.

6.2.3.3.12 Mission Offline Monitoring: MSF

MSF is an off-line element mainly in charge of producing OSPF configuration data (satellite models calibration, Historical SISA…) and generating and/or validating element configuration files (based on external input data, data generated by GCS or by MSF itself).

Furthermore, MSF is used for routine generation and validation of calibration and configuration parameter upgrades for immediate use on the real time system.

MSF also in charge of the mission offline monitoring, providing within the scope of the SFC2.1 the following functionalities :

- Uses TSP generated products (GST-UTC, TAI-UTC and UTC frequency correction data GST monitoring report), and satellite clocks data (OSPF) for the offline analysis of the GST performance(TBC)

6.2.3.3.13 Validation of TSP Products, PTF Monitoring: GACF

Ground Asset Control Facility is in charge of the GMS element control & monitoring and the master archiving of the mission data. Furthermore GACF is within the scope of SFC2.1 also responsible for following operations:

- The integrity check of the TSP data (TAI-UTC & GST-UTC offsets) required by elements such as PTF and OSPF.
- Control/commanding and monitoring of the PTFs. The GACF controls the operating mode of both PTFs, including the selection of master PTF site, the selection of Master GST physical realisation at each PTF site, PTF switching, conversion of the TSP products provided though SPF (TAI-UTC & GST-UTC offsets and UTC frequency correction) into configuration parameters to be implemented by PTF.
- Monitoring and control of redundant PTF/GSS at each PTF site.
- Control/commanding of OSPF and MUCF conversion of the TSP products (TAI-UTC & GST-UTC offsets) into configuration parameters to be implemented by OSPF&MUCF.

6.2.3.3.14 GSS&Satellite Determination/Navigation Batch Generation: OSPF

Orbit and Synchronization Processing Facility is in charge of orbit, GSS & satellite clock determination and the generation of the navigation batches for OS and PRS services. OSPF processes the GSS raw measurements, provided that GST(MC)/PTF-GSS lane configured as master, at the Master PTF to produce the estimates of satellite and GSS clock offset w.r.t. GST(OSPF). The Satellite clock offsets generated by the OSPF can be optionally used by the PTF in the GST generation process, to achieve and extended clock ensemble for GST determination at PTF. This capability is not implemented in the current design, where GST is determined using only PTF clock ensemble. TSP data checked by GACF (TAI-UTC & GST-TAI offsets) are used by OSPF to generate navigation batches (see also section 6.2.2.3.9) as per SIS ICD [AD.6].
The OSPF computes Satellite Clock Correction Parameters by fitting the satellite clock offset estimations to a certain model and for a prescribed time window: those parameters need to be computed with respect to the Galileo System Time as seen by the OSPF (GST(OSPF)). The GST time reference is "*seen*" by OSPF by using the Galileo SIS observables obtained from the GCC-located GSS, please refer also to section 6.2.2.3.4.

The OSPF restitutes GST(OSPF) with high stability by processing the Galileo raw measurements performed by GST(MC)/PTF-GSS lane configured as master in the Master PTF site.
The OSPF is notified by GACF of which PTF site and which GST physical realisation GST(MCx) within that site is assigned as master. Then, the OSPF "locks" on the master GST(MC) by utilising as reference the Galileo raw measurements obtained through the associated master PTF-GSS driven by the master GST(MC). This leads to the determination of the so called "master" GST(OSPF), that is the GST "seen" by the OSPF through Master GST(MC)/PTF-GSS lane. The master GST(OSPF) is used to determine the satellite clock corrections broadcast in the navigation message.

In addition, the OSPF processes the Galileo raw measurements provided by the remaing back up GST(MC)/GSS-PTF lanes, namely

- Slaved GST(MCx)/PTF-GSSx lane at Master PTF site

- Master GST(MCx)/PTF-GSSx lane at Slaved PTF site
- Slaved GST(MCx)/PTF-GSSx lane at Slaved PTF site

to simultaneously determine the offset of the GST seen by the OSPF on those back up lanes (that is GST(OSPFx)) with respect to the master GST(OSPF).

The offset values between GST(OSPFx) and GST(OSPF) depend on the bias inside the relevant PTF-GSSx and the path from GST(MCx) and PTF-GSSx used to inject the GST frequency and time reference signals. The determination of such offsets insided the OSPF is required to implement a proper GST transition law in case of switch of the master GST(MC)/PTF-GSS lane (see also below).

The OSPF pre-processes and validates the Galileo observables every 10 min, the predicted satellite orbits predictions and satellite clock biases estimation are produced by means of an OD&TS Long process (arc length of 48 hours) and an OD&TS short process (arc length of 6 hours). Using those results, the orbit model parameters, the clock prediction parameters, and sets of almanac parameters are generated for all the defined services.

Due to the use of undifferenced pseudo-range measurements, clock errors are introduced in the orbit determination process. This leads to perform both orbit determination and time synchronisation at the same time. To understand the way clocks are handled inside the OD&TS process, the following aspects should be taken into account:

- Time synchronisation process is devoted to estimating the clock differences of all stations and satellites with respect to the system reference clock, which will be based on a master station clock
- The use of undifferenced pseudorange and carrier phase measurements introduces clock values in the process. Clock uncertainties are introduced in the OD&TS filter by means of the measurement equation. Thus, in the selected approach, orbits and clocks are solved for simultaneously.
- Parameter observability is enhanced by the use of undifferenced measurements. Besides, the low noise of carrier phase measurements allows making accurate estimations.
- High accurate dynamic models are used for all forces acting on the satellite.
- Clock values are obtained at every measurement epoch.
- OD&TS parameters are estimated using an weighted least square algorithm with a priori information.
- The clock estimation depends on the pair of frequencies used in the OD&TS process, so a dedicated estimation process is needed for each pair of frequencies.

For details on algorithms used in the OSPF computation of the clock correction parameters, please refer to the ALG NAV Algorithms Detailed Processing Model Volume 2 [GMS-RD.55].

High level definition of the OD&TS algorithms is also provided in the GMS DJF [GMS-RD.3].

A switch of the (master) GST(MC)/ PTF-GSS lane occurs when any one of the following event occurs :

- Failure of PTF-GSS configured as Master or

- Failure of master AHM inside the Master PTF or
- Failure of the whole master PTF

Having two co-located PTF-GSS's at each PTF means that two PTF-GSS measurement dataflows are generated per each PTF. The pseudo range measurements of those PTF-GSS's, even if referenced to only one PTF GST physical realisation, are affected by a bias value that is specific to each PTF-GSS. Therefore this is seen by OSPF algorithms as an additional GST physical realisation, and so the OSPF software needs to be designed to be able to cope with four PTF-GSS measurement dataflows .
A priority order is established in the OSPF configuration file for the selection of the PTF-GSS measurement dataflow to be used as reference for master GST(OSPF) computation In particular, in case the measurements originating from the PTF-GSS currently used by OSPF as master disappears, then the OSPF automatically propagates the existing PTF reference time by using the measurements of the redundant PTF-GSS connected to the same (master) PTF, if those measurements are still available.
If this is not the case, then the OSPF propagates the PTF reference time by using the measurements performed by the PTF-GSS driven by the master GST at remote PTF which becomes the new master PTF.

In order to mitigate the impact on GMS availability following a GST(MC)/PTF-GSS switch, the OSPF generates the clock and orbit products based on the predefined transition law, which is compensating for the difference between the timescales as generated by Master GST(MC)/PTF-GSS lane and back up GST(MC)/PTF-GSS lanes. The effect of the transition law in the case where the raw pseudorange measurement dataflow is available at the output of the master GST(MC)/PTF GSS lane at Slaved PTF site is is outlined in the figure below, which shows the transition from Master PTF to Slave PTF produced by a failure in the GST(MC)/PTF-GSS lane currently configured as master at the Master PTF.

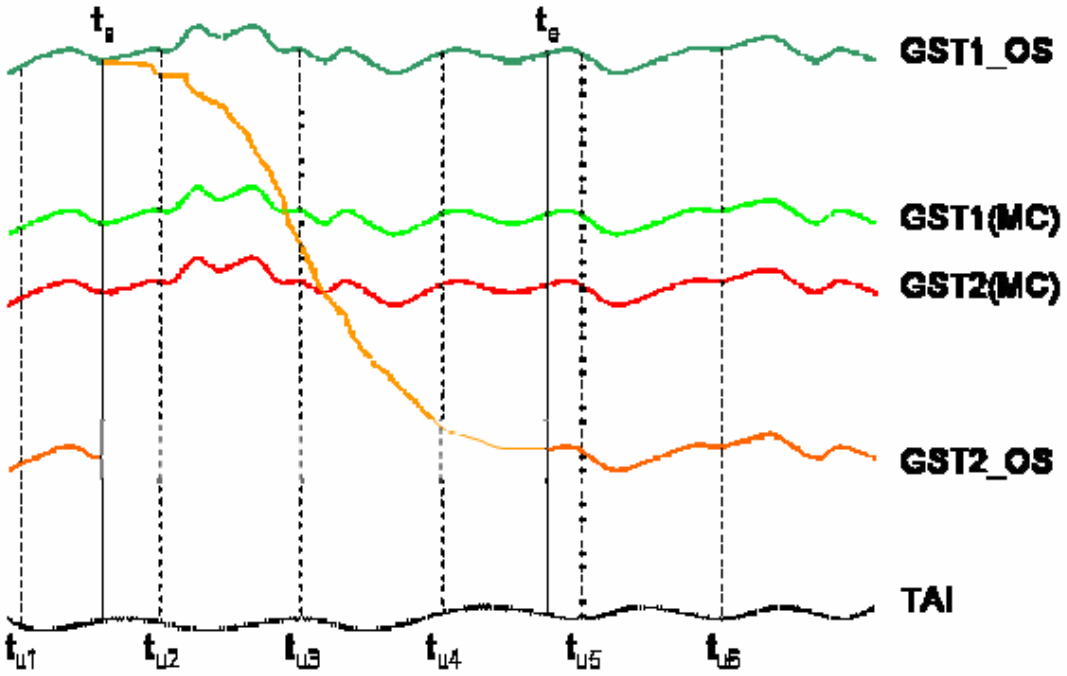


Figure 105 : Transition from Master to Slave PTF triggered by a fail-over switch due to a failure in the GST(MC)/PTF-GSS lane configured as master at the Master PTF.

Legend:
GST1(MC): Physical realization of the GST at the output of the Master AHM at the Slave PTF. In the past the physical realization of the GST was considered at the output of the Master AHM at Master PTF, hence the abbreviation GST(MC).
GST2(MC): Physical realization of the GST at the output of the Master AHM at the Master PTF. In the past the physical realization of the GST was considered at the output of the master clock, hence the abbreviation GST(MC).
GST1_OS: GST1 from the Master AHM at the Slave PTF as seen by OSPF through the processing of the Galileo SiS (Open Service)
GST2_OS: GST2 from the Master AHM at Master PTF as seen by OSPF through the processing of the Galileo SiS (Open Service).

Following the master PTF-GSS failure, the OSPF sees the GST1_OS instead of GST2_OS. As pointed out above to compensate for the difference between the GST1_OS1 and GST_OS2 (to minimize the impact of the PTF switch to the navigation services), OSPF applies the so called transition law. The difference between the GST1_OS and GST2_OS is compensated for smoothly over the predefined transition time.

6.2.3.4 Dynamic Scenarios

Following scenarios have been identified:

- SFC2.1-1: GST Generation Nominal Scenario (Metrological Time keeping): The interface of Master PTF with the TSP is operating according to the interface specification with the TSP products meeting both the required scope and the quality. The Master PTF operates in its nominal mode and generates 2 GST physical realisations (GST(MC1) and GST(MC2)) to support the navigation services. The GST physical realization, configured by GACF as

master (GST(MC)), is steered towards TAI using the TSP generated steering corrections and TSP products are converted into GST-UTC conversion parameters to support the UTC dissemination service

- SFC2.1-2 :GST Generation Contingency Scenario (Navigation time keeping): PTF operates in its nominal mode and generates stable time reference (GST) to support the navigation services. The Master GST(MC) at the Master PTF is however not steered to TAI either because of the malfunctioning of the interface with TSP or because of the rejection of the GST(MC) to TAI steering correction by PTF to protect the navigation services. OSPF generates the GST-UTC conversion parameters however their required performance is not guaranteed.

- SFC2.1-3:Contingency Scenario: GST(MC)/PTF-GSS lane switching due to failure in the Master GST(MC)/PTF-GSS lane at master PTF site: the scenario description is per the metrological timekeeping. During operations, the command for GST(MC)/PTF-GSS lane switching is issued by GACF, based on the detection a critical failure in the GST(MC)/PTF-GSS lane currently configured as master. Following the role exchange between the Master and Slave GST(MC)/PTF-GSS lane, Galileo System Time is derived from a newly appointed Master GST(MC)/PTF-GSS lane timescale. Following the GST(MC)/PTF-GSS lane switch, OSPF generates the clock and orbit products based on the predefined transition law to minimize the impact of the GST(MC)/PTF-GSS lane switching on the Galileo navigation services,
- SFC2.1-4: Leap Second/ERP Configuration Update Scenario:
  This scenario is run on weekly basis with following objectives
  - To check whether the leap second files have to be have to be reconfigured in GMS elements, and eventually do it if applicable;
  - To reconfigure GMS elements with new ERPs, once validated.

The update of the leap second value can occur only up to twice a year (i.e. at the end of the 30th of June and the 31st of December). It is assumed that TSP announces such update at least five months in advance (i.e. an updated TIME_TaiUtcOffset file is received by the 1st of February to announce a leap second change on the 30th of June, or the 1st of August to announce a leap second change on the 31st of December).
It is also assumed that, following the leap second updated announcement, the TSP provides an actualized leap second file.
New ERP values are delivered by GSRP to GMS on a daily basis. These ERPs are generally assumed to be correct, but in order to provide additional validation, the GMS computes the differences between old and new ERPs and compares it to a threshold.

6.2.3.4.1 #SFC2.1-1: GST Generation Nominal Scenario (Metrological Time Keeping)

The MSC for scenario this is provided below:

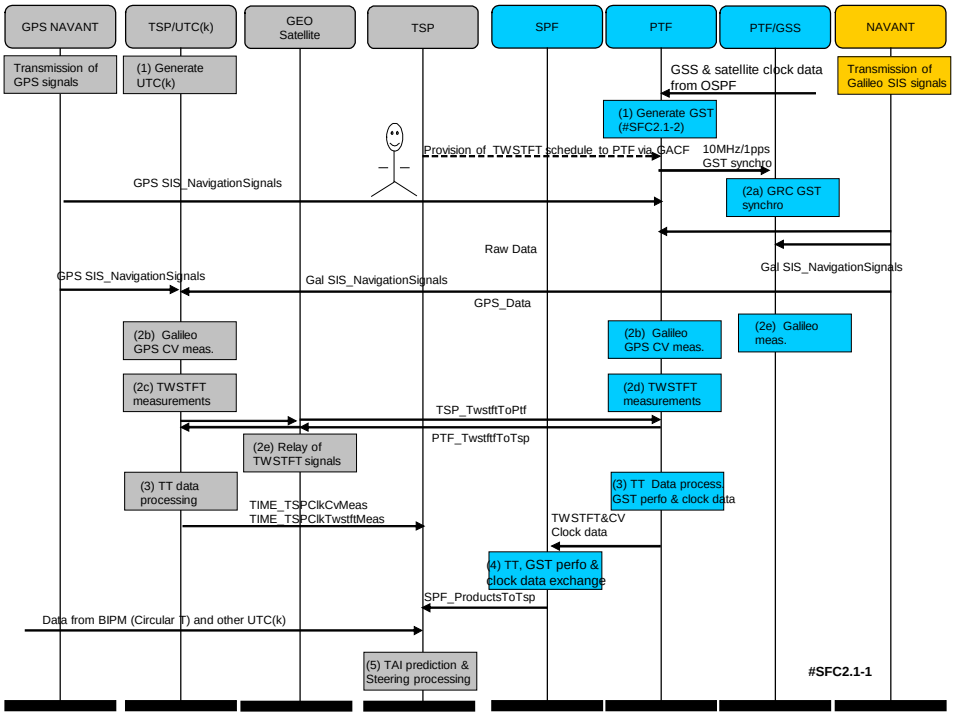


Figure 106 : #SFC2.1-1: GST Generation Nominal Scenario (Metrological Time Keeping)

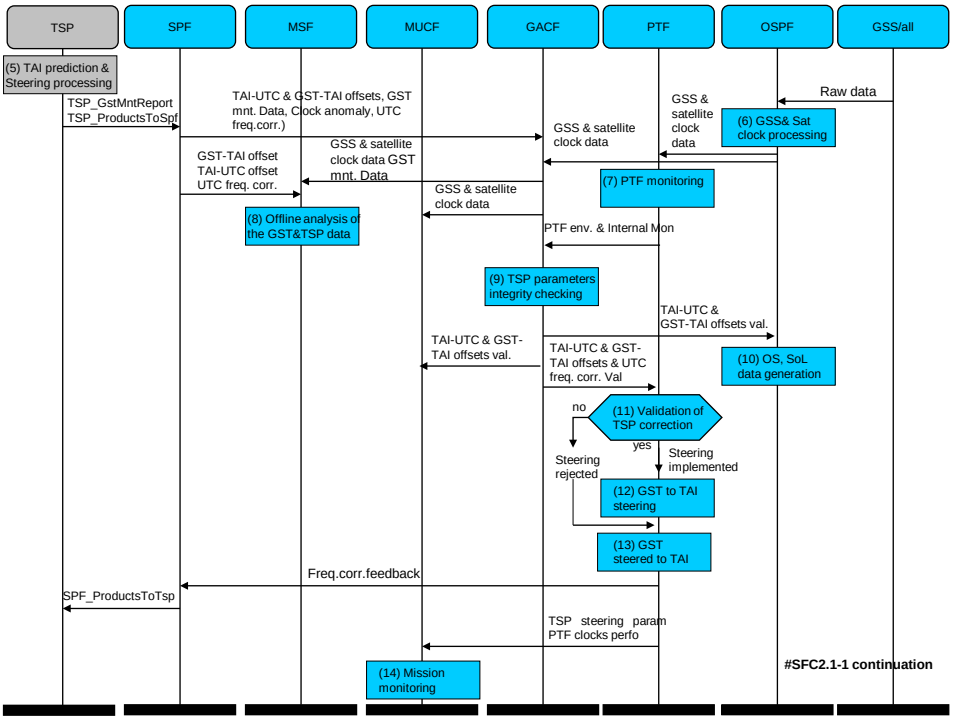


Figure 107 : #SFC2.1-1: GST Generation Nominal Scenario (Metrological Time Keeping) continuation

Scenario Textual Description:
In this mode the Master GST(MC) as generated by the Master PTF is steered towards the UTC modulo 1 second using the steering corrections provided by external Time Service Provider.

The back-up GST physical realisations as generated by the master and slaved PTF's are then automatically slaved to the Master GST(MC) which is steered toward GST(MC).
The aim is to generate the stable GST timescale of which the time offset is kept within the close agreement to UTC.
Galileo system by implementing the Metrological timekeeping will allow Galileo users to utilize the system not only for the navigation, but also for the precise time dissemination. Galileo users will be able to recover UTC through the reception of the Galileo SIS. (Galileo receivers will derive the UTC from the GST and UTC steering information included in the navigation messages).

Note: The navigation functionality of the system must be preserved by providing the necessary GST stability; consequently the steering corrections need to be verified by PTF before they can be applied to ensure that the navigation functionality and its performance are not endangered.
In particular, the steering of GST(MC) at Master PTF to UTC modulo 1 second is performed after validation by GMS of the steering correction provided by TSP.
The steering corrections are applied 1:1 after validation by GMS. Should the received steering correction be larger than a configurable threshold, it is saturated to a threshold value and still implemented. More precisely, the steering is estimated to be not larger than the 24h stability of the master AHM, i.e. ca. 1 to 4E-15 (stochastic component); as a maximum limit, a threshold of 1E-14 is implemented.
So, the Master AHM steering correction saturation has to be understood as follows: in case the TSP steering correction exceeds the PTF AHM steering threshold of 1E-14, the PTF sets the correction to be applied to the threshold value with the sign of the steering correction provided by TSP. This saturated correction is reflected in the signal TIME_FreqCorrApplied sent to the TSP.

The PTF baseline foresees also two additional checks: if the TSP correction ADEV or the difference between the TSP correction and the one computed vs. PTF Cs ensemble exceed configurable thresholds, a warning to GACF is raised, but the corrections are still implemented. PTF is required to report back to TSP on the steers performed to ensure the optimal configuration of the TSP steering algorithm.

Scenario summary:

1) Autonomous generation of the stable GST timescale by PTFs
2) 2 Selection of master PTF site and master GST physical relasisation within both PTF sites. The master GST physical realisation at master PTF site is herunder indicated as GST(MC)
3) Comparison of the GST(MC) against the TSP/UTC(K) timescales to assess the performance and the evolution of the GST versus TAI needed by TSP to generate the steering corrections.
4) Provision of PTF products to TSP for determination of GST(MC) steering to UTC (modulo 1s)
5) Provision of TSP products to GMS.
6) Implementation of the TSP products by PTF, OSPF, MUCF & MSF after their validation by GACF
7) GST(MC) steering to TAI and generation of GST to UTC conversion parameters for the inclusion into the OS and PRS navigation batches. The back-up GST physical realisation generated by the Master PTF is automatically slaved to GST(MC) through an appropriate

GST steering algorithm. Also, the GST physical realisations generated at slave PTF site are slaved to GST(MC) through CV & TWSTFTT links and GST steering algorithm implemented in the slave PTF.

8) Use of the GST and TSP products for the purpose of the mission monitoring by MSF and MUCF.

After configuration by the GACF of the master PTF site and master GST(MC) within that site, the sequential description is detailed below:

1) GST(MC) is generated by the Master PTF clock ensemble which consists of 5 Cesium frequency standards and Master A-HM.
The capability to include the satellite clock offset in the clock ensemble used to determine GST(MC) at PTF is not implemented in the current design.
For more details please refer to scenario #SFC2.1-2. Due to the characteristic properties, of each type of these clocks the best performance can be achieved when they are combined together. The stable reference timescales steered towards TAI are generated also by the national standard timing laboratories all around the world. Each of these laboratories is maintaining so called local/national realization of the UTC.

A number of TSP/UTC(k)laboratories will be used by TSP to assess the 'real time' performance of the GST(MC), representing a necessary input to the TSP steering algorithm.

2) To compare the GST(MC) against other stable timescales as realized by TSP/UTC(k) three time transfer techniques will be used:

- Two Way Satellite Time and Frequency Transfer (TWSTFT)
- GPS Common in View (GPS-CV)
- Galileo Common in View (Galileo-CV)

The three time transfer data will be used by the TSP for the determination of GST(MC) to TAI offset and GST(MC) steering to TAI (modulo 1 s) steering.
All three of the time transfer techniques will be deployed in parallel, TWSTFT will be used as a primary method whereas GPS-CV and Galileo-CV techniques will serve as a backup.

The Galileo and GPS common view data are collected with the GPS-Galileo Combined Receiver deployed at the Master PTF. Likewise the TSP/UTC(k) will be equipped with the TWSTFT, Galileo and GPS receivers in order to collect the data necessary to compare their timescales against the GST(MC). TWSTFT method is based on the simultaneous exchange of the RF signals over a commercial communication satellite. The RF signals contain a timing information which is derived from the local reference clock. Thus TWSTFT likewise the GPS-CV and Galileo-CV is used for the comparison of the distant clocks (PTF to PTF and GST(MC) to TAI steering).

Note: PTF task is to calibrate the signal paths (1PPS/10MHz signals) between the PTF GST(MC) output and PTF Combined Receiver input and the signal path from Combined Galileo-GPS Receiver Antenna to measurement point.

Notes :

- The Common View data and TWSTFT time transfer data collected at PTF will be also used for the purpose of PTF-to PTF synchronisation.
- The PTF Combined Receiver will also provide GGTO computation (see SFC 3.2)

The raw measurements coming from the PTF-GSS driven by GST(MC) are used by the OSPF as time reference for all other GSS measurements.

TWSTFT measurements will be performed a number of times per day in accordance with the TWSTFT schedule which will be provided by TSP to GACF using an operational interface. GACF will provide the schedule to PTF as a configuration parameters.

3) The following PTF products , as generated within the Master PTF , are forwarded to the TSP via the SPF:

- TWSTFT measurements data, prepared according to the format recommended by ITU.
- GPS measurements data in CGGTTS format (TBC)
- RINEX Navigation and Observation data as generated by Galileo receiver at GSS co-located with the PTF required for the calculation of Galileo -CV results.
- Time Offset between the Galileo System Time (GST) and each internal PTF clock measurements contributing to its physical realization including any steering correction applied to individual PTF clocks (other than GST(MC) steering corrections)
- steering corrections actually applied to the master clock at each PTF

In addition to TWSTFT and GPS-CV time transfer data, the Master PTF provides to TSP via SPF also clock data and information on the GST(MC) performance. This information is required by TSP in order to generate the GST(MC) to TAI steering corrections and other timing products including GST monitoring report and the PTF ensemble clock anomaly data. In parallel the time transfer data are also processed by TSP/UTC(k) laboratories and provided to TSP.

4)Time transfer, GST and PTF clock data is provided to TSP via SPF.
Note : Master PTF generated time transfer data from only half of the dataset required by TSP to calculate the time transfer results, therefore TSP needs to receive also the corresponding time transfer data from the TSP/UTC(k).

5) TSP combines the data from Master PTF, TSP/UTC(k), BIPM and other UTC(k) laboratories to obtain the information on the GST(MC) and TAI performance which is needed to generate the TSP products including the GST(MC) to TAI steering corrections. At the end TSP products are made available for the retrieval by SPF.

TSP sends to the Master PTF via the SPF the following data :

- Frequency steering corrections for the alignment of GST(MC) to TAI, in terms of normalized frequency steering correction and time of application
- The GST(MC)-TAI time offset phase and frequency offset prediction and their uncertainties

- UTC-TAI leap seconds: entire number of seconds to recover UTC time scale information, starting from the provided GST-TAI offset.
- Detected PTF data anomalies (clock and time transfer)
- Results of monitoring the Galileo SIS regarding the validation of GST and UTC dissemination performance

In addition , TSP provides the schedule of TWSTFT performed by the TSP/UTC(k) labs and PTF(s)
SPF passes TSP products (TAI-UTC, GST(MC)-TAI, GST monitoring, Clock anomaly, UTC frequency correction data) to GACF for the data archival. Clock anomaly data are used by GACF for the independent validation of the PTF clocks performance and monitoring.

6)OSPF processes the GSS raw measurements to produce the satellite and GSS clocks estimates. Furthermore the satellite clock models are generated. The clock data is passed via GACF to MSF & MUCF. OSPF also processes the raw measurements collected by the back up GST(MC)/PTF-GSS lanes to determine the GST transition law in case of PTF-GSS switch.

7) PTF provides to GACF the information resulting from both the GACF and PTF self initiated monitoring of the PTF and its internal processes.
Based on this information (out of the scope of this scenario) GACF can issue the PTF switch command, for more information please refer to the scenario #SFC2.1-3

8) MSF retrieves from the SPF: GST(MC)-TAI, TAI-UTC and UTC frequency correction data which are together with the GST monitoring, GSS & satellite clocks data retrieved from GACF used by MSF for the offline analysis of the GST performance and validation of the TSP products.

9) GACF performs formal data integrity checks on the TSP data (GST(MC)-TAI, TAI-UTC and UTC frequency correction) before their usage by elements such as PTF and OSPF.

10)OSPF generates based on the TSP products the UTC conversion parameters which are used for the inclusion into the OS, SoL and PRS navigation messages Note the addition of the UTC conversion parameters into the navigation messages enables Galileo users to utilise so called UTC dissemination service.

11) The Master PTF uses the TSP products (checked for integrity by GACF) as configuration parameters for its internal processes.
The PFT applies the GST-UTC frequency correction supplied by TSP, to steer GST(MC) toward UTC(modulo 1s).
The UTC frequency correction is however validated before it is used to steer GST(MC) towards TAI.
GST(MC) steering toward UTC (modulo 1 s)) can based on the outcome of the validation process : decide to apply, apply partially or disregard the corrections in case their application may affect the provision of the navigation services.

12) If the validation process is successful, the TSP steering correction is directly applied by CTT PTF.

13) GST(MC) steered to TAI is available at the output of the Master PTF. The Master PTF sends to SPF the feedback on the actual frequency correction applied. This information is retrieved from SPF by TSP who uses this data to adjust GST(MC) to TAI steering processing.

14)PTF passes to MUCF the TSP steering parameters, including the information on the PTF internal clocks. MUCF uses this information together with the GST(MC)-TAI, TAI-UTC data retrieved from the GACF for the GST and mission online monitoring.

6.2.3.4.2 #SFC2.1-2: GST Generation Contingency Scenario (Navigation time keeping)

The MSC for scenario this is provided below:

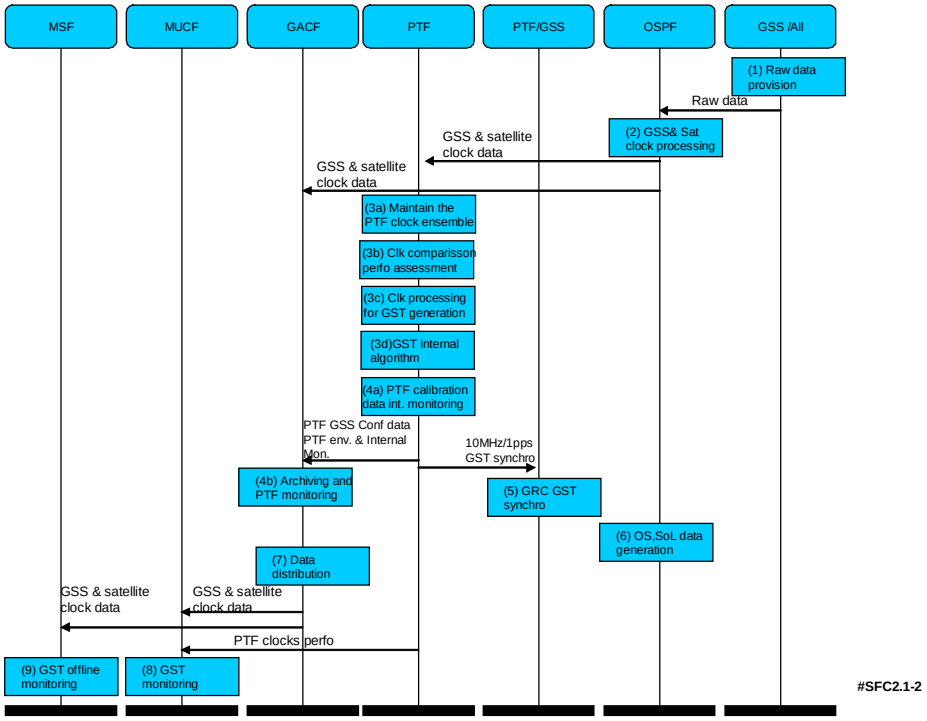


Figure 108 : #SFC2.1-2: GST Generation Contingency Scenario (Navigation time keeping)

Scenario Textual Description:
The aim of the navigation timekeeping is to generate autonomously a stable continuous time scale (so called GST-Galileo System time) to enable the navigation functionality of the Galileo system.
Note for the navigation timekeeping the absolute value of GST-TAI time offset is not crucial as long as synchronization within the system is preserved. As indicated above it is the stability (frequency stability) of the timescale which needs to be ensured to preserve the navigation functionality of the Galileo system.

After configuration by the GACF of the Master PTF (herunder denoted as PTF) and Master GST(MC) within that PTF, the sequential description is detailed below:

1) The network of the globally distributed GSS collects the Galileo and GPS SIS measurements. The GSS raw measurements are provided to OSPF.
2) OSPF processes the GSS raw measurements to produce the satellite and GSS clocks estimates. Furthermore the satellite clock models are generated
Note : in the current design, satellite clock offsets computed by OSPF are not used to augment the clock ensemble for GST determination at PTF
If used, Satellite clock offset data will be also passed to the PTF, where the data can be used for the GST generation. Data is also passed to GACF for their further provision to MUCF& MSF(via GACF)

3) PTF maintains its own clock ensemble formed by Cesium and A-HM clocks and frequency standards. PTF continuously measures the difference between each of the clocks from the PTF clock ensemble. These measurements are processed and fed including the OSPF generated satellite clock data into the PTF internal GST algorithm. The PTF GST algorithm produces the correction to be applied to output of the 2 co-located A-HMs for generation of GST(MC1) and GST(MC2). In this mode (navigation time keeping) the PTFs apply the respective internally generated steering corrections only (By default TSP corrections are not available).

4) PTF task is to calibrate the signal paths (1PPS/10MHz signals) between the PTF GST(MCx), x=1 or 2, output and the GRC (Galileo Receive Chain) input inside the corresponding PTF-GSSx, x=1 or 2. It should be noted that the calibration information is applied by each GSS-PTF when providing the raw measurements to OSPF. In that way, the raw measurements are tagged by the true GST.

The raw measurements coming from the GSS-PTF are then used by the OSPF as time reference for all other GSS measurements.
The PTF provides to GACF also the information resulting from both the GACF and PTF self initiated monitoring of the PTF and its internal processes. Based on this information GACF can issue the GST(MC)/PTF-GSS lane switch command, for more information please refer to the scenario #SFC2.1-3.

5) PTF provides the 1pps and 10MHz reference signals derived from the GST(MCx), x=1 or 2, to the GRC inside the corresponding PTF-GSSx, x=1 or 2.

The data collected by the GRC inside the PTF-GSS configured as master are used by the OSPF as a reference for the orbit and satellite clock calculations, please refer also to the step 2.
The data collected by the GRC inside the PTF-GSS configured as slave at the master PTF site and the GRC's inside the PTF-GSS at the slaved PTF site are used to determine the GST transition law in case of GST(MC)/PTF-GSS lane switch, please refer to the scenario #SFC2.1-3.

6) OSPF generates the satellite clock models and the UTC conversion parameters. These are utilised in the preparation of the OS and PRS navigation messages. Note, by the default TSP parameters are not available in the Navigation timekeeping mode, consequently OSPF has to reuse the data from the last day on which these products were successfully provided to GMS by TSP.

7) GACF passes the OSPF generated GSS & satellite clock data to MUCF who is using the data for the mission online monitoring.

8) PTF passes PTF ensemble clock and GST data to MUCF for the online monitoring of the GST performance.

9) MSF retrieves from GACF GSS & satellite clock data for the offline analysis of the GST performance

6.2.3.4.3 #SFC2.1-3: Contingency Scenario: GST(MC)/PTF-GSS switching due to a Master PTF-GSS failure

The MSC for scenario this is provided below:

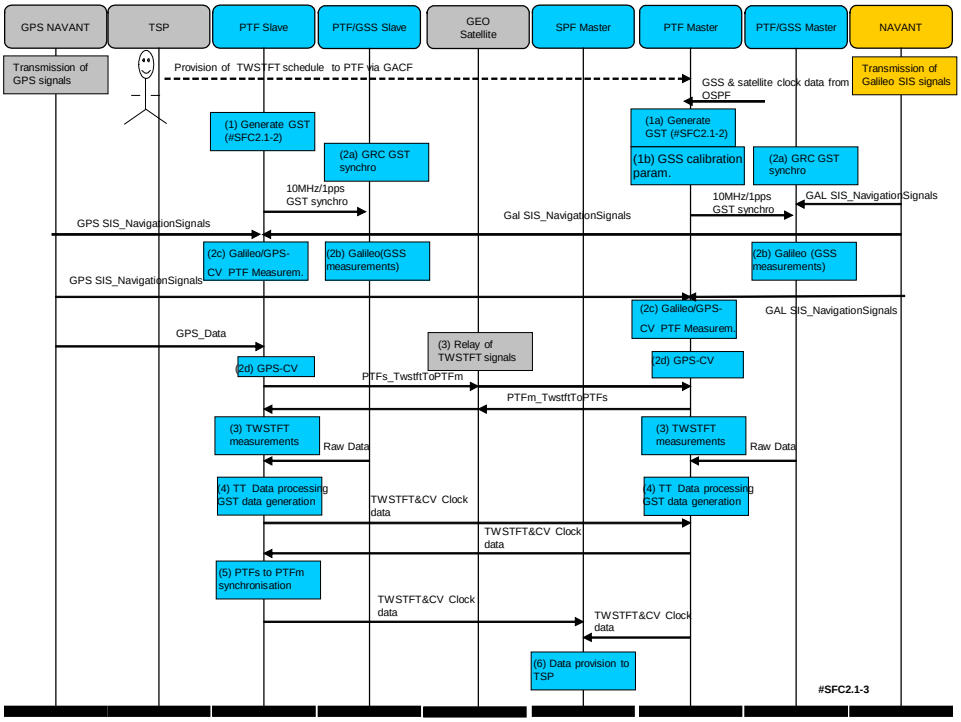


Figure 109 : #SFC2.1-3 Dynamic Scenario (1/3)

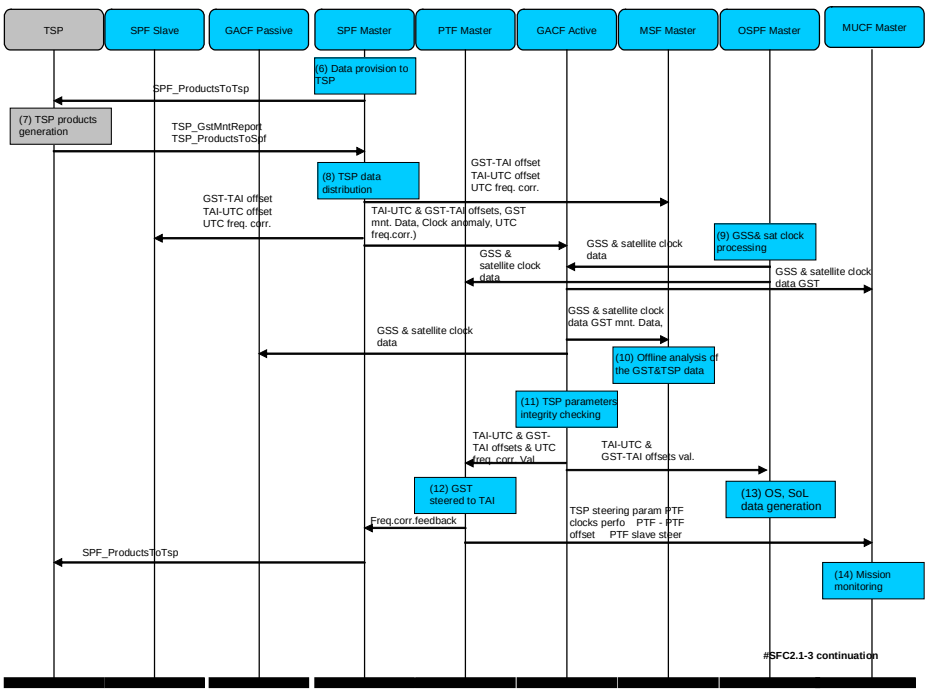


Figure 110 : #SFC2.1-3 Dynamic Scenario (2/3)

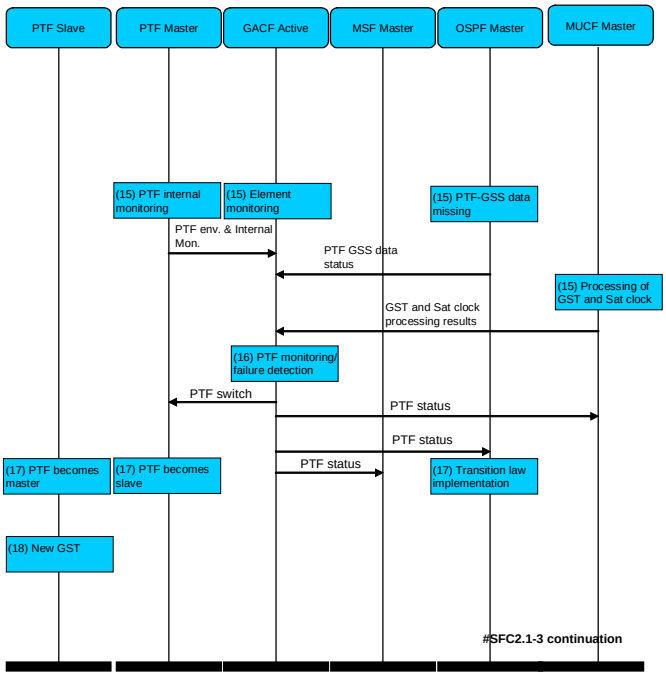


Figure 111 : #SFC2.1-3 Dynamic Scenario (3/3)

Scenario Textual Description:

Due the importance of the GST generation task, and thus the PTF element, there will be two PTFs (present baseline, the number of PTFs might change) working in parallel in the Master-Slave concept. In addition, each PTF generates 1+1 redundant GST physical realisations (GST(MC1) and GST(MC2)), each one driving an associated PTF-GSS (PTF-GSS1 and PTF-GSS2). In total, four GST(MCx)/PTF-GSSx lanes are deployed in the GMS.

The GACF configures the master PTF site and the GST physical realisation within that site to be used as master.
In the case of a failure within the GST(MC)/PTF-GSS lane configured as master, one of the remaing GST(MCx)/PTF-GSSx lanes will become the new Master lane, according to the following priority and depending on the availability of relevant PTF-GSS raw pseudorange measurements:

- GST(MCx)/PTF-GSSx lane co-located with the failed GST(MC)/PTF-GSS lane previously configured as master
- Master GST(MCx)/PTF-GSSx lane at slave PTF site
- Slaved GST(MCx)/PTF-GSSx lane at slave PTF site

In the described dynamic scenario #SFC2.1-3, it is assumed that the raw pseudorange measurements dataflow is not available from the slave GST(MCx)/PTF-GSSx lane co-located with the master GST(MC)/PTF-GSS lane. Consequently, the master GST(MCx)/PTF-GSSx lane at slave PTF site will become master after the detection of a failure within the current master GST(MC)/PTF-GSS lane.
In nominal operations, the master GST(MC) is steered towards UTC (modulo 1 second) using TSP generated steering corrections.
Note that although TSP generates two sets of the GST(MC)-TAI steering corrections as if both PTFs were the Master, the steering corrections are applied only by the Master GST(MC), provided that the navigation functionality is not deteriorated by doing so. The slave GST(MCx) at master PTF site will disregard the TSP steering corrections and will steer itself to the Master to ensure that their respective timescales are kept within the specified time and frequency offset.
The two GST physical realisations generated at slave PTF site will be steered to the Master GST(MC) of the Master PTF by using PTF-to-PTF time transfer measurements (TWSTFT, Galileo & GPS CV) results and the information on the Satellite clocks generated by OSPF to calculate the Slave to Master steering corrections.

The switching from the failed GST(MC)/PTF-GSS lane at Master PTF to the one deployed in the slave PTF will be commanded by GACF based on the results of the PTF monitoring performed by PTF, GACF &MUCF.
The event of GST(MC)/PTF-GSS lane switching represents the worse case scenario following the PTF critical failure, this is also reflected in the PTF design. The functional blocks inside the PTFs will be multiple redundant with most of the critical hardware working in “hot” backup mode, to minimize the number of instances when the actual switching of the PTF needs to be performed.

Scenario summary:

1. Autonomous generation of the stable GST timescale by PTF’s.
2. Selection of the Master PTF and Master GST physical realisation within the that PTF. The GST physical realisation at master PTF site is herunder indicated as GST(MC), while the Master GST realisation at Slave PTF is indicated as GST(MCx).
3. Comparison of the GST(MC) and GST(MCx) to generate the GST(MCx) to GST(MC) steering corrections. This involves CV and TWSTFT links for comparisons involving the GST(MCx) at slaved PTF site

4. Steering of GST(MCx) to a GST(MC)
5. Provision of TSP products to GMS.
6. Implementation of the TSP products by Master PTF, OSPF, MUCF & MSF after their validation by GACF.
7. GST(MC) steering to TAI and generation of GST to UTC conversion parameters for the inclusion into the OS and PRS navigation batches.
8. Use of the GST and TSP products for the purpose of the mission monitoring by MSF and MUCF.
9. GACF commands GST(MC)/PTF-GSS lane switch over based on the results of PTF internal, GACF and MUCF monitoring of PTF and or OSPF.
10. Exchanging of roles between the Master GST(MC)/PTF-GSS lane at Master PTF site and the Slave master GST(MCx)/PTF-GSSx lane at slave PTF site following a detection of a critical failure, or Maintenance of the master PTF.
11. Implementation of the transition law by OSPF.
12. System time generated by newly appointed Master GST(MC)/PTF-GSS lane.

After configuration by the GACF of the master PTF site and master GST(MC) on the same site, the sequential description is detailed below:

1) The Master PTF and Slave PTF generate their Master GST physical realisations indicated as GST(MC) and GST(MCx), respectively. The generation of GST physical realisations at each PTF is based on the respective clock ensembles. For more details please refer to scenario #SFC2.1-2.
A task of each PTF is to calibrate the signal paths (1PPS/10MHz signals) between its own GST(MC) output and the GRC (Galileo Receive Chain) input inside the relevant PTF-GSS. Note that the calibration information is applied at each GSS-PTF when providing the raw measurements to OSPF. In that way, the raw measurements are tagged by the true GST. The raw measurements coming from the master GST(MC)/GSS-PTF lane are then used by the OSPF as time reference for all other GSS measurements

2) The 2 PTF-GSS at each PTF site are synchronized with the 1PPS/10MHz timing signals provided by the respective PTF.
The PTF-GSS receivers collect the Galileo SIS observables, which are processed by OSPF. It should be noted that for ODTS determination only data from Master GST(MC)/PTF-GSS lane are used.
PTFs are equipped with the Galileo-GPS Combined Receiver, which collects observables on Galileo signals (E5a, E5b, AltBOC, E6C and E1C) and GPS signal L1 and L5. These observables will be used to generate the CV data used to establish the difference between the GST(MC) at Master PTF site and GST physical realisation at slave PTF (PTF-PTF synchronisation).
CV technique is also used for the purpose of the GST(MC) to TAI steering.

3) As the third time transfer technique deployed to determine Master-Slave PTF time offset (i.e GST(MC)-GST(MCx) offset required for the PTF synchronization) TWSTFT will be used (refer to #SFC2.1-2).
TWSTFT method is based on the simultaneous exchange of the RF signals over a commercial communication satellite. The RF signals contain a timing information which is derived from the local reference clock. TWSTFT measurements will be performed a number

of times per day in accordance with the TWSTFT schedule which will be provided by TSP to GACF using an operational interface and by GACF further to PTF. TWSTFT technique is also used for the purpose of the GST(MC) to TAI steering.

4) Both Master PTF and Slave PTF process the time transfer data so they can be exchanged between themselves.

5) Slave PTF uses the time transfer measurements results and the information on the Satellite clocks generated by OSPF to calculate corrections for GST(MCx) steering at Slave PTF to GST(Mc) at Master PTF. Slave PTF will apply the generated corrections to steer its timescale to meet the required time and frequency offset between its local timescale and the timescales realized at Master PTF.

6) In line with the concept of the Metrological timekeeping the Master &Slave PTFs provide its timing products (TWSTFT&CV Clock data) to Master SPF for their further provision to TSP, please refer to #SFC2.1-1, steps 3&4 .

7) TSP based on the products received from the SPF generates two sets of the steering and timing products as if each of the PTF was a Master. For the further details on the steering correction generation please refer to #SFC2.1-1.

8) Master SPF passes the products retrieved from TSP to GACF (TAI-UTC & GST-TAI offsets, GST monitoring data, Clock anomaly, UTC frequency correction.). The TSP products (GST- UTC offset, TAI-UTC offset and UTC freq. corr.) are distributed to the Slave SPF & MSF.

9) OSPF processes the GSS raw measurements to produce the satellite and GSS clocks offset estimates. Furthermore the satellite clock models are generated.
The clock data are passed to Master PTF and via GACF to Slave GACF for the further provision to Slave PTF (TBC). Both MUCF and MSF retrieve the clock data from GACF.
Note: OSPF clock products are combined with the PTF internal clocks to generate corresponding GSTs.
Please refer to the step 1 and also to #SFC2.1-2.

10) MSF uses GST-TAI, TAI-UTC and UTC frequency correction data with the GST monitoring report and GSS & satellite clocks data retrieved from GACF for the offline analysis of the GST performance and validation of the TSP products.

11) GACF performs integrity checks on the TSP data (GST-TAI, TAI-UTC and UTC frequency correction) before their usage by elements such as PTF and OSPF.

12) Master PTF uses the TSP products (checked for integrity by GACF) as a configuration parameters for its internal processes. The UTC frequency correction is however validated before it is used to steer GST towards UTC modulo 1 second. Based on the outcome of the validation process, the PTF can decide to apply, apply partially or disregard the corrections in case their application may affect the provision
of the navigation services.

13) If the validation process is successful, the TSP steering correction is directly applied to GST(MC) by CTT PTF. The slaved GST(MCx) at master PTF is in turn steered to GST(MC) through a steering algorithm.

14) OSPF generates based on the TSP products the UTC conversion parameters which are used for the inclusion into the OS and PRS navigation messages.
Note: the addition of the UTC conversion parameters into the navigation messages enables Galileo users to provide the so called UTC dissemination service.

15) The Master PTF passes to MUCF the TSP steering parameters, including the information on the PTF internal clocks, PTF-PTF offset and also the PTF slave steering. MUCF uses this information together with the GST-TAI, TAI-UTC data retrieved from the GACF for the GST and mission online monitoring.

16) Master PTF is monitored by GACF & MUCF and also by its own internal processes. PTF failure can be detected either by the PTF failure detection systems and/or by MUCF & GACF.
Elements which detect the failure (in this scenario it is the PTF itself) send the failure report/status to the GACF. Note that MUCF bases its decision on the PTF “health “ status on the PTF and OSPF generated GST and clock products. Whereas GACF process the information arising from the PTF monitoring and OSPF feedback on the availability of the GCC/GSS data.

17) GACF issues the GST(MC)/PTF-GSS lane switch command based on the PTF, MUCF or its own failure reports indicating the PTF malfunction. GACF distributes the information on the PTF status to MSF, MUCF and OSPF.

18) Following the acknowledgement of the switch command PTFs exchange their roles: Master to Slave, Slave to Master.

19) Following the GST(MC)/PTF-GSS lane switch, the OSPF generates the clock and orbit products based on the predefined transition law, which is compensating for the difference between the timescales as generated by Master and Slave PTF.

6.2.3.4.4 #SFC2.1-4: Leap Second/ERP Configuration Update Scenario

This scenario is run on weekly basis with following objectives

- To check whether the leap second files have to be have to be reconfigured in GMS elements, and eventually do it if applicable;
- To reconfigure GMS elements with new ERPs, once validated.

The update of the leap second value can occur only up to twice a year (i.e. at the end of the 30th of June and the 31st of December). It is assumed that TSP announces such updates at least five months in advance (i.e. an updated TIME_TaiUtcOffset file is received by the 1st of February to announce a leap second change on the 30th of June, or the 1st of August to announce a leap second change on the 31st of December).

It is also assumed that following the announcements above, TSP provides an actualised TIME_TaiUtcOffset file (i.e. “currentOffset” is increased as per announcement, “nextUpdate” is unchanged and kept to the date of applicability of the leap seconds value being applied, “nextChange” is set to zero) on the day following the application of an updated leap seconds value (i.e. the 1st of July or the 1st of January).

New ERP values are delivered by GSRP to GMS on a daily basis. These ERPs are generally assumed to be correct, but in order to provide additional validation, the GMS computes the differences between old and new ERPs and compares it to a threshold.

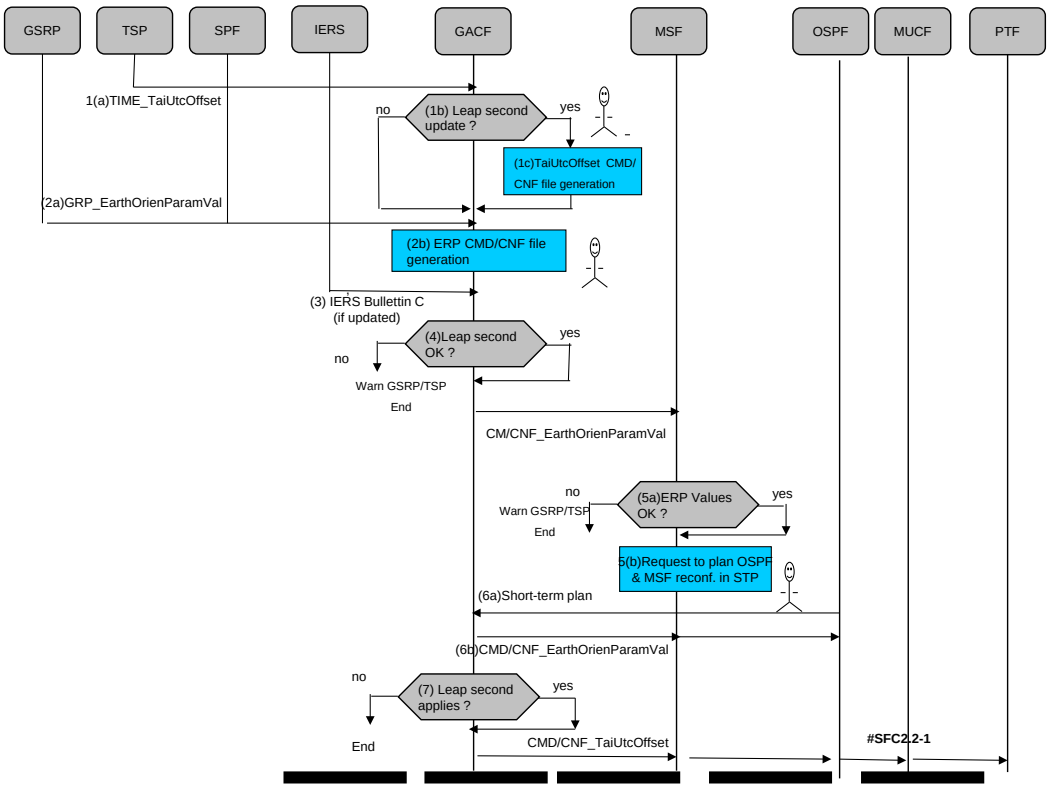


Figure 112 : #SFC2.1-4 Leap Second/ERP Configuration Update Scenario

1) A leap second file (i.e TIME_TaiUtc_Offset) is daily provided by the TSP to the GACF via SPF. However, the GE will update the public configuration file CMD/CNF_TaiUtcOffest stored in GACF/CFGM using the latest received TIME_TaiUtc_Offset file only if the following conditions are verified :
    - o A leap second is announced by IERS, leading TSP to provide the date of applicability of the new leap seconds value, and the actual value itself, by the means of an updated TIME_TaiUtcOffset file;
    - o An updated leap seconds value is actually being applied, i.e. the day after the announced date of applicability.

2) New ERP values are delivered by GSRP to GMS on a daily basis. These ERPs are generally assumed to be correct, but in order to provide additional validation, the GMS computes the differences between old and new ERPs and compares it to a threshold. (2a)

The new ERPs delivered by GRSP (i.e. signal GRP_EarthOrienParamVal) are converted into GMS public configuration file by GACF (i.e. signals CMD / CNF_EarthOrienParamVal), which is subsequently delivered to the MSF for ERP validation. 2b)

3) The GE checks if a new release of IERS bulletin C is available (i.e. twice a year). If this is the case. the Ground Engineer(GE) at GACF retrieves from the IERS website the new release of IERS bulletin C. Otherwise the GE verifies that the latest IERS bulletin C release is available at GMS.

4)The value of the leap seconds is operationally verified by GE at GACF by double operator check, based on the information operationally provided in the IERS Bulletin C, and in the CMD_TaiUctOffset configuration file stored in GACF/CFGM
If the leaps seconds check fails, a warning is raised to GSRP and TSP to perform further analysis, and the procedure stopped. Otherwise the CMD/CNF_TaiUtcOffset file and the new ERP values included in the public configuration file CMD/CNF_EarthOrienParamVal stored file at GACF are to the MSF for ERP validation.

5)The Mission Engineer at MSF executes the ERP validation procedure .
If the new ERPs are not valid or not coherent with the CMD/CNF_TaiUtcOffset files and the Bulletin C, the MSF is re-configured with former ERP values and leap seconds, and a warning is raised to TSP and GSRP for further analysis.
Otherwise, if the ERP validation at MSF is successful, the MSF operator requests to include in the Short-Term Plan the deployment to OSPF and MSF of the new operational (validated) ERP values included in the CMD/CNF_EarthOrienParamVal file .

6) The Short-Term Plan is passed to the GACF that deploys the validated ERP to OSPF and MSF.

7) if updated leap second value applies, the new leap second values included in file CMD/CNF_TaiUTcOfsset file is deployed to MUCF, PTF and OSPF.

6.2.4 SFC2.2 Provide GPS-Galileo Time-Offset

6.2.4.1 SFC2.2 High Level Description

The purpose of this SFC is to:

1. Determine the unique Galileo-GPS Time Offset (GGTO) applicable to OS and PRS navigation services
2. Co-ordinate with USNO the GGTO value and other general info with the GPS (NANU&NAGU).
   Both sides will have to agree during the GGTO coordination process on the common GGTO value which lies within the configurable threshold.

3. Provide the GGTO to the system function generating the navigation batches to be used for OS and PRS navigation services

4. Set the disseminated GGTO to dummy in case the difference between coordinated GGTO value exceeds a configurable threshold or the GGTO is not present at PTF output or an alarm is present in the infrastructure involved in Galileo GGTO determination and monitoring and control.

Notes: 1) NANU: Notice Advisory to Navstar Users
2) NAGU: Notice Advisory to Galileo Users
3) Both NANU and NAGU contain the contingency information related to the respective constellation.
4) DDF2 description captures the baseline described in the latest applicable issue of the GMS-GGTO ICD [AD-15].
The approach for dissemination of dummy GGTO before and after the GGTO computation at USNO is active will be highlighted hereunder .

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6.2.4.2 Physical Static Block Diagram

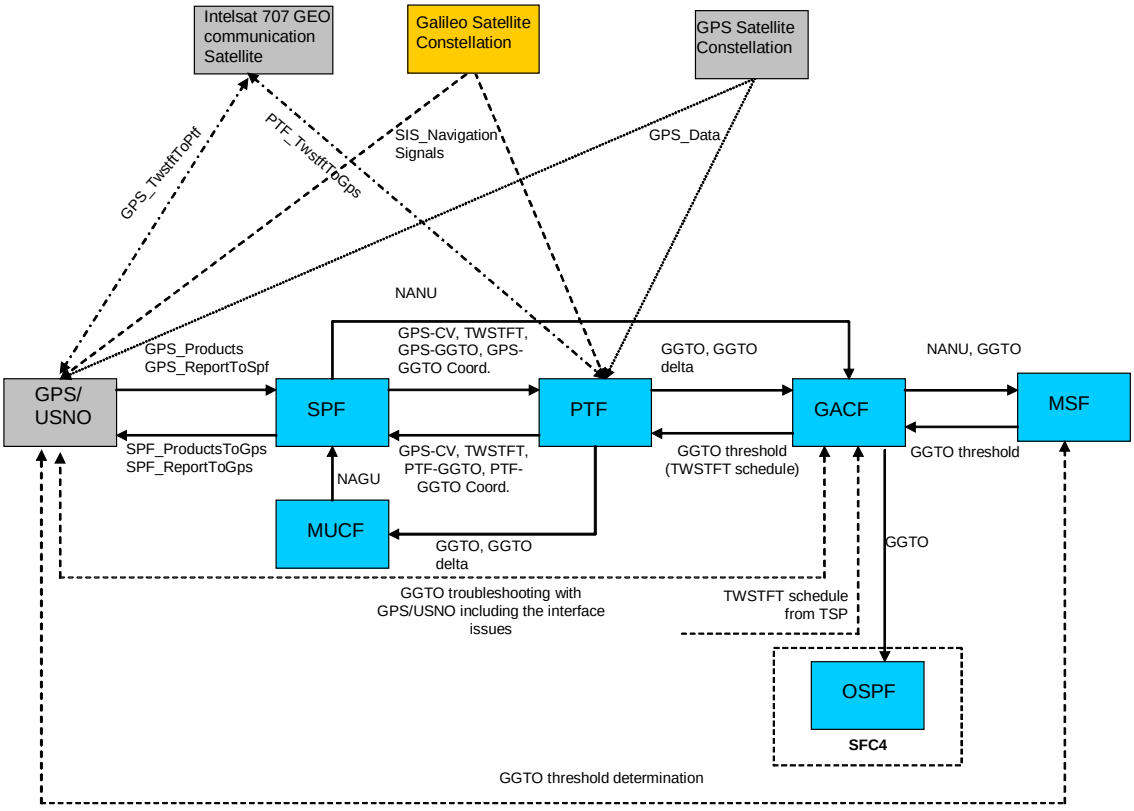


Figure 113 : SFC2.2 Physical Static View

6.2.4.3 Overall Physical Description

Galileo-GPS Time Offset will be determined jointly by Galileo (PTF) and GPS (USNO). The difference between the Galileo and GPS system will be determined in IOV using the TWSTFT and GPS-CV time transfer techniques and in FOC by the means of GPS-Galileo combined receiver.

In the final Galileo configuration, when the GGTO computation will be active, a nominal procedure will be daily executed by a GMS operator before the GGTO values are adopted by GPS and Galileo navigation systems for the dissemination to their respective users. In particular, a non-dummy GGTO will be forward to the OSPF whenever any one of the following conditions occur:

- There is a GGTO is available at output of PTF,and
- There are no critical alarms in the infrastructure involved in the GGTO computation and monitoring and control,and
- The difference between GGTO values computed by Galileo PTF and USNO (TIME_GgtoDelta) is successfully monitored at MUCF and is within a configurable threshold

A dummy GGTO will be forward to the OSPF in case one of the following conditions occur:

- The GGTO is not available at output of PTF, or
- There is a critical alarm in the infrastructure involved in the computation of GGTO and monitoring and control, or
- The difference between GGTO values computed by Galileo PTF and USNO (TIME_GgtoDelta) is not successfully monitored by MUCF.

A contingency procedure will be triggered in case the difference between the GGTO values computed by Galileo PTF and USNO exceeds the configurable threshold.
In this case, after further coordination with USNO, the GGTO forwarded to the OSPF could be set either to the dummy value according to the format specified in SISICD or the GGTO computed by the Galileo PTF not taking into account the exceeding threshold alarm.
The unique GGTO applicable to OS, and PRS services will be provided to OSPF for its inclusion into the corresponding navigation batches.

In the current Galileo design, where the GGTO computation is not active at USNO,
the nominal procedure executed on daily basis by the GMS mission operator for verification of GGTO validity is based on the checking of the absolute value of Galileo GGTO determined by the PTF instead of the difference between the GGTO values computed by Galileo PTF and USNO .
In addition, the checks on PTF critical alarms impacting GGTO are not requested in the current galileo design.

Thus, during GGTO interim policy valid GGTO computed by the PTF (non-dummy GGTO) will be deployed to the OSPF and MSF if any one of the following conditions are verified :

- The GGTO is available at output of PTF,and
- The absolute value of Galileo GGTO determined by the PTF is within a configurable threshold (default value: 100 ns). This condition is indicated by MUCF GMOS parameter “GPS PTF delta T sets to GREEN.

A dummy GGTO will be forward to the OSPF during GGTO interim policy in case one of the following conditions occur:

- The GGTO is not available at output of PTF,or
- The absolute GGTO determined by the Galileo PTF is not monitored at MUCF (MUCF GMOS parameter is set to GREY) or
- The absolute value of Galileo GGTO determined by the PTF exceeds the configured threshold (default value: 100 ns). This condition is indicated by MUCF GMOS parameter “GPS PTF delta T” sets to ORANGE.
  Note : this event triggers the GMS GalileoGpsOffset Contingency procedure GMS-CTG-012, which is forced to deploy a dummy GGTO when the computation at USNO is not active.

During GGTO interim policy GPS still provides TWSTFT and CV measurements. The GGTO determined by GPS is set to a fixed value of 1.5 ns. Finally, the GMS will inform GPS USNO when a dummy GGTO is deployed.

The interfaces deployed between the GPS and GMS will allow also the exchange of the GPS and Galileo constellation contingency information so called NANU and NAGU bulletins.
Notes:
- Based on the available design in FOC, the primary method for GGTO determination will be based on the use of Galileo/GPS combined receiver using the OS signals. Such GGTO will be broadcasted for all services:
- Different processes often require as an input different GST realisation. For instance GGTO, TSP and GST distribution activities require access to the physical realisation of the GST, hence the GST(MC) is used. On the other hand .for the navigation and UTC dissemination services the “paper” realisation of GST referred to as GST or (GST(OSPF)) is employed. It is important that for the difference between these GST realisations it is always accounted for when the final products such as GGTO or GST to UTC conversion parameters are generated. Please refer also to SFC2.1 section “GST reference points”:

6.2.4.3.1 GGTO Calculation: PTF

In addition to the functionalities related to the GST generation, please refer to SFC2.1, PTF is in charge of GGTO determination. PTF will use for this purpose the TWSTFT and GPS-CV techniques already deployed for the purpose of the GST to TAI steering and Slave PTF to Master PTF synchronization as well combined Galileo . In FOC the GGTO determination will be based on the Galileo GPS combined receiver solution. As mentioned above only OS signals will be considered, however obtained GGTO will be used in addition to OS also for and PRS services.

The PTF determines the GGTO on a daily basis.
The PTF performs the following checks:

- The check to verify that the difference between GGTO values computed at PTF and USNO is below a configurable threshold, when the GGTO computation at USNO is active
- The check to verify that the absolute GGTO values computed at PTF is below a configurable threshold, when the GGTO computation at USNO is not active

The relevant GGTO data files are sent by PTF to MUCF and GACF for GGTO monitoring and control including data files for execution of the procedures at GMS level for verification of GGTO validity before GGTO deployment to OSPF for dissemination and update of MSF configuration (GalileoGPSOffsetApplication procedure and GalileoGpsContingency procedure).

6.2.4.3.2 Dissemination of Galileo SiS: Galileo Satellite Constellation (SSeg)

Galileo Satellite Constellation disseminates Galileo Signal in Space, the reception of Galileo SIS is a prerequisite for the GGTO SiS driven determination.

In FOC the Galileo GPS combined receivers deployed at the PTF and GPS/USNO will generate the GGTO based on the simultaneous reception of the respective signals in space. Combined receivers simultaneously collect Galileo and GPS SIS observables.

6.2.4.3.3 Dissemination of GPS SIS: GPS Satellite Constellation

GPS Satellite Constellation disseminates GPS Signal in Space, the reception of GPS SIS is a prerequisite for the GGTO SIS driven determination.
In FOC the Galileo GPS combined receivers deployed at the PTF and GPS/USNO will generate the GGTO based on the simultaneous reception of the respective signals in space. Combined receivers simultaneously collect Galileo and GPS SIS observables.
In IOV the standalone GPS receivers deployed inside PTFs and GPS/USNO will collect the GPS-SIS. The collected pseudo-range measurements and navigation messages will be following their post processing (in line with the Common View practice) used for the comparison of the GPS and Galileo system times, thus GGTO determination.

6.2.4.3.4 TWSTFT GGTO Determination: Geo-Stationary Communication Satellite

Geostationary commercial communication satellites, which are used for the retransmission of RF signals carrying timing information required for the TWSTFT (Two Way Satellite Time and Frequency Transfer) between the GPS/USNO and PTFs. TWSTFT is used likewise the GPS-CV for the comparison of the distant clocks, thus in the PTF-PTF synchronization GST(MC) to TAI steering and also the GGTO determination.

Note: Geostationary commercial communication satellites that will used for Eur-Eur and Eur-USA TWSTFT links are expected change the receive and transmit polarization.
As a result all TWSTFT stations (UTC(k)) had to be equipped with the configurable OMT in order to be able to participate in both EUR and USA sessions. Consequently also PTFs TWSTFT hardware will have to be updated in order to add the configurable OMT.

Note: Following GMS CDR CO the OMT were included into the PTF baseline

Due to a rather frequent change of the GEO satellite, it is not possible to maintain within the core document the references to a specific TWSTFT satellite being used. Consequently all of the references need to be removed and the only TWSTFT GEO satellite specific information will be given in the corresponding Annex. The section 11 provides a list of an out of dated time transfer hardware and as such has a little to none reporting value.

6.2.4.3.5 GGTO Determination & NANU Provision: GPS/USNO

The United States Naval Observatory (USNO) has been designated as GPS agent to construct and operate GGTO-related interfaces between Galileo and GPS. To denote that USNO is acting as an agent for GPS, the notation "GPS/USNO" will be used.

6.2.4.3.6 Galileo Data Provision to GPS/USNO and Vice Versa: SPF

Service Product Facility is serving as a secure interface between GMS and the external entities. Any data (incoming/ outgoing) between GMS and external interfaces for instance GPS/USNO

have to be routed via the SPF. SPF handles also an internal distribution of the data inside GMS. GPS/USNO data are routed from SPF to GACF & PTF.

6.2.4.3.7 Mission Monitoring &NAGU Report Generation: MUCF

Mission Uplink Control Facility is in charge of mission online monitoring. For this purpose MUCF will use the GGTO and GGTO delta data (difference between GPS and Galileo calculated GGTOs) received from Master PTF.

In order to support the dummy GGTO procedures at GMS, the MUCF triggers an alert and a notification to the Mission Controller if the following events occur:

- the difference between the GGTO computed by Galileo and USNO included in the file TIME_GgtoDelta exceed a configurable threshold (default value 10 ns), when the GGTO computation at USNO is active or
- the absolute GGTO determined by PTF exceed a configurable threshold, when the GGTO computation at USNO is not active

The generated alert triggers the execution of GalileoGpsContingency procedure.
To complement the mission monitoring functionality MUCF will generate the NAGU reports based on the information received from GCS.
Remark: *the automated NAGU generation concept is currently not consolidated in the STRB2.0 baseline. It is expected that the content of the signal INF_NoticeAdvisoryToUsers will be refined in close coordination with GSC in the frame of GSC operational concept consolidation. In the meanwhile, the signal INF_NoticeAdvisoryToUsers will not be disseminated to the external entity and the NAGU will be generated manually and sent to the GSC. In both cases (automated and manual NAGU processes) the authorization for dissemination to the public is granted by the corresponding formal process at GSC level.*

6.2.4.3.8 OSPF & PTF monitoring & GGTO Threshold Configuration: GACF

Ground Asset Control Facility is in charge of the GMS element control & monitoring and the master archiving of the mission data.
The GACF is involved in the execution of GGTO application procedure to validate GGTO before deployment to OSPF and MSF.
A GGTO contingency procedure is executed in case the difference between GGTO values determined at PTF and USNO exceeds a configurable threshold.
The GGTO threshold is configured by GACF and passed to the PTF.

Furthermore GACF is within the scope of SFC2.2 also responsible for following operations:

☐ Troubleshooting of GGTO coordination with the GPS/USNO operator via operational interfaces.

☐ Control/commanding and monitoring of the PTF (conversion of TWSTFT schedule and GGTO threshold into configuration parameters to be implemented by PTF).

☐ Control/commanding of OSPF (conversion of the GGTO) into configuration parameters to be implemented by OSPF (generation of navigation batches).

6.2.4.3.9 Mission Offline Monitoring: MSF

Mission Support Facility is in charge of the mission offline monitoring. MSF will for this purpose amongst of other data use also the NANU bulletins provided by GPS/USNO.

6.2.4.3.10 Generation of Navigation Batches and GSS & Satellite Clocks Determination: OSPF

Orbit and Synchronization Processing Facility is in charge of orbit, GSS & satellite clock determination and the generation of the navigation batches for OS and PRS services. OSPF includes the GGTO data defined as per SIS ICD into the navigation batches to support OS and PRS navigation services.

6.2.4.4 Dynamic Scenarios

Following scenarios have been identified:

- SFC2.2-1**:** SIS Based Determination using GPS/Galileo Combined receiver, nominal scenario with GGTO computation at USNO active
- SFC2.2-1 bis SIS Based Determination using GPS/Galileo Combined receiver, nominal scenario with GGTO computation at USNO not active
- SFC2.2-2 Exchange of the NANU & NAGU, nominal scenario
- SFC2.2-3 Determination of the GGTO with TWSTFT and GPS-CV, contingency scenario

6.2.4.4.1 #SFC2.2-1; SIS Based Determination using GPS/Galileo Combined Receiver (FOC baseline)

The MSC for scenario this is provided below:

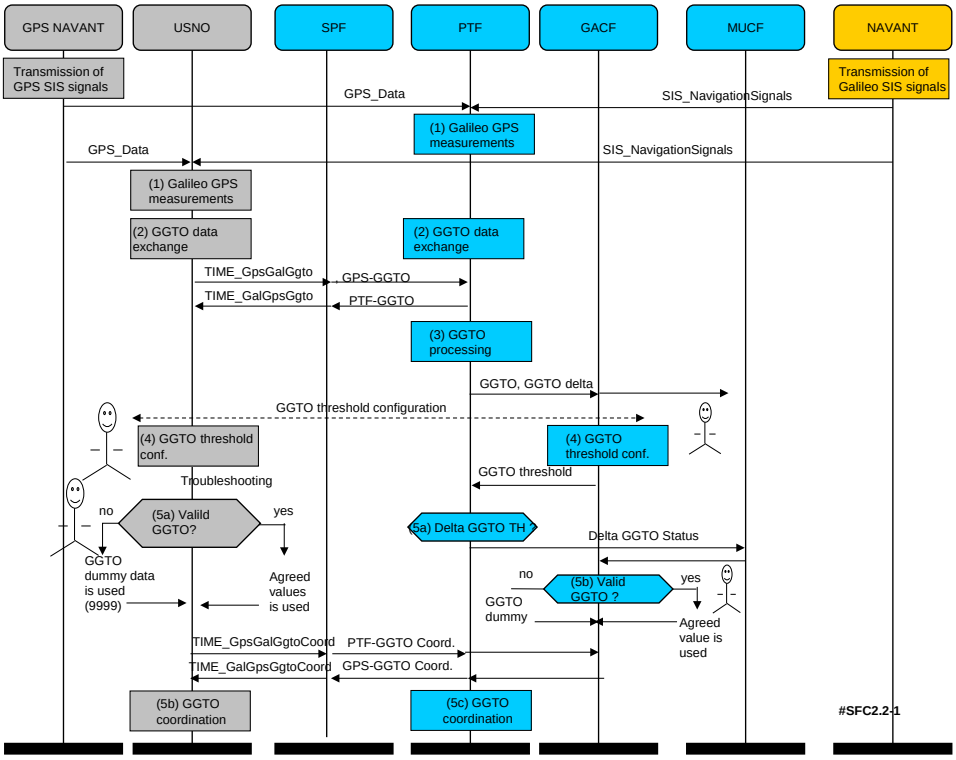


Figure 114 : #SFC2.2-1: SIS Based Determination using GPS/Galileo Combined receiver

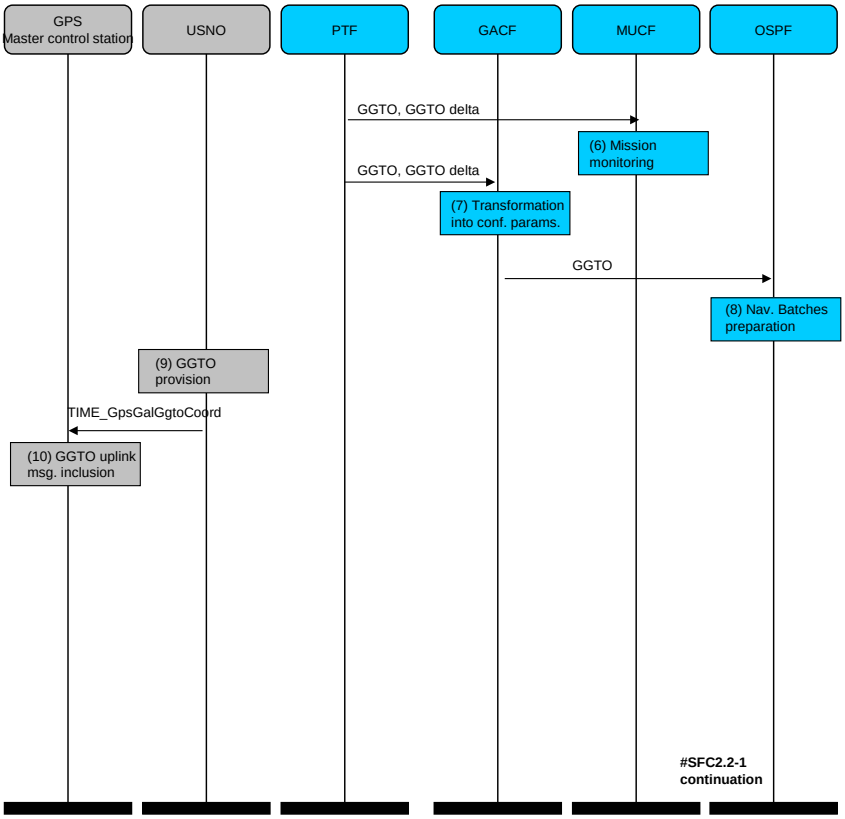


Figure 115 : #SFC2.2-1: SIS Based Determination using GPS/Galileo Combined receiver, continuation

Scenario Textual Description:

In FOC the GGTO determination will be based on the Galileo-GPS combined receiver solution. This scenario applies when the GGTO computation is active at USNO.
This scenario is run under the following assumptions:

- GACF and MUCF are operational
- PTF and OSPF are in running or operational mode

A dummy GGTO will be deployed to OSPF if PTF or MUCF are not in the operational state.

This solution is less complex compared to the time transfer based methods deployed at IOV (TWSTFT and GPS-CV), furthermore it delivers a better performance. The downside of the combined receiver approach is the unavailability of required hardware prohibiting its implementation at the early stages of the project (starting from IOV).
The sequential description is detailed below:

1) The Galileo-GPS combined receivers deployed at the PTF and USNO will generate the GGTO based on the simultaneous reception of the respective signals in space. Combined receivers simultaneously collect Galileo and GPS SIS observables.

The PTF is configured to determine the GGTO in SIS-based mode

2) PTF and USNO extract the GGTO values from their respective receivers and exchange them with each other.

3) PTF calculates GGTO. PTF passes the GGTO and delta GGTO to GACF for its further provision to MUCF, as an input to the GGTO threshold configuration process.

4) Prior to entering the GGTO coordination process the USNO and GACF operators need to agree on the GGTO configurable threshold. This process will be handled operationally upon operator's request.
Upon successful determination of the delta GGTO threshold at GACF, the actual value is converted into configuration parameter and passed to PTF.

5) During the coordination process PTF compares difference between its own GGTO value and GGTO value generated by GPS/USNO against the agreed threshold. The test result is sent to MUCF.
A routine procedure (GMS_NOM_012) is executed at GACF on daily basis by GMS Mission Controller during coordination with USNO, before to configure the OSPF and update MSF configuration.

A valid GGTO is set at the agreed value if any of the following conditions is verified :

- The PTF GGTO data file (TIME_Galileo_Gps_Offset_yyyymmddhhmmss) with the current date is found at GACF.
- The GGTO is successfully monitored at MUCF
- The delta GGTO between PTF and USNO is within the agreed threshold
- No critical alerts are present in the infrastructure used to compute and monitor the PTF GGTO.

If any of above conditions are verified , each side of the interface acknowledges this by sending back the GGTO value received from its counterpart. Thus PTF sends to GPS/USNO the GGTO generated by GPS/USNO and vice versa.

The GGTO is set to dummy value (9999) if one of the following events occur:

- The PTF GGTO data file (TIME_Galileo_Gps_Offset_yyyymmddhhmmss) with the current date is not found at GACF.
- The GGTO is not monitored at MUCF
- TBD alerts are present in the infrastructure used for GGTO computation and monitoring and control.

Note: A dummy GGTO will be deployed to OSPF if PTF or MUCF are not in the operational state.

Should the delta GGTO exceed the agreed threshold, an investigation/troubleshooting is initiated by GACF on an operational level to solve the problem causing the failure.
If the problem cannot be resolved the coordinated signals are filled with the dummy data in the form of “9999” to indicate that the GGTO value is not available.

6) MUCF uses the coordinated GGTO and GGTO delta values received from PTF for the mission monitoring purposes.

7) Actual GGTO and GGTO delta values are provided to GACF. Data is used by GACF for the configuration of the GGTO threshold and to generate OSPF configuration parameters and update of MSF configuration.

8) GGTO values (as per SIS ICD) are converted into the configuration parameters and are sent to OSPF, data is used to prepare the navigation batches applicable to OS and PRS services.
10)Distribution of the GGTO by GPS system. Present GPS satellites do not have a capability to include GGTO into their navigation messages, consequently in the initial stage the GGTO will be broadcasted by GPS to their users by other means e.g. dedicated website etc.

6.2.4.4.2 #SFC2.2-1bis; SIS Based Determination using GPS/Galileo Combined Receiver (FOC baseline) – Before USNO GGTO computation is active

Scenario Textual Description:
In FOC the GGTO determination will be based on the Galileo-GPS combined receiver solution. This scenario applies before the GGTO computation is active at USNO
This scenario is run under the following assumptions

- GACF and MUCF are operational
- PTF and OSPF are in running or operational mode

A dummy GGTO will be deployed to OSPF if PTF or MUCF are not in the operational state.
The MSC for scenario this is provided below:

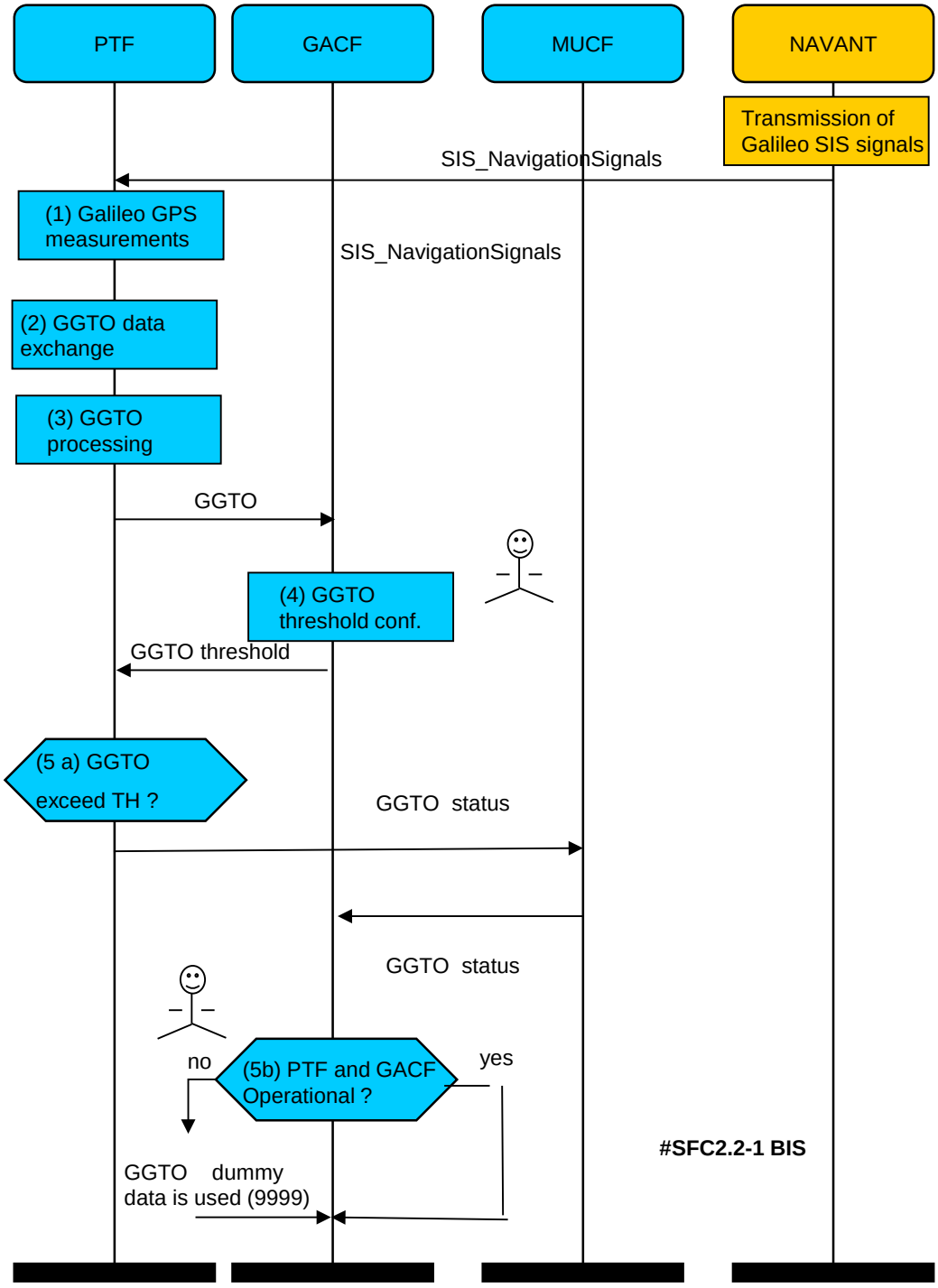


Figure 116 : #SFC2.2-1bis SIS Based Determination using GPS/Galileo Combined receiver (Before GGTO computation at USNO is active)

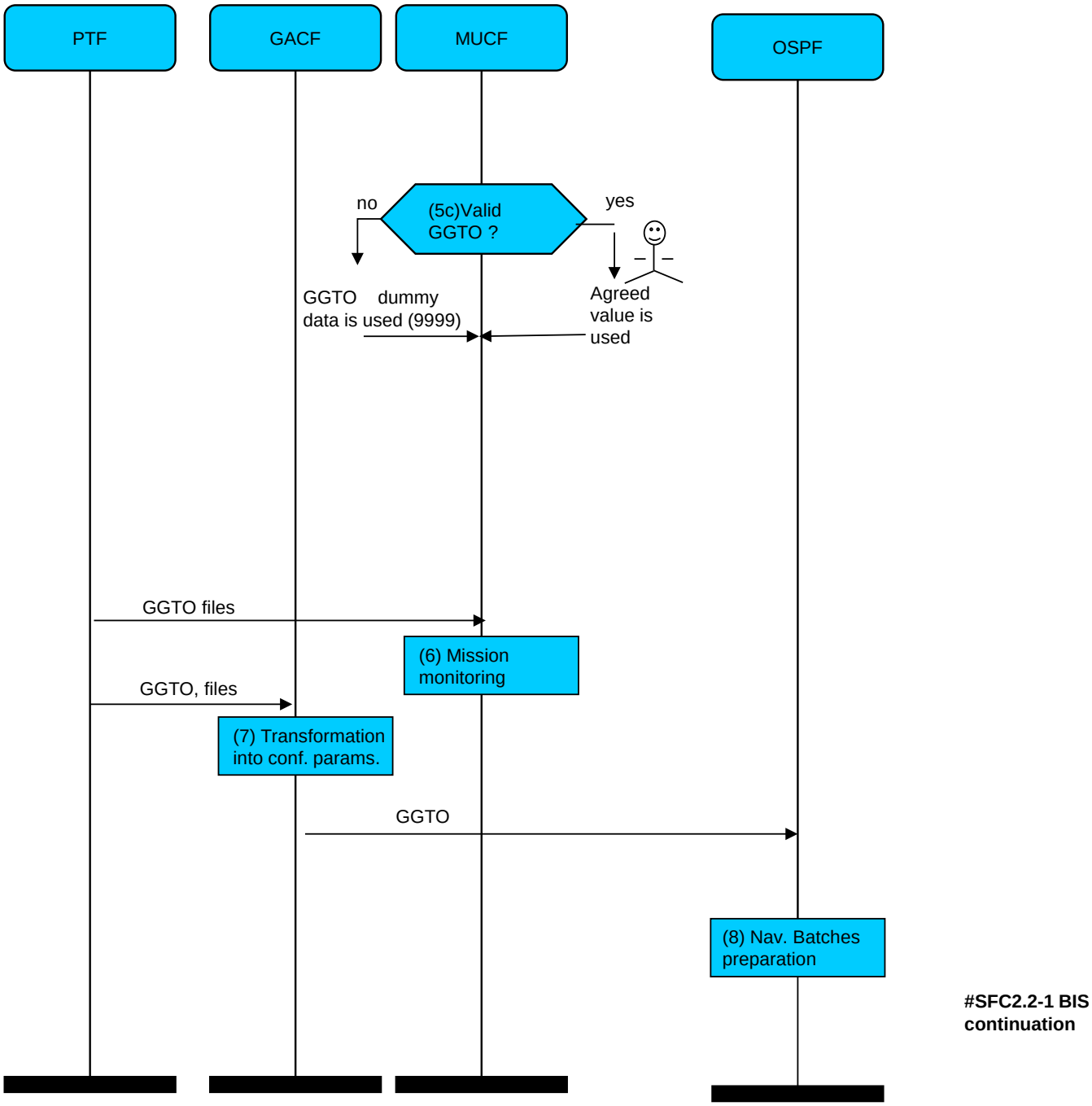


Figure 117 : #SFC2.2-1bis: SIS Based Determination using GPS/Galileo Combined receiver (Before GGTO computation at USNO is active), continuation

Scenario Textual Description:
In FOC the GGTO determination will be based on the Galileo-GPS combined receiver solution. This scenario applies before the GGTO computation is active at USNO.
This solution is less complex compared to the time transfer based methods deployed at IOV (TWSTFT and GPS-CV), furthermore it delivers a better performance. The downside of the combined receiver approach is the unavailability of required hardware prohibiting its implementation at the early stages of the project (starting from IOV).

The sequential description is detailed below:

1)The Galileo-GPS combined receivers deployed at the PTF will generate the GGTO based on the simultaneous reception of the respective signals in space. Combined receivers simultaneously collect Galileo and GPS SIS observables.
The PTF is configured to determine the GGTO in SIS-based mode

Note : GPS USNO will only provide TWSTFT and CV measurements during interim dummy GCTO policy.

2)PTF extracts the GGTO values from its combined Galileo-GPS receiver. PTF passes the GGTO to GACF for its further provision to MUCF as an input to the GGTO threshold configuration process.

- Note : Due to the limitation of the Galileo constellation and calibration issues on the USNO side, USNO does not compute a real GGTO value and provides the fixed GGTO values (as per ESA-USNO coordination):
  - TIME_GpsGalGgto: -1.5 ns
  - TIME_GpsGalGgtoCoord: 1.5 ns

3)The GACF operator set the absolute GGTO configurable threshold (default value :100 ns). This process will be handled operationally upon operator's request.

4)Upon successful determination of the GGTO threshold at GACF, the actual value is converted into a configuration parameter and passed to the PTF.

5) The absolute GGTO is compared with threshold at PTF and the relevant status sent to MUCF.

A routine procedure (GMS_NOM_012) is executed on daily basis by GMS Mission Controller before to configure the OSPF and update MSF configuration.
The GGTO file is filled with valid value (non-dummy value) if any of the following conditions is satisfied :

- The PTF GGTO data file (TIME_Galileo_Gps_Offset_yyyymmddhhmmss) with the current date is found at GACF.
- The GGTO is successfully monitored at MUCF
- The PTF absolute GGTO is within the threshold (the parameter "GPS PTF delta T System" on the MUCF GMOS is set to green)

The GGTO will be set to dummy value (9999) if any one of the above mentioned conditions is not satisfied.

- The PTF GGTO data file (TIME_Galileo_Gps_Offset_yyyymmddhhmmss) with the current date is not found at GACF.
- The GGTO is not monitored at MUCF (the parameter "GPS PTF delta T System" on the MUCF GMOS is set to green)
- The PTF absolute GGTO is within the threshold (the parameter "GPS PTF delta T System" on the MUCF GMOS is set to orange)

.

Notes:

- A dummy GGTO will be also deployed to OSPF if PTF or MUCF are not in the operational state. In case the PTF is under maintenance, the dummy GGTO will be broadcast for five consecutive days.
- In case the absolute GGTO exceeds the configured threshold, the contingency procedure (GMS-CTG-012) will be forced to set the GGTO to dummy value

Note : In the case that a dummy GGTO is broadcast:

a. USNO will be informed that current value broadcast by Galileo is dummy
b. An investigation on the reason shall be launched contacting USNO, if required.

6)MUCF uses the GGTO data files values received from PTF for the mission monitoring purposes.

7)Actual GGTO data files are provided to GACF. Data is used by GACF to generate OSPF configuration parameters and update MSF configuration

8)GGTO values (as per SIS ICD) are converted into the configuration parameters and are sent to OSPF, data is used to prepare the navigation batches applicable to OS and PRS services.

6.2.4.4.3 #SFC2.2-2: Exchange of NANU&NAGU nominal scenario

The MSC for scenario this is provided below:

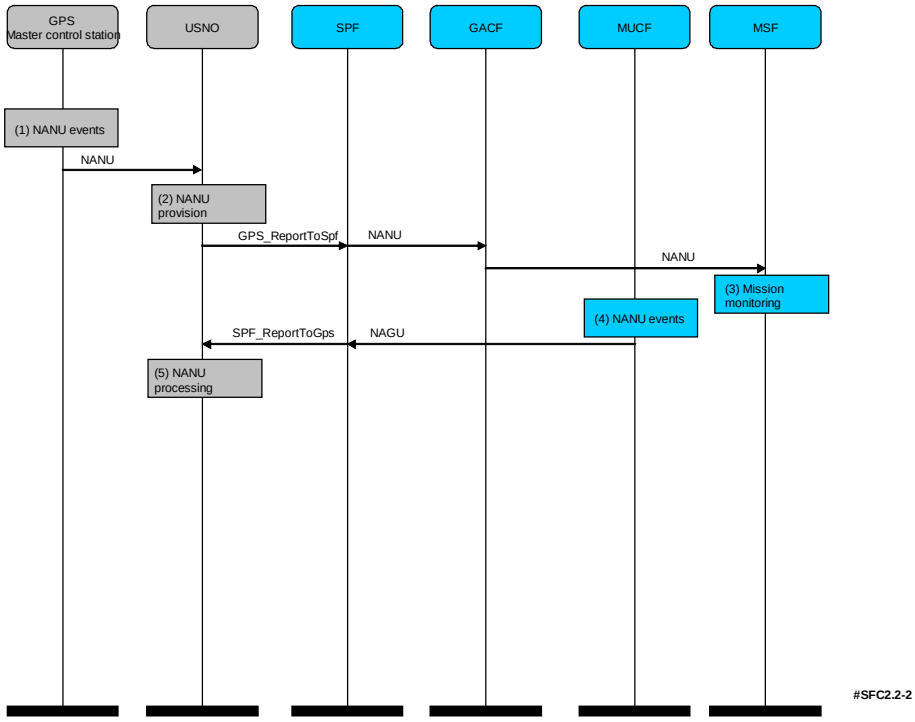


Figure 118 : #SFC2.2-2: Exchange of NANU & NAGU nominal scenario

Scenario Textual Description:

GGTO interface allows also for an exchange of the GPS and Galileo system contingency information which may affect the GGTO determination. Note there will be a NAGU file per event.

The sequential description is detailed below:

(1) USNO retrieves from the GPS control centre upon the occurrence of the contingency event the corresponding NANU bulletin.

(2) USNO provides the NANU data to SPF. SPF passes the NANU data further to GACF for the archiving and also to be checked by GACF operator within the frame of the GPS interface monitoring

(3) MSF retrieves the NANU information form the GACF and uses it for its mission offline monitoring.

(4) MUCF monitors the information from the GCS and other constellation monitoring functions and generates corresponding NAGU report upon the occurrence of the contingency event. This report is forwarded to SPF for the further provision to the GPS/USNO.

Remark: *the automated NAGU generation concept is currently not consolidated in the STRB2.0 baseline. It is expected that the content of the signal INF_NoticeAdvisoryToUsers will be refined in close coordination with GSC in the frame of GSC operational concept consolidation. In the meanwhile, the signal INF_NoticeAdvisoryToUsers will not be disseminated to the external entity and the NAGU will be generated manually and sent to the GSC. In both cases (automated and manual NAGU processes) the authorization for dissemination to the public is granted by the corresponding formal process at GSC level.*

(5) USNO retrieves the NAGU bulleting from the SPF and uses it for the monitoring of the GGTO task/operations.

6.2.4.4.4 #SFC2.2-3: Determination of the GGTO with TWSTFT and GPS-CV (FOC Back up) contingency scenario

The MSC for scenario this is provided below:

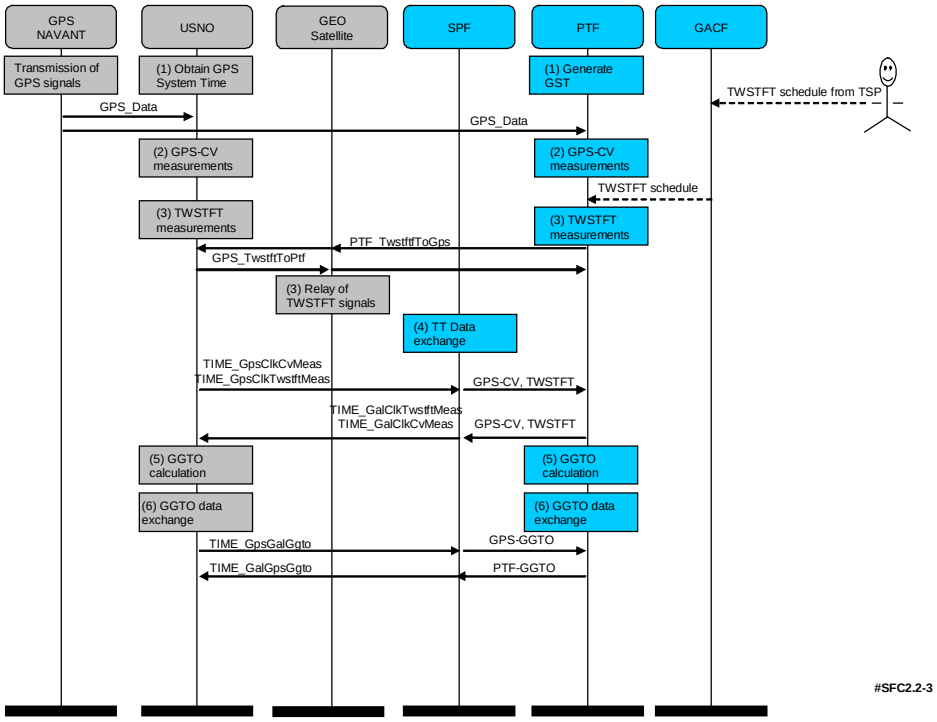


Figure 119 : #SFC2.2-3: TWSTFT and GPS-CV driven GGTO determination scenario

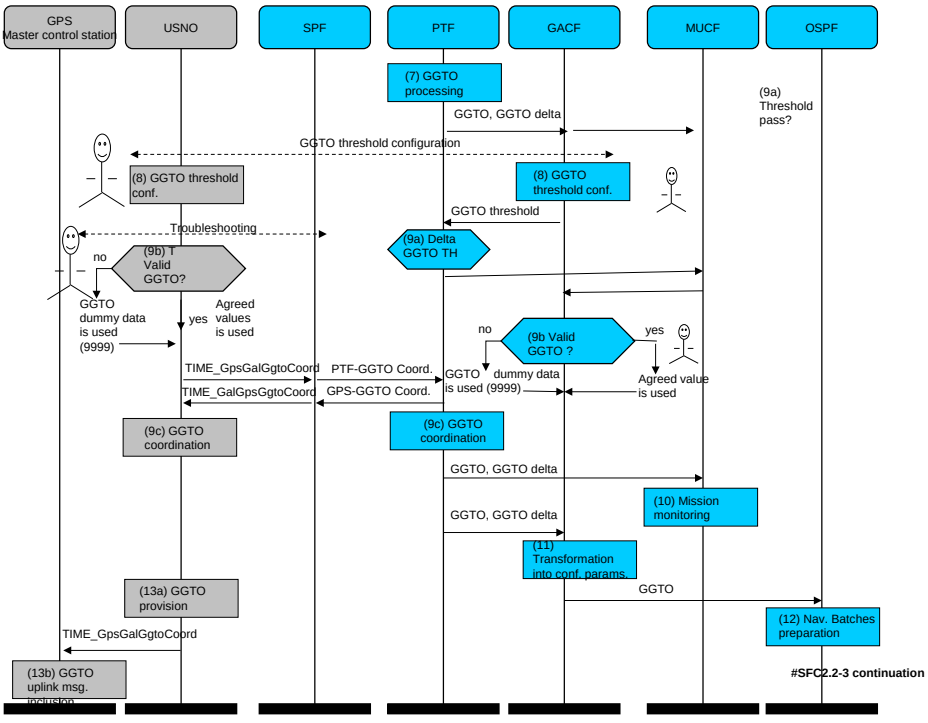


Figure 120 : #SFC2.2-3: TWSTFT and GPS-CV driven GGTO determination scenario

Scenario Textual Description:

In IOV the GGTO determination was be based on the TWSTFT and GPS-CV time transfer methods. In FOC TWSTFT GPS-CV will serve as a backup to the Galileo-GPS combined receiver method.
TWSTFT based GGTO determination is a contingency driven scenario, the TWSTFT method has been selected in case of temporary unavailability of the Galileo-GPS combined receivers in FOC. Note the GGTO performance when determined by TWSTFT or GPS-CV is worse compared to GGTO performance achievable with the combined receiver.

This scenario applies when the GGTO computation is active at USNO.
This scenario is run under the following assumptions:

- GACF and MUCF are operational
- PTF and OSPF are in running or operational mode

A dummy GGTO will be deployed to OSPF if PTF or MUCF are not in the operational state.

The sequential description is detailed below:

(1) The PTF is configured to determine the GGTO in TWSTFT or GPS-CV.
The GPS System Time is derived from the “paper clock”, this makes it difficult to compare GPS system time with GST or any other timescale using the time transfer techniques as there is no physical realization of GPS time as apposed to GST. Consequently USNO needs to access the GPS System Time as a prerequisite for the comparison of the Galileo and GPS System Times (GGTO determination). Military (Y-code) GPS timing receiver installed at the USNO premises will serve for this purpose. It will be the GPS/USNO responsibility to correct for the difference between the GPS System Time and the reference (UTC(USNO)) driving the TWSTFT and GPS-CV time transfer hardware.

(2) Both sides USNO and PTF will use their GPS timing receivers driven by the local timing reference to generate so called GPS-CV measurements. One can calculate the difference between the GPS and Galileo system time by combining these GPS-CV measurements obtained at USNO and PTF. Note the GPS-CV measurements are performed in parallel with TWSTFT measurements.

(3) Both sides perform TWSTFT measurements a number of times per day according to the TWSTFT measurement schedule, provided operationally to GMS by TSP to ensure the smooth operations of TWSTFT. TWSTFT method is based on the simultaneous exchange of RF signals over a commercial communication satellite. The RF signals contain a timing information which is derived from the local reference clock.
Thus TWSTFT likewise the GPS-CV is used for the comparison of the distant clocks (in our case GST and GPS system time).

(4) Both USNO and PTF need to exchange the TWSTFT and GPS-CV measurement files (Time Transfer data) via SPF in order to be able to calculate the results of the time transfer measurements in the form of GPS system time- Galileo system time =GGTO.

(5) The time transfer data are processed by USNO and PTF independently in order to obtain the respective GGTO values.

(6) PTF and USNO exchange their GGTO values based on TWSTFT or GPS-CV measurements via SPF. Note GPS-CV measurements are used as a source data to calculate GGTO only if the TWSTFT data are not available or their quality is poor.

(7) PTF passes the GGTO to GACF for its further provision to MUCF as an input to the GGTO threshold configuration process.

(8) Prior to entering the GGTO coordination process the USNO and GACF operators need to agree on the GGTO configurable threshold. This process will be handled operationally upon operator's request.
Upon successful determination of the GGTO threshold, the actual value is provided converted into a configuration parameter and passed to PTF.

(9) During the coordination process PTF compares difference between its own GGTO value and GGTO value generated by GPS/USNO against the agreed threshold. The test result is sent to MUCF.
A routine procedure (GMS_NOM_012) is executed at GACF on daily basis by GMS Mission Controller during coordination with USNO, before to configure the OSPF and update MSF configuration.

A valid GGTO is set at the agreed value if any of the following conditions is verified :

- The PTF GGTO data file (TIME_Galileo_Gps_Offset_yyyymmddhhmmss) with the current date is found at GACF.
- The GGTO is successfully monitored at MUCF
- The delta GGTO between PTF and USNO is within the agreed threshold
- No critical alerts are present in the infrastructure used to compute and monitor the PTF GGTO.

If any of above conditions are verified , each side of the interface acknowledges this by sending back the GGTO value received from its counterpart. Thus PTF sends to GPS/USNO the GGTO generated by GPS/USNO and vice versa.

The GGTO is set to dummy value (9999) if one of the following events occur:

- The PTF GGTO data file (TIME_Galileo_Gps_Offset_yyyymmddhhmmss) with the current date is not found at GACF.
- The GGTO is not monitored at MUCF
- TBD alerts are present in the infrastructure used for GGTO computation and monitoring and control.

Note: A dummy GGTO will be deployed to OSPF if PTF or MUCF are not in the operational state.

Should the delta GGTO exceed the agreed threshold, an investigation/troubleshooting is initiated by GACF on an operational level to solve the problem causing the failure.
If the problem cannot be resolved the coordinated signals are filled with the dummy data in the form of "9999" to indicate that the GGTO value is not available.

(10) MUCF uses the coordinated GGTO and GGTO delta values received from PTF for the mission monitoring purposes.

(11) Actual GGTO and GGTO delta values are provided to GACF. Data is used by GACF to generate OSPF configuration parameters.

(12) GGTO values (as per SIS ICD) are converted into the configuration parameters and are sent to OSPF, data is used to prepare the navigation batches. applicable to OS, and PRS services.

(13) Distribution of the GGTO by GPS system. Note present GPS satellites do not have a capability to include GGTO into their navigation messages, consequently in the initial stage the GGTO will be broadcasted by GPS to their users by other means e.g. dedicated website etc.

6.2.5 SFC2.3 Contribute to Geodetic Reference Determination

6.2.5.1 SFC2.3 High Level Description

The purpose of this SFC is to:

- Assist GRSP with the generation and the maintenance of the GTRF including the provision of all GTRF related products consumed by MSF, OSPF, MUCF
- Plan & enable the SLR campaigns by ensuring the provision of the required data such as data on the Galileo constellation to ILRS. Process the resulting SLR data for the purpose of off-line ODTS calibration and the independent validation at GMS level of the GTRF provided by GSRP and reference orbit determined at GMS level.
- Enable the GRSP provision of the scientific mainly geodetic products for their use by GMS (OSPF) and GCS (FDF).
- Validate the MSF generated products (orbits, clocks and ERPs) for their use by OSPF.

Remark: It is important to note that the MRD specification of the geodetic service performance and availability are conditioned to the availability and quality of the GRSP products. Therefore, it is essential to ensure that the GRSP product quality and availability is covered by appropriate GRSP Design and Performance requirements in-line with the MRD geodetic availability and accuracy targets.

6.2.5.2 Physical static Block Diagram

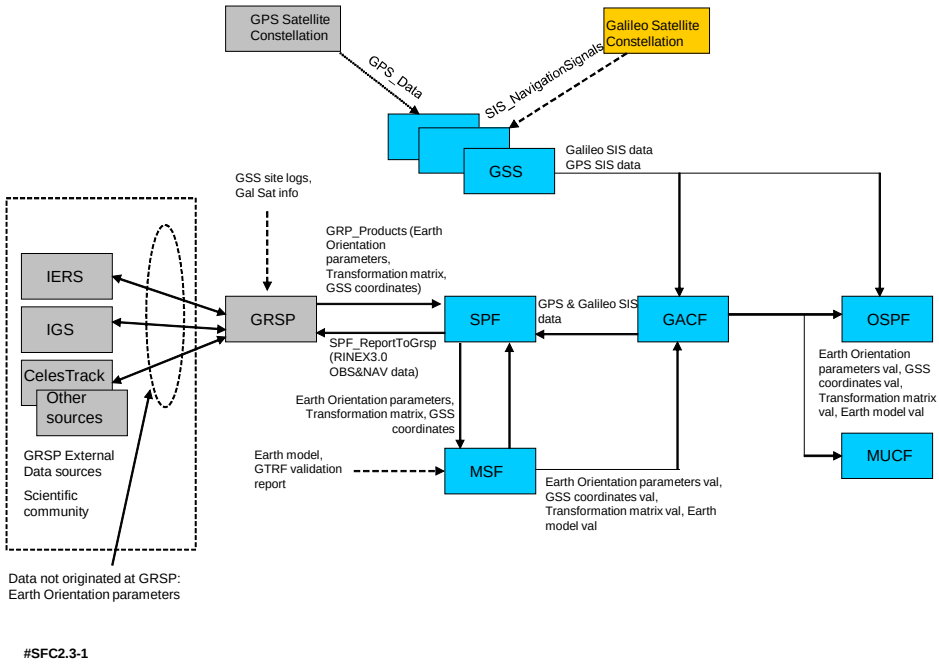


Figure 121 : #SFC2.3-1: Support the generation and maintenance of the Galileo Terrestrial Reference Frame

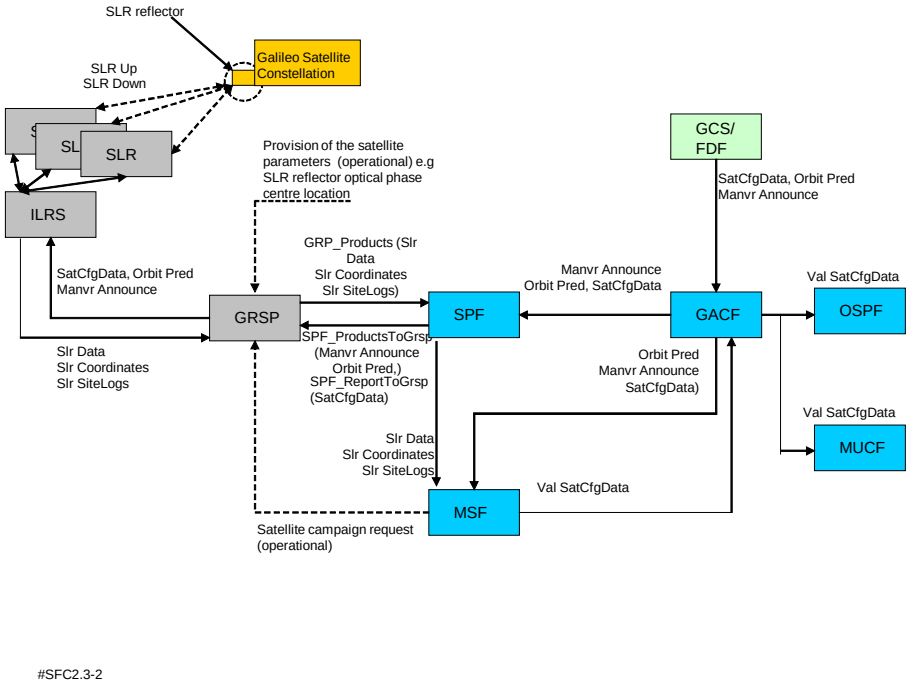


Figure 122 : #SFC2.3-2: SLR Measurements

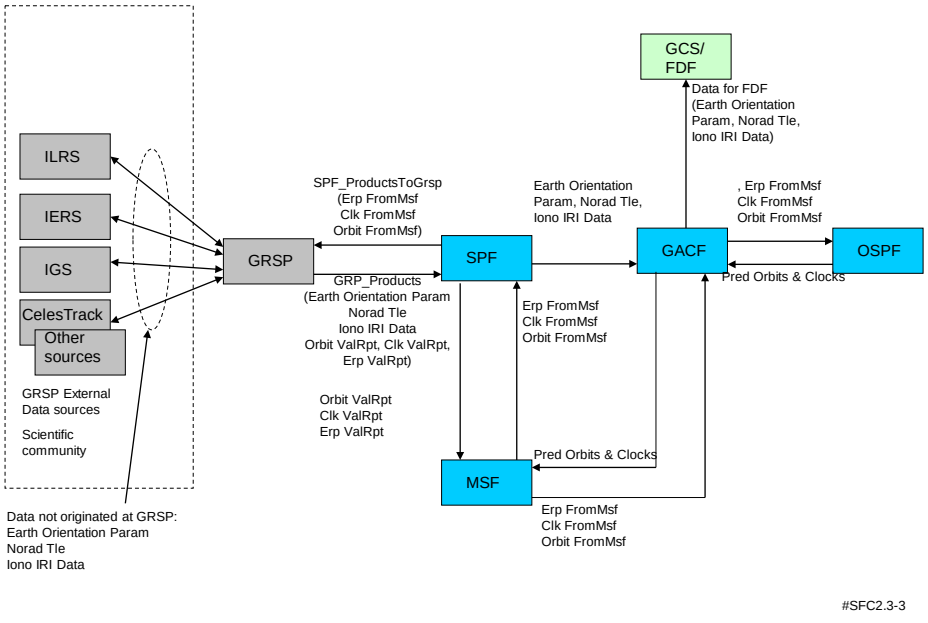


Figure 123 : #SFC2.3-3: Support Data Provision to GCS& GMS& MSF product validation

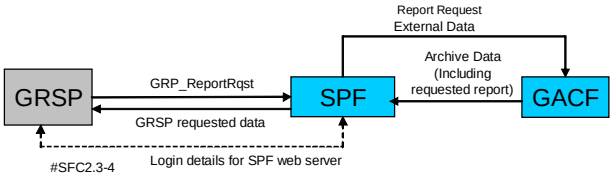


Figure 124 : #SFC2.3-4: Specific Report Dissemination on GRSP request

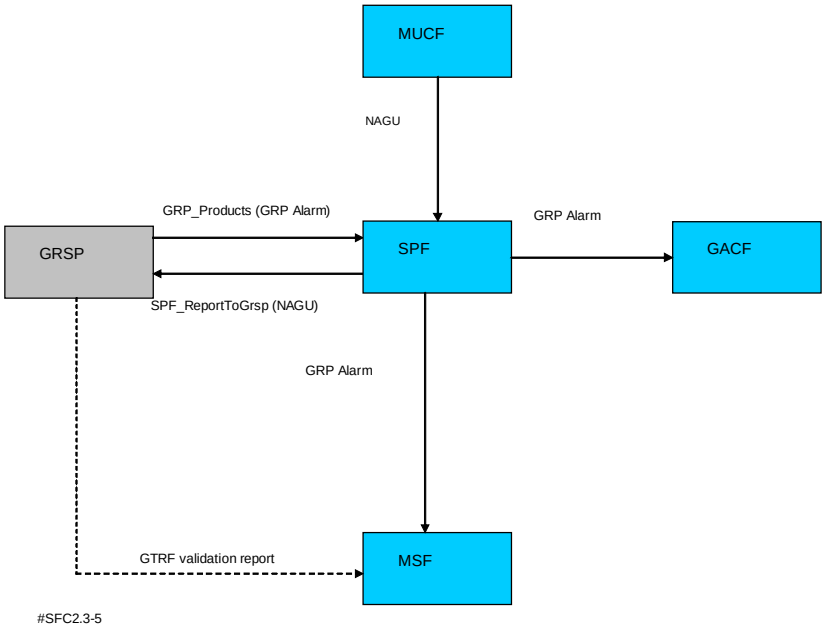


Figure 125 : #SFC2.3-5: Contingency information exchange (NAGU and Alarm provision)

6.2.5.3 Overall Physical Description

The delivery of the GTRF, SLR, scientific data and validation reports (scope for the SFC2.3) relies heavily on the Geodetic Reference Service Provider. GRSP however cannot provide all of the products required themselves, since these are generated by many scientific organizations

around the world. GRSP will have to use its well established links to these scientific communities and if necessary create and maintain new links in order to deliver to Galileo System the required functionality.
The most important functionality provided by this SFC is the realization and maintenance of the GTRF (Galileo Terrestrial Reference Frame). This reference frame will be realized independently by the network of the Galileo GSS stations under the control of Galileo system. Any coordinates calculations done within Galileo will be referenced to GTRF.

6.2.5.3.1 GTRF Generation, SLR and Support Data Provision: GRSP

The mission of the Galileo Reference Service Provider (GRSP) is to:

1) Define and to implement the Galileo Geodetic System. GRSP is providing regularly GMS with a set of data used as the geodetic references (e.g. GMS stations reference positions, earth rotation parameters …) on which the mission service data are built. Note that the term "reference frame", addressed to the Galileo Terrestrial Reference Frame (GTRF), means GTRF is a specific realization of the ITRF defined at a given epoch.

2) Act as an interface between the ILRS and Galileo to enable the SLR measurements and the utilization of the SLR results within the Galileo system.

3) Support the functionality of the GMS and GCS elements by providing:

a. Validation reports for the MSF generated products

b. Geodetic and mission data to GCS/FDF
c. Geodetic and mission data to OSPF, MUCF &MSF

4) Act as an interface between the IERS and Galileo to enable the utilization of EOP data within the Galileo system. :
The EOP data interface provided by GGSRP is as follows:

- normally, the GSRP will send to GMS the EOP data provided by IERS whenever IERS Bulletin A is available.
- If the current IERS Bulletin A is not available, but the age-of-date of latest IERS bulletin A available at GSRP is less than 4 days, then the GSRP will send a Warning to GMS, as well as the EOP data contained in the latest IERS Bulletin A available at GSRP.
- If the current IERS Bulletin A is not available and the age-of-date of latest IERS bulletin A available at GSRP is not less than 4 days, the GSRP will send an Alarm to the GMS, as well as backup EOP data estimated by GSRP itself.

6.2.5.3.2 SLR: ILRS

Satellite Laser Ranging data are provided by the GRSP to GMS automatically on a daily basis. Satellite Laser Ranging (SLR) is a high-precision two-way ranging technique allowing measurement of the distance between a ground SLR station and a spacecraft down to millimeter accuracy. It is produced and used during pre-defined measurement campaigns. The ranging measurements themselves are performed by the ILRS (International Laser Ranging Service), based on information provided by GMS to GRSP. The results of these measurements

are provided by GRSP to GMS for further analysis as e.g. initialization or calibration of OD&TS process.

6.2.5.3.3 GTRF generation: IGS

The International GNSS <http://igscb.jpl.nasa.gov/faqs.html> Service (IGS), formerly the International GPS Service, is a voluntary federation of more than 200 worldwide agencies that pool resources and permanent GPS & GLONASS station data to generate precise GPS & GLONASS products. GRSP will combine the Galileo data with the IGS, IERS & CelesTrak data in order to generate the GTRF and to ensure its alignment to the ITRF (within 3 cm 2σ).

6.2.5.3.4 GTRF Realization: IERS

The GTRF provided by GRSP is linked to the International Terrestrial Reference Frame (ITRF), provided on a multi-yearly basis by the International Earth Rotation Service (IERS) to the community, but also the GRSP.

6.2.5.3.5 FDF Collision and Avoidance Software: CelesTrack

CelesTrack is an organization/service which provides NORAD TLE data. NORAD TLE data are Orbital object Keplerian data to be used with NORAD's SGP4/SDP4 orbital model for the calculation of the position and velocity of the associated object orbiting the earth.

6.2.5.3.6 RINEX conversion, GRSP-GMS Data Exchange: SPF

Service Product Facility is serving as a secure interface between GMS and the external entities. Any data (incoming/ outgoing) between GMS and external interfaces such as the interface to GRSP have to be routed via the SPF.

The SPF in addition to its routing function also generates a RINEX Navigation and Observation data required by the GRSP for the GTRF realization. The SPF needs prior to generating the RINEX files the retrieve of GSS raw measurements from the Galileo archive.
The SPF provides a configuration file (CNF_SvMask) including a correspondence table between identification of the Galileo satellites SVID and CCSDS Identifier.
The file is sent on event, i.e. whenever there is a change in the constellation or in case of test involving a satellite

6.2.5.3.7 Generation of Orbit and Clocks GTRF Data Validation and ERP: MSF

The MSF elaborates and validates the algorithm functions configuration parameters.

It restitutes precise orbits, allows to configure/maintain satellite models configuration data for OSPF, runs the off-line accurate OD&TS computation for determination of calibration data and parameters to support OSPF algorithm.
This includes long term performance analysis for Historical SISA determination.
MSF computations are performed regularly on a routine based scenario using standard GSS measurements and/or Satellite Laser Ranging measurement data from a campaign that has to be planned each time it is needed .
The OSPF calibration data will be inserted in operational configuration parameters/files transferred at last from MSF to GACF (MSF-ERD-0840) in charge of managing the final reconfiguration of the operational element.

The generated operational parameters will be validated off-line as much as possible (on real elements for example) before providing them to the operational chain
The MSF sent its products (reference orbit and clock and ERP) to GSRP for validation (, and it validates the data received from GRSP before on-line configuration of the algorithms

The MSF is capable to use the SLR measurements to provide

- **Offline OD&TS calibration and accuracy monitoring (Use Case 1)**

  The SLR data can be used as an option, as additional input to the MSF orbit determination process to provide solution calibration and monitoring of accuracy of the solution obtained using only GSS measurements. The GSS and SLR data are combined in the off-line orbit determination process by applying a relative weighting between GSS and SLR data. The off-line ODTS solution using only GSS data is obtained by unweighting (i.e setting to dummy) the SLR data.
  The GSS-SLR combined solution can be used to verify GSS coordinates updating and ERP provided by GSRP.

- **Independent validation of GTRF alignment to ITRF through Computation of SLR residuals (Use Case 2)**
  The MSF is capable to provide and independent validation of GTRF any time an updated GRTF is provided by GSRP by using SLR measurements. To this purpose, the new GTRF is applied to obtain a solution and the statistical of SLR residuals (defined in terms of difference between computed satellite range and compensated SLR measurement) is computed. If the computed SLR residual exceed a configurable threshold (default value : 10 cm rms), the GRTF is flagged as not valid. The orbit and clock solution determined by the GMS are sent GSRP for further investigation.

- **Daily validation of MSF reference orbit independently for weekly GSRP validation (Use Case 3)**

  The reference orbits,satellite clocks and ERP determined by MSF are normally validated by GSRP on a weekly basis.
  The GMS is able to send its reference orbits, clocks and ERP solutions to GRSP for validation. However, GRSP takes a week (TBC) for its final products to be finalised.
  The use of SLR measurements, with a latency of one day, allows the MSF to get a daily validation of its reference orbit independently from the weekly validation at GSRP.
  This is still obtained by computing the statistical of SLR residuals.
  If the computed SLR residual exceed a configurable threshold (default value : 10 cm rms), the GRTF is flagged as not valid. The orbit and clock solution determined by the GMS are sent GSRP for further investigation.

  It should be noted that Use cases2 and 3 are not supported by System baseline 1.5.0.
  The triggering frequency of the three SLR Measurement use cases are summarized in the following table.

Use Case 1	Use Case 2	Use Case 3
request. Normally: • A few months after a	• On request. Normally Every time an updated	• Daily , as this use case is foreseen to

new satellite is inserted in GMS solution. • Quarterly for, alignment with what KPI' s are defined for operations.. • After every event that put a satellite out of the GMS solution (maneuver and other satellite outages). • Detection of GMS solution degradation.	GTRF is provided by GSRP.	reduce the latency (one week)of MSF reference orbit validation provided by GSRP.

In order to allow the use of SLR measurements in the current design implementation ,the SLR format has changed from Normal Point
to Consolidated Laser Ranging Data (CRD) Format. The CRD format is indeed already implemented in the MSF, see GAL-DCP-GMV-TGVF-30003.

6.2.5.3.8 Conversion of GRSP and MSF Data Into Configuration Parameters, Provision of GRSP data to FDF:GACF

Ground Assets Control Facility does the monitoring of the hardware infrastructure, including the configuration of the GMS elements (for instance OSPF).GACF is also considered as a gateway for the data (data generated by external entities e.g. GRSP) exchange between the GMS and GCS.

6.2.5.3.9 OD&TS: OSPF

The Orbit and Synchronization Processing Facility uses the configuration parameters as provided by the GRSP and as validated by the MSF to configure its OD&TS processes.

6.2.5.3.10 GTRF realisation & Collection of GPS&Galileo SiS Observables: GSSs

Network of the globally distributed GSS stations will collect the GPS and Galileo SIS observable. The data collected will be provided in the raw format to OSPF (OD&TS processing) and SPF (via GACF).
The raw data will be supplied by SPF into RINEX format and processed by GRSP to generate the GSS antenna coordinates. Note GTRF is defined by the 3D Cartesian coordinates and velocities of the GSS tracking antennas.

6.2.5.3.11 Integrity processing: MUCF

Mission Uplink Control Facility is in charge of mission online monitoring. For this purpose MUCF will also use the GRSP generated, but MSF validated data.
To complement the mission monitoring functionality MUCF will generate the NAGU reports based on the information received from GCS.

Remark: *the automated NAGU generation concept is currently not consolidated in the STRB2.0 baseline. It is expected that the content of the signal INF_NoticeAdvisoryToUsers will be refined in close coordination with GSC in the frame of GSC operational concept consolidation. In the meanwhile, the signal INF_NoticeAdvisoryToUsers will not be disseminated to the external entity and the NAGU will be generated manually and sent to the GSC. In both cases (automated and manual NAGU processes) the authorization for dissemination to the public is granted by the corresponding formal process at GSC level.*

6.2.5.3.12 Configuration of the Collision and Avoidance Software:GCS/FDF

The FDF (Flight Dynamics Facility) at GCS receives the NORAD TLE and ionospheric data from GMS. Those data will be used for flight dynamics and collision avoidance software configuration. The retrieval of those data will be performed periodically by GRSP from the corresponding web-sites of NORAD and NASA and subsequently provided to SPF.
FDF main functionality is the generation of Galileo orbit predictions and spacecraft maneuvers, these are consumed within the system and also by GRSP/ILRS. Note the information on the position of the Galileo satellites including the planned manoeuvres is required by ILRS to enable their tracking using SLR technique.

6.2.5.4 Dynamic Scenarios

Following scenarios have been identified:

- Nominal Routine Operation Scenario: Support the generation and maintenance of the Galileo Terrestrial Reference Frame
- Nominal Routine Operation Scenario: SLR Measurements
- Nominal Routine Operation Scenario: Support Data Provision to GCS& GMS& MSF product validation
- Nominal Routine Operation Scenario: Specific Report Dissemination on GRSP request
- Contingency information exchange (NAGU and Alarm provision)

6.2.5.4.1 #SFC2.3-1: Support the generation and maintenance of the Galileo Terrestrial Reference Frame

The MSC for scenario this is provided below:

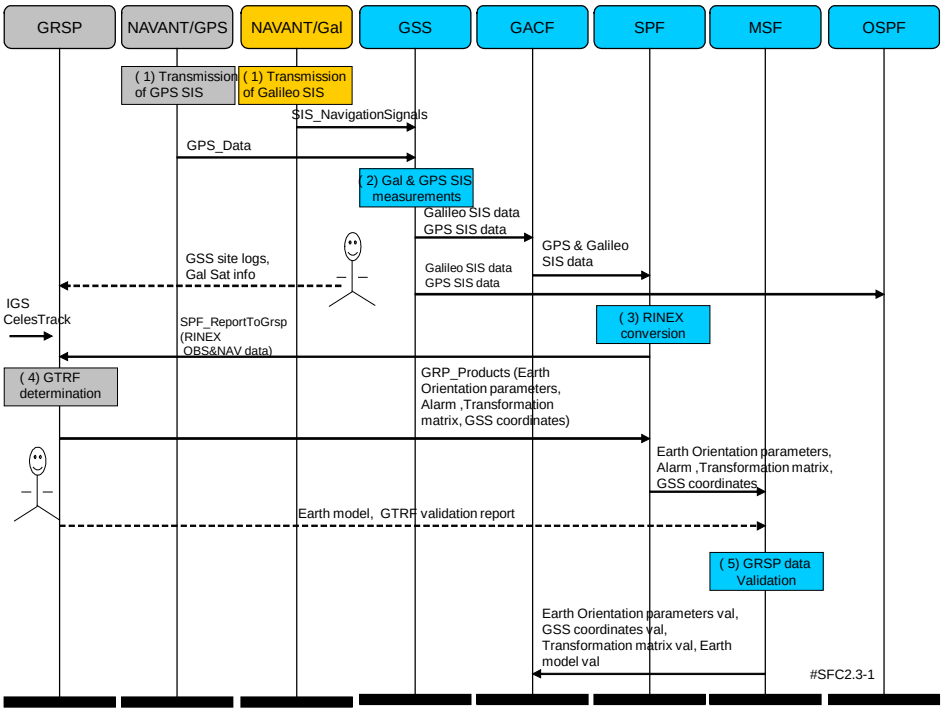


Figure 126 : #SFC2.3-1: Support the generation and maintenance of the Galileo Terrestrial Reference Frame

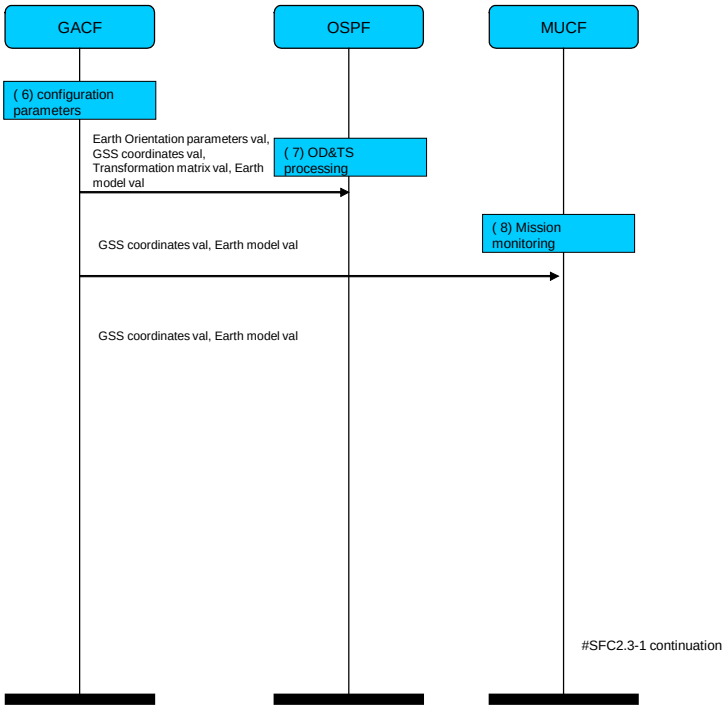


Figure 127 : #SFC2.3-1: Support the generation and maintenance of the Galileo Terrestrial Reference Frame, continuation

Scenario Textual Description:
Galileo Terrestrial Reference Frame is generated by the GRSP (Galileo Reference Service Provider), entity external to Galileo Core.

Inside Galileo it is the GMS/MSF which is supporting the GTRF generation and maintenance.
The GRSP products (Earth Orientation parameters, Alarm, Transformation matrix, GSS coordinates) following their validation by MSF are consumed by MUCF (online missing monitoring) and OSPF (OD&TS processing).
The sequential description is detailed below:

(1) The Galileo and GPS SIS are transmitted by the respective constellations. These signals are collected by global network of GSS.

Note each GSS will be equipped with the GPS and Galileo OS&PRS receivers. The PRS data will not be considered for the GTRF realisation.

(2) Global network of GSS collect Galileo and GPS pseudo-range measurements. The GSS raw measurements are passed to GACF &OSPF. OSPF uses the data for the OD&TS processing.

The GSS measurements are also required by GRSP in order to calculate GSS antenna coordinates, which form the basis for the GTRF. Thus GTRF is realised by the coordinates of the GSS global network.

(3) SPF retrieves the Galileo and GPS raw measurements (Galileo PRS measurements are not used) from GACF and converts them into RINEX format. Two types of files are generated: RINEX Observation file containing the Galileo and GPS pseudo-ranges and RINEX Navigation files (containing the Galileo navigation messages).

Note there will a RINEX Observation file per receiver. There will a RINEX Navigation file per receiver. GPS navigation messages will be retrieved by GRSP from the IGS network.

In addition to the RINEX data GRSP also needs the information on the GSS (GSS site logs), initial GSS coordinates, information on the GSS antenna phase centres offsets, their variations and satellite information. These will be provided to GRSP by an operator via an operational interface. The satellite information will consist of the following items:

- Spacecraft nominal/start mass
- Nominal attitude model/law
- GNSS phase centre offset and phase centre variations for all frequencies
- SLR reflector optical phase centre location
- Location of the spacecraft centre of mass
- Spacecraft dimensions (size of solar panels, and other external surfaces)
- Reflectivity and absorption of the surfaces
- Phase centre variations for the satellite transmitting antennas as a function of nadir angle

(4) GRSP retrieves from the SPF the RINEX files and together with the operationally provided data and the data from the IGS and Celestrack networks combines all data in order to determine the GSS coordinates, basis for the GTRF. GRSP provides 3D Cartesian coordinates and velocities of the GSS tracking antennas in GTRF (GSS coordinates). This set of coordinates and velocities define the GTRF.

Note this data is provided Yearly (first year half-yearly) or whenever GTRF is updated, but can also be requested by GMS operator to GRSP operator. In addition GRSP provides a GTRF validation report every time new GSS coordinates are provided to GMS. This report is provided operationally via e-mail (see AD.14, section 2.4.2.4)

Note the positions of all GSS's are computed/maintained by imbedding the GSS's into a larger network of IGS stations and solving their station coordinates simultaneously with all

the stations of the whole network. A subsequent network alignment will automatically align the GSSs to the ITRF .

The computation are performed on a regular basis, and if differences to the existing GTRF are detected a new GTRF realisation will be provided. If larger errors for single stations occur an alarm will be sent to GMS, please refer to contingency scenario (SFC2.3-5).

GRSP products (Earth Orientation parameters, Alarm, Transformation matrix) are made available for the scheduled download by SPF.

GRSP also provides to MSF a document containing the description of Earth model. This model will be provided only once at the start up of the interface via an operational means and it is assumed to be fixed during the Galileo operational life time.

(5) MSF is in charge of the GTRF maintenance and supporting its generation (validation of GRSP products), retrieves from SPF the GRSP products (Earth Orientation parameters, Alarm, Transformation matrix, GSS coordinates).

MSF validates the GRSP generated data before they are used by other elements such as OSPF and MUCF

MSF also validates the GTRF realisation, however for this purpose also the processing of SLR data is necessary, please refer to scenario SFC2.3-2.

At the end MSF passes the validated products to the GACF for their further distribution of the above mentioned elements.

(6) GRSP products validated by MSF (Earth Orientation parameters val, GSS coordinates val, Transformation matrix val, Earth model val) are converted by GACF into a configuration parameters to be implemented by OSPFand MUCF.

(7) OSPF uses the configuration parameters (Earth Orientation parameters val, GSS coordinates val, Transformation matrix val, Earth model val) commanded by GACF for the OD&TS processing.

(8) MUCF uses the configuration parameters (GSS coordinates val, Earth model val) commanded by GACF for the mission monitoring.

6.2.5.4.2 #SFC2.3-2: SLR Measurements

The MSC for scenario this is provided below:

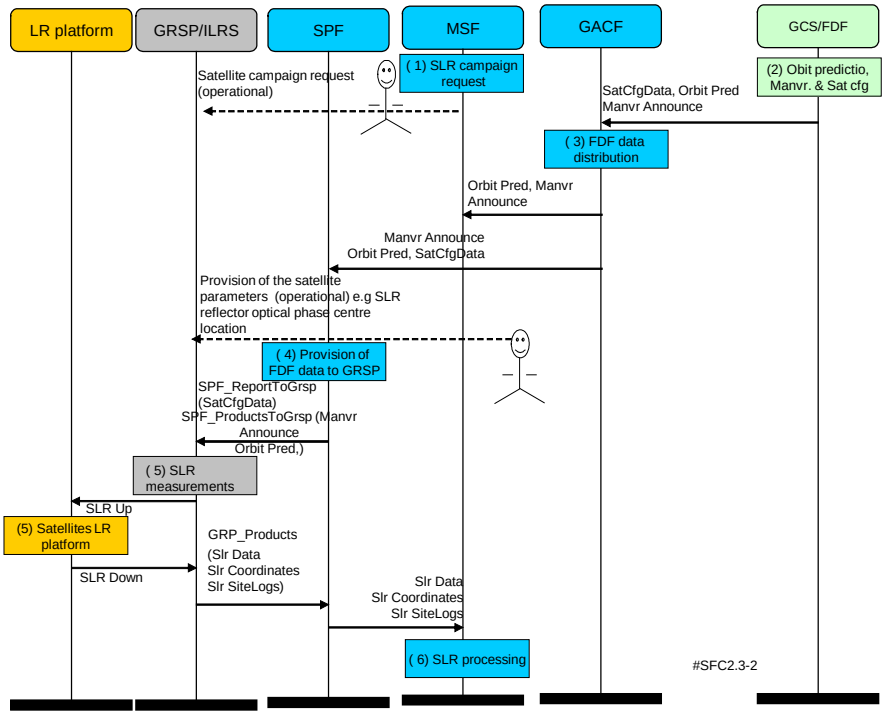


Figure 128 : #SFC2.3-2: SLR measurements

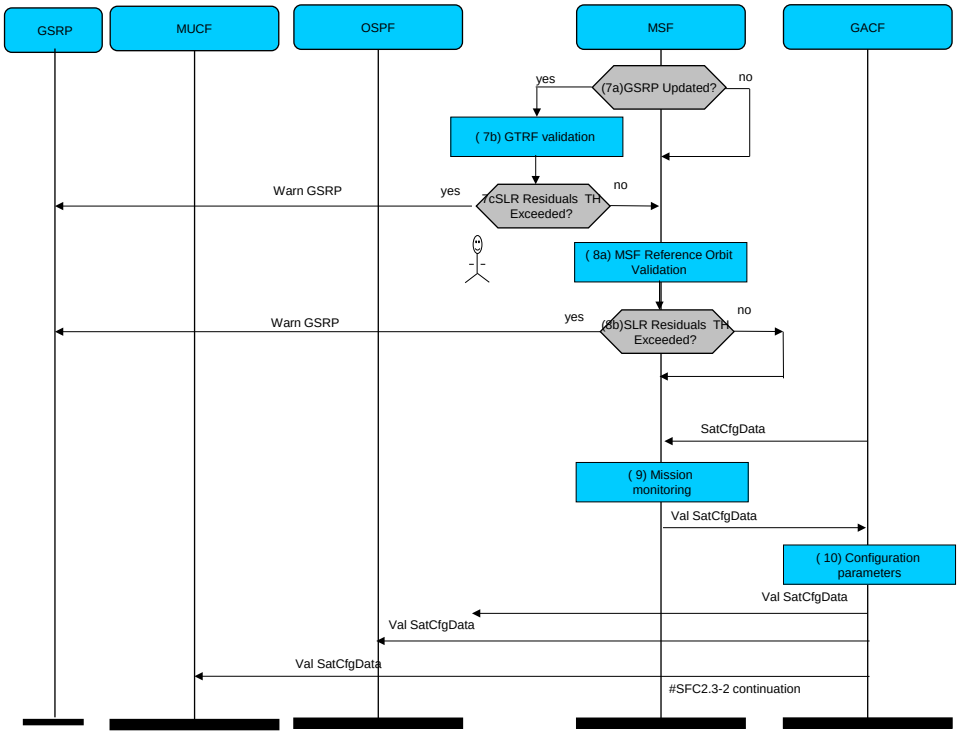


Figure 129 : #SFC2.3-2: SLR measurements, continuation

Scenario Textual Description:
Satellite Laser Ranging is a high-precision two-way ranging technique allowing measurement of the distance between an SLR ground station and a spacecraft down to millimetre accuracy. Its

products are used by ILRS to feed the Scientific Community involved in geodetic studies, as well as the Geodetic Reference Service Provider.

The Dynamic Scenario #2.3-2 described in this section considers the following measurements use cases:

- MSF algorithm calibration and accuracy monitoring using SLR measurements (Use Case#1)
- Independent validation of GTRF alignment to ITRF through Computation of SLR residuals (Use Case# 2)
- Daily validation of MSF reference orbit independently for weekly GSRP validation (Use Case# 3)

It should be noted that Use Cases 2 and 3 are not supported by System baseline 1.5.0.
For Use Case 1, SLR measurements are configured as additional input to MSF algorithms and processed in the same way as GSS measurements.

Satellite Laser Ranging measurements will be used by MSF for independent validation of GTRF realisation. To simplify the diagram the GRSP acts in this scenario as ILRS.
The sequential description is detailed below:

1) MSF which is in charge of the SLR campaign planning and data processing requests the SLR measurements via an operational interface. Request is made to GRSP, who forwards SLR campaign request further to ILRS.

2) FDF generates the the following data:

- orbit predictions (*MISC_OrbitPred*),
- satellite manoeuvres *MISC_ManvrAnnounce*
- satellite configuration *CNF_SatCfgData*

This data are passed to the GACF for their further provision to GRSP/ILRS via SPF and MSF.

3)SPF and MSF retrieve Orbit predictions, satellite manoeuvres and satellite configuration data generated by FDF (Flight Dynamics Facility-GCS) from GACF.

4) SPF makes available orbit predictions, satellite manoeuvres and satellite configuration data needed for the SLR campaign by ILRS to be downloaded by GRSP.
GRSP retrieves the data and passes them to ILRS. Note ILRS in order to be able to track Galileo satellites needs to be able to estimate within certain limit the position and technical parameters/specification of Galileo satellites. ILRS uses for this purpose data generated by FDF including operationally provided satellite parameters such as optical phase centre location of SLR reflector.

5) ILRS performs the SLR measurement campaign during which Galileo satellites are being tracked and the resulting tracking data being collected. The measurements are performed by

the global network of the SLR stations which measure the instantaneous round trip time of flight of ultra-short pulses of light to Galileo satellites equipped with SLR reflectors. ILRS provides the SLR data to GRSP for their further provision to GMS via SPF.

6) **SLR data are made available in GACF archive**. GACF will store the following data received from GRSP:

- GRP_SlrCoordinates (only received in case of ILRS infrastructure change)
- GRP_SlrData (daily signal)
- GRP_SlrSiteLogs (only recieved in case of ILRS infrastructure change)

Remark : it is not needed to request GRSP via an operational interface to deliver the products

7) Stop scenario in case of missing GRP_SlrData
If no signal (GRP_SlrData) was received with data covering the MSF daily computation measurement arc, then the scenario stops and the GRSP shall be alerted via an operational interface.

8) MSF shall receive automatically the following data from GACF :

- GRP_SlrCoordinates (only in case of ILRS infrastructure change)
- GRP_SlrData (daily signal)
- GRP_SlrSiteLogs (only in case of ILRS infrastructure change)

The MSF will able to perform its computation with or without SLR data, depending on the use case.

9) MSF algorithm calibration and accuracy monitoring using SLR measurements (Use Case#1)

MSF can use SLR data as additional input in its off-line precise orbit determination process. To this end the following data are used:

- L-band observation obtained by the GSS network (SIS_RawData)
- GSS coordinates in the GRTF provided by the GRSP (CNF_GSSCoordinatesVal)
- SLR data provided by the GRSP including ;
  - SLR stations coordinates in ITRF coordinates (GRP_SlrCoordinates) (only in case of ILRS infrastructure change)
  - SLR measurements (GRP_SlrData),(daily signal)
  - GRP_SlrSiteLogs (only in case of ILRS infrastructure change)

The GSS and SLR data are combined in the off-line orbit determination process by applying a relative weighting between GSS and SLR data.

The data used by MSF to generate precise orbits are passed to GSRP for validated by GRSP and provided to OSPF, please refer to scenario SFC2.3-3.

10)Satellite Laser Ranging measurements are used at MSF to provide an independent validation of updated GTRF provided by GSRP (Use Case#2)

The GTRF supplied by GSRP is expected to be correct, In order to improve confidence, an additional validation is carried out at MSF any time an updated GTRF is provided by GSRP. The validation is obtained by computing the SLR residuals and comparinhg them against a configurable threshold.
To this end, the following input data are used:

- L-band observation obtained by the GSS network (SIS_RawData)
- New GSS coordinates in theversion) GRTF (new GTRF provided by the GRSP (CNF_GSSCoordinatesVal)
- SLR data provided by the GRSP including ;
  - SLR stations coordinates in ITRF coordinates (GRP_SlrCoordinates) (only in case of ILRS infrastructure change)
  - SLR measurements (GRP_SlrData),(daily signal)
  - GRP_SlrSiteLogs (only in case of ILRS infrastructure change)

The SLR residuals are defined per satellite in terms of difference between SLR measurements converted in GTRF and the computed slant ranges of the SLR stations in GTRF. Upon provision from GSRP of updated GRTF/GSS data (step 10a) , MSF computes the SLR residuals per satellite as follows t (step 10b)

- The SLR station coordinates received from GRSP are converted from ITRF to GTRF
- The ODTS process is run at GMS using as input the new GSS coordinates and GSS observables only.
- The SLR residuals per satellite are computed as difference between SLR measurements converted in GTRF and computed slant range from SLR stations (in GTRF) to the orbit determined using GSS observables only.
- The SLR statistics of SLR residuals are computed per satellite and compared with a configurable threshold (default value : 10 cm rms).

The MSF computation is performed over an arc of 3 days and applying to SLR observables a dummy weight.
Then a global SLR statistics (1-sigma, 2-sigma,3-sigma,…) per satellite is generated using SLR residuals over the measurement arc.
If the computed SLR residuals exceed a configurable threshold (default value : 10 cm rms), the GRTF is flagged as not valid. The orbit and clock solution determined by the GMS are sent GSRP for further investigation (step 10c)

If the SLR residual statistics are within the expected limits, then the validated GTRF/GSS coordinates data are deployed to the OSPF. (step 10d)

11) Indenpendent validation of the MSF products (Use Case#3)
Satellite Laser Ranging measurements can be used at MSF to provide a daily independent validation of reference orbit and clock solution determined at MSF level.
The MSF performs nominal daily computation of its products without using SLR data (step 11a).
The GMS normally sends to GSRP its products (reference orbits&clocks solution) for validation. However, GRSP takes one week for its final products to be finalised,

An independent validation of reference orbit&clocks determined at MSF can be obtained with a latency time of one day by using SLR measurements (step 11b)This is achieved by computing SLR residual statistics per satellite as seen at previous point 10b), but using the GRTF currently available at MSF.

If the SLR residual statistics exceed a configurable threshold (default value: 10 rms), then the orbit and clock solutions determined by the GMS are sent to GSRP via an operational, to warn GSRPthat an issue was found during the MSF computation (step 11c).If the SLR residuals are within the expcted limit, then the MSF product validation is successfully completed.

12)MSF uses orbit predictions and satellite manoeuvres to plan the SLR mission and for the MSF mission offline monitoring purposes.

In addition MSF validates the satellite configuration data for their further usage by OSPF and MUCF

13) MSF provides the validated satellite configuration data to GACF, where the data is converted into configuration parameters, which are commanded to OSPF and MUCF to configure relevant processing performed by these elements.

6.2.5.4.3 #SFC2.3-3: Support Data Provision to GCS& GMS

The MSC for scenario is provided below:

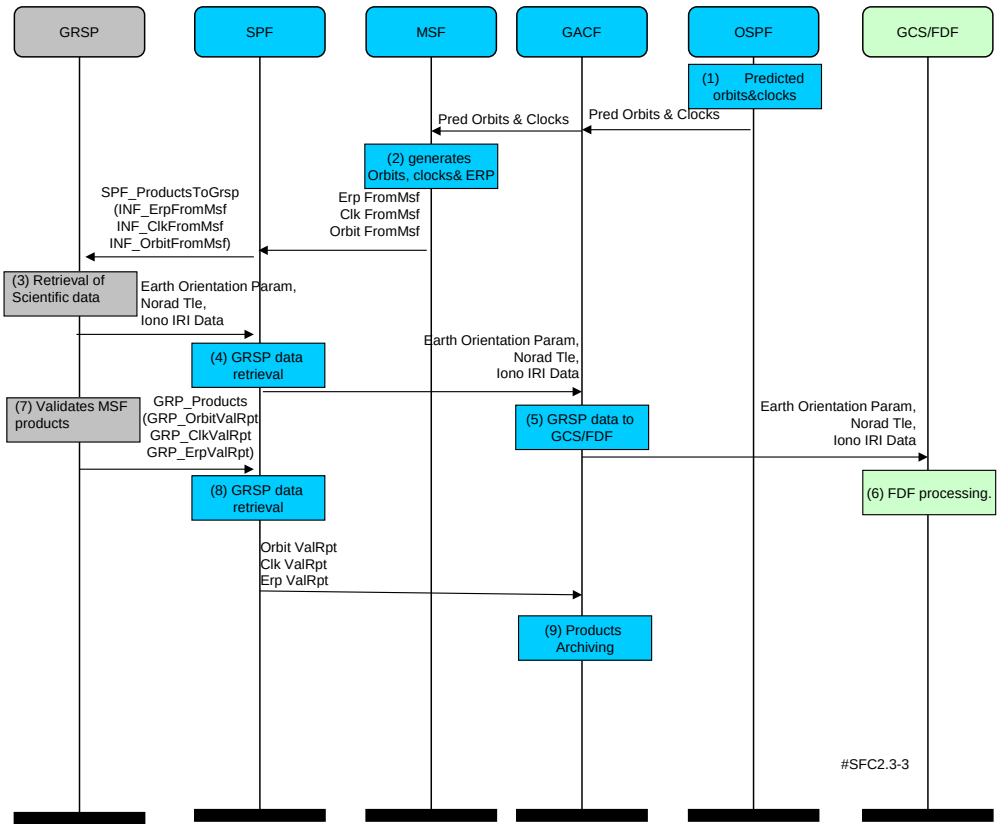


Figure 130 : #SFC2.3-3: Support Data Provision to GCS& GMS

Scenario Textual Description:
GCS (FDF) needs number of external scientific products such as Norad Tle, Iono IRI and Earth orientation parameters for the determination of the position and velocity of associated satellites used in FDF Collision avoidance software. In addition the data is used by FDF to calculate the Ionospheric corrections.
Since the FDF has no access to the outside world (security reasons) it has no way of accessing of this data. To provide this data to FDF it has been agreed that GRSP can play role of the scientific data provider. GRSP to fulfil this task will use its well established links with the scientific community worldwide.

Finally MSF generates based on the OSPF clock and orbit products the precise estimates for the orbits, Galileo satellite and station clocks and estimates of Earth rotation parameter. Following the GRSP validation of the MSF products these will be used for OS and PRS.
The sequential description is detailed below:

1) OSPF generates predicted orbits and clocks and passes the data to GACF for their further provision to MSF
2) MSF process OSPF data retrieved from the Archive and generates the precise estimates for the clocks, orbits and ERP (Earth Rotation Parameters) and passes the data to SPF for their further provision to GRSP
3) GRSP retrieves the needed data (Norad Tle, Iono IRI and Earth orientation for FDF) from the different scientific organisations over the world.
4) SPF retrieves the data from the GRSP and passes them to GACF for their further provision to GCS.

The ERP parameters are sent following the following logic applied daily:

- If available, the current IERS Bulletin A is sent to GMS in the GRP_EarthOrienParam signal.

- If the current IERS Bulletin A is not available and the last available is less than 4 days old, then the latest available IERS Bulletin A and GRP_Alarm signal category WARNING are sent in the GRP_EarthOrienParam signal

- If the current IERS Bulletin A is not available and the last available is not less than 4 days old , then the latest available IERS Bulletin A and GRP_Alarm signal category ALARM are sent in the GRP_EarthOrienParam signal. In this condition the GMS will updated its own ERP daily.
  The GMS updates its own ERP on a daily basis until the GRP_Alarm signal is no longer sent in the GRP_EarthOrienParam signal. This indicates that IERS Bulletin A is available again. In this condition the GMS will updated its own ERP weekly..

5) Norad Tle, Iono IRI and Earth orientation data is made available to GCS/FDF for the retrieval from GACF.
6) FDF retrieves the data and uses them as configuration parameters for its Collision avoidance software and ionospheric corrections determination.

7) GRSP validates the MSF products (orbits, Galileo satellite and station clocks and estimates of Earth rotation parameters) and makes them available for the scheduled retrieval by SPF.
8) SPF retrieves the data from the GRSP for their further provision to MSF
9) GACF retrieves from SPF the orbit, clock and ERP validation reports and after archived the reports
)

6.2.5.4.4 #SFC2.3-4: Specific Report Dissemination on GRSP request

The MSC for scenario this is provided below:

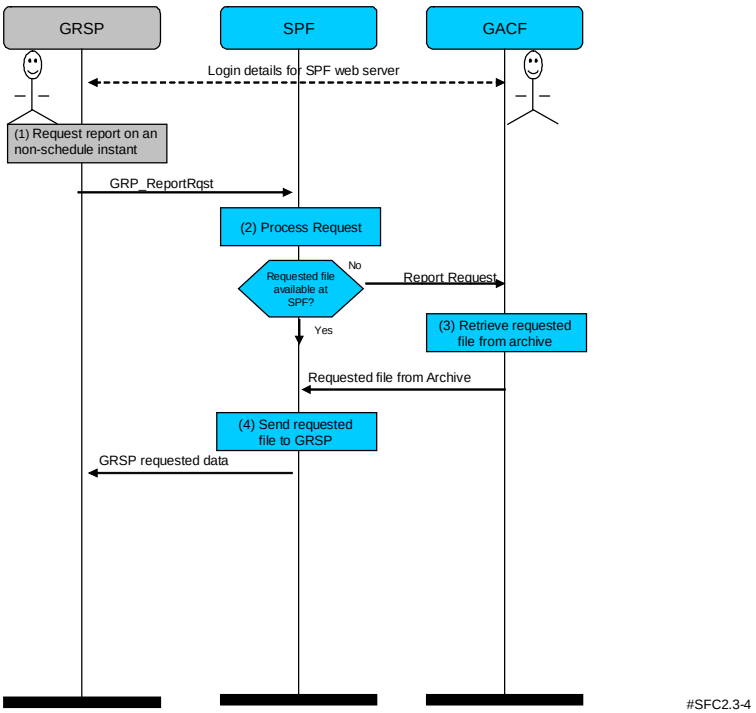


Figure 131 : #SFC2.3-4: Specific Report Dissemination on GRSP request

Scenario Textual Description:
Nominal reports are regularly delivered by the GMS/SPF to the GRSP on the defined periodicity at agreed fixed schedule and on SPF initiative. In parallel, these files are stored on line in the SPF. The on-line storing capability of the SPF will be 7 days i.e. the regular files are available during seven days and are then removed. In addition to the nominal reports GRSP can request from SPF to make available a particular time series of the pre-defined reports. This is to cover the cases where the GRSP was not able to receive the files from the regular process as per above, or the cases where files on GRSP side has been lost (failure case, etc…). If needed, the GRSP operator has then to execute an INF_ReportInfoRqst in connecting to the SPF web server. He will be presented with the list of still available files (less than 7 day old) and the older ones, removed from SPF but still available from the GMS central archive. He will be asked to select the files he wants. He will be able to download file still present on the SPF. The request for the others will be handled internally to the GMS and the files will be delivered to the GRSP once retrieved by GMS from the central archive.
The sequential description is detailed below:

1) GRSP sends the request to SPF to obtain the particular data. Note: A username and password should be required to access the SPF web server. This username/password is used by the SPF to provide a catalogue based on the products available to that external entity (an "account"). It is provided to the external entity by an operational interface.
2) SPF processes the request. Checks if the requested data is available (not older than 7 days). If not (older than 7 days) passes the request to the GACF (Archive)
3) GACF retrieves the requested data and passes them to SPF.
4) SPF sends to GRSP data as per GRSP request.

6.2.5.4.5 #SFC2.3-5: Contingency Information Exchange (NAGU and Alarm)

The MSC for this scenario is provided below:

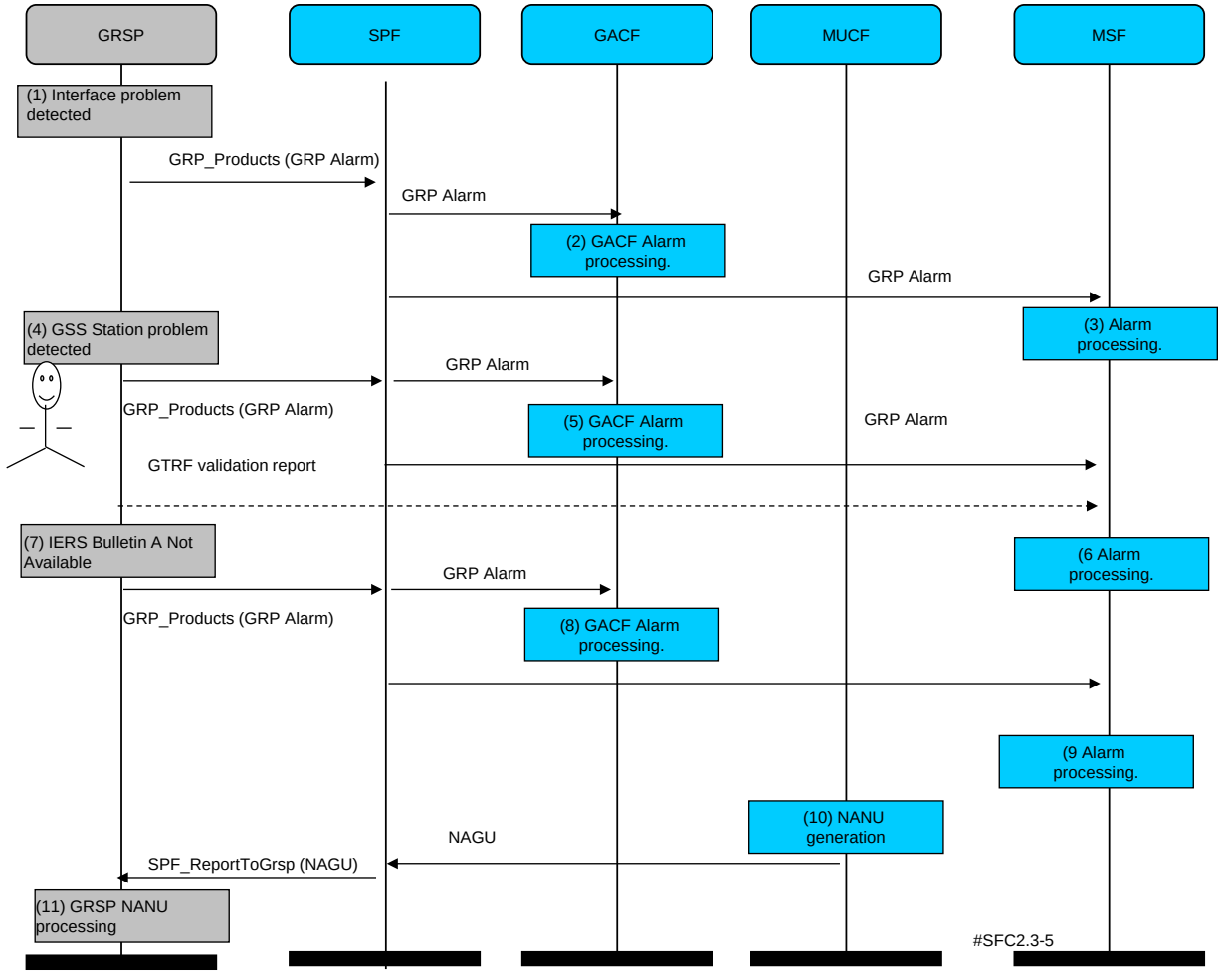


Figure 132 : #SFC2.3-5: Contingency Information Exchange (NAGU and Alarm)

Scenario Textual Description:
The contingency information affecting the GRSP interface will be exchanged between the Galileo System core and GRSP.

Galileo will provide the NAGU information which is going to be used mainly by GRSP/ILRS for the planning/execution of the SLR measurements. GRSP will provide an alarm upon the detection of the problems with the GRSP interfaces.

Remark: *the automated NAGU generation concept is currently not consolidated in the STRB2.0 baseline. It is expected that the content of the signal INF_NoticeAdvisoryToUsers will be refined in close coordination with GSC in the frame of GSC operational concept consolidation. In the meanwhile, the signal INF_NoticeAdvisoryToUsers will not be disseminated to the external entity and the NAGU will be generated manually and sent to the GSC. In both cases (automated and manual NAGU processes) the authorization for dissemination to the public is granted by the corresponding formal process at GSC level.*

GSRP will provide an alarm if an interface problem/failure is detected or the GSS station coordinates solution deviates from the official GTRF solution by more than 10 centimetres in any of the three directions. Should this happen, GRSP will have to regenerate the GTRF and provide corresponding GTRF validation report. The report provision will be handled via an operational interface.

Furthermore, the GSRP will provide an alarm when valid EOP are not vailable from IERS Bullettin A. Should this happen, GSRP will have to generate an estimate of backup EOP. Then GSRP wil provide to GMS the GRP_EarthOrienParam signal containing estimated backup EOP, in addition to the relevant alarm.

The sequential description is detailed below:

1) GRSP generates an Alarm report following the detection of the interface problem/failure. Report is sent to SPF

2) SPF passes alarm report to GACF operator for its inclusion into the ground assets monitoring process (GRSP interface monitoring)

3) MSF retrieves GRP alarm data from SPF. Information provided in the report is processed within the scope of the MSF mission monitoring.
MSF needs to process GAL+GPS data. Therefore, list of unavailable GPS satellites is configured in the MSF using NANU retrieved manually on the internet.

4) GRSP generates an Alarm report following the detection of the GSS station problem (GSS coordinates deviation). Report is retrieved by SPF.

In addition to the issuing of the alarm GRSP needs also to generate a new GTRF validation report which is provided to MSF via an operational interface.

5) SPF passes alarm report to GACF for its inclusion into the ground assets monitoring process.
On reception of the GRP_Alarm notified to the operator by an alert in the GACF AED, the GRP_Alarm file is retrieved and the content of the file is visualized by the operator in order to identified the type of alert and its details."

6) MSF processes the alarm and the new GTRF validation report within the scope of GMS role to support the GRTF generation and maintenance.

7)Normally, the GSRP send to the GMS the EOP data contained in IERS Bulletin A. The GSRP generates a warning or an alarm if the current IERS Bullettin A is not available. If this event occurs:

- The GSRP will send a Warnning to GMS if the age-of-data of the latest available IERS Bulletin A is less than 4 days . In this case the GSRP will send to GMS the EOP data contained in the latest IERS Bulletin A.
- The GSRP will send an Alarm to GMS if the age-of-data of the latest available IERS Bulletin A is not less than 4 days . In this case the GSRP will also send to GMS backup EOP data estimated by GSRP itself.

8) SPF passes alarm report to GACF for its inclusion into the ground assets monitoring process.
On reception of the GRP_Alarm notified to the operator by an alert in the GACF AED, the GRP_Alarm file is retrieved and the content of the file is visualized by the operator in order to identified the type of alert and its details."

9) MSF processes the alarm and uses the new EOP data.. .

10)MUCF monitors the information from the GCS and other constellation monitoring functions and generates corresponding NAGU report upon the occurrence of the contingency event. This report is forwarded to SPF for the further provision to the GRSP.
11)GRSP processes the NAGU information and passes it to other elements which can be affected (e.g. ILRS- Galileo satellites SLR tracking).

6.3 SFC3 MAINTAIN OVERALL SYSTEM TIMING SYNCHRONISATION

The purpose of this section is to describe the physical implementation of the functionalities needed to maintain and overall system synchronization.
Important note: This implementation assumes that the generation and steering of the Galileo System Time (GST) is already provided nominally by the PTF as described in SFC2.1
Then this SFC3 describes the elements and their interaction in:

- The generation and maintenance of the different local timing parameters
- The GST distribution among the system

6.3.1 SFC3.1 Satellite Times Provision and Synchronization

6.3.1.1 Overall Physical Description

The purpose of this section is to describe the physical mapping of the on-board timing parameters generation and maintenance functions.

The objective of the Satellite Times Provision and Synchronisation SFC is to provide all required functionality to:

- generate and maintain the different system timing parameters with associated performances.
- Synchronise the elements involved in different mission chains

The following section describes the physical implementation of the satellite timing parameters provision and synchronisation. It is based on the following documentation:

- ☐ System documents:
  - o System Logical Architecture (SYS-RD.1)
  - o System operational concept: MOCD (SYS-RD.6)
- ☐ GMS documents:
  - o GMS DDF2 (GMS-RD.2): MUCF section and Mission Monitoring section
  - o MUCF DDF (GMS-RD.4), MUCF ERD (GMS-RD.10), MUCF ICD (GMS-RD.11)
- ☐ GCS documents:
  - o SCCF Software Design Document (GCS-RD.8): Correlate Time function
- ☐ SSgt documents:
  - o Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2)
  - o Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1)
  - o CMCU Requirements specification (IOVSS-RD.8): Output Frequency Adjustment section
  - o Galileo IOV PUS Definition (IOVSS-RD.7): Time management service

As described in SYSBSLDEF part 1 (functional architecture), several types of timing parameters are defined on-board each satellite to support different applications:

- Timing parameters for navigation applications:
  - o he on-board payload frequency reference defining a time reference source.
  - o the payload on-board time (or “satellite Local GST”, or NSGU time): this highly accurate time is used to define the Satellite LGST needed to timestamp the natvigation message and to control the transmission start epoch of downlink pages.

Note: an additional time is generated independently and synchronized to the LGST for PRS PRN code generation.

- Timing parameters for Monitoring and Control (TM time stamping and TC time tagging):
  - the ICDU frequency
  - the platform on-board time (also called ICDU on-board time “IOBT”)
- Additional time for security application:
  - PFSU Time
  - PLSU Times

6.3.1.1.1 Physical Mapping of Functional Architecture

This section indicates the functional mapping of SFC3.1 to physical elements. Physically, the following elements are involved in the provision of the timing parameters:

- Payload on-board time parameters provision and management:
  - 2 satellite elements are involved:
    - CMCU: generates and distribute the 10.23 MHz reference signal from the on-board atomic clocks. It is also responsible for applying dedicated frequency step adjustment from TC received from the ground segment and for providing TM to monitor the on-board backup clock
    - NSGU: generates the payload on-board time (LGST) from the CMCU frequency reference (via the FGUU), and maintains the TOW/WN values. It is also responsible for applying dedicated time step adjustment from TC received from the ground segment.
  - 2 GMS elements are involved:
    - OSPF: responsible for generating instantaneous clock offset to GST for the satellite primary clock (based on the raw ranging measurements provided by the GSS stations from the L-Band SIS).
    - MUCF: responsible for
      - on-line (for the primary clock) or off-line (for the backup clock) analyses of the behavior of the clocks, the drift of the payload frequency reference and LGST
      - planning generation including request for on board clock resynchronisation or switch to be forwarded to the GCS
  - 2 GCS elements are involved
    - SCPF: GCS short term plan generation including clock resynchronization
    - SCCF: responsible for:
      - Schedule execution: TC generation and uplink
      - Forward Phase sample TM to GMS for the backup clock monitoring (MM_SatSumStatus)
- Platform time parameters provision and management:
  - 1 satellite element:
    - ☐ ICDU: responsible for generating the platform time, based on an internal clock and the NSGU payload time pps. It is also responsible for transmitting the frequency/time step TCs dedicated to the payload

- o 2 GCS elements are involved:
  - ☐ TTCF: to time stamp the TM received and forward them to the GCS-SCCF
  - ☐ SCCF: to correlate the platform time to GST

- PLSU time provision and management:
  - o PLSU: generates 2 independent times:
    - o 1 time for PRS PRN code generation: this time is directly synchronized to the LGST received from NSGU by pulse per second (PPS)
    - o time for payload STC/STM management
  - o GMS KMF: monitoring and control (synchronisation) of the PLSU time for STC/STM management

- PFSU time provision and management:
  - o PFSU: generates an independent time for platform STC/STM management.
  - o GCS KMF: monitoring and control (synchronisation) of PFSU time

6.3.1.1.2 Physical Static View

The following figure presents the physical architecture of the satellite timing parameters generation and maintenance functions:

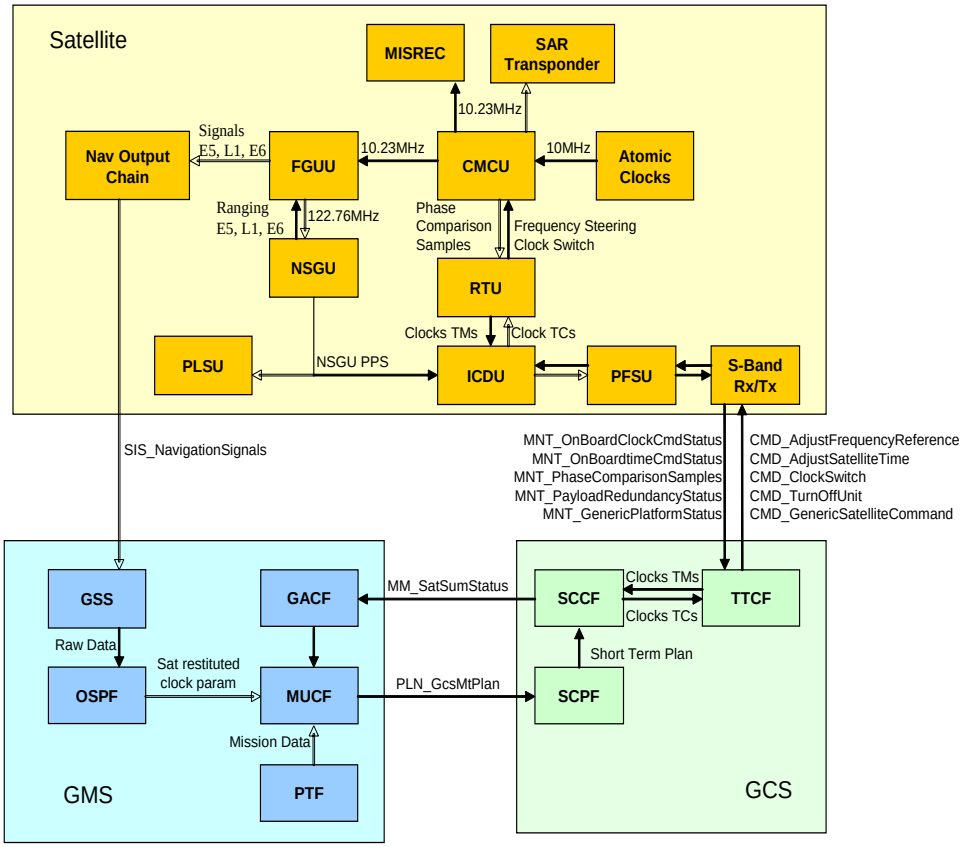


Figure 133: SFC3.1 Physical Static View

6.3.1.2 Payload Time Parameters Provision and Management

The on-board time parameters for navigation application include an on-board frequency reference and a payload LGST derived from this frequency. This payload GST needs to be steered to the on-ground GST within a specified offset. The following section describes the role of the elements involved in the provision and maintenance of these parameters.

6.3.1.2.1 Frequency and Phase adjustment specification

The following table indicates the different frequency adjustment parameters specified in different segments:

Segment	Ref doc	Requirement	Frequency reference	Max frequency step	Elementary step
GMS	AD.3	GMSREQ-11514	10.23MHz	absolute value=102.3 mHz => relative value +/- 10-8	less than 5.44x 10-15 (relative)
GCSGMSICD	AD.8	GCSGMSICD-262	-	step size of an integer multiple of 0.002046 mHz (= in 2x10-13 in relative value)	-
GCS	AD.5	GCSREQ-24048	-	-	-
GCSSSICD	AD.9	-	-	-	-
PUS	SS-RD.7	-	-	-	-
SSgt	AD.4	SSREQ-896 SSREQ-5363	10.23 MHz	-	frequency step: +/-5.44 x 10-15 phase step: 3x10-4 deg
SSgt	SS-RD.8	CMCU-202 CMCU-205	10.23 MHz	1x10-8	1x10-15

Table 72 : Frequency adjustment parameters

6.3.1.2.2 On-Board frequency reference provision: Atomic Clocks and CMCU

As part of the Satellite time synchronisation SFC, the CMCU is responsible for:

a) Frequency reference provision

The CMCU is responsible for generating and keeping a highly stable reference frequency of 10.23MHz to the payload as seen from the ground. The Frequency reference is derived by an internal synthesiser from the 10MHz frequency delivered by the on-board atomic clocks.

b) Frequency reference adjustment

The CMCU is responsible for controlling the frequency reference derived from the primary clock. This control is made when instructed by proper telecommand including a frequency step to be applied on the output frequency reference (see ICDU section):

- Frequency step adjustment size:

  The size of the frequency step is included in a dedicated TC provided by the ICDU.

  The CMCU can apply a maximum relative frequency step of +/- 10-8 with relative frequency step adjustments multiple of +/-5.44e-15.

  Notes:

  o Absolute vs. Relative frequency steps definition: in the system requirement baseline, the frequency steps can be specified in absolute or relative value. The relationship between those values is the following:

$F_{RELATIVE} = \frac{F_{ABSOLUTE}}{F_{REFERENCE}}$ where FReference=10.23MHz

- o This is not in line with CMCU design that allows the following relative frequency step adjustments:
  - ☐ Maximum relative frequency step Df/fout = ± 1x10-8
  - ☐ Elementary relative frequency steps: 1x10-15 in any direction

  This means the CMCU elementary frequency step (10x-15) is not in line with the requested 5.44e-15. However, the difference is considered at system level as a deterministic error that can be negligible considering the fact that the overall error is in the order of 1 elementary frequency step requested (5.44e-15) as specified in [AD.4] (SSREQ-5362)

Depending on the value of the step to be applied, the CMCU output frequency (and thus the satellite) can be considered available to support the mission services or not (coarse or smooth steering method): A relative frequency step of less than 10e-11 produces no degradation in the output frequency stability performance
This has an impact on the operational procedure to be defined, in particular if the satellite needs to be taken out of service before any adjustments (see dynamic scenarios)

- Frequency step adjustment modes:
  The CMCU can apply the frequency adjustment in 3 different modes that can be selected by proper TC:
  - o Standard Frequency Control Mode
  - o Absolute Phase Control Mode (adjustment made in a phase coherent manner)
  - o Continuous Phase Control Mode (adjustment made in a phase coherent manner)

  Note: the type of the mode used is independent from the availability of the satellite for the service. Only the frequency step to be applied impacts this availability.

c) Frequency reference distribution
Distribute the payload frequency reference to:
- o The SAR payload for SAR signal generation
- o The NSGU for LGST generation and navigation signals generation (ranging codes and carriers)
- o The C-Band Receiver for C-Band acquisition

d) Clock redundancy management:
- Redundancy configuration:

The CMCU is responsible for generating frequency reference from two active atomic clocks: one primary and one hot redundant. The Navigation Payload includes two Space qualified Rubidium Atomic Frequency Standards (RAFS) and two Space qualified Passive H-Masers (S-PHM), each type in a double-redundant configuration. Due to Power consumption and dissipation constraints, only one type of each clock is switch on at the same time.
At beginning of satellite life, one RAFS and one Maser clock are switched on (the others being in cold redundancy). After warm-up when full stability is achieved, the S-PHM will act as the primary clock and supply the on-board reference frequency (It will acquire the full frequency stability after 1 week). The RAFS acts as hot redundant clock.

The use of RAFS as primary clock will be allowed during S-PHM primary clock switching

- Monitoring of the frequency reference derived from the hot backup clock: the CMCU is responsible for collecting phase measurement samples, comparing periodically the phase of the frequencies derived from the primary and the hot backup clock. The sampling rate of this phase difference measurement is programmable and can be set to anything between once every 0.1 seconds and once every 1000 seconds, in steps of 0.1 seconds. A high sampling rate is required if the frequency offset between two clocks is high such that 360 phase cycle occurs within the nominal sampling rate (1000s).

These phase comparison are then downlinked to the ground segment via TM.

- The payload clock switching shall be triggered only when instructed by proper TC from ground initiated by GMS request

6.3.1.2.3 Payload Local GST provision: NSGU

As part of the Satellite time synchronisation SFC, the NSGU is in charge of:

a) LGST provision:

- The NSGU is responsible for generating and keeping the satellite LGST. The LGST is derived from a 122.76MHz (120x1.023MHz) intermediate frequency reference provided by the FGUU (derived from the 10.23MHz frequency reference delivered by the CMCU), as shown in the following figure:

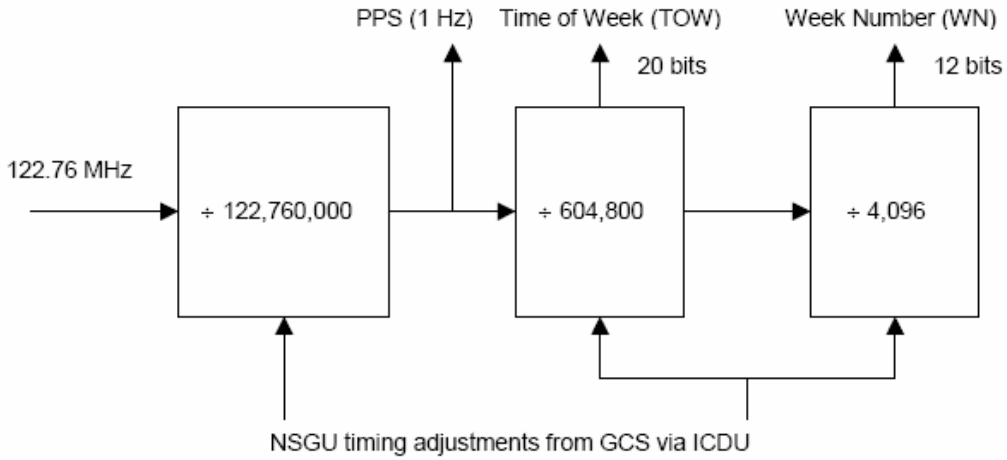


Figure 134: NSGU Time Generation Principle

- In addition, the NSGU generates the timing parameters to be included in the navigation message (WN and TOW): 32-bits timestamp of the LGST

b) LGST control

The NSGU is able to control the LGST value when instructed by proper telecommand:

- a time step can be applied to the LGST. The time adjustment allows to correct +/-0.5sec with a resolution step of 100ns

- a value adjustment can be applied to the ToW/WN values (at initialization only). The ToW/WN adjustment allows to set up values on 32 bits.

c) LGST distribution
The NSGU provides the LGST parameters to the following elements:
- pps signal to PLSU for the PRS PRN code generation (always slaved)
- pps signal for Platform (ICDU) to slave the Platform time (nominally slaved but can be independent)
- ToW/WN values inserted in the navigation message for the users.

d) Downlink page synchronisation
The NSGU is responsible to synchronise the downlink of the pages with the satellite LGST one second epoch

6.3.1.2.4 Payload LGST parameters computation: OSPF

As part of the Satellite time synchronisation SFC, the OSPF is responsible for computing the instantaneous on-board primary clock parameters based on SIS measurements from OD&TS process. These parameters include:

- the clock offset between the on-board primary clock (LGST) and the GST ("instantaneous restituted clock parameters for the primary clock).
- The clock standard Allan deviation (clock behavior)

These parameters are then provided to the MUCF to support the satellite clock monitoring.

Note: 2 kinds of parameters are generated by the OSPF:

- the "restituted" clock parameters that are exactly the parameters of the clock observed at each epoch (derived from the GSS raw data) => these data are generated every 10 minutes and sent to the MUCF in real time for clock monitoring. Note: this offset is an average over 10 minutes
- the "predicted" clock parameters that are the clock models transmitted to the user. These models are a prediction of the clock behavior over a certain period of validity (100minutes for RAFS clocks) and are based on the past 6 hours of historical data (the restituted clock parameters). This model is disseminated to the user for Positioning Service.

6.3.1.2.5 Payload LGST monitoring and control: MUCF

As part of the Satellite time synchronisation SFC, the MUCF is in charge of monitoring and control of the satellite clock behavior. This monitoring is performed by the Galileo Mission On-Line Supervision (GMOS) algorithm, with the support of dedicated tool implemented in the Performance Assessment Tool (PATBox): the Satellite Clock Analysis tool (SCAT). This tool is part of GMPA (Galileo Mission Performance Assessment.

Different processes are identified depending on the clock monitored:

- Primary clock monitoring: on-line process: to monitor the primary clock behavior based on the "restituted" clock parameters received from the OSPF (LGST-GST offset and

Allan deviation). If needed, a time or frequency synchronization, or a switch request is sent to the GCS as part of the mid-term plan. This includes:

- Satellite Primary Clock synchronization request and parameters for on-board clock frequency adjustment;
- Satellite on-board time of transmission synchronization request and parameter for time adjustment;
- Satellite Primary to Backup Clock switch request.

- Hot Backup clock monitoring off-line process: to monitor the hot backup clock based on phase comparison samples received in a dedicated TM via the GCS (SCCF) nominally every 14 hours. These samples are used by the MUCF-SCAT function, to characterise the behaviour of the backup clock (in terms of stability), assuming that the master clock is working correctly. The clock characterisation is carried out by the MUCF-SCAT function in terms of Allan variance.

  The offline backup clock monitoring capability is performed by GMPA-SCAT in two operational modes, namely:
  - "Operational mode": in this mode the hot backp clock monitoring is carried out using on-line telemetry data collected over a trime span up to three days. The operational mode will be for mission needs, including satellite backup clock switching request to be sent to the GCS as part of the mid-term plan.
  - Analysis Mode : in this mode the MUCF-SCAT performs the characterization of hot backup clock using data retrieved from the GMS central archive. In this mode, the analysis is not limited to 3 day , as per the operational mode.
  - Indeed, it will be possible to retrieve and analyse in a single batch historica data from GACF over a time frame which is only limited by the achive size: This capability can be exploited at operational level to perfom hot back up clock trend performance analysis and long-term characterisation, to provide feedback on satellite clock technology development.

6.3.1.2.6 Payload LGST adjustment planning and command: SCPF/SCCF

The SCPF is responsible for building the GCS short term plan including inputs from the GMs (GMS plannings and requests for commands): see SFC5.4
As part of the Satellite time synchronisation SFC, the SCPF is in charge of the identification of the satellite resynchronisation request in the GCS short Term plan based on the GMS mid term plan.
The SCCF is responsible for the Satellite control to execute schedules and uplink commands.
As part of the Satellite time synchronisation SFC, the SCCF is responsible for:

- uplinking proper clock synchronisation or switching commands to the satellite. This is based on the schedule provided by the SCPF. In case the time adjustment has an impact on the mission performance (NSGU time adjustment and frequency coarse steering), the SCCF first ensures that the satellite is taken out of service by the user sending a dedicated TC to trigger the Data Validity Status (DVS) flag to "working without guarantee": see SFC 5.2.1 (step 6 of scenario #SFC5.2.1-1.3 Execution of Satellite Maintenance Activities with Mission Impact). In case of smooth steering, this is not needed.

Note: in case of clock switch, a dedicated operational procedure is in place to define a sequence of TCs in line with the on-board clock configuration constraints. Due to power and thermal constraints on board the satellite, a maximum of 2 clocks can be switched on at the same time, which leads to the following sequence:

o Satellite primary clock switch command:

The sequencing for primary clock switching in case of primary clock S_PMH1 failure is as follows :

- Switch the failed primary clock S-PHM1 to the hot backup RAFS clock (RAFS1) that becomes the new primary clock (Swicth1)
- Switch off the “failed ” clock S-PHM1.
- Switch on the cold backup clock S-PHM2 (that becomes the hot backup clock), for S-PHM2 warm up and stabilisation
- Upon completion of PHM2 stabilisation, switch the current primary clock RAFS1 to S-PMH2 that becomes the new primary clock (Switch 2).

The sequencing for primary clock switching in case of maintenance of cold back up S-PHM2 is as follows :

- Switch the current primary clock S-PHM1 to the hot backup clock RAFS1 that becomes the new master clock (Swicth1)
- Switch off the “old” primary clock (S-PHM1)
- Switch on the cold backup clock S-PHM2 (that becomes hot backup) for maintenance
- Upon completion of PHM2 maintenance, switch off S-PHM2 and switch on S-PHM1 for its stabilisation
- Upon completion of PHM1 stabilisation, switch the primary clock RAFS1 to S-PMH1 that becomes the new primary clock (Swicth 2).

o Satellite hot backup clock substitution command

- Switch off the hot back up clock
- Switch on a cold backup clock
- Switch from the “old” backup clock to the new one

A dedicated TC is uplinked for each step of the sequence

- processing TMs received from the SSgt: in particular the clock phase sampling measurements TMs to be forwarded to GMS for the monitoring of the on-board backup clock.

6.3.1.2.7 On-board frequency and phase step TC generation: ICDU

As part of the frequency reference adjustment, the ICDU is responsible for generating the TC for the CMCU
This generation is based on the frequency step adjustment requested in a dedicated TC received from the GCS.

From this, different on-board TCs can be generated depending on the adjustment method (coarse or smooth steering as defined in System DDF1 SFC3.1.1 Payload Frequency Reference Steering)

- Coarse steering method: consists in adjusting the Payload frequency reference applying a single frequency step: in that case, the ICDU generates a single TC to the CMCU requesting the adjustment of the CMCU frequency in one shot (note: in that case, the satellite needs to be taken out of service before the adjustment). This coarse steering method can be used to:
    - o Initially synchronize the on-board frequency reference to 10.23MHz (with the possibility to optimize this value as indicated in scenario #SFC3.1-1).
    - o Maintain the LGST-GST offset within a specific range.

- Smooth steering method: consists in adjusting periodically the Payload frequency reference value applying in a coherent way small but frequent frequency steps: in that case, the ICDU (on-board control procedures OBCP) is responsible for splitting the overall step in several commands including elementary frequency steps (<10-11), and sending them sequentially to the CMCU (note: in that case, the satellite remains available for the services).

These 2 modes can be selected by TC in the ICDU.

Note: The coarse steering is the baseline today based on the theoretical knowledge (and specification) of the atomic clocks: considering the theoretical clock behavior, the coarse steering method is in line with the overall system performances (availability): see [JT-105] in IOV DJF [SYS-RD.3]. The smooth steering method is considered as a backup solution that can be supported by the design when the real clock will be observed. It is indeed expected that the smooth steering method has no impact on performance. However, this shall be tested during the IOV campaign. From an implementation point of view, the smooth steering method has been discussed during the SEB consolidation and a dedicated CCN has been raised.

6.3.1.2.8 Satellite Clock Switching: Future Evolution

A future evolution is planned for GMS and GCS to provide seamless satellite clock switching avoiding service discontinuity, as well as a refined satellite clock coarse switching concept, to mitigate the impact on service availability wrt satellite (coarse) clock switching concept adopted in IOV.

A possible future implementation the new functionality is depicted in this paragraph.

Thus, two satellite clocks monitoring and control events are planned in Galileo future evolution, namely :

- o Satellite coarse clock switching (“legacy clock switching), and
- o Satellite seamless clock switching,

The following figure provides the satellite functional chain for seamless satellite clock switching.

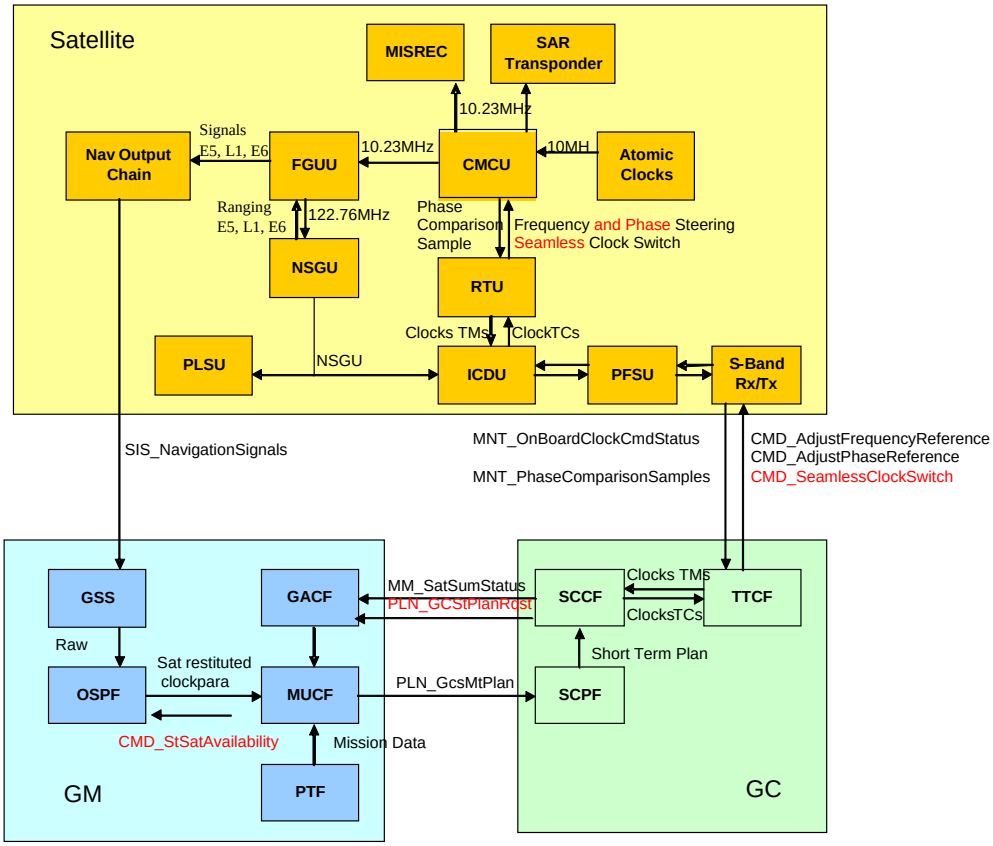


Figure 135: Seamless satellite Clock Switching Functional Chain.

The functionalities at segment level required for seamless satellite clock switching are as follows:

1. GCS: request a seamless switch for the mid-term plan
2. GMS: validate the mid-term plan request for a seamless switch
3. Space Segment: retrieves the phase difference between the Master and Redundant clocks and is capable to implement a seamless switch once they are aligned.
4. GCS: retrieves and processes CMCU phase meter telemetry for telecommand computation and commands the alignment and performs the switch.
5. GMS: flags the event to the user through SISA, and NAGU

The interface with the satellite is not affected, as Galileo satellite capability to support satellite clock switching in a phase coherent and frequency coherent way is already specified in the current baseline, and has been successful tested during IOV test campaign

Satellite clock switching due to satellite cold redundant PHM maintenance, test and primary clock failure are all performed by GCS.
The following main changes are needed at GCS level:

- A change is needed in the satellite clock switching request (PLN_MtPlanRqst) sent by GCS to GMS, to allow GCS to indicate a new type of clock event. In particular, a new event description in the signal PLN_MtPlanRqst to announce the switch and the type, namely:

- Coarse switch for IOV/FOC satellites (that is a renaming of the legacy clock switch request)
- Seamless switch for FOC satellites.

- A new procedure is required at GCS level for computation and triggering of the telecommand for execution of satellite seamless clock switching. The main steps of the relevant procedure are as follows:
  1. Collection of satellite clocks phase meter data during a configurable time (e.g. 20 minutes).
  2. Computation of frequency (*dF*) and delta phase (*dP*) offsets between both primary and hot back up clocks
  3. Check obtained offsets against the threshold
     - o if above threshold, send delta *dF* and *dP* commands and iterate (go to step 1)
     - o if below threshold, swap active clock and finalize procedure.

- A new tool has to be deployed at GCS requirement is needed in GCSREQ in order to provide the operator with the means to compute the telecommands for seamless satellite clock switching.

The following main changes are needed at GMS level to support satellite clock switching evolution

- Following the change in the mid-term plan provided by the GMS,a new event definition is required in the GMS internal signal CMS_StSatAvailability , to indicate the type of clock switching event (seamless or coarse). The internal signal is distributed to MUCF, OSPF and MSF, whose response will change according to the type of clock switching.
- In case of case of seamless clock switching, it will be requested to :
  - o The MUCF to keep the satellite in the ULS contact plan to not interrupt the dissemination of navigation message (as well SAR and CS messages) during the whole event.
  - o The OSPF to maintain continuous generation of the navigation message and monitored SISA during the whole event. The ranging accuracy and the SISA performance are degraded during the seamless clock switching exposure time (100 minutes maximum before event start and 24.5-25.5 hours after the end of the event), During this period the OSPF will set the SISA to the Degraded SISA value.
  - o The MSF to maintain the estimation of the orbit and clock during the entire event..

- In case of case of coarse clock switching, it will be requested to :
  - o The MUCF to remove the satellite for the ULS contact time, to interrupt the dissemination of the navigation message as well SAR and CS messages during the event.
  - o The OSPF stops the generation of navigation message for any epoch within the announced event and to set the SISA to not monitored (NAPA value) for any message generated up to 100 minutes before the start of the event and up to 4 hours after the end of the event. The SISA is set to monitored (not NAPA) starting from 4 hours after the end of the event until the coarse switching exposure time (24.5-25.5 hours) expire. During such time interval, the SISA will be set to the Degraded SISA value.

6.3.1.3 Payload Secure Time Parameter Provision and Management

At payload level, the PLSU is responsible for generating two kinds of time parameters:

6.3.1.3.1 Payload secure time provision for PRS PRN code generation: PLSU

The PLSU is responsible for generating an independent time used to generate the PRS PRN code used for the E6 and L1 signals. This generation is based on:

- an internal oscillator generating a dedicated frequency
- time parameters received from the NSGU: a PPS pulse, and the LGST (WN and TOW on 32 bits).

The PLSU time is synchronized to the LGST thanks to the parameters provided by the NSGU.

6.3.1.3.2 Payload secure time provision for Secure STM/STC management: PLSU

The PLSU generates a specific time needed to execute secure time-tag commands, housekeeping functions and timestamp telemetry (STC, STM) on the payload.
In particular the key management secure telecommands (STC) shall be time-tagged
This time can be synchronized to the LGST by ground command.
Note: this time is independent from the time generated within the PLSU for PRS PRN code generation

6.3.1.3.3 Payload secure time for payload STM/STC M&C: GMS KMF

The setting of the PLSU Time is performed by the GMS KMF from ground via STC.
For more details, see classified annex

6.3.1.4 Platform Time Parameters Provision and Management

6.3.1.4.1 Platform Time Provision for TM/TC stamping/tagging: ICDU

The ICDU is responsible for generating the platform time, based on an internal clock generating a dedicated frequency. This platform time is used to execute time-tag commands, housekeeping functions and timestamp telemetry (attitude and orbit control functions, M&C functions...). The platform time can be slaved to the NSGU PPS time (counter) when the payload is turn on to limit its drift.
The internal representation of the relative on-board time is:

- 32-bit seconds (0… 231-1)
- 16-bit sub seconds (LSB = 1/65535 ~ 15 µs)

6.3.1.4.2 Platform Time Monitoring and control: SCCF

The SCCF is responsible for correlating the satellite platform time with the GST
This correlation between the platform time and the Galileo System Time has an accuracy of ± 5 msec.

Process: In order to elaborate the correlation between platform time and GST, the SCCF relies on a dedicated telemetry: TM (9,2): Time Report with the following parameters:

- Integer Second:32bits, LSB = 1 second
- Sub-seconds: 16 bits, LSB = 1/65536 sec

This correlation can be used for:

- the generation of time corrections to be applied in order to convert a time expressed in the satellite time reference to GST and vice versa. These corrections are included in a dedicated TC: TC(9.128): Set Spacecraft Time Offset to Epoch with the following format:
  - Integer Second:32bits, LSB = 1 second
  - Sub-seconds: 16 bits, LSB = 1/65536 sec

- the tagging of on-board telecommand packets: on ground, the application time of the TCs are defined by operator (as part of operator schedule provided by SCPF) in the Galileo System Time reference. Before uplinking the command, the SCCF is then responsible for converting the application time into the on-board platform time using the correlation time parameters

6.3.1.5 Platform Secure Time Parameter Provision and Management

6.3.1.5.1 Platform secure time provision for platform STM/STC management: PFSU

The PFSU generates independently a specific time needed to execute secure time-tag commands, housekeeping functions and timestamp telemetry (STC, STM) on the platform. In particular the key management secure telecommands (STC) shall be time-tagged
This time is set by ground command independently from any other on-board time.
For more details, see classified annex.

6.3.1.5.2 Platform secure time for platform STM/STC M&C: GCS KMF

The setting and the correlation of the PFSU Time is performed by the GCS KMF from ground via STC independently from any other on-board time.
For more details, see classified annex

6.3.1.6 Dynamic Scenarios

The following scenarios indicates different ways to maintain the satellite LGST steered to the GST. Six scenarios are identified:

-Initial satellite synchronization (LGST and Frequency steering) (scenario #SFC3.1-1)
-Operational satellite LGST steering based on time adjustments (scenario #SFC3.1-2)
-Operational satellite LGST steering based on coarse frequency adjustment (scenario #SFC3.1-3)
-Operational satellite LGST steering based on smooth frequency adjustment (scenario #SFC3.1-4)
-Primary Clock Switch (scenario #SFC3.1-5)
-Hot Backup Clock Switch (scenario #SFC3.1-6)
-Primary Clock Switch (scenario #SFC3.1-7) (to be defined)

6.3.1.6.1 Scenario #SFC3.1-1: Initial satellite synchronization (LGST steering)

6.3.1.6.1.1 General Description

During satellite initialization (after launch or after any payload switch off), three parameters need to be synchronized on board at payload level:

- the LGST 1 second pps epoch wrt GST: the offset between LGST and GST shall be steered within the range [-18ms, +62ms].
- The 1 second epoch value in terms of TOW and WN: adjustment via 32-bits macrocommand with 1s granularity

Note: considering the clock stability performance, the TOW/WN adjustment needs to be done only at satellite initialization.

Following GMS CDR, assumptions have been clarified for initial satellite synchronization in case of PRS mode only. As justified in GMS RFD ref. GAL-RFD-TAS-GMS-A/00222, initial acquisition of PRS PRN signals by GRC-P is possible only if the new commissioned satellite clock is synchronized within +/- 250ms with respect to GST.

Then dedicated System Technical Issue (STI-015) and justification in IOV DJF (SYS-RD.3) JT-123 concluded that GCS and SSgt designs enable to synchronise the Payload LGST to GST or UTC autonomously with an accuracy of less than 10ms.

The satellite initial synchronization baseline is then including the 2 following steps:

1) first step: first GCS to autonomously synchronies “roughly” the NSGU LGST (TOW/WN and LGST 1second epoch to GST) within +/-250 ms

2) 2nd step: GMS is then able to acquire the satellite in any mode (including PRS only mode) and perform accurate satellite synchronization (on-board frequency reference and LGST 1second epoch to GST) within +/- 5 ms

The detailed sequence is presented in the sections after:

6.3.1.6.1.2 Flow Chart

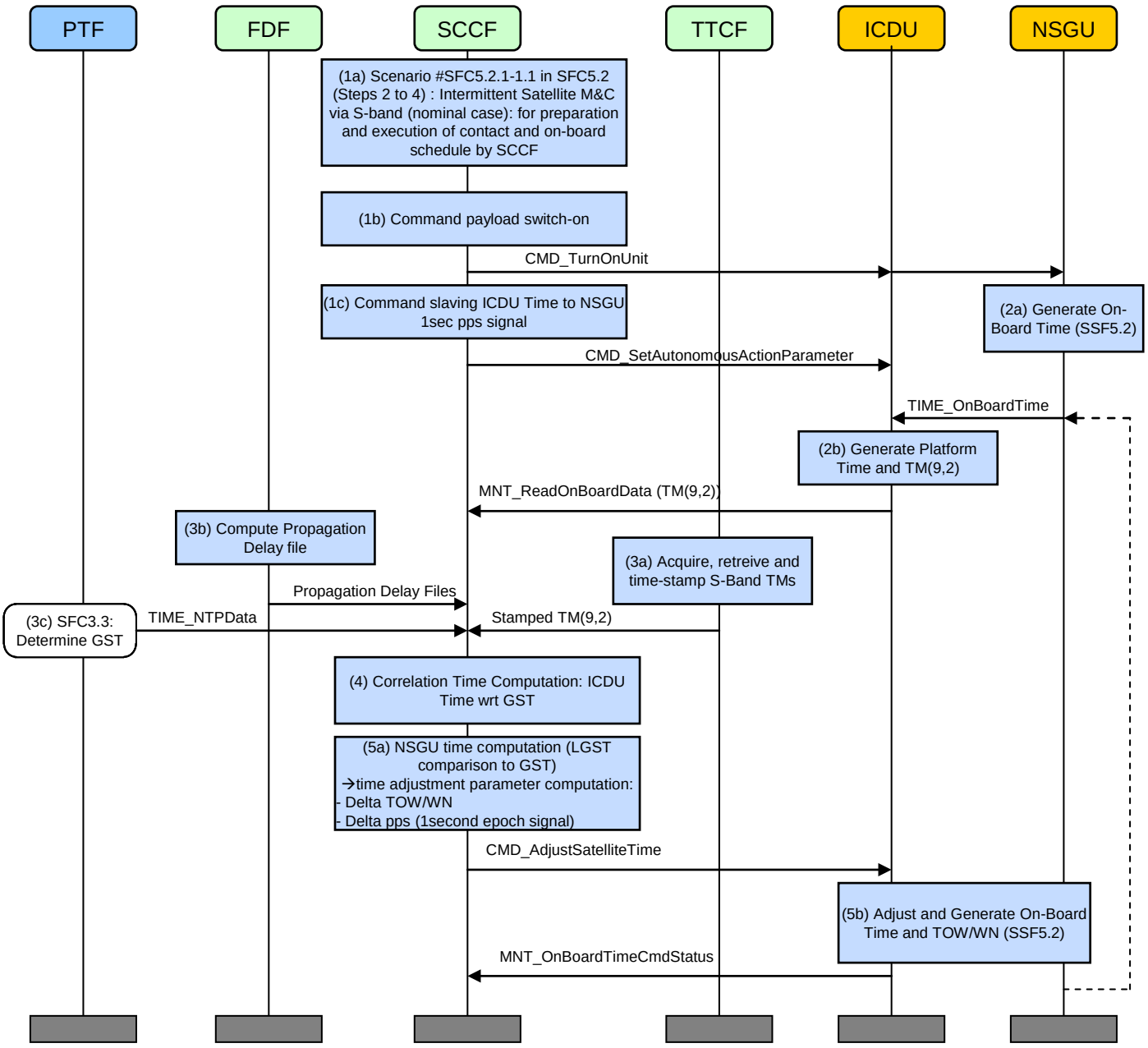
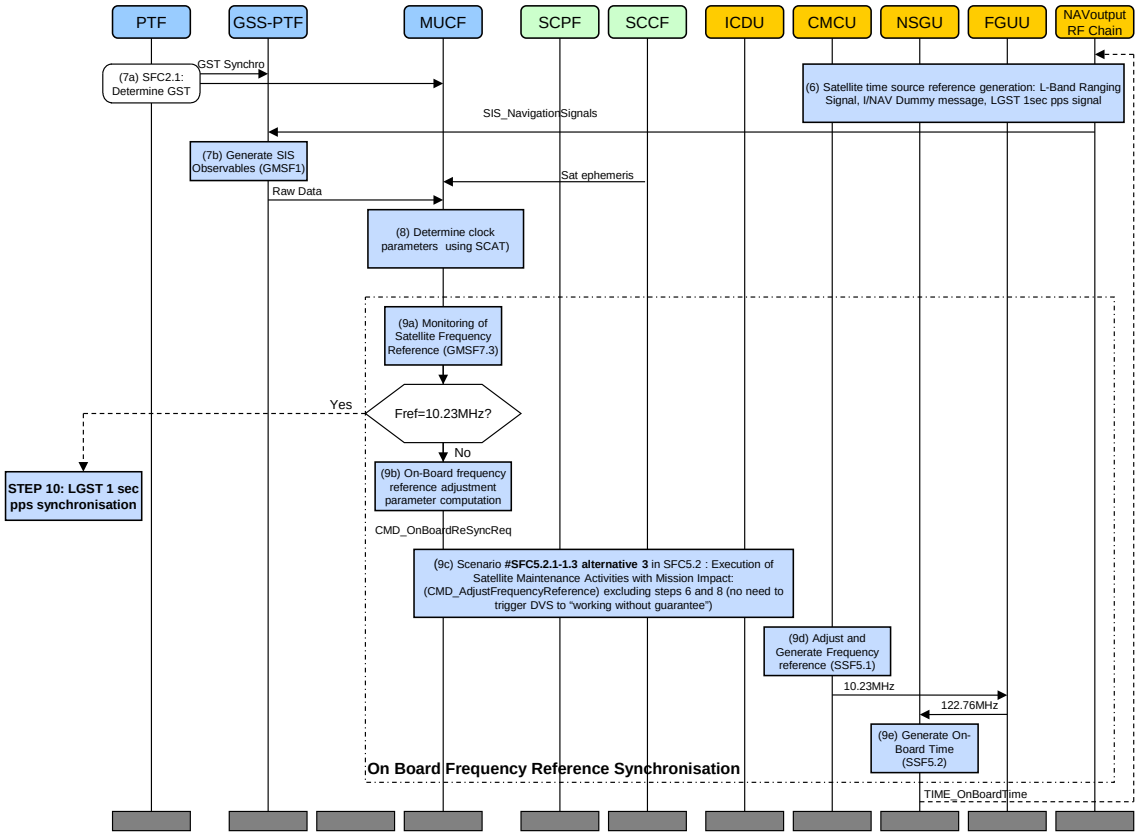


Figure 136: #SFC3.1-1 Dynamic Scenario (part 1/2)


PTF
GSS-PTF
MUCF
SCPF
SCCF
ICDU
CMCU
NSGU
FGUU
NAVoutput RF Chain
(7a) SFC2.1: Determine GST
GST Synchro
(6) Satellite time source reference generation: L-Band Ranging Signal, I/NAV Dummy message, LGST 1sec pps signal
SIS_NavigationSignals
(7b) Generate SIS Observables (GMSF1)
Sat ephemeris
Raw Data
(8) Determine clock parameters using SCAT)
(9a) Monitoring of Satellite Frequency Reference (GMSF7.3)
Yes
Fref=10.23MHz?
No
STEP 10: LGST 1 sec pps synchronisation
(9b) On-Board frequency reference adjustment parameter computation
CMD_OnBoardReSyncReq
(9c) Scenario #SFC5.2.1-1.3 alternative 3 in SFC5.2 : Execution of Satellite Maintenance Activities with Mission Impact: (CMD_AdjustFrequencyReference) excluding steps 6 and 8 (no need to trigger DVS to "working without guarantee")
(9d) Adjust and Generate Frequency reference (SSF5.1)
10.23MHz
122.76MHz
(9e) Generate On-Board Time (SSF5.2)
On Board Frequency Reference Synchronisation
TIME_OnBoardTime

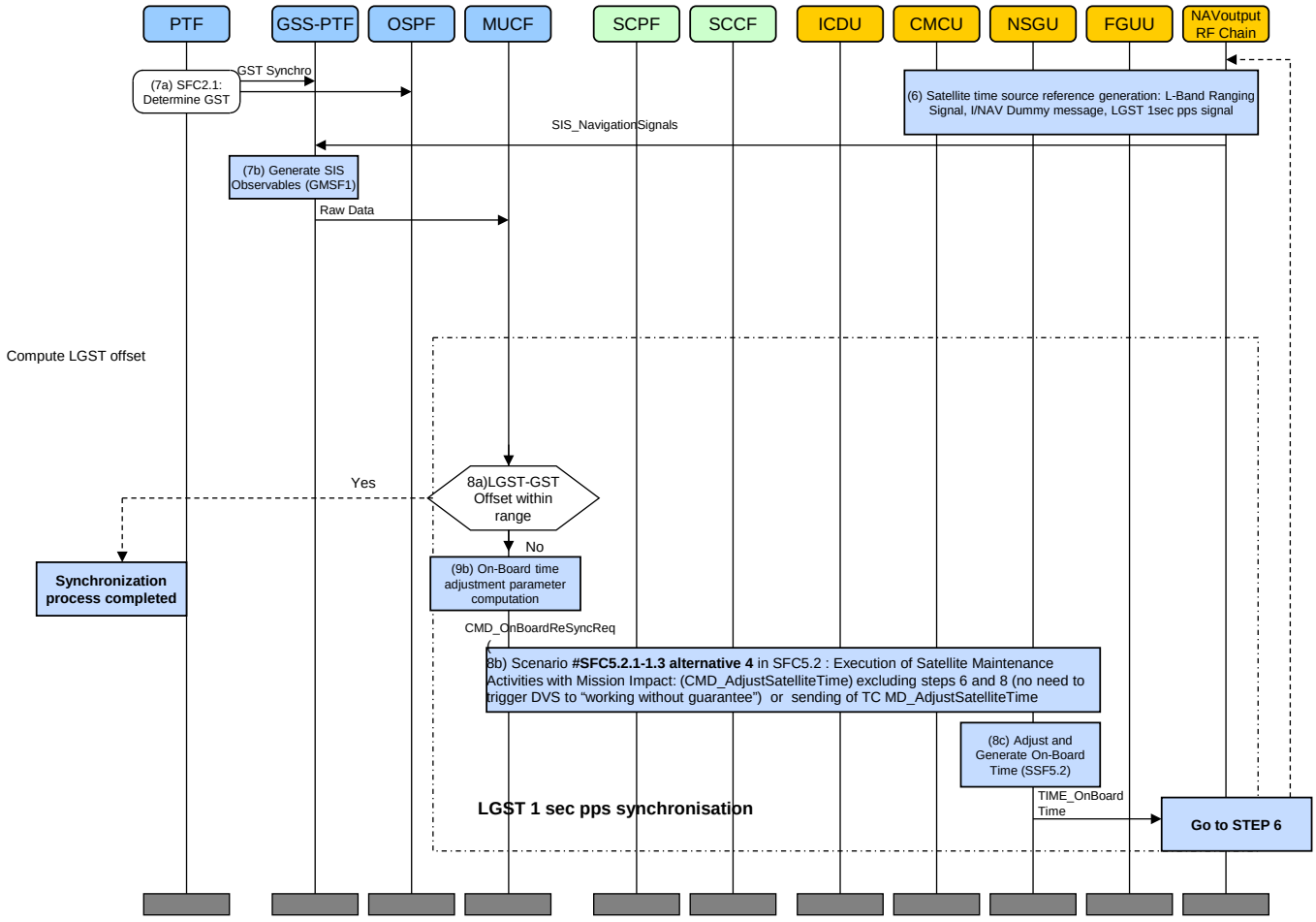


Figure 137: #SFC3.1-1 Dynamic Scenario (part 2/2)

6.3.1.6.1.3 Scenario sequence description

(0) General Description: After payload switch-on and clock stabilisation, the on-board time and frequency reference need to be synchronized. This is performed in two steps:
a) Under GCS responsibility: during initialization, the GCS has the functional capability to synchronize the payload LGST autonomously. 2 parameters can be synchronized by GCS based on S-Band Telemetry:

- the NSGU LGST 1 second pps epoch wrt GST with an accuracy of about +/- 250 ms in
- The 1 second epoch value in terms of TOW and WN: adjustment via 32-bits macrocommand with 1s granularity

This synchronization is achievable based on:

- The on-board capability to slave the ICDU time to NSGU time (LSGT 1 second pps signal), based on dedicated TC. In this way the satellite on-board time in the ICDU can be locked to within approximately 1ms of GST.
- The GCS (SCCF) capability to correlate the ICDU time to GST based on S-Band TM including platform timing information.

This should allow the SCCF to determine indirectly the NSGU LGST.

Notes:

1) The payload frequency reference cannot be steered with the S-Band signals
2) The accuracy of such synchronization should be within +/- 250ms to allow satellite acquisition by GMS in any mode (and in particular in PRS only mode)
3) The different steps at GCS level are all under OPS control during initialization (a dedicated OPS procedure should be is defined)

b) Under GMS responsibility: GMS is then able to acquire the satellite in any mode (including PRS only mode) and perform accurate synchronization. In this step the following parameters can be steered based on L-Band monitoring:

- the on-board frequency reference to be steered to 10.23MHz as seen from the ground.
- the NSGU LGST 1 second pps epoch wrt GST in such a way that the offset between LGST and GST is within the range [-18ms, +62ms] with an uncertainty of -/ 5 ms.

This synchronization is achieved by the MUCF using SCAT algorithm within the GMS, with the support of the GSS co-located in the PTF. Basically, the on-board timing and frequency parameters shall be synchronized with the following sequence:

- First the on-board frequency reference
- Then the NSGU 1second pps epoch

The sequential description is detailed below:

A) Satellite synchronization by GCS

(1) Step 1: Satellite configuration setting: this step is dedicated to configure the satellite to generate ICDU time slaved to NSGU

(1a) The SCCF generates the ground, on-board and contact schedules, and sends contact directives to the TTCF. These include contact start stop times, ranging directives (start/stop time), carrier directives etc.: for more details, see Scenario #SFC5.2.1-1.1 in SFC5.2 (Steps 2 to 4).

(1b) The SCCF generates a command to turn-on the payload, including the payload atomic clocks, CMCU and NSGU. After clock warm-up is completed (2 weeks expected), the NSGU is able to derive a payload LGST time: see step 2a).

(1c) The SCCF generates a command for the ICDU to slave the ICDU generated time to the 1 second pps signal provided by the NSGU.

(2) Step 2: Satellite time source reference generation

(2a) At payload switch-on, the CMCU is responsible for generating the on-board frequency reference based on on-board primary atomic clock (nominally it is expected that this frequency reference should be stabilized after 2 weeks of warm-up of the clocks). From this frequency reference, the NSGU is then responsible for generating the LGST parameters including:

- the LGST 1-second epoch pps signal
- LGST time stamping parameters: TOW and WN (at switch-on, these parameters should be set to zero in principle)

These parameters are distributed to the platform ICDU via the MIL-Bus

(2b) As the ICDU is slaved to the NSGU 1 second pps signal, the ICDU generates a platform time (based on internal clock) synchronized to the NSGU time. This time is included in a dedicated time-stamped telemetry representing the relative on-board time: TM (9,2): Time Report with the following parameters:

- Integer Second: 32bits, LSB = 1 second representing the TOW and WN
- Sub-seconds: 16 bits, LSB = 1/65536 sec, about 15 microseconds accuracy representing the LGST-GST offset within the second.

(3) Step 3: Timing input parameters to GCS

(3a) TTCF station acquires and time stamp the S-Band TM(9,2) including the relative ICDU time parameters wrt GST. The TTCF stamps the received data with the Earth Reception Time (ERT) once the packet has been demodulated and decoded. This timestamp includes also the time for the packet to pass through the TTCF. The accuracy of this stamping is in the order of 1microsecond.

(3b) The FDF estimates the propagation delay from the time the packet was stamped at satellite level (OBT) and the time it was stamped by the TTCF (stamped as the ERT). The output is a Propagation Delay File, consisting in a set of delays from each station to each satellite for each Contact. This file is provided regularly to the SCCF to enable him to correct the propagation delays.

Note: the instantaneous accuracy of the model corresponds to a timing error of approximately 0.5 microseconds. As the model is updated every minutes, the worst propagation error is assumed to be in the order to about 0.3 ms

(3c) The PTF is responsible for providing GST synchronization to the SCCF via NTP network: see dedicated SFC3.3.

Note: during nominal operation, the reference time available at GCS level is the GST provided by GMS. However, if GST is not (e.g. in IOV), the GCS will synchronize to timing delivered from the Hosting Entity (as defined by SUPIOVREQ-346) and the TTCF from GPS time. The usage of UTC time enables however the GCS to achieve the requested synchronization considering UTC-GST synchronization

(4) Step 4: Correlation time computation

Based on these 2 inputs (from 3a and 3b), the SCCF is responsible for correlating the satellite platform time with the GST. This correlation between the platform time and the Galileo System Time has an accuracy of ± 5 msec.

See dedicated JT-108 in IOV DJF [SYS-RD.3].

(5) Step 5: Satellite on-board time parameters adjustments

(5a) The computed correlation time indicates the LGST offset wrt GST with an accuracy of 250ms (see dedicated JT-138 in IOV DJF [SYS-RD.3]). The SCCF is then able to generate proper TC (CMD_AdjustSatelliteTime) included the following timing adjustment parameters

- Time step to synchronize on-board LGST 1second pps
- TOW/WN Parameter adjustment

(5b) The NSGU applies the time step command to drift the LGST pps and adjust the TOW/WN values. The corrected time is used to resynchronize the ICDU time slaved to the NSGU 1second pps time. This enables the GCS to check the correct synchronization of the NSGU time to GST, in addition to the TM confirming the good execution to the TC on-board (MNT_OnBoardTimeCmdStatus)

At this stage, the satellite LGST is synchronised to the GST with an accuracy of about +/- 250 ms.

B) Satellite synchronization by GMS

(6) Step 6: Satellite time source reference generation

The NSGU is responsible for generating:

a. the L-Band ranging signals and the LGST 1-second epoch pps signal based on this frequency reference

b. The following I/NAV dummy message to be downlinked on E5b/L1b frequencies (for further details refer to SISICD [AD.6]), including the TOW and WN (already set up by GCS)

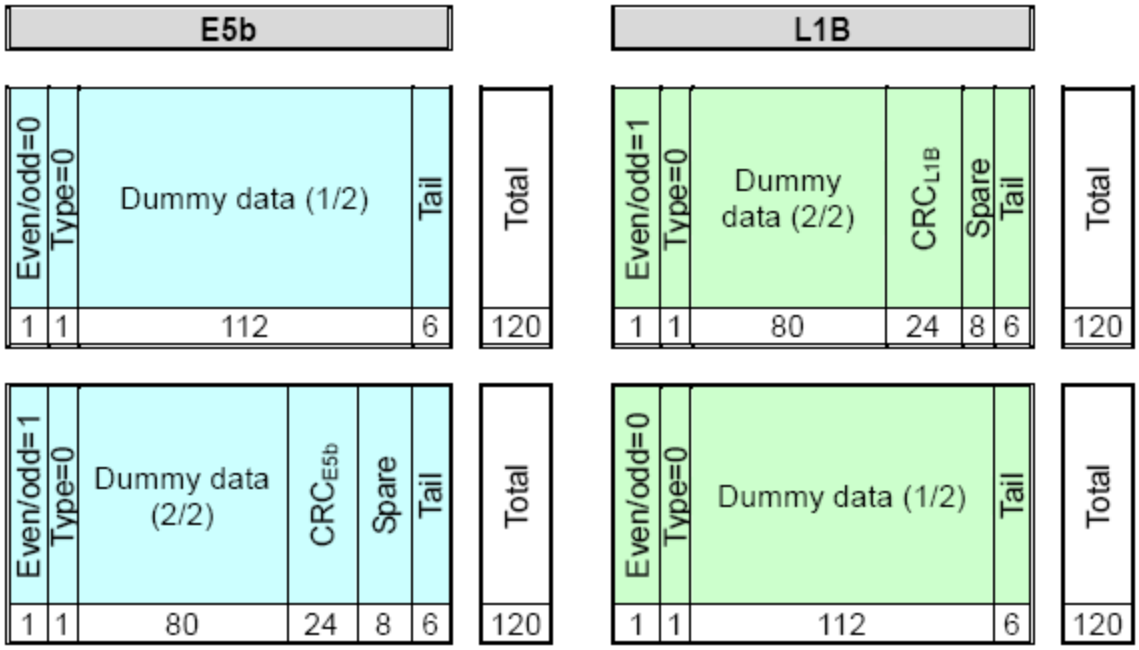


Figure 138: I/NAV Dummy message data page structure

Where the Dummy data includes in particular the TOW and WN formatted according to the following figure:

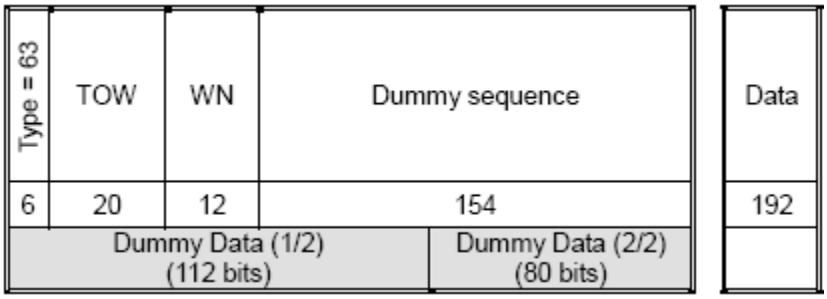


Figure 139: I/NAV Dummy Data Content

(7) Step 7: Timing input parameters to GMS

(7a) The PTF is responsible for providing GST synchronization to the GSS PTF and the MUCF : refer to dedicated SFC2.1.

(7b) As the satellite is already "roughly" synchronized to GST (which means with an accuracy better than +/-250ms), the GSS co-located in the PTF is able to acquire the satellite in any mode (including PRS only mode). This acquisition is based on the L-Band ranging signals.. After acquisition, this GSS provides to the MUCF Navigation Raw Data derived from the SIS Navigation Signals

(

(8) Step 8: On-board LGST 1second pps synchronization (If LGST-GST offset not within specified range, else: end of synchronization process)
(
(8a) OPS Step: The MUCF SCAT function checks that the LGST is within the range [GST-18ms ; GST+62ms].
If this is not the case, the MUCF (SCAT,) is responsible for generating proper time steps to adjust this LGST. When the LGST is synchronsed to the GST, go to step 6 (TOW/WN setting).

(10b) OPS Step: The request for adjustment of LGST can be implemented according to the following operational options:

- Option 1 : the LGST correction is included in the process for execution of satellite maintenance activities with mission impact: this is described in a dedicated scenario of SFC5.2: #SFC5.2.1-1.3 (alternative 4, excluding the steps needed to exclude the satellite from mission services as the satellite is not yet operational for any service during initialization). This includes:
  - Inclusion of the on-board time adjustment request in the Mission Planning and validation by GMS (MUCF)
  - Mission planning exchanged and schedule generation by GCS (SCPF)
    oOn board time adjustment schedule execution including TC uplink to adjust the on-board NSGU time (CMD_AdjustSatelliteTime)
  - TC execution control and feed back

  For all the details, refer to #SFC5.2.1-1.3

- Option 2: the GMS mission operator contacts the GCS operator to send the telecommand CMD_AdjustSatelliteTime, by manually filling the computed LGST correction.

(10c) The NSGU applies the time step command to drift the LGST pps. The corrected time is used to resynchronise the downlink of the dummy navigation page

At this stage, the on-board initial synchronization process within +/- 5 ms is completed.

6.3.1.6.2 Scenario #SFC3.1-2: Operational satellite LGST steering based on time adjustments

6.3.1.6.2.1 General Description

In this scenario, the LGST is steered by ground telecommand applying directly a time step correction to the NSGU time. In that case, the satellite needs to be taken out of service by the user before the application of the time adjustment.
Note: This scenario applies during satellite lifetime and assumes that the LGST has been synchronized during initialization (see scenario #SFC3.1-1). This means in particular that TOW and WN parameters do not need to be adjusted anymore.

6.3.1.6.2.2 Flow Chart

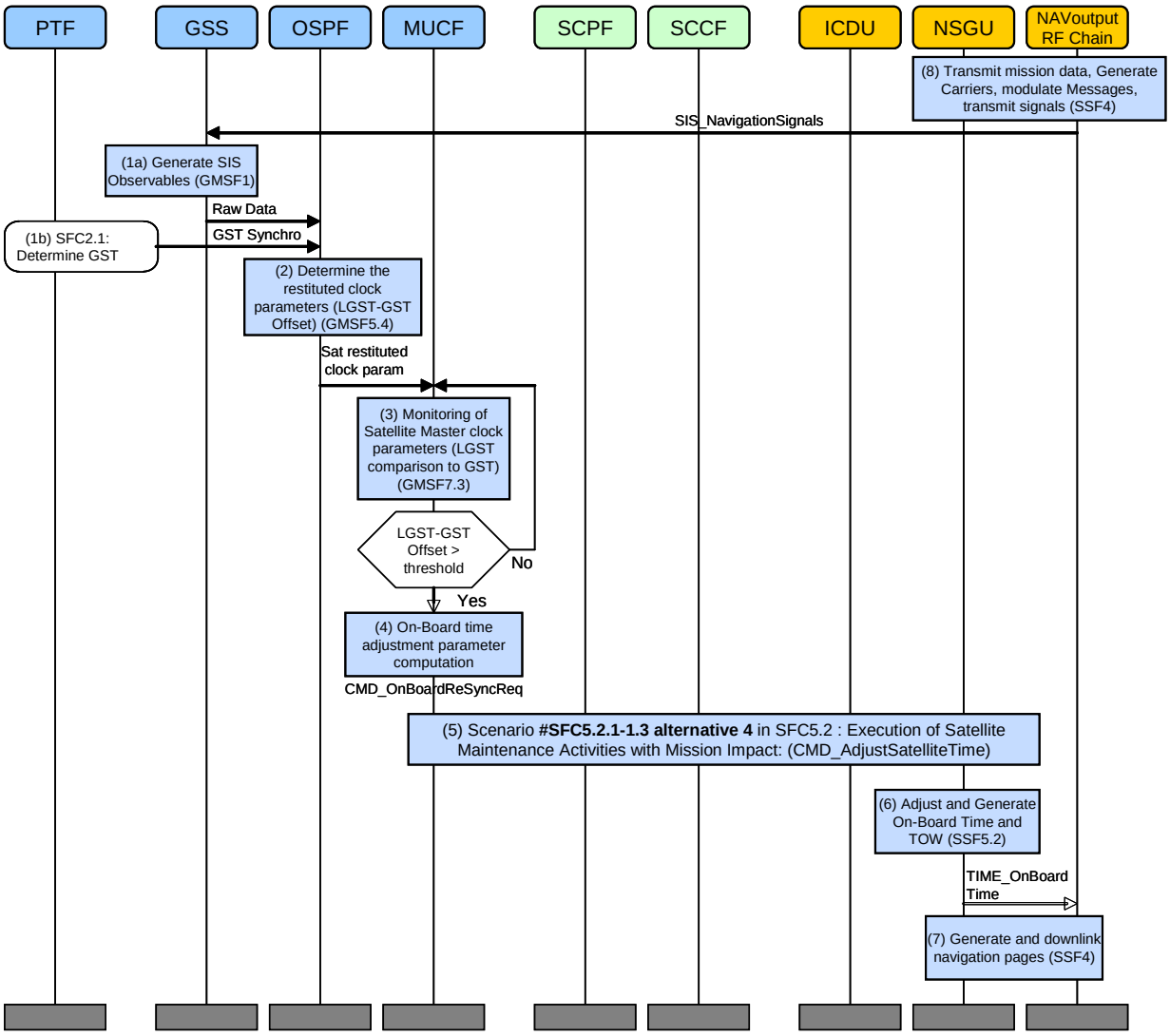


Figure 140: #SFC3.1-2 Dynamic Scenario

6.3.1.6.2.3 Scenario sequence description

(0) General Description: The satellite local GST (LGST) derived from the on-board frequency reference shall be steered to the on-ground GST within the range [-18ms, 62ms]. In this scenario, this steering is performed applying a time step to the satellite LGST directly. Then the clock is in a free running mode. As part of this scenario, the GMS is responsible for monitoring the LSGT-GST offset thanks to clock model computed by the OSPF. In case an adjustment is needed, the GMS shall inform the user that the satellite shall be taken out of service and send a request for time adjustment to the SSgt via the GCS (S-Band TC link).
The sequential description is detailed below:
(1a) The GSS Station provides to the OSPF Navigation Raw Data derived from the SIS Navigation Signals
(1b) The PTF is also responsible for providing Timing Observables and the GST to the OSPF: this is described in dedicated SFC2.1.

(2) Based on these 2 inputs (from 1a and 1b), the OSPF is responsible for computing in real time instantaneous clock parameters indicating for each satellite the offset of the on-board LGST wrt the GST. (Note: this instantaneous model is not the clock model to be provided to the user). These restituted clock parameters are provided to the MUCF. These parameters include the instantaneous LGST-GST phase offset (a0) retrieved from the ranging signal.

(3) The MUCF (GMOS) compares the LGST-GST offset to a specific threshold that is a configuration parameter provided by the GACF. This threshold is set (by OPS as configuration parameter) in such a way that the LSGT remains in the following range [GST-18ms, GST+62ms], with proper margins to consider the operational procedure duration for time adjustment.

(4) OPS Step: As soon as the LGST-GST offset exceed this specified threshold, analyses of the failure are triggered under operator control. When requested, the MUCF (SCAT) is responsible for generating proper parameters to adjust the satellite Local GST. This includes time steps to be applied to the NSGU on board.

Note: the operational procedure for analyzing any failure before triggering any recovery action is still under discussion as part of MUCF CDR. It has to be noted that this analyses can take into account the observed drift of the primary clock in order to optimize the target value of the LGST-GST offset during this synchronization process. For instance, if the drift of the on-board clock lead to a constant acceleration of the LGST wrt GST, the initial relative value of LGST could be set to the lowest value of the offset (e.g. +62ms) in order to maximize the time needed for the next adjustment: see JT-105 in IOV DJF [SYS-RD.3].

(5) OPS Step: This request for adjustment is included in the process for execution of satellite maintenance activities with mission impact: this is described in a dedicated scenario of SFC5.2: #SFC5.2.1-1.3 (alternative 4). This includes:
- Inclusion of the on-board time adjustment request in the Mission Planning and validation by GMS (MUCF)
- Mission planning exchanged and schedule generation by GCS (SCPF)

- On board time adjustment schedule execution including:
  oTC uplink to trigger the DVS to working without guarantee to inform the user prior to the time adjustment (CMD_UpdateOnBoardData)
  oTC uplink to adjust the on-board NSGU time (CMD_AdjustSatelliteTime)
- TC execution control and feed back

For all the details, refer to #SFC5.2.1-1.3.

(6) The NSGU applies the time step command to drift the LGST pps.

(7) The corrected time is used to resynchronise the downlink of the navigation page

6.3.1.6.3 Scenario #SFC3.1-3: Operational satellite LGST steering based on coarse frequency adjustment

6.3.1.6.3.1 General Description

In this scenario, the LGST is steered indirectly by ground telecommand applying one single frequency step correction to the CMCU frequency reference (as the LGST is directly derived from this frequency).
For instance: if the LGST is late wrt GST => LGST needs to be speed up => CMCU frequency reference needs to be increased (and vice-versa).

In that case, the satellite needs to be taken out of service by the user before the application of the time adjustment.
Note: This scenario applies during satellite lifetime and assumes that the Frequency Reference (a1 parameter) has been synchronized at an optimized value during initialization (see scenario #SFC3.1-1).

6.3.1.6.3.2 Flow Chart

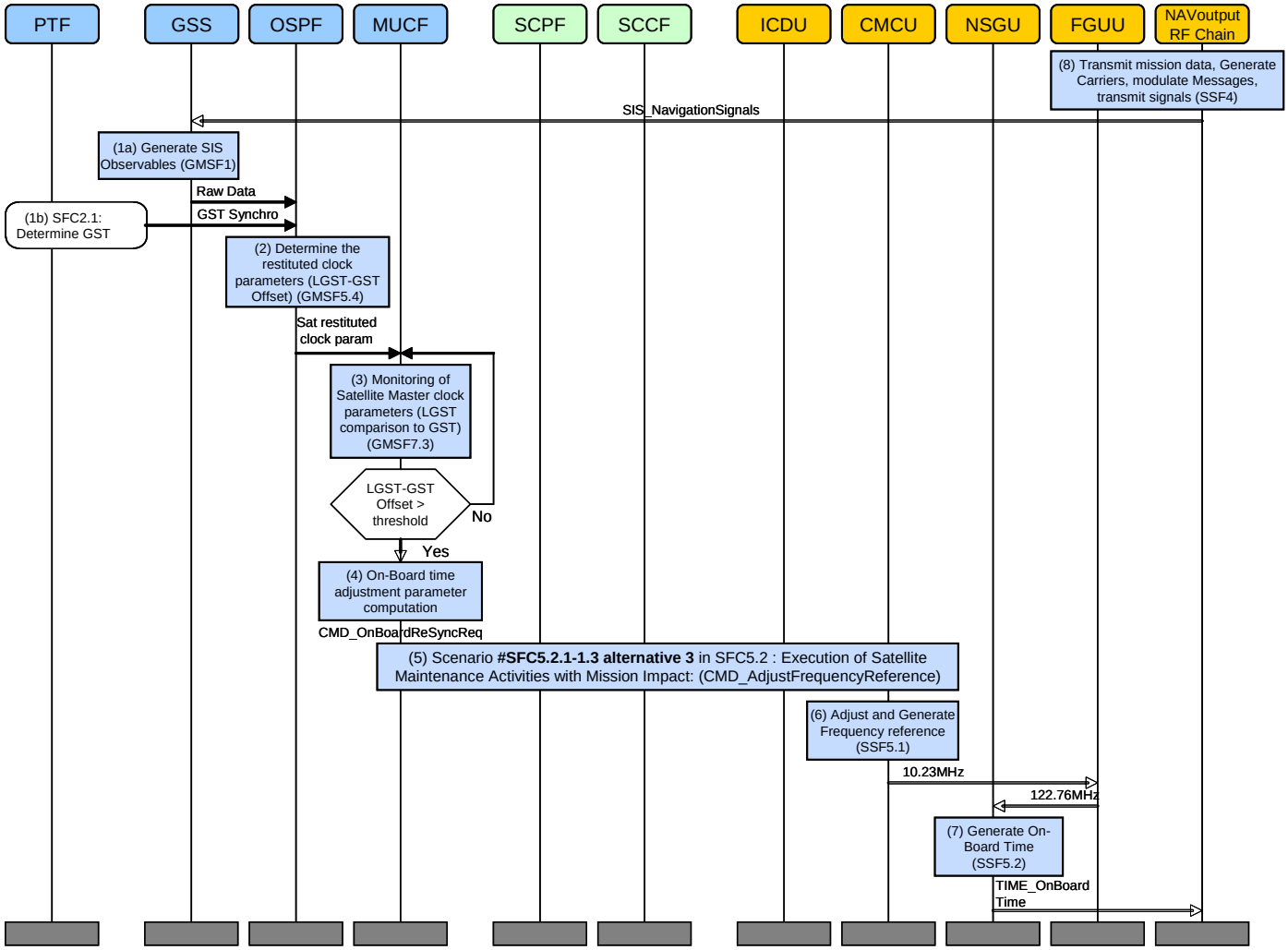


Figure 141: #SFC3.1-3 Dynamic Scenario

6.3.1.6.3.3 Scenario sequence description

0) General Description: The satellite local GST (LGST) derived from the on-board frequency reference shall be steered to the on-ground GST within the range [-18ms, 62ms]. In this scenario, this steering is made applying a frequency step to frequency reference. Then the clock is in a free running mode. . As part of this scenario, the GMS is responsible for monitoring the on-board frequency reference and the LSGT-GST offset thanks to clock parameters computed by the OSPF. In case an adjustment is needed, the GMS shall inform the user that the satellite shall be taken out of service and send a request for time adjustment to the SSgt via the GCS (S-Band TC link).

The sequential description is detailed below:

The steps (1) to (3) related to the monitoring of the on board time is identical to the scenario #SFC3.1-2:

(1a) The GSS Station provides to the OSPF Navigation Raw Data derived from the SIS Navigation Signals
(1b) The PTF provides Timing Observables and the GST to the OSPF
(2) Based on these 2 inputs, the OSPF computes in real time instantaneous clock parameters indicating for each satellite the offset of the on-board LGST wrt the GST. These restituted clock parameters are provided to the MUCF.
(3) The MUCF (GMOS) compares the LGST-GST offset to a specific threshold (configuration parameter provided by the GACF). This threshold is set (by OPS as configuration parameter) in such a way that the LSGT remains in the following range [GST-18ms, GST+62ms], with proper margins to consider the operational procedure duration for time adjustment.

(4) OPS Step: As soon as the LGST-GST offset exceed this specified threshold, analyses of the failure are triggered under operator control. When requested, the MUCF (SCAT) is responsible for generating proper parameters to adjust the satellite Local GST. This includes frequency steps to be applied to the CMCU on board.

Note: the operational procedure for analyzing any failure before triggering any recovery action is still under discussion as part of MUCF CDR. It has to be noted that this analyses can take into account the observed drift of the primary clock in order to optimize the target value of the on-board frequency and LGST-GST offset during this synchronization process. For instance, if the drift of the on-board clock lead to a constant acceleration of the frequency, the initial value of this frequency could be set in order to adjust the LGST to the lowest value of the offset (e.g. +62ms) in order to maximize the time needed for the next adjustment: see JT-105 in IOV DJF [SYS-RD.3].
(5) OPS Step: This request for adjustment is included in the process for execution of satellite maintenance activities with mission impact: this is described in a dedicated scenario of SFC5.2: #SFC5.2.1-1.3 (alternative 3). This includes:
- Inclusion of the on-board frequency adjustment request in the Mission Planning and validation by GMS (MUCF)
- Mission planning exchanged and schedule generation by GCS (SCPF)
- On board frequency adjustment schedule execution including:
  oTC uplink to trigger the DVS to working without guarantee to inform the user prior to the time adjustment (CMD_UpdateOnBoardData)
  oTC uplink to adjust the on-board CMCU frequency reference (CMD_AdjustFrequencyReference)
- TC execution control and feed back

For all the details, refer to #SFC5.2.1-1.3.

(6) The frequency step command applies to the CMCU (transmitted by the ICDU and RTU), responsible to adjust the frequency reference, applying the frequency steps depending on the CMCU mode. The default mode is the Continuous Phase Control Mode. The other modes (Standard Frequency Control Mode and Absolute Phase Control Mode) can be selected by proper TC.
The adjusted frequency reference is then distributed to:
(7) The NSGU: for the generation of the satellite on-board time (LGST) that will be corrected automatically to remain within the specified range.
(8) The RF Output chain to generate the proper carriers and transmit downlink signals.

6.3.1.6.4 Scenario #SFC3.1-4: Operational satellite LGST steering based on smooth frequency adjustment

6.3.1.6.4.1 General Description

In this scenario, the LGST is steered indirectly by ground telecommand applying several frequency step correction to the CMCU frequency reference.
In that case, the frequency steps applied are small enough to avoid impact on the satellite performance. The satellite does not need to be taken out of service by the user before the application of the time adjustment.
Note: this scenario is based on the assumption that the ICDU implement the functionality as described in GAL-CCN-ASTD-AV-A-0104 iss2.1 (ID0004).

6.3.1.6.4.2 Flow Chart

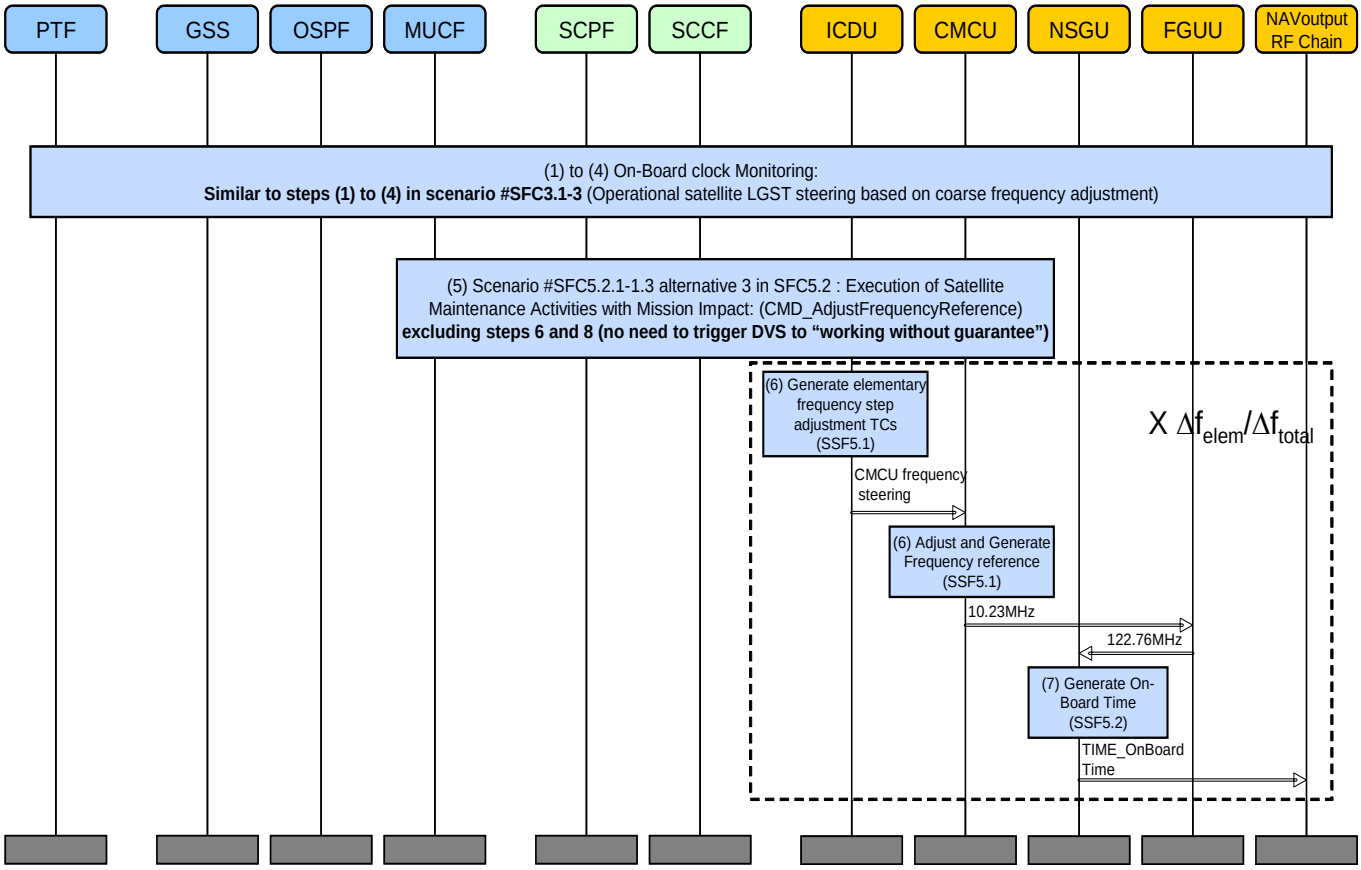


Figure 142: #SFC3.1-4 Dynamic Scenario

6.3.1.6.4.3 Scenario sequence description

(0) General Description: In this scenario, the LGST-GST steering is base on frequency adjustments (as in #SFC3.1-3). However in that case, the steps applied to the CMCU are small enough to avoid degradation of the signal (smooth steering method). Thus, the satellite does

not need to be taken out of service by the user. The ICDU is responsible to split the overall frequency step received in the GCS TC in several elementary small steps (<10-11 relative) to be sent in sequential TCs.
The sequential description is detailed below:
The steps **(1) to (4)** related to the monitoring of the on board time is identical to the scenario #SFC3.1-3

(5) OPS step: This request for adjustment is included in the process described in scenario #SFC5.2.1-1.3 (alternative 3), excluding the excluding steps 6 and 8. Indeed, in that case the satellite should remain available for the mission service and there is no need to trigger DVS to "working without guarantee" (TBC by test).

(6) OPS step: The frequency step command received by the ICDU is split in several elementary frequency steps. Each elementary step is included in a dedicated TC to be sent to the CMCU.

(7) The CMCU applies the frequency step adjustment. This adjustment can be performed in three different modes as described in step (6) of #SFC3.1-3. The adjusted frequency reference is then distributed to:
(8) The NSGU (via FGUU): for the generation of the satellite on-board time (LGST) that will be corrected progressively to remain within the specified range.
(9) The RF Output chain to generate the proper carriers and transmit downlink signals.

The steps (6) to (9) are repeated until the overall frequency steps is performed.

6.3.1.6.5 Scenario #SFC3.1-5: Primary Clock Switch

6.3.1.6.5.1 General Description

This scenario relates to satellite clock switching in case "graceful degradation" of the primary clock

- o Either that the LSGT-GST offset cannot be maintained any more by time or frequency adjustments.
- o And/or that the behavior of the clock is very degraded wrt the nominal model in terms of Allan deviation

In this scenario, the monitoring and control of the primary clock is based on the SIS provided by the satellite.
Upon failure of the Maser, switch-over to the hot-redundant RAFS shall be performed.
This Maser is switched off and the cold Maser shall then be switched on. After warm up completion, this Maser can act as hot redundant clock.
Notes:

- - The switching of the clock due to graceful degradation of satellite primary clock shall always be done on command request (no automatic switch)
- A satellite clock switching will be also
- triggered also for periodic maintenance of the cold redundant satellite PHM

6.3.1.6.5.2 Flow Chart

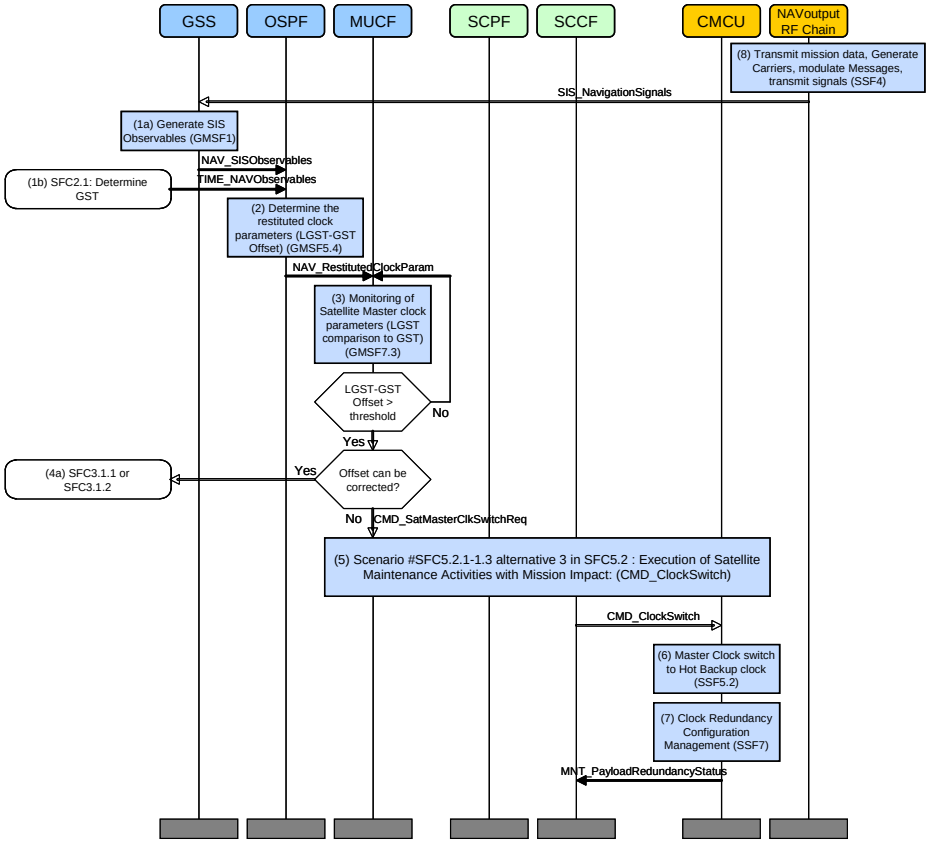


Figure 143: #SFC3.1-5 Dynamic Scenario

6.3.1.6.5.3 Scenario sequence description

0) General Description: The Primary Clock behavior is monitored by the GMS monitoring functions as for SFC3.1.1 and SFC3.1.2. As soon as the LGST-GST offset reaches a predefined threshold, a synchronisation request shall normally be sent. If the monitoring functions observe that a resynchronisation is not possible, it is needed to substitute the primary clock.

The sequential description is detailed below:

(1a) The GSS Station provides to the OSPF Navigation Raw Data derived from the SIS Navigation Signals

(1b) The PTF is also responsible for providing Timing Observables and the GST to the OSPF: this is described in dedicated SFC2.1.

(2) Based on these 2 inputs (from 1a and 1b), the OSPF is responsible for computing in real time instantaneous clock parameters indicating for the primary clock of each satellite:

- the offset of the on-board LGST wrt the GST
- the Allan Deviation parameters

These restituted clock parameters are provided to the MUCF.

(3) The MUCF (GMOS) is responsible for monitoring on-line the clock behavior based on these restituted clock parameters provided by the OSPF.

(4a) OPS step: If this offset can be maintained in the specified range with frequency or time adjustments, the scenarios described for SFC3.1.1 and 3.1.2 can apply.
(4b) OPS step: If the behavior of the clock shows a graceful degradation, a request for clock switch is included and validated in the Mission Planning exchanged with the GCS (SCPF).
(5) OPS step: This request for adjustment is included in the process described in scenario #SFC5.2.1-1.3 (alternative 3). As part of this process a sequence of TCs (as part of CMD_ClockSwitch) is uplinked to switch the primary clock in line with the authorized redundancy configuration.
(6) OPS step: First TC: The CMCU switches the primary clock to the hot backup RAFS. This RAFS becomes the primary clock.
(7) OPS step: Then the on-board clock configuration is managed according to the following sequence (each step corresponding to one uplinked TC):
- the failed “old” primary clock is switched-off by the CMCU
- the cold backup Maser is switched on. After warm up when full stability is achieved (about 1 week), the backup Maser is operational as hot backup.

6.3.1.6.6 Scenario #SFC3.1-6: Hot Backup Clock Switch

6.3.1.6.6.1 General Description

The switch of the hot backup clock occurs in case of graceful degradation. In this scenario, the monitoring and control of the primary clock is based on phase sampling measurements provided by the CMCU and downlinked to the ground segment by Telemetry.
The switch of the hot backup clock occurs in case of “graceful degradation”. This graceful degradation means that the behavior of the clock is very degraded wrt the nominal model in terms of Allan deviation.

In this scenario, the monitoring and control of the hot-backup clock is based on Phase Comparison Samples collected by the CMCU on board combined with L-Band observables indicating the primary clock behavior. The Phase Comparison Samples are downlinked to the GMS by dedicated Telemetry via the GCS. Indeed the hot backup clock frequency cannot be observed directly as not used nominally in the L-Band signal generation.
Upon failure of the hot backup RAFS clock, switch-over to the cold backup RAFS shall be performed.

Notes:
- The graceful degradation of the hot backup clock has no impact on the mission. It is then not needed to take the satellite out of service to perform the switch of this clock
- The switching of the clock shall always be done on command request (no automatic switch)

6.3.1.6.6.2 Flow Chart

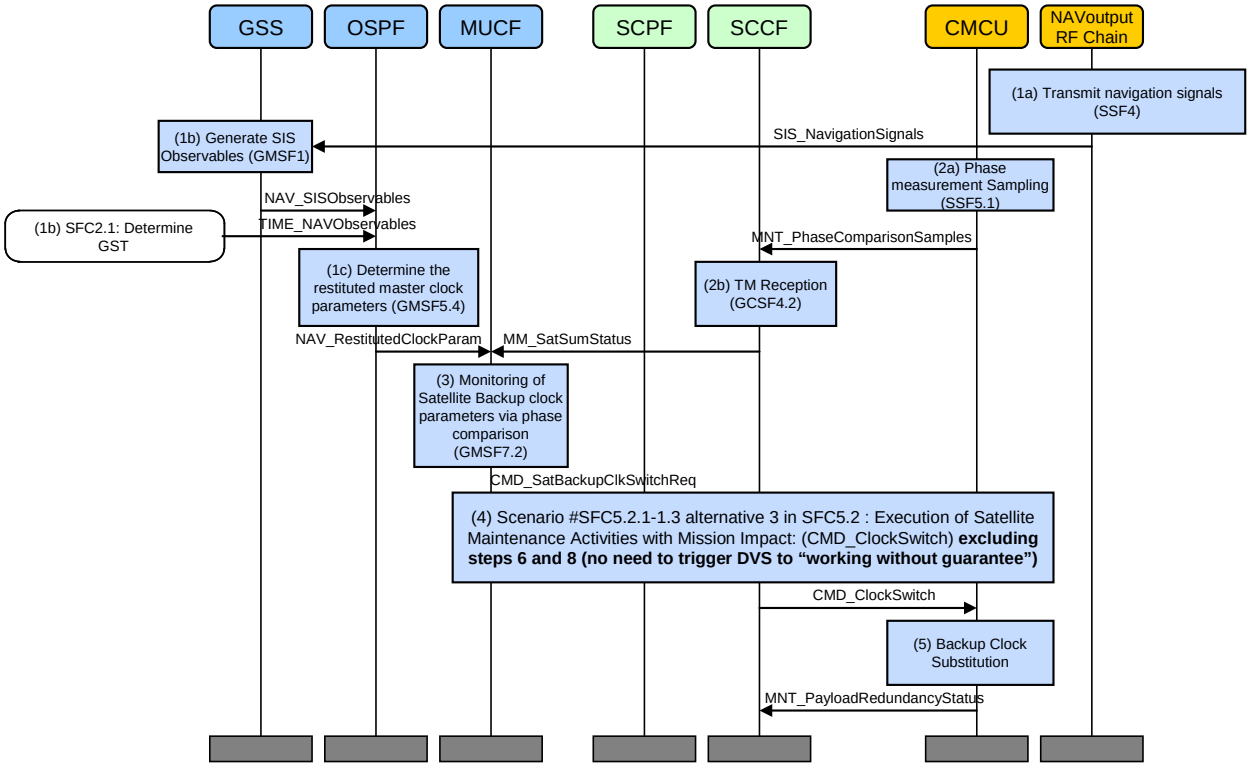


Figure 144: #SFC3.1-6 Dynamic Scenario

6.3.1.6.6.3 Scenario sequence description

0) General Description: The Hot Backup Clock behavior is monitored by the GMS monitoring functions based on Phase Comparison Samples collected on board and downlinked via Telemetry. In case of significant degradation of the hot backup clock (irremediable degradation), the hot backup clock cannot fulfill the hot backup functions and needs to be substituted. A switch request is them transmitted to the GCS and proper TC is sent to the satellite.
The sequential description is detailed below:
(1) Step 1: Collection of L-Band SIS observables and instantaneous Primary clock parameters
(1a) Transmit navigation signals: identical to Scenario #SFC3.1-5 step 1a
(1b) Generate SIS Observables: identical to Scenario #SFC3.1-5 step 1b
(1c) Determine the restituted primary clock parameters: identical to Scenario #SFC3.1-5 step 2
(2) Step 2: Collection of Phase Comparison Samples
(2a) The Phase comparison samples are collected at CMCU level comparing periodically the phase of the frequencies derived from the primary and the hot backup clock. These phase comparison are then downlinked to the ground segment via TM MNT_PhaseComparisonSamples
(2b) The telemetry is received by the SCCF within the GCS and transmitted to the GMS (MUCF): MM_SatSumStatus

(3) OPS step: Based on these inputs (from steps 1 and 2) , MUCF (SCAT in GMPA) is responsible for generating Standard Allan Deviation parameters and frequency parameters for the hot backup clock.
Then if the behavior of the clock shows a catastrophic degradation (wrt to on-ground model), a request for hot backup substitution is included and validated in the Mission Planning exchanged with the GCS (SCPF)
(4) OPS step: This request for adjustment is included in the process described in scenario #SFC5.2.1-1.3 (alternative 3), excluding the excluding steps 6 and 8. Indeed, in that case the satellite remains available for the mission service as the primary clock is not impacted and there is no need to trigger DVS to “working without guarantee”). As part of this process a sequence of TCs (as part of CMD_ClockSwitch) is uplinked to switch the hot backup clock in line with the authorized redundancy configuration.
(5) OPS step: The CMCU will then process to the hot backup clock substitution considering the authorised redundancy configuration, according to the following sequence (each step corresponding to one uplinked TC):
- the CMCU switches-off the failed “old” hot backup clock
- the CMCU switches-on the cold backup clock that becomes hot backup clock
- the CMCU select the new hot backup clock. After warm up when full stability is achieved (about 1 week), the backup Maser is operational as hot backup.

6.3.2 SFC3.2 GMS Times Synchronisation

6.3.2.1 Overall Physical Description

The purpose of this section is to describe the physical mapping of the GMS timing synchronization functions.
Note: this SFC describes only the distribution of the GST with different method from the PTF to the GMS elements. The description of the GST generation is detailed in dedicated SFC2.1.

It is based on the following documentation:

- ☐ System documents:
  - o System Logical Architecture (SYS-RD.1)
  - o System operational concept: MOCD (SYS-RD.6)
- ☐ GMS documents:
  - o GMS DDF2 (GMS-RD.2): Timing signal distribution section 6.5
  - o PTF DDF (GMS-RD.30): GST distribution section 8.2

6.3.2.1.1 Physical Mapping of functional architecture

In general, all the elements within the GMS need to be synchronized to a common time base with an accuracy of 1ms (for M&C purpose). This time base is the GST.
Physically, the PTF is the facility responsible for generating and distributing the GST to all the GMS elements

For some specific applications, some elements need to be synchronized to the GST with a better accuracy (e.g. GSS for navigation signal acquisition). In that case specific means are used for the synchronization process.

6.3.2.1.2 Physical Static View

The following figure presents the physical architecture of the GMS timing synchronization functions:

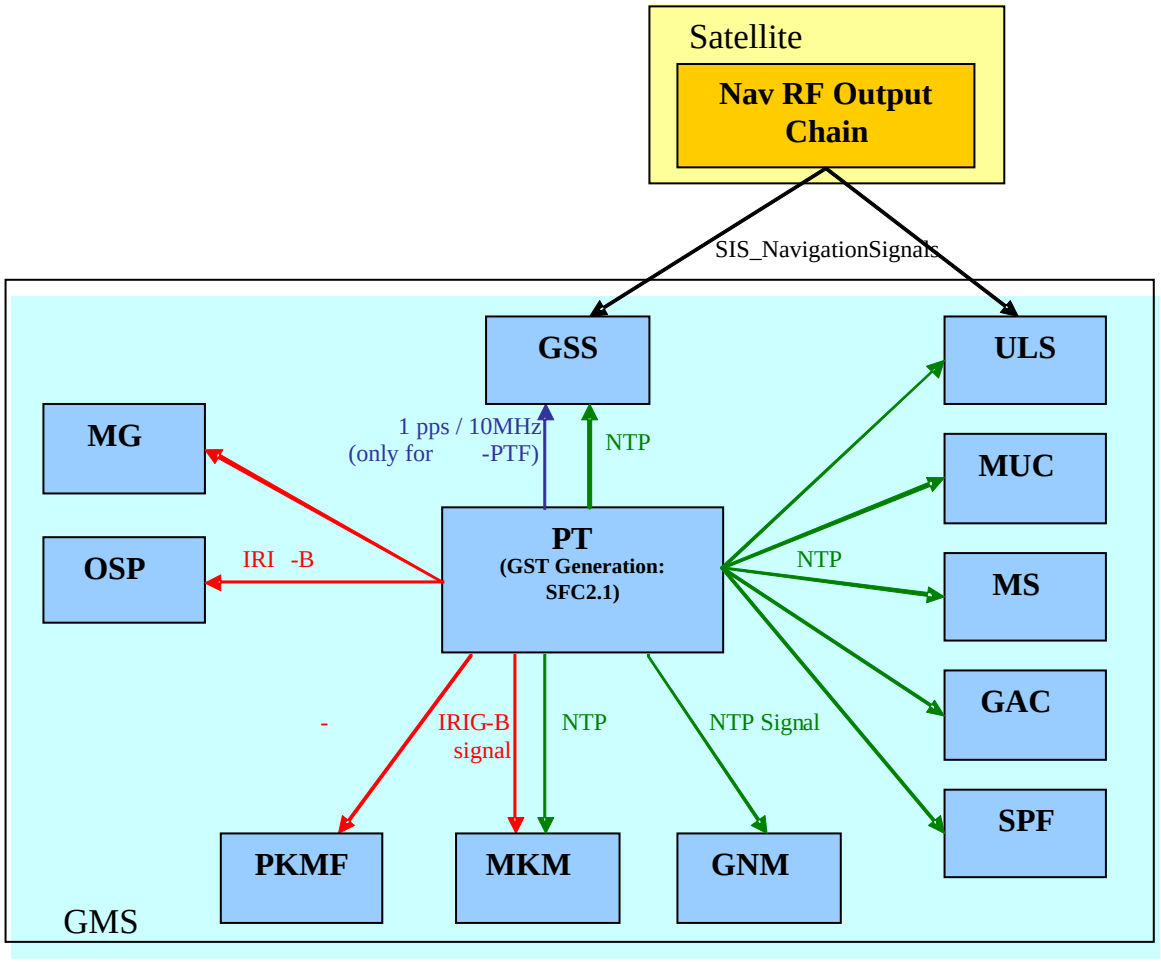


Figure 145: SFC3.2 Physical Static View

6.3.2.2 GST distribution methods from the PTF

The GST distribution from PTF to GMS elements can be performed with 3 different methods depending on the synchronization accuracy needed and the physical link.

6.3.2.2.1 GST signals

2 GST signals are provided:
- 1 pps signal: GST physical representation transferred to a Galileo receiver through a dedicated and calibrated line with TTL signal.
- 10MHz signal: Galileo receiver reference frequency as generated by the PTF.

The accuracy of the GST realization based on the combination of this information is in the order of some ns.

These GST signals are delivered for synchronization of the two GSS co-located in the PTF only.

In the frame of CR12, in order to improve PTF robustness, it is planned to include additional paths to provide redundancy for distribution of the pps signal and 10 MHz frequency reference signal to the two co-located PTF GSS's.

6.3.2.2.2 GST IRIG-B

The IRIG-B is a GST physical representation with an analog time code signal (IRIG-B 126) for SW synchronisation purposes. The accuracy of this representation is in the order of 1ms. The advantage of the IRIG-B is that this accuracy is guaranteed as it is independent from the network traffic (this is not the case with NTP).
The principle is to maintain coherent the time code generator units in the remote element and in the PTF.

The IRIG-B signal is delivered for synchronization to the following elements within the GMS: MGF, OSPF, I, MKMF (for the classified part) and PKMF

In the frame of CR12, in order to improve PTF robustness it is planned to include an additional time distribution circuit to provide redundancy for IRIG-B signal distribution within GMS and GCS.

6.3.2.2.3 GST NTP

The NTP is a time protocol allowing the synchronization of a NTP Client receiving a UDP timing signal provided by a NTP Server transmitted via the network. This allows to achieve an accuracy of about 1 ms through the LAN for the element located in the GCC, and lower accuracy for remote sites (depending on the network traffic).
The GST NTP signal is delivered for synchronization to the following elements within the GMS: GACF, MUCF, MSF, SPF, GNMF, MKMF (for the non classified part), GSS and ULS.

Note: as fully described in GMS DJF [GMS-RD.3] page 63, GMS is justifying the usage of SNTP (Simplified NTP) Client instead of NTP. This choice does not impact any interface (in particular with GCS) as the UDP timing signal remains the same by the NTP Servers. The difference between SNTP and NTP is a simplification of signal selection at element level when provided with several timing signals (SNTP takes the first signal available one whereas NTP perform optimisation algorithm). The rational is to ease certification process for DAL-C elements.

In the frame of CR12, in order to improve PTF robustness it is planned to include additional time distribution circuit to provide redundancy for NTP signal distribution within GMS and GCS.

6.3.2.2.4 GCC site distribution: PTFs synchronisation

The GST is distributed by the PTF to the elements located in the same GCC site (no link with remote elements in other GCCs). However, the PTFs located in different sites are synchronized. The synchronization is performed evaluating the time offset between the 2 PTFs with 2 possible methods:

- Common view receivers
- TWSTFT measurements

See SFC2.1 for details and in particular scenario #SFC2.1-3 (PTF switch).

6.3.2.2.5 GSS and ULS sites synchronization to GST

Nominally, the GSS and ULS remote sites includes Galileo and GPS positioning receivers to synchronise the GST directly using the Galileo SIS. As the position of the ULS and GSS stations are known with sufficient accuracy, only one satellite is needed to achieve proper synchronization (to solve the timing unknown).
The synchronization to GST by NTP relying on MDDN is considered as backup solution.

Note: The realisation of GST Time at GSS level will be synchronised with the True GST 1 second epoch, with an accuracy of :

- 100 microseconds (95%) when the GST reference is computed through PVT on the Galileo satellites.
- ± 100millisecond (95%) when the GST reference is obtained through SNTP.

6.3.3 SFC3.3 GCS Times Synchronisation

6.3.3.1 Overall Physical Description

The purpose of this section is to describe the physical mapping of the GCS timing synchronization functions.
Note: this SFC describes only the distribution of the GST with different methods from the PTF to the GCS elements. The description of the GST generation is detailed in dedicated SFC2.1.

It is based on the following documentation:

- ☐ System documents:
  - o System Logical Architecture (SYS-RD.1)
  - o System operational concept: MOCD (SYS-RD.6)
- ☐ GMS documents:
  - o GMS DDF2 (GMS-RD.2): section 6.5 on Timing signal distribution
  - o PTF DDF (GMS-RD.30): GST distribution section 8.2
- ☐ GCS documents:
  - o GCS DDF (GCS-RD.1)

6.3.3.1.1 Physical Mapping of functional architecture

In general, all the elements within the GCS need to be synchronized to a common time base with an accuracy of 1ms (for M&C purpose). This time base is the GST.
Physically, the PTF is the facility responsible for generating and distributing the GST to all the GCS elements.

6.3.3.1.2 Physical Static View

The following figure presents the physical architecture of the GCS timing synchronization functions:

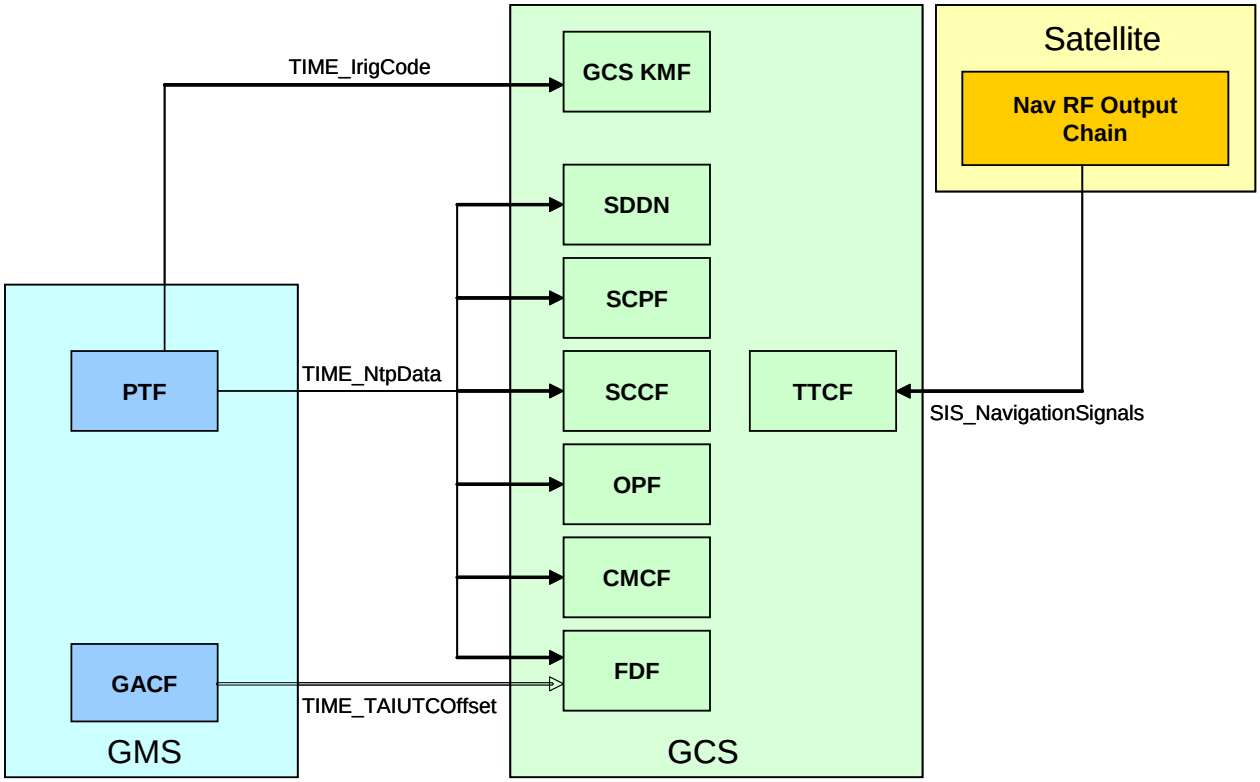


Figure 146: SFC3.3 Physical Static View

6.3.3.2 GST distribution methods from the PTF

The GST distribution from PTF to GCS elements can be performed with 2 different methods depending on the synchronization accuracy needed and the physical link.
Note: this physical view assumes that:

- the PTF generates the GST as described in SFC2.1
- the GACF receives the TAI-UTC offset parameters from the TSP (via SPF) as described in SFC2.1

6.3.3.2.1 GST NTP for GCS non-secure elements

The NTP is a time synchronization protocol transmitted via the network. This allows to achieve an accuracy of about 1 ms but this depends on the network traffic.

The GST NTP signal is delivered for synchronization to all the GCS elements except the KMF: FDF, SCFP, SCCF, OPF, CMCF, SDDN

6.3.3.2.2 GST IRIG-B for GCS KMF

For the GCS KMF in the secure area, separate timing signals are provided by IRIG-B signals. The KMF will be responsible for distributing this timing signal internally.

The GMS·will continuously provide to the GCS KMF IRIG-B time code signals to each and every of the three GCS chains (OPE, VAL and INT).

The IRIG-B is a GST physical representation with a time code signal (IRIG-B 126) for SW synchronisation purposes. The accuracy of this representation is in the order of 1ms. The advantage of the IRIG-B is that this accuracy is guaranteed as it is independent from the network traffic (this is not the case with NTP).
The IRIG-B signal is distributed on one-way coaxial cables, fulfilling the security constraints of the secure area.

6.3.3.2.3 TTCF sites synchronization to GST

The TTCF remote sites includes a Galileo receiver (or a GPS receiver before Galileo signals are available) to synchronise its own timing signal to the GST directly using the Galileo SIS. It implements also its own NTP server for the use of equipment at the TTCF site

Note: The TTCF does not use the SDDN as a NTP server to synchronise with GST.

6.3.3.2.4 TAI-UTC offset provision from GMS (GACF) to GCS (FDF)

The TAI to UTC offset (in terms of leap seconds) is provided to the FDF to enable this element to retrieve the UTC timescale. This is achievable with a sufficient level of performance as the GST available at FDF level by NTP server is steered to the TAI with an offset of less than 50ns.

6.4 SFC4 PROVIDE MISSION SERVICES (OS, PRS)

The purpose of this section is to describe the physical implementation of the SFC4 Provide Mission Services.

The SFC4 is in charge of providing all the means and associated processes needed by the end users to achieve the mission service objectives.

This includes three sub-SFCs corresponding to the three core Mission services provided by the Galileo system: Open Service (OS) and Public Regulated Service (PRS):

- SFC4.1: Open Services (OS), including:
    - SFC4.1.1: OS Positioning Service: responsible for the provision of the ranging signals and navigation data to allow the OS users to perform ranging measurements and compute positioning solution.
    - SFC4.1.2: OS Timing Service: responsible for providing timing data to retrieve the TAI, UTC precise times from GST.
    - SFC4.1.3: OS Signal Quality Monitoring Service: including the following sub SFCs:
        - SFC4.1.3.1: OS Signal-In-Space Accuracy (SISA) provision
        - SFC4.1.3.2: OS Health Status Flags provision.
    - SFC4.1.4: OS Degraded Positioning Service S-band provision.

- SFC4.2: Safety Of Life Integrity Service (SoL): descoped in MRD7.1

- SFC4.3: Public Related Service (PRS), composed of the following sub SFCs, each of them including an encryption layer for access control.
    - SFC4.3.1: PRS Positioning Service: responsible for the provision of the ranging signals and navigation data to allow the PRS users to perform ranging measurements and compute positioning solution, with specific control access.
    - SFC4.3.2: PRS Timing Service: responsible for providing timing data to retrieve the TAI, UTC precise times from GST, with specific access control.
    - SFC4.3.3: PRS Signal Quality Monitoring Service: including the following sub SFCs:
        - SFC4.3.3.1: PRS Signal-In-Space Accuracy (SISA) provision
        - SFC4.3.3.2: PRS Health Status Flags (they are also used for PRS integrity applications) and Satellite Health Status Flags provision.
    - SFC4.3.4: PRS Degraded Positioning Service S-band provision.
    - SFC4.3.5: PRS Integrity Service: descoped in MRD7.1
    - SFC4.3.6: PRS Security service: responsible for managing the access control to the PRS service, providing dedicated security data to the PRS users.

The mission services provision functional chain is mainly driven by the type of data to be generated and the way to disseminate them. Both constraints are directly derived from the service the data are supporting and their associated performances. It is also noted that different types of Galileo satellites have to be supported – at the same time – by SFC4, by proper system configuration (on a satellite per satellite basis).

The following sections present for each service type:

- the data to be provided by the system to the user: the detail description of these data (format, sizing...) is presented in dedicated section 6.5.4.3 "Navigation Message Content and handling"
- the dissemination concepts associated to them.
- the physical provision chain with the role of each element in this chain.
- the handling of the data at user level to be able to achieve the service targets.
- finally, dynamic scenarios are presented

The derivation of the service constraints into data types, associated formats and dissemination concepts justified in the IOV DJF (SFC4 justification) [SYS-RD.3].

Note: this section does not describe the L-Band message structure already presented in dedicated section as part of SISICD description.

6.4.1 SFC4.1 Open Services (OS) Provision

The purpose of this section is to describe the physical implementation of the functionalities needed to provide the Open Services to the users. This includes:

- OS Positioning Service
- OS Timing Service
- OS Signal Quality Monitoring Provision (SISA and DVS/SHS)
- OS Degraded Navigation Data Provision (via S-Band)

The following section describes the physical implementation of the OS service provision functionalities, based on the following documentation:

- System documents:
  - TUSREQ (AD.02)
  - GMSREQ (AD.03)
  - IOV SSREQ (AD.49)
  - FOC SSREQ (AD.04)
  - GCSREQ (AD.05)
  - SISICD (AD.06)
  - GMSSS ICD (AD.07)
  - GCSGMS ICD (AD.08)
  - FOC GCSSS ICD (AD.09)
- GMS documents:
  - GMS Design Definition File (DDF) Part II: Organical Architecture(GMS-RD.2)
  - MUCF DDF (GMS-RD.4)
  - MUCF ERD (GMS-RD.10)
  - MUCF ICD (GMS-RD.11)
  - OSPF ERD (GMS-RD.18)
  - OSPF ICD (GMS-RD.19)
  - MSF ERD (GMS-RD.25)
  - MDDN ERD (GMS-RD.26)
  - GSS ICD (GMS-RD.32)
  - MGF ERD (GMS-RD.38)
  - GSS ERD (GMS-RD.44)
  - ULS ERD (GMS-RD.49)
- SS documents
  - Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1)
  - Payload DDF (IOVSS-RD.3 and FOCSS-RD.02)
  - FDIR Analysis (FOCSS-RD.03)
- GCS documents:
  - GCS DDF (GCS-RD.1)
  - SCCF DDF (GCS-RD.9)
- TUS documents:
  - System Segment Design Document (TUS-RD.1)
  - Annex to TUS System Design Justification File (TUS-RD.3)
  - TUS System Segment Specification Document (TUS-RD.4)

The sections related to each type of OS service are structured according to the following scheme:

- List of the data needed to support each service at user level and its mapping to the physical elements.
- Description of provision concepts associated to these data. This comprises the complete physical provision chain including the role of each element.
- Description of the data handling at user level to achieve the service targets.
- Description of several dynamic scenarios including contingency cases.

The detailed description of the data (i.e. format, sizing...) is included in a dedicated Annex "Navigation message DDJF".
The derivation of the service constraints into data types, associated formats and dissemination concepts are justified in the IOV DJF (SFC4 justification) [SYS-RD.3]

6.4.1.1 SFC4.1.1 OS Positioning Service Provision

6.4.1.1.1 Overall physical Description

This SFC is in charge of providing all the means and associated processes needed by the end user to compute a positioning solution, namely:

- OS ranging signals (E5a-I, E5a-Q, E5b-I, E5b-Q, L1B and L1C) to perform pseudorange measurements. The pseudo-range is the measured distance between the receiver and a satellite used in the navigation algorithm.
- Navigation data (ephemeris, almanacs, time and clock corrections). The navigation data to correct pseudorange measurements (clock corrections) and compute the user position (ephemeris and almanacs). The OS navigation data is transmitted on the E5a, E5b and L1B data channels using the FNAV and INAV message logical structures. The mapping of the navigation data onto message logical structures, carriers and services is described in Figure below.

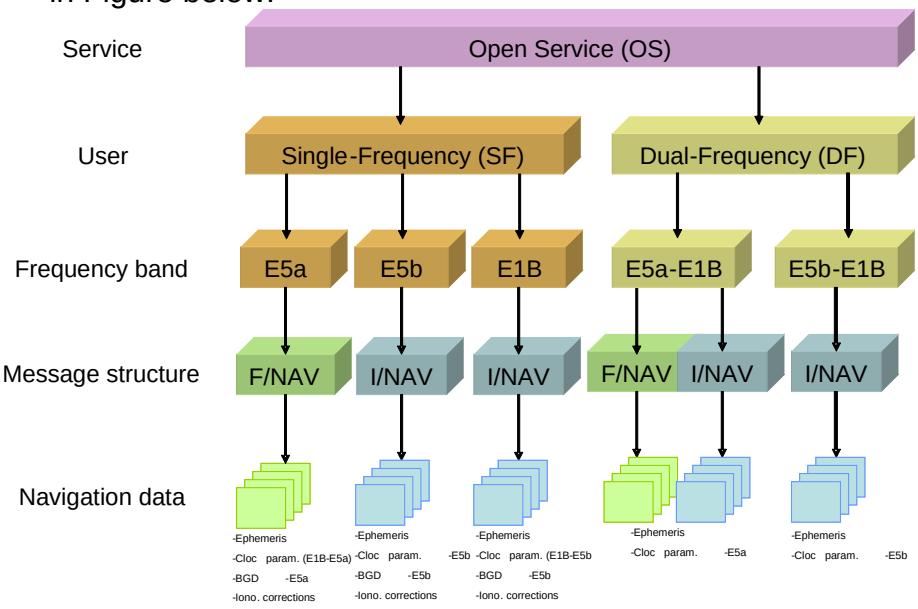


Figure 147 : Navigation data mapping on logical channels, carriers and service

6.4.1.1.2 Physical Static View

This section describes the physical mapping of the OS Positioning Service provision functions. The figure below summarises the physical mapping of the SFC4.1.1:

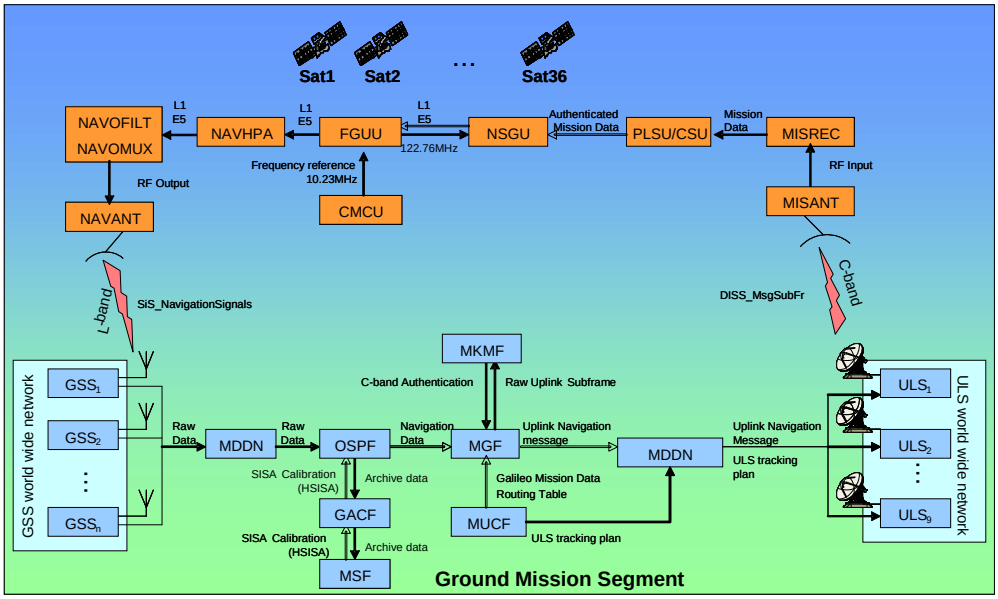


Figure 148: SFC4.1.1 Physical Static View

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.1.1.3 Data needed to support OS Positioning Service and mapping to physical elements

The OS data includes all the information needed by the users to perform positioning, meaning ephemeris and clock corrections (including ionospheric corrections and BGD needed by single frequency users), as well as some complementary data to inform about the Signal quality for each satellite like Signal-in-Space Accuracy (SISA) or Signal Health status (SHS) and Data Validity status (DVS).

Physically, the following segments are involved in the OS Service provision:

- GMS: responsible for generating and disseminating up to the satellites the OS service data.
- SSgt: responsible for L-Band message generation to relay the OS service data to the users.

The table indicates the role of the GMS and SSgt physical element in the provision of each type of OS service data. The role of each element is detailed in next dedicated sections.

	GMS					SSeg			
	GSS	OSPF	MGF	MKMF	ULS	C-Band Antenna / MISREC	PLSU / CSU	NSGU	from FGUU to NAVANT
Ephemeris	Raw data collection	Data generation	C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
Almanacs	-	Data generation	C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
GST parameters	-	-	-	-	-	-	-	Data and L-band message generation	L-band signal generation and Tx.
Clock correction	Raw data collection	Data generation	C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
BGD (*)	Raw data collection	Data generation	C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
Ionospheric parameters (*)	Raw data collection	Data generation	C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
GGTO	-	Data reception	C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

Table 73 : SFC4.1.1 Data and mapping to physical elements

(*) for OS positioning single frequency users only

Note: The following elements are not considered part of this SFC (even if they provide support):

- The MDDN (Mission Data Dissemination Network) is a GMS element that connects the GMS elements outside the GCC to the GMS elements inside the GCC. In the frame of this SFC the MDDN connects the GSSs to the OSPFs and the MGFs to the ULSs.
- The GACF is a GMS element responsible to provide calibration data to the OSPF for OS service data generation. This is addressed in SFC5.3.
- The CMCU is a payload element (SSeg) responsible to provide precise clock reference to the FGUU.
- MUCF is a GMS element responsible to generate the uplink schedule (MGF routing plan and ULS contact plan).

6.4.1.1.4 OS Positioning Physical Detailed Description

6.4.1.1.4.1 Raw Data Computation: GSS

The GSS is responsible for receiving the signals and data transmitted by the satellites and to perform measurements on the signals.
The result of this processing is called raw data (or observables) and is transmitted to the OSPF for navigation data determination every 1s GST.
The raw data contain for each satellite received by the GSS for each OS frequency channel (E5a, E5b, L1B) and also for the CS carrier (E6B):

- Galileo SIS data message
- Galileo SIS measurements: pseudo-range measurements, carrier phase measurements (accumulated Doppler), code-carrier-phase coherency indicators, excessive CCC flags, C/N value, too low C/N flag, signal quality indicator (signal validity, cycle slips

detection/repairing and inter-frequency bias compensation), signal status (receiver channel assignment mode, tracking state, correlator spacing)

All measurements performed by the GSS are relative to the reference time derived from a local atomic clock and synchronised to the GST (see SFC3.2).

6.4.1.1.4.2 Navigation Data Computation: OSPF

6.4.1.1.4.2.1 OS positioning data computation

The OSPF receives the Galileo SIS raw data from all GSS once per second. These observables are filtered and processed to generate:

- Ephemeris and almanacs for each of the satellites in orbit.
- Clock correction parameters for each of the satellites in orbit.
- Broadcast group delay (BGD) for each of the satellites in orbit.
- Ionospheric corrections common to all the satellites.
- Signal-in-Space Accuracy (SISA). See SFC4.1.3.1 for more details.

These data are provided to the MGF every 10 minutes as explained in the next section.

Furthermore the OSPF receives from the GACF:

- The GPS-Galileo time offset (GGTO). See SFC2.2 for more details.
- The GST-UTC time offset which is only used for timing service: see SFC4.1.2

The following figure gives an overview of the generation of the data which go in the navigation message:

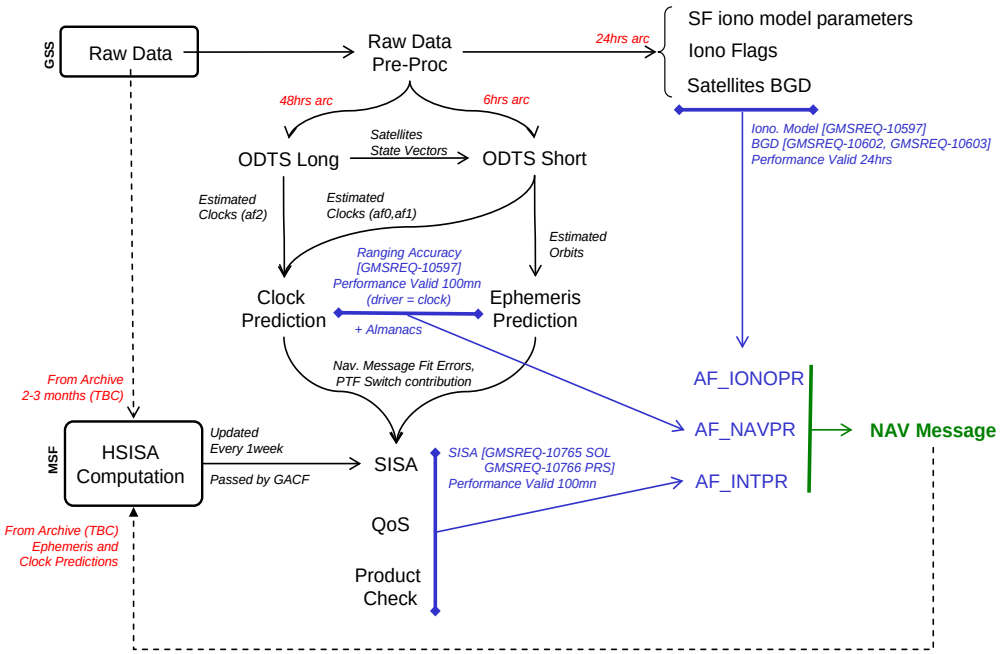


Figure 149 : OSPF Algorithms Overview - Navigation Data Generation

The details for the generation of the above data are as follows:

6.4.1.1.4.2.1.1 Ephemeris

This OSPF computes the satellite orbital parameters estimation (ephemeredes) with respect to the GTRF for all the satellites in orbit (up to a maximum of 36 satellites) for the OS positioning services to cover a 24-hours validity period with several batches (see next section).
An information summary regarding the ephemeris generation is provided below:

- Needed inputs (for more details, please refer to the corresponding IODD 'GAL-IODD-GMV-OSPF-R/030143'):
  - raw data received by the GSS - window of 48 hours
  - GSS positions in the GTRF received by GMS from GRSP
  - Satellite parameters (mass, cross-section area, antenna centre of phase, centre of mass model, yaw steering law)
  - Initial approximate satellite positions received by the GMS-GACF from the GCS-FDF
  - Parameters received from GRSP:
    - Gravitational coefficients validated by the MSF
    - Earth Rotation Parameters validated by the MSF
    -
  - Frequency: generated every 10mn
  - Maximum validity duration to still comply with most demanding UERE budget (see AD.1): 100min

6.4.1.1.4.2.1.2 Almanacs

The OSPF computes the almanacs for each of the satellites in orbit.
An information summary regarding the Almanacs generation is provided below:

- Needed inputs (for more details, please refer to the corresponding IODD 'GAL-IODD-GMV-OSPF-R/030143'):
- Same as for ephemeris and clock correction parameters
- Frequency: generated every 10mn
- Maximum validity duration: N/A since there are no associated performance requirements

6.4.1.1.4.2.1.3 Clock correction parameters

The OSPF estimates the clock correction parameters in order to allow the user compensating for the Galileo satellite onboard clock (LGST) to GST offset.
The clock correction parameters values are computed for frequency pairs:

- E1-C / E5a-Q clock model parameters using E5a-LE ch B/C frequency pair.
- E1-C / E5b-Q clock model parameters using E5b-LE ch B/C frequency pair.

These parameters are dedicated to both single and dual frequency users. Note that the single frequency users need to correct the clock correction parameters with the transmitted BGD and Ionospheric corrections.

An information summary regarding the Clock Correction Parameters generation is provided below:

- Needed inputs (for more details, please refer to the corresponding IODD 'GAL-IODD-GMV-OSPF-R/030143'):
  - Same as for Ephemeris (However, the arc length used to fit the clock model is as follows: af0 is estimated using 1 hour of data, af1 is estimated using 6 hours of data and af2 is estimated using 24 hours of data.).
- Frequency: generated every 10mn

- Maximum validity duration to still comply with most demanding UERE budget (see AD.1): 100min when the satellites use RAFS clocks.

6.4.1.1.4.2.1.4 Broadcast group delay (BGD)

The OSPF computes the Broadcast Group Delay (BGD) exclusively used by single frequency users. The BGD compensates for the errors introduced by the use of the satellite clock correction parameters (computed by frequency pairs) by single frequency users.
The OSPF determines the following specific dual frequency BGD parameters:

- E1-C / E5a-Q clock model parameters using E5a-E1 ch B/C frequency pair.
- E1-C / E5b-Q clock model parameters using E5b-E1 ch B/C frequency pair.

An information summary regarding the BGD Parameters generation is provided below:

- Needed inputs (for more details, please refer to the corresponding IODD 'GAL-IODD-NSL-EOSPF-R/034412'):
  - raw data received by the GSS - window of 24 hours
  - GSS positions in the GTRF received by GMS from GRSP
  - BGD configuration data (calibrated HW biases, height of the Ionospheric layer…)
- Frequency: generated every 10mn (with every new message, although the values are only updated every 24h, after running the corresponding algorithm)
- Maximum validity duration to still comply with most demanding UERE budget (see AD.1): 24h

6.4.1.1.4.2.1.5 Ionospheric corrections

The OSPF is also in charge of the computation of:

- the ionospheric model messages based on GSS measurements and
- Ionospheric disturbance flag or flag storm for 5 predefined regions.

The ionospheric model messages allow computing the effective ionization level of the atmosphere. This will allow single frequency users to compensate the delay introduced by the ionosphere on the measured onboard satellite time [Algorithm described in SYS-RD.73]
The dual frequency users are able to automatically compensate the ionospheric effects (see section 4.1.2.12).
An information summary regarding the Ionospheric Corrections Parameters generation is provided below:

- Needed inputs (for more details, please refer to the corresponding IODD 'GAL-IODD-NSL-EOSPF-R/034412'):
  - raw data received by the GSS - window of 24 hours
  - GSS positions in the GTRF received by GMS from GRSP
  - SFIONO configuration data (NeQuick model parameters, Brent method parameters…)
- Frequency: generated every 10mn (with every new message, although the values are only updated every 24h, after running the corresponding algorithm)
- Maximum validity duration to still comply with most demanding UERE budget (see AD.1): 24h

6.4.1.1.4.2.1.6 GPS-Galileo time offset (GGTO)

The GPS-Galileo time offset is received by the OSPF from the GACF nominally every 24 hours. The OSPF inserts it in the data sent to the MGF.

This offset could be used by combined Galileo-GPS receivers in order to be able to use GPS pseudo-range measurements for position determination.
The generation of the GGTO is described in details in SFC2.2.

6.4.1.1.4.2.2 OS positioning data dissemination: navigation batches

The data are grouped in so-called navigation batches. A navigation batch is composed of the above described navigation data excluding the almanac.
Every 10 minutes, the OSPF produces:

- 8 FNAV Navigation Batches + 1 FNAV Almanac Batch
- 8 INAV Navigation Batches + 1 INAV Almanac Batch

Note: The OSPF also produces GNAV batches but this is addressed in the SFC4.3 related to the PRS service provision.

The 8 navigation batches (for FNAV and INAV) are optimized to be successively broadcast by the satellite with a 3 hours spacing (see ToA subsection). The 4 first batches are uplinked via both C-Band and S-Band, while the 4 last batches are uplinked only via S-Band.
The almanac and navigation batches data are provided to the MGF already formatted as specified in GMSSSICD. The transmission is made nominally every 10mn and retransmission are made every 3mn (tunable parameter).
The Navigation Batches and Almanacs are accompanied by:

- Time-of-Applicability (ToA) parameter: used by the satellites to know when the data can be downlinked. The ToA applies to both the navigation batch and the almanacs. It is not downlinked to the user.
- Issue-of-Data (IOD) parameters: used by the users to recompose the navigation batches in case they lose some pages. There is one IOD (called IODnav) per navigation batch and one IOD (called IODa) per almanac. Quality-of-Service (QoS) parameters: to enable selecting the best OSPF data.

6.4.1.1.4.2.2.1 Time-of-Applicability (ToA)

The ToA is dedicated to the satellite. It is a time reference based on GST and composed by a ToW and a WN corresponding to the epoch when the set of parameters must be downlinked. The ToA is not downlinked in the L-Band message.

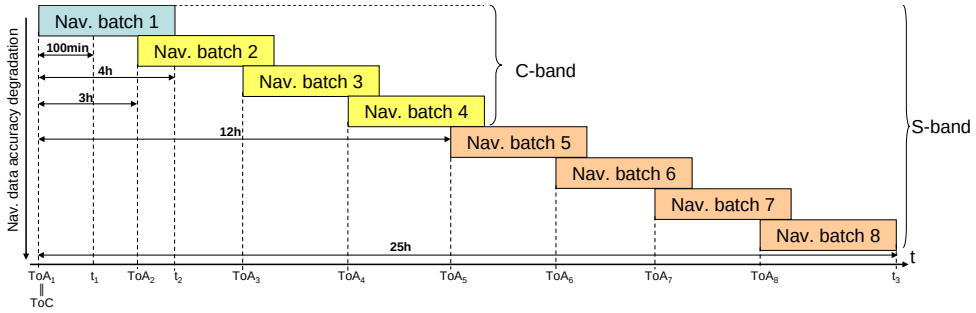


Figure 150 : Navigation Batches Downlink Sequencing

The batches become applicable to be downlined by the satellites one after the other with 3 hours spacing:

- $ToA_2 = ToA_1 + 3hours$
- $ToA_3 = ToA_2 + 3hours$
- $ToA_4 = ToA_3 + 3hours$
- $ToA_5 = ToA_4 + 3hours$
- $ToA_6 = ToA_5 + 3hours$
- $ToA_7 = ToA_6 + 3hours$
- $ToA_8 = ToA_7 + 3hours$

The ToA parameter does not relate to almanacs which become applicable for downlink as soon as received by the satellites.

Note: Some GMS and OSPF documentation mentions a validity duration of the navigation batches of 4 hours. This "validity duration" corresponds only to the ephemeris and can be used only for the first batch. The others are in any case not guaranteeing the accuracy which is needed for nominal service performance.
In practice the duration of validity of the first batch is dimensioned by:

- *The clock and SISA parameters if satellites are operating with RAFS master clock. In such cases the clock and SISA parameters exceed the allocated error budget after 100min (see SISICD-2212).*
- *The ephemeris if the satellites are operating with PHM master clock. Indeed the ephemeris parameters meet the error budgets for 240min while the clock and SISA parameters would be valid for around 8.3 hours (see fault free definition in GMSREQ).*

6.4.1.1.4.2.2.2 Issue-of-Data (IOD)

The IOD is a navigation batch or almanac identifier that allows the users to:

- Differentiate the data from different batches or almanacs and thus to know whether they have received a more recent batch.
- Retrieve the full batch of data even if it misses some pages or start receiving the data somewhere during the transmission

The OSPF generates 2 types of IOD for each computed batch: the IODnav for navigation batches and IODa for the almanacs.

IODnav

The IODnav is generated as follows:

$$IODnav(1,n+1) = IODnav(1,n)+1 \text{ MOD } 128;$$
$$IODnav(k+1,n) = IODnav(k,n)+127 \text{ MOD } 1024$$

Where:

- k ranges from 1 to 7. It corresponds to the rows in Table 78 .
- n is the order on the generation sequencing. It is increased by 1 every 10 minutes. It corresponds to the columns in Table 78

	T0	..	T0+1270 min	T0+1280 min	..	T0+2550 min
IOD nominal batch 1	0	..	127	0	..	127
IOD batch 2	128	..	255	128	..	255
IOD batch 3	256	..	383	256	..	383
IOD batch 4	384	..	511	384	..	511
IOD batch 5	512	..	639	512	..	639
IOD batch 6	640	..	767	640	..	767
IOD batch 7	768	..	895	768	..	895
IOD batch 8	896	..	1023	896	..	1023

Table 74 : IODnav generation

The IODnav sequence maximizes the time distance between two navigation data batches with the same IODnav (this enables to avoid IODnav ambiguity for 1280minutes (roughly 21 hours) and with the same 3 LSB of IODnav (
A new navigation batch is generated every 10mn and therefore the IODnav is incremented as described above every 10mn.

IODa

The IODa is a simple 4-bit cyclic counter which is incremented each time a new almanac is generated by the OSPF:
IODa(n+1)=IODa(n)+1 mod[16]
The frequency of the update of the almanac is left to the GMS design (i.e. no GMSREQ or GMSSSICD specification).

6.4.1.1.4.2.2.3 OSPF Quality-of-Service Indicator (QoS)

The OSPF also generates an associated Quality-of-Service (QoS) indicator common to all navigation batches and almanacs to help the MGF selecting the best OSPF data among all redundant OSPF elements.

6.4.1.1.4.3 C-band Navigation Message Format Generation: MGF

As part of the Navigation Dissemination chain, the MGF receives nominally once every 10 minutes the navigation batches computed by the OSPF (in addition to the QoS used to select the best OSPF among the redundant entities).
The MGF is responsible for building the navigation messages for all services within the Message sub-frame structure (C-band) in terms of packets that are uplinked to the satellite. The MGF generates dummy packets in the Message sub-frame structure when no data is received from the OSPF.
In particular in the frame of the SFC4.1 the MGF is in charge of:

1. Formatting the OS navigation data from the OSPF (4 first batches only) into the navigation messages FNAV and INAV for every satellite.
2. Request the authentication words for each C-band Message sub-frame generated.
3. Formatting the C-band frames dedicated to the ULS by multiplexing as specified in GMSSSICD:
   - Time-of-Applicability (ToA) for OS navigation messages for both FNAV and INAV
   - OS navigation messages for both FNAV and INAV
   - Message sub-frame authentication received from the MKMF.
4. Generation of a QoS indicator to allow the selection of the data generated by several MGF redundant instances by the MDDN.

5. Ensuring the routing of the authenticated Message sub-frames to the ULS antennas in accordance with the ULS to satellite allocation plan generated by the MUCF. The details for the generation of the uplink scheduling are defined in the SFC5.4.

6.4.1.1.4.4 Navigation Message Uplink Schedule: MUCF

The MUCF generates the C-band uplink scheduling that ensures that the navigation data is updated on board the satellite at least once per 100 minutes. See SFC5.4 and GMSF3.5 for more details.
The uplink scheduling plan and the contact plan are respectively disseminated to MGF and ULS stations for execution.

6.4.1.1.4.5 C-Band Uplink Authentication: MKMF

As part of the mission dissemination chain, the MKMF is in charge of the authentication of the C-band data send to the Galileo satellites. This process is made once per second with all the C-band message sub-frames received from the MGF.

6.4.1.1.4.6 Dissemination up to uplink antenna: MDDN

The MDDN is responsible for selecting the MGF with the best QoS to disseminate the best navigation data to the ULS.

6.4.1.1.4.7 Navigation Message Uplink: ULS

The ULS ensures that the navigation sub-frame received from the GCC is transmitted according to the uplink scheduling plan received from MUCF.
The ULS is then in charge of uplinking the mission data for the OS services. In particular:

- To compute the tracking plan based on the contact plan received from the MUCF.
- Point the ULS antennas (up to 4 independently on each ULS site) to the satellites following the ULS according to the satellite tracking plan.
- Check the message received from GCC.
- Build C-band message to be uplinked to the satellites in accordance with the GMSSS-ICD [AD.8] (C-band protocol parameters computation and message modulation).
- Generate and uplink dummy Message sub-frames to keep the satellite receiver locked to the signal when the data generated by the MGF is lost or corrupted. Thus the ULS is responsible for keeping a continuous transmission to the satellite in case of lost of individual Message sub-frames.
- Synchronise the ULS clock to the GST reference. Each ULS site includes a Galileo Receiver used during the FOC phase to synchronise the Local ULS Time to the GST (see SFC3.2 on GMS synchronisation).

Note that no operator control input is needed in routine operation.

6.4.1.1.4.8 Uplink Message Acquisition and Demodulation: MISREC

The MISREC is a six-channel spread-spectrum receiver (1 channel for GMS uplink reception and 5 disabled) that processes the signals from Mission uplinks, including:

1. acquiring and tracking the signals received by the C-band antenna
2. demodulating the signal
3. performing Viterbi decoding and Reed-Solomon check of the uplinked Message sub-frame.

4. providing the recovered Message sub-frames to the Signal Generation Subsystem (NSGU) via the Payload Security Module (PLSU) in IOV satellites and via Common Security Unit (CSU) in FOC satellites.

At the MISREC input it is placed the C-band antenna. This subsystem provides reception of mission uplinks from Galileo .

6.4.1.1.4.9 Uplink Message Authentication: PLSU (IOV satellites), CSU (FOC satellites)

In IOV satellites, the PLSU is responsible for checking the authentication of the Message sub-frames received from the ground in the C-band uplink already demodulated and transmitted by the MISREC. In FOC satellites, the same functionality is allocated to CSU.
If the sub-frame authentication is successful, the page is transmitted to the NSGU, otherwise is rejected and not transmitted.
It is noted that PLSU/CSU can work in a 'clear mode' where the authentication is not checked. This mode can be set by secure TC from the ground. In that case, all received subframes from the MISREC are directly forwarded to the NSGU.

6.4.1.1.4.10 FNAV and INAV L-Band Navigation Message Format and GST Generation: NSGU

In the frame of the navigation data provision, the NSGU is responsible for handling the OS navigation data on-board and its downlink on the corresponding L-band carrier.
The NSGU provides the following functions for the OS navigation message dissemination management:

- Storage management of Mission Data coming from PLSU/CSU (C-Band uplink) and from the ICDU/CDU (S-Band uplink). The total amount of data to be managed by the NSGU for OS is:
  - Navigation Batches:
    - 1 set of 4 navigation data batches for FNAV (received from C-band
    - 1 set of 8 navigation data batches for FNAV (received from S-band)
    - 1 set of 4 navigation data batches for INAV (received from C-band)
    - 1 set of 8 navigation data batches for INAV (received from S-band)
  - Almanacs:
    - 2 almanac for FNAV (1 received from C-band and 1 from S-Band)
    - 2 almanacs for INAV (1 received from C-band and 1 from S-Band)

  A navigation batch is bufferized in the NSGU as ready for downlink only when it has been successfully and entirely received.
-

Generation of the OS L-band messages as per SISICD (FNAV and INAV pages): The page are composed of data received from ground and data generated by the satellite.

- The data received from the ground have been described in the previous sections. Note that the INAV pages also contain data relative SAR (SFC8).
- The service data generated by the satellite are:
  1. ToW and WN with respect to the LGST of the satellite.
  2. Data Validity Status (DVS) flags (see SFC4.1.3.2)

  Note: the satellite health status is added in GNAV and is therefore not mentioned here (see SFC4.3 and SFC5.2).

- The pages are selected for downlink when the satellite LGST becomes larger than the ToA of the navigation and almanac batch with which them have been composed.

CRC, Convolutional encoding, interleaving and formatting of the Navigation Data Message according to the downlink page generation process:

- CRC computation: The checksum, which employs a CRC technique, is used to allow the user to detect potential errors in the downlinked pages. The CRC protects the data coming from the ground as well as the one generated on-board. It is composed of 24-bits and has a fixed position on every message.
- Addition of tail bits.
- Convolutional encoding: The data page is encoded with a convolutional encoder (FEC) in order to detect and correct errors. From a page of p data bits, the output of the encoder is 2xp symbol bits:

  The advantage of encoding each page independently is:
  - to isolate error on each page
  - to enable complete decoding of current page without need to wait for the reception of the other pages.

  The process to generate symbol page starting from a sequence of data bits is described below for the p=120 bits INAV page (the same process is followed for FNAV pages with p=250 bits):

Data (114 bits)										Tail (6 bits)					
6	7	8	9	10	11	12	…	118	119	120	121	122	123	124	125
120 bits															

  120 steps are needed to encode 120 bits of the INAV page, according to the figure below:

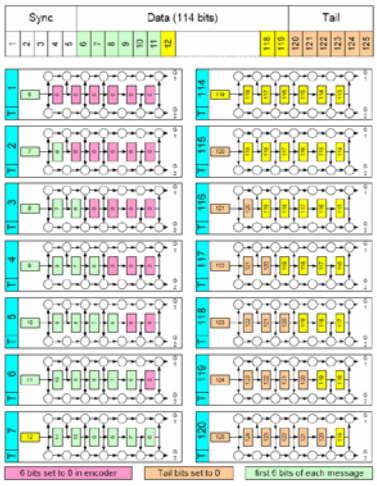


Figure 151 : FEC Encoding Process

o Interleaving of the encoded page: To minimise the effects of short burst errors due to fading, blocking or interference effects interleaving will be applied to the data prior to transmission. In general such a bit error leans on a packet of bits (“burst error”) and rarely

on a single bit. The use of interleaving technics enables to disperse the burst errors in different location to which the convolutional FEC coding is particularly vulnerable.

Then as the Viterbi decoding enables to correct packets of data, the combination FEC encoding/interleaving enables to be more robust to burst of errors (see SYS-RD.3).

The interleaver uses a block interleaver with n columns (where data is written) and k rows (where data is read): before transmission the data is written into a matrix store column by column and is then read out row by row for transmission.

In the Galileo SIS, the number of rows is equal to 8. The number of column depends on the message type.

Note that the FEC symbol page ("interleaver block") shall contain a number of symbol multiple of 8 to be able to interleave in N blocks of 8 bits.

The FEC symbol page is then interleaved according to the following figure:

Generated by Satellite
Interleaved symbols
2xp = Nx8

o Add of synchronisation field: The last step prior to modulate the symbol page with the ranging code is to add synchronisation field of K symbols. The size of the synchronization field shall be compatible with the final downlink page structure, and enables the user receiver to synchronise on the beginning of the page.

Generated by Satellite	
Synchronisation Field	Interleaved symbols
K	2xp = Nx8

At this stage the symbol page is ready to be modulated with the ranging code before being downlinked in L-Band.

- Generation of OS PRN Codes as per SISICD: The Code Generator provides the spreading codes for all OS signal components (E5a-I, E5a-Q, E5b-I, E5b-Q, L1B and L1C).
- Baseband signal generation as per SISICD:
  - o Message to PRN code modulation: This function spreads the navigation message data or pilot signal components with the relevant spreading codes from the code generator.
  - o Signal modulation.

6.4.1.1.4.11 Carrier generation, modulation and transmission: from FGUU to Users

The FGUU element is responsible for:

- Generate all frequencies required for up-conversion of navigation signals in E5, E6 and E1 channels
- Up-convert navigation signals to L-band downlink frequencies

The FGUU output is sent to the L-band antenna for broadcasting.

6.4.1.1.4.12 OS Positioning Process: USER

The standard OS user receiver functions are emulated by the TUS as explained in this section.

The TUS OS positioning process is physically split as followed:

1. The Signals are acquired by the RF part:
   - The **antenna** and its RF front End (preamplifier) composed of the low-noise amplifiers. It also provides the broad-band filter.
   - the ***receiver RF*** composed of:
     - o the RF front end (RFFE) which performs the signal splitting if needed, and filtering
     - o the RF to Intermediate Frequency down-convertor (RFIF) that performs the local oscillator synthesis, the signal down conversion and complementary filtering. Two RFIFs are present in TUS: one for E6/E1 channels, and one for E5a/E5b.

2. The **core module** (CM) performs the digital signal processing and is mainly in charge of performing pseudo-range measurements on the SIS ranging codes and pseudo-phase measurements on the SIS carrier. 16 parallel channels allows to process all the satellite in view.
   The main functions performed are (TUSF1):
   - Carrier Tracking
   - Carrier phase measurement
   - Ranging code tracking
   - Ranging code measurement
   - Multipath mitigation
   - Interference mitigation
   - Cycle-slip detection and correction
   - Signal monitoring

   The core module is also responsible of the recovering of the navigation data transmitted by the Navigation Signals (satellite parameters, integrity flags, SISA, …). To carried out this function the following tasks are needed (TUSF2):
   - Bit synchronization
   - Frame synchronization
   - Frame Demultiplexing
   - Symbol recovery
   - CRC check
   - Viterbi decoding
   - De-interleaving
   - Bit recovery

3. The **Application Unit** (AU) processes all received pseudoranges measurements (from CM) and navigation data recovered from the message to compute the receiver position. The main functions performed are:
   - Raw Data Acquisition and Pre-processing (TUSF5, TUSF6.1)
   - Navigation Message decoding

- Satellite Strategy Acquisition (TUSF6.4)
- PVT solution (TUSF4)

6.4.1.1.5 OS Positioning Service Provision Dynamic Scenario

6.4.1.1.5.1 Scenario #SFC4.1.1-1: Nominal Positioning data provision

6.4.1.1.5.1.1 General Description

This scenario (#SFC4.1.1-1) describes the generation and dissemination up to the users of the OS navigation data used for positioning service (ephemeris, almanacs, time and clock corrections). This navigation data is used to correct pseudorange measurements (clock corrections) and compute the user position (ephemeris and almanacs). The OS navigation data is transmitted on the E5a (for FNAV messages), E5b and L1B (for INAV messages) data channels.

6.4.1.1.5.1.2 Flow chart

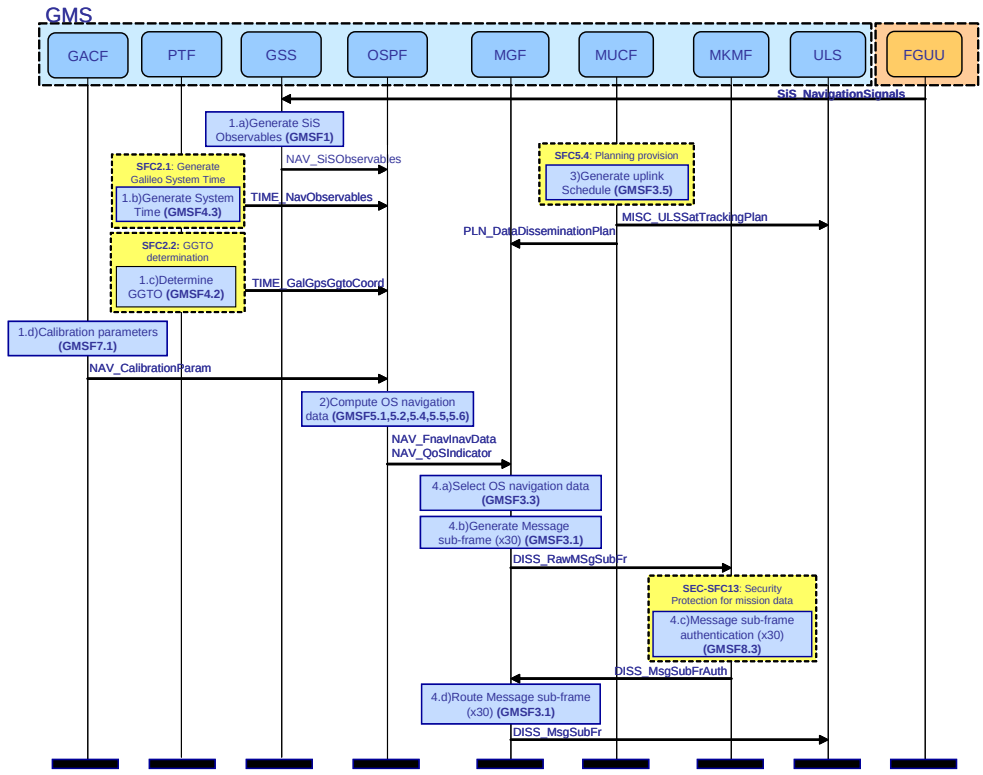


Figure 152 : #SFC4.1.1-1 Positioning Data Provision - part 1/2

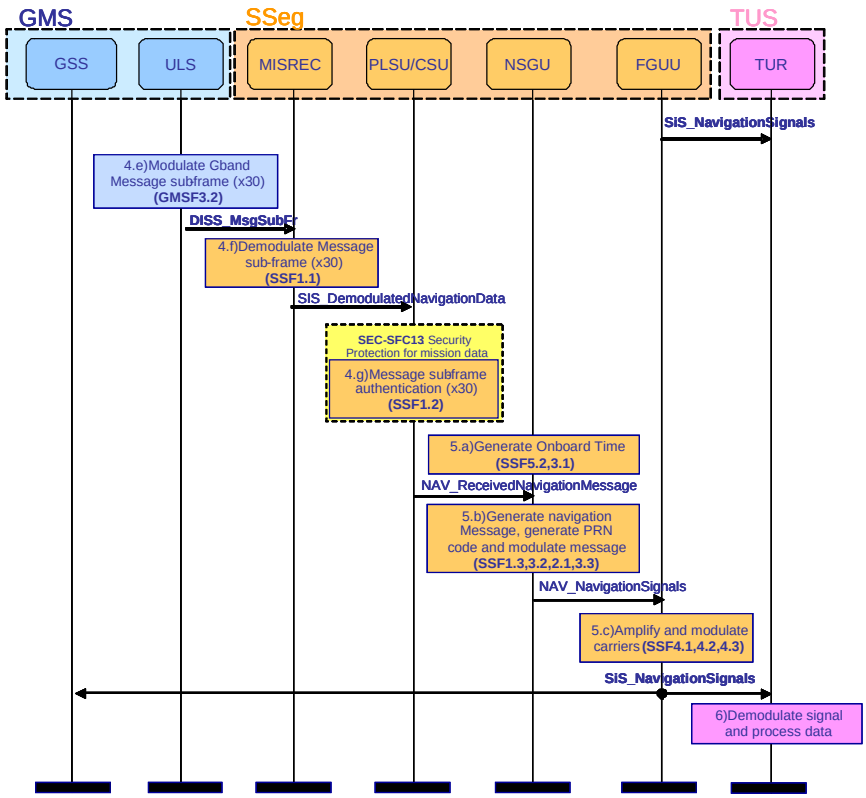


Figure 153: #SFC4.1.1-1 Dynamic Scenario- part 2/2

6.4.1.1.5.1.3 Scenario sequence description

1. Reception of Signal-in-Space (SiS) and generation of time and navigation observables.
   a. Each operational GSS processes every second the signals broadcasted by all the satellites in view. They compute the pseudorange measurements with the satellites and demodulate the signals for all the OS carrier frequencies. The recovered data include pseudorange measurements, raw data and some other data like signal quality parameters. All this is transmitted to the Galileo Control Center once per second. In the frame of the SFC4.1.1 these data is sent specifically to the OSPF.
   b. Synchronously the OSPF receive data from the PTF like the GST reference (see SFC2.1 for more details). This data allows the OSPF to compute for each satellite in orbit:
      - The current deviation of the LGST with respect to the GST reference clocks
      - The prediction parameters of the LGST deviation with respect to the GST for the next period of time.
   c. To help the Galileo interoperability with the other navigation systems (in particular GPS), the GPS-Galileo Time Offset (GGTO) is broadcasted within the navigation message.
      See SFC2.2 for more information about the GGTO generation process.

Contingency scenario (#SFC4.1.1-1/4.c): In case the GGTO is not available, the OSPF would generate a dummy value for the GGTO parameters that would be broadcasted within a valid navigation message.

d. The GACF transmits periodically to the OSPF calibration parameters for the orbital, BGD, clock and ionospheric corrections.

2. OS navigation data generation.
The OSPF process the GSS data and generate once every 10 minutes the following OS navigation data:

- The OS ephemeris and almanacs for each satellite in the constellation. The OS ephemeris are grouped in 8 different batches. The almanacs are grouped in a single batch.
- The onboard clock prediction deviation with respect to the GST reference for each satellite in the constellation.
- Ionospheric model parameters
- Broadcast Group Delay (BGD) for each satellite in the constellation.

Other data like the GPS-GST time offset (GGTO) is also received by the OSPF from the PTF via the GACF to be included in the navigation messages.
Each navigation data batch is identified by an Issue-of-Data (IODnav) also generated by the OSPF. The value of this parameter is synchronized with the start of the GST week. Together with the IODnav the Time-of-Applicability (ToA) is also generated for each navigation batch.
All the generated data have an associated Quality-of-Service (QoS) indicator to help the selection of the best data among all OSPF available in the Galileo Control Center. This is sent to the available MGF elements.

3. Uplink scheduling computation.
In parallel the MUCF generates the uplink and tracking schedules and distributes them to the MGF (contact plan) and to the ULS stations (tracking plan).
See SFC5.4 for more details about the uplink scheduling plan generation.

4. OS navigation message format and uplink.
   a. The MGF receive every 10 minutes the data from all operational OSPF available and selects the one with the best QoS. In case of all OSPFs have the same QoS value the MGF selects the so called master OSPF as configured by the GACF
      Contingency scenario (#SFC4.1.1-1/4.a): In the case the MGF do not receive valid data from any OSPF after 10 minutes from the last reception, the current OS navigation data is considered not valid and C-band dummy packets within the Message sub-frame are uplinked. This data will be discarded by the satellite on the step 5.a below.
   b. Each MGF, based on the MUCF uplink plan, format the OS navigation data in Message sub-frames that will be uplinked on the C-band links. This process is made once every second.
   c. The MGF prior to the dissemination, sends the Message sub-frames to the MKMF for uplink protection (authentication).

See SEC-SFC13 for more details about the Message sub-frames uplink protection .

   d. Once the Message sub-frames have been authenticated are routed to the ULS stations according to the contact plan received from the MUCF (see step 3).
   e. Each time the ULS receives a valid Message sub-frame, it is modulated and uplinked to the satellite according to the computed tracking plan.

Contingency scenario (#SFC4.1.1-1/4.e): in the case a Message sub-frame is not received on time according to the tracking plan or it is corrupted, the ULS has the capability to generate a

Dummy Message sub-frame that allows the C-band satellite receiver to keep tracking the signal until a new message sub-frame is received.

f. Each Message sub-frame transmitted within the C-band is demodulated by the satellite Mission Receiver (MISREC).
g. Each C-band demodulated Message sub-frame is sent to the PLSU/CSU for security protection check. (see SEC-SFC13 for more details)

Contingency scenario (#SFC4.1.1-1/4.g): If the authentication fails nothing is sent to the NSGU.

Note: In the case where the PLSU/CSU works in 'clear mode' the C-band sub-frames authentication is not checked and the sub-frames are directly passed through the PLSU/CSU towards the NSGU. Nominally the PLSU/CSU works in 'security mode'.

5. Onboard data processing and dissemination.
   a. The NSGU is responsible for generating the LGST in terms of Time-of-Week (ToW) and Week Number (WN).
      See scenario #SFC4.1.1-2 for more details on the generation process.
   b. Once the complete data is received, the NSGU compares the Time-of-Applicability (ToA) of the navigation data to its local GST (LGST).
      Every second (for INAV messages) or every 10 seconds (for FNAV) it generates the data to be transmitted on the L-band. This process includes the generation of some other data to be added in the L-band navigation message like CRC computation.
      After that process the data is modulated on baseband with the PRN code before it is sent to the FGUU.
   c. The FGUU modulates the signal to the correct frequency band and sends it to the L-band antenna for its dissemination.
6. Data reception and processing by the user receivers.

6.4.1.1.5.2 Scenario #SFC4.1.1-2: LGST provision

6.4.1.1.5.2.1 General Description

This scenario (#SFC4.1.1-2) describes the generation and the dissemination up to the users of the Satellite Local realization of Galileo System Time (LGST) parameters. These parameters are generated by the satellite which is synchronized with the on-ground GST. Both LGST parameters (ToW and WN) are included in the navigation messages (FNAV and INAV).

6.4.1.1.5.2.2 Flow chart

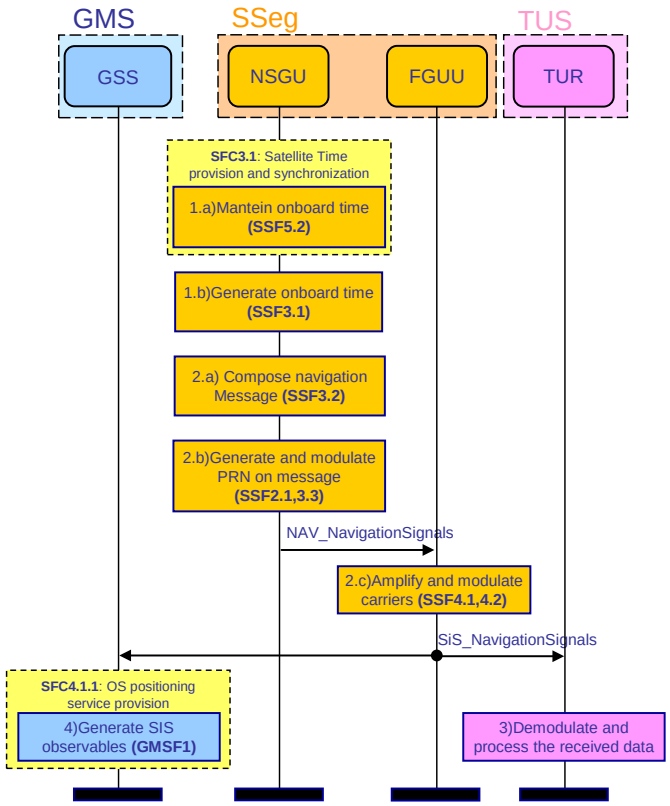


Figure 154: #SFC4.1.1-2 Dynamic Scenario

6.4.1.1.5.2.3 Scenario sequence description

1. Onboard time generation
   a. The LGST is maintained on board the satellite and sychronized with the on ground GST: see SFC3.1 for more details.
   b. The NSGU generates the LGST parameters ToW and WN that will be transmitted in the navigation message. The ToW and the WN indicates the 1-second value during which the page starts to be downlinked.

Contingency scenario (#SFC4.1.1-2/1.b): In case the ToW or WN are not available a dedicated FDIR is identified and the DVS flag is set to "Working without guarantee".

2. Message generation
   a. The ToW and WN parameters are inserted in the navigation messages (FNAV and INAV) received from ground.

2.b) and 2.c) The NSGU is also responsible for generating the PRN codes and modulate the navigation messages with them.

3. Demodulate and process the received data by the user receivers

4. Reception and processing on ground of the data received.
   See SFC4.1.1 for more details.

6.4.1.2 SFC4.1.2 OS Timing Service Provision

6.4.1.2.1 Overall Physical description

6.4.1.2.1.1 Physical Static View

The overall physical description of the OS Timing service provision is summarised below.

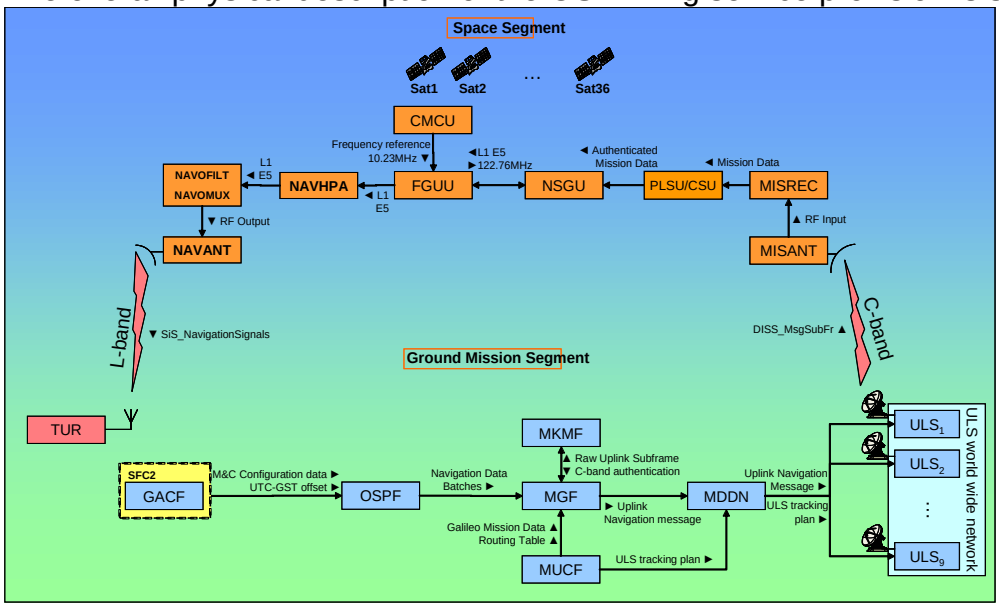


Figure 155: SFC4.1.2 Physical Static View

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.1.2.1.2 Data needed to support OS Timing Service and Mapping to the Physical Elements

The table indicates the role of the GMS and SSgt physical element in the provision of each type of OS service data. The role of each element is detailed in next dedicated sections.

	GMS					SSeg			
	GACF	OSPF	MGF	MKMF	ULS	C-Band Antenna / MISREC	PLSU / CSU	NSGU	from FGUU to NAVANT
			Dissemination to users (Common to the SFC4.1.1)						
UTC-GST offset	GST-UTC Data Collection	Data formatting	C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

Table 75 : SFC4.1.2 Data and mapping to physical elements

The data needed to support the OS Timing Service are the following:

- Navigation Protocol parameters: synchronisation field to perform signal acquisition
- Clock Corrections
- Timing Parameters:
  - GST
  - GST-UTC conversion parameters

6.4.1.2.2 OS Timing Service Physical Detailed Description

6.4.1.2.2.1 Data formatting: OSPF Element

The OSPF computes the GST-UTC conversion parameters from the GST-TAI offset and TAI-UTC offset periodically received from the GACF (see SFC2.1). The OSPF then formats the data to fit the navigation message and transmits the data to the MGF.

6.4.1.2.2.2 TIMING parameters Dissemination: from MGF to USERS

The GST-UTC offset parameters are disseminated together with the OS positioning service parameters described in the SFC4.1.1. It is inserted within each navigation data batch generated for both C-band and S-band links and follow the same provision process.

6.4.1.2.2.3 OS Timing User Process

The TUS OS timing process is physically split as the TUS OS positioning process plus an extra function (TUSF7) within the Application Unit:
The OS user receivers will compute GST with high precision and process the GST-UTC offset parameters according to the formula described in the SISICD and TUSREQ.
The UTC-GST offset update is not associated to a change on the IODnav transmitted within the navigation data batches and therefore the user receivers must check any variation on the offset on each message reception.

6.4.1.2.3 OS Timing Service Provision Dynamic Scenarios

6.4.1.2.3.1 Scenario #SFC4.1.2-1: GST-UTC time offset provision

6.4.1.2.3.1.1 General Description

This scenario (#SFC4.1.2-1) describes the dissemination up to the users of the UTC-GST offset.

6.4.1.2.3.1.2 Flow chart

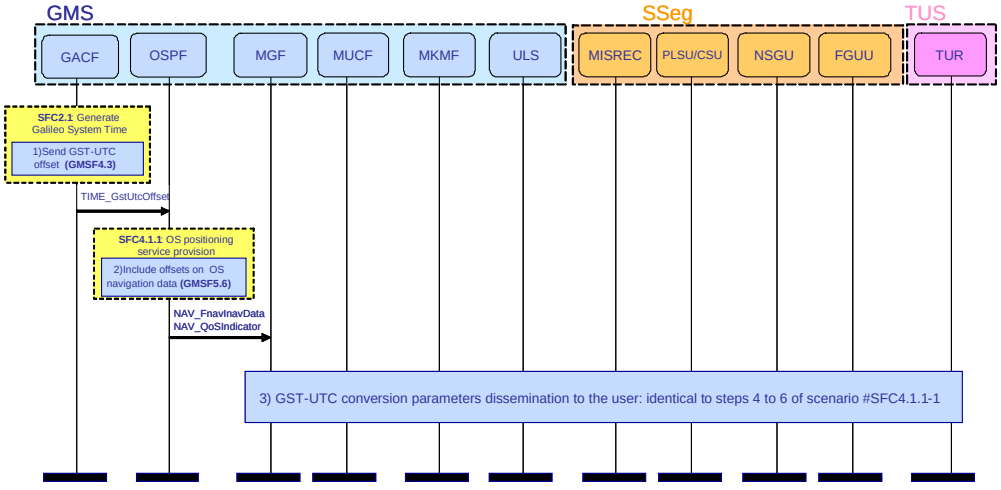


Figure 156 : #SFC4.1.2-1 Dynamic Scenario

6.4.1.2.3.1.2 Scenario sequence description

1) GST-UTC offset prediction computation

The GACF archives the GST-UTC offset predictions from the TSP and send the conversion parameters to the OSPF element. See SFC2 for more details about the GST-UTC offset generation.

2) GST-UTC conversion parameters inclusion in OS navigation data batches.

The GST/UTC conversation parameters are included by the OSPF in the navigation batches generated every 10 minutes for each satellite in the constellation and sent to MGF for formatting and dissemination

3) GST-UTC conversion parameters dissemination to the user

The GST-UTC conversion parameters are disseminated as part of the navigation messages, in the FNAV and INAV messages. Thus the formatting and dissemination processes from GMS to the user via the Galileo satellites follow exactly the same sequence as the one described from step 4 to 6 in the scenario #SFC4.1.1-1 (Positioning data provision): refer to this scenario for more details.

6.4.1.3 SFC4.1.3 Signal Quality Monitoring Provision

On-top of positioning and timing service information, the system provides the Open Service users with the following information about the signal quality:

- Signal in Space Accuracy (SISA) parameter - see SFC4.1.3.1: . The SISA is a prediction of the minimum standard deviation (1-sigma) of the unbiased Gaussian distribution, which over-bounds the Signal-In-Space Error (SISE) predictable distribution for all possible user locations. When no prediction is available the SISA is set to 255 by the GMS.
-
- **Signal Health Status (SHS):** The SHS will be used to inform the user of the status of the signal in space from a given satellite for each message type (i.e. F/NAV, G/NAV or I/NAV) and signal type (E5a-I, E5b-I, E6-A, E1A, E1-B). The SHS shall be set in two ways in the GMS:
  - Manual setting by operator
  - Automatic setting as results of planned events.

- Data Validity Status (DVS) flag - see SFC4.1.3.2: This flag informs the user about the whether a given satellite can be considered by the user as offering reliable performance (fault free) and whether the navigation data on board the satellite have been recently refreshed from the ground.

6.4.1.3.1 SFC4.1.3.1 OS SISA Provision

6.4.1.3.1.1 Overall Physical description

6.4.1.3.1.1.1 Physical Static View

This section describes the physical mapping of the OS SISA provision functions. The figure below summarises the physical mapping of the SFC4.1.3.1:

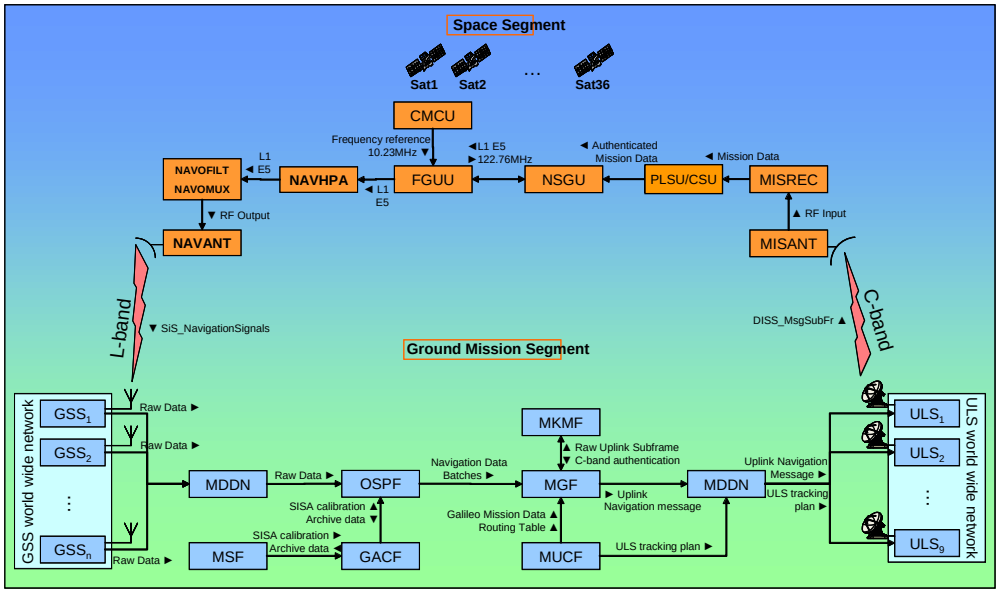


Figure 157: SFC4.1.3.1 Physical Static View

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.1.3.1.1.2 Data needed to support OS Positioning Service and mapping to physical elements

	GMS						SSeg			
	GSS	OSPF	MSF GACF	MGF	MKMF	ULS	C-Band antenna / MISREC	PLSU / CSU	NSGU	from FGUU to NAVANT
				Dissemination to users (Common to the SFC4.1.1)						
SISA	Raw data collection	Data generation	SISA calibration parameter computation	C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

Note: The following elements are not considered part of this SFC (even if provide support):

- The MDDN (Mission Data Dissemination Network) is a GMS element that connects the GMS elements outside the GCC to the GMS elements inside the GCC. In the frame of this SFC the MDDN connects the GSSs to the OSPFs and the MGFs to the ULSs.
- The GACF is a GMS element responsible for providing configuration data to the OSPF for SISA data generation.
- The CMCU is a payload element (SSeg) responsible for providing precise clock reference to the FGUU.
  MUCF is a GMS element responsible to generate the uplink schedule (MGF routing plan and ULS contact plan).

6.4.1.3.1.2 OS SISA Provision Physical Detailed Description

The OS SISA is provided in free-access on Open Service Signals. .

6.4.1.3.1.2.1 Raw Data Computation: GSS

Same process as for SFC4.1.1 (OS positioning): see section 4.1.2.1

6.4.1.3.1.2.2 Calibration SISA parameter Computation (HSISA): MSF

The MSF computes off-line the HSISA parameter (Historical SISA) based on archived data retrieved from the GACF computed by the OSPF and the GSS pseudo-range measurements.
This parameter is sent as a configuration parameter to the OSPF via the GACF to calibrate the SISA.
An information summary regarding HSISA generation is provided below:

- Needed inputs:
  - raw data received by the GSS - window of 2 months minimum
  - GSS positions in the GTRF received by GMS from GRSP
  - Satellite parameters (mass, cross-section area, antenna centre of phase, centre of mass model, yaw steering law)
  - Initial approximate satellite positions
  - Gravitational coefficients
  - Earth Rotation Parameters
  - Navigation Messages generated by OSPF
- Frequency: updated every 1 week
- Maximum validity duration to still comply with most demanding UERE budget (see AD.1): 1 week

6.4.1.3.1.2.3 SISA Computation: OSPF

Space Accuracy (SISA). This computation is based on the SIS observables received from the GSS and the Historical SISA (HSISA) computed by the MSF.
Determination of SISA parameters is based on the following pairs of frequencies. They are the same as per the satellite clock correction parameters:

- E1-C / E5a-Q using E5a-E1 ch B/C frequency pair.
- E1-C / E5b-Q using E5b-E1 ch B/C frequency pair.

Each computed navigation data batch has a SISA value associated for each pair of frequencies. Therefore SISA is computed every 10 minutes simultaneously with the navigation data.
An information summary regarding SISA generation is provided below:

- Needed inputs (for more details, please refer to the corresponding IODD 'GAL-IODD-NSL-EOSPF-R/034413'):
  - ODTS process on-line contributions to SISA (Navigation Message fitting and discretization error, and PTF Switch contribution)
  - HSISA (from MSF)
- Frequency: updated every 10min
- Maximum validity duration to still comply with most demanding UERE budget (see AD.1): 100min

Note: the SISA can also be set to the value 255 (SISA not available) based on satellite planned events (e.g maneuvers, clock adjustements….) as identified in the Mission Planning: See SFC5.2.

6.4.1.3.1.2.4 SISA Dissemination: From MGF to Users

The dissemination strategy is identical to the navigation data batches.

6.4.1.3.1.2.5 OS SISA User Process

This parameter enables to indicate to the user the quality of the SIS broadcast by satellites and in particular the quality of the ODTS outputs.

6.4.1.3.1.3 Scenario #SFC4.1.3-1: OS SISA provision

6.4.1.3.1.3.1 General Description

This scenario (#SFC4.1.3-1) describes the dissemination up to the users of the SISA parameters.

6.4.1.3.1.3.2 Flow chart

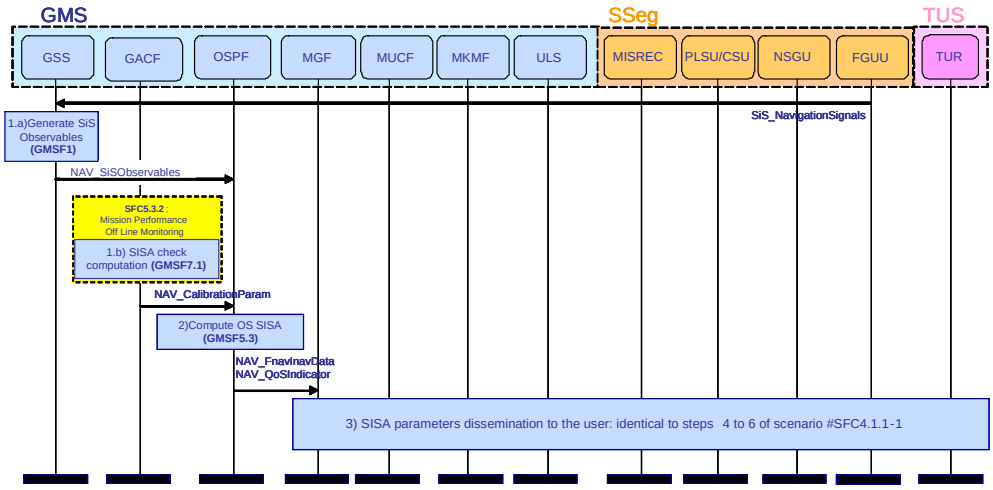


Figure 158: #SFC4.1.3.1-1 OS SISA Provision

6.4.1.3.1.3.3 Scenario sequence description

1) SISA parameter input collection

The computation of the SISA parameter is based on SIS observables from the GSSs and HSISA calibration parameter from the MSF (via GACF)

1.a) Collection of raw data measurement by the GSSs and provision to the OSPF: same process as for step 1.a in scenario #SFC4.1.1-1

1.b) HSISA calibration parameters computed by the off-line mission monitoring function at MSF level: see SFC5.3.2 for detail. This calibration parameter is provided to the OSPF via the GACF.

2) SISA computation

The SISA parameter depends directly on the navigation data generated by the OSPF. Therefore the SISA are generated every 10 minutes, as part of each OS navigation data batch.

Two values of SISA are disseminated, one for each of the following frequency pairs:

- L1B / E5a
- L1B / E5b

The computed SISA parameters are sent to the MGF together with their associated OS navigation data batch for formatting and dissemination

3) SISA parameters dissemination to the user

The SISA parameters are disseminated as part of the navigation messages, in the FNAV and INAV messages. Thus the formatting and dissemination processes from GMS to the user via

the Galileo satellites follow exactly the same sequence as the one described from step 4 to 6 in the scenario #SFC4.1.1-1 (Positioning data provision): refer to this scenario for more details.

6.4.1.3.2 SFC4.1.3.2 OS Health Status Parameters Provision

This section describes the implementation of the functional chains in charge of providing to the user the following status flags:

-
- The SHS will be used to info the user of the status of the signal in space from a given satellite for each message type (i.e. F/NAV, G/NAV or I/NAV) and signal type (E5a-I, E5b-I, E6-A, E1A, E1-B).

  The SHS can be set in two ways in the GMS:
  - o Manual setting by operator
  - o Automatic setting as results of planned events.
- Data Validity Status (DVS): This flag informs the user about the whether a given satellite can be considered by the user as offering reliable performance (fault free) and whether the navigation data on board the satellite have been recently refreshed from the ground.

6.4.1.3.2.1 Signal Health Status (SHS)

6.4.1.3.2.1.1 Overall physical Description

6.4.1.3.2.1.1.1 Physical Static View

This section describes the physical mapping of the OS Signal Health Status provision. The figure below summarises the physical mapping of the SFC4.1.3.2 for the Signal Health Status flag (SHS):

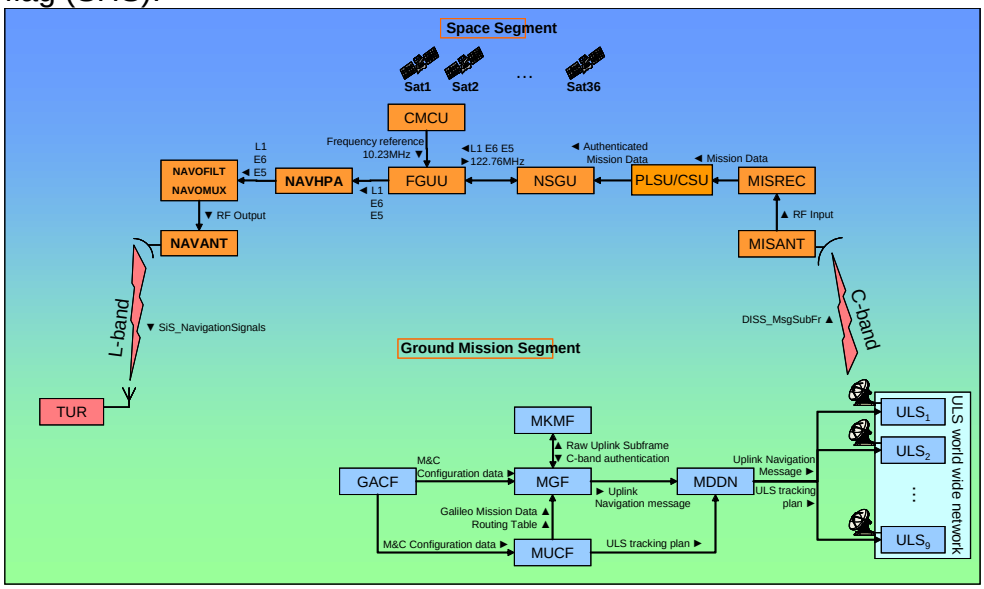


Figure 159 : SFC4.1.3.2 (SHS) Physical Static View

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.1.3.2.1.1.2 Data needed to support OS Positioning Service and mapping to physical elements

The table indicates the role of the GMS and SSgt physical element in the provision of the SHS flag. The role of each element is detailed in next dedicated sections.

	GMS				SSeg			
	MUCF GACF	MGF	MKMF	ULS	C-Band Antenna / MISREC	PLSU / CSU	NSGU	from FGUU to NAVANT
			Data dissemination (common to SFC4.1.1)					
Signal Healt Status (SHS)	Plan and Schedule generation	SHS setting / C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

6.4.1.3.2.1.1.2 OS Signal Health Status Parameter Provision physical Detailed Description

The SHS will be used (STRB 1.5.1) to info the user of the status of the signal in space from a given satellite for each message type (i.e. F/NAV, G/NAV or I/NAV) and signal type (E5a-I, E5b-I, E6-A, E1A, E1-B).

The SHS can be set in two ways in the GMS:
- Manual setting by operator
- Automatic setting as results of planned events.

In order to accomplish the above reported functionality the MGF needs to be upgraded. For planned events the MGF will be upgraded to have the capability to set the SHS per satellite, per service and per signal according to the planned events as notified by the GCS.
For what concern the manual setting the MGF will be upgraded in order to allow the operators to perform the following steps:

SHS configuration: A MMI shall be provided to the operator which shall offer some specific capabilities in order to ease the manual change of the SHS.
When opening the MMI, the last configured SHS values shall be displayed. The operator may modify the current displayed configuration, and then finally save the updated configuration using a new file version, then stored in GACF database.
When modifying the SHS, the MMI shall have the following ergonomic:
- First level (the most general use case): set the SHS for a specific satellite, all message types and signals included
- Second level (less frequent): set the SHS for a specific message type of a specific satellite
- Third level (rare events): set the SHS for a specific signal of a specific satellite

SHS application: the operator shall deploy the previously generated configuration file using GACF functionality. This file can be deployed while GMS elements are in operational mode using an online command.

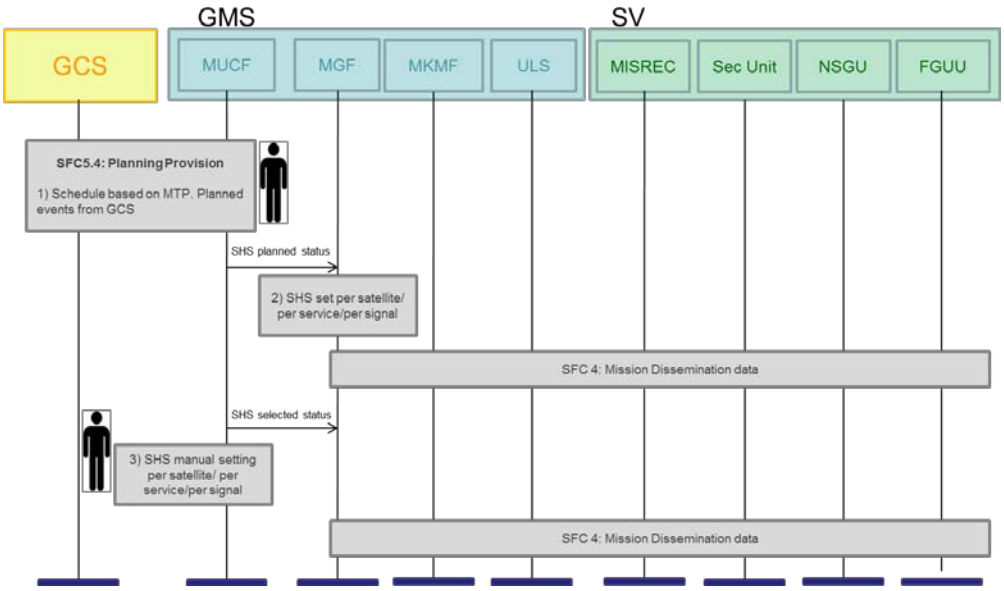
SHS uplink: as soon as the SHS configuration is applied, the GMS dissemination chain will uplink the new SHS.

Moreover the operator will be able to overwrite the SHS set automatically during a planned even.

It is noticed that what above reported is still TBC as no input from the Segments are currently available since the requirement introduced by STRB 1.5.1 are not yet applicable to them

Scenario #SCF4.1.3.2-1-1: SHS provision

The following dynamic scenario explains the SHS provision evolution according to the functionality introduced by STRB 1.5.1



1) GMS and GCS provide, through several data exchanges, an agreed Mid-term planning. The agreed planned tasks that have impact on the mission performances are sent to the MUCF to prepare a suitable uplink schedule.
2) The SHS is automatically set according to the MTP
3) The SHS can be overwritten manually by the operator via a dedicated MMI in MGF or GACF (TCB)

It is noticed that what above reported is still TBC as no input from the Segments are currently available since the requirement introduced by STRB 1.5.1 are not yet applicable to them.

6.4.1.3.2.2 Data Validity Status (DVS)

6.4.1.3.2.2.1 Overall physical Description

6.4.1.3.2.2.1.1 Physical Static View

This section describes the physical mapping of the DVS flag provision. The figure below summarises the physical mapping of the SFC4.1.3.2 for the Data Validity Status flag (DVS):

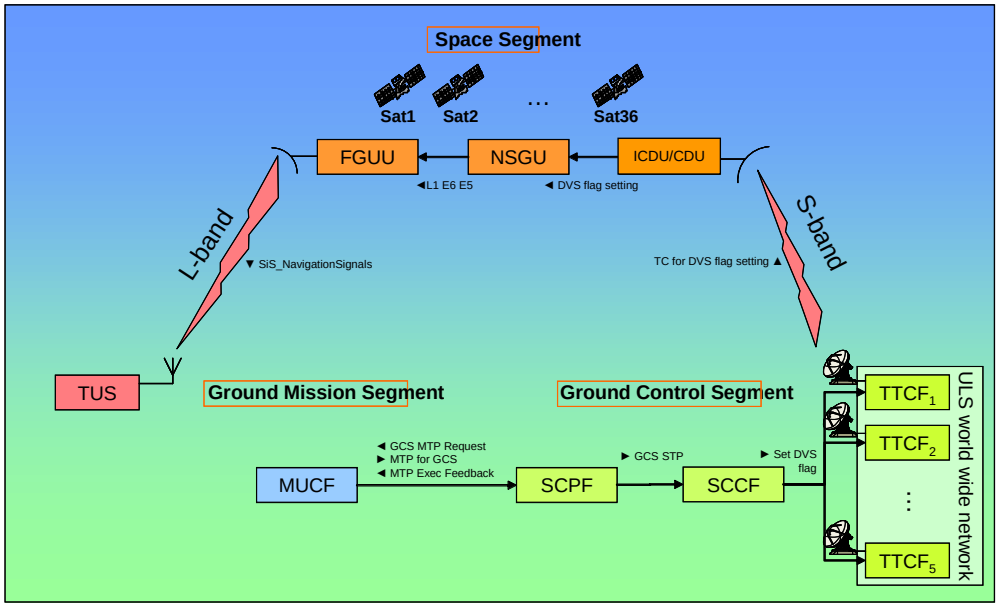


Figure 160 : SFC4.1.3.2 (DVS) Physical Static View

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.1.3.2.2.1.2 Data needed to support OS Positioning Service and mapping to physical elements

The table indicates the role of the GMS and SSgt physical element in the provision of the Data Validity Status flag. The role of each element is detailed in next dedicated sections.

	GMS	GCS			SSeg			
	MUCF	SCPF	SCCF	TTCF	S-band Antenna / Receiver	ICDU / CDU	NSGU	from FGUU to NAVANT
Data Validity Status (DVS)	Plan generation	Plan generation	DVS constraint setting	S-band message uplink	TC reception	TC process and DVS generation	L-band message generation	L-band signal generation and Tx.

6.4.1.3.2.2.2 OS Data Validity Status flag Provision physical Detailed Description

6.4.1.3.2.2.2.1 Mission planning: MUCF, SCPF

The MUCF is responsible for the Long/Mid-Term Planning Inputs Collection, in particular:

- The MUCF collects overall status and configuration information on the Galileo system from the GACF and orbits events from the GCS.
- The MUCF collects planning requests from GMS planning operators.
- The MUCF collects planning requests from the GCS-SCPF.

A Galileo operator will set the DVS flag by a dedicated TC when any critical maintenance on a satellite results in the satellite no longer being reliable for SOL (e.g. manoeuvre, clock switch, OBSW update…). These maintenances are planned in advance in the frame of the mid-term plan.

The planning preparation and execution details are detailed in the SFC5.4 on Planning provision.

6.4.1.3.2.2.2.2 DVS setting: ICDU (IOV satellites), CDU (FOC satellites)

In IOV satellites the ICDU is the onboard element responsible for setting the DVS to “Working without guarantee” in the following conditions:

- Navigation data is older than a predefined time. This predefined time can range from 100 minutes to 10 orbits.
- FDIR event occurred impacting the navigation performances.
- Satellite is in ISM (Intermediate Safe Mode)
- When instructed by an on-ground TC as described in the previous section.

The same functionality is carried out by CDU in FOC satellites.
The DVS is sent to the NSGU for it insertion in the navigation messages.

6.4.1.3.2.2.2.3 Navigation Message generation: NSGU

The nominally the NSGU includes the value of the DVS received from the ICDU/CDU in the navigation messages.
Only when the link between ICDU/CDU and NSGU is lost due to an internal anomaly for more than 2 seconds it is automatically set to “Working without guarantee” until the link is restablished.

6.4.1.3.2.2.2.4 Data Validity Status Parameters User Process

Health status parameters are used to exclude measurements from the PVT solution

6.4.1.3.2.2.3 Data Validity Status Parameters Provision Dynamic Scenarios

6.4.1.3.2.2.3.1 Scenario #SFC4.1.3.2-2: Data Validity Status provision

6.4.1.3.2.2.3.1.1 General Description

This scenario (#SFC4.1.3.2-2) describes the generation and dissemination to the users of the OS Data Validity Status (DVS) flag. The OS DVS flag is transmitted in all OS navigation signals every 15s to indicate to the user the “mode” of the satellite according to the following definitions: ‘Navigation Data valid’ or ‘Working without guarantee’.

6.4.1.3.2.2.3.1.2 Flow chart

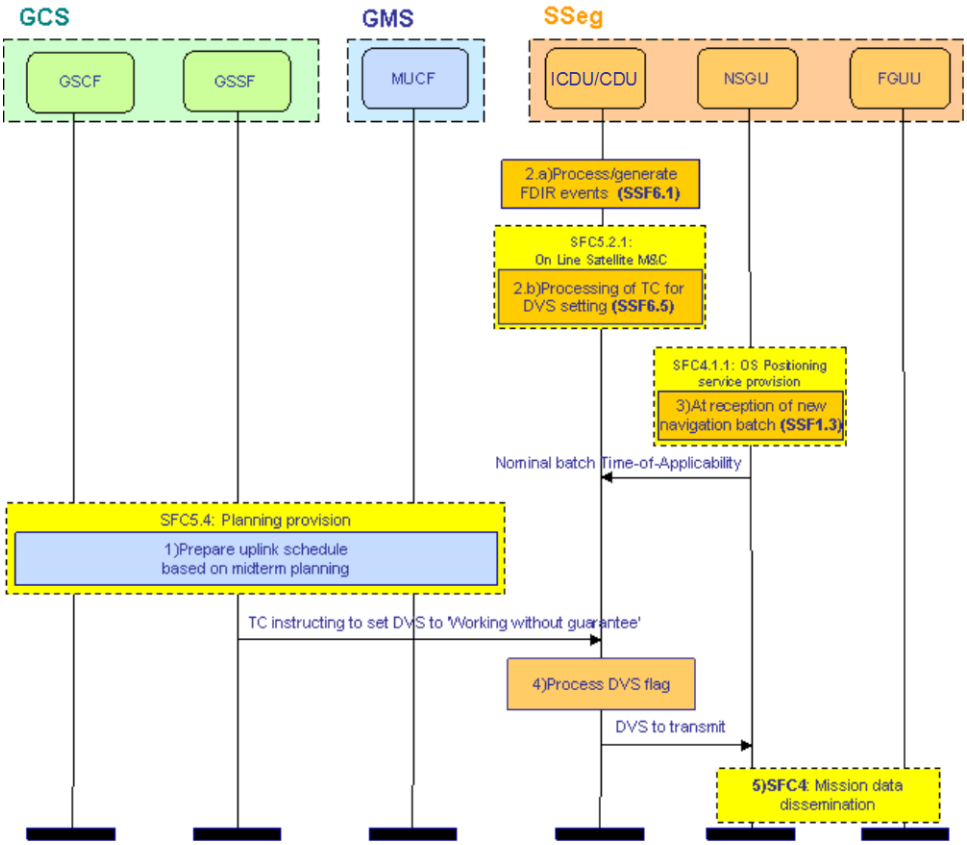


Figure 161 : #SFC4.1.3.2-2 Data Validity Status Provision

6.4.1.3.2.2.3.1.3 Scenario sequence description

1) GMS and GCS provide, through several data exchanges, an agreed Midterm planning. The planned tasks that have an impact on the system Mission performances are preceded by the setting by a TC of the DVS flag to 'Working without guarantee' set by an operator. The reseting of the DVS flag can only be managed by an operator decision through a new TC.

2) The ICDU/CDU automatically sets on-board the spacecraft through automated Failure Detection, Isolation and Recovery (FDIR) functionality. The DVS flag is nominally set to "Navigation Data Valid" but is automatically* updated to "Working Without Guarantee" by the on-board FDIR under the following conditions:
   - When the S-Band TT&C link has been lost for longer that a pre-set period (configurable between 1 and ten orbits).
   - When specific events are detected on board which impact navigation service performance.
   - Due to a spacecraft mode change (either in response to on-board FDIR action or due to telecommand from ground) from normal mode to any other spacecraft mode

   Once the DVS flag is automatically set to "Working without guarantee" by any of the above conditions the resetting to "Navigation Data valid" can only be done by telecommand from ground by operator decision.

note: the mentioned feature is available for IOV whilst it is not available for OHB (GS-OSW-OHB-RFW-0002) and will be implemented by a future version of ASW.

3) At each reception at the NSGU of new navigation data batches, the Time-of-Applicability (ToA) of the nominal batch is sent to the ICDU/CDU for the setting of a 'navigation timer'. Based on that, the ICDU/CDU will set the DVS flag to 'Working without guarantee' once a predefined time is elapsed with respect to that ToA. The predefined elapsed time to change the DVS flag status can range from 100 minutes to 10 orbits. This time is determined by a TC.
4) The ICDU/CDU computes the correct DVS flag depending on the inputs provided by the above steps 1), 2) and 3) and sends it to the NSGU to be included in the navigation message.

Contingency scenario #SFC4.1.3.2-1/4: When the link between ICDU/CDU and NSGU is lost due to an internal anomaly for more than 2 seconds it is automatically set to "Working without guarantee" until the link is established.

5) The dissemination process until the user reception is equivalent to the OS navigation data (#SFC4.1.1-1).

6.4.1.4 SFC4.1.4 OS Degraded Navigation Data Provision

6.4.1.4.1 Overall Physical Description

This section describes the mapping of the functions involved in the provision of OS Navigation Data via S-band for use in degraded mode conditions to the physical architecture.
The purpose of this system functional chain (SFC) is to support the graceful degradation of the OS navigation service. In support of this aim, OS navigation data and OS almanacs are uplinked to each spacecraft so that this data is available for use when more up-to-date data is not available on the satellite (ie via the nominal C-band uplink).
The generation of the navigation data for uplink via S-band follows the same scheme as the nominal data. The key difference is that the MGF transmits the 8 batches to the GCS for uplink (not 4 as for the mission uplink).

6.4.1.4.1.1 Physical Static View

The mapping of this SFC to the physical architecture is illustrated in the figure below.

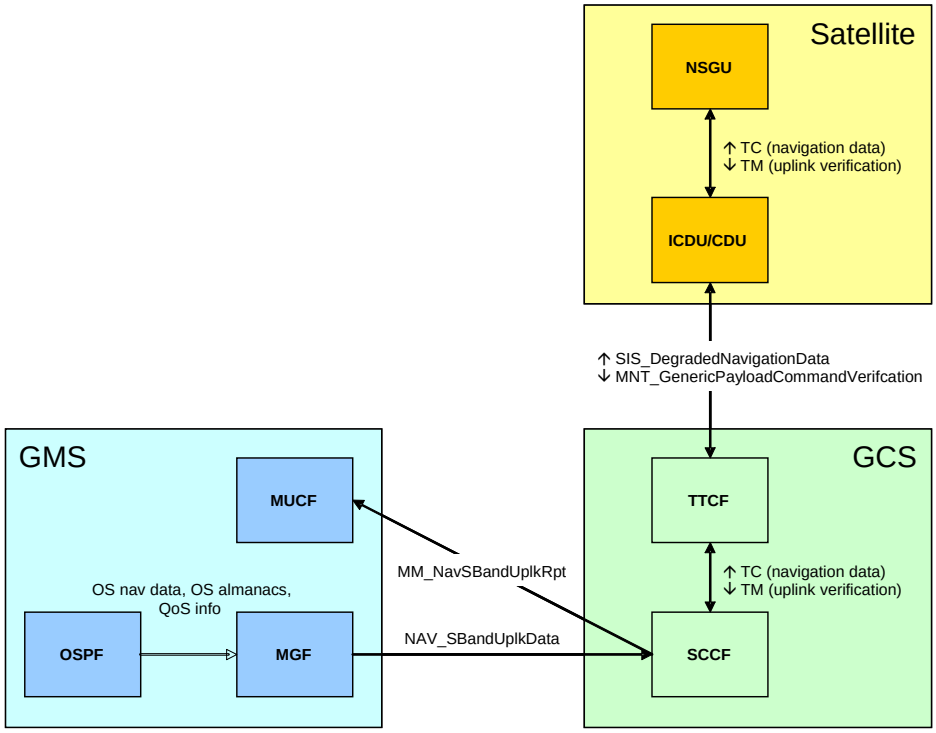


Figure 162 : SFC4.1.4 OS Degraded Navigation Data Provision

6.4.1.4.1.2 Data needed to support OS S-band Positioning Service and mapping to physical elements

The table indicates the role of the GMS, GCS and SSgt physical elements in the provision of each type of OS service data through the S-band. The role of each element is detailed in next dedicated sections.

	GMS			GCS	SSeg		
	OSPF	MUCF	MGF	SCCF	ICDU/CDU	NSGU	from FGUU to NAVANT
Ephemeris (see SFC4.1.1 for generation details)	Data generation	S-band link report reception	S-band message generation	TC and GMS reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
Almanacs (see SFC4.1.1 for generation details)	Data generation	S-band link report reception	S-band message generation	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
Clock correction (see SFC4.1.1 for generation details)	Data generation	S-band link report reception	S-band message generation	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
BGD (see SFC4.1.1 for generation details)	Data generation	S-band link report reception	S-band message generation	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
Ionospheric parameters (see SFC4.1.1 for generation details)	Data generation	S-band link report reception	S-band message generation	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
GGTO (see SFC2.2 for generation details)	Data formatting	S-band link report reception	S-band message generation	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
UTC-GST time offset (see SFC4.1.2 for generation details)	Data formatting from PTF	S-band link report reception	S-band message generation	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
SISA (see SFC4.1.3.1 for generation details)	Data generation	S-band link report reception	S-band message generation	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
SHS flag (see SFC4.1.3.2 for details)	-	S-band link report reception	S-band message generation	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.

Table 76: SFC4.1.4 Data and mapping to physical elements

Note: The following elements are not considered part of this SFC (even if provide support):

- The GSS observables data is generated and sent to the OSPF. See SFC4.1.1 for details.
- The MDDN (Mission Data Dissemination Network) is a GMS element that connects the GMS elements outside the GCC to the GMS elements inside the GCC. In the frame of this SFC the MDDN connects the GSSs to the OSPFs and the MGFs to the ULSs. See SFC4.1.1.
- The SDDN is the GCS element that connects GCS elements outside the GCC to the GCS elements inside the GCC.
- PTF (Precise Time Facility) is responsible for generating the UTC-GST time offset and provide it to the OSPF. More details can be found in the frame of the SFC2.1 and SFC4.1.2.

6.4.1.4.2 OS Degraded Navigation Data Provision Physical Detailed Description

6.4.1.4.2.1 Batches generation: OSPF

See SFC4.1.1 section on OSPF.

6.4.1.4.2.2 S-band data Message formatting: MGF

The MGF will receive once every 10 minutes 8 OS navigation data batches as described in detail on the OSPF section in SFC4.1.1 and summarized in SFC4.1.4.
The SFC4.1.1 describes how the first 4 of those batches are formatted and uplinked on the C-band link.
This SFC describes how the MGF formats all 8 batches and routes them on the S-band link.
The main tasks carried on by the MGF in the frame of this SFC are:

1. Data reception: The MGF receives periodically 8 OS navigation data batches from the OSPF.
2. Data selection: Each delivery from each OSPF includes OSPF QoS information that allows the MGF to select the best data to use (note that the MGF receives batches from each of the OSPFs in the local GCC).
3. Data collection and formatting: All navigation data batches from OSPF are formatted according to AD.8 prior to sending to the SCCF in the local GCC. This includes as well:
   a. GST-UTC time offset (see SFC4.1.2 for details)
   b. SISA values associated to each batch (see SFC4.1.3.1 for details)
   c. Signal Health Status (see SFC4.1.3.2 for details).

   This is done for every satellite.
4. Data routing: Nominally the MGF receives new batches from the OSPF every 10 minutes. As defined in AD.8, however the onward message delivery rate to SCCF is configurable between 10 minutes and 10 hours in steps of 10 minutes. The nominal delivery rate to the SCCF is once every 3 hours. The data are sent to both SCCF as specified in the GCSGMSICD.
5. Quality-of-Service (QoS) generation: The MGF includes in each delivery sent to the SCCF a QoS parameter which is applicable to the entire OS navigation data delivery (FNAV and INAV). This parameter provides information to the SCCF to allow the sectioin of the best quality data to uplink to the satellite. This is necessary because the SCCF will receive regular deliveries from each of the MGFs in the local GCC (nominally 2 MGFs per GCC).

6.4.1.4.2.3 S-band data Uplink report reception: MUCF

The MUCF is responsible for receiving the S-band uplink report generated by the SCCF once the S-band navigation data generated by the MGF is uplinked to the satellite.

6.4.1.4.2.4 TC generation for navigation data uplinking: SCCF

The following constraints apply to the SCCF for this SFC:
- uplinking of OS data at each nominal contact with each spacecraft
- selection of data to be uplinked based on QoS parameter
- formatting of data in TC packets
- provision of uplink status reports to GMS

The SCCF is required to transmit a complete delivery of OS navigation data to each satellite in the constellation at each TT&C contact. During normal operations a TT&C contact will take place with each spacecraft once per orbit (nominally every 14 hours).
The SCCF will receive regular deliveries of navigation data from each MGF in the local GCC (nominally one delivery from each of the MGFs in the local GCC every 10 minutes up to 3 hours). In order to identify which of these deliveries should be uplinked to the satellite the SCCF will use the QoS information provided by the MGF with the constraint that all of the data uplinked must originate from a single MGF delivery.
Following each S-band uplink, the SCCF will provide an uplink report to the GMS (MUCF) to report the status of the entire data uplink. This report will be provided to both MUCFs in both GCSs simultaneously to allow a synchronized view of status to be maintained. The report itself will be generated on the basis of telemetry provided by the NSGU.

6.4.1.4.2.5 TC processing and TM generation: ICDU (IOV satellites), CDU (FOC satellites)

In IOV satellites the ICDU will receive the TC and will generate the TM messages related to the reception of the S-band navigation data by the NSGU.
The contents of the TC related to the navigation data will be sent to the NSGU for storage and dissemination when needed.
In FOC satellites the same functionalities are covered by CDU.

6.4.1.4.2.6 L-band Message generation: NSGU

The NSGU will store the OS data received via S-band ready for use in degraded conditions. The S-band data will only be used once all available C-Band batches have already been downlinked and no new data has been received via the C-band link. Once the last available C-band batch has been downlinked, the NSGU will select the S-band batch to be transmitted on the basis of the ToA. S-Band batches with a ToA prior to that of the last C-Band batch will not be transmitted.
The NSGU will provide confirmation to the ground via telemetry that the telecommands carrying the OS data have been received and processed.

6.4.1.4.2.7 Carrier generation, modulation and trasmission: from FGUU to Users

The FGUU element is responsible for:
- Generate all frequencies required for upconversion of navigation signals in E5, E6 and E1 channels
- Upconvert navigation signals to L-band downlink frequencies

The FGUU output is sent to the L-band antenna for broadcasting.

6.4.1.4.2.8 User data Processing

The user is not aware from which link (C-band or S-band) the data has been uplinked to the NSGU. The data will be processed by the users as described in the SFC4.1.1.

6.4.1.4.3 OS Degraded Navigation Data Provision Dynamic Scenarios

6.4.1.4.3.1 Scenario #SFC4.1.4-1: Uplink of OS Data via S-band

6.4.1.4.3.1.1 General Description

This MSC (#SFC4.1.4-1) describes the generation and dissemination up to the users of the OS data (ephemeris, almanacs, SISA, Signal Health Status, Data Validity Status and time and clock corrections) through the S-band (OS S-band data). This data is available onboard to support a degraded mode of operation for the OS when no more up-to-date navigation data is available on the satellite (i.e. via the nominal C-band uplink).
This navigation data is used to correct pseudorange measurements (clock corrections) and compute the user position (ephemeris and almanacs). The OS navigation data is computed to be transmitted on the E5a (for FNAV messages), E5b and L1B (for INAV messages) data channels.

6.4.1.4.3.1.2 Flow chart

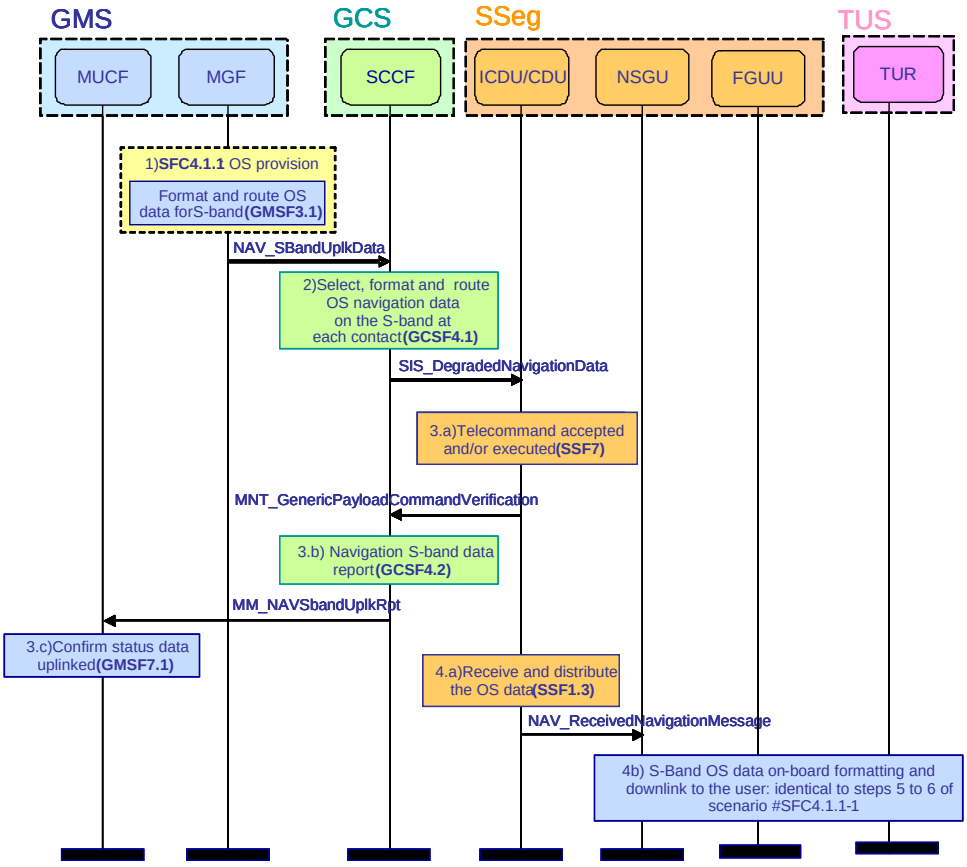


Figure 163: #SFC4.1.4-1 OS Degraded Navigation Data Provision

6.4.1.4.3.1.3 Scenario sequence description

1) Generation of OS S-band data
   The OS data that is uplinked on the S-band is generated following the same process described for the SFC4.1.1. The OS data that will be uplinked on the S-band link provides the following services:
   - OS positioning service: Ephemeris, almanacs, clock and time corrections (ionospheric and BGD) and GGTO. This is described in SFC4.1.1.
   - OS timing service: UTC-GST offset. This is described in SFC4.1.2.
   - OS Signal Quality Monitoring Service. This is described in the SFC4.1.3.

   All this data is generated for both FNAV and INAV kind of messages and for each satellite in orbit once every 10 minutes. It is divided in 8 different batches of data each of them a validity of 4 hours. It is noted that through the C-band link only the first 4 batches are used (see SFC4.1.1 for details).
2) OS S-band data transmission
   The MGF (including the redundant MGFs) send the OS data to the SCCF (GCS) with a Quality-of-Service indicator (QoS). The parameter helps the SCCF to select the best data that must be uplinked. The data is sent on a text file format.
   The latest OS data received is uplinked at each S-band link contact for each satellite through the TTCF. To be uplinked the data is reformatted in several telecommands (TC) and then processed by the ICDU/CDU.
3) OS S-band data reception reporting
   3.a) The ICDU/CDU reports on ground the reception and execution of the TC containing OS S-band data.
   3.b) The report is processed by the SCCF (GCS) and forwarded towards the GMS.
   3.c) The MUCF is the GMS element responsible for keeping the OS S-band data reports.
4) OS S-band data dissemination
   4.a) The TCs contents are transmitted to the NSGU for storage in memory.
   4.b) The OS S-band batches of data are disseminated sequentially according to their Time-of-Applicability (ToA). They are only used when all stored C-band navigation data have been downlinked and that no updated navigation batch has been received via C-Band.

The OS S-band data formatting and dissemination on board follow exactly the same sequence as the one described for OS C-Band data from step 5 to 6 in the scenario #SFC4.1.1-1 (Positioning data provision): refer to this scenario for more details.

6.4.2 SFC4.3 Public Regulated Service (PRS) Provision

The purpose of this section is to describe the physical implementation of the functionalities needed to provide the PRS Services to the users. This includes:

- PRS Positioning (SFC4.3.1)
- PRS Timing (SFC4.3.2)
- PRS Signal Quality Monitoring Provision (SFC4.3.3)
- PRS Degraded Navigation Data Provision (SFC4.3.4)
- PRS Security (SFC4.3.6)

The following section describes the physical implementation of the PRS service provision functionalities, based on the following documentation:

- ☐ System documents:
  - o TUSREQ (AD0.2)
  - o GMSREQ (AD.03)
  - o IOV SSREQ (AD.49)
  - o FOC SSREQ (AD.04)
  - o GCSREQ (AD.05)
  - o SISICD (AD.06)
  - o GMSSS ICD (AD.07)
  - o GCSGMS ICD (AD.08)
  - o FOC GCSSS ICD (AD.09)
- ☐ GMS documents:
  - o GMS Design Definition File (DDF) Part II: Organical Architecture (GMS-RD.2)
  - o MUCF DDF (GMS-RD.4)
  - o MUCF ERD (GMS-RD.10)
  - o OSPF ERD (GMS-RD.18)
  - o OSPF ICD (GMS-RD.19)
  - o PTF ERD (GMS-RD.24)
  - o MSF ERD (GMS-RD.25)
  - o MDDN ERD (GMS-RD.26)
  - o GSS ICD (GMS-RD.32)
  - o MGF ERD (GMS-RD.38)
  - o GSS ERD (GMS-RD.44)
  - o ULS ERD (GMS-RD.49)
  - o PKMF ERD (GMS-RD.57)
- ☐ SS documents
  - o Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1)
  - o Payload DDF (IOVSS-RD.3 and FOCSS-RD.02)
  - o FDIR Analysis (FOCSS-RD.03)
- ☐ GCS documents:
  - o SCCF DDF (GCS-RD.9)
- ☐ TUS documents:
  - o System Segment Design Document (TUS-RD.1)
  - o Annex to TUS System Design Justification File (TUS-RD.3)
  - o TUS System Segment Specification Document (TUS-RD.4)

6.4.2.1 SFC4.3.1 PRS Positioning Service Provision

6.4.2.1.1 Overall physical Description

This SFC is in charge of providing all the means and associated processes needed by the end user to compute a positioning solution, namely:

- PRS ranging signals (E6A and E1A) to perform pseudorange measurements. The pseudo-range is the measured distance between the receiver and a satellite used in the navigation algorithm.
- Navigation data (ephemeris, almanacs, time and clock corrections). The navigation data to correct pseudorange measurements (clock corrections) and compute the user position (ephemeris and almanacs). The PRS navigation data is transmitted on the E6A and E1A data channels using the GNAV message logical structure. The mapping of the navigation data onto message logical structure, carriers and services is described in the following figure.

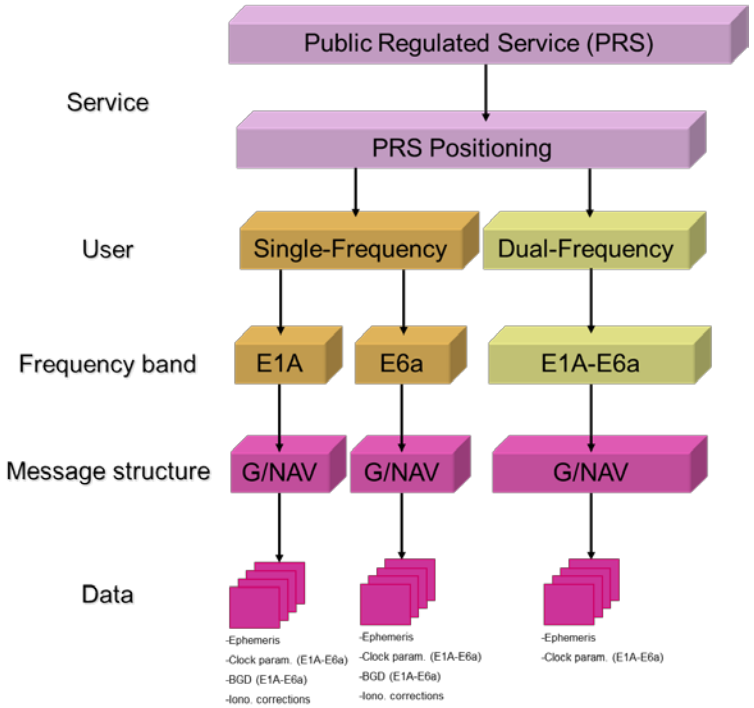


Figure 164: Navigation data mapping on logical channels, carriers and service

6.4.2.1.1.1 Physical Static View

This section describes the physical mapping of the PRS Positioning Service provision functions. The figure below summarises the physical mapping of the SFC4.3.1:

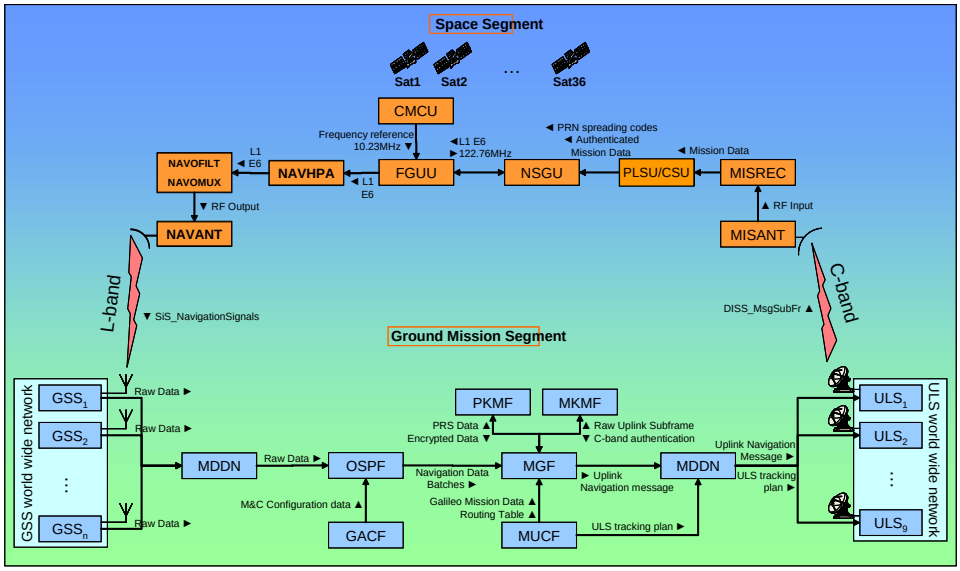


Figure 165 : Overall physical description SFC4.3.1

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.2.1.1.2 Data needed to support PRS Positioning Service and mapping to physical elements

The PRS data includes all the information needed by the users to perform positioning, meaning ephemeris and clock corrections (including ionospheric corrections and BGD needed by single frequency users), as well as some complementary data to inform about the Signal quality for each satellite like Signal-in-Space Accuracy (SISA) or Signal Health status (SHS) and Data Validity status (DVS).

Physically, the following segments are involved in the PRS Service provision:

- GMS: responsible for generating and disseminating up to the satellites the PRS service data.
- SSgt: responsible for L-Band message generation to relay the PRS service data to the users

The table indicates the role of the GMS and SSgt physical element in the provision of each type of PRS service data. The role of each element is detailed in next dedicated sections.

	GMS						SSeg			
	GSS	OSPF	MGF	PKMF	MKMF	ULS	C-Band Antenna / MISREC	PLSU/CSU	NSGU	from FGUU to NAVANT
Ephemeris	Raw data collection	Data generation	C-band message generation and dissemination	Message encryption	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
Almanacs	-	Data generation	C-band message generation and dissemination	Message encryption	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
GST parameters	-	-	-	-	-	-	-	-	Data and L-band message generation	L-band signal generation and Tx.
Clock correction	Raw data collection	Data generation	C-band message generation and dissemination	Message encryption	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
BGD	Raw data collection	Data generation	C-band message generation and dissemination	Message encryption	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
Ionospheric parameters	Raw data collection	Data generation	C-band message generation and dissemination	Message encryption	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.
GGTO	-	Data reception	C-band message generation and dissemination	Message encryption	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

Table 77 : SFC4.3.1 Data and mapping to physical elements

Note: The following elements are not considered part of this SFC (even if they provide support):

- The MDDN (Mission Data Dissemination Network) is a GMS element that connects the GMS elements outside the GCC to the GMS elements inside the GCC. In the frame of this SFC the MDDN connects the GSSs to the OSPFs and the MGFs to the ULSs.
- The GACF is a GMS element responsible to provide calibration data to the OSPF for PRS service data generation. This is addressed in SFC5.3.
- The CMCU is a payload element (SSeg) responsible to provide precise clock reference to the FGUU.
- MUCF is a GMS element responsible to generate the uplink schedule (MGF routing plan and ULS contact plan).

6.4.2.1.2 PRS Positioning Service Physical Detailed Description

6.4.2.1.2.1 Raw Data Computation: GSS

The GSS is responsible for receiving the signals and data transmitted by the satellites and to perform measurements on the signals with the same characteristics already described in the SFC4.1.1 for each PRS frequency channel (E6A and E1A).

6.4.2.1.2.2 PRS Navigation Data Generation: OSPF

6.4.2.1.2.2.1 PRS positioning data computation

The OSPF receives the Galileo SIS raw data from all GSS once per second. These observables are filtered and processed to generate:

- ☐ Ephemeris and almanacs for each of the satellites in orbit.
- ☐ Clock correction parameters for each of the satellites in orbit for the pair of frequencies E1A-E6A.
- ☐ Broadcast group delay (BGD) for each of the satellites in orbit for the pair of frequencies E1A-E6A.
- ☐ Ionospheric corrections common to all the satellites.

☐ Signal-in-Space Accuracy (SISA). See SFC4.3.3.1 for more details.

These data are provided to the MGF every 10 minutes and generated following the same process as already explained in the SFC4.1.1.

Furthermore the OSPF receives from the GACF:

☐ The GPS-Galileo time offset (GGTO). See SFC2.2 for more details.

☐ The GST-UTC time offset which is only used for timing service: see SFC4.3.2.

6.4.2.1.2.2.2 PRS positioning data dissemination: navigation batches

The data are grouped in so-called navigation batches. A navigation batch is composed of the above described navigation data excluding the almanac.

Following the same process as described in the SFC4.1.1 every 10 minutes, the OSPF produces 8 GNAV Navigation Batches + 1 GNAV Almanac Batch.

6.4.3.1.2.2.2.1 OSPF Quality-of-Service Indicator (QoS)

The OSPF also generates an associated Quality-of-Service (QoS) indicator common to all navigation batches and almanacs to help the MGF selecting the best OSPF data among all redundant OSPF elements.

6.4.2.1.2.3 C-band Navigation Message Format Generation: MGF

As part of the Navigation Dissemination chain, the MGF receives every 10 minutes the navigation batches computed by the OSPF (in addition to the QoS used to select the best OSPF among the redundant entities).

The MGF when managing the PRS navigation data has two main tasks detailed below before and after the PRS navigation data has been encrypted by the PKMF.

6.4.2.1.2.3.1 MGF tasks before PRS data encryption by PKMF

The MGF is responsible for formatting the PRS navigation data in G/NAV messages to be encrypted by the PKMF. In this process the MGF will:

- Generate a set of G/NAV messages numbered from G/NAV-1 to G/NAV-5 for each navigation batch received from the OSPF (4 first batches only for the C-band plus 4 extra batches for S-band).
- Generate 36 G/NAV messages, each called G/NAV-6, to allocate the 36 satellite almanacs provided by the OSPF.

Each of these messages (5x8+36=76 in total) is then sent to the PKMF for encryption.

6.4.2.1.2.3.2 MGF tasks after PRS data encryption by PKMF

The MGF is responsible to fill in the encrypted PRS navigation messages from PKMF within the Message sub-frame structure (C-band) in terms of packets that are uplinked to the satellite. The MGF has the capability to generate dummy packets in the Message sub-frame structure when no data was initially received from the OSPF.

In particular at this point the MGF is in charge for:

1. Formatting the C-band frames to be uplinked to the satellite in Message sub-frames where the PRS navigation messages (G/NAV) are multiplexed.
2. Request the authentication words for each C-band Message sub-frame.
3. Formatting the C-band frames dedicated to the ULS by multiplexing:
   - Time-of-Applicability (ToA) for PRS navigation messages
   - PRS navigation messages

- Message sub-frame authentication received from the MKMF.

4. Generation of a QoS indicator to allow the selection of the data generated by several MGF redundant instances by the MDDN.
5. Ensuring the routing of the authenticated Message sub-frames to the ULS antennas in accordance with the ULS to satellite allocation plan generated by the MUCF. The details for the generation of the uplink scheduling are defined in the SFC5.4.

No operator control input is needed in routine operation to manage the processes above.

6.4.2.1.2.4 Navigation Batch Encryption: PKMF

The PKMF is in charge of encrypting the batches for the G/NAV message with an additional COMSEC encryption layer based on a specific key. Each G/NAV message generated by the MGF is sent to the PKMF for encryption and sent back to the MGF once performed.
The encryption keys are changed periodically and consequently a specific process is put in place on the PKMF to guarantee that the satellite downlinks the G/NAV messages encrypted with the right key at the right time when the key changes.
The management of the encryption keys (and in particular the epoch when a key is changed, t_{KC}) is managed for the navigation data and almanacs as follows:

6.4.2.1.2.4.1 PRS Navigation messages encryption (G/NAV-1 to G/NAV-5)

At the epochs when an encryption key needs to be changed in the middle of the applicability period of a batch (in the period that ranges from ToC to ToC+24hours), implies that a batch will need to be encrypted twice:

- once with the previous key (key-1) and
- once with the new key (key-2).

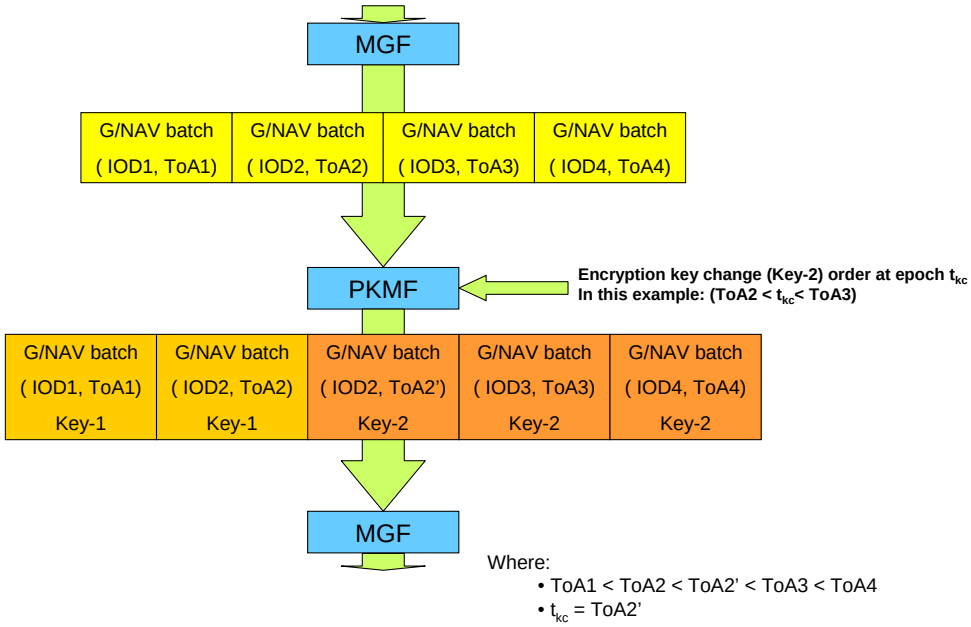
At the epoch of key change, the batch encrypted with the new key needs to be downlinked. For this, an additional Time of Applicability is introduced.
As the key change and the encryption of each batch is made by the PKMF this ToA is set by the PKMF. Note however that the batch with 2 different encryptions will still have a unique IOD.
Finally, for the PRS navigation service, 5 batches of navigation data will be uplinked and stored on board the satellite (9 batches for the data uplinked on the S-band).
In order to clarify the issue, an example is given where the key change is foreseen at ToA2 + 2hours = ToA2'.
Then:

- the G/NAV batches 1 and 2 shall be encrypted with the old key (Key-1)
- the G/NAV batches 2, 3 and 4 shall be encrypted with the new key (Key-2)

The PKMF produces 5 encrypted batches and an additional Time of Applicability: ToA2' equal to the epoch of Key change t_{KC}. Note that the validity period of the 5 batches remains 12 hours. However the period of transmission of each batch is now:

- G/NAV batch 1: between ToA1 and ToA2 (3hours): unchanged
- G/NAV batch 2: between ToA2 and ToA2' (2hours)
- G/NAV batch 2': between ToA2' and ToA3 (1 hour)
- G/NAV batch 3: between ToA3 and ToA4 (3hours): unchanged
- G/NAV batch 4: between ToA4 and ToA4+3hours (3hours): unchanged

This process is reflected in the figure below:



Note: if during the validity period of the initial 4 batches there is not key change foreseen, the PKMF will produce only 4 encrypted batches.

Note that the same process is followed for the encryption and key change of the data to be uplinked on the S-band. In that case 9 batches are uplinked instead of the 8 generated by the OSPF (see SFC4.3.4 for details on the S-band dissemination of the PRS data).

6.4.2.1.2.4.2 PRS Almanacs messages encryption (G/NAV-6)

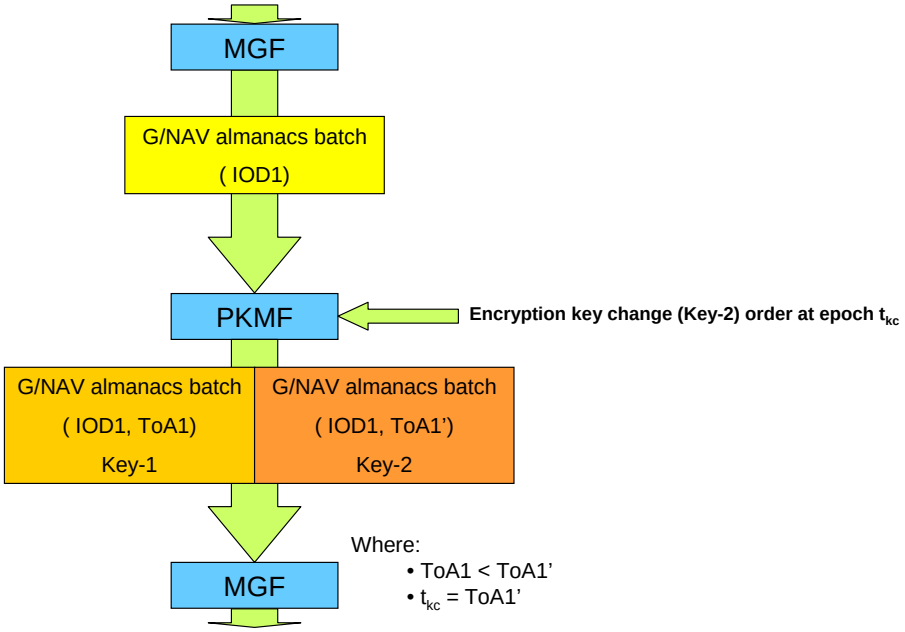
At the epochs when an encryption key needs to be changed the G/NAV almanac batch will need to be encrypted twice:

- with the previous key (key-1) and
- with the new key (key-2).

At the epoch of key change, the batch encrypted with the new key needs to be downlinked. For this, a Time of Applicability (ToA) is introduced. As the key change and the encryption of each batch is made by the PKMF this ToA is set by the PKMF. Note however that the batch with 2 different encryptions will still have a unique IOD.
Finally, for the PRS navigation service, 2 batches of almanacs data will be uplinked and stored on board the satellite.
In order to clarify the issue, an example is given where the key change is foreseen at ToA1 + 2hours = ToA1'.
The PKMF produces 2 encrypted batches and an additional Time of Applicability: ToA1' equal to the epoch of Key change t_{KC}:

- G/NAV almanac batch 1: between ToA1 and ToA1' (2hours)
- G/NAV almanac batch 2: from ToA1'

This process is reflected in the figure below:



6.4.2.1.2.5 C-Band Uplink Authentication: MKMF

The C-band uplink authentication follows the same process described in SFC4.1.1 MKMF section.

6.4.2.1.2.6 Message Uplink Schedule: MUCF

The Message uplink schedule follows the same process described in SFC4.1.1 MUCF section.

6.4.2.1.2.7 Dissemination up to uplink antenna: MDDN

The Dissemination up to uplink antenna follows the same process described in SFC4.1.1 MDDN section.

6.4.2.1.2.8 Message Uplink: ULS

The Message uplink follows the same process described in SFC4.1.1 ULS section.

6.4.2.1.2.9 Uplink Message Acquisition and Demodulation: MISREC

The message acquisition and demodulation follows the same process described in SFC4.1.1 MISREC section.

6.4.2.1.2.10 Payload security Unit: PLSU (IOV satellites), CSU (FOC satellites)

See SFC4.1.1 for authentication of C-band received data from MISREC.
In IOV satellites the PLSU is responsible for generating the PRS PRN spreading codes and transmit them to the NSGU for modulation. In FOC satellites, the same functionality is allocated to CSU.

6.4.2.1.2.11 G/NAV L-Band Navigation Message Format: NSGU

In the frame of the navigation data provision, the NSGU is responsible for handling the PRS navigation data on-board and its dissemination on the corresponding L-band signals.

The NSGU provides the following functions for the PRS navigation message dissemination management:

- Storage of Mission Data coming from PLSU/CSU and switch between the nominal or alert dissemination modes when needed.
- Generation of the PRS L-band messages
- Spectrum Spreading: This function spreads the navigation message data or pilot signal components with the relevant spreading codes from the code generator (PLSU/CSU).

The total amount of data to be managed by the NSGU in the frame of this SFC is:

- 1 set of 5 navigation data batches for G/NAV (received from C-band)
- 1 set of 9 navigation data batches for G/NAV (received from S-band)
- 2 batch of almanacs (up to 36 satellites) for G/NAV (received from C-band)
- 2 batch of almanacs (up to 36 satellites) for G/NAV (received from S-band)

The general process to generate a PRS downlinked page is detailed hereafter:

1. All page data bits shall be assembled based on the data received from ground and the data generated by the satellite as follows:

Packet received from ground	Added by the satellite		
283	Onboard generated data	PRS Region Status*	Tail
	27	8	6
	41		
324 bits			

*: The *PRS Region Status* is received from ground (C-band only).

In order to initialize the FEC encoder buffer, the satellite shall add 6 tail bits at the end set to 0.

At this stage, the page is complete at data level with a total of 324 data bits but still needs to be channel encoded.

2. The data page is encoded with a convolutional encoder (FEC) in order to detect and correct errors. From a page of 324 data bits, the output of the encoder is 2x324 symbol bits according to the following figure:

FEC symbols
2x324=648

The process to generate symbol page starting from a sequence of data bits is described in the SFC4.1.1 on the NSGU section.

3. Interleaving of the encoded page: To minimise the effects of short burst errors due to fading, blocking or interference effects interleaving will be applied to the data prior to transmission. In general such a bit error leans on a packet of bits (“burst error”) and rarely on a single bit. The use of interleaving technics enables to disperse the burst errors in different location to which the convolutional FEC coding is particularly vulnerable.

Then as the Viterbi decoding enables to correct packets of data, the combination FEC encoding/interleaving enables to be more robust to burst of errors.

The FEC symbol page is then interleaved resulting still on 648 symbols:

Interleaved FEC symbols
2x324=648

4. The resulting Interleaved FEC symbols are divided in 6 blocks with 108 symbols each.

Page(1/6)	Page(2/6)	Page(3/6)	Page(4/6)	Page(5/6)	Page(6/6)
108	108	108	108	108	108
648 symbols					

5. Add Initial field (two symbols): To each page(k/6) two symbols are included to indicate to the PRS users the nominal transmission of the PRS or alert transmission. In nominal transmission will be set both to zero.

1st block transmission		2nd block transmission		3rd block transmission		4th block transmission		5th block transmission		6th block transmission	
Initial field	Page(1/6)	Initial field	Page(2/6)	Initial field	Page(3/6)	Initial field	Page(4/6)	Initial field	Page(5/6)	Initial field	Page(6/6)
2	108	2	108	2	108	2	108	2	108	2	108

Each block is transmitted on the corresponding L-band lasting each 1 second.

6. Data batch refreshment: A set of navigation batches from C-band covers a period of 12 hours and are all generated to, once reached its ToA, replace the previous valid navigation data batches. All four batches are uplinked in several seconds partially formatted within C-band Message sub-frames (GMSSSICD) in this way it is probable that the C-band link could be lost during the reception of a new generated set of batches. To keep the consistency of the navigation data stored on board all five batches shall be made applicable together when all have been completely received without errors and belong to the same set of data.

6.4.2.1.2.12 Carrier generation, modulation and transmission: from FGUU to NAV Antenna

See SFC4.1.1 for carrier generation and modulation process.

6.4.2.1.2.13 PRS Positioning Process: USER

The TUS PRS positioning process is physically split as followed:

1. The Signals are acquired by the RF part:
   - ☐ The **antenna** and its RF front End (preamplifier) composed of the low-noise amplifiers. It also provide the broad-band filter.
   - ☐ the ***receiver RF*** composed of:
     - ☐ the RF front end (RFFE) which performs the signal splitting if needed, and filtering
     - ☐ the RF to Intermediate Frequency down-convertor (RFIF) that performs the local oscillator synthesis, the signal down conversion and complementary filtering. Two RFIFs are present in TUS: one for E6/E1 channels, and one for E5a/E5b.
2. The **core module** (CM) performs the digital signal processing and is mainly in charge of performing pseudo-range measurements on the SIS ranging codes and pseudo-phase measurements on the SIS carrier. 16 parallel channels allows to process all the satellite in view. The main functions performed are (TUSF1):
   - ☐ Carrier Tracking
   - ☐ Carrier phase measurement

- ☐ Ranging code tracking
- ☐ Ranging code measurement
- ☐ Multipath mitigation
- ☐ Interference mitigation
- ☐ Cycle-slip detection and correction
- ☐ Signal monitoring

The core module is also responsible of the recovering of the navigation data transmitted by the Navigation Signals (satellite parameters,
integrity flags, SISA, …).
To carried out this function the following tasks are needed (TUSF2):

- ☐ Bit synchronisation
- ☐ Frame synchronisation
- ☐ Frame Demultiplexing
- ☐ Symbol recovery
- ☐ CRC check
- ☐ Viterbi decoding
- ☐ De-interleaving
- ☐ Bit recovery

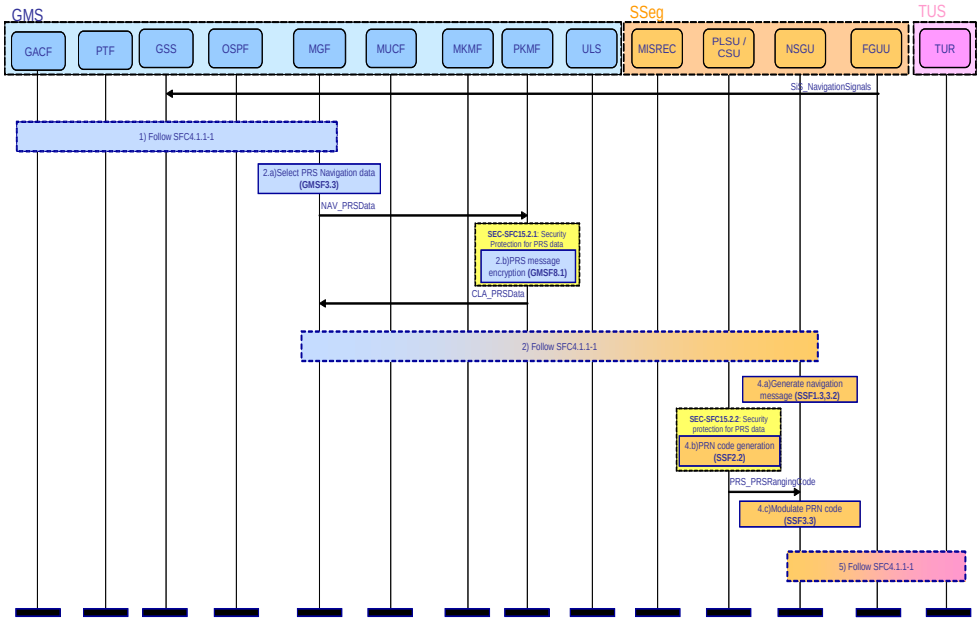
3. The **Crypto Module** processes the NAVSEC function and a crypto software module in the Application Unit (PC) to manage the keys and processes the COMSEC function. (classified).
4. The **Application Unit** (AU) processes all received pseudoranges measurements (from CM) and navigation data recovered from the message to compute the receiver position. The main functions performed are:

- ☐ Raw Data Acquisition and Pre-processing (TUSF5, TUSF6.1)
- ☐ Navigation Message decoding
- ☐ Satellite Strategy Acquisition (TUSF6.4)
- ☐ PVT solution (TUSF4)

6.4.2.1.3 PRS Positioning Service Provision Dynamic Scenarios

6.4.2.1.3.1 SSFC4.3.1 Scenario - Ephemeris, almanacs and Clock correction provision

This scenario (#SFC4.3.1-1) describes the generation and dissemination up to the users of the PRS navigation data (ephemeris, almanacs, time and clock corrections). This navigation data is used to correct pseudorange measurements (clock corrections) and compute the user position (ephemeris and almanacs). The PRS navigation data is transmitted on the E6A and E1A (G/NAV messages) data channels.
The provision of the pseudoranging codes is intrinsic to the navigation data dissemination and modulation within the L-band.



1) Process is as described on scenario SFC4.1.1-1 until MGF element.
2) PRS navigation message format and uplink.

2.a) The MGF receive every 10 minutes the data from all operational OSPF available and selects the one with the highest QoS.

Contingency scenario (#SFC4.3.1-1/4.a): In the case the MGF do not receive valid data from the OSPF after 10 minutes from the last reception, the current PRS navigation data is considered not valid and C-band dummy packets within the Message sub-frame are uplinked. This data will be discarded by the satellite on the step 5.b below.

2.b) All the PRS data uplinked to the satellite is encrypted. After the encryption the number of PRS navigation data batches to be uplinked can be increased from 4 up to 5 to indicate the satellite about encryption key changes.

See SEC-SFC15.2.1 for more details about the PRS data encryption.

3) Process is as described on scenario #SFC4.1.1-1.
4) Onboard data processing and dissemination.

4.a) Once the complete data is received the NSGU checks the Time-of-Applicability (ToA) of the navigation data.

It generates the data to be transmitted on the L-band. This process includes the generation of some other data to be added in the L-band navigation message.

4.b) The PRS PRN codes are encrypted and generated by the PLSU/CSU. The codes are passed to the NSGU for the modulation.

See SEC-SFC15.2.2 for more details about the PRS PRN codes generation.

4.c) The PRN codes are modulated with the PRS navigation messages.

Contingency scenario (#SFC4.3.1-1/4.c): In the case the NSGU does not receive from the PLSU/CSU the PRN code or receives a default PRN code, the NSGU switch to silence mode a do not transmit the PRS signal.

5) Process is as described on scenario #SFC4.1.1-1.

6.4.2.2 SFC4.3.2 PRS Timing Service Provision

6.4.2.2.1 Overall Physical Description

The GST-UTC conversion parameters are handled by the GACF within the GMS, based on information provided by an external entity (Time Service Provider). These parameters are disseminated together with the Navigation data.

6.4.2.2.1.1 Physical Static View

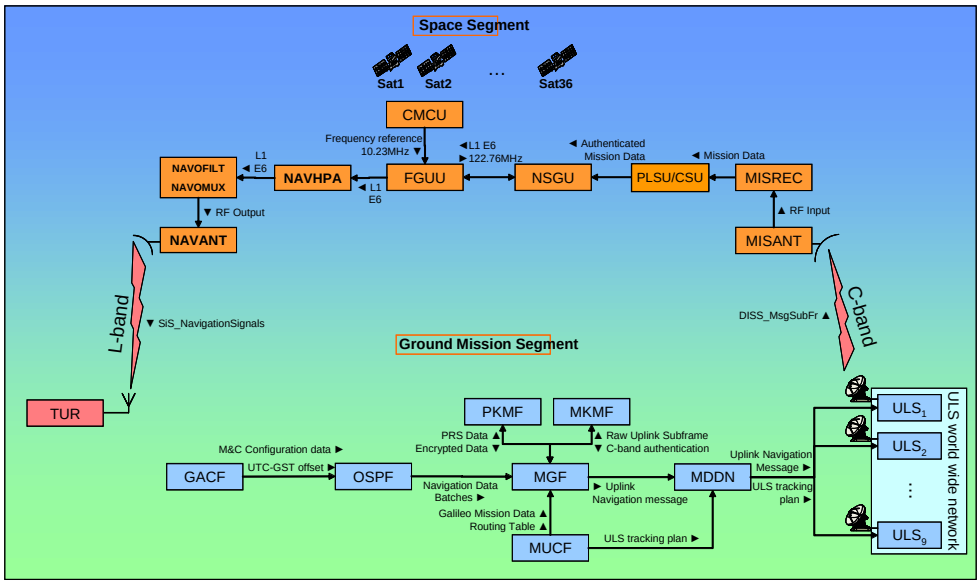


Figure 166: Overall physical description SFC4.3.2

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.2.2.1.2 Data needed to support PRS Timing Service and Mapping to the Physical Elements

The table indicates the role of the GMS and SSgt physical element in the provision of each type of PRS service data. The role of each element is detailed in next dedicated sections.

	GMS					SSeg			
	GACF	OSPF	MGF	PKMF/ MKMF	ULS	C-Band Antenna / MISREC	PLSU / CSU	NSGU	from FGUU to NAVANT
			Dissemination to users (Common to the SFC4.3.1)						
GST-UTC offset	GST-UTC Data Collection	Data formatting	C-band message generation and dissemination	PRS encryption C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

Table 78: SFC4.3.2 Data and mapping to physical elements

The data needed to support the PRS Timing Service are the following:

- Navigation Protocol parameters: synchronisation field to perform signal acquisition
- Clock Corrections
- Timing Parameters:
  - o GST
  - o GST-UTC conversion parameters

6.4.2.2.2 PRS Timing Service Physical Detailed description

6.4.2.2.2.1 Data formatting: OSPF Element

See SFC4.1.2.

6.4.2.2.2.2 Timing parameters Dissemination: from MGF to USERS

The GST-UTC offset parameters are disseminated together with the PRS positioning service parameters described in the SFC4.3.1. It is inserted within each navigation data batch generated for both C-band and S-band links.

6.4.2.2.2.3 PRS Timing User Process

The TUS PRS timing process is physically split as the TUS PRS positioning process plus an extra function (TUSF7) within the Application Unit (AP).
The PRS user receivers will compute GST with high precision and process the GST-UTC offset parameters according to the formula described in the SISICD and TUSREQ.
The GST-UTC offset update is not associated to a change on the IODnav transmitted within the navigation data batches and therefore the user receivers must check any variation on the offset on each message reception.

6.4.2.2.3 SFC4.3.2 Scenario 1: GST-UTC time offset provision

6.4.2.2.3.1 SFC4.3.2 Scenario - Nominal

This MSC (#SFC4.3.2-1) describes the dissemination up to the PRS users of the GST-UTC offset.

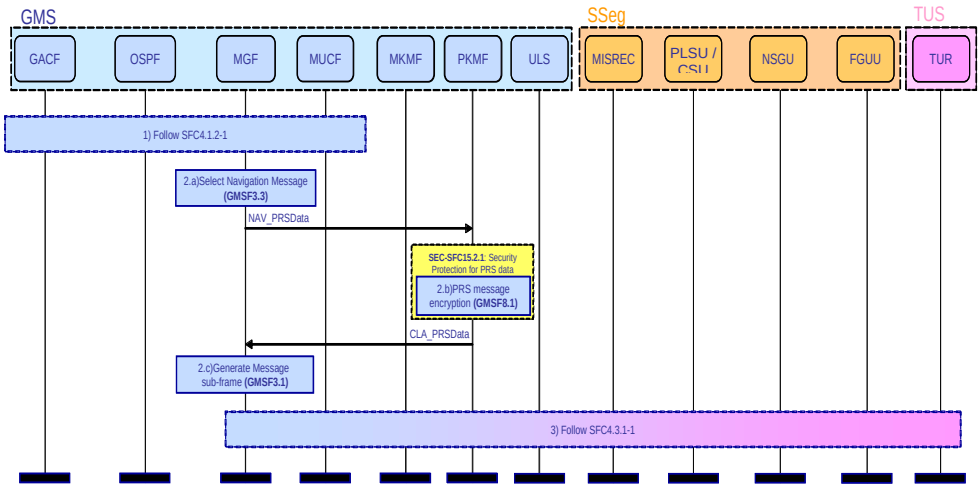


Figure 167 : SFC4.3.2 Dynamic Scenario 1

1) Follow scenario 4.1.2-1.
2) PRS navigation message format and uplink.
2.a) The MGF receive every 10 minutes the data from all operational OSPF available and selects the one with the highest QoS.
2.b) Each PRS navigation message is encrypted by the PKMF before inserted in the Message sub-frames and disseminated to the users.
See SEC-SFC15.2.1 for details about PRS message encryption.
2.c) Each MGF, based on the MUCF dissemination plan, format the PRS navigation data in Message sub-frames that will be uplinked on the C-band links. This process is made once every second.
3) Follow scenario 4.3.1-1.

6.4.2.3 SFC4.3.3 PRS Signal Quality Monitoring Provision

On-top of positioning and timing service information, the system provides the PRS users with the following information about the signal quality:

- Signal in Space Accuracy (SISA) parameter - see SFC4.3.3.1: The SISA is a prediction of the minimum standard deviation (1-sigma) of the unbiased Gaussian distribution, which over-bounds the Signal-In-Space Error (SISE) predictable distribution for all possible user locations. When no prediction is available the SISA is set to 255 by the GMS.

- The SHS will be used to info the user of the status of the signal in space from a given satellite for each message type (i.e. F/NAV, G/NAV or I/NAV) and signal type (E5a-I, E5b-I, E6-A, E1A, E1-B).

  The SHS can be set in two ways in the GMS:
  - o Manual setting by operator
  - o Automatic setting as results of planned events.

- Data Validity Status (DVS) flag - see SFC4.3.3.2: This flag informs the user about the whether a given satellite can be considered by the user as offering reliable performance (fault free) and whether the navigation data on board the satellite have been refreshed from ground.

6.4.2.3.1 SFC4.3.3.1 PRS SISA provision

6.4.2.3.1.1 Overall Physical description

This section describes the physical mapping of the PRS SISA provision functions.

6.4.2.3.1.1.1 Physical Static View

The figure below summarises the physical mapping of the SFC4.3.3.1:

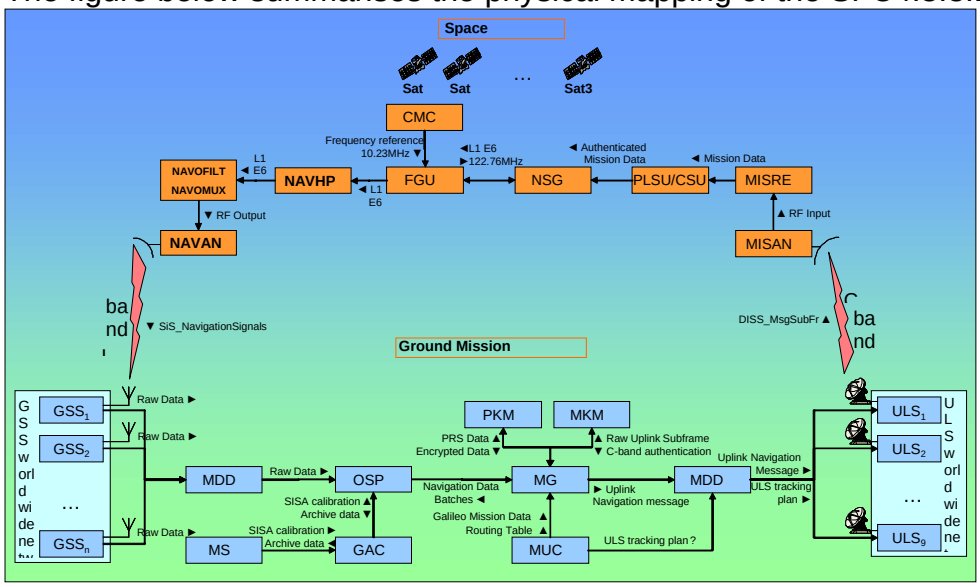


Figure 168 : Overall physical description SFC4.3.3.1

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.2.3.1.1.2 Data needed to support PRS Positioning Service and mapping to physical elements

The table indicates the role of the GMS and SSgt physical elements in the provision of SISA. The role of each element is detailed in next dedicated sections.

	GMS						SSeg			
	GSS	OSPF	MSF	MGF	PKMF /MKMF	ULS	C-Band antenna / MISREC	PLSU / CSU	NSGU	from FGUU to NAVANT
				Dissemination to users (Common to the SFC4.3.1)						
SISA	Raw data collection	Data generation	SISA calibration parameter computation	C-band message generation and dissemination	PRS encryption /C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

Table 79 : SFC4.3.3.1 Data and mapping to physical elements

Note: The following elements are not considered part of this SFC (even if provide support):

- The MDDN (Mission Data Dissemination Network) is a GMS element that connects the GMS elements outside the GCC to the GMS elements inside the GCC. In the frame of this SFC the MDDN connects the GSSs to the OSPFs and the MGFs to the ULSs.
- The GACF is a GMS element responsible to provide configuration data to the OSPF for SISA data generation. This is addressed in SFC5.3.
- The CMCU is a payload element (SSeg) responsible to provide precise clock reference to the FGUU.
- MUCF is a GMS element responsible to generate the uplink schedule (MGF routing plan and ULS contact plan).

6.4.2.3.1.2 PRS Positioning Service Physical Detailed Description

As specified in GSRD, the PRS SISA is provided in free-access on PRS Signals.

6.4.2.3.1.2.1 Raw Data Computation: GSS

Same process as for SFC4.1.1.

6.4.2.3.1.2.2 Calibration SISA parameter Computation (HSISA): MSF

Same process as for SFC4.1.3.

6.4.2.3.1.2.3 SISA Computation: OSPF

Same process as for SFC4.1.3.
Determination of SISA parameters is based on the following pairs of frequencies. They are the same as per the satellite clock correction parameters:

- ☐ E1A / E6A SISA parameter to be used by E1A-E6A dual-frequency users.

Note: the SISA can also be set to the value 255 (SISA not available) based on satellite planned events (e.g manoeuvres, clock adjustments….) as identified in the Mission Planning: see SFC5.2.

6.4.2.3.1.2.4 PRS SISA Dissemination: From MGF to Users

The dissemination schema to disseminate the SISA values is identical to the ones described in for the navigation data batches described in the SFC4.3.1 from the MGF to the user receivers.

6.4.2.3.1.2.5 PRS SISA User Process

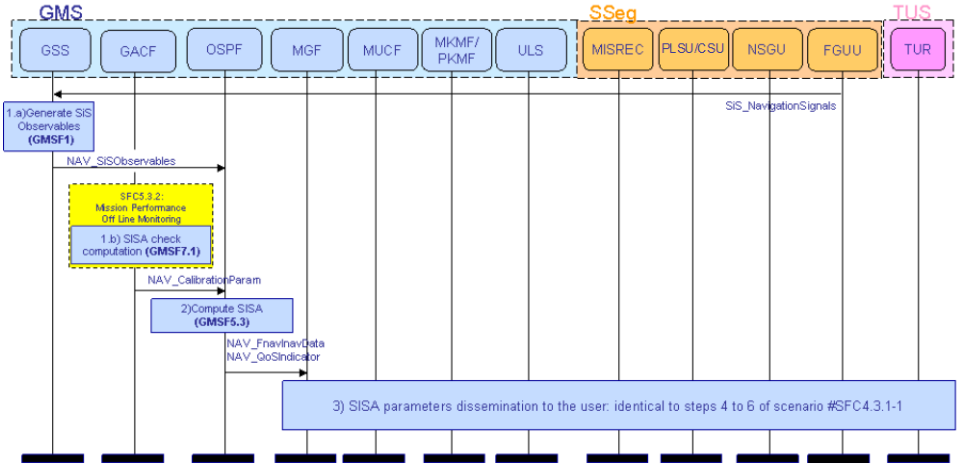
.

This parameter enables to indicate to the user the quality of the SIS broadcast by satellites and in particular the quality of the ODTS outputs.

6.4.2.3.1.3 PRS SISA Dynamic Scenario

6.4.2.3.1.3.1 SFC4.3.3.1 Scenario - Nominal

The figure below illustrates the nominal uplink scenario for PRS Signal Qualtity Monitoring provision via C-band.



1) SISA parameter input collection

The computation of the SISA parameter is based on SIS observables from the GSSs and HSISA calibration parameter from the MSF (via GACF)

1.a) Collection of raw data measurement by the GSSs and provision to the OSPF: same process as in scenario #SFC4.3.1-1.

1.b) HSISA calibration parameters computed by the off-line mission monitoring function at MSF level: see SFC5.3.2 for detail. This calibration parameter is provided to the OSPF via the GACF.

2) SISA computation

The SISA parameter depends directly on the navigation data generated by the OSPF. Therefore the SISA are generated every 10 minutes, as part of each PRS navigation data batch. One values of SISA are disseminated: E1A / E6A.

The computed SISA parameter is sent to the MGF together with their associated PRS navigation data batch for formatting and dissemination.

3) SISA parameters dissemination to the user

The SISA parameters are disseminated as part of the G/NAV navigation messages. Thus the formatting and dissemination processes from GMS to the user via the Galileo satellites follow exactly the same scenario #SFC4.3.1-1 (Positioning data provision): refer to this scenario for more details.

6.4.2.3.2 SFC4.3.3.2 PRS Health Status Parameters Provision

This section describes the implementation of the functional chains in charge of providing to the user the following status flags:

- The SHS will be used to info the user of the status of the signal in space from a given satellite for each message type (i.e. F/NAV, G/NAV or I/NAV) and signal type (E5a-I, E5b-I, E6-A, E1A, E1-B).

  The SHS can be set in two ways in the GMS:
  - o Manual setting by operator
  - o Automatic setting as results of planned events.

- Data Validity Status (DVS): This flag informs the user about the whether a given satellite can be considered by the user as offering reliable performance (fault free) and whether the navigation data on board the satellite have been recently refreshed from the ground.

6.4.2.3.2.1 Signal Health Status (SHS)

6.4.2.3.2.1.1 Overall physical Description

6.4.3.3.2.1.1.1 Physical Static View

This section describes the physical mapping of the PRS Signal Health Status provision. The figure below summarises the physical mapping of the SFC4.3.3.2 for the Signal Health Status flag (SHS):

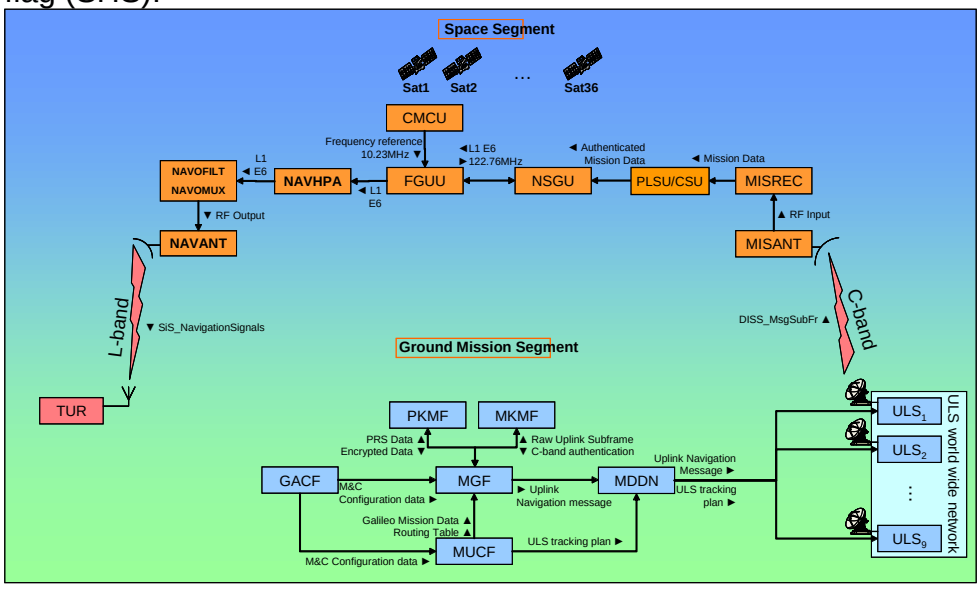


Figure 169: Overall physical description SFC4.3.3.2 (SHS)

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.3.3.2.1.1.2 Data needed to support PRS Positioning Service and mapping to physical elements

The table indicates the role of the GMS and SSgt physical element in the provision of the SHS flag. The role of each element is detailed in next dedicated sections.

	GMS				SSeg			
	MUCF GACF	MGF	MKMF	ULS	C-Band Antenna / MISREC	PLSU / CSU	NSGU	from FGUU to NAVANT
			Data dissemination (common to SFC4.3.1)					
Signal Health Status (SHS)	Plan and Schedule generation	SHS setting / C-band message generation and dissemination	C-band message protection	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

Table 80 : SFC4.3.3.2 (SHS) Data and mapping to physical elements

6.4.2.3.2.1.2 PRS Signal Health Status Parameter Provision physical Detailed Description

The SHS will be used to info the user of the status of the signal in space from a given satellite for each message type (i.e. F/NAV, G/NAV or I/NAV) and signal type (E5a-I, E5b-I, E6-A, E1A, E1-B).

The SHS can be set in two ways in the GMS:
- Manual setting by operator
- Automatic setting as results of planned events.

In order to accomplish the above reported functionality the MGF needs to be upgraded. For planned events the MGF will be upgraded to have the capability to set the SHS per satellite, per service and per signal according to the planned events as notified by the GCS.
For what concern the manual setting the MGF will be upgraded in order to allow the operators to perform the following steps:

SHS configuration: A MMI shall be provided to the operator which shall offer some specific capabilities in order to ease the manual change of the SHS.
When opening the MMI, the last configured SHS values shall be displayed. The operator may modify the current displayed configuration, and then finally save the updated configuration using a new file version, then stored in GACF database.
When modifying the SHS, the MMI shall have the following ergonomic:
- First level (the most general use case): set the SHS for a specific satellite, all message types and signals included
- Second level (less frequent): set the SHS for a specific message type of a specific satellite
- Third level (rare events): set the SHS for a specific signal of a specific satellite

SHS application: the operator shall deploy the previously generated configuration file using GACF functionality. This file can be deployed while GMS elements are in operational mode using an online command.

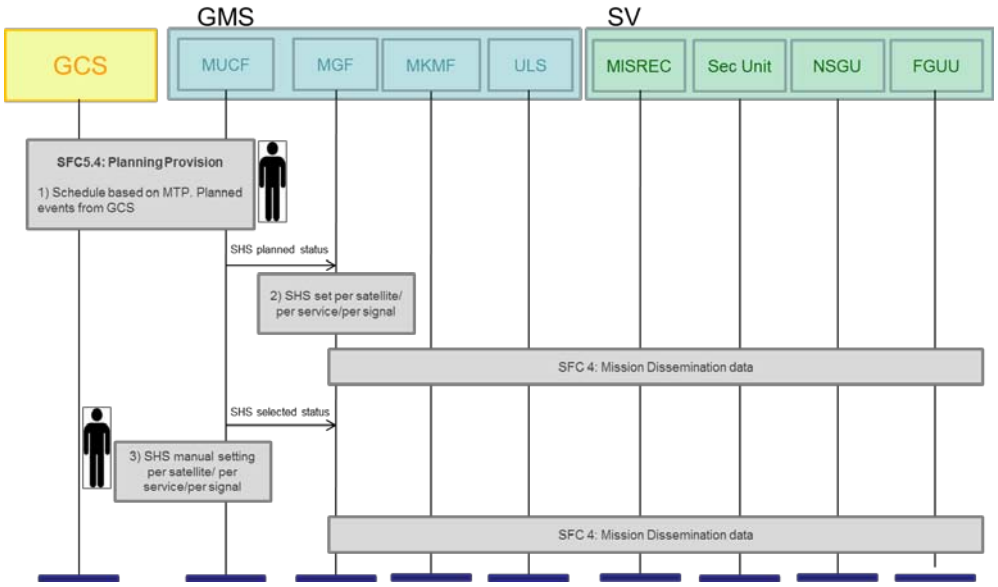
SHS uplink: as soon as the SHS configuration is applied, the GMS dissemination chain will uplink the new SHS.

Moreover the operator will be able to overwrite the SHS set automatically during a planned even.

It is noticed that what above reported is still TBC as no input from the Segments are currently available since the requirement introduced by STRB 1.5.1 are not yet applicable to them

Scenario #SCF4.1.3.2-1-1: SHS provision evolution for TRB 1.5.1

The following dynamic scenario explains the SHS provision evolution according to the functionality introduced by STRB 1.5.1



4) GMS and GCS provide, through several data exchanges, an agreed Mid-term planning. The agreed planned tasks that have impact on the mission performances are sent to the MUCF to prepare a suitable uplink schedule.
5) The SHS is automatically set according to the MTP
6) The SHS can be overwritten manually by the operator via a dedicated MMI in MGF or GACF (TCB)

It is noticed that what above reported is still TBC as no input from the Segments are currently available since the requirement introduced by STRB 1.5.1 are not yet applicable to them.

6.4.2.3.2.2 Data Validity Status (DVS)

6.4.2.3.2.2.1 Overall physical Description

6.4.3.3.2.2.1.1 Physical Static View

This section describes the physical mapping of the DVS flag provision. The figure below summarises the physical mapping of the SFC4.3.3.2 for the Data Validity Status flag (DVS):

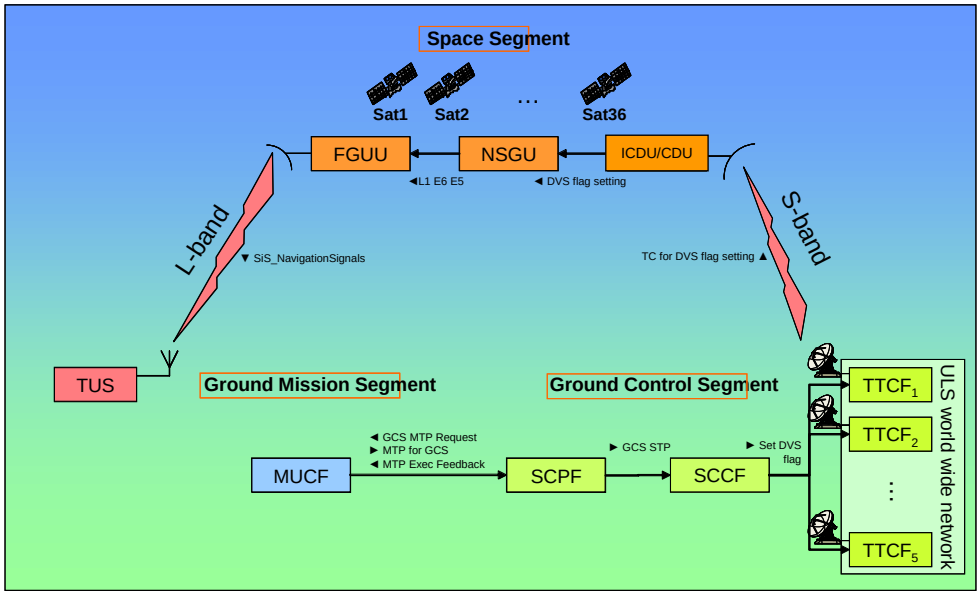


Figure 170: Overall physical description SFC4.3.3.2 (DVS)

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.3.3.2.2.1.2 Data needed to support OS Positioning Service and mapping to physical elements

The table indicates the role of the GMS and SSgt physical element in the provision of the Data Validity Status flag. The role of each element is detailed in next dedicated sections.

	GMS	GCS			SSeg			
	MUCF	SCPF	SCCF	TTCF	S-band Antenna / Receiver	ICDU / CDU	NSGU	from FGUU to NAVANT
Data Validity Status (DVS)	Plan generation	Plan generation	DVS constraint setting	S-band message uplink	TC reception	TC process and DVS generation	L-band message generation	L-band signal generation and Tx.

Table 81 : SFC4.3.3.2 (DVS) Data and mapping to physical elements

6.4.2.3.2.2.2 PRS Data Validity Status flag Provision physical Detailed Description

6.4.3.3.2.2.2.1 Mission planning: MUCF

The MUCF is responsible for the Long/Mid-Term Planning Inputs Collection, in particular:

- ☐ The MUCF collects overall status and configuration information on the Galileo system from the GACF and orbits events from the GCS.
- ☐ The MUCF collects planning requests from GMS planning operators.
- ☐ The MUCF collects planning requests from the GCS-SCPF.

A Galileo operator will set the DVS flag by a dedicated TC when any critical maintenance on a satellite results in the satellite no longer being reliable for Integrity users (e.g. manoeuvre, clock switch, OBSW update…). These maintenances are planned in advance in the frame of the mid-term plan.

The planning preparation and execution details are detailed in the SFC5.4 on Planning provision.

6.4.3.3.2.2.2.2 DVS setting: ICDU (IOV satellites), CDU (FOC satellites)

In IOV satellites the ICDU is the on-board element responsible for setting the DVS to “Working without guarantee” in the following conditions:

- ☐ Navigation data is older than a predefined time. This predefined time can range from 100 minutes to 10 orbits.
- ☐ FDIR event occurred impacting the navigation performances.
- ☐ Satellite is in ISM (Intermediate Safe Mode)
- ☐ When instructed by an on-ground TC as described in the previous section.

The same functionality is carried out by CDU in FOC satellites.
The DVS is sent to the NSGU for it insertion in the navigation messages.

6.4.3.3.2.2.2.3 Navigation Message generation: NSGU

The nominally the NSGU includes the value of the DVS received from the ICDU/CDU in the navigation messages.
Only when the link between ICDU/CDU and NSGU is lost due to an internal anomaly for more than 2 seconds it is automatically set to “Working without guarantee” until the link is re-established.

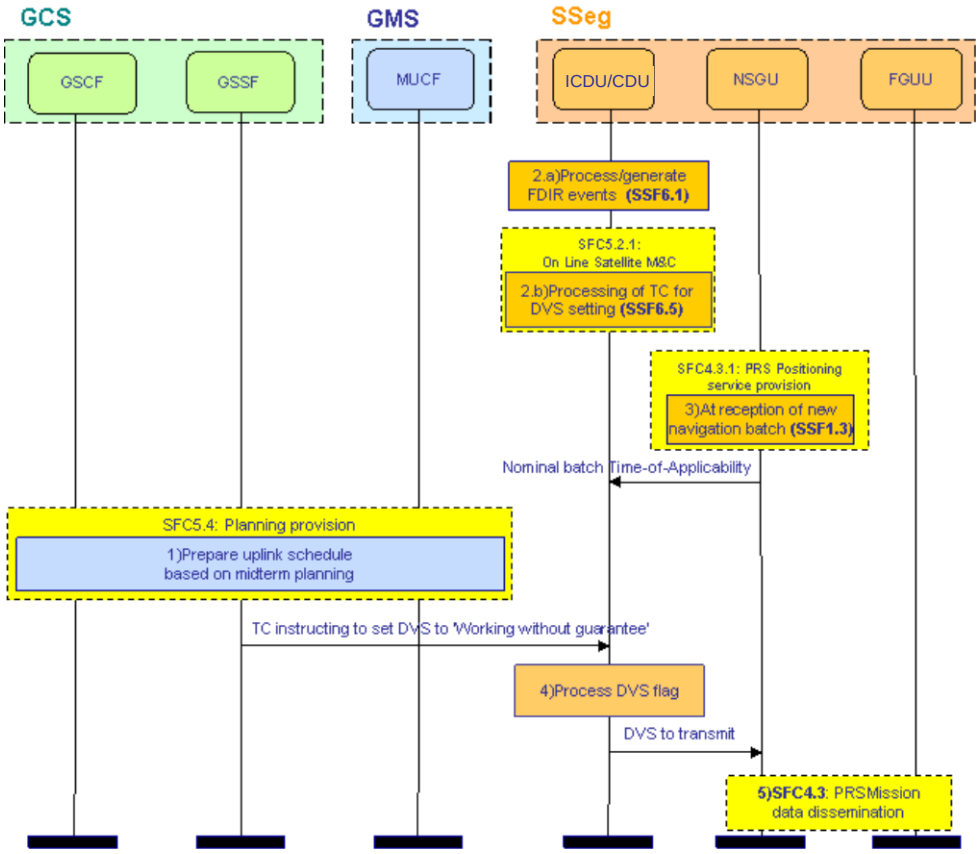
6.4.3.3.2.2.2.4 Data Validity Status Parameters User Process

Health status parameters are used to exclude measurements from the PVT solution

6.4.2.3.2.2.3 Data Validity Status Parameters Provision Dynamic Scenarios

6.4.3.3.2.2.3.1 Scenario #SCF4.3.3.2-2: DVS provision

This scenario (#SFC4.3.3.2-2) describes the generation and dissemination to the users of the PRS Data Validity Status (DVS) flag. The PRS DVS flag is transmitted in all PRS navigation signals every 15s to indicate to the user the “mode” of the satellite according to the following definitions: ‘Navigation Data valid’ or ‘Working without guarantee’.



1) GMS and GCS provide, through several data exchanges, an agreed Midterm planning. The planned tasks that have an impact on the system Mission performances are preceded by the setting by a TC of the DVS flag to 'Working without guarantee' set by an operator.. The resetting of the DVS flag can only be managed by an operator decision through a new TC.

2) The ICDU/CDU automatically sets on-board the spacecraft through automated Failure Detection, Isolation and Recovery (FDIR) functionality. The DVS flag is nominally set to "Navigation Data Valid" but is automatically updated to "Working Without Guarantee" by the on-board FDIR under the following conditions:

- o When the S-Band TT&C link has been lost for longer that a pre-set period (configurable between 1 and ten orbits).
- o When specific events are detected on board which impact navigation service performance.
- o Due to a spacecraft mode change (either in response to on-board FDIR action or due to telecommand from ground) from normal mode to any other spacecraft mode

Once the DVS flag is automatically set to "Working without guarantee" by any of the above conditions the resetting to "Navigation Data valid" can only be done by telecommand from ground by operator decision.

3) At each reception at the NSGU of new navigation data batches, the Time-of-Applicability (ToA) of the nominal batch is sent to the ICDU/CDU for the setting of a 'navigation timer'. Based on that, the ICDU/CDU will set the DVS flag to 'Working without guarantee' once a predefined time is elapsed with respect to that ToA. The predefined elapsed time to change the DVS flag status can range from 100 minutes to 10 orbits. This time is determined by a TC.

4) The ICDU/CDU computes the correct DVS flag depending on the inputs provided by the above steps 1), 2) and 3) and sends it to the NSGU to be included in the navigation message.

Contingency scenario #SFC4.3.3.2-2/4: When the link between ICDU/CDU and NSGU is lost due to an internal anomaly for more than 2 seconds it is automatically set to “Working without guarantee” until the link is established.
5) The dissemination process until the user reception is equivalent to the PRS navigation data (#SFC4.3.1-1).

6.4.2.3.2.3 Satellite Health Status (SatHS) and C-band Link Status

The SatHS and the C-band link status are disseminated as part of the PRS navigation data batch received from ground (see SFC4.3.1 for details).

6.4.2.3.2.3.1 Overall physical Description

6.4.3.3.2.3.1.1 Physical Static View

The diagram below summarises the physical mapping of the SFC4.3.3.2 for the Satellite Health Status (SATHS) and C-band link status:

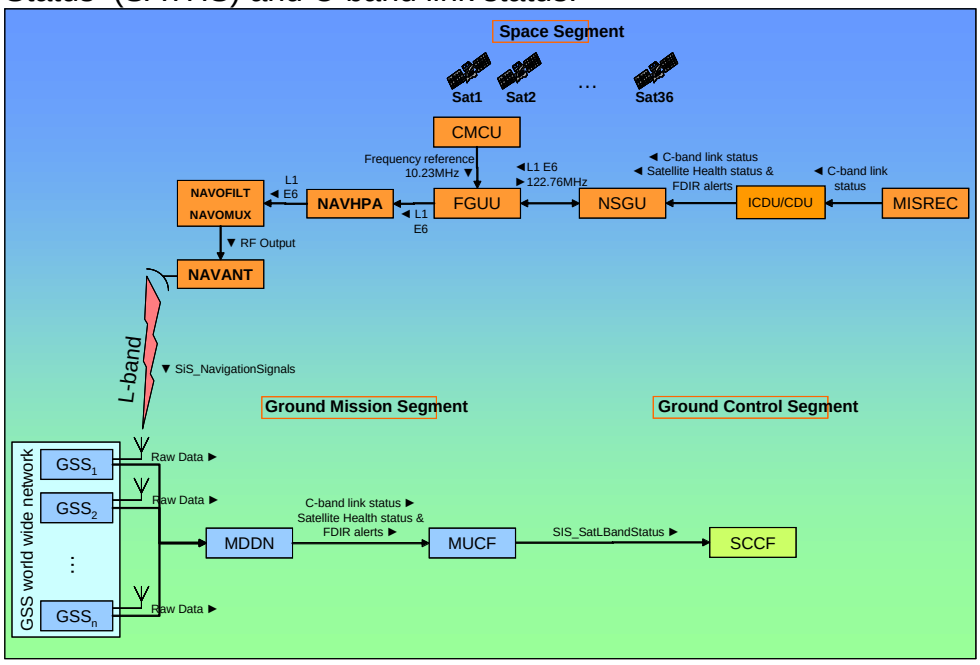


Figure 171 : Overall physical description SFC4.3.3.2 (SATHS)

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.3.3.2.3.1.2 Data needed to support PRS Health Status dissemination and mapping to physical elements

The table indicates the role of the GMS and SSgt physical element in the provision of the Satellite Health status and C-band link status. The role of each element is detailed in next dedicated sections.

	SSeg				GMS		GCS
	MISREC	ICDU/CDU	NSGU	from FGUU to NAVANT	GSS	MUCF	SCCF
Satellite Health Status (SatHS)	-	Data generation	L-band message generation	L-band signal generation and Tx.	Satellite data reception	Data formatting	Data contents processing
C-band link status	Data generation	Data processing	L-band message generation	L-band signal generation and Tx.	Satellite data reception	Data formatting	-

Table 82 : SFC4.3.3.2 (SATHS) Data and mapping to physical elements

6.4.2.3.2.3.2 PRS Data Validity Status flag Provision physical Detailed Description

6.4.3.3.2.3.2.1 C-band receiver: MISREC

The MISREC is responsible for generating the C-band link status which is sent to the ICDU/CDU once per second.

6.4.3.3.2.3.2.2 Payload control: ICDU (IOV satellites), CDU (FOC satellites)

In IOV satellites the ICDU process the C-band link status from the MISREC to forward it to the NSGU.
The ICDU is responsible for setting the Satellite Health Status and forward it to the NSGU.
In FOC satellites the same functions are allocated to CDU.

6.4.3.3.2.3.2.3 Message generation: NSGU

The NSGU inserts into the G/NAV message both C-band link status flag and Satellite Health Status and broadcast them. The process is as described in SFC4.3.1.

6.4.3.3.2.3.2.4 Navigation Message dissemination: from FGUU to Users

See SFC4.3.1 for details.

6.4.3.3.2.3.2.5 Health Status Parameters User Process

There is no use foreseen at user level for the SatHS and the C-band link status.

6.4.2.3.2.3.3 Health Status Parameters Provision Dynamic Scenarios

6.4.3.3.2.3.3.1 SFC4.1.3.2 Scenario 3 - SatHS and C-band link status provision

This MSC (#SFC4.3.3.2-3) describes the generation and dissemination up to the users of the:

- Satellite Health Status. It comprises the following two components: Satellite health status alert (2 bits) and Satellite health status telemetry summary (30 bits). The transmission of the Satellite Health Status is synchronized with the onboard clock and disseminated at a rate of 2 bits every 3 seconds. The dissemination is interrupted in case of dissemination of a PRS integrity alert (see c*ontingency scenario* below).

  It is not foreseen the use by the PRS user receivers but only for system operational purposes.

- C-band link status. It comprises the C-band Mission Receiver (MISREC) channel status for the GMS channel It informs about the frame/bit synchronisation of the C-band frames and/or the carrier/code lock of the signals.

It is not foreseen the use by the PRS user receivers but only for system operational purposes.

The transmission of these parameters is only protected with NAVSEC encryption and not with COMSEC encryption. See SEC-SFC15.2.2 for more details about NAVSEC encryption.

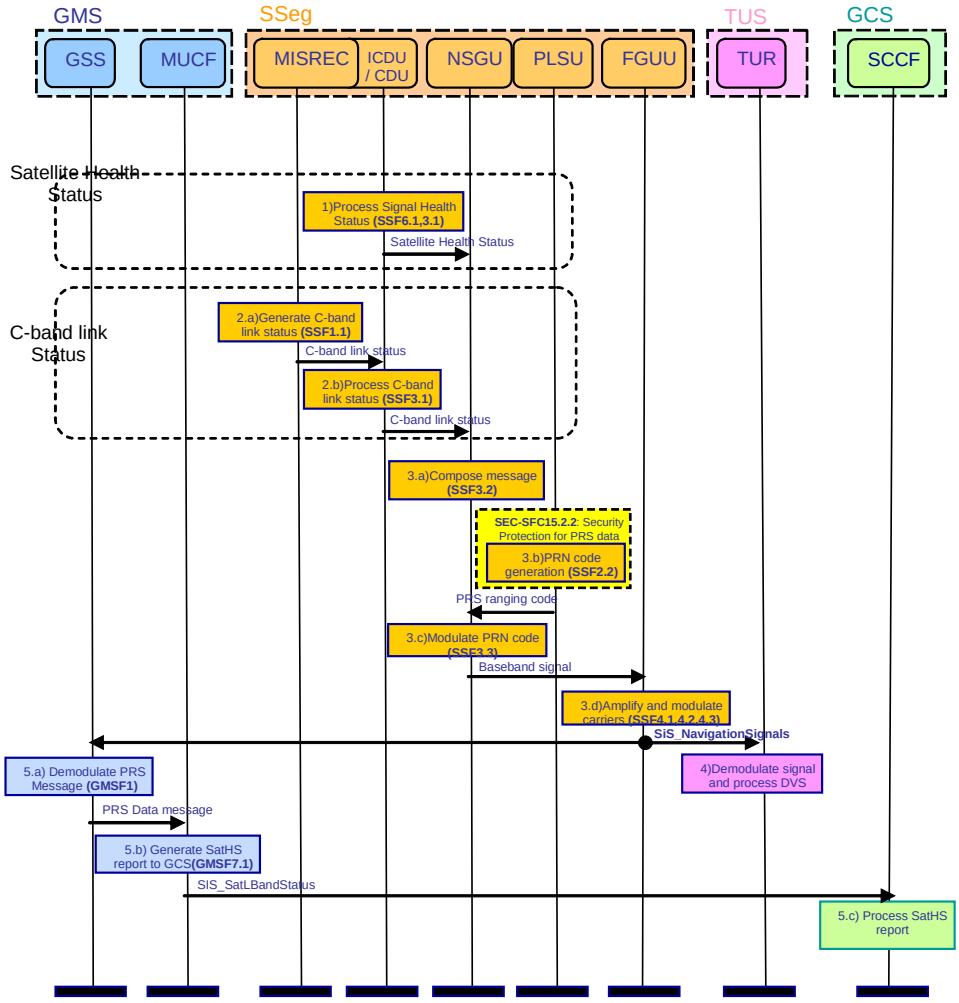


Figure 172 : SFC4.3.3.2-3 Dynamic Scenario 3

1) Onboard data processing and setting of the Satellite health status

The Satellite Health Status is generated by the ICDU/CDU and sent to the NSGU for its insertion in the G/NAV navigation messages.

2) Onboard data processing and setting of the C-band link status

2.a) The C-band link status for all 6 C-band channels (1 GMS is generated by the MISREC and sent to the ICDU/CDU.

2.b) The ICDU/CDU sends the C-band link status received from the MISREC directly to the NSGU.

3) PRS DVS flag, Satellite Health Status and C-band link status dissemination

3.a) The PRS navigation messages (G/NAV) are generated and modulated with the corresponding PRN code.

Contingency scenario (#SFC4.3.3.2-3/3.a): In case the NSGU does not receive a valid DVS flag from the ICDU for more than 2 seconds the DVS flag is set to “Working without guarantee” status.

Contingency scenario (#SFC4.3.3.2-3/3.a): In case the NSGU does not receive a valid C-band link status flag from the MISREC the flag is set to “No signal” status.

3.b) The FGUU modulates the signal to the correct frequency band and send it to the L-band antenna for its dissemination.

4) Data reception and processing by the user receivers.

5) GMS to GCS reporting on Satellite Health Status (SatHS)

5.a) The GSS demodulates the PRS message and sends the contents to the MUCF.

5.b) The MUCF collects the received data from the GSS on the SatHS field and reports a summary to the GCS (SCCF) once per minute for troubleshooting investigation.

5.c) The SCCF stores the SatHS report from the MUCF. An operator will check the contents and react in case of troubles.

6.4.2.4 SFC4.3.4 PRS Degraded Navigation Data Provision

6.4.2.4.1 Overall Physical Description

The physical architecture for this SFC is very similar to that for the OS case (see SFC4.1.4). The main difference is that an additional link to the PKMF is included. The PKMF performs encryption of the G/NAV navigation data and G/NAV almanacs, and also generates the G/NAV-9 data. Note that the functionality of the PKMF in this context is described in more detail in SEC-SFC15.3.

6.4.2.4.1.1 Physical Static View

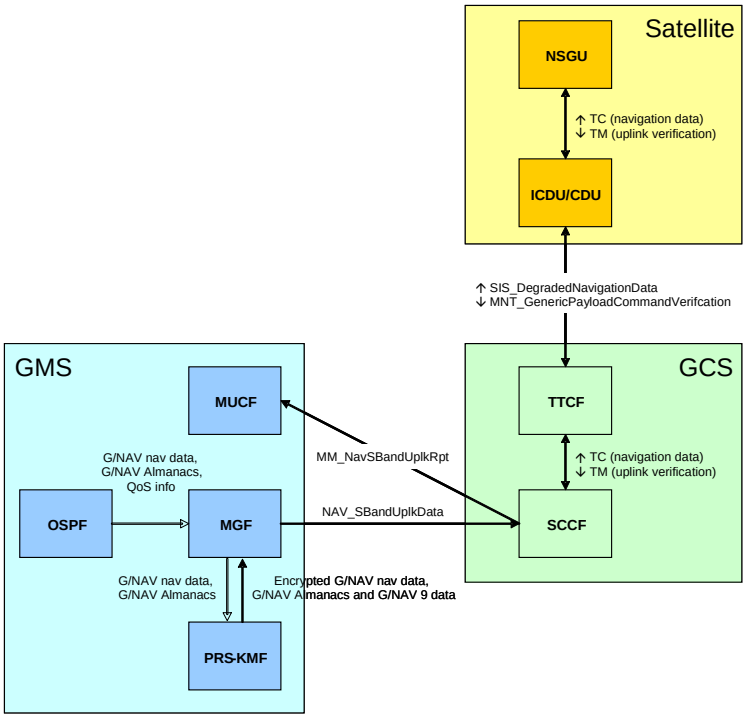


Figure 173 : Overall physical description SFC4.3.4

6.4.2.4.1.2 Data needed to support PRS S-band Positioning Service and mapping to physical elements

The table indicates the role of the GMS, GCS and SSgt physical elements in the provision of each type of PRS service data through the S-band. The role of each element is detailed in next dedicated sections.

	GMS			GCS	SSeg		
	OSPF	MUCF	MGF /PKMF	SCCF	ICDU/CDU	NSGU	from FGUU to NAVANT
Ephemeris (see SFC4.3.1 for generation details)	Data generation	S-band link report reception	S-band message generation and encryption	TC and GMS reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
Almanacs (see SFC4.3.1 for generation details)	Data generation	S-band link report reception	S-band message generation and encryption	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
Clock correction (see SFC4.3.1 for generation details)	Data generation	S-band link report reception	S-band message generation and encryption	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
BGD (see SFC4.3.1 for generation details)	Data generation	S-band link report reception	S-band message generation and encryption	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
Ionospheric parameters (see SFC4.3.1 for generation details)	Data generation	S-band link report reception	S-band message generation and encryption	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
GGTO (see SFC2.2 for generation details)	Data formatting	S-band link report reception	S-band message generation and encryption	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
GST-UTC time offset (see SFC4.3.2 for generation details)	Data formatting from PTF	S-band link report reception	S-band message generation and encryption	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
SISA (see SFC4.3.3.1 for generation details)	Data generation	S-band link report reception	S-band message generation and encryption	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.
SHS flag (see SFC4.3.3.2 for details)	-	S-band link report reception	S-band message generation and encryption	TC and reception report generation	TC processing and TM report generation	L-band message generation	L-band signal generation and Tx.

Table 83 : SFC4.3.4 Data and mapping to physical elements

Note: The following elements are not considered part of this SFC (even if provide support):

- The GSS observables data is generated and sent to the OSPF. See SFC4.1.1 for details.
- The MDDN (Mission Data Dissemination Network) is a GMS element that connects the GMS elements outside the GCC to the GMS elements inside the GCC. In the frame of this SFC the MDDN connects the GSSs to the OSPFs and the MGFs to the ULSs.
- The SDDN is the GCS element that connects GCS elements outside the GCC to the GCS elements inside the GCC.
- GACF is responsible for managing the GST-UTC time offset and provide it to the OSPF. More details can be found in the frame of the SFC2.1 and SFC4.3.2.

6.4.2.4.2 PRS Positioning Service Physical Detailed Description

6.4.2.4.2.1 SFC4.3.4 Related Physical Constraints: OSPF

As in the OS case, the OSPF generates 8 batches of navigation message data as well as the almanac data for the constellation and sends this to the MGF along with QoS information. See SFC4.3.1 section on OSPF.

6.4.2.4.2.2 SFC4.3.4 Related Physical Constraints: MGF

As in the OS case, the MGF selects the best data to use based on the QoS information provided by each OSPF. The OSPF then sends this data to the PRS-KMF for further processing. See SFC4.3.1 section on MGF.

6.4.2.4.2.3 SFC4.3.4 Related Physical Constraints: PRS-KMF

The MGF receives the encrypted G/NAV navigation data batches, G/NAV almanacs and also the G/NAV 9 data and includes this in the next delivery to be sent to the GCS for uplinking to the satellite. See SFC4.3.1 section on PKMF.

6.4.2.4.2.4 SFC4.3.4 Related Physical Constraints: SCCF

As for OS case (see SFC4.1.4).

6.4.2.4.2.5 TC processing and TM generation: ICDU (IOV satellites), CDU (FOC satellites)

As for OS case (see SFC4.1.4).

6.4.2.4.2.6 SFC4.3.4 Related Physical Constraints: NSGU

The S-band received data will be downlinked on the L-band once the ToA of the 4th batch received from the C-band is reached and one of the S-band batches is more up-to-date.
For the dissemination it is as for SFC4.3.1 section on NSGU.

6.4.2.4.3 SFC4.3.4 Dynamic Scenario

6.4.2.4.3.1 SFC4.3.4 Scenario - Nominal Uplink of PRS Data via S-band

This MSC (#SFC4.3.4-1) describes the generation and dissemination up to the users of the PRS data (ephemeris, almanacs, SISA, Signal Health Status, Data Validity Status and time and clock corrections) through the S-band (PRS S-band data). This data is available onboard to support a degraded mode of operation for the PRS when no more up-to-date navigation data is available on the satellite (i.e. via the nominal C-band uplink).
This navigation data is used to correct pseudorange measurements (clock corrections) and compute the user position (ephemeris and almanacs). The PRS navigation data is computed to be transmitted on the E6A and E1A (G/NAV messages) data channels.

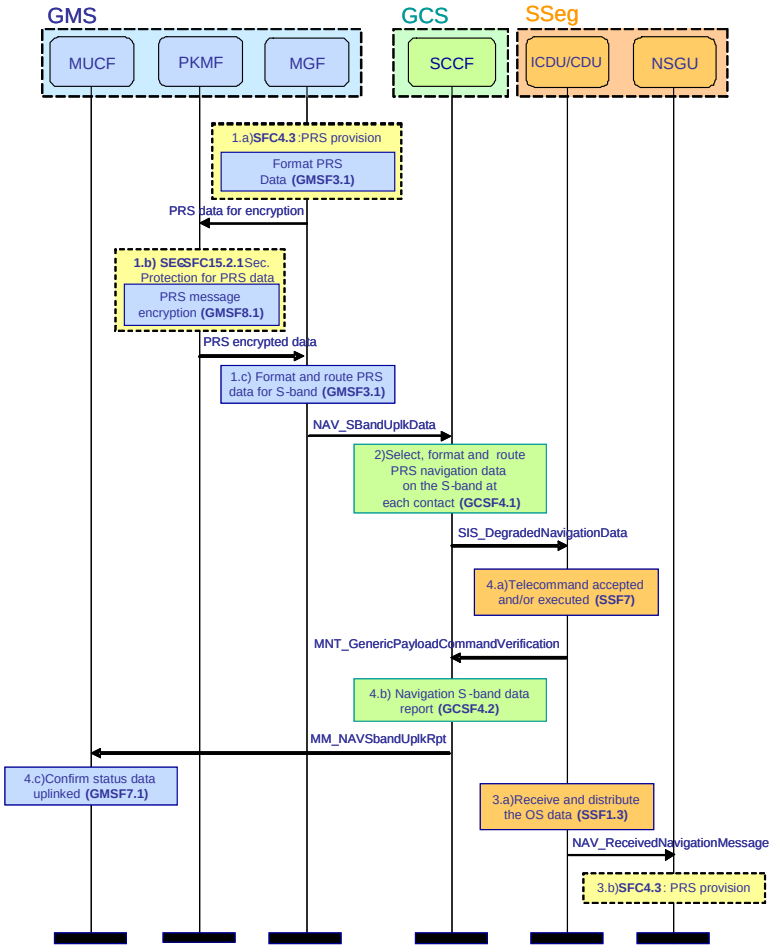


Figure 174 : SFC4.3.4 Dynamic Scenario 1

The sequential description is detailed below:

1) Generation of PRS S-band data

The PRS data that is uplinked on the S-band is generated following the same process described for the SFC4.1. The PRS data that will be uplinked on the S-band link provides the following services:

- PRS positioning service: Ephemeris, almanacs, clock and time corrections (ionospheric and BGD) and GGTO. This is described in SFC4.3.1.
- PRS timing service: GST-UTC offset. This is described in SFC4.3.2.
- PRS Signal Quality Monitoring Service. This is described in the SFC4.3.3.
- PRS security data service (G/NAV-9). This is described in the SFC4.3.6.

All this data is generated for G/NAV kind of messages and for each satellite in orbit once every 10 minutes. It is divided in 9 different batches of data each of them a validity of 4 hours. It is noted that through the C-band link only the first 5 batches are used (see SFC4.3.1 for more details).

2) PRS S-band data transmission

The MGF (including the redundant MGFs) send the PRS data to the SCCF (GCS) with a Quality-of-Service indicator (QoS). The parameter helps the SCCF to select the best data that must be uplinked. The data is sent on a text file format.

The latest PRS data received is uplinked at each S-band link contact for each satellite through the TTCF. To be uplinked the data is reformatted in several telecommands (TC) and then processed by the ICDU/CDU.

3) PRS S-band data dissemination

3.a) The TCs contents are transmitted to the NSGU for storage in memory.

3.b) The PRS S-band batches of data are disseminated sequentially according to their Time-of-Applicability (ToA). However they are only used when the validity of all stored C-band navigation data is finished. That is when the transmission of the 4th C-band navigation batch is reached.

The PRS S-band data is disseminated on the L-band following the same process as per PRS C-band data.

4) PRS S-band data reception reporting

4.a) The ICDU/CDU reports on ground the reception and execution of the TC containing PRS S-band data.

4.b) The report is processed by the SCCF (GCS) and forwarded towards the GMS.

4.c) The MUCF is the GMS element responsible for keeping the PRS S-band data reports. However it is not foreseen any functionality to process the contents of the report

6.4.2.5 SFC4.3.5 PRS Integrity Service Provision

Section Deleted

6.4.2.6 SFC4.3.6 PRS Security Service Provision (Uncla)

6.4.2.6.1 Overall Physical Description

The dissemination of the Security data (G/NAV-9) is made together with the G/NAV navigation messages from the MGF but are generated by the PKMF instead of the OSPF. Only one batch of G/NAV-9 data is generated and disseminated at a time with one associated Time-of-Applicability (ToA). This ToA corresponds to the epoch at which the satellite shall start the downlink of the G/NAV-9 message.

The G/NAV-9 message size is not fixed but depends on the security operational needs. It can contain up to 520 packets, common to all the satellites.

6.4.2.6.1.1 Physical Static View

The GMS and SSeg designs implement this functional chain as shown in the following figure:

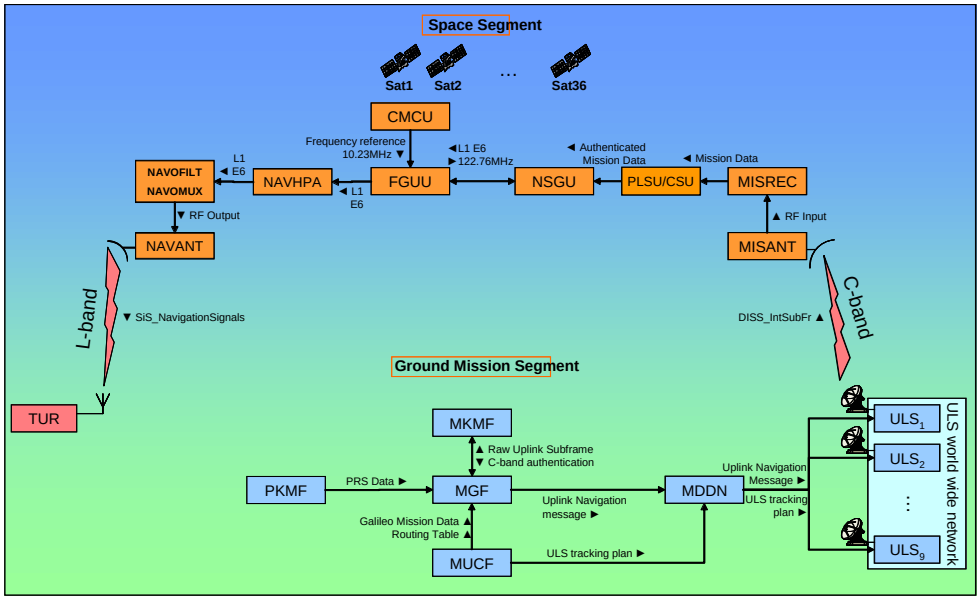


Figure 175 : Overall physical description SFC4.3.6

Note: this static view address a maximum expandability scenario in general; its purpose is not to show the applicable physical architecture which is specified in SYS-RD.74

6.4.2.6.1.2 Data needed to support PRS Security Data Dissemination Service and mapping to physical elements

The table indicates the role of the GMS and SSgt physical elements in the provision of Security data. The role of each element is detailed in next dedicated sections.

	GMS					SSeg			
	PKMF	MGF	MKMF	MUCF	ULS	C-Band Antenna / MISREC	PLSU / CSU	NSGU	from FGUU to NAVANT
Security data (G/NAV-9)	Data generation and encryption	C-band message generation and dissemination	Dissemination to users (Common to the SFC4.3.1)						
			C-band message protection	Uplink scheduling plan	C-band message uplink	C-band message reception	C-band message protection	L-band message generation	L-band signal generation and Tx.

Table 84 : SFC4.3.6 Data and mapping to physical elements

Note: The following elements are not considered part of this SFC (even if provide support):

- The MDDN (Mission Data Dissemination Network) is a GMS element that connects the GMS elements outside the GCC to the GMS elements inside the GCC. In the frame of this SFC the MDDN connects the GSSs to the OSPFs and the MGFs to the ULSs.

6.4.2.6.2 Security Data Provision Physical Detailed Description

6.4.2.6.2.1 Security data generation: PKMF

The PKMF is in charge of:

- generating the security data G/NAV-9 periodically or on operator request,
- encrypt such data and
- send it to the MGF for uplinking on both C-band and S-band links.

It is remarked that the amount of security data varies depending on the operational needs (see classified DDF).

6.4.2.6.2.2 Message formatting: MGF

The MGF is responsible for generating the Message subframes that contain the G/NAV-9 messages and route them to be uplinked on both C-band and S-band.

6.4.2.6.2.2.1 C-band message formatting

The MGF is responsible for formatting the G/NAV-9 data being uplinked to the satellite on the C-band. The G/NAV-9 message is uplinked in the uplink message subframe in blocks of 26 packets every 30 seconds. If the total amount of packets is not multiple of 26, the MGF will generate dummy packets that will not be stored on board the satellite.
The uplink is synchronised in a way that at a given epoch the same G/NAV-9 packets are uplinked to all the satellites connected.

6.4.2.6.2.2.2 S-band message formatting

The MGF is also responsible for formatting the G/NAV-9 data to be uplinked to the satellite on the S-band. In this case always 520 packets are sent to the GCS for uplinking. In the case less than 520 packets were generated by the PKMF the MGF will fill in the missing packets with dummy data (set to all zeros) that will be stored on board the satellite.
The S-band dissemination process is detailed in the SFC4.3.4.

6.4.2.6.2.3 Uplink Scheduling Plan: MUCF

The MUCF is in charge of the Uplink scheduling generation. It is responsible for ensuring the uplink of the complete Security data batch to all the satellites before it becomes applicable. This is ensured taking into account the following constraints:

- the time needed to upload all 520 packets is approximately 11 minutes.
- all data is uplinked at least once to every satellite with a probability of about 99%.

6.4.2.6.2.4 C-band dissemination: from ULS to PLSU/CLS

The Security data (G/NAV-9) are uplinked in C-band within the uplink message sub-frame. This process is already described in the SFC4.3.1.

6.4.2.6.2.5 L-band message generation: NSGU

The G/NAV messages are stored in the NSGU and downlinked with the following process:

- in case the satellite is connected to the ground: two G/NAV-9 data messages are downlinked every 30 seconds in the pages allocated to security data (as specified in SISICD).
- in case the satellite is not connected: the G/NAV-9 messages are then also downlinked as part of G/NAV-7 and G/NAV-8, normally reserved for real time transmission of data uplinked from ground. In that case the dissemination rate is increased up to 4 G/NAV-9 messages every 30 seconds.

Note:
- The storage of the packets is independent from the way they are received (C-band or S-band): as soon as a new packet is received, it replaces the packets stored with the same ToA.
- The NSGU will downlink nominal and dummy G/NAV-9 packets received in S-Band (constant amount of 520 packets) without distinction.

6.4.2.6.2.6 PRS Users processing

See classified DDF.

6.4.2.6.3 SFC4.3.6 PRS Security Service Provision Dynamic Scenarios

6.4.2.6.3.1 General Description

This MSC (#SFC4.3.6-1) describes the dissemination up to the users of the PRS security data (G/NAV-9). This data is used to transmit security data to the users. The generation of the PRS security data is described in the SEC-SFC-15.2.1.

6.4.2.6.3.2 Flow chart

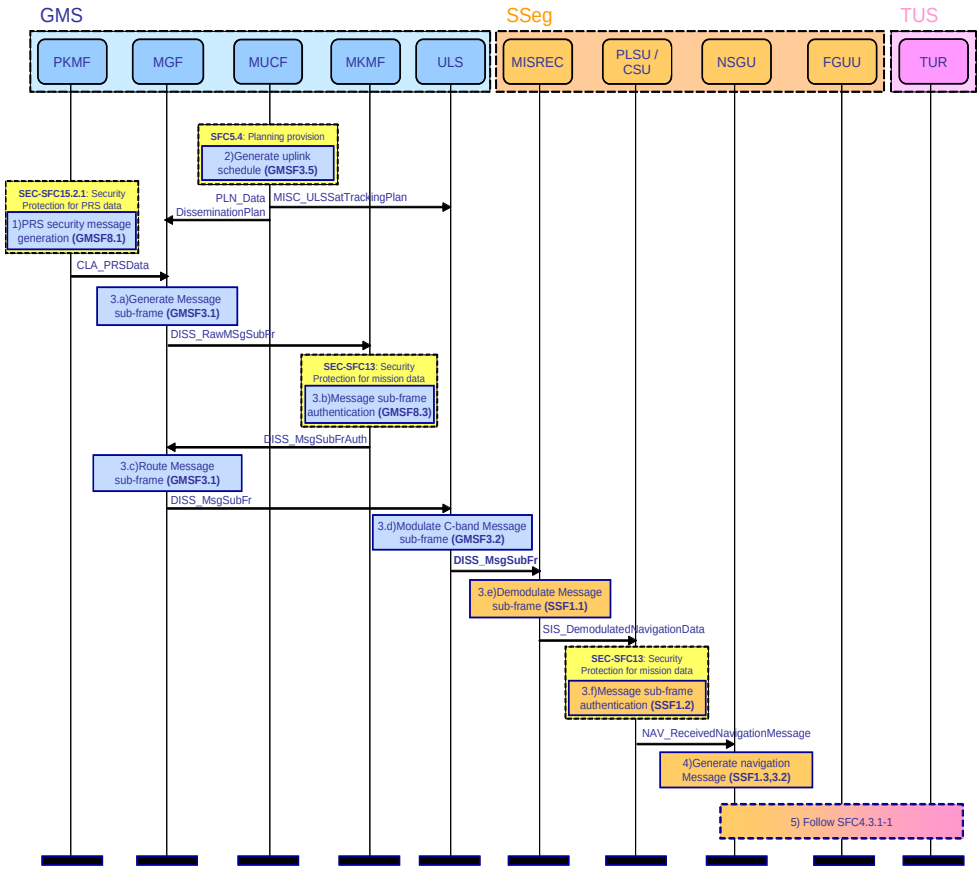


Figure 176: SFC4.3.6 Dynamic Scenario 1

6.4.2.6.3.3 Scenario sequence description

1) PRS security data generation
A batch of PRS security data is periodically generated by the PKMF. It is composed from 1 up to 520 packets. These packets are transmitted, already encrypted, to the MGF for its formatting and uplink to the satellite.
The generated batch has an associated Time-of-Applicability (ToA) to indicate the satellite the starting validity of the data.
The PRS security data generation is classified and it is described in the SEC-SFC-15.2.1.
2) Uplink scheduling computation
In parallel the MUCF generates the uplink and tracking schedules and distributes them to the MGF (uplink plan) and to the ULS stations (tracking plan).

See SFC5.4 for more details about the uplink scheduling plan generation.

3) PRS security message uplink

3.a) The PRS security data is received from the PKMF and inserted within the C-band Message sub-frames. The PRS security data is composed from 1 up to 520 packets, however they are uplinked on the C-band in blocks of 26 packets every 30 seconds of contact. In the cases were the number of generated packets is not multiple of 26, the uplinked Message sub-frames are filled with dummy packets. The dummy packets will not be broadcasted by the satellite.

3.b) The MGF previously to the dissemination the Message sub-frames are sent to the MKMF for its authentication.

See SEC-SFC13 for more details about the Message sub-frames authentication on ground.

3.c) Once the Message sub-frame have been authenticated are routed to the ULS stations according to the dissemination plan.

3.d) Each time the ULS receives a valid Message sub-frame is modulated and uplinked to the satellite that the tracking plan indicates.

3.e) Each Message sub-frame transmitted within the C-band is demodulated by the satellite Mission Receiver (MISREC).

3.f) Each C-band demodulated Message sub-frame is sent to the PLSU/CSU for the authentication.

See SEC-SFC13 for more details about the Message sub-frame authentication on board the satellite.

4) Follow scenario SFC4.3.1-1

Once the complete data is received the NSGU checks the Time-of-Applicability (ToA) of the batch data.

Every three seconds it generates the data to be transmitted on the L-band. This process includes the generation of some other data to be added in the L-band navigation message.

5) Data reception and processing by the user receivers

6.5 SFC5 MONITORING AND CONTROL (GROUND, SATELLITE, MISSION AND PLANNING)

The purpose of this section is to describe the physical implementation of the SFC5 Monitoring and Control

The SFC5 is in charge of providing all the means needed to perform the monitoring and control, and planning of the Galileo system. This includes ground assets M&C (including network), satellite M&C, Mission monitoring, and planning provision.

6.5.1 SFC5.1 Ground Asset Monitoring And Control

6.5.1.1 Introduction

The purpose of this section is to describe the physical mapping of this SFC wrt Galileo ground elements.
It is based on the following documentation:

- System documents:
  - System Logical Architecture (SYS-RD.1)
  - GCSGMSICD (AD.8)

- GMS documents:
  - GMS DDF2 (GMS-RD.2): MDDN section, MUCF section
  - MUCF ICD (GMS-RD.11)
- GCS documents:
  - GCS DDF (GCS-RD.1), CMCF, TTC and GCS Networks sections

6.5.1.2 SFC5.1.1 - Ground Elements Monitoring and Control

6.5.1.2.1 Physical Static View

6.5.1.2.1.1 Overall Physical Description

The functional organization of the ground assets monitoring and control functional chain, described in the DDF1 section, is implemented in the physical architecture, as shown in the following picture and described after:
The physical organization of the ground assets and networks monitoring and control chain is shown in the following figure:

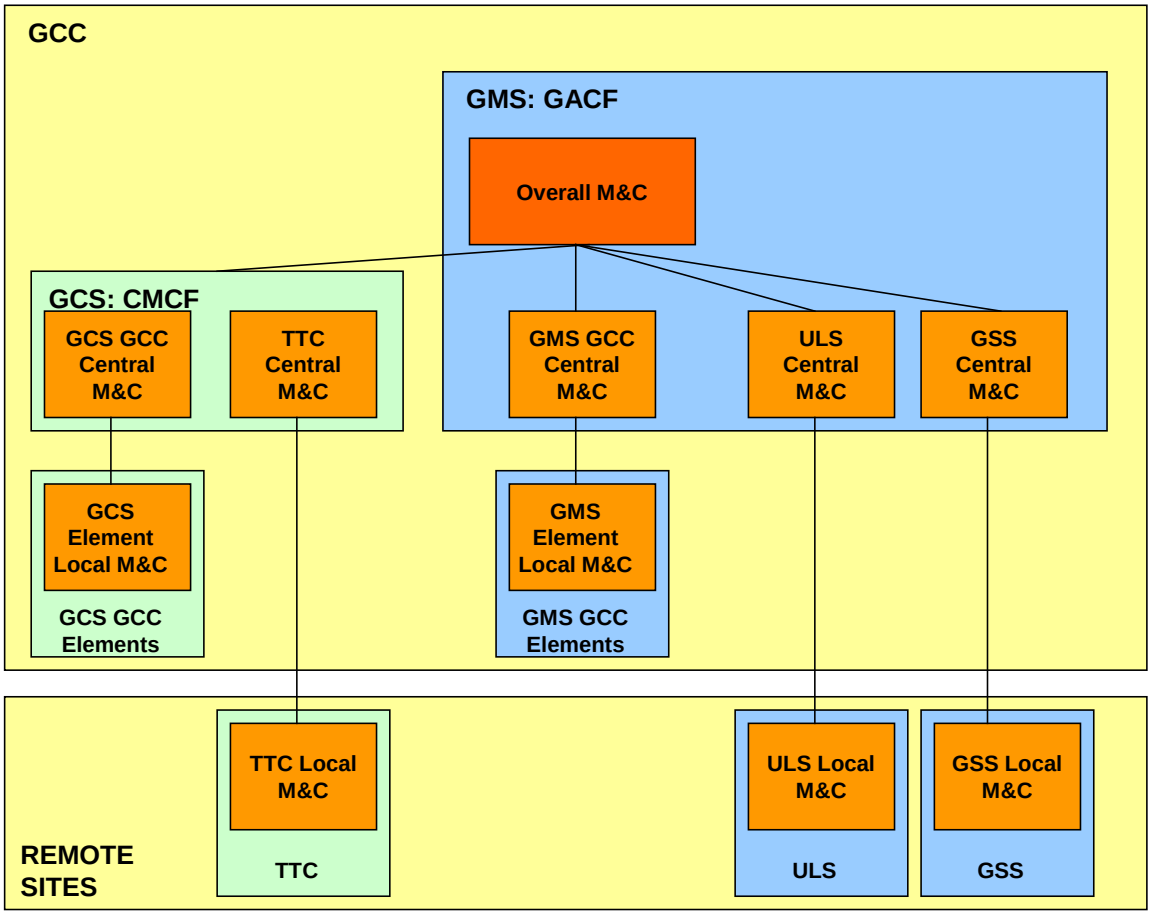


Figure 177 : Ground asset M&C physical organization

- The overall monitoring and control function is supported by the GMS-GACF (Ground Asset Control Facility), which is the ground monitoring and control equipment for the entire Galileo system.
- The central monitoring and control functions are supported by two elements, one in each ground segment:
  - the GMS-GACF, which performs the monitoring and control of the GMS segment
  - the GCS-CMCF (Central Monitoring and Control Facility), which performs the monitoring and control of the GCS segment

- The element monitoring and control function is supported by all ground elements, and more precisely by the M&C component / process inside the elements:
  - GMS-ULS, Up-Link Station, and more specifically the monitoring and control computer of the element
  - GMS-GSS, Galileo Sensor Stations, and more specifically the GSS core computer
  - the following GMS elements installed in the GCC: OSPF, PTF, MGF, MUCF, MSF, MTPF, SPF and MKMF (unclassified subsystem only)
  - GCS-TTCF, Tracking, Telemetry and Command facility
  - the following GCS elements in GCC: CMCF, SCCF, SCPF, OPF & FDF.

The proposed mapping between elements and function is strictly limited to functions described in this SFC. For example, the GACF has other functions than monitoring and control, but they will not be addressed in the frame of this SFC.

The periodical monitoring is performed as follows:

- The local Monitoring and Control instance of each element (at remote site or in the GCC, for GMS and GCS segments) gathers elements' statuses and health information and generates a monitoring message containing all elements' data.
- For GMS, these data are transferred to the GACF for GMS central monitoring, whereas for GCS, these data are sent to the CMCF for GCS central monitoring.
- The CMCF then builds a GCS status summary message, to be sent to the GACF for overall monitoring and control function.
- The GCS summary status together with the GMS status is processed by the GACF to build an overall Ground Segment summary status. This message is sent to the GMS MUCF element, and used for GMS short term planning and integrated mission monitoring.

The control of the ground assets is nominally performed through the short term planning and schedule execution routines:

- For GMS, the GACF receives the short term plan and generates and executes the associated schedules.
- For GCS, this task is performed by the SCCF.
- The schedules are derived into low levels commands to be sent for execution to the elements.

Note that this capability towards a specific TTCF is not possible during a satellite pass for the TTCF following the contact.

The following figure shows the monitoring and control static diagram, describing the links between all GCS and GMS elements.

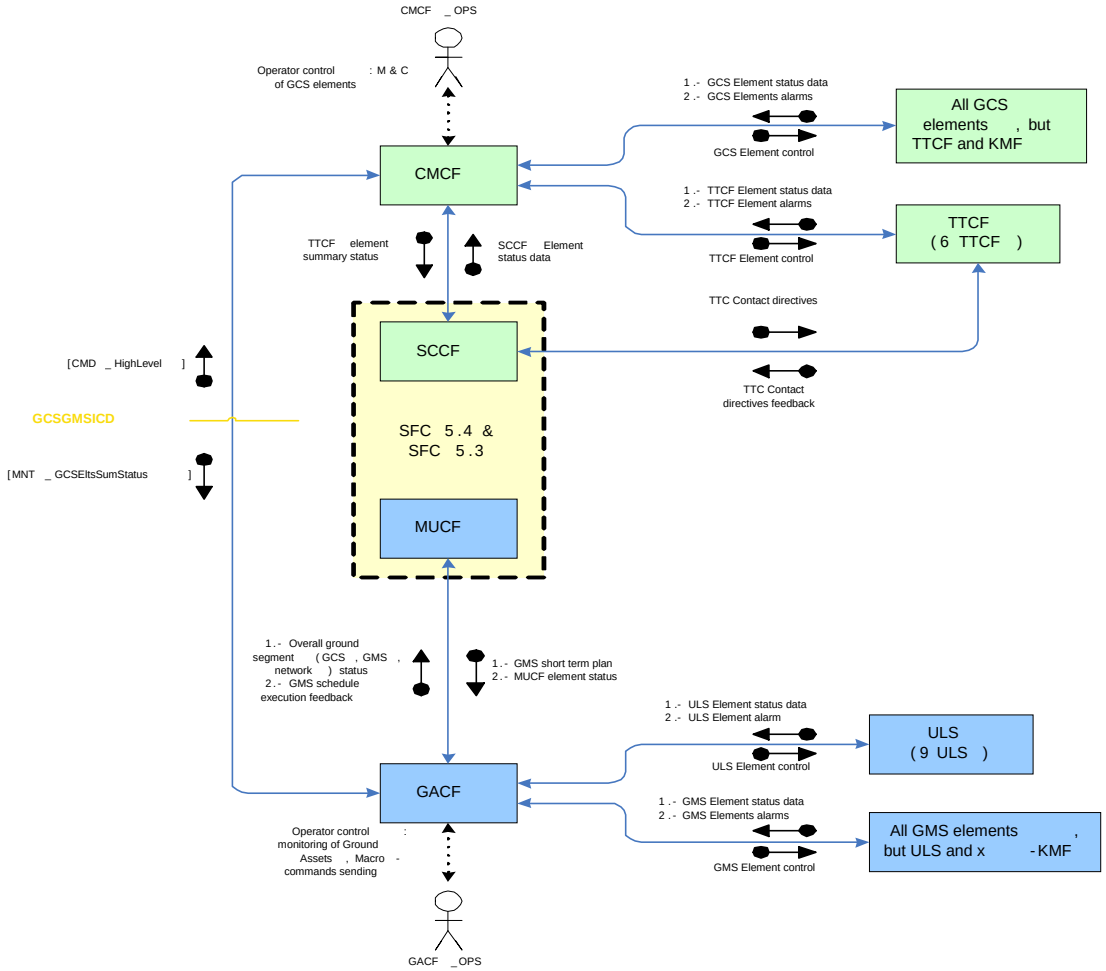


Figure 178 : SFC5.1.1 Physical Static View

Only inter-segment flows are detailed in this diagram. Intra segment flows are described in the relevant segment documentation.

6.5.1.2.1.2 SFC5.1.1 Related Constraints: General considerations

The following constraints have been identified wrt this SFC:

- Use of SNMP standard for M&C data exchange and of associated monitoring and control database for all elements
- Use of the NRT LAN in GCC for all M&C data exchange

The SNMP protocol will be used for the monitoring and control exchanges, whereas file exchanges will be in TCP, FTP or XML formats.
For all ground elements, the monitoring function allows all parameter information to be accessible through the M&C systems SNMP agent: each element implements a local M&C agent allowing stand alone level-0 operation of the element, and remote controlled levels-1 and -2 operation of the element. This local M&C agent handles all the internal events (warnings, failures and monitoring internal parameters).
The following schemes show the monitoring and control processes (monitoring, alarm notification and command), managed by the SNMP protocol.

Monitoring process

The central / overall monitoring and control elements (i.e. CMCF and GACF) monitor all ground elements upon request, sending the SNMP Get request and receiving the SNMP GetResponse message from each element.

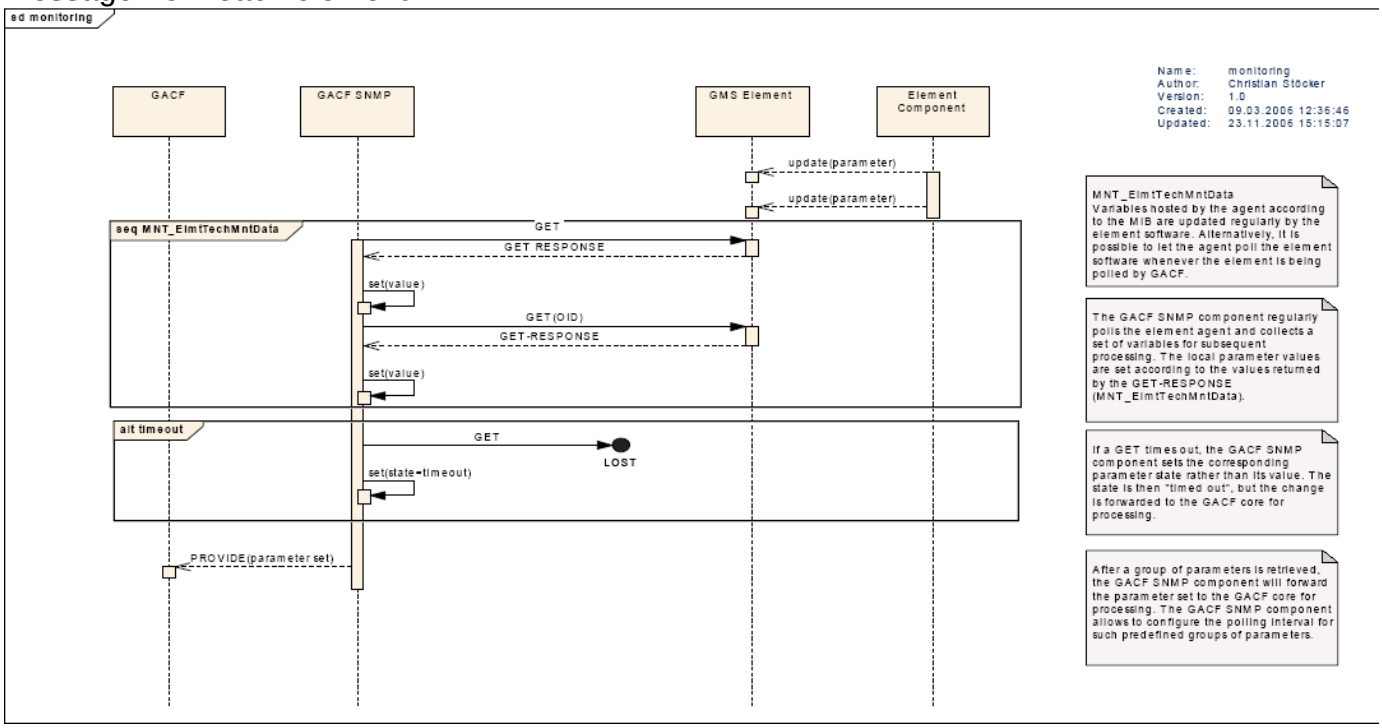


Figure 179 : SNMP monitoring process

Alarm reporting (asynchronous monitoring)

Ground elements can also generate traps asynchronously when they need to notify their respective central M&C of specified events, e.g. alarms.

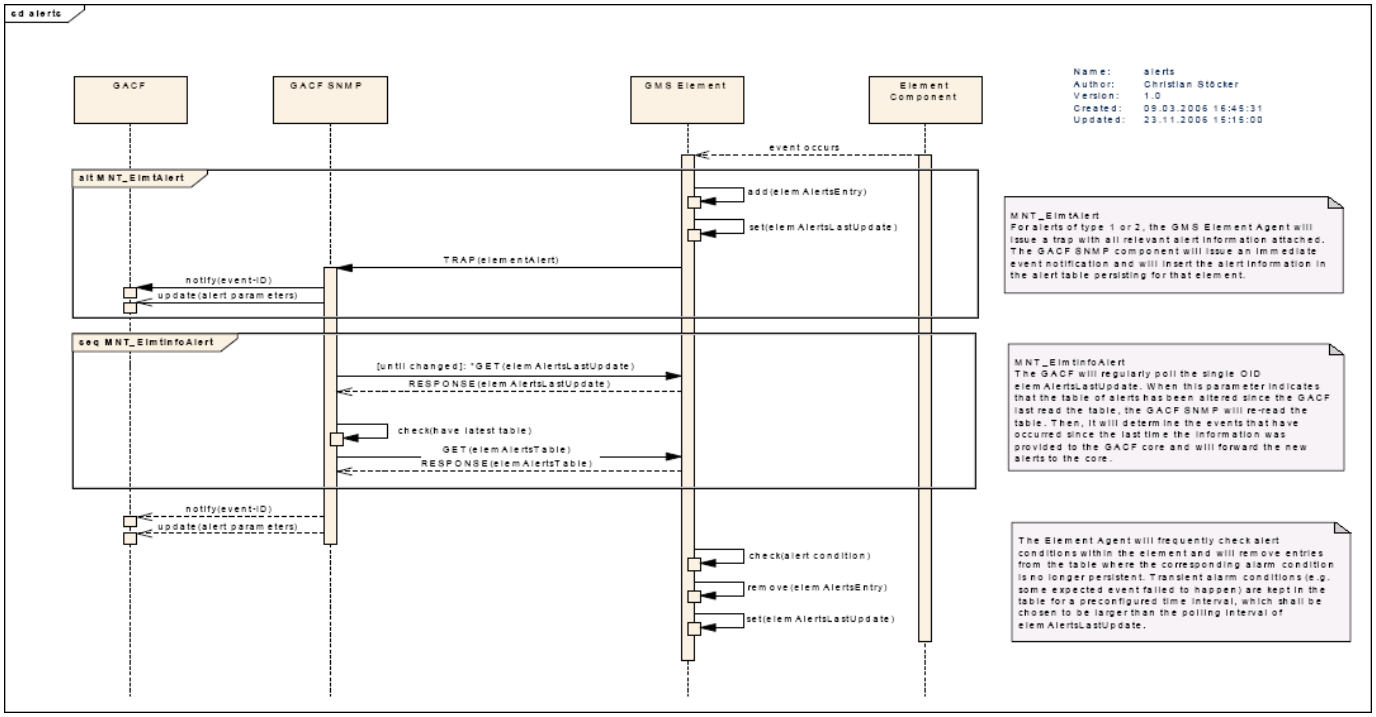


Figure 180 : Alarm notification

Central M&C then associate every event and data coming from elements to the appropriate identifier or description text of the M&C database.

Control process

Controls will be performed using the following SNMP messages:

- SNMP Set commands that contain the necessary commands.
- SNMP Get requests in order to follow the execution state of the commands
- Note that SNMP Trap can be sent once the command is executed successfully, or in the case of failure

For the specific case of GCS control through GMS, GACF also maintains the GCS macro-commands in a command database. These macro-commands allow GCS control from GMS, for example (but not limited to):

- assignment / de-assignment of GCS resources to the activities to be performed,
- requests for changes, simulations and/or verification of the TTC contact plans,
- enabling/disabling detailed monitoring of the execution of the scheduled operations

The ground asset monitoring and control process will use the near real time (nRT) LAN, both in GCC and in remote elements, such as GSS, ULS and TTCF.

6.5.1.2.1.3 Overall M&C and schedule execution: GACF

The GACF is responsible to manage level-1 operations for GMS elements monitoring and control (excepting PKMF and the classified part of the MKMF), as well as level-2 operations for all the Galileo ground assets.
In detail, GACF is responsible for:

- GMS technical monitoring, i.e. periodical polling of the GMS elements in order to refresh the elements MIB, display of the status of the GMS elements' equipment and processes, generation and display of the logbook, handling of equipment or software failure occurrences (alarms)
- GMS technical control, allowing a special protocol for safety critical elements commanding: equipment commands for on-line or off-line reconfiguration, software upgrades
- Reception of the GMS short term plan from MUCF, production and execution of the associated schedules, i.e. transform the short term plan instructions received into macro-commands for GMS elements
- Galileo ground assets overall monitoring, i.e. display of summary information about the status of both ground segments (GCS and GMS)
- Archiving the remote monitoring data and to deliver these external installations data to the GALILEO External Entities through the GMS NRT External Interface

For this purpose, the GACF interacts with:
- all GMS elements (or more precisely the elements' M&C equipment or process) for M&C purpose through their SNMP agent:
    - ULS and GSS stations via the MDDN network
    - All GMS elements in the GCC via the GCC LAN
- the CMCF in GCS for GCS summary statuses reception
- the MUCF for:
    - Transfer of GMS and GCS ground assets summary status for integrated mission monitoring
    - GMS summary status transfer
    - GMS short term planning instruction reception and acknowledge

The GACF is located within the Galileo Control Centre (GCC) and is permanently manned by staff from the GCC Operation Team.
From operational point of view, overall M&C function allocated in the GACF provides:
- Synthetic and detailed displays of the overall ground segments (GCS, GMS and networks) configuration and status. Such information is supplied with different levels of abstraction, allowing operators «to navigate» from the top levels, where information for the overall system is synthesized, to functional chains and down to isolated facilities, where their summary status can be depicted.
- Capability to control the mode transition for each individual element (e.g. to remove / add one element from / to an operational chain, to declare one element in test mode for maintenance, etc.…).

6.5.1.2.1.4 GCS Monitoring and control: CMCF

The GCS function GCSF5.2 is entirely allocated to the CMCF. Therefore CMCF is responsible for GCS M&C level-1 operations, i.e.:
- M&C of the GCS Hardware and software processes within the local GCC (including raising alarms) but ILS tools and GCS KMF
- Monitor of all TT&C stations (including raising alarms) and control of the TT&C stations to which it is assigned

- Support the use of a control token mechanism: the control token for a TTCF can be assigned to CMCF, TTCF local MMI or to none of the elements. CMCF token management MMI allows to see who the owner of the token is at any given time and to acquire the token only when it is assigned to “none”.

In addition, CMCF role consists also in the consolidation of low-level monitor data into high-level data for transmission to the GACF, as well as deriving high-level commands received from into low level commands for Element control.
The consolidated status includes overall GCS Elements’ health at the local GCC and consolidated TT&C stations health. This information is passed to the GACF and to the SCCF.
In the meantime, CMCF transfers a summary status of the TTCF to SCCF, in order to elaborate the contact execution directives.
For this purpose, the CMCF interacts with:

- the GACF in GMS for GCS summary statuses transfer
- SCPF, OPF and FDF (or more precisely the elements’ M&C equipment / process) in the GCC for M&C purpose through their relevant SNMP agent via the GCC LAN
- the TTCF via the SDDN network for:
    - TTCF M&C through the TTCF local M&C agent
    - M&C Control token management, between CMCF and TTCF local M&C
- the SCCF for:
    - SCCF M&C through its SNMP agent
    - TTCF summary status transfer

The CMCF sits in a hierarchy of M&C Elements, acting as a mid level manager within the overall ground segment network management architecture.
Therefore the CMCF is normally unmanned, but may be manned when required to carry out GCS specific monitoring and maintenance tasks or when the GACF is not available.
From operational point of view CMCF provides for the GCS ground assets:

- Capability to have synthetic and detailed visualizations of the configuration and status per type of elements
- Capability to control GCS elements when they are not involved in on-going operations
- Capability to test GCS element, e.g. launch of built-in test inside the element for maintenance or troubleshooting.

6.5.1.2.1.5 GCS Schedule execution: SCCF

For this SFC, GCS function GCSF1 is allocated to the SCCF.
SCCF is responsible for:

- Execution of the GCS schedules received from SCPF
- TTC contact directives (nominal or back-up, depending on operator choice) transmission to the TTCF elements

- monitoring of operations progress, i.e. monitoring the status of operations executed both from ground and on-board the spacecraft and compares this to the active plan

For this purpose, the SCCF interacts with:
- the CMCF for:
    - transfer of the SCCF own monitoring data
    - reception of a TTCF element summary status
- the TTCF for:
    - contact plan transmission through a specific interface

6.5.1.2.1.6 GMS short term plan production and mission monitoring: MUCF

The MUCF is the GMS element in charge of:
- the generation of GMS short term plan: the GMS short term plan are produced by MUCF and are passed to GACF for deriving these short term plans into schedules and for execution of these schedules.
- integrated mission monitoring
- ULS up-link schedule (as described in SFC5.4)

For this purpose, MUCF interacts with:
- GACF for:
    - MUCF remote monitoring and control through its SNMP agent
    - Transfer of the GMS short term planning to be processed by the GACF, reception of associated feedback
    - Transfer of the ground segments (GMS and GCS) summary status, to include the ground assets status in the integrated mission monitoring function

6.5.1.2.1.7 Contact directives execution: TTCF

In the frame of this SFC, TTCF is responsible for its own level-0 operations, i.e.:
- provide a local M&C function that interfaces to the CMCF for the exchange of monitor and control data
- provide a token exchange mechanism that allows only one controlling entity to have access to the local M&C function at each station

For this purpose, TTCF interfaces with:
- SCCF for:

- contact plan reception

- CMCF for:
  - transfer of the control data from the CMCF, and the return of monitor data from the TTCF
  - token exchange function in order to allow only one CMCF to be in control of any one TT&C station at a time

- Remote site for:
  - transfer of some infrastructure monitoring data (intrusion or fire / smoke alarms) from the TT&C local M&C to the site provider
  - transfer of some infrastructure monitoring data (physical alarms) in order to allow sites' visibility for remote operators

During normal operations control is performed from the SCCF located at the GCC. Local control is possible by means of either a local terminal located at the TT&C station, or remotely from a login facility at the GCC.
The local M&C holds the macro commands used to configure the TT&C station for the various tasks it has to perform. Such macros include those to configure the TT&C station to initiate a contact, and to terminate a contact.
Note that the token mechanism also extends to a local operator, who would need access during periods of maintenance or testing.

6.5.1.2.1.8 Element Monitoring and control: all the other elements

All other elements (either GMS or GCS elements) include a monitoring and control equipment or process that allows local M&C at element level (level-0 operations) and M&C data transfer to the respective central monitoring and control (CMCF or GACF).
Note: GMS-PKMF and GMS-MKMF (classified subsystem) are not monitored nor controlled by GACF. Only independent visible/audible alarms in the Main Control Room of the GCC will happen in case of problems.
Note that no M&C is performed on GCS-KMF and ILS tools by the CMCF.

6.5.1.2.2 Dynamic Scenarios

6.5.1.2.2.1 Scenario #SFC5.1.1-1: Nominal Ground Elements Monitoring and Control

The first operational scenario consists in ground assets (GMS and GCS) nominal monitoring and control scenario.
General description:
The Ground assets monitoring and control function is performed at three different levels: at element levels, at a central level in each segment and at an overall ground segment level, with interactions between GMS and GCS segments.
This dynamic scenario shows how each segment is performing its monitoring and control and how both segments are linked together.
This dynamic scenario describes only the nominal operations.
The sequential description is detailed below:.

Ground segment monitoring process:
(1a) The local Monitoring and Control instance of GCS each element (at remote site or in the GCC) gathers elements' statuses and health information. It then generates a monitoring message containing all elements' data.
This corresponds to segment functions GCSF5.1.
These data are transferred (message GCS Element Status) to the CMCF for GCS central monitoring.
Ground segment monitoring process:
(1a) The local Monitoring and Control instance of GMS each element (at remote site or in the GCC) gathers elements' statuses and health information. It then generates a monitoring message containing all elements' data.
This corresponds to segment functions GMSF7.4.1.
These data are transferred (message GMS Element Status) to the GACF for GMS central monitoring.
(1b) For GMS, the monitoring information coming from elements is received at GACF level and processed for display to the Ground Segment operator. This corresponds to the function GMSF7.4.2.
For GCS, the monitoring information coming from elements is received at CMCF level and processed to build a GCS status summary message. This message is sent (message MNT_GCSEltsSumStatus) to the GACF for overall monitoring and control function.
The CMCF also sends a TTCF summary status (message TTCF summary status)to the SCCF for inclusion in the GCS schedule generation.
This corresponds to the function GCSF5.2.
(1c) The GCS summary status and GMS status are processed by the GACF to build an overall Ground Segment summary status (messages Overall ground segments (GCS, GMS and networks) status). This message is sent to the GMS MUCF element, and used for:

- GMS short term planning function
- and integrated mission monitoring function

This corresponds to the function GMSF7.4.3.
This process is automatic and periodic.
Ground Segment control process:
In the nominal scenario, control of the equipment is performed via the short term planning in each segment. Coherence of these short term schedules is performed in SFC5.4.2 - Short term planning.
Direct control from GACF or from CMCF is reserved for elements which are not concerned by the schedule (maintenance purpose for example, see next operational scenario).
(2a) For GMS, MUCF sends the GMS short term planning to the GACF (message GMS Short term plan). GACF is responsible to derive this short term plan into a task schedule.

(2a) For GCS, SCCF is responsible to perform the implementation of the contact schedules received from SCPF. These contact schedules include nominal and back-up contact plans. The nominal contact plan is activated automatically, unless the SCCF operator selects the back-up the back-up one. SCCF therefore sends to:

- TTCF element (TTCF local monitoring and control instance) the derived TTC contact directives. This is represented by the message TTC Contact directives.
- CMCF some GCS schedule instructions, if any, is transmitted to CMCF for execution through operator actions. This is represented by the message GCS Schedule operator actions.

(2b) The instructions received are then derived by CMCF into low levels commands for elements. This function is achieved by segment functions GMSF7.4.2. Note that some GMS schedule instructions can be performed by an operator action. These commands are sent to elements' monitoring and control instance.
(2b) The instructions received are then derived by GACF into low levels commands for elements. This function is achieved by segment functions GCSF5.2. These commands are sent to elements' monitoring and control instance.
(2c) The execution of the elements' commands is performed by function GMSF7.4.1 at element level.
The execution of the short term planning is verified through the automatic monitoring process described above.
(2c) The execution of the elements' commands is performed by function GCSF5.1 at element level.

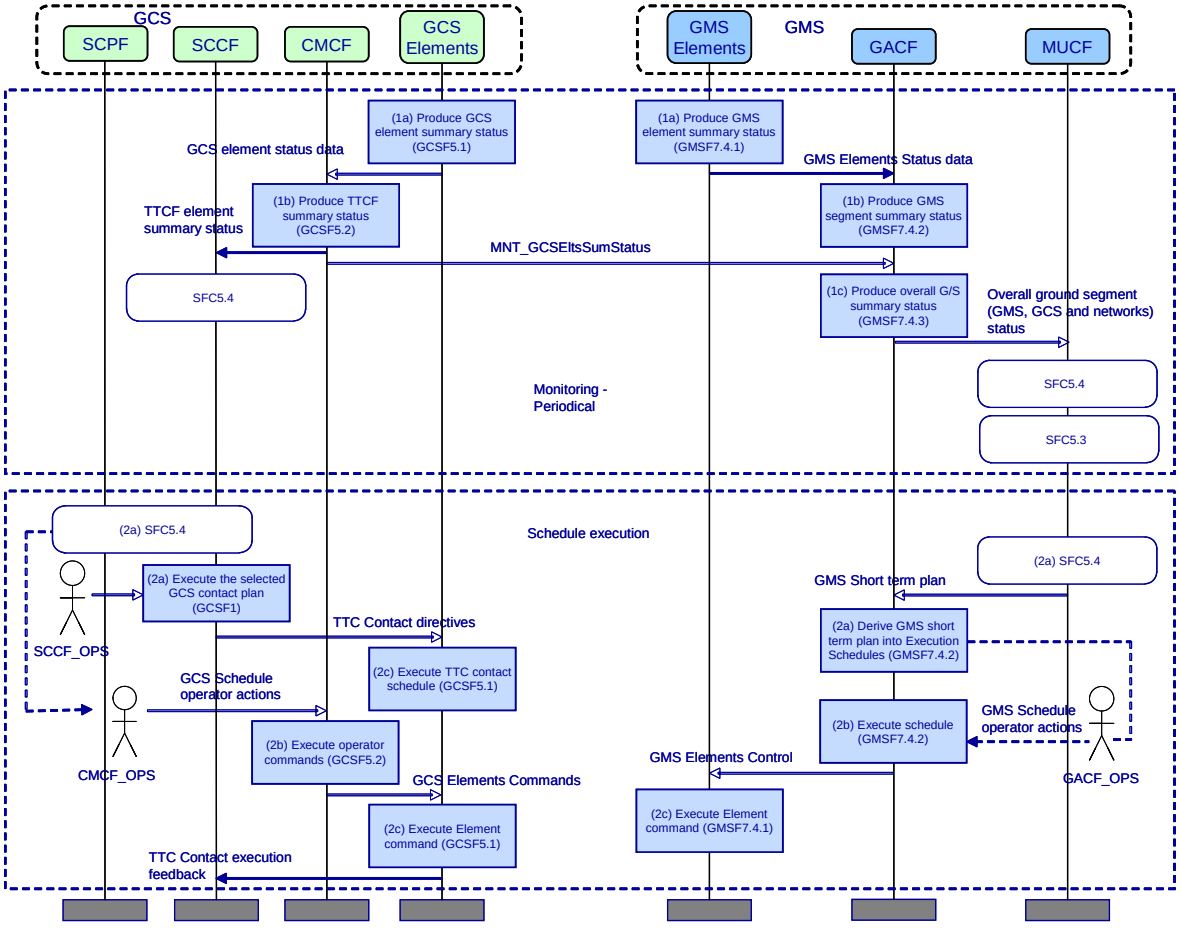


Figure 181 : SFC5.1.1 - operational scenario 1 - Ground assets nominal M&C

6.5.1.2.2.2 Scenario #SFC5.1.1-2: Ground Elements Monitoring and Control Operational Scenario 2: GCS control by GMS

This second operational scenario consists in controlling the GCS by the GMS.
General description:

Although in the nominal operations, control of the equipment is performed via the short term planning as described in the previous operational scenario; it is possible to command directly the all the ground assets from GMS.
Note that this operational scenario is reserved for elements which are not concerned by the schedule (maintenance purpose for example) or in case of contingency.
Note that control over the CMCF by the GACF is not possible during a satellite pass.
The sequential description is detailed below:

GCS segment control by GMS segment:

(1) The GACF is able to generate some high level commands, in order to command the overall GCS segment. The operator selects then the high level command to be sent to the GCS. GACF sends this high level command to the CMCF (message CMD_HighLevel) for execution. This corresponds to segment functions GMSF7.4.3.

(2a) Once the high level command has been received by CMCF, CMCF is in charge to derive it into several elementary commands to be sent to each GCS element. This corresponds to GCS function GCSF5.2.

(2b) The execution of the elements' commands is performed by GCS elements' local monitoring and control instances. This corresponds to the GCS function GCSF5.1 at element level.
Once the derived elements' commands have been properly executed, CMCF sends a command execution message to the GMS. If those elements' commands have not been properly executed, CMCF sends a negative command execution message containing the reason for command failure.
The here-above described acknowledge protocol is included in the complex signal "CMD_HighLevel" (SNMP MIB file) and is therefore not represented in the drawing.

GCS segment monitoring by GMS segment:

Steps 3a, 3b and 4 are identical to the nominal monitoring scenario, as described in the previous dynamic scenario.
The following dynamic view shows how is performed the GCS ground asset control by the GMS.

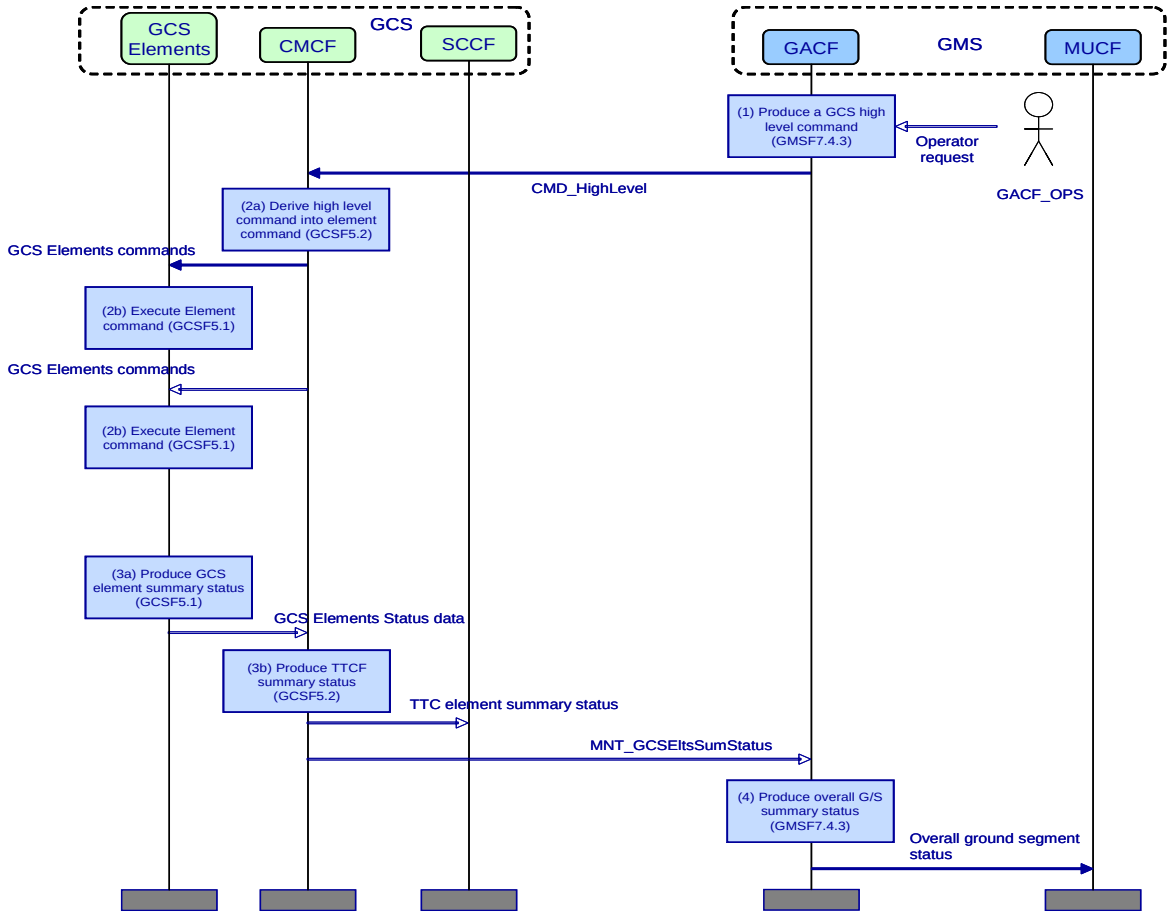


Figure 182 : SFC5.1.1 - operational scenario 2 - GCS control from GMS

6.5.1.2.2.3 Scenario #SFC5.1.1-3: Ground Elements Monitoring and Control Operational Scenario 3: Alarm Management and Troubleshooting

General description:

This dynamic scenario shows how each segment is reporting the element alarms or error to the overall monitoring and control function.

This operational scenario consists in:

- managing the alarms generated by elements
- performing troubleshooting if necessary

The sequential description is detailed below:

Ground segment monitoring process:

(2a) When an equipment in an element becomes faulty, the associated element monitoring and control function generates an alarm and sends it to the relevant central M&C function. This is the purpose of messages GCS elements alarms and GMS elements alarms, respectively generated by GMS or GCS elements' local monitoring and control instances and corresponds to function GMSF7.4.1 and GCSF5.1.

(2b) For GCS, the alarms generated by GCS elements are received by the CMCF and included in the GCS overall status message (message MNT_GCSEltsSumStatus).

This message is sent to SCCF and to GACF so that operator and integrated mission monitoring function can be aware of this alarm.

This corresponds to function GCSF5.2.

For GMS, the alarms generated by GMS elements are received by the GACF and included in the GMS overall status message. The alarms are then reported to the overall M&C function and transferred to mission monitoring and GMS short term planning function.
This corresponds to functions GMSF7.4.2 and GMSF7.4.3.

GMS troubleshooting:

The following process for GMS troubleshooting is implemented:
(3a) As explained in the logical architecture part, GACF is responsible for level-1 operations and therefore troubleshooting operations. In case an alarm is generated by an element, operator, through GACF, can send a status request (message Element Status Request) to this element to get further details on the faulty equipment. This
request allows:

- to focus on specific relevant element parameters through re-configuration of the implemented monitoring mask
- to change the monitoring cycle to allow a more precise monitoring

(3b) Upon reception of this request, the element produces the data, including the critical requested parameters and send them to the GACF, as per normal element monitoring (message GMS Element Status data).
This corresponds to the GMS function GMSF7.4.1.
(3b) Upon reception of this request, the element produces the data, including the critical requested parameters and send them to the GACF, as per normal element monitoring (message GMS Element Status data).
This corresponds to the GMS function GMSF7.4.1.

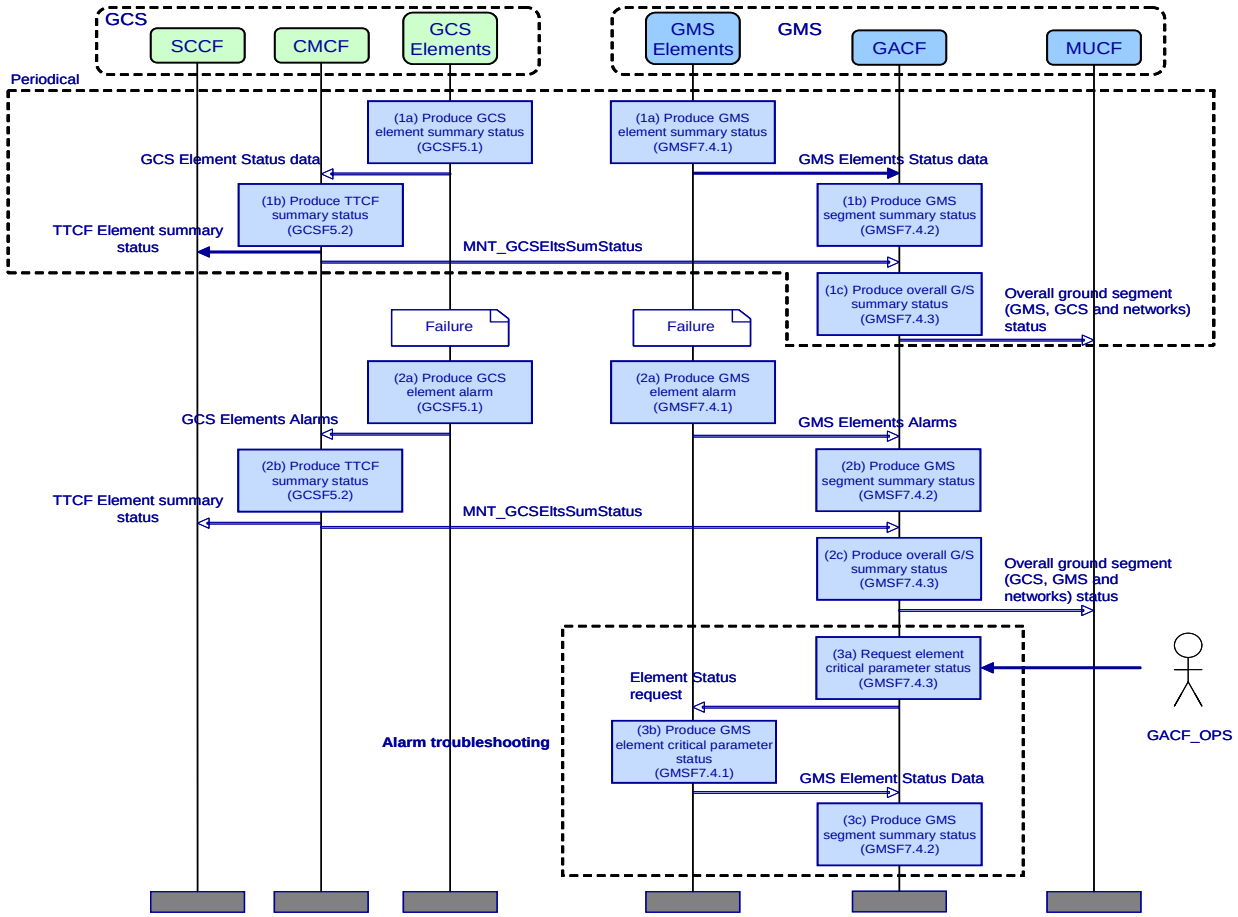


Figure 183 : SFC5.1.1 - operational scenario 3 - Alarms management and troubleshooting

6.5.1.2.2.4 Scenario #SFC5.1.1-4: GCS planned and unplanned handover

This dynamic scenario describes the GCS handover functionality based on the GCC split operational concept described in SYS-RD.76.

Complete GCS Handover

According to the GCC split concept, one single GCS is able to perform all the functions necessary to perform its mission, in case one the GCC site is no longer available. In such a case, the GCS available in the remaining GCC is able to perform all the re-planning and reconfiguration activities to take over in less that 1 hour. Once replanning is performed the currently planned operations of the GCC which failed can be resumed within 15 minutes for the TTC station. Any active TTC contact started by the GCS which failed has to be completed before being taken over by the remaining GCS.

Partial GCS Handover

The GCS is able to share some tasks across the sites (partial hand-over) in order to:

- Avoid complete hand-over in case one site completely fails
- Cope with the unavailability of qualified operator in one site
- Optimize workload between the two sites
- Facilitate the performance maintenance operation

In particular the following tasks can be shared between the sites:

- o Responsibility of the constellation, each GCS being responsible for a number of satellites.
- o Responsibility of the TTC contacts (note: in the event of the failure of a TTC site, the GCS enables the activation of a pre-defined back-up contact plan).

The GCS ensures that the needed context and configuration is kept synchronized on both sites in order to smoothly resume any function in case an unplanned hand-over is needed.

6.5.1.2.2.5 Scenario #SFC5.1.1-5: GMS planned and unplanned handover (UNCLA)

This dynamic scenario describes the operational steps to execute the GMS handover functionality based on the GCC split operational concept described in SYS-RD.76 and GMS input GMS-RD.59.

The objective of this scenario is to transfer the monitoring and control of the mission from one GCC to the other, while minimizing the impact on the other system functional chains for the navigation, timing, SAR and CS users.

Planned handover: (example from GCC-I to GCC-D)

Step 1 (Before the handover, nominal GCC-split operations):

- All GMS elements in GCC-D are switched ON. The detail of GMS element configuration configuration is provided in SYS-RD.76.
- Navigation determination chain in GCC-D (OSPF, MGF) is up and running and in RUNNING mode. Its reference GST is the master GSS-PTF in GCC-D.
- Global status of elements in GCC-D are remotely monitored from GACF-I and MUCF-I.
- Functional and Performance status of navigation determination chain in GCC-D is performed remotely from GCC-I with checks as frequent as for local elements in GCC-I.
- No MSF daily operations are executed on MSF-D.

Step 2 (Preparation of the handover):

- From GCC-I, command the OSPF-D to switch from GSS-PTF-D to GSS-PTF-I.
- If not already executed, start the synchronisation of GACF-D/GNMF-D/GNMF-s-D with GCC-I.
- If not already executed, start the synchronisation of MUCF-D/SPF-D with GCC-I.

Step 3 (Handover execution, requires OPS team relocation in GCC-D):

- SPF handover to GCC-D : Reconfigure SPF-D (currently slave) as master to reconnect external entities.
- OSPF-D shall be loaded with GGTO and TAI-GST files related to PTF-I.
- Put GCC-I navigation determination chain in RUNNING.
- Put GCC-D navigation determination chain in OPERATIONAL.
- Switch GACF and MUCF master role from GCC-I to GCC-D.
- Change the MNE-GCC mode in RUNNING in GCC-I.
- Change the MNE-GCC mode in OPERATIONAL in GCC-D.
- If no current plan is available at MUCF GCC-D, compute a backup plan and distribute to ULS.

Step 4 (Post-handover execution)

- Start MSF-D daily computations
- PTF master handed-over: command PTF to switch master role for PPTO from PTF-I to PTF-D). PTF-D as master.
- From GCC-D, command OSPF-D to switch from GSS-PTF-I to GSS-PTF-D.
- OSPF-D shall load GGTO and TAI-GST files related to PTF-D.
- Execute nominal planning operations from MUCF-D.

Unplanned handover: (example from GCC-I to GCC-D)
Step 1 (Before the handover, nominal GCC-split operations): as per planned handover

Step 2 (Preparation of the handover): skipped since it is an unplanned handover

Step 3 (Handover execution, requires OPS team relocation in GCC-D):
As soon as conditions are met to operate from GCC-D in terms of OPS personnel:
- Send a command from GCC-D so that GACF-D takes the master role..
- Perform sanity checks to verify that all remote elements can be controlled from GCC-D.
- If no current plan is available at MUCF GCC-D, compute a backup plan and distribute to ULS.
- Change the MNE-GCC-D mode in OPERATIONAL.
- Put the navigation determination chain in OPERATIONAL.

Step 4 (Post-handover execution)
- Start MSF-D daily computations
- Execute nominal planning operations from MUCF-D.

Partial handover: A partial switch is possible for the following functional chains:
- MUCF offline analyses (GMPA): can be allocated to the second GCC
- Navigation determination chain (OSPF,MGF). It is possible to switch the OSPF+MGF chain from one GCC to the oter while maintaining the M&C in the master GCC.
- Timing chain (PTF or GSS-PTF)
  - Following a GSS-PTF failure, it is not mandatory to handover the PTF. Assuming a GSS-PTF-I failutre in the masterGCC-I, the OSPF-I will be commanded to switch from GSS-PTF-I to GSS-PTF-D while maintaining the loading of GGTO and TAI-GST files related to PTF-I. The performance degradation will be minor and linked to PPTO accuracy.
  - However, in case of non-recoverable PTF failure, it is necessary to handover the complete timing chain (PTF+GSS-PTF).

6.5.1.3 SFC5.1.2 Network management

6.5.1.3.1 Physical Static View

6.5.1.3.1.1 Overall Physical Description

The functional organization of the Ground Network management, described in the DDF1 section, is implemented in the physical architecture, as shown in Figure 202 and described below:

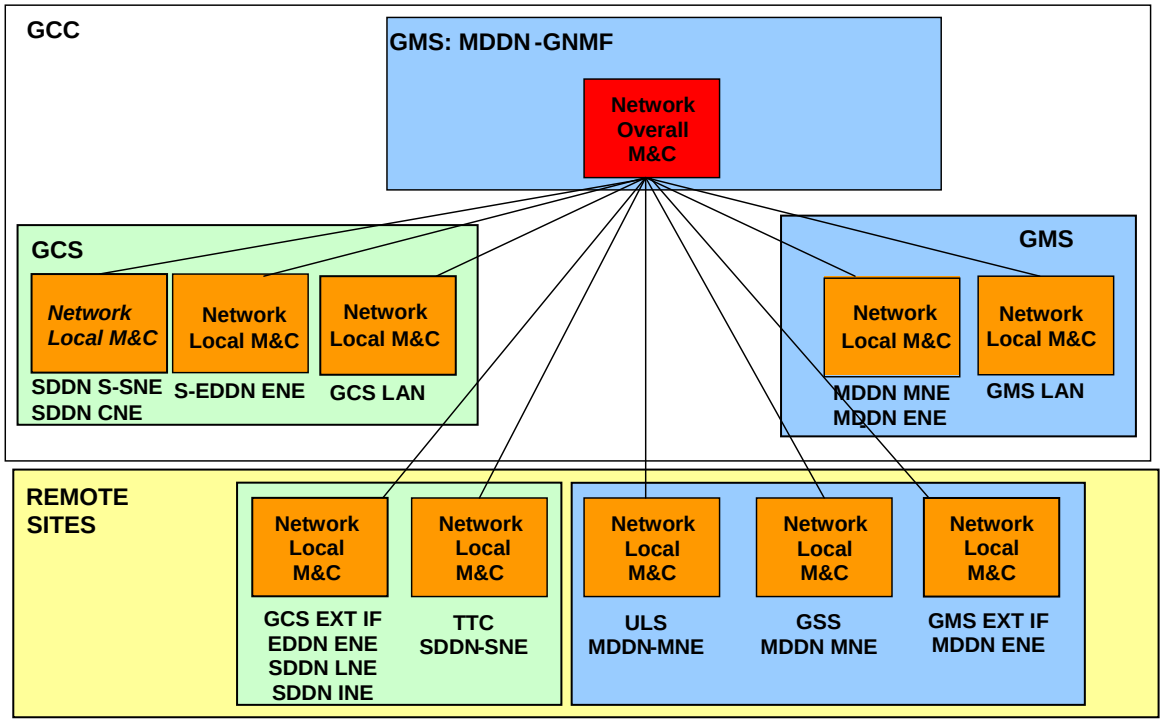


Figure 184 : Network M&C physical organization

- The network central monitoring and control function is supported by the MDDN-GNMF: Galileo Network Management facility, in charge of the overall network management (MDDN, M-EDDN, SDDN and S-EDDN).

The network element monitoring and control function is supported by the following elements:

- The GMS-MDDN, Mission data Dissemination Network, linking the GCC and all the remote sites. In addition to the GNMF, MDDN is composed of:
  - The MNEs (Mission Network Equipment) are making the interface between the GMS elements and the WAN network (GDDN).
  - The M-ENEs (Mission External Network Equipment) are making the interface between the GCC and the GMS external entities.

- SDDN, Satellite Data Dissemination Network, which is the network linking GCC and the remote elements (TTCF, SATMAN, TGVF). The SDDN is composed of:
  - S-SNE (Secure Satellite Network Equipment, located at the GCCs) and SNE (Satellite Network Equipment, located at remote sites) providing the interface between the GCS elements and the WAN network (GDDN).
  - CNEs(Common Network Equipment, located at the GCCs) , providing the interface with the Inter GCCs link

- S-EDDN, the Satellite External Data Distribution Network, providing links for the transfer of files and databases using HTTPs, comprising S-ENEs (Satellite External Network Equipment) to support the interface between the GCC and GCS external entities (SATMAN, ENOC, IOT sites, TGVF).

- GDDN, Galileo Data Dissemination Network, which is in fact the WAN part of the MDDN, SDDN and S-EDDN.

The network management functional chain also impacts the following external entity:
- CSP, Communication Service Provider, responsible of the GDDN management and GDDN service level assessment.

Note that elements' LAN are monitored by the elements' relevant M&C process.

The GMS and GCS networks (MDDN, M-EDDN, SDDN and S-EDDN) are managed for fault, configuration, performance and security aspects, by the GNMF component of the MDDN. The GNMF is also in charge of the reconfiguration of the networks connectivity.

This information is passed to:
- the GACF, which is able, on operator request to display the current technical monitoring data related to the connection between two elements

- the MUCF, in order to include the network status inside the mission monitoring function
- the CMCF for GCS network configuration
- the CSP (external interface) for GDDN communication service level assessment.

The following figure shows the monitoring and control static diagrams, describing the links between all GCS and GMS elements as well as with CSP external entity.

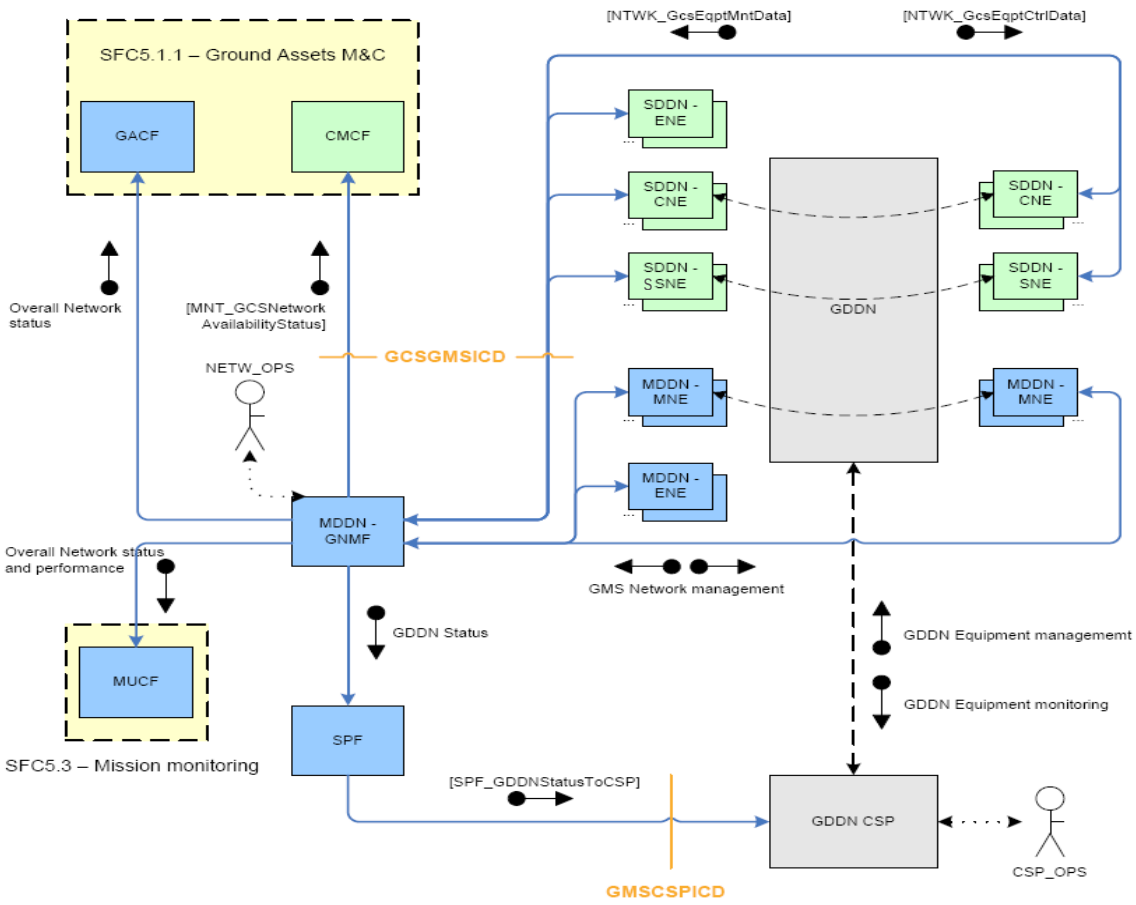


Figure 185 : SFC5.1.2 static view

High level Signal	GCSGMSICD Signal	Protocol	From	To
NTWK_GcsEqptMntData	CMD_CmdNeAck CMD_CmdComplete CMD_CmdNeStatus	SNMP / MIB	GCS S-SNE GCS SNE GCS CNE GCS LNE GCS INE GCS ENE GCSLAN	GMS-GNMF
	MNT_ElmtAlert MNT_ElmtNotif	SNMP / MIB	GCS S-SNE GCS SNE GCS CNE GCS LNE GCS INE GCS ENE GCSLAN	GMS-GNMF
	CNF_NeCommonConf	SFTP / XML	GCS S-SNE	GMS-GNMF

High level Signal	GCSGMSICD Signal	Protocol	From	To
	CNF_NeMiscConf ARCH_NeLog		GCSSNE GCS CNE GCS LNE GCS INE GCS ENE GCSLAN	
	CMD_ElmtCmd	SNMP / MIB	GMS-GNMF	GCS S-SNE GCS-SNE GCS-CNE GCS LNE GCS INE GCS ENE GCSLAN
NTWK_GcsEqptCtrlData	MNT_ElmtMntDataReq	SNMP / MIB	GMS-GNMF	GCS S-SNE GCS-SNE GCS-CNE GCS LNE GCS INE GCS ENE GCSLAN
	ARCH_NeLog CNF_NeCommonConf CNF_NeRtGwConf CNF_NeMiscConf	SFTP / XML	GMS-GNMF	GCS S-SNE GCS-SNE GCS-CNE GCS LNE GCS INE GCS ENE
	CNF_NeRtGwConf	SFTP / XML	GMS-GNMF	GCS-CNE

Table 93: M&C signals between GCS / Network Equipments and GMS / MDDN
Only inter-segment flows are detailed in this diagram. Intra segment flows are described in the relevant segment documentation.

6.5.1.3.1.2 General considerations: network performance measurement

The network performances are measured by the MNEs and overall performance is assessed by the GMS through its GNMF element.

The Performance Monitoring agents at the MNEs will collect performance data of all RT traffic streams going across the GDDN and generate QoS stamps that will be used by the GNMF to derive end-to-end performance data and check the fulfilment of the CSP Service Level Agreement (SLA).
Three types of measurements are made as follows.

- Delay (latency):

- The traffic latency is measured by the MNEs, taking into account the time at exit from the sending site with that when entering the receiving site and using synchronization of all Galileo sites to the GST. This latency measurement is of particular importance for operational flows and associated budgets.
- If the delay is out of limit this leads to the generation of an alarm, which is forwarded at the end of the M&C chain to the GNMF.

- Traffic integrity, or packet loss measurement
  - The M- and SNEs at each side of the links are regularly comparing the amount of packets of each type. Moreover, by means of a CRC check the received packets are controlled on errors.
  - If packet loss is over the limit, this leads to the generation of an alarm, which is forwarded at the end of the M&C chain to the GNMF.

- Network availability
  - Using the detected packet loss and time-stamps of received packet as described above, actual reception and sequence of each critical packet is checked.
  - As a result, even interruptions that last only a few seconds can be detected, and a highly precise availability can be calculated.

This availability control mechanism focuses on the WAN network performances provided by the CSP, since the tagging and evaluation are done right at the WAN endpoints.

It has to be noted that the GNMF transfers:
- The GCS network performance to the CMCF, i.e. mainly the GCS network links state and mode

- The GDDN performance assessment to the CSP

6.5.1.3.1.3 Network management: GNMF (part of MDDN)

The MDDN-GNMF is MDDN component in charge of:
- Network HW component monitoring and control (for any network equipment, i.e. M-EDDN, MDDN, SDDN and S-EDDN including the GCC LAN). This task consists in:

- Fault Management, i.e. management of the status of the network equipment (gathered from receiving SNMP traps (alarms) and performing regular polling of relevant equipment status information). A summarized fault management information is forwarded to GACF.
- Network M&C, i.e. network administration (traffic priority selection), discovery & topology, inventory management, fault management, status monitoring, troubleshooting, event console, etc …
- Network performance monitoring by permanent monitoring of the MDDN network flows between GSS-MNE and GCC-MNE, between GCC-MNE and ULS-MNE, across GCCs and between MDDN and M-EDDN.
  The GNMF provides key indicators on availability, latency, packet/message loss and derived continuity for the critical integrity and navigation communication services, as explained above.

For this purpose, the MDDN-GNMF interfaces with:
- MDDN MNEs for reception of the links performance measurement and equipment status
- GCS SNE, S-SNE, INE, LNE and ENE s for reception of equipment status

- GACF for transfer of the overall network availability status, monitored from the MNEs and SNEs and required as part of the overall monitoring and control (note that no control of the networks from GACF is possible)
- MUCF for transfer of some network performance assessment and network availability status required for integrated performance monitoring
- CMCF for transfer of the GCS network availability status, since GCS networks is not directly managed by the CMCF
- SPF for transfer of the GDDN WAN performance assessment, as performed by the GNMF

6.5.1.3.1.4 Network monitoring: GACF

The MDDN interface with the GACF to:
- allow the physical interconnection of the GACF Element to the RT LAN for data capture and to the NRT LAN for data exchange for other GMS Elements.
- provide to the GACF the monitoring and control of the GNMF part of the MDDN.

The MDDN-GNMF interfaces with the GACF for transferring the overall network availability status, monitored from the MNES and SNEs and required as part of the overall monitoring and control (note that no control of the networks from GACF is possible)

6.5.1.3.1.5 Network performance monitoring: MUCF

The MUCF is in charge of the overall mission performance assessment and receives from MDDN information on:
- the MGF selected for message dissemination
- MDDN and SDDN Networks high-level status.

6.5.1.3.1.6 GCS Networks monitoring: CMCF

Function Deleted. GCS network monitoring is performed from GNMF.

6.5.1.3.1.7 Interface with external entity (CSP): SPF

The SPF is the GMS element in charge to interface with the external entities. Therefore, it has to interact with the CSP for GDDN performance assessment transfer.
For this purpose, the SPF interacts with:
- the MDDN-GNMF in GMS for GDDN performance assessment reception
- CSP to transfer the information received by the GNMF

6.5.1.3.2 Dynamic Scenarios

6.5.1.3.2.1 Scenario #SFC5.1.2-1: Nominal network management

This operational scenario consists in managing the overall Galileo ground network.
• **Networks Monitoring process:**

(1a) GCS networks are monitored (performance measurement as well as network equipment monitoring) by GCS function GCSF5.3, i.e. inside SNEs. GMS networks are monitored (performance measurement as well as network equipment monitoring) by GMS function GMSF7.4.4, inside MNEs.
(1b) GCS networks monitoring information is transferred from GCS SNEs to GMS GNMF (message NTWK_GcsEqptMntData), which is in charge to combine these information with GMS network ones in order to assess the Galileo ground network performance.
This corresponds to GMS function GMSF7.4.4.
(2) Note that the GCS networks availability is taken into account in the short term plan elaboration processing information given by GMS in the mid term plan.
The network summary status is also passed to the GACF, for overall monitoring and control and transferred to MUCF for mission monitoring.
The network monitoring process is continuous.
This is described in dynamic scenario “Nominal M&C”.
(3a) Using the results of the network performance measurement performed by the MNEs, the GNMF is also in charge to assess the GDDN performance and required service fulfillment. The result of this assessment is transferred to the SPF.
This corresponds to the GMS function 7.4.4.
(3b) The SPF receives the GDDN network performance information and manages the interface with the CSP for data transfer. Once communication has been setup, the SPF transfers the GDDN performance, as assessed by the GNMF (message SPF_GDDNStatusToCSP).
This corresponds to the GMS function GMSF12.
(3c) Upon reception of the GDDN network performance data, the CSP Network Monitoring System (NMS) shall assess the level of the service achieved by the GDDN. It includes production of reports,…
This corresponds to the CSP function CSPF1.

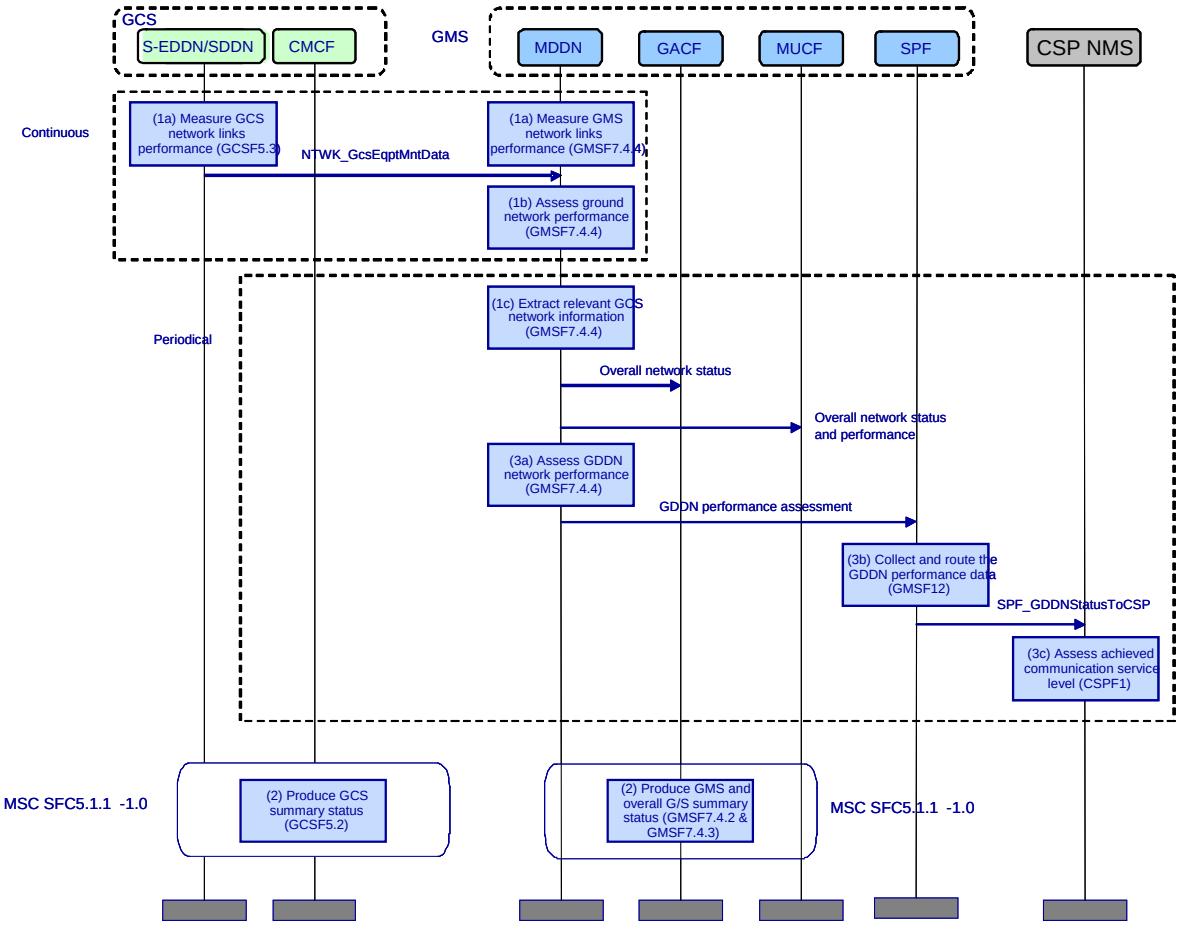


Figure 186 : SFC5.1.2 - operational scenario 1 - Network nominal management

6.5.2 SFC5.2 Satellite Monitoring and Control

6.5.2.1 Introduction

The objective of the Satellite Monitoring and Control SFC is to enable both on-line and off-line satellite monitoring and control functions for both routine and special operations.
The following section describes the physical implementation of the SFC. It is based on the following documentation:

- System documents:
  - System operational concept: MOCD (SYS-RD.6)
  - System concept for the use of signal status flags (SYS-RD.10).

- GCS documents:
  - GCS DDF (GCS-RD.1)
  - GCS Design Justification File (DJF) (GCS-RD.2)
  - SCCF to TTCF ICD (GCS-RD.21)
- SSeg documents:
  - IOV Space Segment Design Description and Design Justification File (SS-RD.1)
  - Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1)
  - Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2)
  - Avionics DDJF (IOVSS-RD.4 and FOCSS-RD.07)
  - Galileo IOV PUS Definition (IOVSS-RD.7)
  - FDIR Analysis (FOCSS-RD.3)

- GMS documents:
  - GMS DDF1 (GMS-RD.1)
  - GMS DDF2 (GMS-RD.2)
  - MUCF DDF 1 (GMS-RD.4)
  - GACF Element Requirement Document (GMS-RD.9)
  - MUCF ICD (GMS-RD.11)
  - GACF ICD (GMS-RD.12)
  - GSS ICD (GMS-RD.32)

6.5.2.2 SFC5.2.1 On-line Satellite M&C

6.5.2.2.1 Physical Static View

6.5.2.2.1.1 Overall Physical Description

An overview of the On-line Satellite M&C SFC implementations is provided in the following figures (IOV and FOC Design).

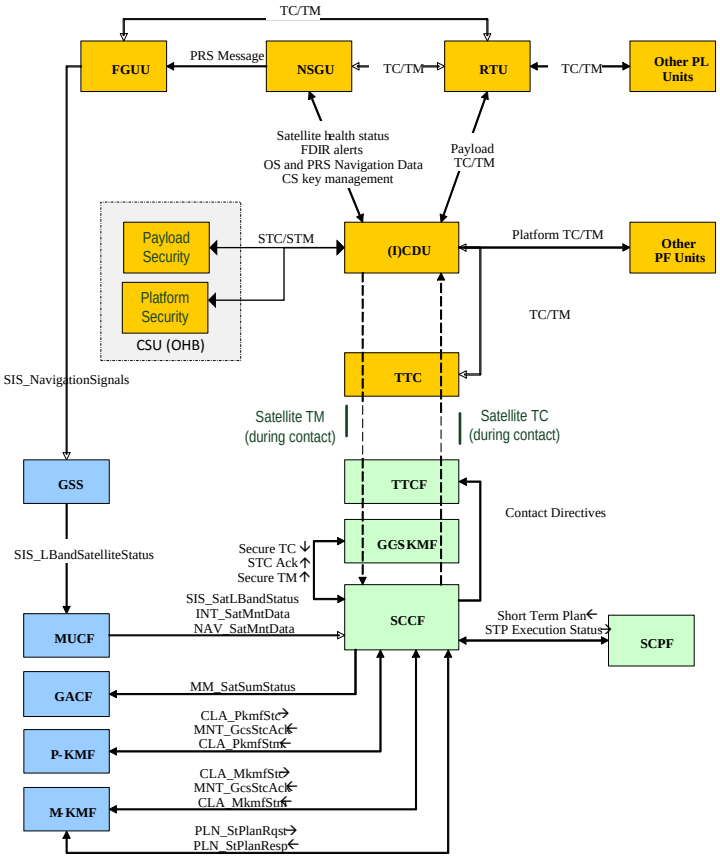


Figure 187 : SFC5.2.1 Physical Static View

6.5.2.2.1.2 Schedule Generation: SCCF

Upon receipt of the GCS Short Term Plan (which includes the TTCF contact schedule) from the SCPF, the SCCF is responsible for generating the following executable schedules:

- TT&C station contact directives. These are used by the SCCF automation system to initiate and terminate individual contacts as required.
- Ground asset configuration schedule to be executed by the SCCF automation system (this includes the schedule of both fully automatic and manual operations)
- Spacecraft command schedule to be uplinked to each satellite. The on-board schedule generated by the SCCF is a SCOS-2000 command stack file generated from command sequences defined in the TM/TC database, and with time-tags associated with each command according to timing data provided in the Short Term Plan.

Normally these schedules will run autonomously (except where manual intervention is preplanned). However provides the functionality to allow these planned operations to be interrupted to allow manual intervention.

6.5.2.2.1.3 Satellite TC / TM, STC / STM and Health Status Processing: SCCF

The SCCF will provide real-time TC and TM processing including:

- Preparation and transmission of TCs, verification of command execution, and on-board command queue management
- S-band Telemetry processing including reception, interpretation, calibration and out-of-limits checking.

- Reception and processing of L-band satellite health status information extracted from the PRS navigation message by the GMS (MUCF) and forwarded to the SCCF.
- Reception and processing of data provided by MUCF related to ODTS processes in the GMS (see SFC5.3 for more details).
- Reception and acknowledgement of STCs from both the GCS KMF and also the GMS KMFs for uplinking to the spacecraft as part of the normal TC stream.
- Reception of STM from each spacecraft downlinked in the normal TM stream and forwarding to the relevant KMF.

6.5.2.2.1.4 Operations Progress Monitoring: SCCF

The SCCF provides a real-time display of the status of the planned operations for each spacecraft in the constellation, together with the summary status of each satellite and ground station. It monitors the status of operations executed both on the ground and on-board the spacecraft and compares this to the active plan to create an overview of the progress of constellation operations.

6.5.2.2.1.5 Constellation Status: SCCF

The SCCF generates a consolidated overview of the status of all satellites in the Galileo constellation and provides this to both the operator locally and also to the GMS (GACF) for input to mission monitoring and planning functions.

6.5.2.2.1.6 Ground S-Band TT&C Uplink and Downlink: TTCF

The TTCF provide the remote TT&C stations that provide S-band contact with each spacecraft in the constellation. In the context of this SFC, the TTCF is responsible for :

- Providing an interface to the SCCF for the exchange of Telecommands and Telemetry.
- Providing the S-band Uplink transmission of Telecommands and Downlink reception of telemetry for the management of the Galileo satellite constellation. In particular:
  - Modulation, up conversion and amplification of all telecommands.
  - Amplification, down conversion, demodulation and transmission decoding of all received telemetry.
- Performing two-way range measurements for use in the calculation of spacecraft orbits, including support for the following ranging modes:
  - Spread Spectrum ranging using the PN spreading code
  - ESA code ranging when operating in Standard Mode
- Providing auto-tracking antennas supporting operation in both spread spectrum and PM modes.
- Supporting the CCSDS SLE protocol on the interface to SCCF in order to receive telecommands (ie F-CLTUs) from the SCCF and to send telemetry (ie RAFS) to the SCCF.

- Receiving contact directives from the SCCF which provide the instructions that control TT&C Station activities in order to acquire, track, and range satellites. The directives include for example start contact, activate carrier, activate Doppler measurement, configure station for Phase Modulation or Spread Spectrum etc.

6.5.2.2.1.7 Satellite S-Band Uplink and Downlink: TTC

The satellite TT&C subsystem supports the two way transfer of data between the satellite and the ground. In particular it provides an S-band antenna and transponder that support the following:

- Uplink reception of telecommands from ground including:
  - Tracking and acquisition of the uplink RF signal
  - TC demodulation
- Forwarding of TCs to PFSU (for IOV satellites) or CSU (for FOC satellites)
- Transmission of telemetry to ground including:
  - Reception of TM data stream from the PFSU (for IOV satellites) or CSU (for FOC satellites)
  - Convolutional coding of the TM data
  - Modulation of this data onto the downlink RF carrier
- Operation in the following RF modes:
  - Standard PM mode
  - Spread spectrum mode
- Two way ranging including the following modes:
  - ESA standard code ranging (in PM mode)
  - Tone ranging (in PM mode)
  - Spread spectrum ranging (in spread spectrum mode)

6.5.2.2.1.8 Ground COMSEC TC / TM: GCS-KMF

The GCS-KMF supports this SFC by providing COMSEC protection for TC/TM in terms of:

- Encryption/decryption;
- Integrity check application/verification;
- Authentication check application/verification;
- Anti-replay check application/verification.

These functions are supported by the GCS-SM which is the physical cryptographic device.

6.5.2.2.1.9 Satellite COMSEC TC / TM: PFSU (CSU)

The PFSU (for IOV satellites) or CSU (for FOC satellites) supports this SFC by providing COMSEC protection for TC/TM at transfer frame level in support of a PROTECTED mode on the S-band link which addresses the following security objectives:

- Confidentiality protection
- Anti-replay protection
- Authentication Protection

In particular the PFSU (for IOV satellites) or CSU (for FOC satellites) provides:

- Decryption of all telecommands prior to transmission to the ICDU/CDU
- Encryption of all telemetry prior to transmission to the TTC

Note that the PFSU (for IOV satellites) or CSU (for FOC satellites) also implements a CLEAR mode supporting processing of TC and TM without COMSEC protection. .

6.5.2.2.1.10 On-board Automation: ICDU/CDU and Avionics Software

The ICDU (for IOV satellites) / CDU (for FOC satellites) and Avionics Software support the automation of the satellite monitoring and control functions through the following:

1. Implementation of the master time line (MTL) which allows time tagged telecommands to be stored on-board. The ICDU/CDU stores each time tagged TC and sends it to the relevant

spacecraft unit when the time tag is due (with 200 ms resolution on the on board execution time).

2. Supporting on-board control procedures (OBCPs) which consist of a list of TCs separated by delays. OBCPs allow a sequence of commands to be executed by a single command. Note that OBCP execution can also be triggered by on-board events (e.g. initialisation sequence).
3. Providing the core of the on-board Failure Detection Isolation and Recovery (FDIR) functionality including:
    - Failure detection through software monitoring (eg of house-keeping parameters, consistency checks etc), processor system traps, built in software error detection and hardware alarms (eg temperature limit)
    - Failure isolation through processing of monitored parameters, consistency checks down to equipment, function, process or interface level
    - Failure recovery through hardware reconfiguration (eg switch to redundant unit or part), software reboot, processor restart or change to satellite mode (eg transition to safe-mode)
4. Provision of a redundant safeguard memory to store default on-board system configuration data
5. Automated routing of TC/TM and STC/STM to and from the NSGU, -PLSU (IOV only) RTU, PFSU (IOV only) and CSU (FOC only) via the Mil-Std 1553B bus.
6. Automated management of all satellite units including in particular:
7. Platform units (including Thermal Control, Power, TT&C and PFSU (IOV only) / CSU (FOC only)
8. Payload units (including PLSU (IOV only) / CSU (FOC only) NSGU and RTU)
9. Automated attitude and orbit control (based on earth sensors, sun sensors, gyros, reaction wheels and magnetic torquers) and propulsion sub-system (including 8 thrusters).

6.5.2.2.1.11 On-board Observability: ICDU/CDU and Avionics Software

The ICDU (for IOV satellites) / CDU (for FOC satellites) and avionics software provide the ground with the means to monitor and analyse the satellite behaviour during its life time and all mission phases. This is achieved through the provision of all necessary telemetry.
Observability of satellite unit status during out of contact periods is ensured through the provision of on-board packet stores in the ICDU/CDU (TM-memory).
The TM memory can store telemetry data collected in 2 orbits and supports both routine telemetry download or a flexible selection and retrieval of detailed telemetry for ground analysis.
L band TM data are provided as part of the encrypted G/NAV message and collected by the GSSs equipped with GRC-P receivers. L band TM data are provided as soon as the payload is switched on, when dummy G/NAV messages are transmitted (before the PRS service is operational).
L band TM data supporting satellite M&C functions consist of:

- C band uplink acquisition status, reporting for each of the six different C band uplink channels (only one enabled) the RX channel status (No signal/ Bit lock/Code lock/Frame synch). Two bits are allocated for each channel, for a total of 12 bits. This information is refreshed every minute and is available at the MUCF level (MUCF_SatStatusForMkmf_SIS).
- o A reduced set of satellite TM parameters (Health Status data) regarding:
    - o on board Subsystems (e.g., AOCS Mode, TTCF mode),
    - o TC execution status,

- a discrete set of High level alarms provided by the on board FDIR for a total of 16 bits,
- 16 programmable alarm bits based on board TM parameter checks to be defined by ground controllers via PUS Service 12.

These data (see table below) are refreshed every minute and transferred from the MUCF to the common constellation domain at SCCF level (SIS_SatLbandStatus)

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6.5.2.2.1.12 IOV Satellite Health Status Summary: ICDU and Avionics Software

For IOV satellites the ICDU (is responsible for generating the satellite health status summary and FDIR alerts and sending these to the NSGU for inclusion in the PRS message.
The Health Status data (32 bits in total) is sent by the (I)CDU to the NSGU via the MILBUS interface.
The parameters included in the satellite health status words are summarised in the following table:

Satellite Health Status Contents	**Size (bits)**
Satellite Health Status Alert (current satellite mode)	2
Payload Summary FDIR Status	1
PHM Clock Health Status	1
RAFS Clock Health Status	1
CMCU Phase Comparison	1
PFSU Health Status	1
PLSU Health Status	1
Mil-bus Health Status	1
EPS Health Status	1
TCS Health Status	1
AOCS Health Status	1
Summary Telecommand Processing Status	1
Reconfiguration Memory Status	1
TT&C Transponder Mode (ie Spread Spectrum or STD mode	1
Spare bit	1
Programmable Alerts (based on PUS service 12)	16
Total	**32 bits**

The health status word includes 16 programmable alerts that can be configured using PUS service 12. This service provides on-board parameter monitoring and allows parameters contained in the on-board data pool to be monitored against a specified limit or an expected status value and react with an event report if the parameter is out-of-limit.
The satellite health status word is transmitted from the ICDU/CDU to the NSGU every second.

6.5.2.2.1.13 FOC Satellite Health Status Flags

As per approved GS-OSW-OHB-RFW-0002, version V1 of the on board SW of FOC (OHB) satellites will provide only a subset of the 32 monitoring bits to be transmitted on L band, i.e.:

- The 12 bits allocated to the C band uplink acquisition status, reporting the on board receiver status for each of the six different C band uplink channels.
- One bit for the S Band Transponder Mode (Standard /Spread spectrum mode)
- One bit for the TC processing status, reporting whether:
  - all TC in the past second have been processed successfully or
  - S band receiver is unlocked or at least one TC in the last second has been rejected.
- The 16 programmable alert bit based on check defined by the Operations team on on board TM parameters via PUS service 12.

The additional on board subsystem FDIR alarms will be only available in SW Version V2.

6.5.2.2.1.14 Satellite Health Status Summary: NSGU

The NSGU is responsible for including the satellite health status information in the PRS message. The 32 bits provided by the ICDU/CDU (see above) are multiplexed by the NSGU onto a 2 bit field within the G/NAV message format. For this multiplexing, a 60s repeat cycle is used where by in total 40 bits of health status information can be transmitted to the ground each minute (of these 40 bits only 32 are currently defined to carry meaningful health status data and 8 bits are spare). In this way the full 32 bits of meaningful health status information are transmitted to ground every minute.

6.5.2.2.1.15 Providing PRS Message Data to MUCF: GSS

The Galileo Sensor Station (GSS) receives and demodulates the four carrier frequencies (L1, E5a, E5b & E6) carrying the four navigation services (OS, SOL, CS & PRS) transmitted by each Galileo spacecraft.
The GSS obtains the SIS observables and navigation data from these signals (including the PRS message data) and distributes this to the GCC via the MDDN. One of the recipients of this data within the GCC is the MUCF.
As the GSS transmits the raw PRS message data once per second (as received from the spacecraft), the GSS will send to the MUCF the 32 bits of health status data spread across multiple individual messages each carrying 2 bits of data derived from the SIS.

6.5.2.2.1.16 Extraction of Satellite TM for Delivery to GCS: MUCF

The MUCF extracts the satellite L-band telemetry from the PRS (G/NAV) message data (provided to the MUCF by the GSS) and sends it to the GCS (SCCF).
To do this the MUCF will extract the 2 bits of health status data from the individual PRS messages received from GSS, consolidate these over a 60s period, add 8 bits and send the 40 bits to the SCCF as a single file.
The health status bits transported in the G/NAV message will be available also before the PRS service is operational, i.e., when G/NAV dummy messages are transmitted from the satellites.
The MUCF will then transmit the health status information (ie of 40 bits) to the SCCF every once every minute. Note that if the reception of the PRS message is interupted by a G/NAV integrity alert, the validity status of the message sent to SCCF is set to “Error” to ensure that no erroneous data is processed by SCCF.

6.5.2.2.2 Dynamic Scenarios

6.5.2.2.2.1 #SFC5.2.1-1.1 Intermittent satellite M&C via S-band (nominal case)

This is the nominal scenario where satellite monitoring and control is performed via the S-band TT&C link according to the GCS Short Term Plan

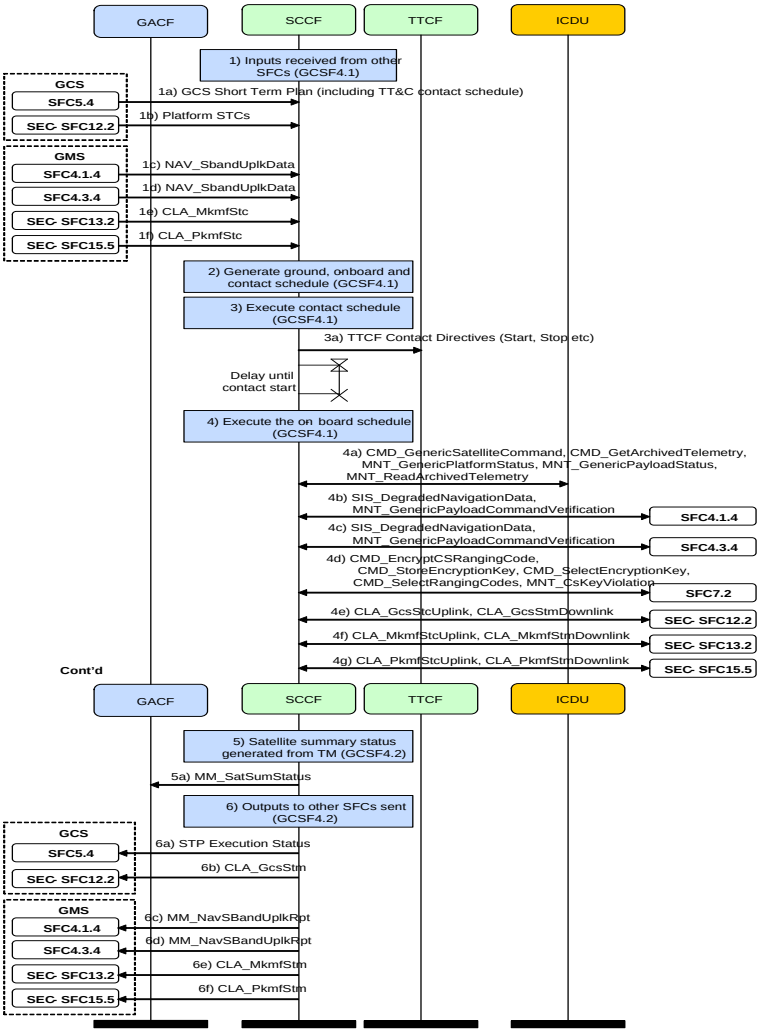


Figure 188 : #SFC5.2.1-1.1 Intermittent Satellite M&C via S-band (nominal case)

This scenario illustrates the nominal satellite M&C performed intermittently via the S-band TT&C link.

1. A number of inputs are sent to the SCCF from other SFCs. These include:
   a) The GCS Short Term Plan (STP) sent by the SCPF (see also SFC5.4). The STP may contain OPS manual inputs.
   b) STCs to be sent to the platform security unit sent by the GCS-KMF (see also SEC-SFC12.2). OPS prepare the STC in advance.

   c) OS navigation data to be sent to the NSGU (sent by the MGF - see SFC4.1.4)
   d) PRS navigation data to be sent to the NSGU (sent by the MGF - see SFC4.3.4)
   e) STCs to be sent to the Payload security unit (PLSU / CSU)- mission part (sent by the M-KMF - see SEC-SFC13.2). OPS prepare the STC in advance.
   f) STCs to be sent to the Payload security unit (PLSU / CSU) - PRS part (sent by the P-KMF - and SEC-SFC15.4). OPS prepare the STC in advance.

2. The SCCF generates the ground, on-board and contact schedules based upon the constraints defined in the STP.
3. The SCCF executes the contact schedule and sends contact directives to the TTCF. These include contact start stop times, ranging directives (start/stop time), carrier directives etc.
4. Once the contact start time has been reached, the SCCF executes the on-board schedule, uploading telecommands to the spacecraft and processing telemetry received from the spacecraft. IThis scenario captures some of the key exchanges from a system perspective (not exhaustive):
    a) Commands are encrypted by the GCS-KMF (providing L1 protection), then sent to the ICDU/CDU to perform platform and payload management.
    This includes requesting the dump of all TM archived on board in between contacts. Satellite platform and payload health status telemetry is downlinked;

    b) OS navigation data for use in degraded mode is sent to the NSGU with successful reception confirmed via telemetry (see SFC4.1.4);
    c) PRS navigation data for use in degraded mode is sent to the NSGU with successful reception confirmed via telemetry (see SFC4.3.4);
    d) CS key management commands to and related TM from the NSGU (see SFC7.2);
    e) STC / STM exchanges with the - PFSU / CSU (see SFC12.2);
    f) STC / STM exchanges with PLSU / CSU (mission part - see SFC13.2)
    g) STC / STM exchanges with PLSU / CSU (PRS part - see SFC15.4)

5. The SCCF generates a consolidated view of the status of the spacecraft based on the TM data received during the contact and provides this summary to the GMS (GACF). Note that this summary is sent periodically during the contact and also following the completion of the contact.
6. The SCCF generates a number of outputs that are in turn inputs to other SFCs including:

    a) Feedback on execution status of the STP to the SCPF (see SFC5.4)
    b) Forwarding STM received from the CSU to the GCS-KMF (see SFC12.2)

    c) Sending reports indicating the uplink status of the OS navigation data to the MUCF (see SFC4.1.4)
    d) Sending reports indicating the uplink status of the PRS navigation data to the MUCF (see SFC4.3.4)
    e) Forwarding STM received from the CSU (mission part) to the M-KMF (see SFC13.2)
    f) Forwarding STM received from the CSU (PRS part) to the P-KMF (see SFC15.4)
    OPS check the contents of the STC acknowledgements

6.5.2.2.2.2 #SFC5.2.1-1.2 Continuous satellite health status monitoring via L-band (nominal case)

This is the nominal case where the health status of the satellite is continuously monitored via data contained in the L-band navigation messages.

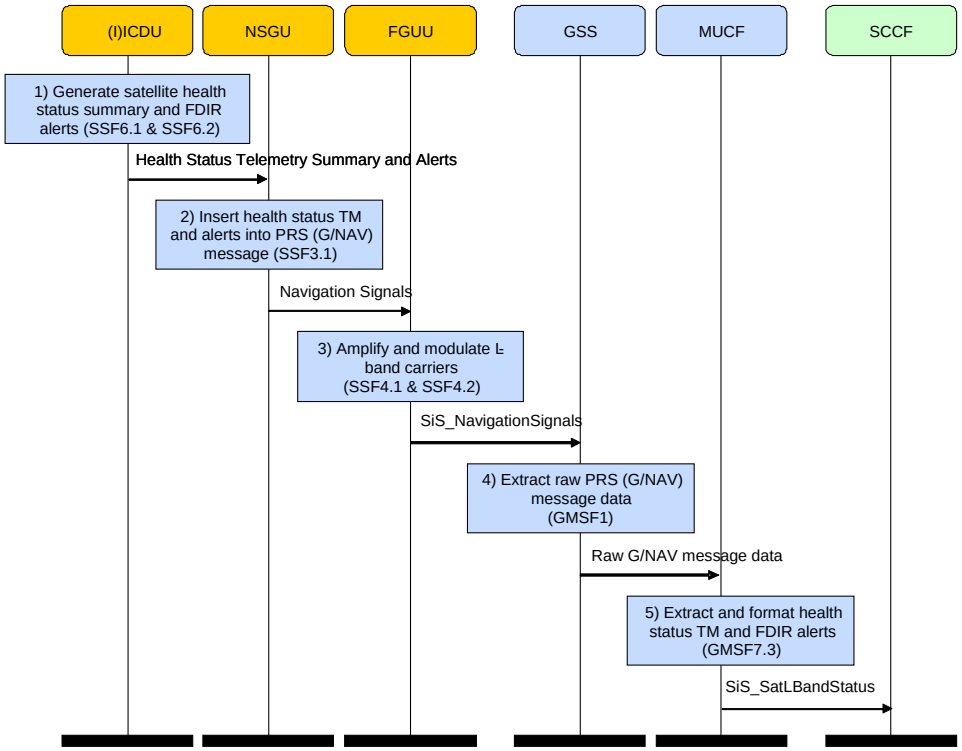


Figure 189 : #SFC5.2.1-1.2 Continuous satellite health status monitoring via L-band

This scenario illustrates the provision of satellite health status information provided continuously via L-band as part of the PRS service.

1) The ICDU/CDU collates health status information from all platform and payload units, generates the satellite health status words (4 x 8 bit words, 32 bits in total) and sends this information to the NSGU.
   These 32 bits of information will be transmitted from the ICDU/CDU to the NSGU once every second.

2) The NSGU receives the health status words includes them in the PRS (G/NAV) message. The 32 bits sent by the ICDU/CDU are multiplexed by the NSGU onto a 2 bit field within the G/NAV message. For this multiplexing, a 60s repeat cycle is used where by in total 40 bits are transmitted to ground each minute (of these 40 bits, 32 carry meaningful health status data and 8 bits are spare). In this way the 32 bits of health status information are transmitted to ground every minute.

Contingency scenario #SFC5.2.1-1.2/2:

If no valid data is received from the ICDU/CDU, the NSGU sets the health status bits in the navigation message to ‘all ones’.

3) The PRS message is amplified and modulated onto the L-band carriers by the FGUU and transmitted from the satellite.
4) GSSs equipped with the GRC-P receive the PRS message on the ground, extract the raw PRS message data and send this to the MUCF(*SIS_SatLBandStatus)*
   Due to multiple reception of each Satellite Signal by all the GSS in view, a process for filtering the redundant information is implemented at MUCF level.
5) The MUCF processes the raw message data, extracts the satellite health status information, generates the health status word and sends this to the SCCF where the status information is displayed to the operator. The MUCF transmits the health status

information (ie of 40 bits) to SCCF once every 60s. OPS monitor continuously the satellite health status. If the health status bits are set to “all ones”, OPS initiates an anomaly investigation to determine the reason why the NSGU is not receiving valid data from the ICDU/CDU.

Contingency scenario #SFC5.2.1-1.2/5: If a G/NAV integrity alert is received during the construction of the SIS_SatLBandStatus message, the message validity flag within the signal will be set to ‘error’ by the MUCF. In this case, OPS GMS notify OPS GCS that the presence of dummy data is due to an integrity alert and not to an on-board failure.

6.5.2.2.2.3 #SFC5.2.1-1.3 Execution of Satellite Maintenance Activities with Mission Impact

This scenario illustrates the system functionality available to allow the execution of satellite maintenance activities that directly impact satellite availability in a manner to minimise the impact on service performance.

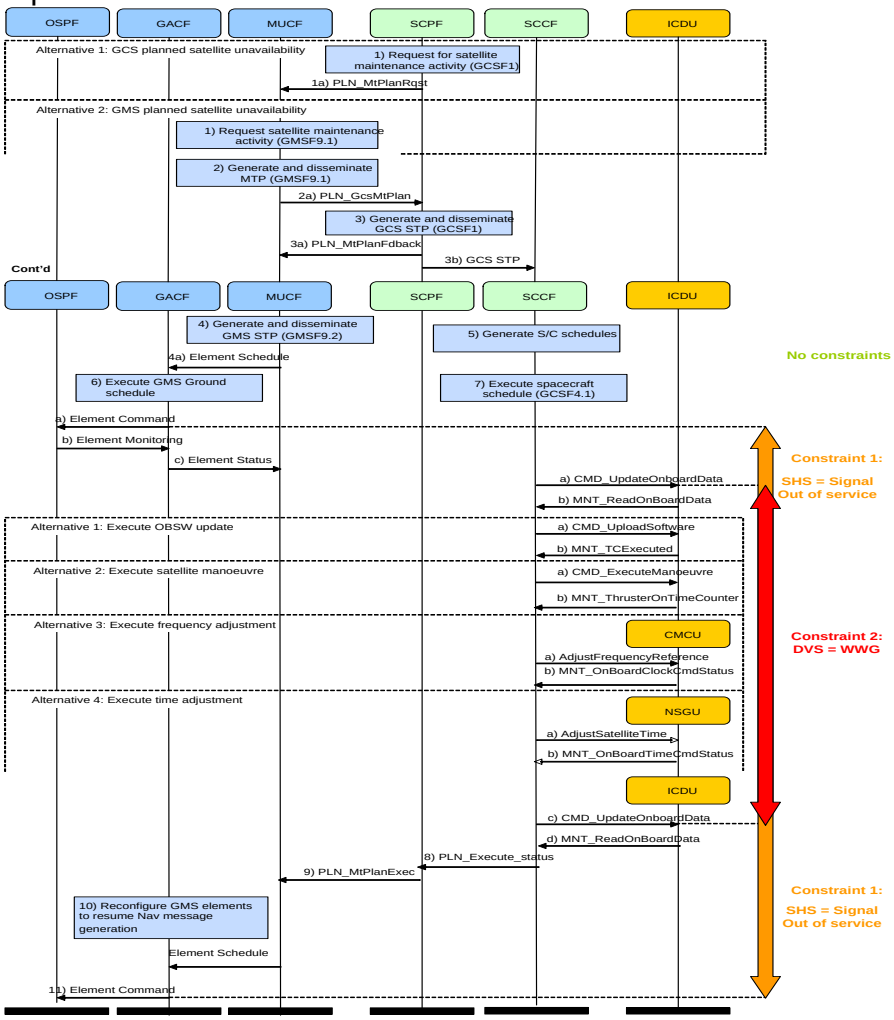


Figure 190 : #SFC5.2.1-1.3 Execution of satellite maintenance activities with mission impact

1. The OPS Team identifies the need to perform a satellite maintenance activity that will impact satellite availability.

Alternative 1:

A satellite maintenance activity is identified within the GCS domain (e.g. satellite manoeuvre, HM maintenance, OBSW update).

a. In this case the maintenance activity is included as a specific task by the SCPF in a MTP planning request to the MUCF.

Alternative 2:

A satellite maintenance activity is identified within the GMS domain (e.g. frequency or time adjustment - see SFC3.1.1 or SFC3.1.2 respectively).

1. The maintenance activity is included as a task in the MTP by the MUCF.
   As part of the MTP the SHS is planned to be set to “Signal “Signal is out of service” in the navigation messages to be broadcasted during the maintenance activity
2. The MUCF generates the GCS MTP and sends it to the SCPF.

3. The SCPF generates the GCS STP from the MTP
   The GCS STP is elaborated to include preliminary GCS activity is scheduled to configure the Data Validity Status (DVS) flag for each message to the value “Working without guarantee” (WWG) before the satellite maintenance is started;
   The GCS STP will include the activity pre-allocated in the MTP.
   a. The SCPF sends the GCS STP to the SCCF.
   b. The SCPF sends planning feedback to the MUCF regarding the implementation of the MTP in the GCS STP. This will provide confirmation to the MUCF that all of the GCS activities specified in the MTP have been correctly scheduled in the GCS STP
4. The MUCF generates the GMS STP based upon the MTP.

   a. The MUCF sends details of the element configuration schedule derived from the STP to the GACF. This signal will include the start and end time of the planned satellite unavailability.
5. The SCCF generate the S/C schedules

6. At the scheduled time The GACF translates this schedule into element configuration commands.:
   a) The GACF translates this schedule into element configuration commands. For the purposes of this scenario, the SHS[1] value for the impacted satellite is set to “Signal Out of Service” for all signals. This constraint will apply to all of those navigation messages whose validity time enters the window during which the task is planned to occur. This setting will ensure that the user receivers exclude that satellite throughout the maintenance operation.

   b) The OSPF monitoring data reports to GACF that the change in configuration has been implemented.

   c) The GACF sends a report to the MUCF providing feedback on the implementation of the GMS STP.

Note [1] Automatic setting of SHS on planned events is foreseen by DCN GAL-DCN-TASI-SYST-X-0908 v1.0, improving the capability foreseen by GAL-DCN-TASI-SYST-X-0881 v1.0 in which the SHS is only set, for the specific satellite, manually by operators.

7. At the due time the SCCF executes the schedule, uploading telecommands to the spacecraft and processing telemetry

   Before the actual maintenance operation is started operators get confirmations that the SHS and the DVS are both set as required. At this point, the planned satellite operations can continue knowing that this activity has been announced to the users of the affected satellite via both the SISA and DVS flags.

   Contingency scenario #SFC5.2.1-1.3/6b:
   The operators don't receive confirmation of either the DVS flag setting or the SHS setting. In this case, the planned maintenance task (schedule for manual execution), is suspended / postponed and an anomaly investigation is started.
   Assuming nominal conditions, a number of different Tasks having mission impact can be envisaged here.
   The following alternatives are illustrative examples:

   Alternative 1:
   Execution of an OBSW update (see #SFC5.2.2-1.1 for more details)

   a. The SCCF sent commands to perform the OBSW patch and after to download the updated memory content
   b. Confirmation of the actual OBSW patch operation is provided by the ICDU/CDU via telemetry.

   Alternative 2:
   Execution of a satellite manoeuvure (see #SFC5.2.2-1.4 for more details)
   a. The SCCF send the commands to perform all necessary activities required to perform a manoeuvre.
   b. Telemetry regarding the execution of the commands is downlinked by the ICDU/CDU

   Alternative 3:
   Execution of payload frequency steering (see SFC3.1.1 for more detail)
   An adjustment of the payload frequency reference of the on-board atomic clocks is performed
   a. The SCCF commands the CMCU to adjust the on-board frequency reference according to the associated task definition in the MTP.
   b. The execution of the command is confirmed to SCCF via dedicated TM.

   Alternative 4:
   Execution of satellite LGST steering (see SFC3.1.2 for more detail)

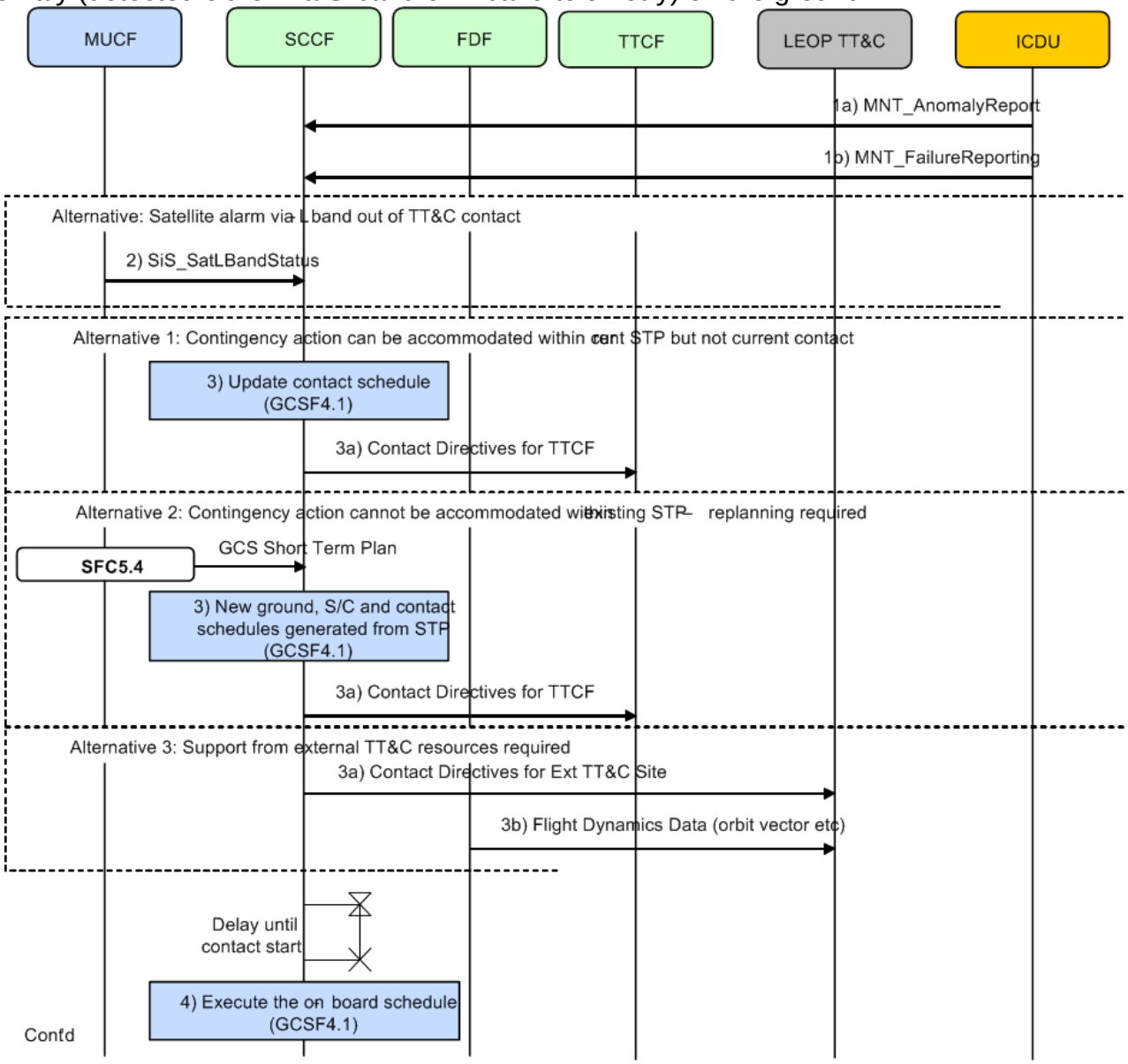
   a. The SCCF commands the NSGU to adjust the on-board time according to the associated task definition in the MTP.
   b. The execution of the command is confirmed to SCCF via dedicated TM.
8. On completion of the satellite maintenance task, the DVS setting is restored to "Valid" by TC and verification is performed in the TM. The SCCF provides the TCs execution Status to the SCPF.

9. The SCPF in turn provides the execution status feedbacks to the MUCF

10. In accordance to the planned schedule, ground will reconfigure the GMS segment to resume services from the affected satellite updating if necessary the Satellite ephemerides for the GSSs and OSPF in order to resume the processing of the satellite observables
11. At the suitable time, when the Navigation message for the satellite still out of service is newly available for uplink, the GACF sends the command to re-configure the SHS for the affected satellite to " Signal OK" for all signals and services
    A maximum of 100 minutes after the message with the new SHS value has been uplinked by GMS, the SHS value will be received by users.

6.5.2.2.2.4 #SFC5.2.1-2.1 Satellite failure / anomaly reported via S-band or L-band

This scenario illustrates the functionality provided by the system to handle a satellite failure / anomaly (detected either via S-band or L-band telemetry) on the ground.



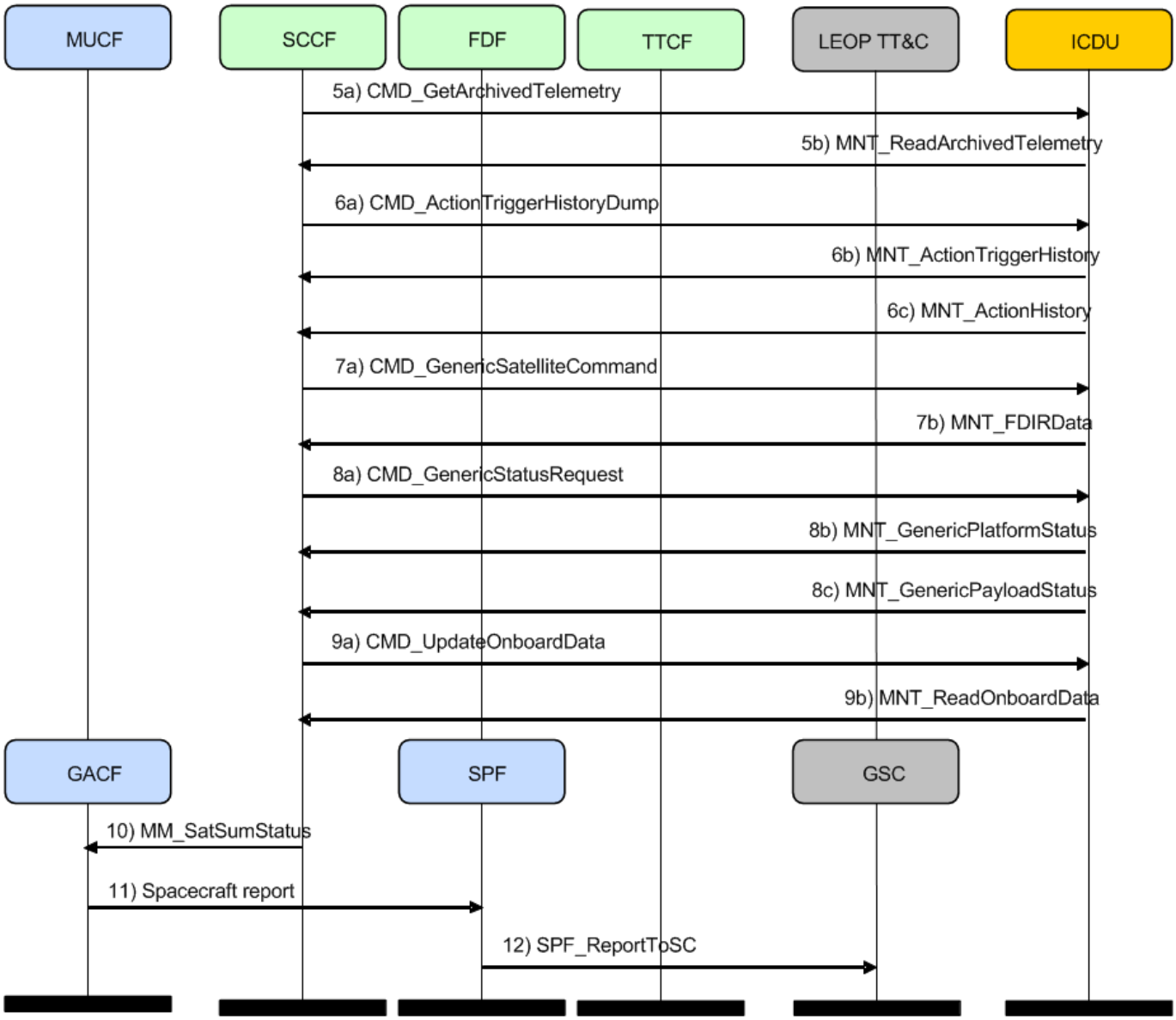


Figure 191 : #SFC5.2.1-2.1 Satellite failure / anomaly reported via S-band or L-band

This scenario illustrates the functionality provided by the system to handle a satellite failure / anomaly (detected either via S-band or L-band telemetry) on the ground.

1) An anomaly (1a) or failure (1b) is reported via telemetry during an on-line S-band TT&C contact. During routine TM monitoring, OPS detect an anomaly
2) Alternatively, if the satellite is not in TT&C contact via S-band, the indication of the failure or anomaly is provided via the satellite health status information received via L-band. During routine Satellite Health Status monitoring, OPS detect an anomaly

At this point, the anomaly is assessed by the operations team on the ground. If the anomaly is identified to be sufficiently serious, a special operations team may be assigned to address this anomaly.

The ability for the ground operations team to investigate the failure / anomaly is dictated by a number of considerations on ground. Three alternative scenarios that are important from a system perspective are considered in detail here:

Alternative 1:
There is sufficient contingency margin available in the current STP to handle the anomaly / failure investigation (i.e. through activation of the back-up contact plan).

3) The SCCF activates the back-up contact and sends new contact directives to the TTCF to establish the back-up contact.

Alternative 2:
The anomaly / failure investigation cannot be accommodated in the existing STP so re-planning is required. OPS initiate the necessary re-planning activities.

3) On reception of the new STP, the SCCF generates new ground, spacecraft and contact schedules. The SCCF sends the updated contact directives to the TTCF.

Alternative 3:
The failure is such that detailed investigation is required that cannot be accommodated using the available Galileo TT&C station resources. In this (highly exceptional) circumstance, the system supports the capability to make use of external LEOP TT&C facilities.

3) OPS decides on the required characteristics of the necessary data to be transferred. The GCS provides the following inputs to the LEOP TT&C stations:
   a) Satellite orbit vectors
   b) Contact start/stop time
   c) Equipment configuration data (frequency, polarization etc).
4) Once the contact time has been reached, the SCCF will execute the spacecraft schedule and transmit TCs to the satellite to investigate the anomaly / failure in more detail. To illustrate the functionality provided by the system, the following exchanges are possible (note that one, some or all of these may be executed depending on the circumstances):
5) A dump of the on board TM buffer to retrieve all data stored since the last TT&C contact.
6) A dump of on board TM buffer to retrieve only specific TM type , including the history of the defined events occurring before and after the detection of the failure condition
7) A dump of the full FDIR TM history log to provide full satellite observability for investigation.

8) Commands to request the provision of more detailed TM health status reporting from both platform and payload units (eg over-sampled TM).
9) Updating (and reporting of) SW parameters or settings used on board in an attempt to recover nominal spacecraft operation.
10) Based upon the TM received during the contact, the SCCF generates a consolidated view of the status of the satellite and sends this to the GACF.
11) The GACF in turn forwards the spacecraft summary information to the SPF.
12) The SPF sends the spacecraft summary to the Galileo Service Centre. In this way, the Galileo Operating Company is kept aware of the detailed current status of the spacecraft.

6.5.2.2.2.5 #SFC5.2.1-2.2 Ground TT&C Station Failure

This scenario illustrates the functionality provided by the system in order to handle a ground TT&C station failure.

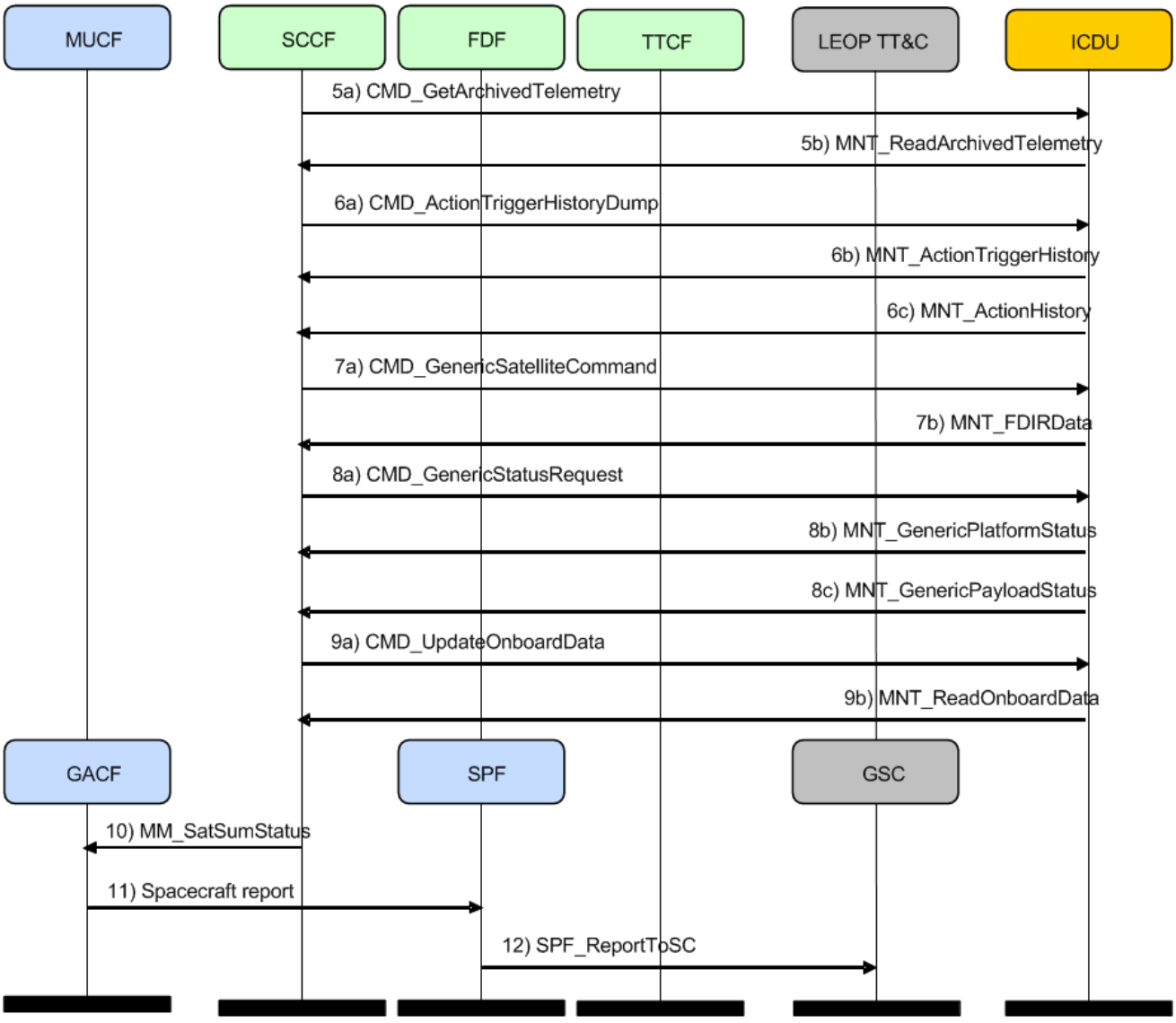


Figure 192 : #SFC5.2.1-2.2 Ground TT&C station failure

This scenario illustrates the functionality provided by the system in order to handle a ground TT&C station failure.

1) A TT&C site failure is detected by the ground monitoring and control functions (see SFC5.1.1) and a report detailing this status is sent to the SCCF. The OPS Team assesses the problem and determines how to respond.

A number of alternative scenarios can be envisaged. Some key scenarios from a system point of view are described here.

Alternative 1:

The failure can be accommodated through use of the back-up contact plan included in the original GCS STP.

2) The SCCF sends contact directives defined within the back-up contact plan to the TTCF.

Alternative 2:

The failure cannot be accommodated in the existing STP and so re-planning is required. OPS initiate the necessary re-planning activities.

2a) A new short term plan is generated (see SFC5.4) and sent to the SCCF.

2b) The SCCF generates new ground, spacecraft and contact schedules.

2c) The SCCF sends contact directives to the TTCF in readiness for the contact.

Alternative 3:

The system resources are over-stretched and the failure cannot be accommodated using the existing Galileo TT&C resources. In this (highly exceptional) circumstance, use of external TT&C facilities may be considered. OPS contact the ESCC operators and notify them of the need for support. In order to facilitate this, the GCS is able to perform the following:

2a) The SCCF generates contact directives for the ESCC (note these may be delivered off-line).

2b) The FDF generates the necessary flight dynamics inputs (eg orbit vectors) for the ESCC in the same way as for the Galileo TT&C stations (note these may be delivered off-line TBC).

3) At the start of the contact, the SCCF can then send and receive TC and TM as normal.

6.5.2.3 SFC5.2.2 Off-line Satellite Monitoring and Control

6.5.2.3.1 Physical Static View

6.5.2.3.1.1 Overall Physical Description

The following figure shows the off-line satellite monitoring and control static diagram, describing the links between GCS and GMS elements :

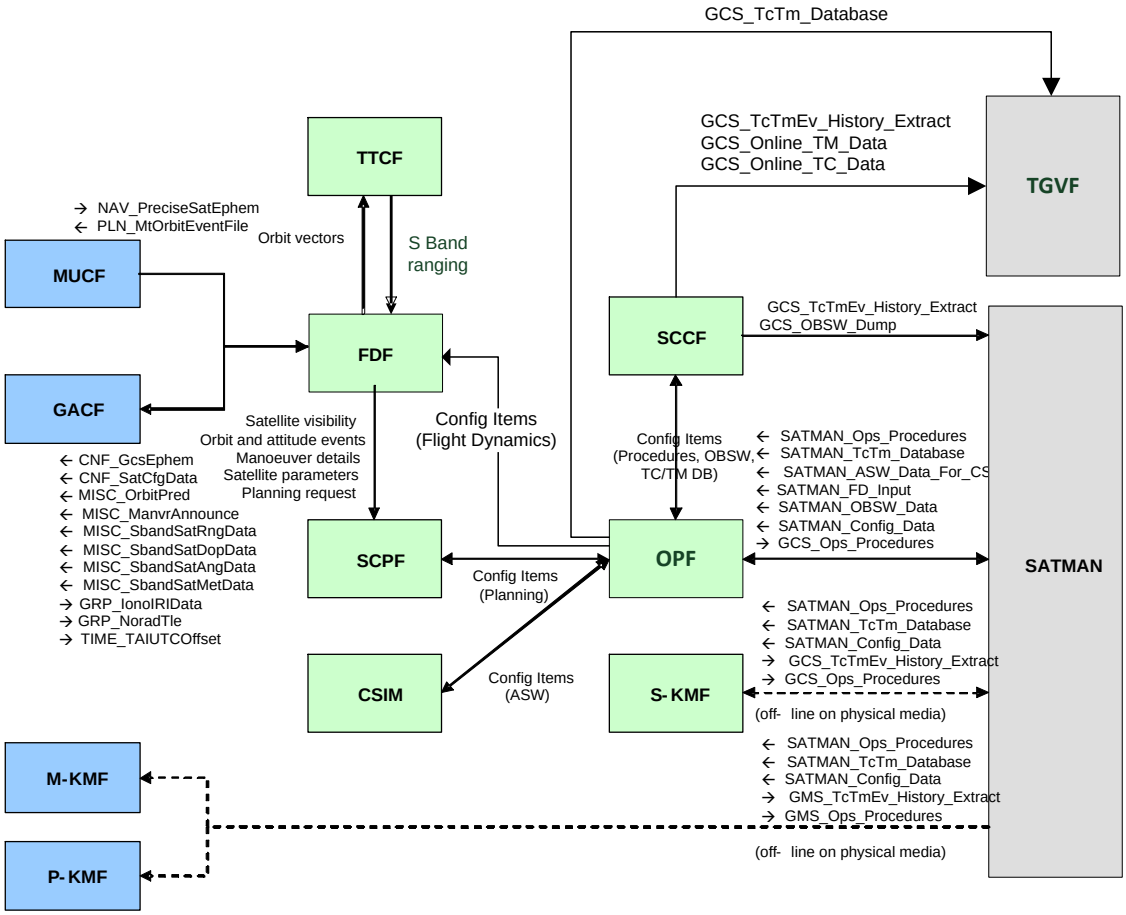


Figure 193 : SFC5.2.2 Physical Static View

6.5.2.3.1.2 Flight Dynamics Analyses: FDF

The FDF is responsible for:

- Performing orbit determination based on S-band two-way ranging data provided by TTCF
- Performing orbit propagation in order to determine furture orbital events and spacecraft visibility for all TT&C stations
- Performing manoeuvre planning and generation of command parameters (eg thruster firing times) to maintain each satellite within the correct orbit within the constellation
- Performing attitude determination and propagation based on spacecraft telemetry and the spacecraft attitude law.
- Providing predicted orbital events to MUCF for input to mid-term planning
- Providing the following input data to GACF:
  - o orbit predictions and manoeuvre data in support of Laser Ranging campaigns
  - o spacecraft ephemeris data
  - o satellite configuration data

  - o S-band 2-way ranging data (including ranging, Doppler and meteorological data as well as antenna azimuth and elevation angles)
- Providing the following inputs to SCPF:

- Satellite visibilities
- Orbit events (including eclipse and sensor blinding times)
- Planning requests and associated command data (for manoeuvres etc)

6.5.2.3.1.3 Provision of S-band ranging data: TTCF

The TTCF is responsible for providing raw S-band ranging data to the FDF for use in orbit determination.

6.5.2.3.1.4 Provision of Flight Dynamics Data to FDF: GACF

The GACF is responsible for providing the following inputs to the FDF:

- Earth orientation parameters (from GRSP - see SFC2.3 for more details)
- Ionospheric data (from GRSP - see SFC2.3 for more details)
- NORAD two line element data (from GRSP - see SFC2.3 for more details)
- Leap-second data (from PTF - see SFC2.1 for more details)

6.5.2.3.1.5 Provision of Flight Dynamics Data to FDF: MUCF

The MUCF is responsible for providing precise orbit vectors to the FDF (see SFC5.3).

6.5.2.3.1.6 Provision of Satellite Input Data and Databases: SATMAN

SATMAN is responsible for providing the following inputs to the GCS. Note that these inputs are provided for all new spacecraft (ie at spacecraft handover) and also as required operationally during the system lifetime (eg following updates due to anomaly investigation, design improvements etc):

- To OPF (note: via secure web-server on the VAL chain):
  - Satellite operational procedures
  - Satellite TC/TM databases
  - Avionics software images for input to CSIM
  - Flight dynamics data
  - On-board software (OBSW) patches (for platform and payload)
  - Configuration data for operational procedures, TC/TM databases and OBSW.
- To GCS-KMF for management of CSU (off-line on physical media due to security classification):
  - Classified satellite operational procedures
  - Classified satellite STC/STM databases)
  - Configuration data for classified operational procedures and STC/STM databases.
- To GMS KMFs for management of CSU (C-band part to M-KMF, PRS part to P-KMF). Also delivered off-line on physical media due to security classification.
  - Classified satellite operational procedures
  - Classified satellite STC/STM databases
  - Configuration data for classified operational procedures and STC/STM databases.

6.5.2.3.1.7 Management and Distribution of Satellite Data: OPF

The OPF is responsible for:

- Configuration control of all data within the GCS (including on-board software, spacecraft procedures, spacecraft databases).

- Spacecraft database editing to allow modification of the spacecraft databases used by SCCF
- Spacecraft procedure development including the editing of procedures for routine and contingency operations.
- Data consistency checking
- Providing configuration controlled items to other GCS elements:
  - SCCF (in particular operational procedures, on-board software patches, spacecraft TC/TM databases)
  - SCPF (planning database)
  - FDF (flight dynamics database)
  - CSIM (avionics software)
- Importing data provided by SATMAN (by the OPF on the VAL chain), including:
  - Satellite operational procedures
  - Satellite TC/TM databases
  - Avionics software images for input to CSIM
  - Flight dynamics data
  - On-board software (OBSW) patches (for platform and payload)
- Exporting data to SATMAN including:
  - Satellite operational procedures

6.5.2.3.1.8 Management of Classified Satellite Data: KMFs

The KMFs are responsible for receiving classified deliveries from SATMAN (off-line on physical media), managing them under configuration control and providing facilities for editing them. The deliveries will include:

- Classified satellite operational procedures
- Classified satellite STC/STM databases)
- Configuration data for classified operational procedures and STC/STM databases.

The KMFs are also responsible for providing the following data to SATMAN in support of anomaly investigations (off-line on physical media):

- Extracts from the STC/STM and Event packet history for each spacecraft
- Classified operational procedures for each spacecraft

6.5.2.3.1.9 Provision of Data in Support of Anomaly Investigation: SCCF

The SCCF is responsible for providing the following data to SATMAN in support of satellite anomaly investigation:

- Extracts from the TC/TM and Event packet history for each spacecraft
- Dumps of on-board software

6.5.2.3.2 Dynamic Scenarios

6.5.2.3.2.1 #SFC5.2.2-1.1 On-board Software Update

This scenario illustrates how an update to the on-board software may be executed by the GCS.

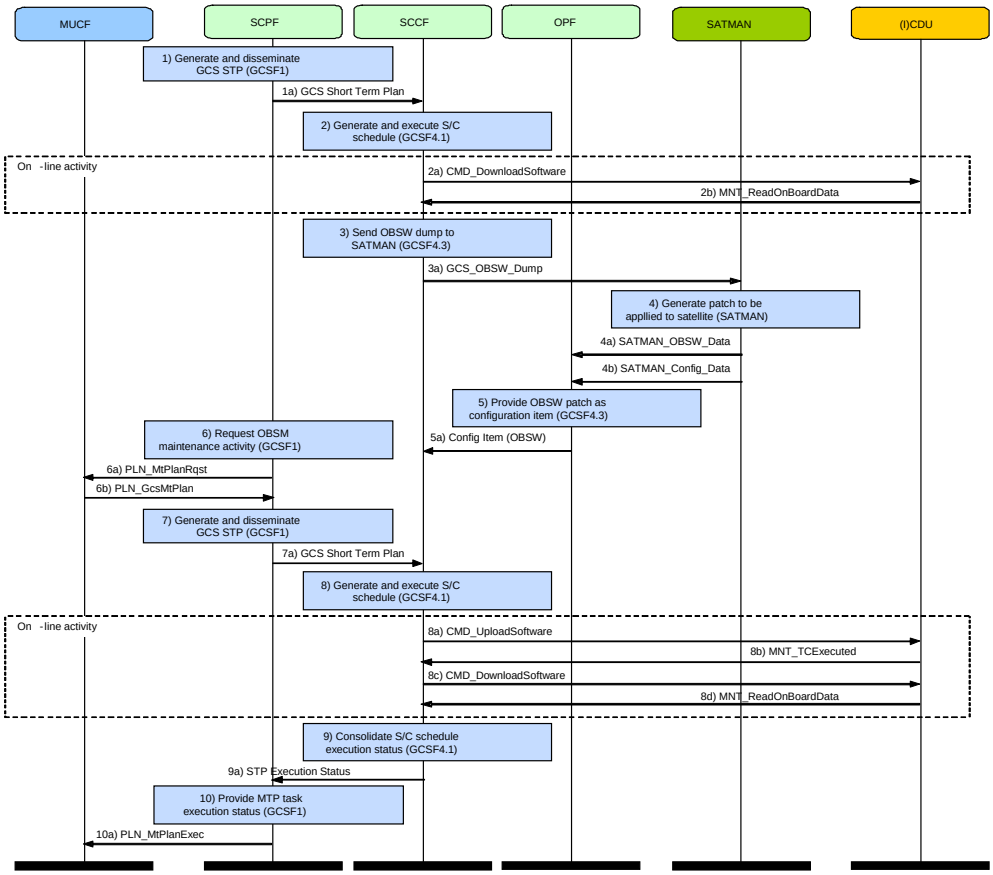


Figure 194 : #SFC5.2.2-1.1 On-board software update

1) Upon identification of a OBSW issue, a command to dump the affected OBSW area is included in the GCS STP:
   a) The SCPF sends the GCS STP to the SCCF.
2) The SCCF generates the spacecraft schedule and executes it at the next contact:
   a) The SCCF sends a command to the satellite to generate a dump of the affected OBSW memory area (typically this will be the avionics software).
   Note that this is an on-line activity performed as part of a nominal TT&C contact (see SFC5.2.1).
   b) The ICDU/CDU responds with telemetry containing the requested OBSW memory dump.
   3) The OBSW memory dump is sent from the SCCF to SATMAN for analysis. OPS manually initiate the transfer

4) SATMAN analyse the dumped OBSW memory and generate a validated memory patch to be uplinked to the satellite.
   a) SATMAN send the OBSW patch to OPF.
   b) SATMAN also send to the OPF on the VAL chain a configuration data file with the patch and expected the OBSW memory image, defining which versions of the TC/TM database and Operational Procedures this patch has been validated against.
5) The patch is stored as a configuration item within the OPF on the VAL chain.

a) OPS rehearse the OBSW patch operation with the CSIM (patch, dump and compare)
b) After validation with the CSIM, the OPS team transfer the Configuration Item on the OPF on the OPE chain and from there to the SCCF.

6) The execution of the OBSW patch may have mission impact the OBSW update task must be included in the MTP.
a) The SCPF includes the OBSW update as a task in a MTP request sent to MUCF.
b) The MUCF includes the OBSW update task in the MTP and sends the MTP to the SCPF. Note that the MTP will also include the necessary tasks to ensure that the forthcoming satellite unavailability is announced to the users. This is described in detail in SFC5.2.1 (see MSC #SFC5.2.1-1.3).
7) The SCPF generates the GCS STP and sends this to the SCCF.
8) The SCCF generates the spacecraft schedule and executes it at the next contact. OPS compare the OBSW dump with the OBSW patch and report the result to SATMAN
a) The SCCF sends a command to the satellite (ICDU/CDU) to apply the OBSW patch. Note that this is an on-line activity.
b) The ICDU/CDU responds with confirming the execution of the patching.
c) The SCCF sends a command to request a dump of the updated OBSW
d) The ICDU/CDU responds with telemetry containing the requested OBSW dump.
9) The SCCF provides feedback to the SCPF on the execution status of the task as identified in the STP (9a).
10) In turn, the SCPF provides feedback to the MUCF on the execution status of the task as identified in the MTP.

Note that the patching of OBSW related to the on-board security units will be co-ordinated between the classified and unclassified security domains by means of a dedicated operational procedure.

6.5.2.3.2.2 #SFC5.2.2-1.2 Anomaly Investigation with Support from SATMAN

This scenario illustrates how SATMAN may provide support to spacecraft anomaly investigation.

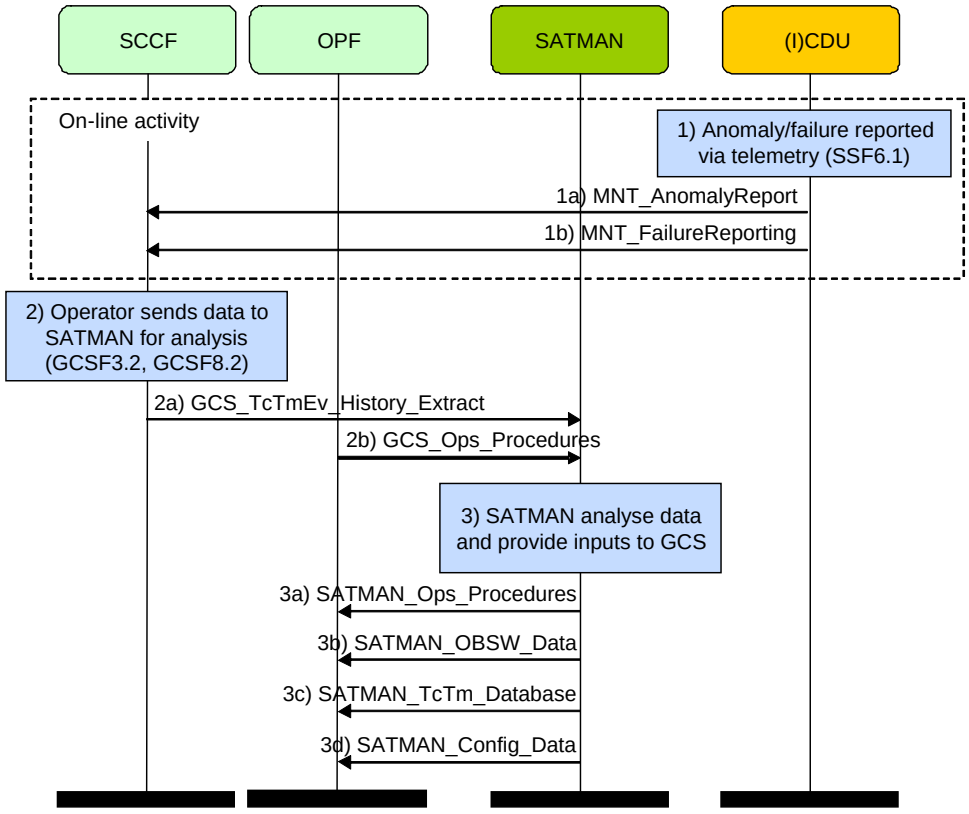


Figure 195 : #SFC5.2.2-1.2 Anomaly investigation with support from SATMAN

1) An anomaly (1a) or failure (1b) is reported to the ground via telemetry during a nominal TT&C contact with the satellite (see SFC5.2.1). OPS detect the anomaly through routine TM monitoring

2) The operators determine that the anomaly/failure cannot be resolved without support from SATMAN. To allow SATMAN to carry out the investigation, GCS is able to send the following data:
   a. The SCCF generates an export of TC, TM and event packet data for a defined period to SATMAN (for example either side of or just before the anomaly).
   b. The OPF generates an export of all of the operational procedures that are currently in use with for this satellite.
3) Following their own investigation of the anomaly, SATMAN may respond with one or more of the following exchanges:
   a. New or updated operational procedures for the spacecraft.
   b. An update (i.e. patch) to the OBSW.
   c. An updated TC/TM database (or database parameters)
   d. A configuration data file describing the configuration status of the items a, b and c above (provided for all of a, b and c even if only one of these sent).

Note that all of the file exchanges described in points 2) and 3) above will be preceded by some form of communication between the Operators in the GCC and SATMAN (eg email, voice, fax etc). In support of these exchanges, the GCS provides a secure web-server in the DMZ of the EDDN (see [AD.12]).
Files can then either be uploaded or downloaded (via HTTP-S) by SATMAN as agreed during the prior communication between the parties.

6.5.2.3.2.3 #SFC5.2.2-1.3 Orbit Determination and Two-way Ranging

During nominal operations the Galileo system will make use of accurate orbit vectors generated within the GMS (OSPF). However, the system also provides the means to generate orbit vectors within the GCS using S-band two-way ranging data as input. In this way the system is able to perform accurate orbit determination for spacecraft that are not active within the Galileo mission, or when the GMS orbit determination functions are not available or cannot be used (eg during initial operations or satellite graveyard phase).
This scenario describes how orbit determination (including S-band two-way ranging) can be performed by the GCS.

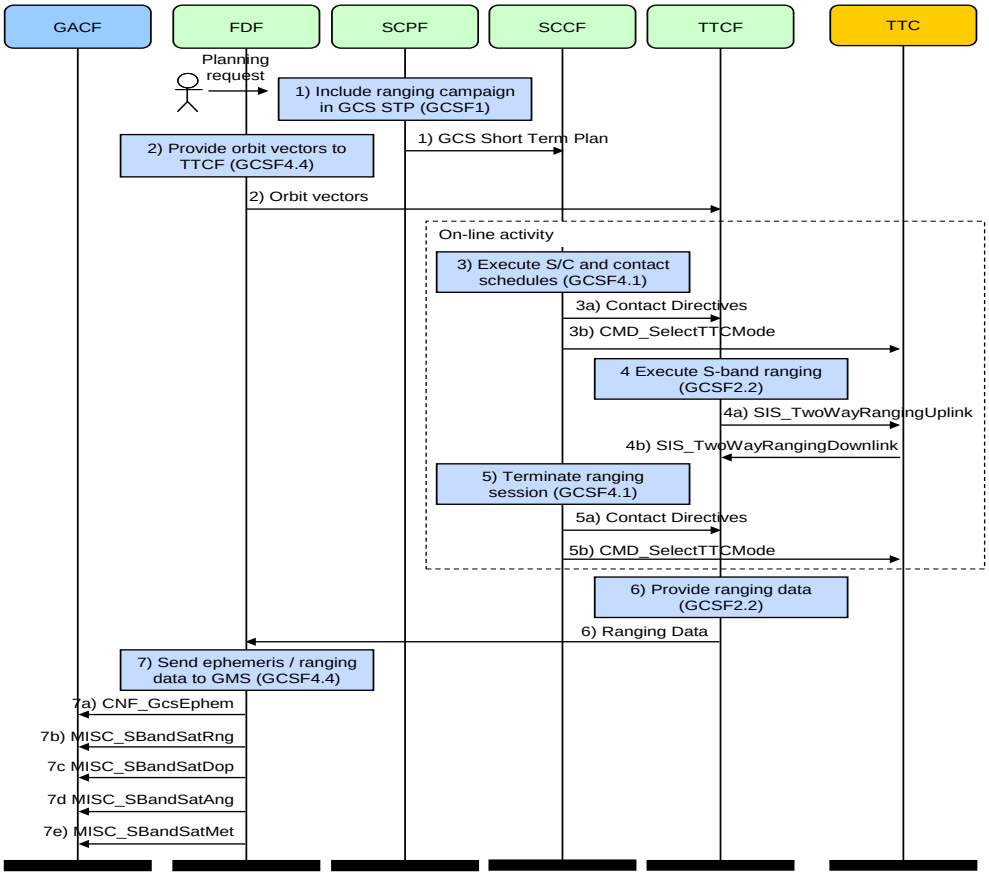


Figure 196 : #SFC5.2.2-1.3 Orbit determination and two-way ranging.

1) OPS generate a planning request for the SCPF to include a ranging campaign in the GCS STP. Depending of the purpose of the tracking campaign (e.g. generation of quick ephemerides for restart of GMS operations or accurate orbit determination to validate Station keeping precise positioning), up to 5 TTCF can be involved over a period of up to 4 orbits.
   The SCPF includes a task in the STP and sends the updated plan to the SCCF.
2) The FDF provides orbit vectors directly to each TTCF (from which TTCF generates antenna pointing data to acquire and track spacecraft during TT&C contact).
3) The SCCF processes the STP and generates executable ground, spacecraft and contact schedules.
   a) The SCCF executes the contact schedule and sends contact directives to the TTCF (start/stop times etc).

b) The SCCF executes the spacecraft schedule and sends a TC to the TTC transponder on the spacecraft to configure it into the correct RF mode and also into coherent mode.

4) The TTCF starts the ranging session according to the contact directives. To this the TTCF:
   a) Transmits the uplink RF carrier including the required ranging signal.
   b) Receives and processes the downlink RF carrier following reception and re-transmission of the uplink signal by the TTC transponder on-board the spacecraft.
5) According to the STP, the SCCF then terminates the ranging session by:
   a) Sending contact directives to the TTCF to stop the ranging session.
   b) Sending a TC to the TTC to re-configure the transponder into the nominal operating mode.

Note that a ranging campaign will typically involve performing ranging measurements from several TT&C stations in sequence, with each station performing continuous ranging measurements during contacts. As a result steps 3 - 5 can be expected to be repeated a number of times over multiple contacts.

6) The TTCF sends the ranging data recorded during the contact to the FDF so that it can be used as input to the orbit determination process.
7) The FDF performs orbit determination based on the two-way ranging data. It also provides a number of outputs to the GMS (GACF - see SFC5.3.2 for more details):
   a) Spacecraft ephemeris
   b) Ranging data (including calibration)
   c) Doppler data
   d) TTC station antenna data (azimuth and elevation angles)
   e) TTC station meteorological data

6.5.2.3.2.4 #SFC5.2.2-1.4 Orbit Manoeuvre Execution

This scenario describes a typical scenario that can be supported for the execution of an orbit manouevre.

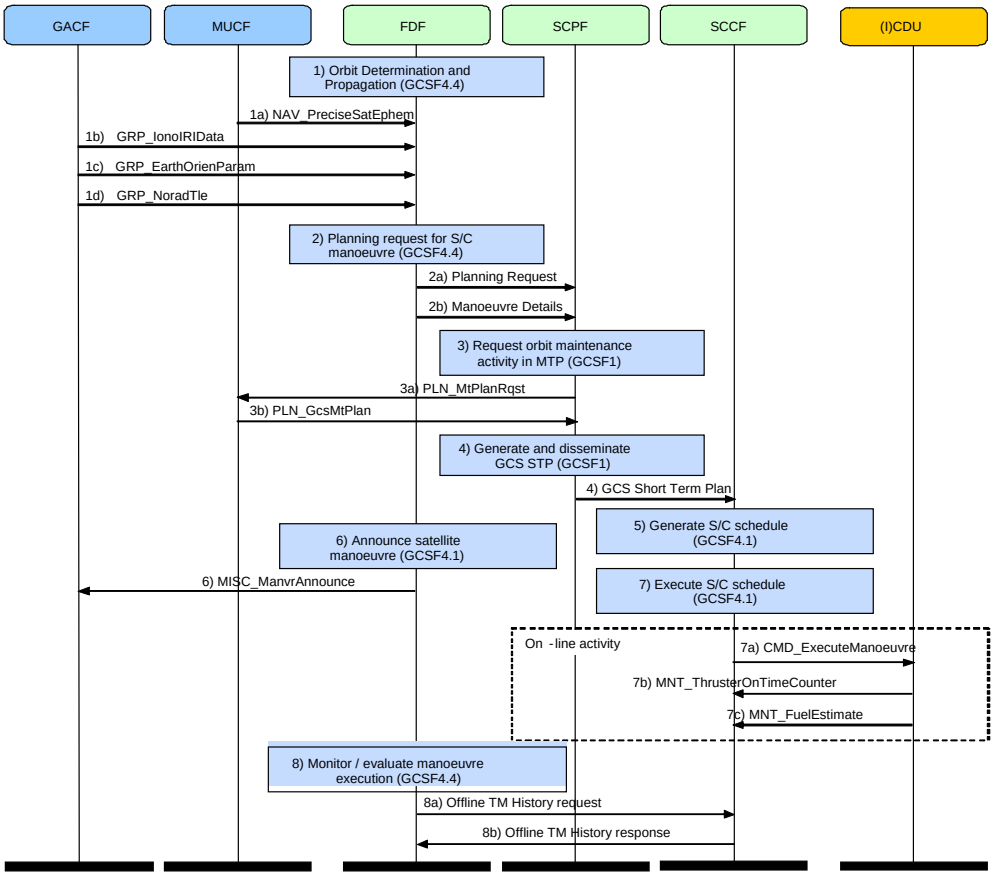


Figure 197 : #SFC5.2.2-1.4 Orbit manoeuvre execution

1) The FDF performs orbit determination and propagation for all Galileo spacecraft. Key inputs to this process from the GMS are:
   a) Precise satellite ephemeris data calculated by the OSPF and provided to the FDF by the MUCF.
   b) Ionospheric data (provided by the GRSP via the GMS - see SFC2.3)
   c) Earth orientation parameters (provided by the GRSP via the GMS - see SFC2.3)
   d) Two-line element data from NORAD (provided by the GRSP via the GMS - see SFC2.3)
2) From FDF outputs, OPS identify the need to perform a manoeuvre. The FDF calculates all necessary parameters and sends:
   a) A planning request for the manoeuvre to the SCPF (including details of the manoeuvre).
   b) Manoeuvre command parameters to the SCCF for use during the execution of the manoeuvre.
   Note that before the manoeuvre request is sent by the FDF, the operator will typically validate the manoeuvre and associated operational procedures and commands on the GCS VAL chain using the SCCF and CSIM. The exchanges associated with this activity are not shown explicitly in the scenario diagram.
3) As manoeuvres have an impact on the Galileo mission, OPS include, using the SCPF, a dedicated satellite maintenance task for the manoeuvre in a mid-term planning request to the MUCF.
   Note that the functionality provided by the system to ensure that this task does not impact the mission is described in detail in a dedicated scenario (see #SFC5.2.1-1.3 - Execution of satellite maintenance activities with mission impact).

a) The SCPF includes the manoeuvre as a specific task in a MTP planning request to the MUCF.
b) The MUCF sends the MTP containing the planned manoeuvre back to the SCPF.

4) The SCPF generates the GCS STP from the MTP received from the MUCF and sends the STP to the SCCF.

5) The SCCF generates the spacecraft schedule according to the STP.

6) Prior to the execution of the manoeuvre, the FDF announces the manoeuvre to the GMS (GACF). This allows the GMS (via the SPF) to provide details of the manoeuvre to the GRSP in support of laser ranging campaigns (see SFC2.3 for more details).

7) The SCCF executes the spacecraft schedule during the next contact. Note this is an on-line activity.
   a) The SCCF sends the TC(s) to the satellite to execute the manoeuvre.
   b) The satellite provides telemetry that provides feedback on the execution of the manoeuvre. This includes:
   c) Telemetry defining the accumulated 'thruster_on times' independently for each thruster
   d) Telemetry providing data to the ground for the estimation of the propellant level
   e) OPS analyses the telemetry.
8) The FDF monitors and evaluates the execution of the manoeuvre based on telemetry retrieved from the SCCF. To do this the following exchanges are required:

   a) The FDF sends a TM history request to the SCCF requesting the TM data required.
   b) The SCCF responds with the data requested.

Note that full evaluation of the manoeuvre can only be achieved once the new spacecraft orbit has been determined. This will typically be achieved by the FDF based upon a S-band two-way ranging campaign (see scenario #SFC5.2.2-1.3).

6.5.3 SFC5.3 Mission Performance Monitoring

6.5.3.1 Introduction

The mission performance monitoring is in charge of providing the means to collect, display and report mission performance analysis, of elaborating algorithm functions configuration parameters. The mission performance monitoring is supported by on-line mission monitoring and off-line mission monitoring.

6.5.3.2 SFC5.3.1 On-Line Mission Monitoring

6.5.3.2.1 Physical Static View

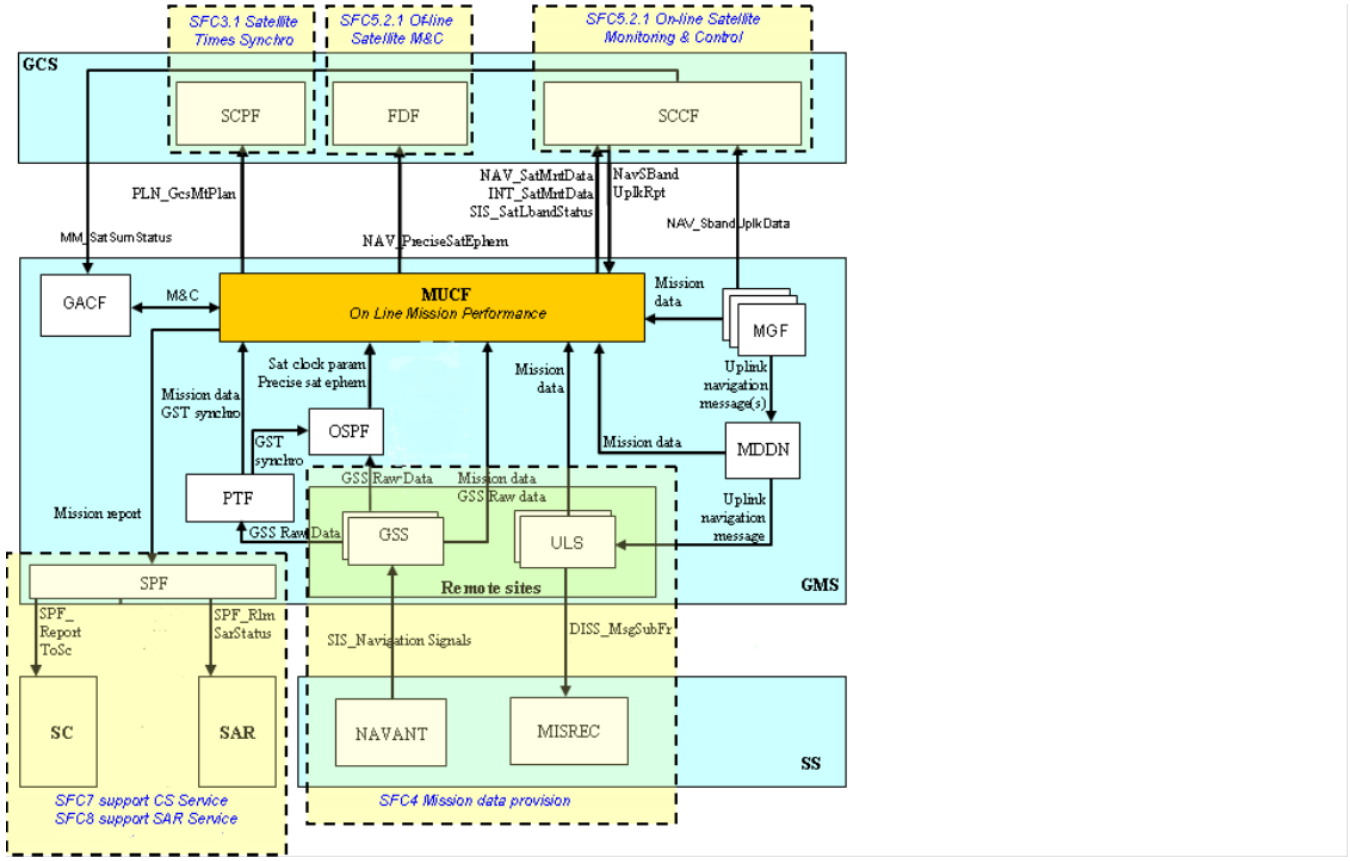
6.5.3.2.1.1 Overall Physical Description

The On-line Mission Monitoring in GMS is responsible for performing in real time:

- Galileo services Mission Performance as provided to each relevant user Group (OS, PRS);
- On line monitoring of the mission data (OS, and PRS) provision chain using relevant assets monitoring data extracted from Real-time M&C of ground assets. For SAR? CS data dissemination chain monitoring refer to SFC 8 and 7 respectively.
- On line monitoring of the Satellite Status data (including possibly alerts to spacecraft anomalies detected through TM monitoring by the GCS) , see SFC5.2.1 On-line Satellite Monitoring and Control.

During routine operations this task is limited to pure supervision of the performance

The following figure presents the physical architecture of the on-line mission monitoring functional chain:



6.5.3.2.1.2 On line verification of GALILEO services Mission performance

6.5.3.2.1.2.1 SFC5.3.1 Related Tasks: MUCF

6.5.3.2.1.2.1.1 Service performance on-line monitoring

Service performance monitoring

The MUCF monitors the online performances w.r.t the Navigation Service aspects on the basis of data extracted from the GSS Raw Data.

The mission performances used for the end-to-end monitoring are the following

- position accuracy performance in terms of horizontal and vertical position error at 95% percentiles, for the Open, and Public Related Services at each point of a configurable deg grid (down to 1 deg) representing the service area.
- time accuracy performance in terms of time error at 95% percentile, for the Open, and Public Related Services at each point of a configurable deg grid (down to 1 deg) representing the service area.

Service on-line performance predictions:

The MUCF is in charge to perform the service performance predictions using the following input:

- on-line mission performance monitoring described above
- information about the planned availability/unavailability of satellites and ULS
- services for which predictions shall be done (OS, PRS, SAR, CS)
- satellites signal components Health status,
- satellites service data health status
- satellite ephemeris prediction

Service performance predictions are computed in the MUCF and compared against monitoring threshold on world grid representing the service areas for:

- OS Position accuracy availability
- OS Time accuracy availability
- PRS Position accuracy availability
- PRS Time accuracy availability
- SAR Return-Link availability
- CS availability

The MUCF mission controller decides of the event (planned and unplanned outages) to be sent to users via the Service Centre in bulletins called NAGU.

Remark: *the automated NAGU generation concept is currently not consolidated in the STRB2.0 baseline. It is expected that the content of the signal INF_NoticeAdvisoryToUsers will be refined in close coordination with GSC in the frame of GSC operational concept consolidation. In the meanwhile, the signal INF_NoticeAdvisoryToUsers will not be disseminated to the external entity and the NAGU will be generated manually and sent to the GSC. In both cases (automated and manual NAGU processes) the authorization for dissemination to the public is granted by the corresponding formal process at GSC level.*

6.5.3.2.1.2.1.2 Service performance alerts and display

The MUCF collects inputs for the Mission Monitoring of the overall Mission:

- Satellite functional and performance status,
- Ground assets functional and performance status,
- Mission monitoring, prediction and trend analyses data.

Based on these inputs the MUCF Mission Monitoring function provides through a MMI interface, a synthetic view of satellite, contacts and ground resources used, with the related status and the mission progress in a window with the relevant past, present and planned events.

6.5.3.2.1.2.1.3 Recovery procedures

The MUCF

- provides the technical means to recognise a specific Mission anomaly or a status change.
- generates visual monitoring alarms when an event occurs (and audible, when operators have switched on such feature for the event), together with information to help the operators in locating and repairing the anomaly.
- provides to the operator the technical means to generate a dedicated problem report and operations log in the case of contingency, relevant to mission performance monitoring and prediction, describing the problem and any action taken to solve it.

6.5.3.2.1.2.1.4 On line verification of GALILEO services Mission performance Archive policy

The MUCF host a local archive in charge of temporary storing:

- all input data need at the on-line performance analysis aged of less of 3 days.
- all data output data resulting of MUCF processing (analysis results, error and event log, ...) before central online archiving to the GACF.
- all MUCF configuration data

6.5.3.2.1.2.2 SFC5.3.1 Related Tasks: GSS

The GSS provide to the MUCF:

- o the satellite HealthStatus including Satellite L-band telemetry data by processing G/NAV navigation signal and filtering GSS redundant information.
- o Monitoring data provided by the GSS regarding mission-related issues, particularly the quality and characteristics of the signal in space received from the Galileo satellites
- o Raw data messages (see below)

The GSS provide via the MDDN network to the OSPF, PTF and MUCF raw data messages: raw measurements and data from the SIS message (navigation non PRS and PRS, commercial services, SAR); with time stamping and parameters concerning the SIS measurement quality.

6.5.3.2.1.3 On line check of the on-ground mission data provision chain

6.5.3.2.1.3.1 SFC5.3.1 Related Tasks: MUCF

6.5.3.2.1.3.1.1 Mission data determination chain product quality on-line monitoring

The MUCF monitors the quality of the mission data determination chain products based on Quality of Service indicators with an automatic and periodic (configurable down to 1 min) refreshment of the results:

- o From PTF mission data, the MUCF monitors the PTF clocks stability
- o From OSPF mission data, GMOS monitors ODTS algorithms and GSS status

6.5.3.2.1.3.1.2 Mission data provision chain configuration on-line monitoring

The MUCF monitors the quality of switching mechanisms inside the GMS with an automatic and periodic (configurable down to 1 min) refreshment of the results.

- o From GSS mission data, the MUCF monitors the correct tracking of satellites
- o From MGF mission data, the MUCF monitors the OSPF source selection
- o From MGF mission data, the MUCF monitors the C-Band multiplexing functions
- o From GNMF mission data, the MUCF monitors the MGF source selection
- o From MGF mission data, the MUCF monitors that the MGF use the correct U/S
- o From ULS mission data, the MUCF monitors that the ULS use the correct U/S
- o From ULS mission data, the MUCF monitors the handover duration of antennas
- o From MUCF mission data, the MUCF monitors the correct distribution of the U/S to the MGF and ULS
- o From GCS data, the MUCF monitors the correct TTC S-Band up-linking

6.5.3.2.1.3.1.3 Mission data provision chain configuration on-line monitoring Archive policy:

The MUCF host a local archive in charge of temporary storing:

- o all input data need at the on-line performance analysis aged of less of 3 days.
- o all data output data resulting of MUCF processing (analysis results, error and event log, ...) before central online archiving to the GACF.

- all MUCF configuration data

The MUCF periodically sends local data for migration to GACF long term archive. The MUCF builds the complete archive identification (control file) and computes MD5 transfer signature(s). The MUCF. transfers archive to GACF and notifies transfer final status to GACF.

6.5.3.2.1.3.2 SFC5.3.1 Related Tasks: GSS

The GSS provide to the MUCF:

- the satellite Health Status including Satellite L-band telemetry data by processing G/NAV navigation signal and filtering GSS redundant information.
- Monitoring data provided by the GSS regarding mission-related issues, particularly the quality and characteristics of the signal in space received from the Galileo satellites
- Raw data messages (see below)

The GSS provide via the MDDN network to the OSPF, PTF and MUCF raw data messages: raw measurements and data from the SIS message (navigation non PRS and PRS, commercial services, SAR); with time stamping and parameters concerning the SIS measurement quality.

6.5.3.2.1.3.3 SFC5.3.1 Related Tasks: OSPF

The OSPF receives all SiS observables (raw data) from all GSS once per second. The observables are filtered and processed to generate:

- Ephemeris and almanacs for each of the satellites in orbit.
- Clock correction parameters for each of the satellites in orbit.
- Broadcast group delay (BGD) for each of the satellites in orbit.
- Ionospheric corrections common to all the satellites.
- Signal-in-Space Accuracy (SISA)

Then the OSPF provide the following data to the MUCF:

- Mission Monitoring data including:
  - OSPF estimated residues on the orbits &clock for each satellite,
  - Hardware bias estimations for the GSS within the ODTS process,
  - Ionospheric data (Vertical delays estimations within the ODTS process),
  - General information about the navigation function processes,
  - Navigation satellite Monitoring data derived from the ODTS (OK, NOK),
  - Troposphere vertical delay estimations for the GSS,
  - GSS number of measurements accepted and reason of rejection,
  - List of barriers activated (barrier code) and barriers result.
- Navigation data products for Fnav, Gnav and Inav messages:
  - Batch of parameters for satellite ephemeris prediction,
  - Clock model parameters,
  - Almanac parameters,
  - Ionospheric correction parameters,
  - Broadcast Group Delay parameters,
  - Parameters for the conversion of GST to the UTC,
  - GST/GPS Time Offset Parameters,

- SISA Index
- IOD associated to each batch of data

- Quality of Service (QoS) associated to the OSPF data relevant to Fnav, Gnav and Inav messages

6.5.3.2.1.3.4 SFC5.3.1 Related Tasks: IPF

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6.5.3.2.1.3.5 SFC5.3.1 Related Tasks: PTF

The PTF is responsible for the following main functions:
- Generation of the GST time scale and its physical realisation (based on atomic clock measurements)
- Distribution of the GST externally and within the GCC
- Support to the GST steering to UTC modulo 1 second (interaction with Time Service Provider)
- Computation of the Time offset between GST and GPS time scale (GGTO)
- Ensuring the steering of the 2 GST time scales from the 2 GCCs.
- Support to Timing mission monitoring functions

Then the PTF provides the following data to the MUCF:
- Mission Monitoring data (including algorithms internal information):
- Estimations on time:
  - Running estimate PTF clocks stability vs GST for each clock (Cesium and Master) in terms of Allan Deviation
  - PTF - PTF offset as computed internally by the PTF on the basis of TWSTFT and CV data
  - Clock ID and weighting in the GST algorithm,
  - PTF slave to PTF master steering parameters applied by PTF slave,
  - TSP steering parameters applied for the PTF master
- Time data products:
  - Delta between GGTO coordinated calculated by Galileo and GGTO coordinated calculated by GPS-USNO,
  - Agreed time offset between Galileo (PTF) and GPS (USNO).

6.5.3.2.1.3.6 SFC5.3.1 Related Tasks: MGF

The Message Generation Facility (MGF) is the meeting point for all the data streams that make up the C-Band Uplink Messages. Its mission is to elaborate messages as specified in the GMS to Space ICD by multiplexing those data that allow providing the OS, CS, , PRS, SAR services.

The MGF interface with the Mission and Uplink Control Facility (MUCF): the MUCF sends the up-link schedules to the MGF in order to allow mission data dissemination. MGF sends to MUCF some relevant information for mission monitoring data:
- Status of the received CS data since last epoch,
- Status of the CS data that are being uplinked,
- Current active MGF and last contact plan received,

- Status of disseminated data:
  - MGF MSF QoS (used by MNE for MGF selection message subframe),
  - selection code based on QoS indicators (for OSPF),
  - OSPF source (for OS, and PRS nav data),
  - Result of the MGF safety barriersflag,
  - Satellite CRC Message subframes and Int sub-frames,
  - Security Modules error codes for MKMF and PKMF,
  - Navigation data for each satellite including ULS contact which address MSF ID,
  - PRS alert data raised by MGF including Sat data, alert type and integrity tables,
  - SAR messages triggered by the a new RLM event (request or repetition),
  - SAR monitoring data including request, pre-processing status, RLM transmission and uplink status,
  - Signal component health status for each satellite,

6.5.3.2.1.3.7 SFC5.3.1 Related Tasks: ULS

The ULS stations are responsible for uplink messages to the satellites.

The ULS stations interface with the Mission and Uplink Control Facility (MUCF): the MUCF sends the up-link schedules to the ULS in order to allow satellite tracking. ULS sends to MUCF some relevant information for mission monitoring data:

- MGF sub-frames reception status and message transmission
- ULS traking plan ID and MGF contact plan ID received at
- And for each antenna: antenna ID, Status, RF power status, Pointing angles and time, RepointingTime, AcquisitionTime.

6.5.3.2.1.3.8 SFC5.3.1 Related Tasks: MDDN

The Mission Data Dissemination Network (MDDN) support the Galileo Mission ground segment data circulation data:

The MDDN interface with the MUCF for mission monitoring data provision:

- from MNE to inform MUCF about MGF selection at each Epoch,
- from GNMF to inform MUCF about MDDN&SDDN Networks high-level status.

6.5.3.2.1.4 On line monitoring of the satellite status

6.5.3.2.1.4.1 SFC5.3.1 Related Tasks: MUCF

6.5.3.2.1.4.1.1 Satellite on-board master clock monitoring

The MUCF performs the analysis of the satellite clocks performance assessment. This function is used to decide a clock switching need

See SFC3.1 Satellite time synchronisation

The MUCF elaborates synchronisation requests to the GCS (PLN_GcsMtPlan) for adjustment of on-board clock and to request clock switching issued by a specific operational procedure and activated on demand.

See SFC3.1 Satellite Times provision and Synchronisation for details.

From GCS S-Band uplink reports and ULS C-Band uplink report, MUCF-GMOS verifies the number of batches in the satellites and thus monitor the satellite autonomy.

See SFC4 Mission data provision for details.

6.5.3.2.1.4.1.2 General satellite status from GMS to GCS

The MUCF provide periodically the GCS with:

- the satellite Health Status including the Satellite L-band telemetry data transmitted by the Galileo Satellites inside the G/NAV Signal In Space and acquired continuously by the GSS network, after processing for filtering the redundant information, see SFC5.2 Satellite on-line M&C.

- NAV_SatMntData, the satellite status information (OK, NOK) obtained from the internal OD&TS processes determination.The status "OK" implies that the GMS has correctly performed orbit & clock determination and that the product check has been passed. This "OK" status does not include SISA checks (e.g. a spacecraft with this flag set to OK could have the SISA set to Not Monitored). The status of "NOK" implies that at least one of these three steps have been unsuccessful : orbit, or clock determination, or product check.

6.5.3.2.1.4.1.3 Satellite Precise Ephemeris on-line provision to GCS

The MUCF provides to the GCS (NAV_PreciseSatEphem), with the precise Galileo satellites constellation ephemeris prediction and orbits vectors, computed and provided by OSPF.

6.5.3.2.1.4.1.4 Satellite status On line monitoring Archive policy

The MUCF host a local archive in charge of temporary storing:

- all input data need at the on-line performance analysis aged of less of 3 days.
- all data output data resulting of MUCF processing (analysis results, error and event log, ...) before central online archiving to the GACF.
- all MUCF configuration data

The MUCF periodically sends local data for migration to GACF long term archive. The MUCF builds the complete archive identification (control file) and computes MD5 transfer signature(s). The MUCF. transfers archive to GACF and notifies transfer final status to GACF.

6.5.3.2.1.4.2 SFC5.3.1 Related Tasks: GSS

The GSS provide periodically to the GCS the satellite HealthStatus including Satellite L-band telemetry data by processing G/NAV navigation signal and filtering GSS redundant information.

The GSS support the Mission Monitoring by providing data input extracted from the GSS Raw data flow Satellite Summary Status:

- Satellite ID
- Satellite overall operational mode
- Summary status of SAR payload]
- Summary status of the navigation payload including clock status (phase comparison measurements between PHM and RAFS, parameters on on-board clock resynchronisation

6.5.3.2.1.4.3 SFC5.3.1 Related Tasks: OSPF

The OSPF supports the On Line Monitoring of the satellite status by providing data inputs to the MUCF:

- Service data:
  - GSS (including PTF-GSS) and satellite clocks w.r.t. GST and associated Allan Standard Deviations.
  - Almanac data
  - Troposphere estimation at GSS sites
  - Comparison between predicted and determined orbits and clocks.
  - Navigation Quality of Service indicators
  - Station status from a navigation point of view
  - Predicted ephemeris and clocks
- OSPF processes:
  - Status: Nominal, Degraded, Faulty
  - OSPF indicators characterising the behaviour (e.g., GSS data rejection statistics)
- Satellite clock:
  - SV clock status
  - SV clock synchronisation data

6.5.3.2.1.4.4 SFC5.3.1 Related Tasks: IPF

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6.5.3.2.1.4.5 SFC5.3.1 Related Tasks: SCCF

The Galileo Ground Control Segment is responsible for the monitoring and control of the entire Galileo satellite constellation.

The SCCF compute the Satellite Summary Status (MM_SarSumStatus) for each satellite in contact from satellite TM data and provided it to GMS-MUCF.

After uplink of the NAV/PRS data, the SCCF generates a NAV/PRS data uplink report (MM_NavSBandUplkRpt) and send it to the GMS-MUCF identifying the success or failure of the uplink.

6.5.3.2.2 Dynamic Scenarios

List of system dynamic scenario defined and detailed in this document for the Mission Performance monitoring:

Scenario number	Type of Monitoring	System scenario name	Description
#SFC5.3.1-1	On-line Mission Monitoring	On-line Mission Monitoring Operational Scenario	This scenario describes the mission on-line monitoring performed by the GMS-MUCF
#SFC5.3.1-2	On-line Mission Monitoring	On-line provision of precise Sat ephemeris to GCS for FD purpose	This scenario describes the provision of the precise satellite ephemeris from the GMS-OSPF to the GCS-FDF
#SFC5.3.2-2	Off line Mission Monitoring	OSPF parameters calibration	This scenario describes the following actions driven by the MSF: the collection of navigation data and raw measurements, the off line calibration of OD&TS parameters and the final delivery of these parameters to the OSPF

Table 85 : List of System dynamics scenarios

6.5.3.2.2.1 Scenario #SFC5.3.1-1: On-line Mission Monitoring

6.5.3.2.2.1.1 Flow Chart

This scenario describes the sequences identified to support the mission on-line monitoring.

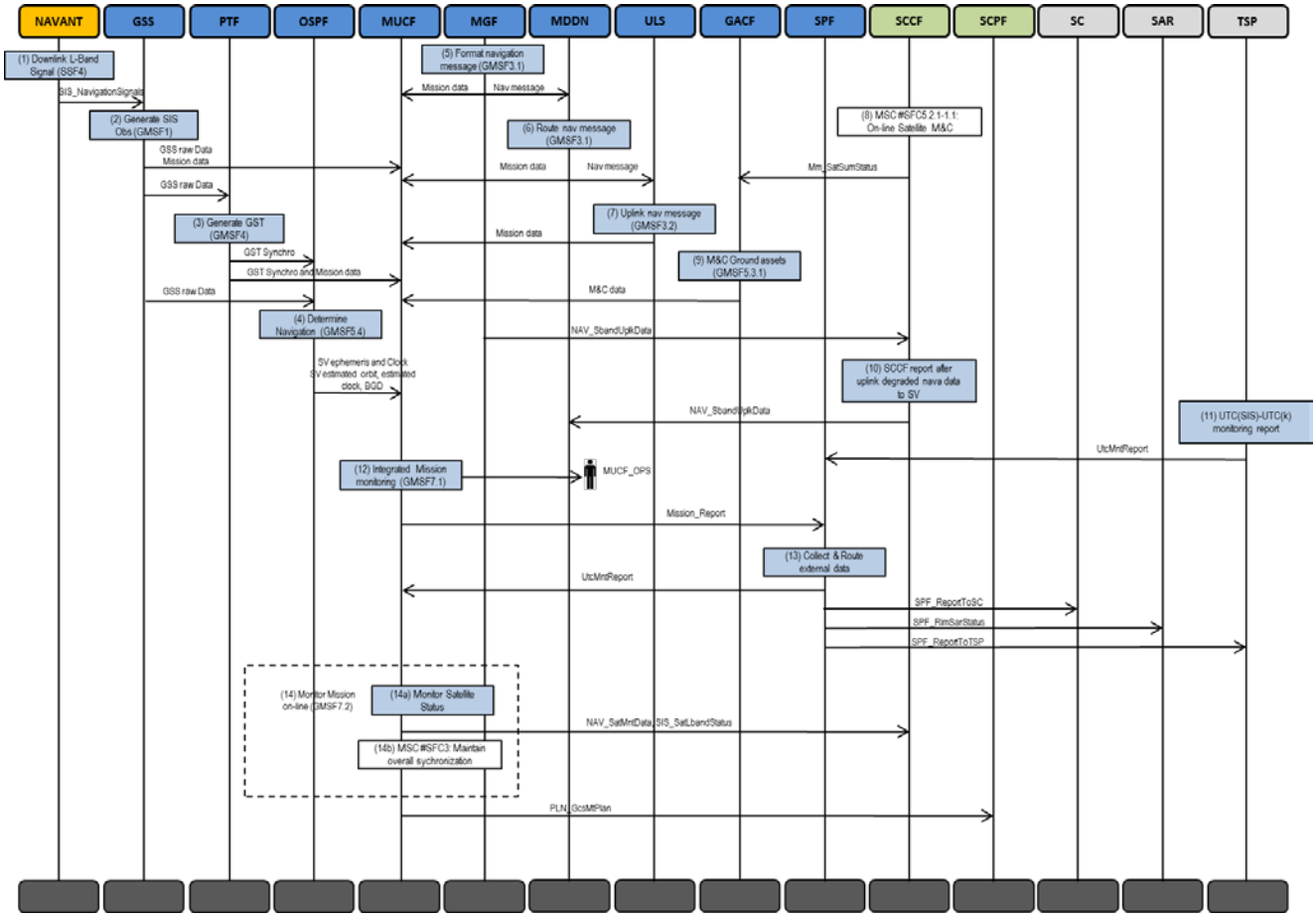


Figure 198 : #SFC5.3.1-1 flow-chart

6.5.3.2.2.1.2 Scenario sequence description

(1) Each GSS processes the SIS broadcasted by all the satellites in view. They compute the pseudorange measurements and demodulate the signals to collect the navigation messages.

(2) The GSS Stations provide to the OSPF, the PTF and the MUCF the raw measurement data and the content of the navigation messages (C/NAV, I/NAV, F/NAV and G/NAV).derived from the SIS Navigation Signals.

(3) The PTF generates the GST and provides it to the OSPF and MUCF. The PTF provides also to the MUCF:

- GGTO offset and the difference between the GGTO coordinated offsets determined by the PTF and the GPS USNO.
- Mission monitoring data including PTF clock stability, PTF to PTF synchronization, clock weight in the GST, PTF steering parameters.

(4) Based on the GST and on the raw data the OSPF is responsible for computing in real time precise satellite ephemeris and time synchronization to the GST for GSS and satellite clocks. Upon completion of Orbit Determination and Time Synchronisation short batch, mission monitoring (SISE, biases, Ionospheric and tropospheric data, navigation and GSS status) and

navigation data including QoS for each service (OS, PRS) are sent periodically by the OSPF to the MUCF.

(5) The MGFs generate the navigation messages and send them through the network with quality indicators for each message. The MGFs send to the MUCF mission monitoring data including the exchange activity and the generated products of the MGF.

(6) A selection of the message is done by some MDDN equipment based on the quality of service indicators. The MDDN equipment provide to the MUCF the status about the MGF selected by the MNE at each Epoch and send a MDDN&SDDN Networks high-level status to the MUCF.

(7) The ULS send a message to the MUCF containing a high level status about the ULS (including mission monitoring, antenna position, RF mode, acquisition time, …) and the status of each uplinked navigation message to the satellites.

(8) The SCCF compute a Satellite Summary Status to alert the GMS of spacecraft anomalies detected through TM monitoring. These data are encapsulated in the MM_SatSumStatus sent by the SCCF to the GACF.

(9) Mission Monitoring data are produced by GACF based on facility state/mode changes. The GACF sends these Mission Monitoring data to the MUCF including the Satellite Summary Status from GCS.

(10) A degraded navigation message is also regularly provided by the MGF to the SCCF as NAV_SbandUplkData to be uplinked to the satellite. The SCCF send to the MUCF the MM_NavSBandUplkRpt as report of the S-Band Uplink Navigation Service Data, with the result of the data up-link performed.

(11) The TSP will compute the offset between UTC(SIS) and the UTC(k) and a statistical analysis of this difference over the past 24 hours (Mean value, 1σ, ADEV, MDEV, HDEV, TDEV). The TSP will then forward the last hourly values and the statistical analysis to the GMS on an hourly basis by means of the signal TIME_UtcMntReport.
It is noticed that this point is still TBC as no input from the Segments are currently available since the requirement introduced by STRB 2.0 (DCN-986 and DCN-13922) are not yet applicable to them.

(12) Based on all the data collected in the previous steps defined below the integrated mission monitoring inside the MUCF provides:
1- to the operators via a MMI:
- synthetic views on the mission & assets status with past, present & future events
- periodically & automatically services performance indicators & relative mission alarms
- performance indicators & parameters on Data Determination & Dissemination chains
2- Mission reports to SPF for further distribution
(13) Distribution of reports to external entities:
- to RLSP and on the SAR, & CS services performances and status
- to the Service Centre (SC) about the CS services performances monitored
(14) Monitor Mission on line

(14a) Monitor satellite status

1- From the OSPF mission monitoring, the MGF mission monitoring, the on-line mission monitoring function of the MUCF monitors the satellite status, from ODTS, point of view analysis and report regularly to GCS the satellite status (NAV_SatMntData,).

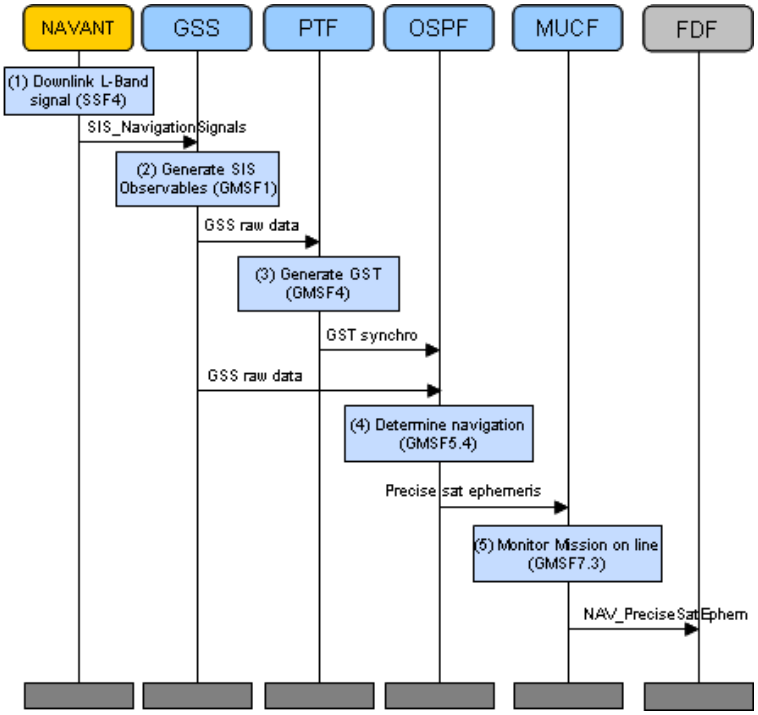
2- From G/NAV frame, the MUCF gets the L-Band telemetry and provides it to the SCCF in the SIS_SatLbandStatus.

(14b) Maintain overall synchronisation

Based on the Satellite clock status from the GCS MM_SatSumStatus, the clock performances computed by OSPF, the MUCF make on-board clock resynchronisation / switch request when necessary; these requests are performed through the planning request made by the MUCF to the SCPF (GCS_MtPlan).

6.5.3.2.2.2 Scenario #SFC5.3.1-2: Provision Of Precise Ephemeris To GCS-FDF

6.5.3.2.2.2.1 Flow Chart



6.5.3.2.2.2.2 Scenario Sequence Description

(1) Each GSS processes the SIS broadcasted by all the satellites in view. They compute the pseudorange measurements and demodulate the signals to collect the navigation messages.

(2) The GSS Stations provide to the OSPF and the PTF the raw measurement data and the content of the navigation messages (C/NAV, I/NAV, F/NAV and G/NAV).derived from the SIS Navigation Signals.

(3) The PTF generates the GST and provides it to the OSPF.

(4) Based on the GST and on the raw data the OSPF is responsible for computing in real time precise satellite ephemeris and to provide it to the MUCF.

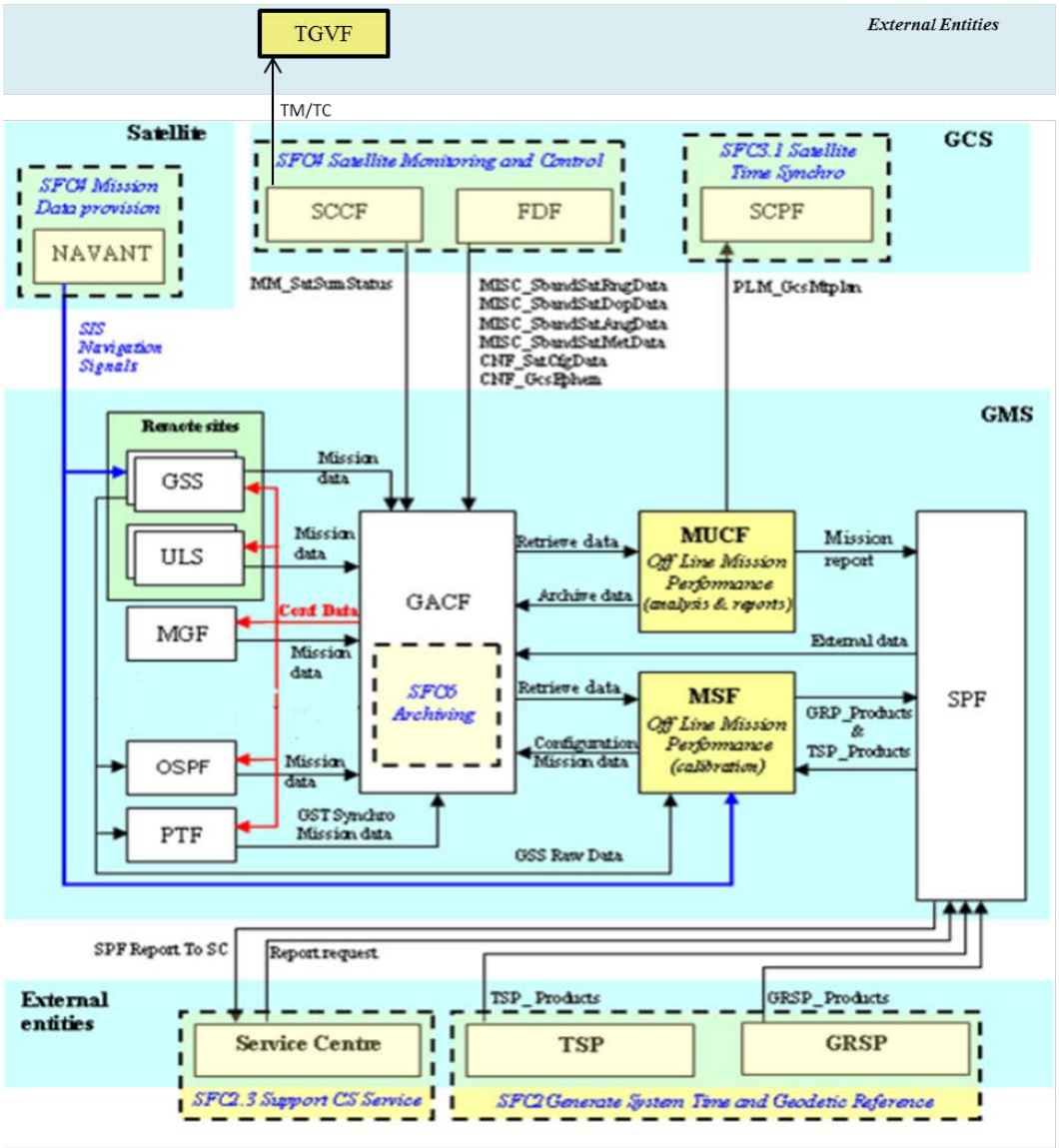
(5) The Monitor Mission on line function in the MUCF provides the precise satellite ephemeris to the GCS-FDF (NAV_PreciseSatEphem) every 10 minutes.

6.5.3.3 SFC5.3.2 Off-Line Mission Monitoring

6.5.3.3.1 Physical Static View

6.5.3.3.1.1 Overall Physical Description

The purpose of this section is to describe the physical mapping of the off-line mission monitoring functions. The following figure presents the physical architecture of the off-line mission monitoring functional chain:



The offline mission performance monitoring is shared between:
- the statistical mission performance analysis and generation of reports
- the navigation data calibration and SIS RF analysis.

6.5.3.3.1.2 Statistical mission performance analysis & reports generation

6.5.3.3.1.2.1 SFC5.3.2 Related Tasks: MUCF

6.5.3.3.1.2.1.1 Mission service performances statistical analyses

The MUCF performs the following statistical mission performance and analysis:
- o Analysis relevant to GST stability performance,
- o Analysis relevant to GST-UTC offset accuracy performance,
- o Analysis relevant to the verification of the correct computation of SISA,
- o Algorithms internal performances
- o Analysis relevant to Galileo satellites atomic oscillator behaviour
- o PTF, ODTS

6.5.3.3.1.2.1.2 Performance Trend analyses of Galileo elements

The MUCF perform trend analysis to get statistical analysis reports relevant:
- o ULS tracking performance.
- o GSS interference and multipath characteristics
- o GSS internal performances (data rejection statistics after-pre-processing, GSS and Satellite HW Biases estimates, tropospheric error estimates, GSS clock synchronisation parameters)

6.5.3.3.1.2.1.3 Generation of reports to external world

The MUCF provides display allowing to request a report from the result analysis described above. According to the kind of analysis performed, a list of possible reports is displayed:
- o GST stability performance
- o GST-UTC offset accuracy performance
- o Verification of the correct computation of SISA

- o Long term of statistics of integrity performance
- o Long term of statistics of continuity performance
- o GSS signal reception anomalies
- o GSS interference and multipath characteristics
- o PTF, ODTS o Interval algorithms performances
- o ULS tracking performance

The relevant analysis report(s) are provided to Mission Engineer or SPF for further distribution.

6.5.3.3.1.2.1.4 General satellite status from GCS to GMS

The satellite summary status (including parameters on on-board clock re-synchronisation) is provided by the GCS (MM_SatSumStatus) to the GACF and finally routed to the MUCF. The

SCAT function determine the parameters for changing the frequency of the on-board time reference and issue corresponding synchronisation requests to the GCS for adjustment of on-board clock, considering:

- o A maximum change of the frequency of the on-board time reference of 10.23 mHz.
- o A step size of an integer multiple of 0.002046 mHz.

See SFC3.1 Satellite Times provision and Synchronisation for details.

6.5.3.3.1.2.2 SFC5.3.2 Related Tasks: GACF

6.5.3.3.1.2.2.1 Data archive and retrieved by the MUCF for Off-line mission monitoring

MUCF perform archive migration to the GACF archive of the following mission monitoring data:

- o Mission events
- o Nav data extraction
- o Planning data
- o Uplink schedules
- o Reports

To perform the off-line mission monitoring, the MUCF need to perform archive retrieval from the GACF archive of the following mission monitoring data:

Retrieved data by the MUCF from the GACF archive	Archived in GACF by
Received CS_Data since last epoch and status of the CS_data that are being uplinked:	MGF
Active and last contact plan received at MGF	MGF
Status of dissiminated data including MGF QoS (MSF), OSPF ID source for selected nav data(OS, PRS), , selection code based on QoS indicators (for OSPF) associated to the switching mechanisms, MSF ID and ISF ID Information for each satellite (up to 36) including CRC of the satellite Message Subframes (MSF) CRC, MSF timestamp, result of the MGF safety barriers flag forMSF generation,	MGF
Information on the message transmission, including reception and uplink status of message	MGF
List of barriers activated (barrier code) in the last 10 minutes and barriers result.	OSPF
MGF navigation data generation information for each satellite addressed in the epoch (up to 36):	MGF
Satellite status information derived from the ODTS processes (1 flag per spacecraft).	OSPF
Monitoring data for each satellite when the MGF raise an alert	MGF
Message triggered by the reception of a new RLM request or by the scheduling of an RLM repetition:	MGF
Data for received RLM requests since last epoch (0 to 27).	MGF

The Satellite Summary Status from Satellite TM data	GCS
Tropospheric vertical (zenith) delay estimations for the GSS within the GSTS process.	OSPF
ULS traking plan ID and MGF contact plan ID received at ULS	ULS
File containing the Uplink Antenna positions provided to the MUCF during a configurable period	ULS
For each processed satellite, contain the three predicted predicted clock prior to ephemeris message fitting.	OSPF
Message containing SAR Service bulletin.Contents	MUCF
Information regarding the GSS channel status derived from the ODTS determination processes.	MUCF
Routing information for the Commercial Service (CS). Earth grids defining which satellites should be accessed to disseminate information on a particular CS region.	MUCF
Information on the dissemination of regional integrity through direct uplink to Galileo satellites. Plan defining time periods of connection for an external region to particular satellites.Content	MUCF
ULS to Satellite connection plan.	MUCF
Tracking plan for one ULS antenna.Content :	MUCF
Information received from the non-PRS receiver of each GSS at 1 second GST epoch rate.	GSS
Information received from the PRS receiver of each GSS at 1 second GST epoch rate.	GSS
Information on PTF internal clocks comparison, sampled hourly with respect to GST(MC)	PTF
Signal containing information on the frequency corrections applied by PTFs	PTF
GPS-CV measurements, collected at PTF and sent to the External Enities via the Master SPF.	PTF
PTF TWSTFT measurements data.In line with the Data Format for TWSTFT measurements data	PTF
Signal containaing GGTO measurements results obtained by PTF, using both TWSTFT and CV techniques.	PTF
GGTO coordination results obtained by PTF, using both GalGpsGgto values computed in the GMS and in the GPS/USNO.	PTF
Information regarding the deviation of satellite clocks with respect to GST and for each service (OS / PRS).	OSPF

Table 86 : Data Retrieved by MUCF from the archive for off-mission monitoring

6.5.3.3.1.2.2.2 Data from GCS

The GACF is in charge to provide to the MUCF the satellite summary status (including parameters on on-board clock re-synchronisation) provided by the GCS (MM_SatSumStatus).

6.5.3.3.1.2.3 SFC5.3.2 Related Tasks: SPF

The MUCF build and send reports to the Service Centre (SC) in 3 cases :

- o automatically and daily
- o once new almanac is available
- o on SC request

The SC perform report requests via the SPF web server. The SPF then processes the request to determine which files are to be retrieved and where from (either the SPF or GACF) If the file is not stored at the SPF, the GACF is requested to retrieve the file from the archive. The SPF do not initiate connections to elements within the GCC, so there is an SNMP agent in SPF which is polled by GACF in order to know if there are request in SPF, in case there is a request in SPF for GACF this is notified through SNMP protocol and then GACF initiates and FTP-Get in order to receive the query by FTP, then GACF send the data to SPF with an FTP-Put.

6.5.3.3.1.2.4 SFC5.3.2 Related Tasks: SCCF

The SCCF is responsible to send the TM/TC data to TGVF. The Time and Geodetic Validation Facility (TGVF) processes TM and TC packet data relating to specific spacecraft domains to correlate against environmental conditions on-board the spacecraft when validating orbit and clock estimations. The TGVF also requires the operational spacecraft TM/TC databases to provide the definitions necessary for processing the TM/TC packet data provided.

6.5.3.3.1.3 Navigation Data calibration

6.5.3.3.1.3.1 SFC5.3.2 Related Constraints: MSF

The MSF major functions are summarized as follows:

- ☐ Offline generation of the reference orbit and clocks for the GMS.
- ☐ Long term performance analysis of the OSPF for calibration purposes
- ☐ Generation of operational configuration data.
- ☐ Analysis of SiS digital and RF characteristics.

6.5.3.3.1.3.1.1 Reference orbits and clocks

The long term performance analysis of the satellite ephemeris and clocks is one of the primary functions of the MSF. It is performed regularly on a routine based scenario using standard GSS measurements and Satellite Laser Ranging data from dedicated campaigns.

6.5.3.3.1.3.1.2 Support to the OD&TS algorithms calibration

The long term analysis of the satellite ephemeris and clocks are oriented towards the calibration of operational algorithms. More particularly the MSF is in charge of providing calibration data such as:

- ☐ Satellite Model Parameters (SMC) to OSPF,
- ☐ Historical SiS Accuracy (HSISA).to OSPF,

These calibration data are inserted in operational configuration parameters/files and transferred from the MSF to the GACF, which is in charge of managing the final reconfiguration on the operational Element.

6.5.3.3.1.3.1.3 Support to calibration of parameters used in the navigation chain

The MSF collects configuration data from various sources:

- ☐ Galileo external service providers,
- ☐ the internal MSF calibration function,
- ☐ the Galileo Control Centre,
- ☐ the operator,
- ☐ the GCS

The MSF checks the provision of these operational configuration files according to a predefined set of algorithms.

These data are inserted in operational configuration parameters/files and transferred from the MSF to the GACF, which is in charge of managing the final reconfiguration on the operational Element (OSPF, PTF, MUCF, etc).
See table below for more details

These data provided includes:

- ☐ Satellite Configuration data (mass & center of mass position),
- ☐ Earth Orientation Parameters (EOP),
- ☐ GSS coordinates (position & velocity),
- ☐ Element configuration parameters,
- ☐ Solar Flux data for ionosphere propagation delay in L-Band
- ☐ ITRF-GTRF transformation matrix

Data	Origin	Via	To	MSF role	Description
Differential Group Delay	MSF	GACF	OSPF	generation	Differential Group Delay both for satellites and stations (retrieved in the from the MTPF)
Earth model	SPF	GACF & MSF	OSPF MUCF	validation	Physical model of the Earth (Earth) gravity field
EOP	SPF	GACF & MSF	OSPF	validation	Default configuration file for the predicted and estimated Earth Orientation Parameters as provided by the IERS bulletin A.
GCS ephem	GCS	GACF & MSF	OSPF GSS MUCF	validation	GCS Computed S/C Ephemeris file for all the S/Cs in ITRF reference coordinates

GSS coordinate	SPF	GACF & MSF	OSPF MUCF	validation	GSS coordinates
GST UTC Offset	SPF	GACF & MSF	OSPF MUCF PTF	validation	GST UTC offset
Historical SISA	MSF	GACF	OSPF	generation	Configuration file of the Historic SISA for each satellite, as computed by the off-line analysis function
Priority Set	MSF	GACF	MGF	generation	Defines the OSPF priorities for the MGF
MGF conf Param	MSF	GACF	MGF	generation	Configuration parameters for the MGF
Sat conf data	GCS	GACF & MSF	OSPF MUCF	validation	S/C configuration data provided by the GCS: Satellite mass (subject to evolution) and Centre of Mass Evolution.
Solar Flux (F10.7)	MSF	GACF	OSPF	validation	Configuration file validated by MSF and including application-level CRC. Contents: Solar flux data (F10.7) needed by ODTS
SRP coefficients	MSF	GACF	OSPF	validation	Solar Radiation pressure coefficients
Station-satellite data	MSF	GAFC	OSPF GSS ULS PTF MUCF		Identification of the Galileo satellites and codes to be considered by the GSS, OSPF (ODTS), and MUCF (uplink scheduling algorithms), GCS interface with CCSDS codes, ULS uplink codes, etc.., and the status (mask) for each satellite.
TUC TAI offset	MSF	GACF	OSPF PTF MUCF	validation	Off line configuration file containing leap seconds, separating TAI time scale from UTC time scale values.
ITRF-GTRF matrix	MSF	GACF	OSPF	validation	Contains the information on transfer procedure between GTRFcoordinates and the ITRF 2005 reference frame.
ULS conf data	MSF	GACF	ULS		ULS configuration parameters

Table 87 : List of data configuration processed in MSF

6.5.3.3.1.3.2 SFC5.3.2 Related Constraints: GACF

6.5.3.3.1.3.2.1 Data retrieved by the MSF for Off-line mission monitoring

The GACF supports the MSF for the off-line mission monitoring with retrieval of the following data:

Retrieved data by the MSF from the GACF archive	Archived in GACF by
Manoeuvre announce files (MISC_ManvrAnnounce) and orbit prediction (MISC_OrbitPred)	GCS
Satellite Clock predictions (with the three predicted clock prior to ephemeris message fitting)	OSPF
Satellite Orbit predictions (with the three predicted orbits prior to ephemeris message fitting)	OSPF
Description of the GMS short-term activities	MUCF
Information regarding the deviation of GSS clocks with respect to GST and for each service (OS / SoL / PRS).	OSPF
Information regarding the deviation of satellite clocks with respect to GST and for each service (OS / SoL / PRS).	OSPF

Table 88 : Data Retrieved by MSF from the archive for off line mission monitoring

6.5.3.3.1.3.2.2 Support to MSF configuration files distribution

The GACF receive configuration files dedicated to GMS facilities (i.e: OSPF, PTF, MUCF, ULS and MGF) from the MSF. These files are collected by the Archive functions, via FTP protocol. The commanding is done by the M&C function using SNMP.

6.5.3.3.1.3.3 SFC5.3.2 Related Constraints: GSS

The GSS provide to the GACF file migration from local storage with provision of:
- Mission Monitoring data
- Raw measurements and data from the SIS messages

6.5.3.3.1.3.4 SFC5.3.2 Related Constraints: OSPF

The OSPF receive the following configuration files from the GACF:
- ☐ Configured Synchronisation Data
- ☐ GSS Network configuration data
- ☐ GSS Measurements rejection angle
- ☐ GTRF calibration data
- ☐ PRN mask configuration
- ☐ Quality of Service Target Locations
- ☐ Galileo satellites configuration data
- ☐ ODTS Algorithm configuration data
- ☐ Ionospheric model data file
- ☐ GSS receiver Interfrequency Bias calibrated data
- ☐ Historic SISA configuration data
- ☐ SRP model

6.5.3.3.1.4 SIS RF analysis

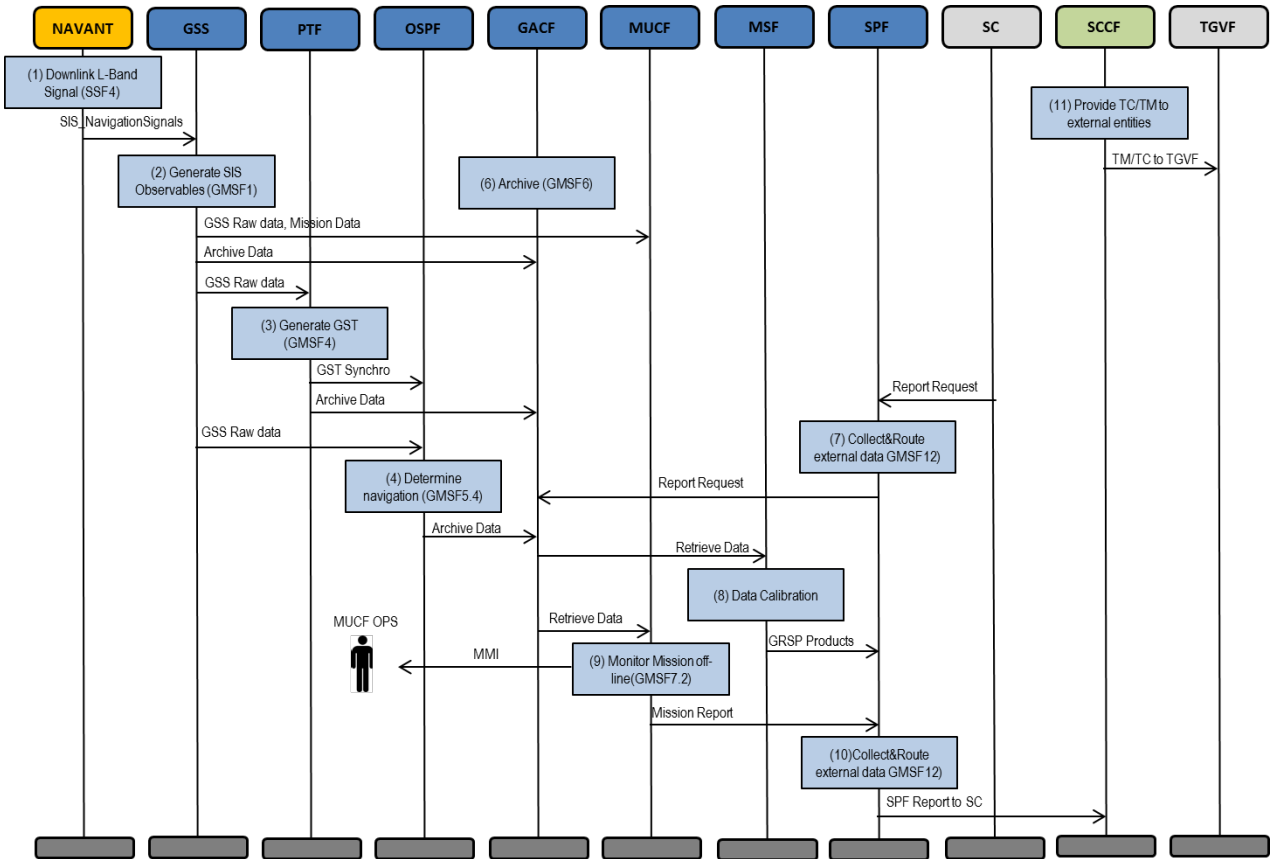
6.5.3.3.1.4.1 SFC5.3.2 Related Tasks: IOT facility

The IOT facility performs the SIS analysis with equipment that contains a Vector Signal Analyser. The instrumentation will allow the operator to check diverse signal characteristics, such as carrier frequency, bandwidth, chip length, etc.
Note: The signal analysis tool is located in the IOT facility.

6.5.3.3.2 Dynamic Scenarios

6.5.3.3.2.1 Scenario #SFC5.3.2-1 General Off-line Mission Monitoring

6.5.3.3.2.1.1 Flow Chart



6.5.3.3.2.1.2 Scenario sequence description

(1) Each GSS processes the SIS broadcasted by all the satellites in view. They compute the pseudorange measurements and demodulate the signals to collect the navigation messages.
(2) The GSS Stations provide to the OSPF, the PTF and the MUCF the raw measurement data and the content of the navigation messages (C/NAV, I/NAV, F/NAV and G/NAV).derived from the SIS Navigation Signals. The MUCF also provides these data to GACF for archive.
(3) The PTF generates the GST and provides it to the OSPF and MUCF. The PTF provides also to the MUCF:

- GGTO offset and the difference between the GGTO coordinated offsets determined by the PTF and the GPS USNO.
- Mission monitoring data including PTF clock stability, PTF to PTF synchronization, clock weight in the GST, PTF steering parameters.

(4) Based on the GST and on the raw data the OSPF is responsible for computing in real time precise satellite ephemeris and time synchronization to the GST for GSS and satellite clocks. Upon completion of Orbit Determination and Time Synchronisation short batch, mission monitoring (SISE, biases, Ionospheric and tropospheric data, navigation and GSS status) and navigation data including QoS for each service (OS, PRS) are sent periodically by the OSPF to the MUCF. The MUCF also provides these data to GACF for archive.
(5) Removed (integrity related)
(6) All the mission data are archived in the GACF and are available to be retrieved on request.
(7) Report requests from SC via the SPF web server by operators. The SPF processes the requests to determine which files are to be retrieved from the archive.

(8) Based on the data collected from the GACF archive, the MSF performs the calibration of some navigation data (ephemeris, clock, earth rotation parameters) for further distribution to the GRSP.

(9) Based on all the collected data the mission monitoring function inside the MUCF provides:
- to the operators via a MMI services performance indicators
- Mission reports to SPF for further distribution

(10) Distribution of reports to the Service Centre (SC).

(11) Provision of TM/TC data to TGVF for external system monitoring

6.5.3.3.2.2 Scenario #SFC5.3.2-2: ODTS calibration

6.5.3.3.2.2.1 Flow Chart

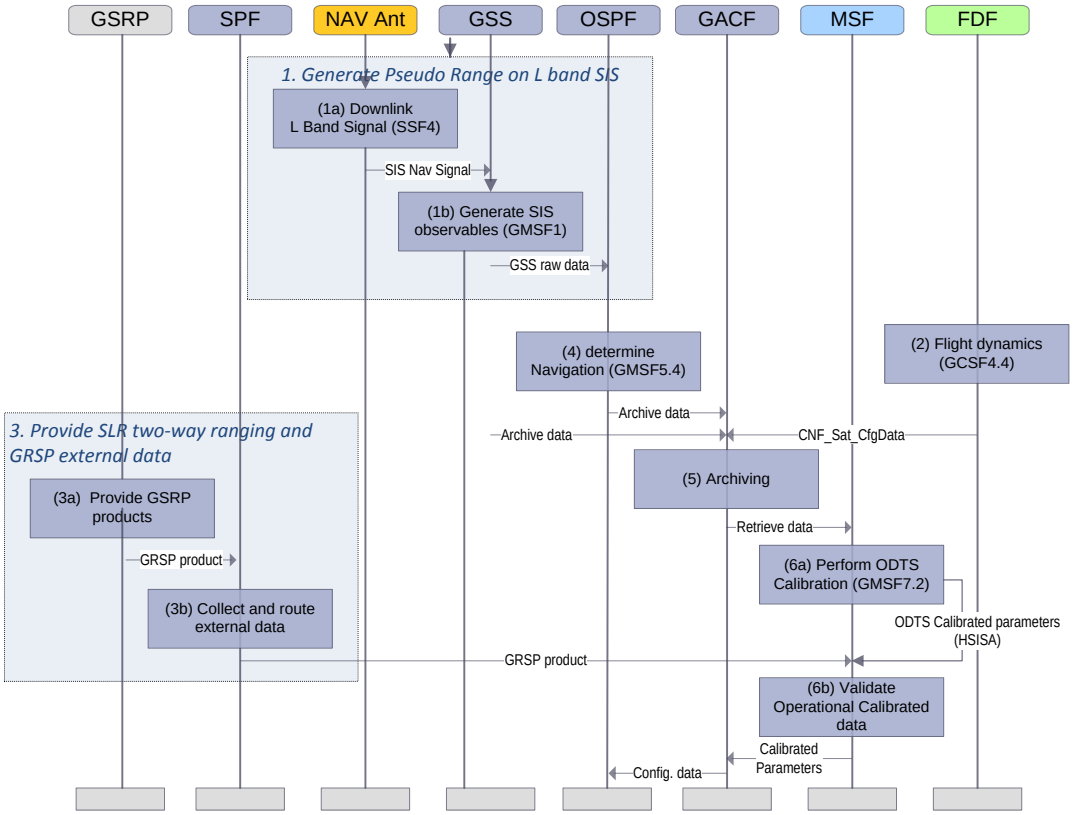


Figure 199 : #SFC5.3.2-2 Dynamic Scenario

6.5.3.3.2.2.2 Scenario sequence description

(0) This scenario describes the following actions driven by the MSF: the collection of navigation data and raw measurements, the off line calibration of OD&TS parameters and the final delivery of these parameters to the OSPF

(1) *Perform pseudo-ranging on L-Band signal*

(1a) The L-Band SIS is broadcasted by the Galileo satellites

(1b) The GSS compute the pseudorange measurements from the L-Band SIS for provision to the OSPF and GACF

(2) *Provision of Satellite Configuration data*

The GCS FDF provides updated Satellite configuration data to the GMS-GACF

(3) *Provide GRSP products*

(3a) The GRSP provides to the GMS-SPF:

- Satellite Laser Ranging data performed on Galileo satellites by the ILRS
- GSS stations coordinates
- Earth Orientation Parameters
- Solar Flux data for ionosphere propagation delay in L-Band
- ITRF-GTRF transformation matrix

(3b) These are sent by the SPF to the MSF

(4) Based on pseudo measurement data, the OSPF compute OD&TS and provide the following data to the GACF:

- restituted satellite orbits and clocks
- restituted GSS clocks and coordinates
- estimated tropospheric corrections

(5) The data collected by the GACF are provided to the MSF

(6) Calibration of data and distribution

(6a) Based on the data collected from the GACF, the MSF performs the calibration of the following parameters:

- ☐ Historical SiS Accuracy (HSISA, computational details are reported in SYS-RD.4).

(6b) Based on the data collected from the GACF, the SPF and the OSPF parameters internally calibrated by itself, the MSF performs a validation of the calibrated parameters before further distribution:

- ☐ Satellite Model Parameters (SMC),
- ☐ Satellite Configuration data (mass & center of mass position),
- ☐ Earth Orientation Parameters (EOP),
- ☐ GSS coordinates (position & velocity),
- ☐ Historical SiS Accuracy (HSISA),
- ☐ and OSPF configuration parameters.

The MSF sends the calibrated parameters to the GACF for distribution to the OSPF

6.5.3.3.2.3 Scenario #SFC5.3.2-3: Integrity Parameter Calibration

Section deleted

6.5.3.3.2.4 Scenario #SFC5.3.2-4: SIS OFF Line monitoring Scenario

Section deleted

6.5.4 SFC5.4 Mission Planning Provision

The objective of the Mission Planning provision SFC is to enable planning both routine and special system operations for the following activities:

- Constellation management and satellite control
- Ground assets monitoring and control
- Mission service management

The mission planning includes the production of executable schedules.

The following section describes the physical implementation of the mission planning. It is based on the following documentation:

- System documents:
  - System Logical Architecture (SYS-RD.1)
  - System operational concept: MOCD (SYS-RD.6)
- GMS documents:
  - GMS DDF2 (GMS-RD.2): MUCF section and Mission Planning section
  - MUCF ERD (GMS-RD.10), MUCF ICD (GMS-RD.11)
- GCS documents:
  - GCS DDF (GCS-RD.1), SCPF, SCCF and Planning Concept sections
  - SCPF-ERD (GCS-RD.17)

6.5.4.1 Physical Static View

6.5.4.1.1 Overall Physical Description

6.5.4.1.1.1 Definitions

This section recalls the main definition of planning and schedules as per the operational concept

Long-Term Plan: The long-term plan is the high level planning which provides input to lower, more detailed levels of the planning process, used to derive both the mid-term plan (plan refinement) and the short term plans. It typically covers a period of 1-6 months, with periodic updates, but can be extended to cover up to 12 months. The long term plan considers activities such as new launches, replenishment, G/S and S/C highlevel maintenance. Due to the long period to be planned, some inconsistencies or conflicts between activities may coexist (for example, exact start/end dates may not be known and longer temporal. slots could be reserved/specified for an activity).

Mid-Term Plan: The mid-term plan is essentially a refinement of an existing long-term plan which specifies a greater level of detail. It covers a period of up to 4 weeks, where activities planned are conflict-free (all the inconsistencies and overlaps are resolved).
The mit-term plan will in particular

Short-Term Plan: The Short Term plan is a refinement of the mid-term plan. It consists in a time ordered and conflict free plan with a fine granularity of the tasks to be executed based on the activities planned in the Mid Term plan, and integrating all tasks and events required to complete the planning of these activities. The short-term plan period is expected to be in the range of 1 to 10 days.

Schedules: Schedules are generated from the short-term plans and represent the last element in the planning chain prior to execution. The schedules can be divided in 3 main categories:

- ☐ Manual Operations: Timelines & Operators Tasks,
- ☐ Automatic Tasks: Executable Schedule for the Overall, Central and Element M&C functions,
- ☐ Ground to space contact schedules (ULS to satellite and TTC).

6.5.4.1.1.2 Mission Planning Implementation

The main tasks of the mission planning process are described below:

- Task 1 - Long/Mid-Term Planning inputs collection:
  - The GMS-MUCF collects system configuration information. These information come from the GMS-GACF) and from the GCS-FDF on constellation orbit events (PLN_MtOrbitEventFile) - e.g. eclipses, sensor blinding, etc.
  - The GMS-MUCF collects Mid/Long-Term planning requests from GMS and GCS Mission Planning Operators. The GCS operators have to submit requests to the GMS-MUCF from the GCS-SCPF (PLN_MtPlanRqst) in case these requests have an impact on the mission (see JT-86 in IOV DJF [SYS-RD.3]).
    Planning request having mission impacts are those which could cause some impact on the Mission service provision for the affected satellite. To minimise the effect on the level of service provided to the user it is necessary to ensure that only a single satellite at a time can undergo such a kind of operations, maintaining coordination trough a common plan at GCS and GMS level. Example of such tasks are provided in the table below:

Planning request	Impact on affected satellite
Orbit manoeuvre	SIS Suspended for the duration of the manoeuvre, message generation processing to re-convergence after operation
On Board SW patch	SIS may be suspended for the duration of the operation
CMCU Clock Switch	For FOC satellites: temporary negligible performance degradation after operation FOR IOV Satellites: message generation processing to re-converge after operation
Clock /frequency correction	message generation processing to re-convergence after operation
Other Payload reconfiguration	SIS interruption for the duration of the operation, message generation processing may have to re-convergence after operation

Table 89. Example of Operations with mission impact

Note: The satellite operation requests originated from GCS-FDF operators are processed by the GCS-SCPF before being forwarded to the GMS-MUCF [GCS-RD.1].

- Task 2 - Verification of Mid-Term Plan compatibility with the system mission performance and provision of service bulletin:

- The GMS-MUCF is in charge of validating that the Mid-Term Plan is compatible with the mission performance.

- The validation of the Short Term plans is provided by the GACF and the MUCF and edited as part of the Mid Term Plan validation. The Mid Term Plan and Short Term Plan become operational and are transmitted to other entities for application. In this process of validation, several levels of feedback can lead to invalidate the proposed plan and to force a replanning (see evolution section 6.5.4.1.1.2).

- The GMS-MUCF also produces system status and service performance bulletins/reports via the GMS-SPF to:
    - the GOC (in SPF_BulletinToSc: INF_maintenancebulletin, INF_maintenancebulletin; and in SPF_ReportToSc: INF_SrvcPerfoPredictions , INF_NoticeAdvisoryToUsers),
    - the RLSP (SPF_SarServiceBulletin)

- Task 3 - Provision of Mid-Term Plan to the GCS and GMS Short-Term Planning functions:
    - The GMS-MUCF provides an extract of the Mid-Term Plan containing all planning requests involving the GCS to the GCS-SCPF (PLN_GcsMtPlan).

- Task 4 -Elaboration of Short-Term Plan and executable schedules by GCS and GMS Mid-Term Plan Refinement:
    - The GMS-MUCF produces the GMS Short-Term Plan which is a refinement of the Mid-Term Plan for GMS tasks. The GMS Short-Term Plan nominally covers a period of 10 days (MUCF-ERD-1910, 2520, 7390) and is updated every 3-6 days (MUCF-ERD-1760, MUCF-ERD-1770). The Short-Term Plan is validated to ensure that it is compatible with the mission performance.
    - The GMS produces executable schedules from the GMS Short-Term Plan as follows:
        - The GACF produces (cf. [GMS-RD.2] section on mission planning):
            - Automatic and manual command schedules for the GMS elements.
            - Detailed maintenance activity schedule and maintenance order (text file containing detailed description of operator activities).
        - The MUCF produces:
            - Mission C-Band uplink schedules for operational and test dissemination which are passed to MGF and ULS. The C-Band uplink schedule is composed of an ULS to satellite allocation plan (SCH-OverallUlsContactPlan) sent to the MGF and a ULS to satellite tracking plan including the satellite almanacs sent to the ULS (SCH-UlsTrackPlan). The MUCF ensures that both the MGF and the ULS use the same uplink schedules as explained in [GMS-RD.11] (see also SYS-RD.1 SFC5.4 on future evolution of the tracking plan distribution).
            - Predictions for the Satellite Heath Status (SHS) based on the tasks planned in the MTP originating by GMS internal requests or by GCS requests.
              The following table provide a list of tasks originating from GCS planning requests which affects the SHS setting:

MTP Task requested by GCS	SHS value set during

	planned task
Non-routine payload configuration	Out of service
Satellite Master clock synchronization	Out of service
Satellite on-board time transmission synchronization	Out of service
Satellite Master Clock Switch	Out of service
Satellite hot backup to cold backup clock switch	OK (FOC SV) Out of service (IOV SV)
Orbit manoeuvre	Out of service
OBSM (requested by GCS)	Out of service
TTCF Station unavailability	OK
SAR unavailability	OK
Satellite unavailability	Out of service
SIS unavailable	Out of service

Table 90. SHS values vs GCS planned tasks

Note that the planned SHS setting can be manually overwritten on a per satellite/service/signal basis during actual operations by Operators, however only in the direction of selecting a more conservative SHS value (see table below), due to the possible interference with other planned activities.

SHS planned value	**Allowed SHS manual override values**		
OK	Will be out of service	Out of service	In Test
Out of service	In Test		
In Test	*<No override allowed>*		

Table 91. SHS Allowed manual override values

The possibility always exists to replace a plan with new one, therefore overriding all previous planned SHS setting, but such an action, due to the necessary time to approve and distribute the new plan before it becomes operational, cannot produce observable effects on the short term.

- The GCS-SCPF takes the mission level Mid-Term Plan, and combines this with specific requests for GCS and spacecraft operations to produce the GCS Short-Term Plan.

- The GCS produces executable schedules from the GCS Short-Term Plan as follows:
  - The SCPF produces schedules for GCS elements manual operations. The manual operations are then executed by operators at CMCF or SCCF, which is not remotely controlled from the CMCF.
  - The SCPF produces the TTC contact schedule based on satellite visibility data provided by the FDF and on specific planning requests. The TTC contact schedule is passed to the SCCF for implementation (sending directives to the TTCF) and to the GMS-MUCF to allow the GACF operator to be aware of the planned TT&C stations usage.
  - Primary and back-up contacts are scheduled for each routine satellite contact, the back-up contact uses a different TT&C station at a later time than the prime contact, it is also forwarded to the GMS for co-ordination purposes.
  - The SCCF produces the ground based configuration and operations schedule (this includes the schedule of both automatic and manual operations).
  - The SCCF produces the on-board command schedules to be uploaded to each satellite.

- Task 5 - Collection of Feedback on Short-Term Plan Execution:
  - The GMS-MUCF collects feedbacks on the execution of the Short-Term Plan from:
    - The GMS-GACF on the execution of the GMS Short-Term Plan [GMS-RD.11].
      Note: This feedback is only for the execution of automatic tasks and does not cover the execution of manual tasks by operators.
    - The GMS-MGF and GMS-ULS on the execution of the C-Band uplink schedule. This consists in the monitoring by the MUCF that the MGF and ULS apply consistent uplink schedules [GMS-RD.11].
      Note: ULS antenna chain failure is not considered as part of the routine feedback management as in this situation the Mission Planner (during working hours) or the on-duty Mission Controller (during nights and weekends) will need to rebuild a schedule immediately as soon as the failed ULS antenna chain is identified.
  - The GCS-SCPF on the execution of the GCS Mid-Term Plan tasks (PLN_MtPlanExec).
  - The GCS-SCPF receives feedback on the execution of the GCS Short-Term Plan from the GCS-SCCF.

6.5.4.1.1.3 Security Mission Planning Implementation

This section provides unclassified details on the implementation of mission planning of security activities related to the management of:

- ☐ The satellites Platform Security Units (PFSU) for IOV satellites, Common Security Unit (CSU) for FOC satellites by the GCS-SKMF.
- ☐ The satellites PLSU/CSU by the GMS-MKMF (C-Band authentication) and GMS-PKMF (PRS encryption).
- ☐ The Commercial Service (CS) ranging code encryption by the GMS-MKMF.
- ☐ Ground infrastructure security.

6.5.4.1.1.3.1 Planning of PFSU/CSU Activities

The planning of PFSU/CSU activities will be managed by a local planning function in the GCS-SKMF. The secure telecommands (STC) are then directly fed into the GCS-SCCF sufficiently in advance using the 15kbyte bandwidth per TTC contact booked for this purpose. However it is possible for KMF operators to make specific planning requests in the GCS-SCPF in particular in cases where the security operation needs a near-real-time access the satellite to deal with contingency recovery (constraint on TTC contact schedule).
More details on the PFSU/CSU monitoring and control can be found in the SEC-SFC12 in the classified DDF [SYS-RD.8].

6.5.4.1.1.3.2 Planning of PLSU/CSU Activities

The planning of the satellite PLSU/CSU operations can be split in two parts:

1. Planning of operations relative to the C-Band Protection functionality: A local planning function within the GMS-MKMF plans the release of STC to the GCS-SCCF sufficiently in advance so that the time-tagged STC are uplinked to the satellite well before their execution. A bandwidth of nominally 15kbyte per contact is booked by GCS-SCCF for this purpose. More details on the management of the C-Band protection function of the PLSU can be found in the SEC-SFC13 in the classified DDF [SYS-RD.8].

2. Planning of operations relative to the PRS ranging code management: A local planning function within the GMS-PKMF plans the release of STC to the GCS-SCCF sufficiently in advance so that the time-tagged STC are uplinked to the satellite well before their execution. A bandwidth of nominally 15kbyte per contact is booked by GCS-SCCF for this purpose. More details on the management of the PRS ranging code function of the PLSU can be found in the SEC-SFC15 in the classified DDF [SYS-RD.8].

In case of contingency requiring the planning of a dedicated TTC contact, the KMF operators will make a specific contact request in the GCS-SCPF.

6.5.4.1.1.3.3 Planning of CS Ranging Code encryption Activities

The CS ranging code encryption is managed by the NSGU on-board the satellite and is controlled from the GMS-MKMF. Two main tasks are identified:

1. Upload new CS ranging code encryption keys to the NSGU:

- The MKMF makes a short term planning request to the GCS-SCPF so that the needed tasks are included in the GCS short-term plan. The GCS-SCCF will send the TC uploading the new keys according to the GCS STP.

2. Activation/Deactivation of the ranging code encryption:
   - The enabling / disabling of the CS ranging code encryption can be only planned in the GCS SCPF, via a manual planning request in GCS-SCPF by a GCS operator. This request will be managed as a GCS planning request without mission impact and therefore managed directly in the GCS STP, without passing through the mid-term plan exchanges with GMS.

6.5.4.1.1.3.4 Planning of Ground Infrastructure Security Activities

The planning of operations relative to the protection of the overall Galileo network (including GCS network) will be managed by a local planning function in the GMS-MDDN (GNMF-SEC). More details on the network security monitoring and control can be found in the SEC-SFC11 in the classified DDF [SYS-RD.8].

6.5.4.1.1.4 Static Block Diagram

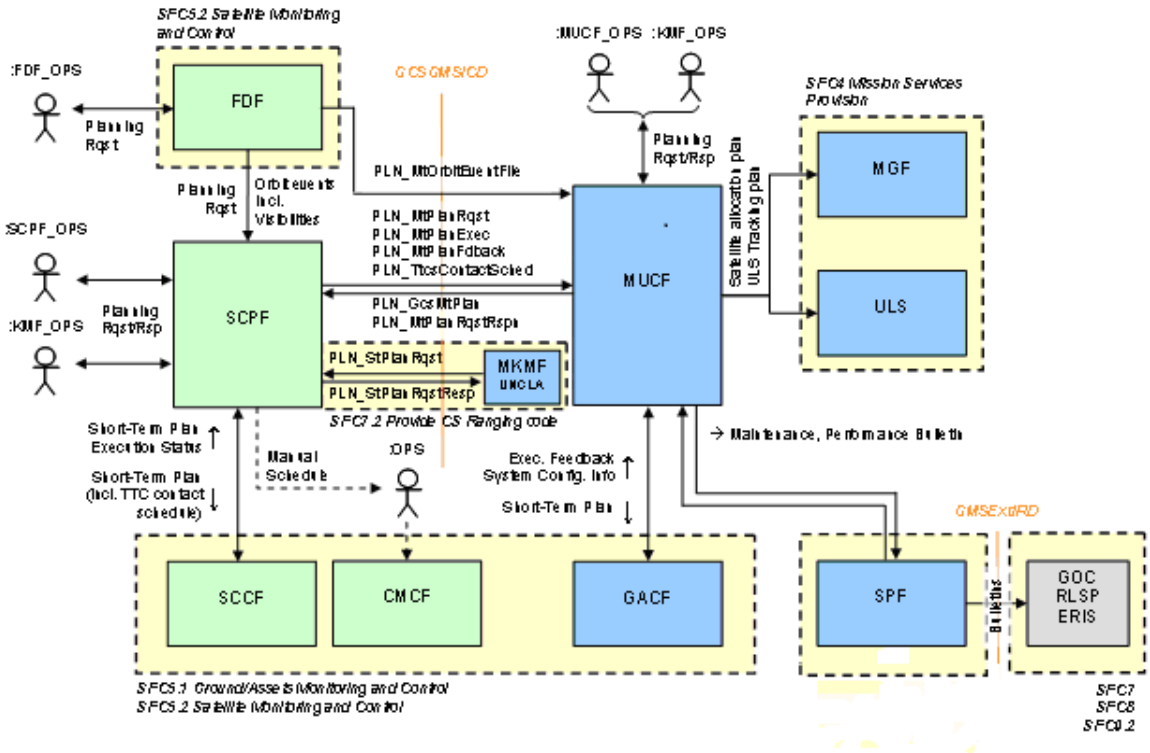


Figure 200 : SFC5.4 Mission Planning Physical Static View

The following subsection summarizes the involvement of the elements in the Mission Planning element-by-element.

6.5.4.1.2 GMS Schedule Production: GACF

The GACF is responsible for:

- Producing schedules from the GMS short-term plan [GMS-RD.2].
- Providing feedback on the execution of the short-term plan to the MUCF [GMS-RD.11].

6.5.4.1.3 Spacecraft Manoeuvres Planning: FDF

The FDF provides the SCPF with (see [GCS-RD.1]):

- Orbit event file, with information like eclipse periods where specific manoeuvres could be forbidden.
- Spacecraft to TTC visibility files in order to allow the production of the TTC contact schedule.
- Manoeuvres planning requests, which contains types of manoeuvre and window for their execution.

The FDF provides the MUCF with:

- Orbit event file, with information like eclipse periods where specific payload operations could be forbidden.
- GCS computed ephemerides (CNF_GcsEphem), to allow the GMS to resume pseudorange data processing after a satellite has been put out of service for maintenance.

6.5.4.1.4 System Long/Mid-Term Planning and GMS Planning: MUCF

The MUCF is responsible for (see [GMS-RD.2]):

- Maintaining the Long/Mid-Term Planning including the processing of requests and planning feedback received from the GCS-SCPF, and the production of Mid-Term planning requests for the GCS-SCPF.
- Planning the Signal Health Status (SHS) value with respect to the GMS MTP planned tasks.
- Producing the GMS Short-Term Planning to be passed the GMS-GACF for executable schedule derivation.
- Validating the Long/Mid-Term and GMS Short-Term Plans compatibility with the mission performance.
- Producing C-Band Uplink Scheduling.
- Processing Mid-term plan execution feedback received from GCS-SCPF and GMS short-term plan execution feedback received from GMS-GACF.
- Generation of maintenance and performance bulletin to be provided to the external entities via the GMS-SPF.

The Mid-Term Plan Performance Validation inside the MUCF consists in providing the following service level reports:

- Service performance bulletin: the reports indicate the period of time (and locations) where the SoL, OS and PRS performance are not met with the planned service availability. A set of configurable threshold is used to identify the service performance unavailability.
- Notice for Advisory to Galileo User: the list of events that have impact on the mission and service availability. This list is obtained by a comparison between the input events of the Mission Performance Prediction and the performance bulletin.

6.5.4.1.5 GCS Schedules Production: SCCF

The SCCF is responsible for (see GCS-RD.1):

- Producing executable schedules from the GCS Short-Term Plan and TTC contact schedule received from GCS-SCPF., which are then internally executed at the due time.

- Providing execution feedback to the GCS-SCPF.

6.5.4.1.6 GCS Planning: SCPF

The SCPF is responsible for (see GCS-RD.1):

- Issuing Long/Mid-Term planning requests to the GMS-MUCF.
- Processing the GCS Mid-Term planning received from the GMS-MUCF.
- Producing the GCS Short-Term Planning to be passed to GCS-SCCF for executable schedule derivation.
- Producing the TTC contact schedules to be passed to GCS-SCCF for execution and to GMS-MUCF for information.
- Collecting GCS short-term plan execution feedback from the GCS-SCCF and providing GCS Mid-Term Plan execution feedback to the GMS-MUCF.

6.5.4.2 Dynamic Scenarios

6.5.4.2.1 Scenario #SFC5.4-1: Nominal Mission Planning

6.5.4.2.1.1 General Description

(0) General Description: This scenario describes the nominal sequence of mission planning.

6.5.4.2.1.2 Flow Chart

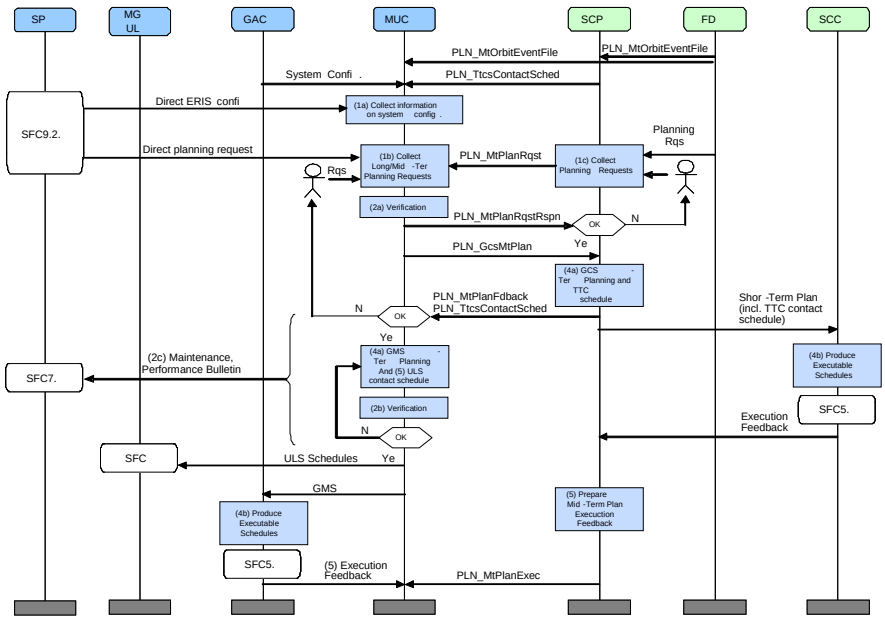


Figure 201 : #SFC5.4-1 Nominal Mission Planning MSC

6.5.4.2.1.3 Scenario Sequence Description

(1) Long/Mid-Term Planning Inputs Collection

(1a) The MUCF collects overall status and configuration information on the Galileo system from the GACF (see [GMS-RD.11]: GACF_GmsConfig_MM, GACF_SystCurrentStatus_MM) and orbits events from the GCS (PLN_TtcsContactSched, PLN_MtOrbitEventFile).

(1b) The MUCF collects planning requests from GMS planning operators

(1c) The MUCF collects planning requests from the GCS-SCPF (PLN_MtPlanRqst).

(2a) MTP verification: The MUCF verifies that the Mid-Term Plan is compatible with the mission performance [GMS-RD.2]. If it is not the case, the operator can make adjustment in the Mid-Term Plan. In case a problematic request was made by the GCS-SCPF operators, a negative feedback is sent from MUCF to the SCPF (PLN_MtPlanRqstRspn). [*Note: this feedback is positive in case the GCS request can be accommodated in the Mid-Term Plan*]

(2c) The MUCF produces system status and maintenance bulletin which are sent to external entities via the SPF ([GMS-RD.11]: MUCF_BulletinToSc_INF, see also SFC7.3).

(3) The MUCF provides to the SCPF an extract of the mid-term plan containing GCS mid-term plan tasks submitted by the GMS (PLN_GcsMtPlan).

(4) Production of the Short-Term Plan (STP) and Executable Schedules.

(4a) The MUCF and SCPF derive Short-Term Plan from the Mid-Term Plan respectively for GMS [GMS-RD.2] and GCS [GCS-RD.1]. The MUCF produces the ULS contact plan, while the SCPF produces the TTC contact plan [GCS-RD.1].

(2b) The MUCF validates that the GMS STP is compatible with providing the mission performance [GMS-RD.2].

(4b) The GACF produces the GMS executable schedules based on the GMS STP provided by the MUCF [GMS-RD.2]. The SCCF produces the GCS executable schedules based on the GCS STP provided by the SCPF [GCS-RD.1].

(5) The GACF and SCCF provide feedback on the execution of the STP respectively to the MUCF (see [GMS-RD.11]) and the SCPF (see [GCS-RD.1]). On the basis of this information, the SCPF provides an Mid-Term Plan execution feedback to the MUCF (PLN_MtPlanExec).

6.5.4.2.2 Scenario #SFC5.4-2: CS Key Up-Load and Activation Planning

6.5.4.2.2.1 General Description

(0) General Description: This scenario describes process to plan the upload of new CS keys on-board the satellites and to activate them.

1.1.1.1.1.1 6.5.4.2.2.2 Flow Chart

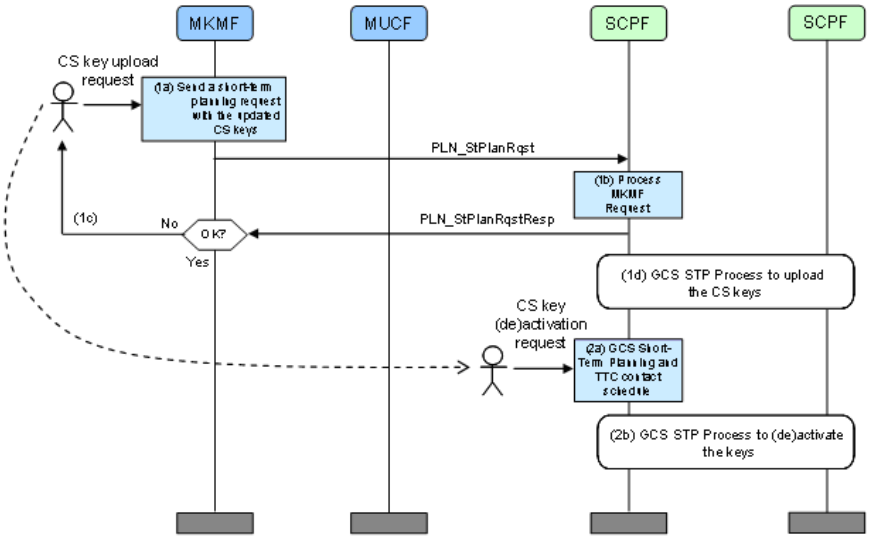


Figure 202 : #SFC5.4-2 CS Keys Upload and Activation Planning MSC

6.5.4.2.2.2 Scenario Sequence Description

(1) CS Keys Upload Request

(1a) The MKMF makes a request to the SCPF to plan the upload of CS keys in the GCS Short-Term Plan (PLN_StPlanRqst).

(1b) The SCPF processes the MKMF request and sends a response to the MKMF (PLN_StPlanRqstResp).

(1c) In case the answer of SCPF is negative the MKMF operators have to prepare a new request.

(1d) In case the MKMF request can be accommodated in the GCS STP, the nominal GCS STP process follows (i.e. the SCCF translates the STP in executable schedule and the needed TC are sent to the satellites at the planned TTC contact).

(2) CS (de-)activation Request

(2a) The MKMF operator communicates to the SCPF operator the parameters of a request to activate/deactivate CS keys. The SCPF operator request is treated at GCS STP level (i.e. is considered as having no mission impact, and therefore is not treated via MTP)

(2b) The nominal GCS STP process takes care of uploading the activation/deactivation TC.

Note: The current implementation is not in line with the GMSREQ-10313. The GCSGMSICD does not allow to plan such a task in the GCS mid-term plan exchanges between the MUCF and the SCPF. GMS deviate from the nominal scenario and requirement GMSREQ-10313 "Clear Mode for CS Third Carrier Ranging Code Protection Function". GMS propose that planning and commanding of the CS ranging mode are managed operationally between KMF and GCS Operators. This design implies that commanding and status of the CS Clear Mode is by GCS.
Resolution of this issue is currently under discussion and it is planned for the next STRB.

6.6 SFC6 ARCHIVING

The objective of the Archive SFC is to describe the physical exchanges between mission segment elements, ground control segment elements and satellite to perform data archiving migration and retrieval.

The following section describes the physical implementation of the archive. It is based on the following documentation:

- System documents:
  - System Logical Architecture (SYS-RD.1)
  - GCSGMS ICD (AD.08)
- SS documents:
  - SS DJF (SS-RD.2)
- GMS documents:
  - GMS DDF2 (GMS-RD.2): Archive section and GCS to mission segment section
  - GMS DJF (GMS-RD.3): GACF and Archive sections
  - GACF ERD (GMS-RD.9): Archive section
  - GACF ICD (GMS-RD.12): Archive section
- GCS documents:
  - GCS DDF (GCS-RD.1): FDF, SCCF, SCPF. OPF & CMCF context sections
  - GCS DJF (GCS-RD.2)
  - GCS CLA DDF (GCS-RD.28) for the GCS event centralisation description
  - FDF ERD (GCS-RD.5): Data store section

6.6.1 Physical Static View

6.6.1.1 Overall physical Description

Two functional levels of archive are defined:

- Data Archived locally in short and medium term storages within ground assets elements and in satellite
- Data Archived permanently in a long-term Galileo data storage for the entire lifetime of the system.

Local short and medium data archive within ground asset can be found:
On board the satellite: the (I)CDU is responsible to store locally all the telemetry to be downlink to the GCS to be stored in the SCCF via the TTCF. The contact nominally occurs once per orbit (once every 14h) and the (I)CDU has a 2 orbits storage capacity.

In GCS:

- the CMCF, SCCF, SCPF, FDF and the OPF have a medium term storage capacity, with up to 3 years of storage capacity. These storage areas act as GCS Central archives supporting intermediate storage for local and quick access of configuration, databases and historical data for one or more GCS elements.
- the TTCF stations have only a short term archive (one week)

In GMS:

- the PTF, GSS, OSPF, MGF, ULS, GNMF, and MUCF have a short term archive, also called local archive, with a storage capacity up to several weeks.
- The Central archive is managed by a single element: the GACF, which collects all GMS local archives with a mid-term storage capacity, up to few months.
- The Global Long Term Archive (LTA), also called permanent archive, is managed by the GACF, which stores for the entire life of the project the GCS element central archives and the GACF central archive.

The following figure presents the physical architecture of the archiving functional chain:

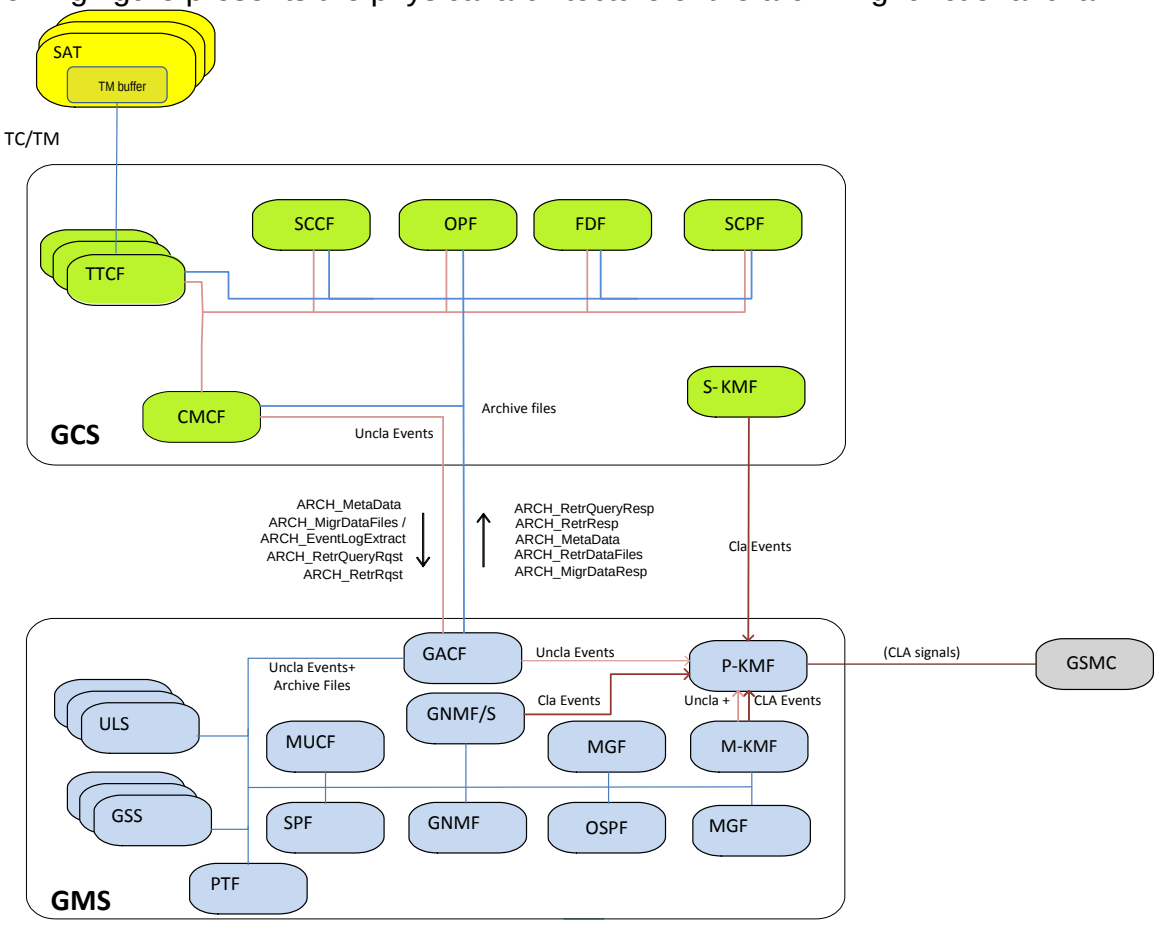


Figure 203 : SCF6 Physical static view

6.6.1.2 LOCAL archive on-board satellite: (I)CDU

Data archived and duration of archive in (I)CDU: The (I)CDU on board the satellite stores locally telemetry to be downlinked to the GCS/TTCF during S-Band contact:

- the contact nominally occur once per orbit (once every 14hours),
- there is storage of selected TM data during between two ground contacts, with TM packets managed by TCs according to PUS service 15,
- the TM is downlinked either as the result of a direct telecommand from the ground or by a time activated command stored on board the spacecraft,
- the on-board buffer is deleted on TC request or there is an automatic on-board override process which is enabled or disabled on TC request.

- the size of the archive capacity on-board (for a total of 4 Gb of physical memory) is robust to one loss of contact (meaning able to store TM for 2 orbits).

The TT&C facility transmits in real time the TM to the GCC Spacecraft Constellation Control Facility (SCCF).

6.6.1.3 Data Archived Locally in GCS Elements

Data are archived in GCS :

- in Local Archives, providing local working storage area for configuration data and database. Local archive can be also used for Short / Mid term data storage of historical M&C data
- in GCS Central Archives, providing storage areas for configuration data and database pertaining several GCS elements. In some Central archive are also used as Short / Mid-term data storage for M&C data originating from several GCS elements

Details per elements are provided in the sections below.

6.6.1.3.1 Local/Central archiving for elements in GCS

Each GCS element will store locally any data needed with fast direct access. Each element is responsible for its own local archiving policy. Each facility will also be responsible for replicating this history to the alternate GCC site.
The CMCF acts as central server for the GCS events collection from all GCS elements. The OPF acts as a Central archive for Configuration Items used by GCS elements
All data will be transferred to the permanent archive in accordance with mechanisms specified in the GCSGMS ICD [AD.8].
The holding period in the local archive is defined by each element. Data will be stored locally within the GCS for the short to medium term in the corresponding GCC Facility of the GCS.

6.6.1.3.2 Local Archiving of data in TTC Station

Data archived locally in TTCF:

- All Ranging data perform by the TTCF (Ranging, Doppler, Angular and meteo) that can be retrieved by the FDF,
- All M&C data of the TT&C that can be retrieved by the CMCF,
- TM frames are also locally stored in the TTCF for possible retransmission to the SCCF.

Duration of archive in TTCF:
The capacity is sized to enable a minimum of 7 days of this data to be stored.

6.6.1.3.3 Local/Central Archiving of GCS Assets M&C Data: CMCF

Data archived and duration of archive in CMCF:
The CMCF is responsible to collect and archive the GCS element M&C data in its local archive for a minimum of three years.
The CMCF is also the central server for the collection of unclassified event log files from all GCS elements. Events are all categorised in accordance to a common Security Event Dictionary and common xml format.
A local agent in each GCS element interacts with the CMCF, which periodically extracts unclassified event logs based on specific filtering criteria retrieval. From the CMCF the events are transferred to the GACF according to two different priorities, based on the event category,

supporting configurable polling intervals comprised between 0 minutes (immediate) and 24 hours.

Archive Synchronisation policy in CMCF:
As there is no CMCF data synchronisation between GCCs, data is archived independently at both GCC sites and periodically migrated to the local GACF Global Long Term Archive (which, via the GACF LTA synchronisation process, will replicate the CMCF archived data into the remote GACF LTA).
In nominal conditions, each CMCF will be able to retrieve the available data from the local GACF LTA.
Configuration of the CMCF will ensure that the local archive is migrated automatically and regularly.to the GACF LTA.
Locally archived data already migrated to the GACF LTA can be purged automatically when the local storage capacity is exhausted.

6.6.1.3.4 Local/Central Archiving of GCS Flight Dynamics Data: FDF

Data archived in FDF: The FDF archives the dynamic data:

- Orbit file,
- Manoeuvre file,
- Manoeuvre performance factors file,
- Attitude file,
- Mass evolution file,
- FDF input information files:
  - Archived telemetry
  - Archived tracking data (raw and pre-processed)
  - Plan response files for each task send to the SCPF

Duration of archive in FDF:
The FDF has a short/medium term storage archive area.
The FDF interfaces to the GMS for regular migration of the local archive to the GACF LTA.
Archive Synchronisation policy in FDF: The FDFs element in the two GCCs works according to a Master / Stanby role.
Despite the Stanby FDF can still perform all necessary computations, only the Master FDF can distribute FDF products.
FDF products are automatically copied to the Standby FDF in near real time to support the FDF role switch-over in case of need. As the FDF of both GCC receive the same data, GCS maintains the FDF instances present in the two GCCs fully synchronised.

6.6.1.3.5 Local/Central Archiving of TM/TC and Events DATA: SCCF

Data archived in SCCF:
The SCCF holds in its local archive historical TM/TC packets, Automation History, Procedures and Schedules as separate data sets to support the Performance Evaluation and Reporting function of the SCCF.
The GCS Telemetry archived for each satellite is stored in chronological order with respect to the on board S/C time.
All sent TCs are archived with the following additional information:

- parameters of the telecommand packet (raw and calibrated)

- the time the command was uplinked
- the source of the telecommand
- the destination of the archived telecommand data
- the time of reception and status of Acceptance & Execution
- the time when the command was executed

Duration of archive in SCCF:
The SCCF stores the TM, TC out-of-limit history locally within the SCCF for immediate access for at least three years.
Data form the SCCF local archive are regularly migrated to the GACF LTA.
Once migrated to the GMS LTA, the SCCF data can be maintained in the local archive until the available capacity allows it.
Archive Synchronisation policy in SCCF:
When the SCCF at one GCC is Prime for a spacecraft (and it is responsible for the relevant housekeeping operations), the SCCF at the other GCC performs no operations for that spacecraft.
The SCCF can be configured to share the Constellation Operations across GCCs in flexible manner, allowing all operations to be conducted from a single GCC only.
In order to support a possible SCCF handover across GCC it is necessary to maintain the SCCF local archive synchronised in both GCCs. The SCCF archive synchronisation is based on 2 strategies:

- "in-contact" synchronisation: TM is received and processed at each GCC. During contact, event & TC history are synchronised between GCCs in "near-real-time". Thus is expected to be only occasionally used
- "end-of-contact" synchronisation: At the end of each contact the synchronisation process will copy all TC, TM and Event history data from the responsible GCC to the other GCC, overwriting any existing TM. Thus is the default strategy adopted by GCS.

The GCS (SCCF) allows selecting the internal synchronisation strategy on manual configuration. Both strategies ensure that the data sets held in each SCCF local archive is identical. After migration to the GACF LTA, the GACF LTA will not need to synchronise these data which are already stored in the local GACF-LTA .

6.6.1.3.6 Local/central Archiving of Planning data: SCPF

Type of data archived & archive Synchronisation policy in SCPF:
The SCPF is responsible for archiving the GCS short plan data.
Planning Data are provided to the SCPF instances of both GCCs on distribution of the GCS STP by the Master SCPF. This includes: planning requests; contact plans; spacecraft operations plans.
Replication is performed once all processing within the Master site is completed, maintaining the local archive synchronised across GCCs.
Duration of archive in SCPF:
The SCPF will maintain a store of all generated STPs created in the previous 12 months.
SCPF data will be sent regularly to the GACF LTA, which will not need further synchronisation of the data (already available in the local GACF LTA).

6.6.1.3.7 Local/Central Archiving of Database & Configuration DATA: OPF

Data archived in the OPF:

The OPF is responsible to manage central archiving of all GCS Databases (Satellite, Flight Dynamics, M&C databases, Procedures) and in general all Configuration Item (CI). Configuration Items are purged from the local archive based on an obsolescence policy. The local archive maintains a reference to the purged CI.
The OPF archive does not hold historical M&C data

Duration of archive in OPF:

The OPF holds in its Central Archive all Configuration Items needed to support the operations, based on the available capacity
The OPF transfers the data from its Central Archive regularly to the GMS Long Term Archive, according to a configurable repeat cycle.

Archive Synchronisation policy in OPF:

The OPF elements present the GCCs work according to a Master / Stanby role, in which only the Master OPF is responsible for the distribution of Configurable Items.
The Central archives of all OPF instances are kept synchronised by the Master OPF which will transfer new Item to the Standby element once these are created.

6.6.1.3.8 Local/Central Archiving of security DATA: SKMF

The GCS KMF manages its own archiving separately from the data of other unclassified GCS elements, as well as GCS KMF event logs which are transferred to the PKMF.

6.6.1.4 Data Archived Locally in GMS elements

Data are stored locally in each GMS Element in the short term storage archive of each GMS element. Archive-related functionalities are similar between MSF, SPF, PTF, GSS, OSPF, MGF, ULS, GNMF, and MUCF. The need for local archiving in those elements is a pure element choice (not driven by GACF needs but rather by efficiency needs); the following type of data are locally archived (raw un-processed data, detailed technical information on status for M&C purposes, detailed results of routine test, etc.) for a duration up to several weeks.
Periodically, the GACF archives unclassified Event log files from GCS and GMS elements, with the exception of MKMF Unclassified data, which are archived together with the classified data in the MKMF local archive and backed up there.
The GMS collects automatically all security event logs from all GMS elements and GCS and send them to the GSMC. This is performed in the following steps:

1. Events logs from GCS and GMS unclassified elements are regularly archived into the GACF LTA: GCS Unclassified event s via archive migration, GCS unclassified network events via the GMS MDDN (GNMF).
2. Events from Unclassified elements (with the exception of unclassified MKMF) are gathered by GACF via automatic (and on operator request) retrieval of event log files from the GACF LTA.
3. Event logs are filtered to generate security events in the GACF. The filtering capability allows filtering by sources, security events flag, event types and criticality.
4. Classified GMS and GCS Security Events are transferred directly to the PKMFs on the nRT LAN
5. The GACF in each GCC exports the unclassified security events to all PKMFs in both GCC, but only the Master PKMF processes the data. In case of need, upon operator request, the GACF can re-transmit events to the PKMF.
6. The Master PKMF forward both CLA and uncla events to the GSMC using Classified security channels (refer to CLA DDF)

6.6.1.5 Galileo Global Long Term Archive (LTA)

6.6.1.5.1 GACF: Global Long Term Archive (LTA)

The GACF is responsible for the implementation of the global Long Term Archive (LTA) supporting data archival and retrieval of the following type of data:

- Constellation Management and Satellite Control data: including Databases, TM/TCs, Operational Data, Flight Dynamics
- Navigation Processing and Control data
- Services Performance Monitoring (and Prediction) data and results,
- M&C data including:
  - Database
  - Event and message logs
  - Operator Actions log-files
  - Ground assets and network M&C data
  - All main processes related data
- All the data exchanged between GMS elements across the GMS networks, unless already recorded otherwise in the central repository,
- All the data exchanged with other external systems across the GMS networks, unless already recorded otherwise in the central repository
- Special data needed to implement the search and rescue function
- Configuration data and software parameters
- Reports and Audit Files

The GACF is also in charge of the maintenance of a catalogue of products allowing products retrieval

The GACF LTA provides the function to archive all the data generated by GCS and GMS elements, including e.g. GSS raw measurements, configuration files and many other data. The archiving process that is nominally completely automated is:

- Archiving based on Data capture flow transiting on real-time LAN, this capture (1 MB/second) dominates total the required archive capacity (> 80% of the total data)
- Archiving of all data items send directly by GCS and GMS elements to and from the GACF

The GACF LTA will only store data relevant to the elements belonging to the Operational Chains.

GACF LTA data recorded from Operational Chain elements can be transferred on separate archives (e.g. Validation Chain archives) used for testing and validation purposes.

GACF does not automatically capture data flows on the Non Real Time (NRT) LAN. NRT Data exchanged between elements or between an element and external entities where GACF is not involved are archived locally in element archives and automatically migrated to GACF where they are finally archived. The element originating the data flow is responsible for providing the data via the local archive to the GACF LTA.

For the migration of data to the permanent archive, the GACF has direct interfaces with the GCS elements (SCCF, SCPF, OPF, FDF, and CMCF).

The GACF interface with the GCS CMCF is used specifically for the transfer of GCS unclassified event log files into the GMS domain and for the retrieval of event logs from the GMS Global Archive.

The Global GACF LTA supports permanent data archival and retrieval functions in accordance to a modular and scalable architecture entirely supported by hard disks.

However, in order to minimise risks of storage technology obsolescence, the capacity required to archive data for a period of 20 years (730 TB, based on a peak archive data volume of 110 GB per day) is procured in a staggered approach, establishing check points in which the need for additional capacity is assessed based on the GACF LTA storage filling ratio.
The current GACF LTA is based on components providing in each GCC a total of 256 TB of locally redundant capacity, which is expected to cover the archival needs until 2019.

6.6.1.5.2 Data archived in the GACF global archive

The table below shows the data archived by automated processes:

- automated archive migration from local element archives for NRT data
- data capture for RT data,
- automated archiving of all NRT data flows with GACF as source or destination

Nominally, no operator interaction (e.g. through a catalogue on the element) is needed. Whereas real time data archiving is instantaneous, the archival of NRT data may be completed within 24 hours.

Source	NRT Data Archived from element data archive ingestion	RT Data Archived	NRT data archived directly by GACF
GNMF	Event Logs Test Port Logs HW Description File SW configuration GDDN Status	Mission Monitoring data (MNE only)	Polled MIB content Command + Migration Responses
GSS	Event Logs Test Port Logs HW DescriptionFile SW configuration	Galileo SIS measurement	Polled MIB content Command + Migration Responses GPS SIS measurements
MGF	Event Logs Test Port Logs HW DescriptionFile SW configuration Clock Ephemeris Iono Predicted Data (nav) Navigation S-band Uplink Messages (sent to GCS)	Dissemination subframes SAR flow control info Mission Monitoring data	Polled MIB content Command + Migration Responses
MSF	Event Logs Test Port Logs HW DescriptionFile SW configuration MSF offline ODTS MSF computed ERP	None	Polled MIB content Command + Migration Responses Archive retrieval requests MSF generated or validated configuration files Upload results report
MTPF	Event Logs HW DescriptionFile SW configuration	None	Polled MIB content Command + Migration Responses Archive retrieval requests
MUCF	Event Logs Test Port Logs HW Description File SW configuration Security Event Logs CS Dissemination Report Satellite Almanacs files Performance Report Performance Bulletin Assessment report Maintenance Bulletin Notice Advisory To Users RLM Notification Mnt	Uplink schedule	Polled MIB content Command + Migration Responses Archive retrieval requests Upload results report

Source	NRT Data Archived from element data archive ingestion	RT Data Archived	NRT data archived directly by GACF
	Performance predictions Satellite MonData Eris Uplink Report S&R System Status Precise Satellite Ephemeris GCS Mid Term Plan Long Term Plan GMS Mid Term Plan GMS Short term Plan SAR Service Bulletin Satellite & GSS planned Availability CS Grid SAR Grid Eris Direct Contact Plan Eris planned visibility Uls Contact Plan Uls Tracking Plan SIS Satellite Lband Status		
OSPF	Event Logs Test Port Logs HW DescriptionFile SW configuration	Mission Monitoring Data Navigation batches (Fnav, Gnav,Inav + Qos)	Polled MIB content Command + Migration Responses Predicted Orbits Predicted Clocks Restart Data
PTF	Event Logs Test Port Logs HW DescriptionFile SW configuration Time provider data	Mission Monitoring Data	Polled MIB content Command + Migration Responses GgtoData from time provider
SPF	Event Logs Test Port Logs HW DescriptionFile SW configuration CS Data GRP data User Anomaly Notification Gps Notice Advisory New Product Rqst Eris Status TSP AnomalyData GST data Tai to Utc Offset Utc Frequency Correction	None	Polled MIB content Command + Migration Responses Archive retrieval requests Upload results report GACF query and retrieval notifications
ULS	Event Logs Test Port Logs Test Loop Result HW DescriptionFile SW configuration	Mission Monitoring Data	Polled MIB content Command + Migration Responses
MKMF			None Polled MIB content Command + Migration Responses
GACF	Not applicable	None	Commands send to: GNMF, GSS, MGF, MKMF, MSF, MTPF, MUCF, OSPF, PTF, SPF, Config files send to: ULS, GSS, MGF, MSF, MTPF, MUCF, OSPF, PTF, SPF, ULS Requests/Responses for archive migration & log file extraction from:

Source	NRT Data Archived from element data archive ingestion	RT Data Archived	NRT data archived directly by GACF
			GCS-FDF, GNMF, GSS, MGF, MTPF, MSF, MUCF, OSPF, PTF, SPF, ULS Data sent to the test port Polled MIB content (other GACF) File transfer upload result reports to MSF, MTPF, OSPF, SPF Feedback on STP to MUCF CFGM content (holding all GMS configuration information, software and configuration files), including generated by GACF operators
RLSP	Not applicable	SAR return link requests and ACK	Not applicable
ERIS	Not applicable	ERIS data	Not applicable
FDF	Event Logs Orbit file, Manoeuvre file, Manoeuvre performance factors file, Attitude file, Mass evolution file, FDF input information files: Archived telemetry Archived tracking data (raw and pre-processed) Plan response files for each task send to SCPF	Not applicable	Data files and metadata associated to automated Migration request
SCCF	Event Logs Constellation TC/TM data	Not applicable	Data files and metadata associated to automated Migration request
SCPF	Event Logs STP data: planning requests; contact plans; spacecraft operations plans	Not applicable	Data files and metadata associated to automated Migration request
OPF	Event Logs GCS Databases (Satellite, Flight Dynamics, M&C databases) and conf data	Not applicable	Data files and metadata associated to automated Migration request
CMCF	Event Logs Ground M&C data	Not applicable	Data files and metadata associated to automated Migration request

Table 92 : Data archived GACF global archive

The following archive limitations apply:

- RINEX data output from SPF are not archived in GACF: RINEX files are created in a deterministic manner based on GSS receiver raw output data which are already stored on GACF.
- OSPF Restart Data are partially archived in GACF: All the files are available for 48 hours in the local GACF repository/drop box, to be used for OSPF restart purpose if any. A subset of restart data files are systematically long term archived, these are Restart Navigation Data files. The other parts – the Restart State Vectors - are only long time archived cyclically (with configurable time cycle), or in case of a restart point structure is transferred from GACF towards OSPF or on operator demand.

- Several files flowing from GCS to MUCF regarding S band Uplink report and Mid Term Plan exchanges are not captured by GACF archive. Being such data produced by GCS elements, they are expected to be part of the GCS daily-migrated files to GACF, under GCS control. The files involved include MM_NavSBandUplkRpt, PLN_MtPlanExec,PLN_MtPlanFdback, PLN_MtPlanRqst, PLN_MtOrbitEventFile and PLN_TtcsContactSched.

6.6.1.5.3 ARCHIVE SYNCHRONISATION POLICY IN THE GACF GLOBAL ARCHIVE

In FOC, one GACF LTA per GCC is deployed. The GACF LTA in each GCC archives independently the locally received data. In all cases where the same data are provided in parallel to both GCCs, the local GACF LTA will store the same data.
There are however data that are different in each GCC: to ensure consistency of the GACF LTA in each GCC a synchronisation process is foreseen to allow the exchange of such data over the inter GCC link.
The synchronisation between the two GACF LTA is configurable with the following choices:

- Manual synchronisation, which may be required to recover data from one GACF LTA which are missing at one side (after a failure or a planned GACF LTA maintenance activity) F)
- Automatic synchronisation performed at configurable interval (nominally every 24 hours)

Completeness of data archiving in the GACF permanent archive is ensured by design for the nominal archiving as well as some contingency cases depending the way the archiving of data is handled. The following different types of data are considered:

- Real time data
- NRT data exchanged without GACF involvement
- NRT data exchanged between elements and GACF

Real time Data
All real time data flowing on the GCC RT LAN are archived by the local GACF. As the GSS network feeds in parallel both GCCs these data flows are captured independently in each GCC.
Also mission monitoring data from elements in one GCC are exchanged on the RT LAN. In this case the monitoring data flows are cross strapped across GCCs, this means that these data flows are duplicated in each of the two GACF archive.
NRT data exchanges with GACF. This includes;

- M&C data flows transferred via SNMP between GMS elements and GACF. The GACFs of both GCCs poll independently all GMS elements for M&C information and event traps will be sent to both GACF in parallel. The monitoring data of GMS elements from both GCCs will be hence archived by both GACF simultaneously.
- NRT data flows exchanged between GACF and any other element. These data are replicated in the two GCCS via the GACF LTA synchronization process

NRT data exchanges without GACF involvement
NRT Data exchanged between elements or between an element and external entities where GACF is not involved are archived in the elements local archives and automatically migrated to GACF LTA. The element originating the data flow is responsible for providing the data via the local archive to the GACF LTA.
These data are hence replicated across GCCs via the GACF LTA synchronisation mechanism.
Note that a lossless compression is applied on data files exchanged over the GMS internal interfaces.
Specific measures are taken to avoid data duplication in the GACF LTA:

- In case of multiple file transfers of same file between different elements only a single copy (sent by the first element) is stored in the GACF LTA.
- In cases the same file is present in each GACF LTA, the file is not copied by the GACF LTA synchronisation process

For reliability reasons, for each GACF LTA it is foreseen the replication of the archived data in separate physical storage media located in separated rooms of each GCC.
The following figure describes the data archive process in one GCC.

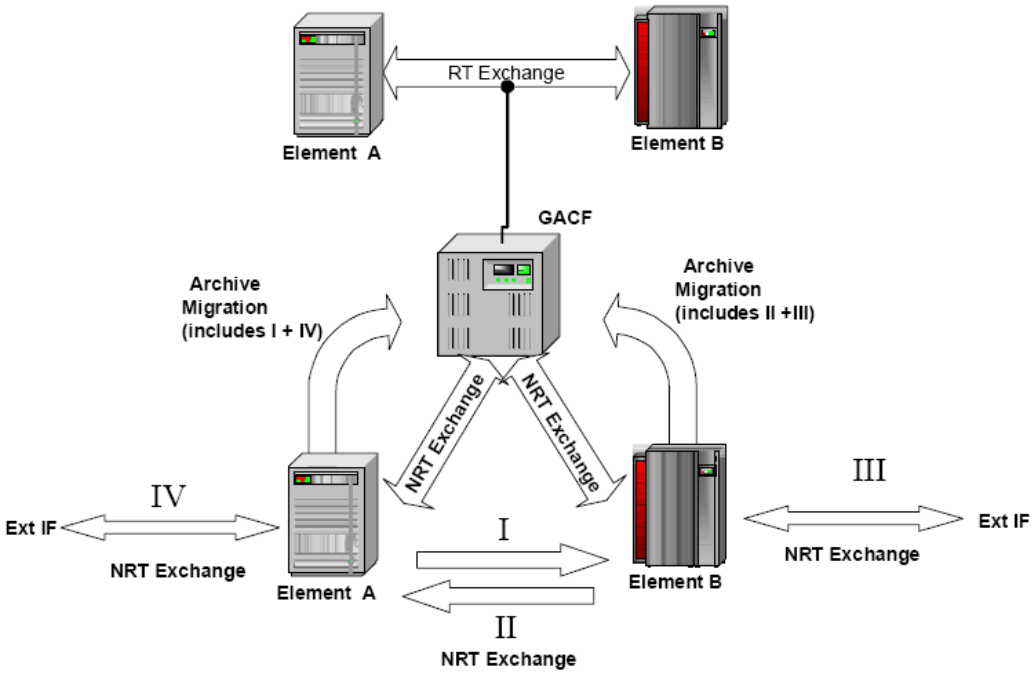


Figure 228: Data Flow capture for Archiving in one GCC

The table below gives for each data type the archive data policy and the associated potential loss of data:

Data type	GACF archive policy	Potential loss of data
Real time data	Real-time exchanges on LAN are archived by the local GACF: - GSS real-time data transferred to both GCCs at the same time are captured by both GACF archives. - RT outputs of the GCC located elements are only locally archived. - Mission monitoring data (transferred on RT LAN) of elements in one GCC are cross-exchanged to the other GCC and hence are archived in both GACF without synchronization.	In case of a GACF failure or shutdown, RT data will not be recorded until the GACF LTA is recovered. RT input data (GSS RT data and Mission monitoring data) can be recovered manually from the remote GACF LTA, while RT output data produced within the GCC during the period the GACF LTA was down are definitely lost t.
NRT data exchanged without GACF involvement	GACF does not automatically capture data flows on the Non Real time LAN. NRT Data exchanged between elements where GACF is not involved are archived in the elements local archives. These data are migrated in the GACF LTA under control of the source element. Migrated data are then replicated in each GACF LTA via the synchronisation process	In case of shutdown/failure of one GACF archive, the data are temporarily kept in the local archive elements and then can be send on the other GACF archive.
NRT data exchanged	GACF archives all data flows exchanged between GACF and any element.	In case of shutdown/failure of one GACF archive: the other GACF can be

between elements and GACF	These data are synchronized between GACF archives, after the synchronization, these data are available in both archives.	configured to request the archive migration from all elements and both archives synchronised on operator request.

Table 93 : Archive data policy

6.6.1.5.4 Data Retrieved from the GACF global archive

Data from the GACF Global archive can be retrieved:
- in GCS by some elements (SCCF, SCPF, OPF, FDF, CMCF) following a specific retrieval process initiated by any element which needs some archived data to perform analysis.
- In GMS only by MSF, MUCF and SPF:.
  - MSF and the MUCF to support the off-line analysis (GSS raw data for the operation of GMS elements in factory, for performance analysis, for exercising prototype, algorithms, for satellite clocks performance analysis, user /SIS /Algorithm /mission monitoring archived data for performance analysis…).
  - SPF to support data retrieval for external entities.

Catalogue products are prepared by the MUCF, the MSF, the GACF and the GCS elements.

The elements in the table below can access the GACF archive to retrieval data GMS-RD.2, GCS-RD.1):

Elements	GACF Global archive
GACF	operator/automatic
MUCF	operator (in analysis mode)
MSF	operator/automatic
MTPF	Operator
SPF	Automatic
FDF	Operator
SCPF	Operator
SCCF	Operator
OPF	Operator
CMCF	Operator

6.6.2 Dynamic Scenarios

6.6.2.1 Scenario #SFC6-1 Archiving Scenario 1: Data Migration

This scenario describes the archive process for the data migration part.

Scenario 1 Textual Description:

0) General Description:
The GACF is the Global archive of Galileo for the whole life of the system.

Each element in the GMS or in the GCS performs file migration independently.
The migration process is different in GCS and GMS but is the same for each element in the same segment.

1) GCS archive process:
The sequential description is detailed below for files to be archived for a GCS element (CMCF, SCCF, SCPF, OPF, and FDF):

1- The list of files to be uploaded is described in the GCS Archive Metadata file(s) sent to the drop box of the GMS/GACF assigned to the GCS Element; One metadata file is sent to the GACF per file to be archived.
2- The GCS Element uploads the data file(s) to the assigned drop box in the order defined in the Metadata file.
3- GMS archives the data files in the Global Archive.
4- When the archive migration is finished (success or failure), the GMS provides an Archive Migration response file.
5- GCS gets the migration response file within 24 hours.
6- GMS will delete all the files related to the request 24 hours after creation of the migration response.

2) Satellite TM archive:
The sequential description is detailed below for satellite TM archived by the ground segment (TTCF and SCCF):

1- On operator commands, specific Satellite telemetry packets are stored on board the satellite within the (I)CDU TM buffer.
2- The TM packets are transferred on TC request to the ground control segment during TT&C contacts
3- The on-board buffer can be deleted on TC request or managed autonomously according to a storage policy selected from ground.

3) GMS archive process
GMS elements (SPF, MUCF, MSF, MDDN/GNMF, PTF, ULS, GSS, OSPF, and MGF) archive data files in the Global Archive in the GACF:

- Navigation Processing and Control data
- M&C data covering Database, event logs, ground assets and network
- All data exchanged via network
- External Entities data

The GMS archive process is the following: data of each element are archived in the Global Archive within the GACF by automated processes (automated archive migration from local element archives for NRT data and data capture for RT data, automated archiving of all NRT data flows with GACF as source or destination).

A storage equipment is implemented for permanent off-line backup.

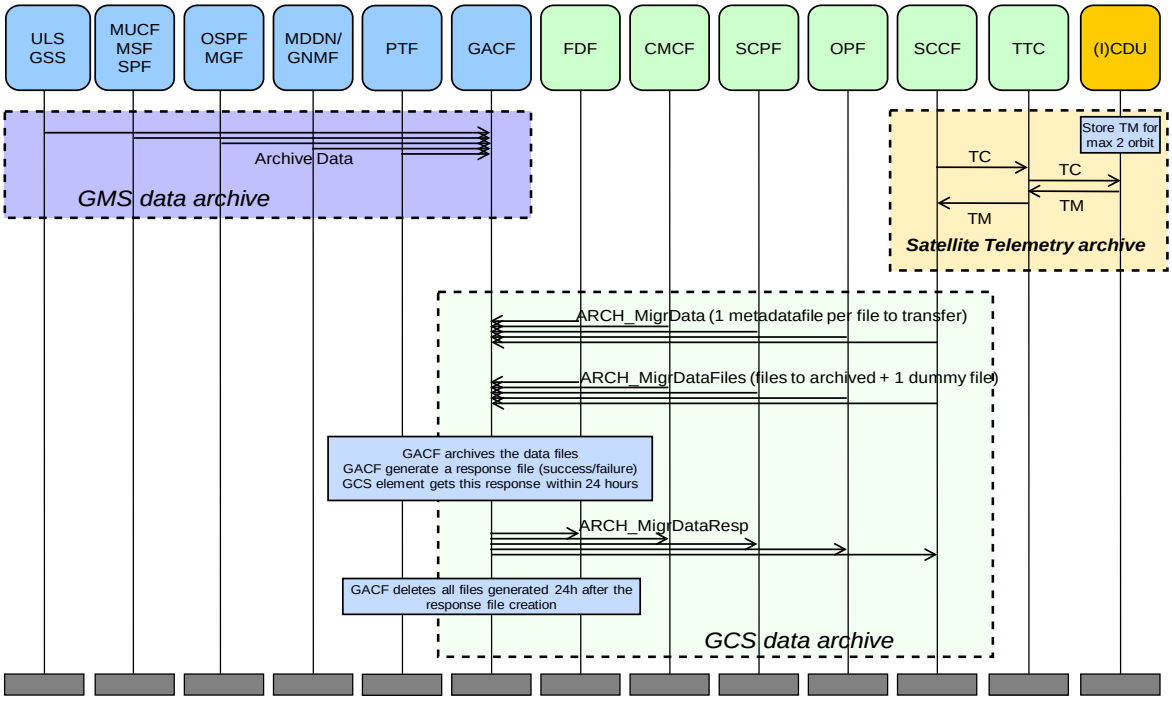


Figure 204 :Scenario #SFC6-1 Archiving Scenario 1: Data Archival

6.6.2.2 Scenario #SFC6-2 Archiving Scenario 2: Data Retrieval

This scenario describes the archive process for the data retrieval part.

Scenario 1 - Textual Description:

0) General Description:

The GACF is the Global archive of Galileo for the whole life of the system.
Each element in the GMS or in the GCS performs file retrieval independently
The retrieval process is different in GCS and GMS but is the same for each element in the same segment.

1) GMS retrieval process

In the GMS, the GACF provides on operator request the MSF, the SPF, the MUCF and the GACF with data to support the off-line analysis based on specific retrieval request (GSS raw data for the operation of GMS elements in factory, for performance analysis, for exercising prototype, algorithms, for satellite clocks performance analysis, user /SIS /Algorithm /mission monitoring archived data for performance analysis).

2) GCS retrieval process

Sequential description for files to be retrieved by a GCS element (CMCF, SCCF, SCPF, OPF, FDF) on operator request:

1. The GCS sends to the GMS the Archive Retrieval Query Request (ARCH_RetrQueryRqst). This file includes a Boolean Shortcut Flag used as follow:

- Flag=0 => the query is made using some field(s) from the metadata provided during the migration to the GACF (no shortcut in the retrieval process),
- Flag=1 => all files matching the query shall be returned to the requesting element immediately (shortcut in the retrieval process of step 2 to 4 described below).

2. The GMS responds with a Query Result file (ARCH_RetrQueryResp) which is made available in the assigned GACF drop box of the GCS Element.
3. The GCS gets the Query Result file (ARCH_RetrQueryResp) within 24 hours of its creation; after this time the file is deleted by GMS.
4. The GCS sends a Retrieval Request file (ARCH_RetrRqst) to the GMS/GACF specifying the desired files to be made available to the drop box.
5. The GMS/GACF places the requested data file(s) into the drop box of the GCS Element.
6. A separate metadata file (ARCH_MetaData) is generated per data file retrieved from the Global Archive.
7. After the last file has been provided by the archive, the GMS/GACF places Retrieval Response file (ARCH_RetrResp) in the drop box of the GCS Element to inform the GCS Element that the file(s) are available.
8. The GCS gets the Retrieval Response file (ARCH_RetrResp).
9. The GCS gets the data file(s) which are listed in the response file (ARCH_RetrResp) from the GMS/GACF within 24 hours of their retrieval. After this time the GMS may delete all the files generated for the specific Retrieval Request; including the metadata file and the Retrieval Response. The size of the retrieval drop-box will be limited to10 GB per GCS element instance. If the retrieval drop-box exceeds this capacity, retrieval is no longer possible and the GACF operator is alerted.

6.6.2.3 Scenario #SFC6-3 Event Log collection and forwarding to GSMC

This scenario describes the Event Log collection and forwarding of the data to the GSMC

Scenario 1 - Textual Description:

0) General Description:

Event log files are collected into the GACF from all elements and for two purposes:

1. Permanent storage into the Galileo Long term Archive (LTA)
2. Forwarding of even log files to the GSMC with configurable latency.

Each GCS and GMS element archives event logs independently, as part of the general archive process described in scenario #SFC6-1.

For the transmission of event files to GSMC, the following design choices have been made:

- GMS retrieves, at configurable intervals, uncla event log files from all uncla elements using the GACF buffer used for the archive management.
- The GMS P-KMF is the CLA gateway with the GSMC. The P-KMF collects
  - CLA event log files directly from GMS CLA and GCS CLA elements,
  - uncla events logs from the GACF.
- The GCS CMCF acts as GCS central server for:
  - Collecting and filtering all GCS uncla Event files;
  - transferring the files to the GMS GACF. To this end, the CMCF uses the same protocol used for the GCS files migration in the GACF Archive.

1) Event log are generated by all CLA and Uncla ground elements, as well as from the Satellites (Event TM).

   CLA events generated by CLA elements (GCS KMF, GMS MKMF, GMS GNMF/s) are provided to the GMS PKMF.

Unclassified GCS element event collection is performed by CMCF, via agents installed in each GCS element, allowing the CMCF to extract source events generated by each GCS element. Events can be selected according to predefined criteria (e.g. category, criticality).

2) On command, Uncla event logs from all elements (and stored in their local archives) are ingested by the GMS GACF with the signal ARCH_EventLogExtract, as part of a standard archive migration request (ARCH_MigrData).

   Two different priority classes are defined to control the latency associated to the migration of GCS event logs to GMS, based on the event classification. For each priority level the polling cycle is configurable from 0 minutes (immediate) and 24 hours

3) The GACF responds to migration requests notifying the operation success / failure (ARCH_MigrDataResp), which allows deleting the local event log file.
4) The GACF forwards all event logs to the P-KMF using a secure protocol (see CLA DDF)
5) The P-KMF forward CLA and Uncla events to the GSMC using protected channels (see CLA DDF)
6) The GSMC responds back to the P-KMF using protected channel (see CLA DDF).

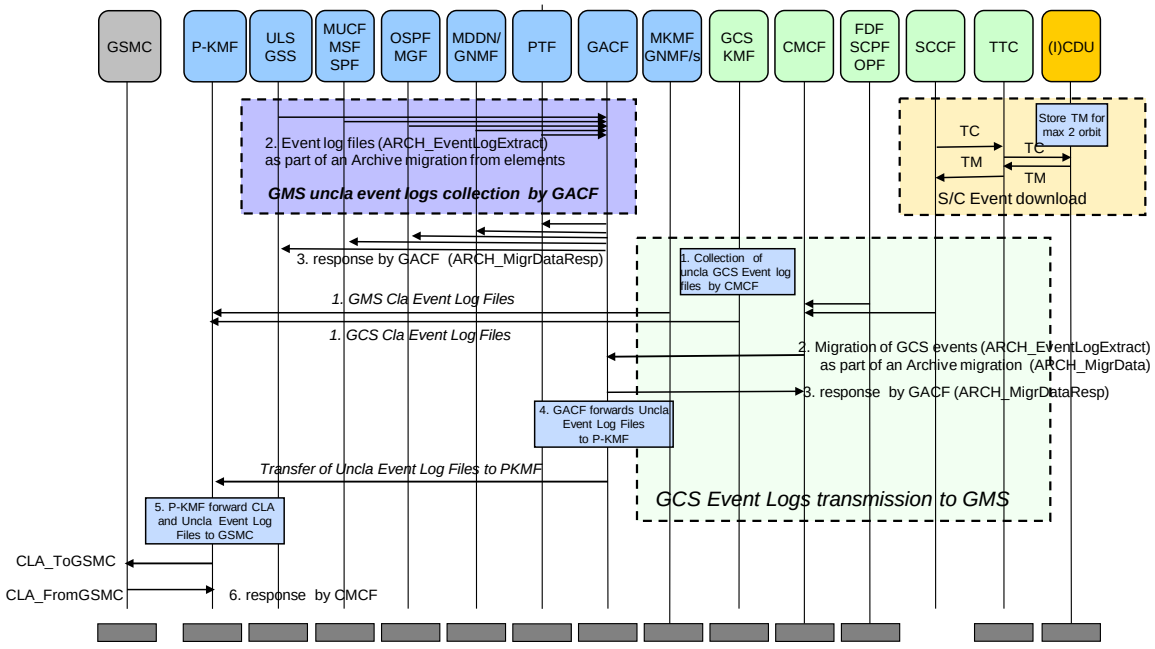


Figure 205 : Scenario #SFC6-3 Event Log collection and forwarding to GSMC

6.7 SFC7 SUPPORT COMMERCIAL SERVICE

6.7.1 Introduction

The objective of the Support to CS Support SFC is to provide the GSC capability to:

- Disseminate via the Galileo constellation the CS data in agreed regions (SFC7.1).
- Management and dissemination of CS ranging code keys for use of authorized users (SFC7.2).
- Provide scheduled Galileo products like reports and bulletins to GSC, and enable the request of new and archived products and the reception of anomaly reports sent by GSC to GMS, provide overall status, service performance reports, NAGU, maintenance and forecast bulletins to GSMC, and provide multiple products to GALSEE (SFC7.3).

Remark: *the automated NAGU generation concept is currently not consolidated in the STRB2.0 baseline. It is expected that the content of the signal INF_NoticeAdvisoryToUsers will be refined in close coordination with GSC in the frame of GSC operational concept consolidation. In the meanwhile, the signal INF_NoticeAdvisoryToUsers will not be disseminated to the external entity and the NAGU will be generated manually and sent to the GSC. In both cases (automated and manual NAGU processes) the authorization for dissemination to the public is granted by the corresponding formal process at GSC level.*

The following section describes the physical implementation of the SFC. It is based on the following documentation:

- ☐ System documents:
  - o System Logical Architecture (SYS-RD.1)
  - o GMS-SC ICD (AD.17)
- ☐ GMS documents:
  - o GMS DDF2 (GMS-RD.2)
  - o GACF ERD (GMS-RD.9)
  - o MUCF ERD (GMS-RD.10)
  - o MUCF ICD (GMS-RD.11)
  - o GACF ICD (GMS-RD.12)
  - o SPF ICD (GMS-RD.16)
  - o MSF ERD (GMS-RD.25)
  - o SPF ERD (GMS-RD.36)
  - o MGF ERD (GMS-RD.38)

6.7.2 SFC7.1 CS Data Dissemination

6.7.2.1 SFC7.1 High level description

Galileo provides support to GSC in terms of commercial service data dissemination; this means that GMS shall be able to receive from GSC commercial data to be included in the C/NAV data stream broadcasted through the C/NAV navigation signal.

6.7.2.2 Physical Static Block Diagram

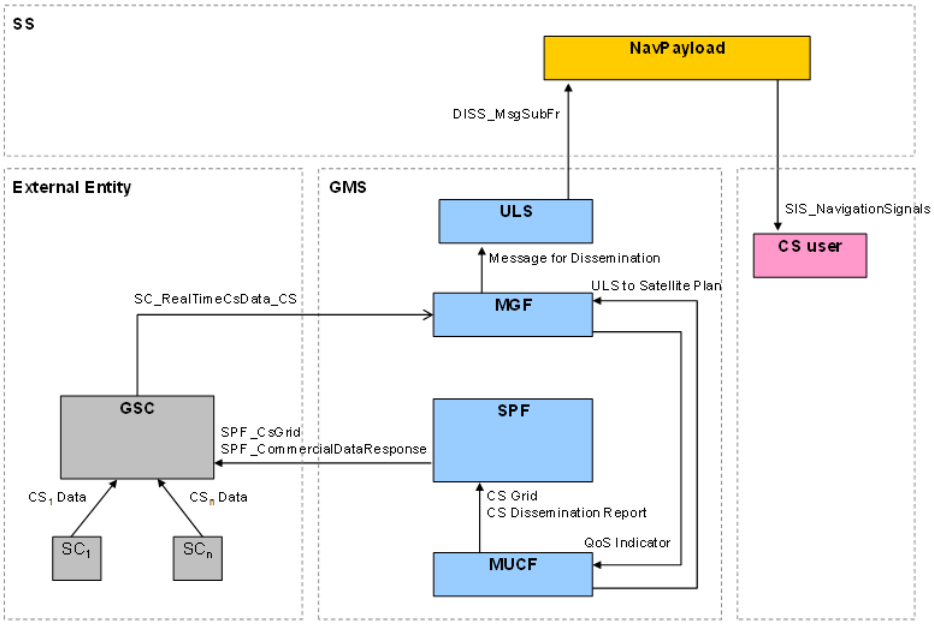


Figure 206: SFC7.1 Physical Static View

6.7.2.3 Overall Physical Description

The purpose of this SFC is to allow CS Data dissemination through the Galileo SIS to Commercial Service Users. In order to do this the GSC gathers all the CS Data of SC providers and based on CS Grid received from SPF send back the Commercial Data to be disseminated to MGF in a RT per satellite basis, MGF includes C/NAV information in the Message Subframe and sent to ULS where is included in the uplink message subframe for uplink to the satellites., The SS Navigation Payload receives and downloads it in the Navigation Signals to CS Users. MUCF receives CS Gen Data from MGF to know the status of the uplinked CS Data and sends a Dissemination report to SPF to be relayed to GSC. MUCF also sends to SPF the CS Grid containing satellite allocation to region, to be passed to GSC. In the sections below are detailed the main contributions of all physical elements involved.

6.7.2.3.1 CS Data generation: GSC

Galileo Service Centre is the external entity which acts as only interface between the Galileo Core Infrastructure and the user community but SIS. It manages requests from other SC centralizing them to GMS. For the purpose of this SFC7.1 the GSC acts as the collection point for CS data. It provides a site for the CS providers to upload CS data streams to the Galileo System.

The Service Centre will sell commercial data bandwidth to broadcast to fixed regions on the earth (say Africa, Asia, North America, South America, Oceania and Europe). The Service Centre will make the mapping between earth region and satellite. The commercial data uploaded to the GMS will therefore be tagged by satellite.

Based on the region selected by the individual service providers and the CS Grid provided by GMS, the GSC will perform an allocation of CS data to individual satellites (as described further below in detail) and will provide the data together with a satellite identifier to the GMS.

Each CS_Data message will contain information for up to 36 satellite
For each satellite , 448 bits of Commercial Data plus 14 bits of page type information will be included as follow:

CS Data Packet (1/2) 261 bits (page Type: 14 bits + Commercial data 247 bits)
CS Data Packet (2/2) 201 bits

Each satellite based information will contain a Dissemination report identifier

6.7.2.3.1.1 CS Data structure

The CS data message is preformatted in order to facilitate its uplink to the Galileo satellites. The CS data must be grouped into packets of 14+247 bits or 201 bits maximum to be incorporated in message sub-frames as follows.

Commercial Data (1/2) is 14+247 bits
Commercial Data (2/2) is 201 bits.

The message identifies the number of pages being transmitted, and the content or of each page.
The CS_Data (1/2) and CS_Data (2/2) packets are joined together in the satellite to be downloaded in the same page of C/NAV.

Each satellite based information will contain a Dissemination report identifier
Content of message between GSC and -MGFFor each satellite, up to 36 satellites:

a. Satellite ID
b. Dissemination report Identifier
c. Content of page :
   CS Data Packet (1/2): 261 bits (Page Type: 14 bits + Commercial Data: 247 bits)
   CS Data Packet (2/2): 201 bits

The CS data will be provided from GSC to GCC on a RT basis to each GCC using UDP multicast datagrams to the expected multicast group.
The CS data is made available at MDDN between GST-1s and GST-700s to be uplinked between GST and GST+1.

User Level	CS_RealTimeData Message											
	< 25 kB											
SIS Level	CS_Data Satellite 1			CS_Data Satellite 2			…	CS_Data Satellite 36				
	PT	CD		PT	CD		…	PT	CD			
	14	448 bits		14	448 bits			14	448 bits			
Uplink Level / GOC SC Level	CS_Data Packet (1/2)		CS_Data Packet (2/2)	CS_Data Packet (1/2)		CS_Data Packet (2/2)	…	CS_Data Packet (1/2)		CS_Data Packet (2/2)		
	PT	CD	CD	PT	CD	CD	…	PT	CD	CD		
	14	247	201	14	247	201		14	247	201		

Table 94: CS_Data Message Format (without headers)In previous table are used the following acronyms: PT for Page Type and CD for Commercial Data.

The Page Type (14 bits) content is defined by Commercial Service Provider/GSC to identify service data for Commercial Data (448 bits) in the same page.
In order to fit GMS uplink format, the GSC shall provide CS data in grouped packets of 261 bits (14 bits for Page Type +247 bits for Commercial Data) and 201 bits for Commercial Data), plus an identifying header (used for routing on the ground, but not for uplink). The CS Data may be encrypted using the CS service keys, and may include key management messages for this purpose. The content of the CS Data is not relevant to the GMS, which will simply strip the CS Data packets out and insert it into the uplink.

Note that CS data packet (1/2) and CS Data Packet (2/2) shall be considered as part of the same single page (containing a total of 14 bits for Page Type description and 448 bits of Commercial Data), this means that both packets are merged in space segment and included as a single page (disseminated in 1 second) in the C/NAV.

6.7.2.3.2 CS Data relay: SPF and MGF

The Service Product Facility is the only GMS element which has a NRT network interface with the GSC, and mainly gathers all the reports to be sent from and to the GSC and the information related to CS Data distribution.

Also send back to GSC every 30 minutes feedback of the status of Commercial Data previously uplinked, this feedback is based on the information provided from MUCF.
The Message Generation Facility is the only GMS element which has a RT network interface with the GSC. MGF receives Commercial Data from GSC to be uplinked by listening to the expected multicast ports

The MGF will be able to receive CS Data following the ASN1 schema for up to 36 satellites (or only the CS Data messages related to a subset of satellites.)

The MGF receives the CS data from GSC including the information of which satellite should be used for the dissemination of that CS Data.

6.7.2.3.3 Message generation : MGF

Message Generation Facility generates C-band uplink message**.** Gather CS Data from GSC based on listening to the expected multicast group depending on MGF element mode.

The allocation to the corresponding uplink message is performed on the basis of the satellite allocation as provided by the GSC together with the CS-data.
The Commercial Service data to be disseminated toward each of the satellite as requested in the CS data shall be routed by MGF by determining the ULS Antennas allocated to those satellites using the ULS Antennas to Satellite Allocation Plan received from MUCF.
The dissemination to the target region is then performed by the corresponding satellites identified at GSC level.
The MGF shall start disseminating the Commercial Service data just after reception of CS data from GSC in RT format. Data received between GST(T)-1 sec and GST(T)-700 msec will be included in the Message Subframes to be uplinked in the epoch T (i.e from Galileo time (GST) T to T+1 sec)

No data repetition logic is expected in Commercial Service provision
After the successful uplink of CS Data a report is sent to MUCF included in the CS Gen Data.

6.7.2.3.4 CS Grid Generation and Provision: MUCF

Mission and Uplink Control Facility generates the CS-grid.
The CS Grid includes the allocation of connected satellites per region (7 regions defined: Global, Europe, North America, Africa, Asia, South America, and Oceania) in predefined time frames.
The MUCF shall determine a grid for regional dissemination of CS data on 6 CS regions and for global dissemination of CS data over all user locations.
The global dissemination mode for CS data makes use of all satellites used for integrity dissemination, and is exclusive from regional dissemination mode.
The grid for regional dissemination of CS data shall be built on the basis of the following constraints:

- ☐ Only connected satellites are used for CS data dissemination.
- ☐ Each connected satellite could be allocated to more than one CS region (a satellite cannot disseminate on 2 different CS regions at the same time, so GSC should decide the region where is finally allocated).
- ☐ The allocation time window shall have a time granularity of 2 min (The time granularity of 2 minutes corresponds to the 120 sec of a CS frames sequence).
- ☐ A CS region is considered as completely covered if every user location of this CS region receives CS data disseminated by at least 1 connected satellite at any time.
- ☐ Regions are ranked by priority. The allocation process shall allocate satellites to each region in turn along this priority level. In case not enough satellites remain available to cover completely a region, a partial coverage shall be retained for the grid.

This CS-grid is provided to the SPF for further dissemination to the GSC for the allocation of CS data to individual satellite.

MUCF generates the ULS Antennas to Satellite Allocation Plan to be sent to MGF.
MUCF generates also the CS Dissemination Report message which is provided to SPF at a regular interval of 30 minutes, this allows detecting a problem in the uplink, note that this report do not close the loop via the sensor stations.
The dissemination report is based in the CS Gen Data provided by the MGF.

6.7.2.3.5 CS Message uplink: ULS

Up-link Stations provides baseband modulation (with DS-SS using O-QPSK) of the carrier and transmitting the C-band RF signals. The ULS takes the messages from MGF as prepared for each individual satellite (including the CS-data) and uplinks them to the corresponding satellite.

6.7.2.3.6 CS Message relay: SS

The Galileo Space Segment navigation satellites transmit Galileo SIS (Signal in Space). For the purpose of SFC7.1 includes the Commercial Data to be disseminated in C/NAV by a selected satellite. In case there are no CS-data uplinked, the individual satellite fills the corresponding parts of the SIS with predefined dummy data.

6.7.2.3.7 CS Data USERS

Galileo receiver extracts the CS Data information from SIS, also with the appropriate codes can use three-carrier phase-ambiguity resolution (TCAR) techniques through E6 signal to get more accurate positioning.

6.7.2.4 Dynamic Scenarios

6.7.2.4.1 SFC7.1-1 Nominal CS Data Dissemination

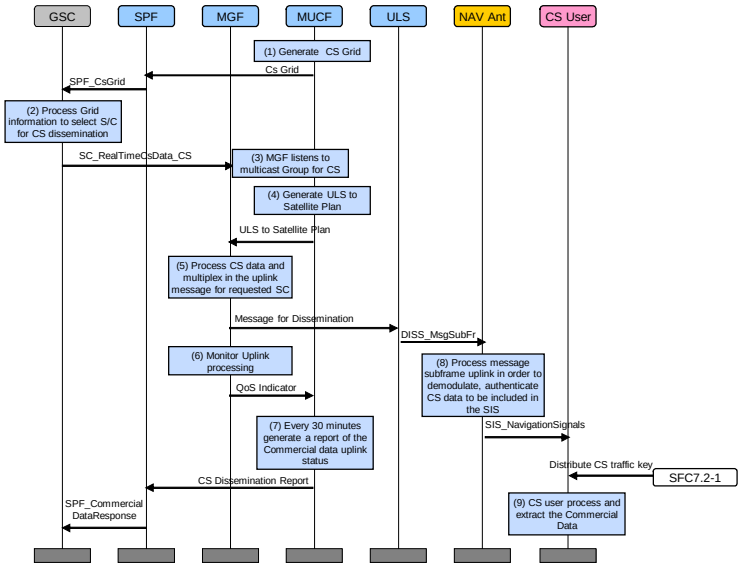


Figure 207: SFC7.1-1 Nominal CS Data Dissemination

Scenario Textual Description:
(0) General Description: The GSC commercial service data dissemination
The sequential description is detailed below:

(1) The MUCF generates CS Grid passed daily to the GSC via the SPF. This Grid changes when there is a modification on the ULS to satellite allocation plan (for contingency reasons).
(2) The GSC processes Grid information to select S/C for CS dissemination: information sent to SPF jointly to the CS data
(3) MGF, depending on the internal element mode, listens to the multicast group where CS in RealTime shall arrive. MGF do a time window reception check on this data

(4) The MUCF Generate ULS to Satellite Plan and delivers it to MGF.
(5) The MGF process CS data on a satellite basis and multiplex in the uplink message for requested satellite.. If no Commercial Data is available for a satellite ~~in SPF~~, dummy data is generated by MGF for this satellite
(6) The MGF monitor Uplink process generates the CS Gen Data with the CS Data uplink status.
(7) The MUCF every 30 minutes generates a report of the Commercial data uplinked based on CS Gen Data received from MGF.
(8) The navigation antenna in SS process message subframe uplink in order to demodulate, authenticate CS data to be included in the SIS. If no Commercial Data is available Dummy data is generated in S/C.
(9) CS user process and extract the Commercial Data based in SIS and CS traffic keys distributed as per SFC7.2.

6.7.2.4.2 SFC7.1-2 Contingency Scenario (CS Data Dissemination)

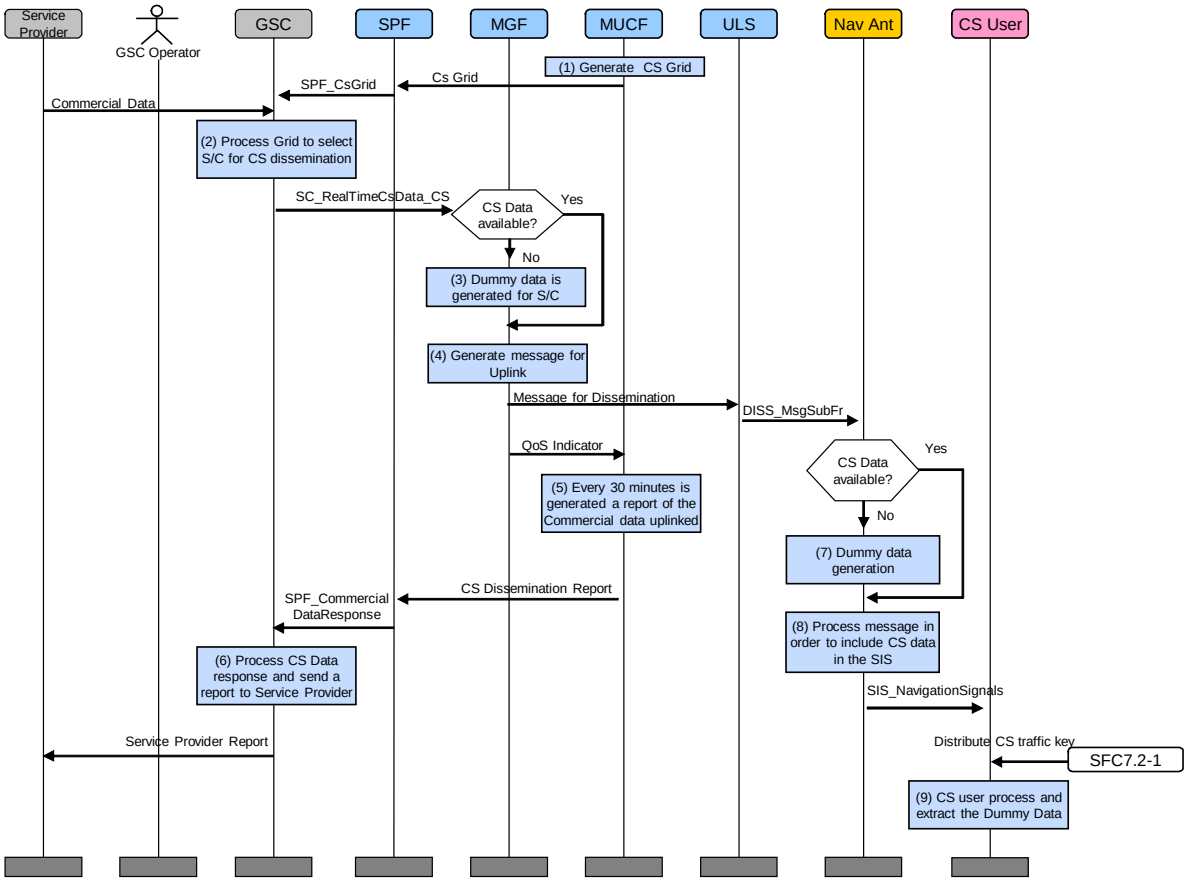


Figure 208: SFC7.1-2 Contingency Scenario (CS Data Dissemination)

The sequential description is detailed below:

(1) MUCF generates CS Grid and sends it to SPF to be relayed to GOC-SC.

(2) GOC-SC Process Grid information to select S/C for CS dissemination.

(3) MGF listens to the multicast group (GCC dependant) where Commercial Service data is coming. MGF checks:

Window reception time
Data Header information
CRC
Message structure

In case of any problem during reception, MGF shall raise a warning to GACF operator.

(4) MGF generates message for Uplink with Dummy Data instead of CS Data in case no CS Data available from GSC or data is rejected. .

(5) MUCF every 30 minutes generates a report of the Commercial data uplinked based on the information contained in CS Gen Data, received from MGF.

(6) GSC process CS Data response and send a report to Service Provider. In case of dissemination chain failure GSC receives the SPF_CommercialDataResponse with the failure status. GSC will decide to react accordingly resending the CS service data or not.

(7) In case no CS Data is available in the uplink message the Galileo satellite generates Dummy data to replace the CS Data in the SIS. This situation could happen during satellite initialization, or when no valid C/NAV messages are available, or when satellites not connected the Space Segment, then generate dummy data on-board. The dummy data field (462 bits) shall be generated containing the following random sequence: Page type (14 bits): shall be generated containing the sequence: '00 1010 1100 1001' (binary). This sequence means no commercial data available. The rest of commercial data page (448 bits) shall be generated containing the following random sequence: [AF3B C301 3D9C 46FE 184D C21A 84B0 53E1 F01E D31A A937 812F 3AE1 84B0 D31A 46FE AF3B 3AE1 AF3B 53E1 46FE C301 D31A 184D 812F FD21 1234 CD21] hex. Other combinations are to be defined and managed by GSC and CS-provider.

(8) Galileo Satellite process message in order to include CS Data (Dummy Data) in the SIS.

(9) CS user process and extract the Commercial Data (Dummy Data) based in SIS and CS traffic keys distributed as per SFC7.2.

6.7.3 SFC7.2 Provide CS Ranging Code

Disclaimer : Please note that this section is pending to reflect STI002 and DCN for FOC2 not yet contracted at segment level

6.7.3.1 Overall Physical Description

Galileo provides support to users of the commercial service by broadcasting the C/NAV navigation signal by means of a ranging code.

In order to control user access to the commercial service, Galileo has the capability to encrypt the CS ranging code.

The purpose of this section is to describe the physical mapping of the CS Ranging Code provision and management functions including activation/deactivation of the ranging code encryption and the corresponding key management.

1. Perform the L-Band CS signal protection including:
    - CS Ranging Code generation and the L-Band CS signal encryption
    - Activation/deactivation of CS Protection in the L-Band signal
    - Manage the Encryption Status in the L-Band signal
2. Perform the remote management of the NSGU with regards to CS ranging code protection
3. Manage the CS Keys including:
    - Interface with GSC for CS key provision
    - CS key distribution up to satellite level

The following section describes the physical implementation of CS ranging code management. It is based on the following documentation:

- ☐ System documents:
    - o System Logical Architecture (SYS-RD.1)
    - o System operational concept: MOCD (SYS-RD.6)
    - o GMS to SC ICD (AD.17)
    - o Annex D (CS) of Cryptographic Algorithms Requirements and Definition (SYS-RD.12)
- ☐ GMS documents:
    - o GMS DDF2 (GMS-RD.2)

- ☐ SSgt documents:
    - o Satellite DJFs (IOVSS-RD.2 and FOCSS-RD.1)
    - o Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2)
    - o NSGU documentation (SS-RD.10)

6.7.3.1.1 Physical Mapping of functional architecture

This section indicates the functional mapping of SFC7.2 to physical elements. Physically, the following elements are involved in the management of CS ranging code:

- 2 elements at GMS level:
    - o MKMF (non-cla): to generate and manage all the CS keys
    - o SPF: to provide an interface with the GSC for key exchange
- 3 elements at GCS level: SCPF (with direct interface with MKMF), SCCF and TTCF for the scheduling and TC generation and uplink for CS key distribution and activation
- 1 element (mainly) at SSgt level:
    - o NSGU: for the CS ranging code generation and navigation message generation (CS encryption status flag in the I/NAV message) and to process the CS key management telecommands.

6.7.3.1.2 Physical Static View

The following figure presents the physical architecture of the satellite timing parameters generation and maintenance functions:

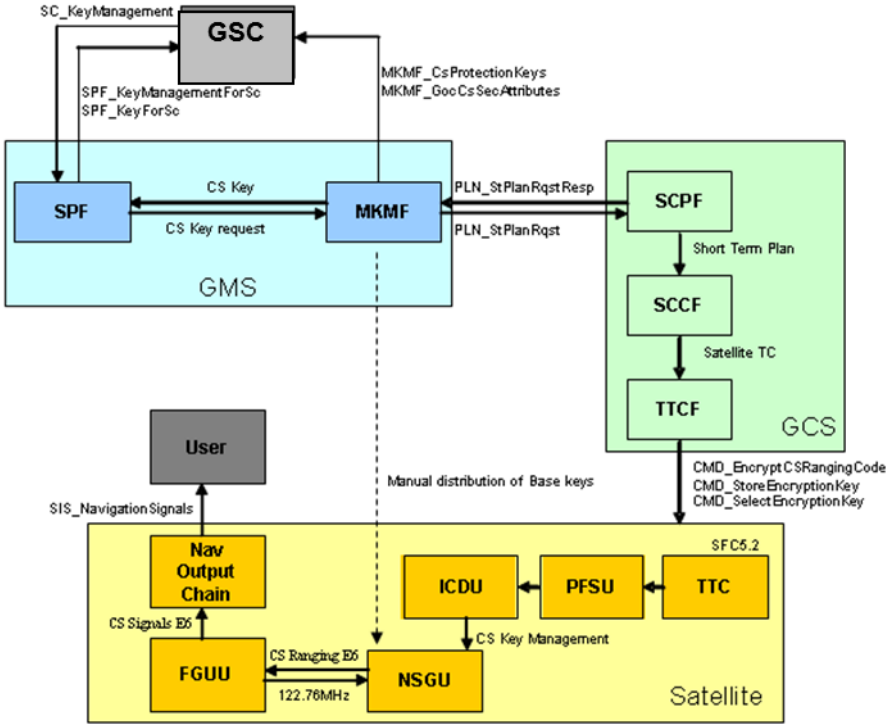


Figure 209: SFC7.2 Physical Static View

6.7.3.2 Protection of CS Ranging Code

The ranging codes are long codes of certain code periods which are built from so-called primary and secondary codes of certain respective code-length by using a tiered code construction.

The signal components dedicated to the commercial service are the ones on E6-B and E6-C.
The E6-B and E6-C primary codes are pseudo-random memory code sequences. For each signal component E6-B and E6-C, overall 50 different codes are defined (each satellite will use a different code from this range).
For E6-C (a pilot channel), the secondary code is a 100 bit pseudo-random memory code, overall 50 codes are defined. For E6-B (carrying the C/NAV navigation data stream) there is no secondary code (this is not a tiered ranging code).

If the encryption of the CS ranging code is disabled, the CS ranging code is produced by the NSGU using these pseudo-random memory code sequences.
If the encryption is enabled, the E6-B and E6-C ranging codes are encrypted using the CS traffic key before the CS signals are broadcast.

All satellites in the constellation make use of the same CS traffic key. CS user receivers will need the same CS traffic key in order to decrypt the E6-B and E6-C ranging codes and thus acquire the signals.

6.7.3.3 Remote NSGU Management

This functionality concerns the secure monitoring and control of the NSGU on board the Spacecraft with regards to functions related to CS ranging code protection.
This monitoring and control takes place from the MKMF Unclassified and is routed through the GCS which communicates with the Space Segment over the S-band.
The NSGU is managed through telecommands (TC) and provides telemetry (TM). Note that there are no secure telecommands (STC) or secure telemetry (STM) related to the NSGU, because it is not classified as a security unit.
To provide an additional layer of protection on top of the standard S-band protection (layer 1), the TC related to CS ranging code protection contain some cryptographic protection. This applies to the following commands:

- CS traffic key change
- CS protection key change

These CS commands are protected with data integrity protection (CRC), with encryption and with data authentication (MAC) as defined in [SYS-RD.12].
A CS command is provided directly from the MKMF (non-cla) to the SCPF in the GCS for uplinking to the satellites. Based on this input, the SCCF will uplink each command to the satellites with a dedicated TC (non Cla). The CS related TCs follow the same scheme as other TCs as described in SFC5.2.
The GCS provides a distribution status report directly to the MKMF once receipt of the instruction has been acknowledged by the satellite or in case of dissemination failure.

6.7.3.4 Key Management for CS

6.7.3.4.1 CS Key types

Several types of CS keys are generated depending on the need:

- "CS traffic keys": Key generated by GMS (MKMF) for CS ranging codes encryption (see section 6.7.3.2). This key is distributed (with proper CS protection key) to:
  - NSGU (for CS ranging code encryption) via the GCS in a dedicated telecommand
  - GSC (for CS users to be able to decrypt CS ranging code)
  - GSS (to be able to decrypt CS ranging code)

- "CS protection keys": Key generated by GMS (MKMF) used to protect the dissemination of CS traffic keys from MKMF to NSGU via GCS (see section 6.7.3.3) and to GSC and GSS. This key is distributed to:
  - NSGU (for CS traffic key retrieval) via the GCS in a dedicated telecommand (protected with a CS base key, see section 6.7.3.3)
  - GSC (for CS traffic key retrieval) via an on-line message through SPF (protected with a CS base key) or via off-line distribution
  - GSS (for CS traffic key retrieval) via remote management command (protected with a CS base key) or via off-line distribution

- "CS Base Key": Key generated by GMS (MKMF) used to protect the dissemination of CS protection keys from MKMF to NSGU via GCS (see section 6.7.3.3) and to GSC and GSS. This key is distributed via off-line media to:
  - NSGU (for CS protection key retrieval) pre-loaded on-board the satellites (in the NSGU) before launch and not changed.

- GSC (for CS protection key retrieval) via off-line distribution
- GSS (for CS protection key retrieval) via off-line distribution

These key types form a key hierarchy as shown in Figure below:

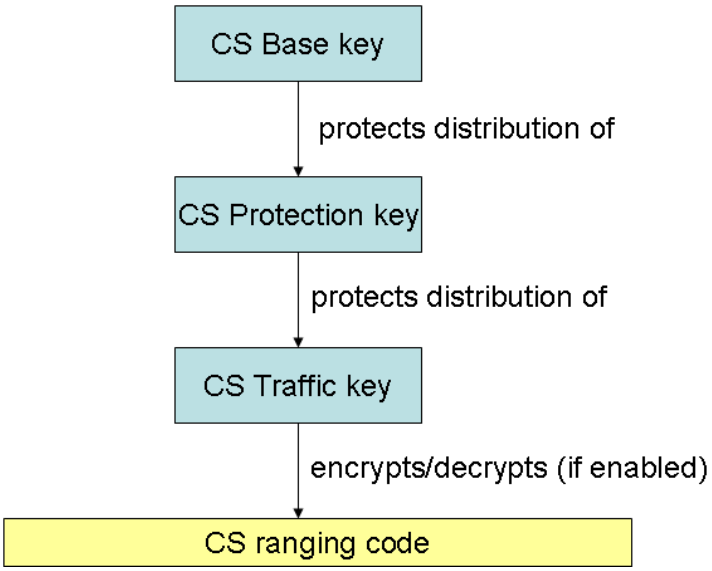


Figure 210: CS key hierarchy

Note: all satellites and authorised CS receivers need to share the same CS traffic key in order to allow access to the CS ranging code. The CS base keys and CS protection keys however are not shared in common, these are uniquely dedicated to each communication path. For example, the GSC does not know the CS base key or CS protection key of a NSGU.

6.7.3.4.2 CS traffic keys

CS traffic keys are periodically changed. This means new keys need to be generated, distributed to all parties involved and then activated at a preset time.

6.7.3.4.2.1 CS traffic keys generation

The CS traffic keys are generated by the MKMF (non-classified).
CS traffic keys are periodically changed. The period length and date and time of change, must be agreed beforehand with the CS providers.

6.7.3.4.2.2 CS traffic key distribution to CS users: GSC

The CS traffic keys are disseminated to GSC via the SPF and the EDDN (protected with dedicated CS Protection key).
Alternatively, the CS traffic key can be distributed via off-line media.
The GSC acts as a distribution point for the CS traffic keys to the Commercial Service (CS) providers. The GSC maintains a distribution list of CS providers that are entitled to receive the

Keys. When the GMS issue a new key the GSC forwards it to the members of the distribution list. The keys are sent out electronically or otherwise. The GSC does not have use of the keys themselves, since it is just a courier of the key material.

6.7.3.4.2.3 CS traffic key distribution to GSS

The CS traffic keys are disseminated to the GSS via remote management command (protected with dedicated CS Protection key).

6.7.3.4.2.4 CS traffic key distribution to satellite

The CS traffic keys are disseminated to Satellite via the GCS in a secure manner (protected with dedicated CS Protection key).

A key change instruction which includes the encrypted keys and the time of activation is provided from the MKMF to the Satellite via Remote NSGU Management (see section 6.7.3.3).

6.7.3.4.2.5 CS ranging code encryption based on CS Traffic keys: NSGU

The NSGU is the unit responsible to generate the CS encrypted ranging codes. In case the satellite receives a key change instruction, it is validated by the PFSU and stored in the ICDU for future activation.
When the time of activation has arrived, the NSGU will replace the currently active CS ranging code encryption key by the decrypted key from the key change instruction, and trigger the encryption engine to use the new key.

6.7.3.4.2.6 CS traffic keys change request

The CS traffic key change can be initiated by either MKMF or GSC:

- Key change request form GSC: GSC may request that the CS ranging code encryption key be changed. Such a request will be implemented within 30 days.

- Key change notice from MKMF: GMS can issue a key change notice 7 days in advance of a change to the CS Keys .This Key change notice is sent to GSC and forwarded onto the CS providers.

6.7.3.4.3 CS Protection Key

6.7.3.4.3.1 CS Protection keys generation

The CS Protection keys are generated by the MKMF (non-classified).
They are used for key distribution of CS traffic keys to the GSC, GSS and to the satellite.

6.7.3.4.3.2 CS protection key distribution to GSC

The CS protection keys can provided to the GSC in 2 different ways:
- with direct interface through the SPF (protected with dedicated CS Base key)
- with manual interface (CD-ROM) directly from MKMF operator to GSC

6.7.3.4.3.3 CS protection key distribution to GSS

The CS protection keys are disseminated to the GSS via remote management command (protected with dedicated CS Base key).

6.7.3.4.3.4 CS protection key distribution to satellite

The CS protection keys are disseminated from MKMF to the NSGU via from the MKMF to the Satellite via Remote NSGU Management (see section 6.7.3.3).The protection keys are encrypted and authenticated using base keys which are loaded in each NSGU before launch.

The satellite will receive the protection key change instruction, validate its data integrity and authenticity and, if these are in order, decrypt and store the protection key ready to be used for future traffic key decryption and authentication.

6.7.3.4.4 CS Base Key

The base keys are generated by the MKMF. Dedicated base keys are provided to each NSGU, the GSC and each GSS.

6.7.3.4.4.1 CS base key distribution to GSC

CS base keys are distributed from MKMF to GSC via off-line media .
These keys are distributed in black form in a CD-ROM encrypted under the CS Protection Manual Key

6.7.3.4.4.2 CS base key distribution to GSS

CS base keys are distributed from MKMF to GSS via off-line media (GMS internal operational interface).

6.7.3.4.4.3 CS base key distribution to satellite

The base keys are disseminated from MKMF to the satellite by off-line media (e.g. by CD-ROM). The base keys are loaded in each satellite before launch.

The satellite will store the base keys, ready to be used for future CS protection key decryption and authentication.

6.7.3.4.5 Encryption Status Change

On request of the GSC, the CS ranging code encryption can be activated or deactivated. Such an encryption status change needs to be requested sufficiently in advance to allow for dissemination planning. When triggered by the GSC, the MKMF will generate an encryption status change instruction for each satellite and provide these to GCS for dissemination.
In order to minimize discontinuity of service, the CS providers may provide advance notice of the encryption status change to the CS users via the CS message data or through other means.

GCS will disseminate the encryption status change instruction to the satellite using a telecommand. In case of communication failure, GCS will resend the telecommand.The GCS provides a distribution status report to the MKMF once receipt of the instruction has been acknowledged by the satellite or when it is apparent the dissemination is not possible.

The satellite will receive the encryption status change instruction, validate its data integrity and authenticity and, if these are in order, store the instruction for future activation.

When the time of activation has arrived, the satellite will set the encryption status for the CS ranging code in accordance with the instruction (i.e. encrypted ranging code or ranging code in clear).
At the same time, the NSGU will flag the new encryption status to the CS users via the CS encryption status parameter in the I/NAV message.

6.7.3.5 Dynamic Scenarios

There are six scenarios described for this SFC:

- Scenario #SFC7.2-1: CS traffic key distribution
- Scenario #SFC7.2-2: CS protection key distribution to the spacecraft
- Scenario #SFC7.2-3: CS protection key distribution to GSC
- Scenario #SFC7.2-4: CS base key distribution to satellite
- Scenario #SFC7.2-5: CS base key distribution to GSC
- Scenario #SFC7.2-6: CS encryption status change

6.7.3.5.1 Scenario #SFC7.2-1: CS traffic key distribution

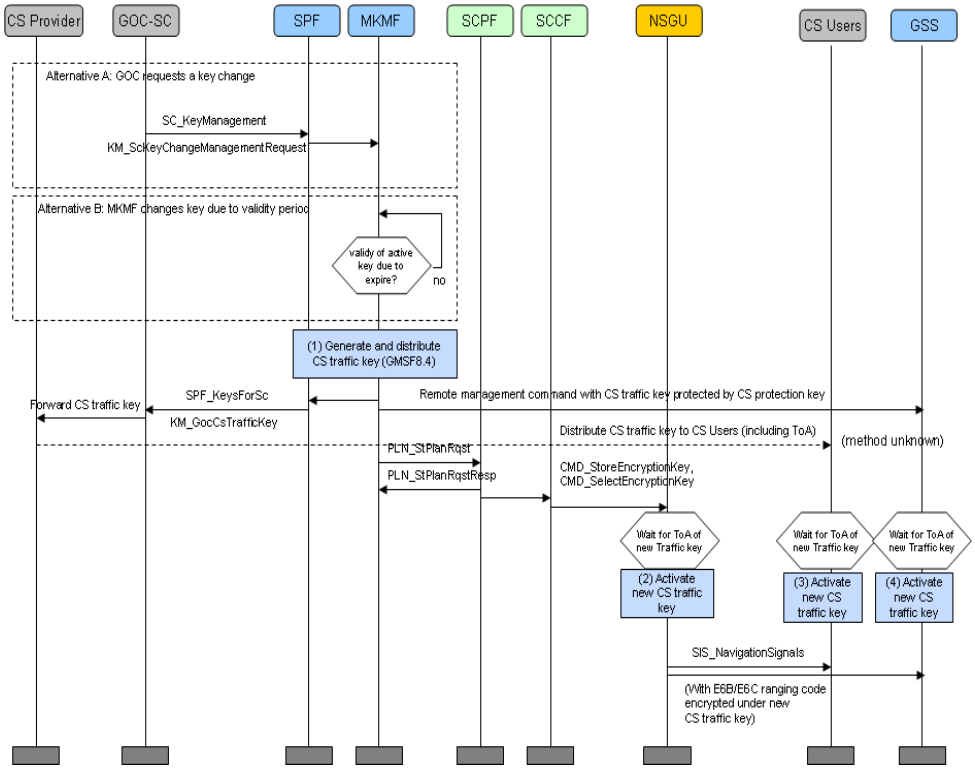


Figure 211 : Scenario #SFC7.2-1 CS traffic key distribution

Scenario Textual Description:
General Description: Either at request of GSC or due to a pending expiry of the active key, the MKMF will generate a new CS traffic key and distribute this in parallel to the GSC and to the NSGU (via telecommand from GCS). The GSC distributes the key to the CS users. The NSGUs and the CS receivers will activate the new CS traffic key at the pre-specified Time-of-Activation.

The sequential description is detailed below:

- (1) Either at request of GSC or based on the expiry date of the currently active traffic key, MKMF will generate a new traffic key and distribute it to GSC via SPF and to the NSGU via GCS. (This dynamic scenario shows distribution to GSC done on-line by SPF instead of the alternative of manual distribution)

- (2) On receipt of a new traffic key by telecommand, the NSGU will store it to be activated at the associated time-of-activation.

- (3) On receipt of a new traffic key from the CS provider, the CS User receiver will store it to be activated at the associated time-of-activation.

- (4) On receipt of a new traffic key from MKMF by remote management command, the GSS CS receiver will store it to be activated at the associated time-of-activation.

6.7.3.5.2 Scenario #SFC7.2-2: CS protection key distribution to the spacecraft

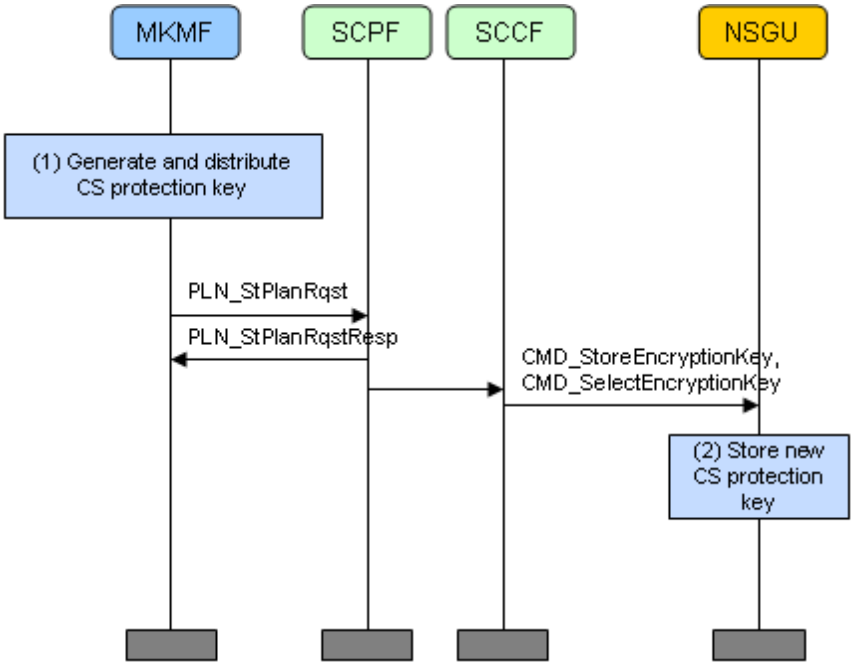


Figure 212 : Scenario #SFC7.2-2: CS protection key distribution to the spacecraft

Scenario Textual Description:
General Description: CS protection keys are generated by the MKMF and distributed to the NSGU via GCS and a telecommand to the spacecraft.

The sequential description is detailed below:
(1) MKMF generates a new CS protection key and distributes it to the NSGU via GCS.
(2) On receipt of a new CS protection key by telecommand, the NSGU will store it to be used to decrypt and authenticate future CS traffic key change commands.

6.7.3.5.3 Scenario #SFC7.2-3: CS protection key distribution to GSC

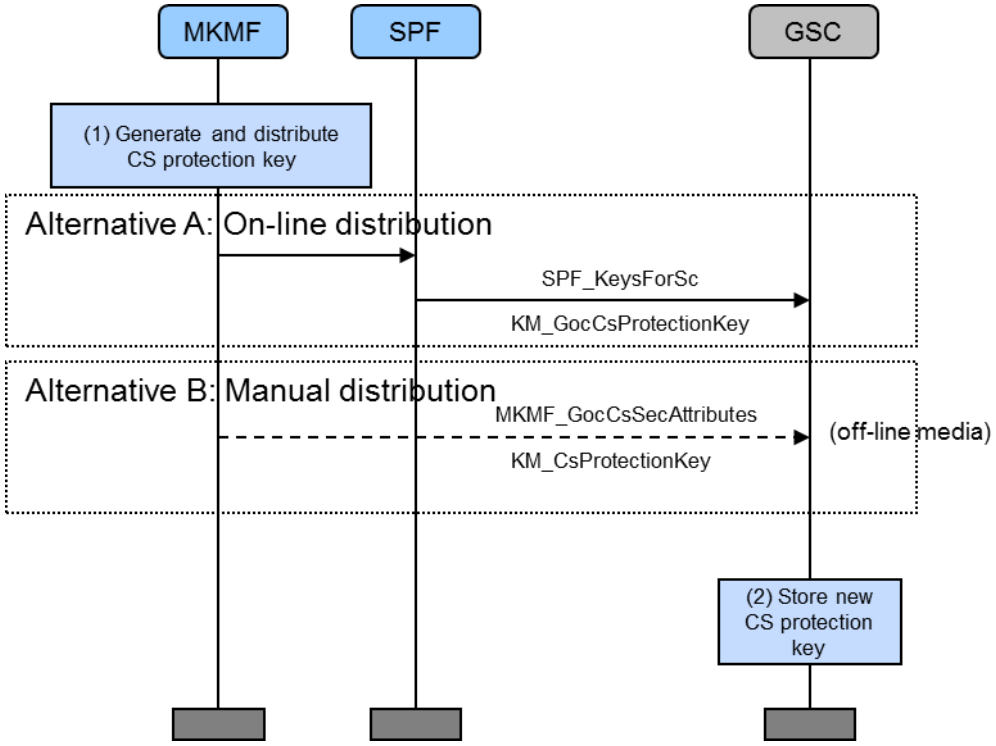


Figure 213 : Scenario #SFC7.2-3: CS protection key distribution to GSC

Scenario Textual Description:
General Description: CS protection keys are generated by the MKMF and distributed to the GSC via off-line media.

The sequential description is detailed below:

- (1) MKMF generates a new CS protection key and distributes it to the GSC. Alternative A is distribution via SPF and the EDDN network (protected by the CS Base key shared between MKMK and GSC), alternative B is manual distribution via off-line media.
- (2) On receipt of a new CS protection key, the GSC will store it to be used to decrypt and authenticate future CS traffic key change commands.

6.7.3.5.4 Scenario #SFC7.2-4: CS base key distribution to satellite

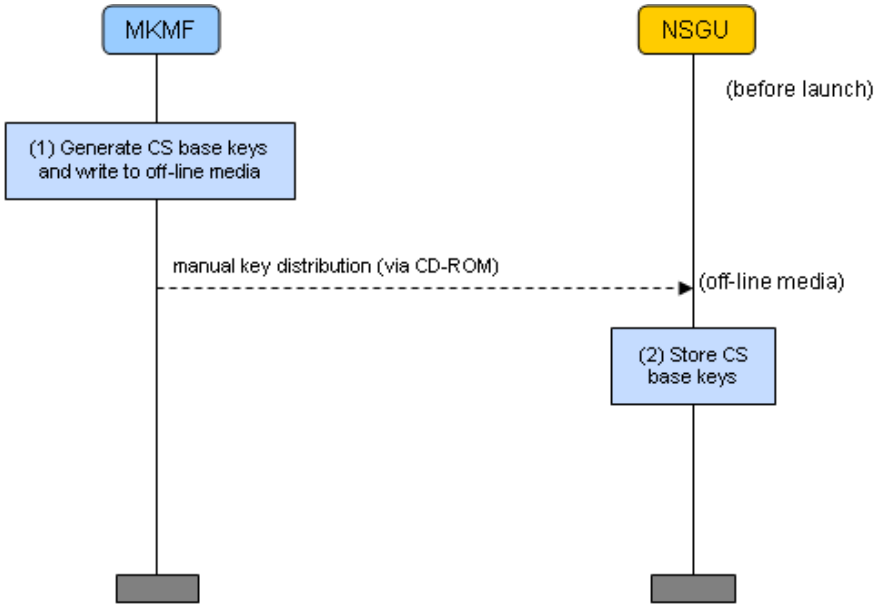


Figure 214 : Scenario #SFC7.2-4 CS base key distribution to satellite

Scenario Textual Description:
General Description: CS base keys are generated by the MKMF and distributed to the NSGU before launch via off-line media.

The sequential description is detailed below:
(1) MKMF generates CS base keys for a NSGU and writes these to off-line media
(2) The off-line media is provided to the NSGU for storage before launch.

6.7.3.5.5 Scenario #SFC7.2-5: CS base key distribution to GSC

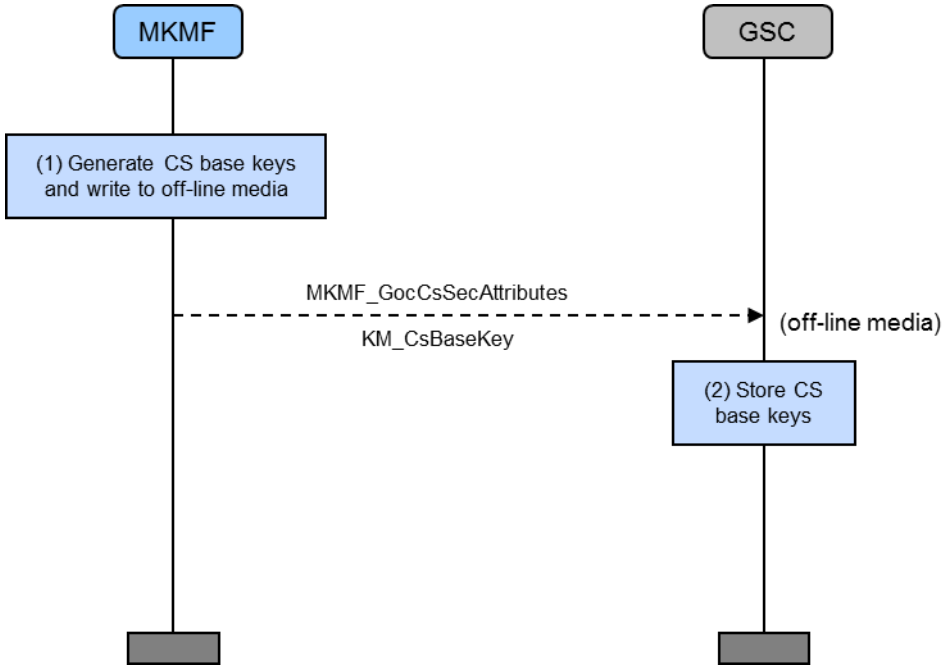


Figure 215 : Scenario #SFC7.2-5: CS base key distribution to GSC

Scenario Textual Description:
General Description: CS base keys are generated by the MKMF and distributed to the GSC via off-line media.

The sequential description is detailed below:

- (1) MKMF generates CS base keys for GSC and writes these to off-line media
- (2) The off-line media is provided to the GSC and stored for future use.

6.7.3.5.6 Scenario #SFC7.2-6: CS encryption status change

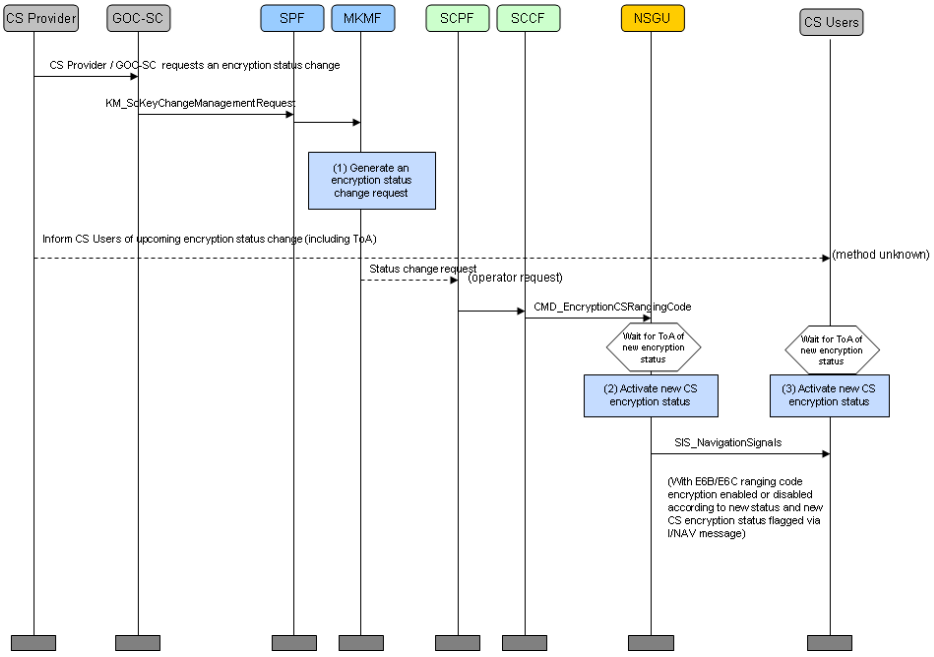


Figure 216 : Scenario #SFC7.2-6: CS encryption status change

Scenario Textual Description:
General Description: A change of CS encryption status is triggered by the GSC, the MKMF plans the status change and forwards this to all NSGU. Meanwhile the CS provider shall notify the CS user terminals.

The sequential description is detailed below:

(1) At request of GSC, the MKMF initiates a CS encryption status change. Note: the status change command is not foreseen in the GMS to GCS ICD. A manual procedure is foreseen with the MKMF operator requesting the status change to GCS.

(2) On receipt of the encryption status change request, the NSGU will store the request for activation at the specified time of activation.

(3) On receipt of the encryption status change notification, the CS user terminal will store the request for activation at the specified time of activation (Alternatively, the CS user terminal will adapt the CS encryption status based on the status flags in the I/NAV navigation message).

6.7.4 SFC7.3 Provide Information to Galileo Service Centre, GSMC and GALSEE

A) The Galileo Core Infrastructure will provide support to GSC providing the following information:
Providing reports to GSC, which contains the following information:

- ☐ Daily Service Performance Reports (INF_DailySrvcPerfoRpt)
- ☐ Long Term Service Assessment Reports (INF_LTSrvcAssessmtRpt)
- ☐ Service Performance Predictions (INF_SrvcPerfoPredictions)
- ☐ Daily System Performance Report (INF_DailySystemPerfoRpt)
- ☐ Long Term System Assessment Report (INF_LTSystemAssessmtRpt)
- ☐ Almanac Data (INF_AlmanacData)
- ☐ Notice Advisory to Users (INF_NoticeAdvisoryToUsers)
- ☐ Galileo Overall Status (INF_GalileoOverallStatus)
- ☐ Clock Ephemeris Ionospheric Predicted Data (INF_ClkEphemIonoPredData)
- ☐ Clock Ephemeris Ionospheric Restituted Data (INF_ClkEphemIonoRestData)
- ☐ Estimated (restituted) clocks (INF_ClkFromMsf)
- ☐ Estimated (restituted) orbits (INF_OrbitFromMsf)
- ☐ RINEX Galileo Observables Data (INF_RinexGalData)
- ☐ RINEX Galileo Navigation Message Data (INF_RinexGalNavMsg)
- ☐ Real Time raw data related to GPS and Galileo receivers from GMS
- ☐ Galileo ULS Pointing Data (only for Svalbard) (INF_UlsPointingData)

Providing bulletins to GSC, which contains the following information:

- ☐ Maintenance Bulletins (INF_MaintenanceBulletin)
- ☐ Forecast Performance Bulletins (INF_ForecastPerfoBulletin)

On the other hand GMS can also receive from GSC Report Requests, which contains the following information:

- ☐ Request of Information Reports (INF_ReportInfoRqst)
- ☐ Request of New Products (INF_NewProductRqst)

GMS can also receive from GSC Standard Notifications, which contains the following information:

- ☐ User Anomaly Notifications (INF_AnomalyNotif)

B) The Galileo Core Infrastructure will provide support to GSMC providing the following information:

- ☐ Galileo Overall Status (INF_GalileoOverallStatus)
- ☐ Daily Service Performance Report (INF_DailySrvcPerfoRpt)
- ☐ Service Performance Predictions (INF_SrvcPerfoPredictions)
- ☐ Notice Advisory to Users (INF_NoticeAdvisoryToUsers)
- ☐ Maintenance Bulletin (INF_MaintenanceBulletin)
- ☐ Forecast Performance Bulletin (INF_ForecastPerfoBulletin)
- ☐ Classified Data from PKMF (Classified information see classified SEC-SFC 13.1)

On the other hand GMS can also receive from GSMC the following information:

- ☐ Classified Data to PKMF (Classified information see classified SEC-SFC 15)

C) The Galileo Core Infrastructure will provide support to GALSEE providing the following information:

- ☐ Almanac Data (INF_AlmanacData)
- ☐ Clock Ephemeris Ionospheric Predicted Data (INF_ClkEphemIonoPredData)
- ☐ RINEX Galileo Observables Data (INF_RinexGalData)
- ☐ RINEX Galileo and GPS Observables Data (INF_RinexData30s)
- ☐ RINEX Galileo Navigation Message Data (INF_RinexGalNavMsg)
- ☐ RINEX PRS Galileo Observables Data 1 sec (INF_RinexGalData_S)
- ☐ RINEX PRS Galileo Observables Data 30 sec (INF_RinexData30s_S)
- ☐ RINEX PRS Galileo Navigation Message Data (INF_RinexGalNavMsg_S)
- ☐ GMS archive files in GACF format including the following information : MM_RawData, NAV_RawDatra, NAV_SBandUplkDataFile,DISS_RawData and CNF_SvMask
- ☐ Estimated restituted orbits and clocks for all–Galileo SVs (INF_ClkFromMsf, INF_OrbitFromMsf)
- ☐ Measurements files containing the ΔTi=Ti -GST information as available from PTF (TIME_GstTaiOffset, TIME_GalClkTwstftMeas, TIME_GalClkCvMeas)
- ☐ Measurements files from TWSTFT containing the ΔT= GST-UTC(USNO) as available from PTF (TIME_GalGpsGgto, TIME_GpsClkTwstftMeas, TIME_GpsClkCvMeas)
- ☐ On-board clocks phase difference,Telemetry data (MM_SatSumStatus)
- ☐ Orbits data provided by FDF (MISC_OrbitPred, MISC_ManvrAnnounce)
- ☐ S-Band ranging data provided by TTC Stations (MISC_SBandSatRngData, MISC_SBandSatDopData, MISC_SBandSatAngData, MISC_SBandSatMetData)
- ☐ Galileo Satellites configuration data (CNF_SatCfgData) and default static parameters configuration (CNF_SatCfgDataStatic)

☐ 3D coordinates and velocities of the GSS tracking antennas to the GCC for enabling ODTS functions (CNF_GssCoordinatesVal)
☐ Meteo monitoring data on GSS site (MNT_SiteData
☐ Short Term Contact planning for both ULS and TTC antennas (SCH_OverallUlsContactPlan, SCH_UlsTrackPlan, PLN_TtcsContactSched)
☐ S-Band navigation Uplink Report generated by GCS regarding the uplinking of navigation data in S-Band using the TTC stations
☐ An operational interface (email exchange) between operators allowing to report data exchange problems (missing data, I/F failures..)

6.7.4.1 Physical Static Block Diagram

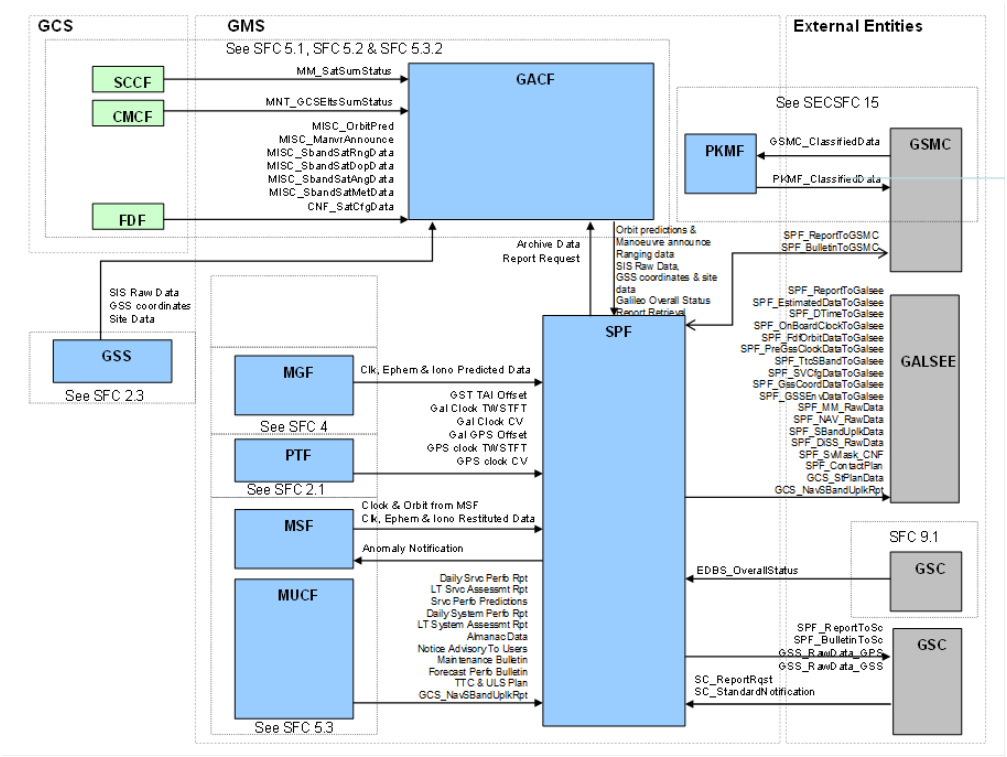


Figure 217 : SFC7.3 Physical Static View for GSC, GSMC and GALSEE

6.7.4.2 Overall Physical Description

The purpose of this SFC is to exchange information (or reports) between GMS and the GSC and GSMC and also allow GSC to request for new and old reports. In the sections below are detailed the main contributions of all physical elements involved.

6.7.4.2.1 Satellite summary status: SCCF

SCCF shall provide to GACF the Satellite Summary status.

6.7.4.2.2 GCS summary status: CMCF

CMCF shall provide to GACF the summary status of the elements of GCS.

6.7.4.2.3 S-Band Uplink Report: SCCF

SCCF shall provide to MUCF with the S-Band Uplink Navigation Service Uplink Report after S-Band uplink per each satellite. This information is provided from MUCF to SPF in order to relay it to GALSEE.

6.7.4.2.4 TTCS Contact Schedule: SCPF

SCPF shall provide to MUCF with the TTCS Contact Schedule each time Short Term Plan is updated. This information is provided from MUCF to SPF in order to relay it to GALSEE.

6.7.4.2.5 Flight Dynamics: FDF

FDF shall provide to GACF Orbit predictions, manoeuvre announcements, ranging data provided by the TTC stations, and satellite configuration data.

6.7.4.2.6 Archive and Overall Status: GACF

GACF shall provide to SPF the Galileo overall status, the SIS raw data and archived reports older than 7 days that SPF could be requested. Shall provide also Orbit predictions and manouvre announcements,, ranging data, and GSS coordinates information and GSS site data. GMS operator should also assess new product requests received through SPF and GACF.

6.7.4.2.7 Mission reports generation: MUCF

MUCF shall generate mission monitoring reports, service level reports, maintenance bulletins, top level spacecraft reports, forecast bulletins, and notice advisory users. All this information is passed to SPF in order to relay it to GSC and also partially to GALSEE (SPF_ReportToGalsee).

6.7.4.2.8 Short Term Plan provision: MUCF

MUCF shall provide, on a daily basis and after re-planning, the SPF with the Contact Plan (SCH_OverallUlsContactPlan) and Tracking Plan (SCH_UlsTrackPlan).This information is provided to SPF in order to relay it to GALSEE.

6.7.4.2.9 Predicted data generation: MGF

MGF shall generate Clock, Ephemeris and Ionospheric Predicted Data to be sent to SPF.

6.7.4.2.10 Restituted data generation: MSF

MSF shall generate Clock, Ephemeris, and Ionospheric Data to be sent to SPF MSF also receives anomaly notifications relayed by SPF from GSC users.

6.7.4.2.11 External Entities interface and RINEX data generation: SPF

6.7.4.2.11.1 Report and request interface:SPF

Forward accordingly reports, bulletins to GSC, GSMC, and GALSEE and receive standard notifications and report request from GSC.

6.7.4.2.11.2 RINEX observables generation and relay information: SPF

Shall transform Galileo observables and navigation message in RINEX to be provided to GSC and GALSEE (SPF_ReportToGalsee).

6.7.4.2.12 SFC7.3 Related Constraints: GSC

6.7.4.2.12.1 Reception of Galileo reports: GSC

Shall receive SPF reports and bulletins in order to relay them and provide value added services to other service providers.

6.7.4.2.12.2 Provision of anomaly reports: GSC

Shall provide anomaly notifications to SPF from filtered user anomaly reports.

6.7.4.2.12.3 New product requests: GSC

Shall provide to SPF request for inclusion of new products in the catalog of products offered to GSC

6.7.4.2.12.4 Report requests: GSC

Shall provide to SPF requests of products already existing in the catalog.

6.7.4.2.13 Key Management: PKMF

Exchange classified information with GSMC (see SEC SFC15)

6.7.4.2.14 Clock Offset: PTF

Provides to SPF measurement files containing TWSTFT and CV measurements and time offsets information.

6.7.4.2.15 Security management: GSMC

Receives from SPF reports with Galileo overall status, service performance reports, NAGU, maintenance bulletins, forecast bulletins and exchange classified information with PKMF (see SEC SFC15).
NAGU provision can be activate/de-activate by GMS operator.

6.7.4.2.16 Verification Platform: GALSEE

Receives from SPF, the agreed Galileo System information in order to do the verification for all the system and element verification activities, performed during the Galileo System pre-operational phases.

6.7.4.3 Dynamic Scenarios

In the following dynamic scenarios figures are described how the different reports and bulletins are sent form Galileo Core Infrastructure to GSC and GSMC, or in the other way how GSC is sending requests and notifications to Galileo Core Infrastructure.

6.7.4.3.1 SFC7.3-1 Information exchange with GSC

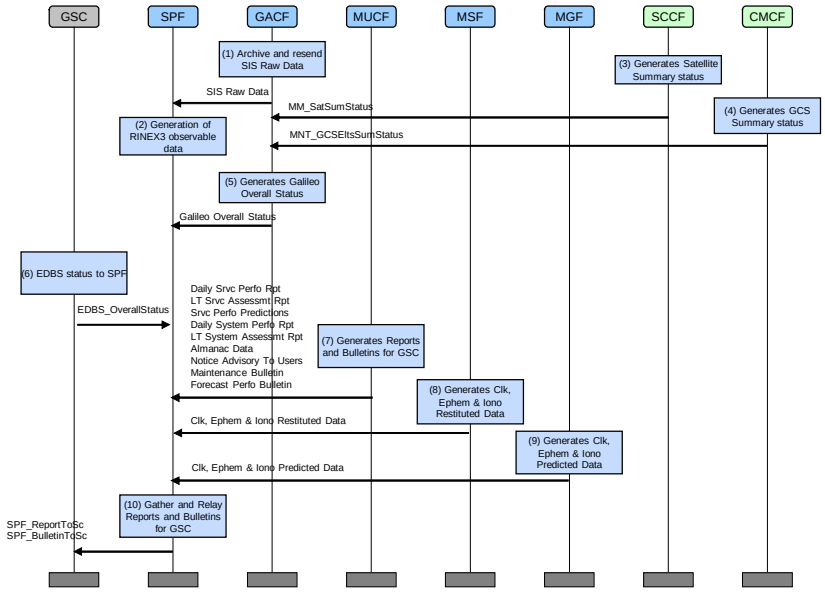


Figure 218 : SFC7.3-1 Dynamic scenario Information exchange with GSC

Scenario Textual Description:
(0) General Description:
This scenario describes the way Galileo overall status, Rinex data, clock, ephemeris, ionospheric restituted and predicted data, and all the standard reports and bulletins are generated in GCC and distributed to GSC.
The sequential description is detailed below:

(1) GACF archives and resends SIS Raw Data to SPF.
(2) SPF makes the conversion to RINEX3 format of SIS Raw data and provides it regularly to GSC.
(3) SCCF Generates Satellite Summary status and provides it to GACF.
(4) CMCF Generates GCS Summary status and provides it to GACF.
(5) GACF compiles the previous GCS and Sat status and add GMS status to Generate Galileo Overall Status and provide it regularly to SPF to be relayed regularly to GSC.
(6) GSC provides EDBS status to SPF
(7) MUCF Generates Reports and Bulletins for SPF which contains:

- Daily Srvc Perfo Rpt
- LT Srvc Assessmt Rpt
- Srvc Perfo Predictions
- Daily System Perfo Rpt
- LT System Assessmt Rpt
- Almanac Data
- Notice Advisory To Users
- Maintenance Bulletin
- Forecast Perfo Bulletin

(8) MSF Generates Clk, Ephem & Iono Restituted Data and provides it to SPF.
(9) MGF Generates Clk, Ephem & Iono Predicted Data and provides it to SPF.

(10) SPF Gathers and relay previous reports and bulletins for GSC.

6.7.4.3.2 SFC7.3-2 : GSC Report request

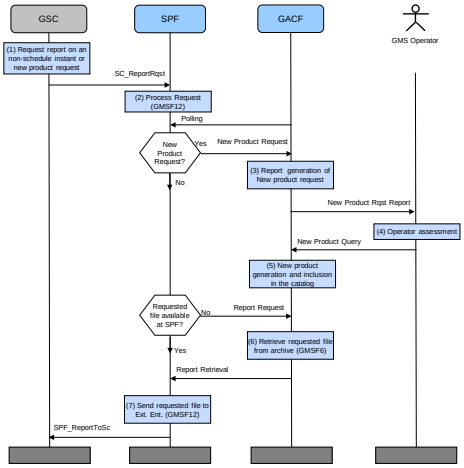


Figure 219 : SFC7.3-2 Dynamic scenario GSC Report request

Scenario Textual Description:

0) General Description: The External entity may request a standard report is transmitted at a non-scheduled instant. Reports are normally produced at scheduled instants but can be requested at other times. This scenario describes the process that take place to retrieve the requested file and send to the external entity.

The sequential description is detailed below:

(1) The external entity operator fills an INF_ReportInfoRqst online via the SPF web server to request the file outside of the scheduled delivery. The external entity can browse the SPF web server for files currently on the SPF, product types currently unavailable on the SPF, but available in the GACF. Note that a list of individual files available on the GACF archive is not provided as such a list would be unreasonably large and difficult to browse.

(2) The SPF then processes the request to determine which files are to be retrieved and where from (either the SPF or GACF). In case of a request of a new product, i.e. product not in the SPF catalog, automatically jumps to process (3) Note that the SPF must never initiate connections to elements within the GCC, so there is an SNMP agent in SPF which is polled by GACF in order to know if there are request in SPF, in case there is a request in SPF for GACF this is notified through SNMP protocol and then GACF initiates and FTP-GET in order to receive the query by FTP, then GACF send the data to SPF with an FTP-PUT.

(3) A report with a new product request is generated by GACF based on the new product request received from SPF, in order to be assessed by a GMS operator.

(4) Operator assessment of new product request in order to decide if the product is generated and included in the catalog or not.

(5) In case of positive assessment from GMS Operator the new product is generated and included in the GACF archive and sent to SPF catalog. *Assumption: GACF archive located in the GMS Central archiving of GACF.

(6) If the file is in the catalog, but not stored at the SPF, the GACF is requested to retrieve the file from the archive. *Assumption: For new product request the GACF after receive the request, as per procedure described before, and sends the new product in the same way FTP-PUT to SPF.

(7) Once the file is retrieved, the SPF makes an FTP put of the requested file/s to the address specified by the external entity. The request is then considered closed with no further action.

6.7.4.3.3 SFC7.3-3 Commercial Users Anomaly Report Provision

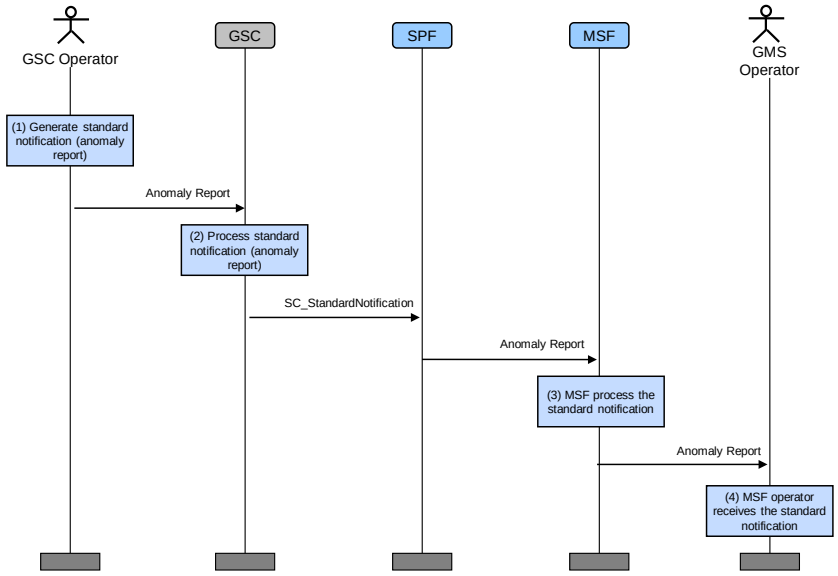


Figure 220 : SFC7.3-3 Commercial Users Anomaly Report Provision

Scenario Textual Description:
(0) General Description:
GMS provision of Anomaly Notifications generated from the collection and filtering of User Anomaly Reports by the operator of GSC.
The sequential description is detailed below:

(1) GSC operator generate the standard notification, this standard notification includes anomaly report which is a summary of anomaly reports provided by commercial service users to GSC.

(2) GSC process the standard notification (anomaly report) and is sent to SPF which relays this anomaly report to MSF.

(3) MSF process these reports and send the information to GMS operator.

(4) GMS operator receives the anomaly report and process this standard notification according to operational procedures.

6.7.4.3.4 SFC7.3-4 Information exchange with GSMC

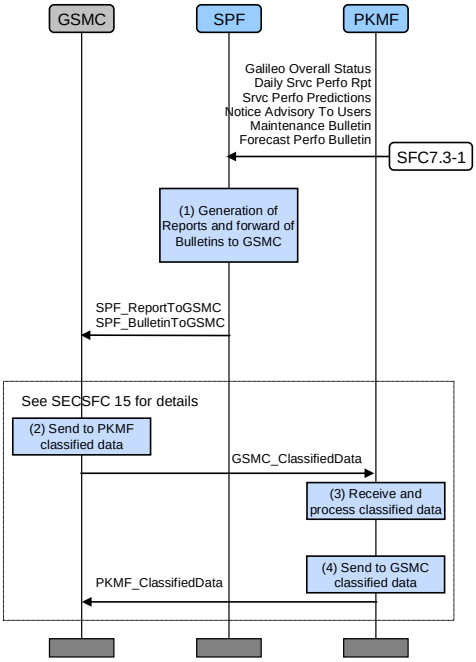


Figure 221 : SFC7.3-4 Information exchange with GSMC

Scenario Textual Description:
(0) General Description:
GSMC receives Galileo overall status Reports from GMS and send and receive classified information from PKMF.
The sequential description is detailed below:

(1) Generation by SPF of Report to GSMC including Galileo Overall Status, Daily Service Performance Report, Service Performance Predictions, Notice Advisory To Users. And forward of Bulletins to GSMC including Maintenance Bulletin, Forecast Performance Bulletin.
(2) GSMC sends to PKMF classified data.

(3) PKMF receive and process classified data.

(4) PKMF sends to GSMC classified data (Refer to SEC-SFC 15 for further details of the PKMF to GSMC interface).

6.7.4.3.5 SFC7.3-5 Information Exchange with GALSEE

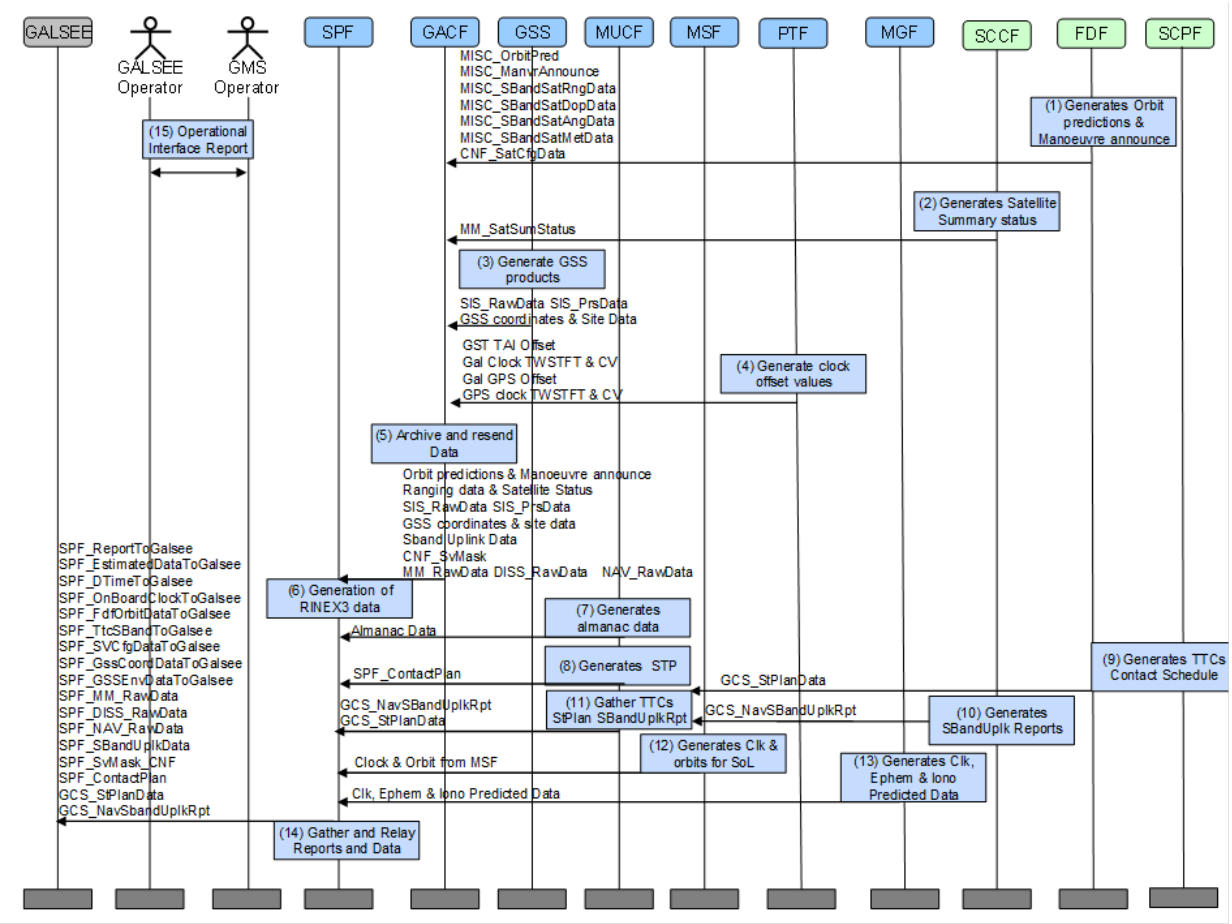


Figure 222: SFC7.3-5: Information Exchange with GALSEE

Scenario Textual Description:
(0) General Description:
GALSEE receives Galileo Reports and Products from GMS.
The sequential description is detailed below:

(1) Generation by FDF of Orbit predictions & Manoeuvre announcements, which are sent to GACF.
(2) SCCF sends to Satellite summary status to GACF.
(3) GSS generate the associated products, SIS Raw Data,SIS_PrsData, GSS coordinates and Site Data to send to GACF and GALSEE via the MDDN ENR RT.

(4) PTF generates the clock offset parameters and send them to GACF.

(5) GACF archive and forward to SPF in a schedule way data produced by (1) (2) (3) (4) ,
GACF archive and forward to SPF S Band Uplink Data from MGF migration,
GACF archive and forward to SPF CNF_SvMask
GACF archive and forward to SPF in a schedule way RT data captured using GACF-DCA (MM_RawData, NAV_RawData, DISS_RawData)

(6) SPF generate based on received data from GSS the RINEX3 data to be provided to GALSEE after (14).
(7) MUCF generates almanac data to be provided to SPF.
(8) MUCF provides SPF with Contact Plan information on a daily basis/each time a replanning is executed.
(9) SCPF sends the TTCS Contact Schedule to MUCF each time a short Term Plan is updated
(10) SCCF generates Navigation S-Band Uplink Report after every S--Band uploink with a satellite

(11) MUCF gathers all the information provided from (9) and (10) and redirect it to SPF
(12) MSF generates estimated clock and orbits and provides them to SPF.
(13) MGF generates Clock, Ephemeris, and Iono predicted data to be provided to SPF.
(14) SPF gathers all the information provided from (5), (6), (7), (8),(11),(12), and (13) to be provided to GALSEE.
(15) An Operational Interface (e-mail exchange between operators) is in place to allow interface and data exchange problem report.

6.8 SFC8 PROVIDE SAR SERVICE

6.8.1 Introduction

The objective of the Support to SAR SFC is to provide support to the COSPAS-SARSAT system with regards to:

- The forwarding of SAR Forward Link Distress messages in support of the MEOSAR Forward Link Service
- The dissemination of SAR Return Link Messages in support of the MEOSAR Return Link Service.

The following section describes the physical implementation of the SFC. It is based on the following documentation:

- System documents:
  - System Logical Architecture (SYS-RD.1)
  - GMS-RLSP ICD (AD.20)
- SSeg documents:
  - Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2)
- GMS documents:
  - GMS DDF1 (GMS-RD.1)
  - GMS DDF2 (GMS-RD.2)
  - MUCF ICD (GMS-RD.11)
  - SPF ICD (GMS-RD.16)
  - MGF ICD (GMS-RD.27)
  - GMS to Ops concept (GMS-RD.41)
- SGS documents:
  - SGS DDJF (SYS-RD.78)
  - SAR Ground Segment SGS Definition file (SYS-RD.82)
- SAR documents:
  - SAR RLSP-GMS ICD (AD.20)

6.8.2 SFC8 High Level Description

The Galileo Core Infrastructure will provide Search and Rescue Services in two ways:

- providing SAR Forward Link Service
- providing SAR Return Link Service

Following an overview of the SAR/Galileo architecture, as based on COSPAS-SARSAT documentation and reflected in the COSSARGALICD.

In order to achieve these two targets Galileo will provide the Search And Rescue services by equipping its satellites with a SAR Payload that relays up to 5 beacon distress signals transmitted from the Forward Link coverage area.
Relayed signals from the different satellites are received by SAR/Galileo Medium-altitude Earth Orbit Local User Terminals (MEOLUTs) located in European hosting sites.

The MEOLUT is in charge of determining the location of the beacon, either by demodulating the beacon message or by processing the times of arrival and Doppler shifts of the received signals. European MEOLUTs can be coordinated by MTCF (MEOLUT Tracking Control Facility) element. It elaborates the tracking plan.
The MEOLUT then sends the estimated beacon position and other relevant data (e.g. Beacon ID, list of satellites used to compute the position) to a Mission Control Centre (MCC), which communicates with the Rescue Coordination Centre (RCC) and the Return Link Service Provider (RLSP). The RCC is in charge of launching a rescue operation.

A Galileo Return Link Service allows for the provision of messages back to suitably equipped beacons (with Galileo receivers). Return Link Messages (RLMs) are generated based on information originating at the RLSP (acknowledgement) and/or the RCCs (acknowledgement and additional information), and are forwarded to the beacon by the Galileo satellites.

The Figure below shows the different components involved in the SAR/Galileo Services:
1. The beacon transmits a distress signal, which is received by several Galileo satellites.
2. The distress signal is amplified and frequency-converted by the satellite transponder (SAR Functions) and down-linked to the MEOLUT(s).
8S. The RLSP requests to SPF (Service Products Facility) to obtain particular data/reports from Galileo related with spacecraft orbital data for all Galileo satellites.
8M. The RLSP generates an acknowledgement. This can be included to a message generated from the COSPAS/SARSAT and the RCCs (RLM). RLMs are sent to the Galileo Ground Mission Segment, to disseminate them to particular satellites within visibility of the distress beacons or to cancel the dissemination of a previously requested RLM.

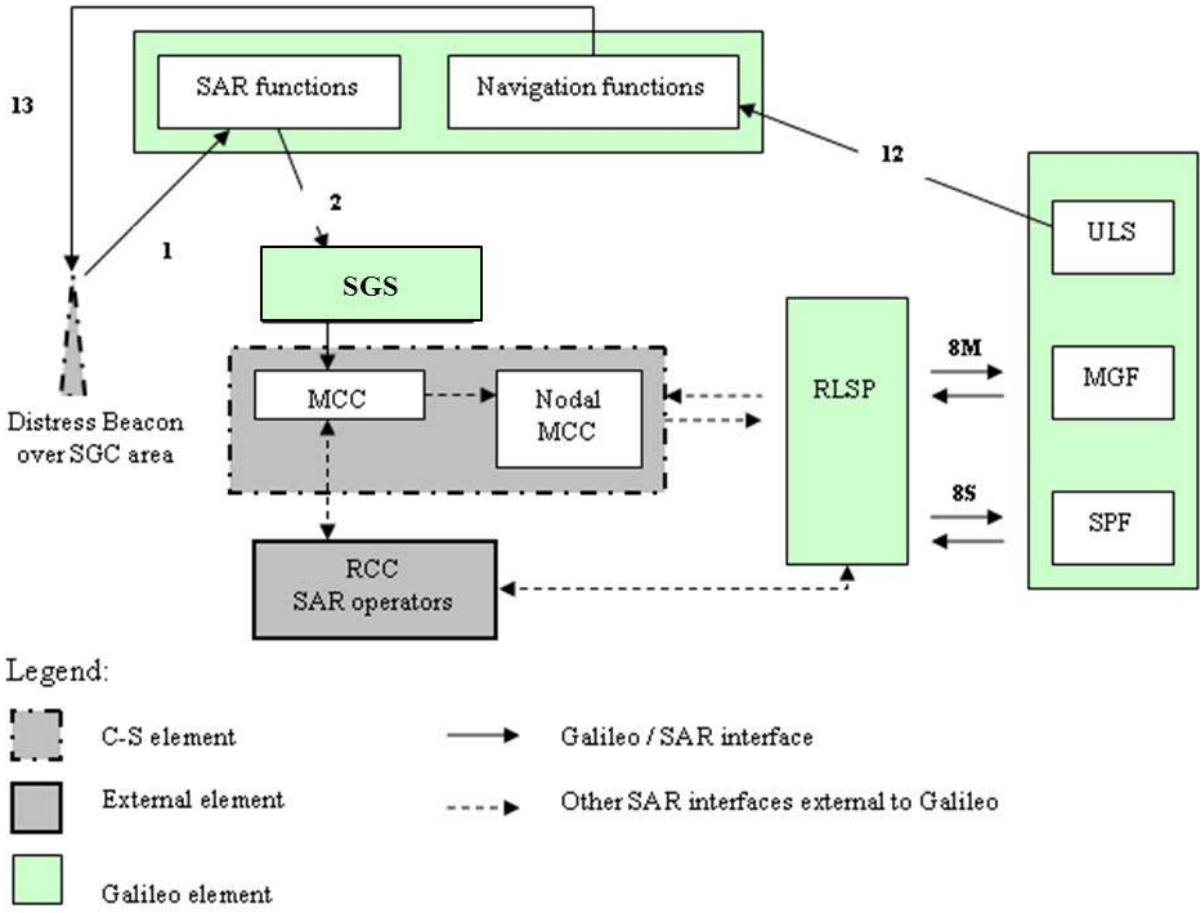


Figure 223 : SAR/Galileo architecture

12. The Galileo Navigation Messages have a number of bits per second reserved for SAR RLM.

13. RLMs are incorporated into the Galileo Navigation Messages and down-linked to the Galileo-equipped beacons.

Additionally, some interfaces exist in order to allow the exchange of status information between the different subsystems involved in SAR/Galileo.

BS. Operational Status Messages are sent by the Galileo Ground Mission Segment (SPF) to the RLSP containing the status of different SAR/Galileo components and the Ephemeris Data of Galileo Satellites.

BM. The Galileo Ground Mission Segment (MGF) sends to the RLSP warning information regarding capacity restrictions to disseminate RLM, or saturation is reached in a particular area or GMS is no longer able to process RLM requests.

The SAR/Galileo Ground Segment shall archive statistical data about the Search and Rescue Process during the entire lifetime of the system.

6.8.3 SFC8.1 Provide SAR Forward Link

6.8.3.1 High Level Description

The purpose of this SFC is to relay the SAR distress message received from the SAR beacon laying on the Forward Link Service area through the SAR Payload on board each one of the Galileo Satellites to the user MEOLUT that process and localize the beacon emissions and send the information to the COSPAS SARSAT operational network.

6.8.3.2 Physical Static Block Diagram

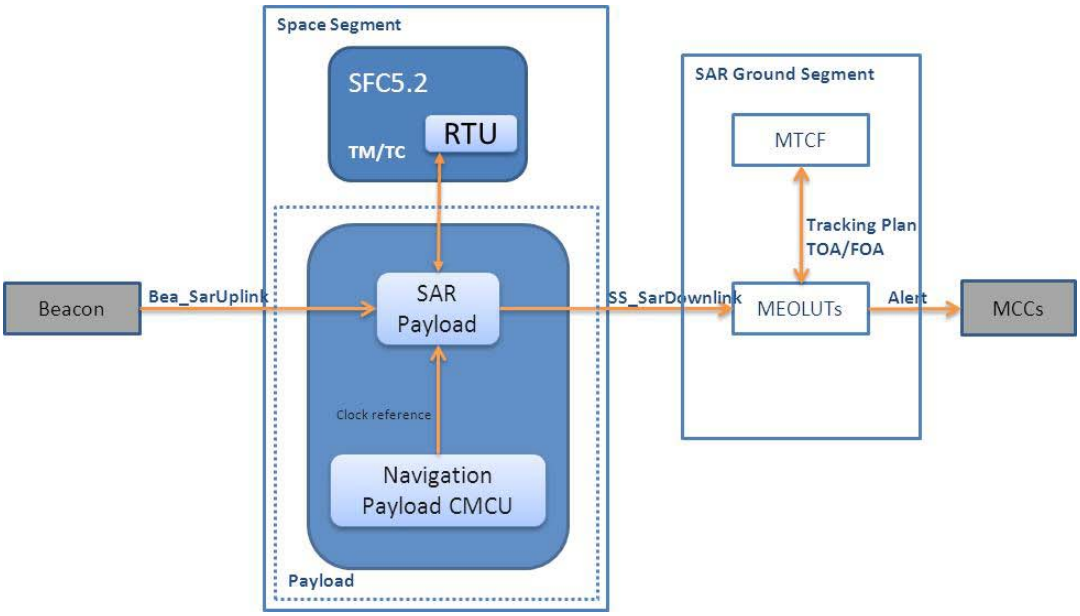


Figure 224: SFC8.1 Physical Static View

6.8.3.3 Overall Physical Description

The SAR payload performs on board the following tasks for the Provision of the SAR Forward Link:

- Receives the signals coming from up to 5 simultaneous SAR Beacons in different frequencies in the 406 MHz band (BEA_SarUplink)
- Receives the satellite payload reference Frequency generated by the CMCU (TIME_OnBoardClock) in order to achieve the required frequency accuracy and stability for the downlink.
- Performs the uninverted frequency translation from the uplink band in 406 MHz to the 1544 MHz downlink band.
- Transmits immediately through the L-Band antenna and without any onboard processing of the data the up converted signals to the whole Earth visible from the satellite towards the MEOLUT (SS_SarDownlink).

The SAR Payload is only monitored by GCS in the form of telemetry. Therefore, in case of SAR payload maintenance or failure the GCS informs GMS which in turn informs RLSP via the GMS-RLSP interface, further described in SCF8.2.

MEOLUTs perform the following tasks for the provision of SAR Forward link Service:

- track the satellites according to their own tracking plan (Stand Alone mode) or to the tracking plan received by the MTCF (Networked Mode)
- perform the localization process through DOA method using the TOA/FOA estimated locally and /or exchanged with other MEOLUTs or exploiting the position data that may be embedded within the beacon distress messages
- exchange TOA/FOA with other MEOLUTs and with the MTCF
- disseminate the Alert message of located or unlocated beacons to the associated MCC (part of Cospas Sarsat operational network)

6.8.3.3.1 SAR Distress Message Forwarding: SAR Payload

The SAR payload operates independently from most of the navigation payload. It receives distress signals in UHF (406.05 MHz) from SAR beacons, and converts this to L-band (1544.1 MHz) on board for forward transmission to the MEOLUTs. To achieve this, the SAR payload consists of two equipments:

- the SAR Transponder (SART)
- the SAR Antenna (SARANT - consisting of both the UHF receive antenna and the L-band transmit antenna).

The SARANT provides the “air interfaces” for the SART both at the receive end and the transmit end, while the SAR Transponder performs the necessary low-noise amplification, frequency up-conversion, RF power amplification and filtering, in order to transform the received UHF distress signal to the L-band transmit signal for the MEOLUTs, at the power level required.
In terms of RF functionality, the only dependency of the SAR payload on other Payload units is for the highly-stable 10.23 MHz frequency reference from the CMCU (Reference Clock Section). The SART uses this frequency source to synthesise LO frequencies for the frequency conversion from UHF to L-band.

6.8.3.3.2 Provision of Frequency Reference Signal: Navigation Payload Clock Monitoring and Clock Unit (CMCU)

The CMCU in the payload shall provide a 10.23 MHz reference signal to the SAR Payload. This signal will be used by the transponder to perform the frequency translation.

6.8.3.4 Dynamic Scenarios

6.8.3.4.1 #SFC8.1-1 Nominal Scenario - SAR Forward Link Service

The Forward link Service has only one nominal scenario represented in the figure 2 below.
In this scenario the SAR Payload will relay the messages received from distress beacons in UHF-Band towards L-Band MEOLUT terminals using a transparent transponder.

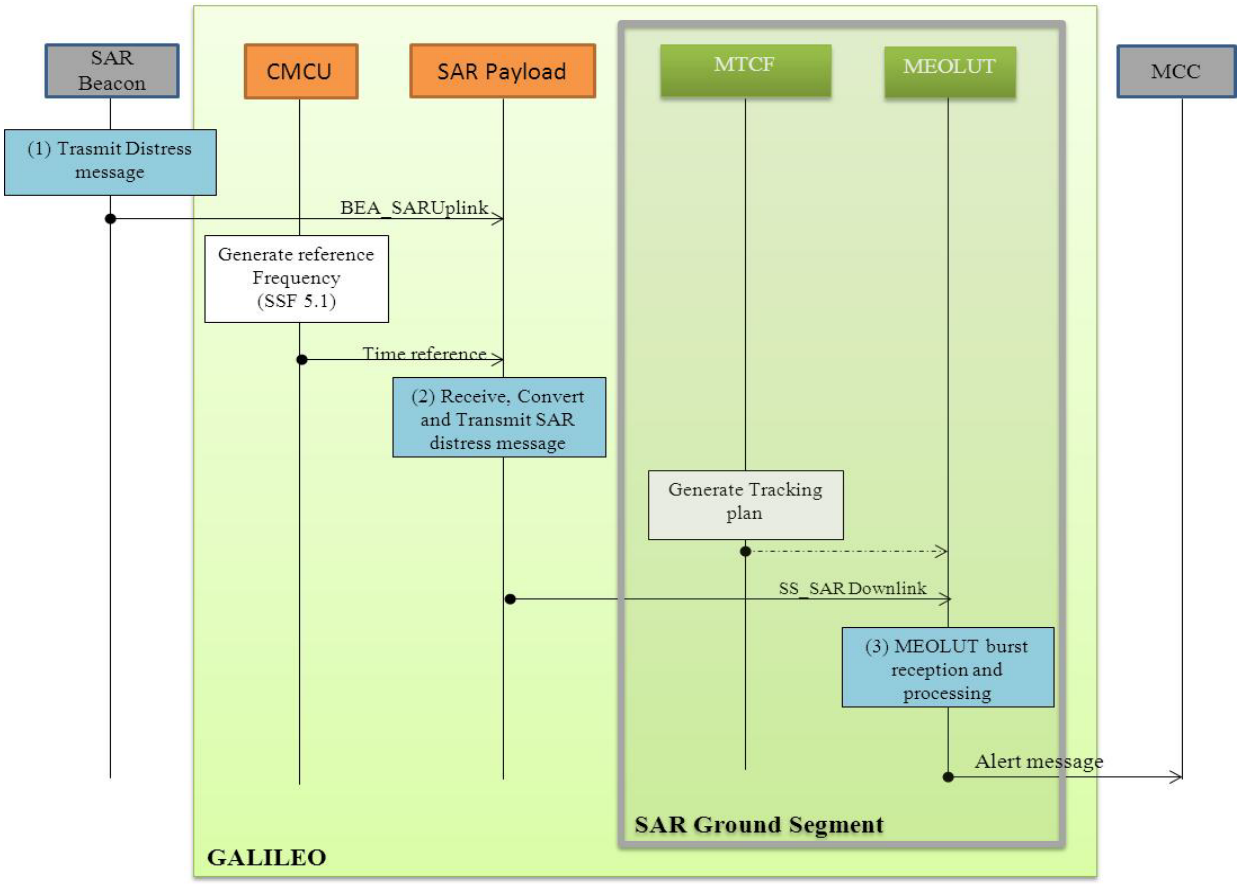


Figure 225: SFC8.1-1 Nominal Scenario - SAR Forward Link service

Scenario Textual Description:

0) General Description: The MEOSAR system allows the forwarding of SAR distress messages via Galileo satellites. This dynamic scenario describes the process carried out by the Galileo core system from the emission of the SAR beacon burst up to the transmission of the alert to the Cospas Sarsat Operational Network

The CMCU, generates the onboard frequency and sends the on board frequency to the SAR payload to ensure the SAR payload can achieve a stable frequency conversion of the distress message. It is shown in the above figure ("white box" indicating an action performed independently from the "SFC 8.1.1" scenario).

The sequential description is detailed below:

(1) The SAR Beacon is activated automatically or by a user in distress and transmits a signal to the MEOSAR satellites.

(2) The SAR payload receives the SAR distress message at 406MHz via the SAR antenna; frequency converts the message, without inverting or any processing and transmits in the SAR downlink frequency of 1544MHz.

(3) MEOSAR satellites will be tracked according to a tracking Plan generated by the MEOLUT (Local) or by the MTCF (Coordinated). The MEOLUT ground station in visibility to the satellite receives the distress message along with the same distress message in view of the beacon and MEOLUT and processing the received signals to calculate the location of the distress and to inform the rescue services to allow a rescue to take place.

Contingency Scenario:

In case the SAR Payload does not transmit the SAR Distress message, a level of redundancy is ensured in the Galileo constellation as other Galileo satellites in view of the SAR Beacon will also transmit the same SAR Distress message. This will ensure that the SAR Forward link is not lost due to a single loss of SAR Payload function.

6.8.3.4.2 #SFC8.1-2 Contingency Scenario - SAR Payload Failure/Anomaly Reporting

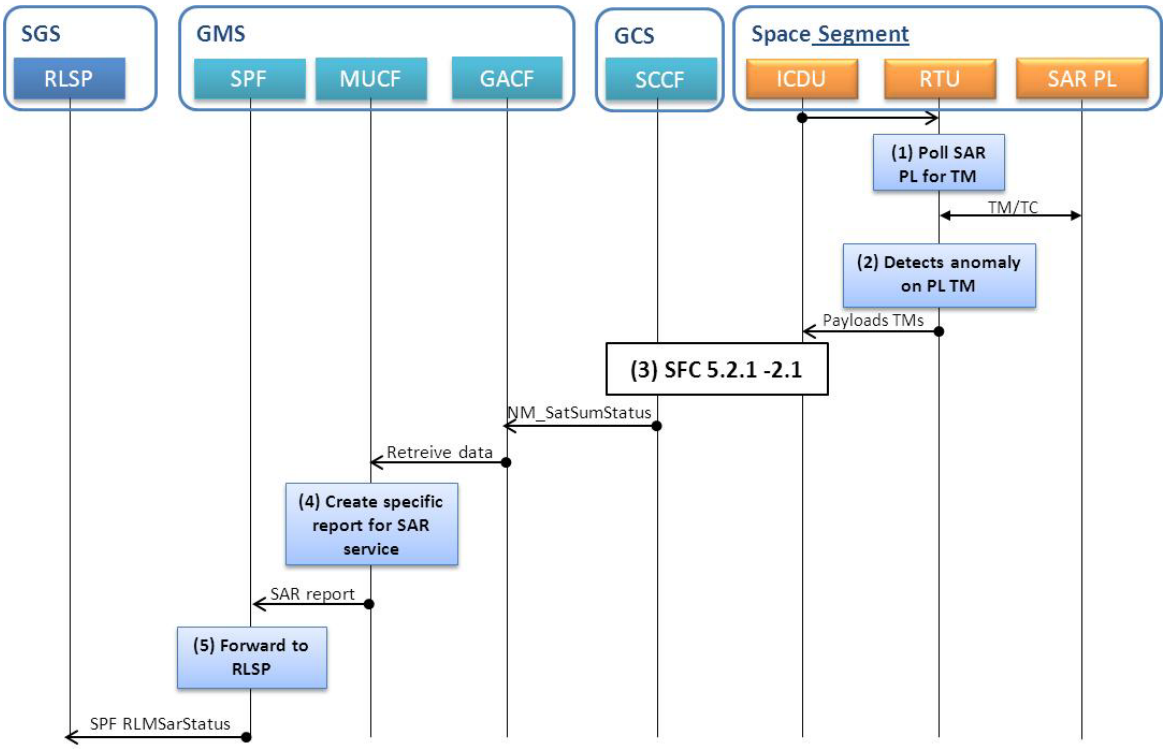


Figure 226: SFC8.1-2 SAR Payload Failure/Anomaly Reporting

Scenario Textual Description:
0) General Description: This contingency scenario shows the process that takes place to inform the RLSP there is a failure or anomaly on a SAR Payload.

Note: Limited visibility on the actual SAR status is possible because:
- SAR PL status is monitored only during S-Band contact via TMs (e.g. about 1 hour every 14 nominally).
- S-band TM cannot provide full evidence of the Transponder function.

Nevertheless it is foreseen that the SAR Ground Segment can be used in the frame of ad hoc investigations in case of anomaly on specific payload

The sequential description is detailed below:

1. The ICDU (Integrated Control and Data handling Unit) commands the RTU (Remote Terminal Unit) to collect telemetry from the Payload Equipments including the SAR payload. The SAR payload returns no telemetry or telemetry stating an anomaly.

2. The RTU forwards to the ICDU the telemetry received from the payload units

3. The process involved between the ICDU and the SCCF (Spacecraft and Constellation Control Facility) is described in scenario SCF5.2.1-2.1 Satellite failure/anomaly reported via S-Band or L-Band. [Nb The SAR status is only delivered via S-Band telemetry and not via the satellite health status in the L-Band, therefore the ground station only gets the SAR failure information upon S-Band contact (every 14 hours). If the SAR status would have been included in the satellite health status this delay would be reduced dramatically.]

3b. OPS perform a failure analysis, determine the necessary recovery actions and put them in place at GCS

4. When the GACF (Ground Asset Control Facility) has received the satellite status from the SCCF it forwards it on to MUCF (Mission and Uplink Control Facility).

Handling of SAR S-Band TM including SAR status between GCS-SCCF and GMS-GACF: OPS process.

5. The MUCF creates a specific report for the RLSP including information on the GMS/SAR status, and the information regarding the SAR Payload failure/anomaly.

OPS interface between GMS and RLSP for SAR status reporting GMS/RLSP OPS interface to be performed in cooperation with COSPAS-SARSAT OPS Team.

6. The SPF forwards the MM_SarSystemStatus message inside the SPF_RlmSarStatus signal to the RLSP including the SAR payload status and informing RLSP of the failure/anomaly.

6.8.4 SFC8.2 Provide SAR Return Link

6.8.4.1 SFC8.2.3 SAR Return Link Interface to Cospas/Sarsat

6.8.4.1.1 SFC8.2.3 High Level Description

This SFC is responsible for the interface with the Cospas/Sarsat in particular for the reception of Return Link Message Request and cancellation and the transmission of Satellite Ephemeris and SAR/Galileo and GALILEO status reports.

6.8.4.1.2 Physical Static Block Diagram

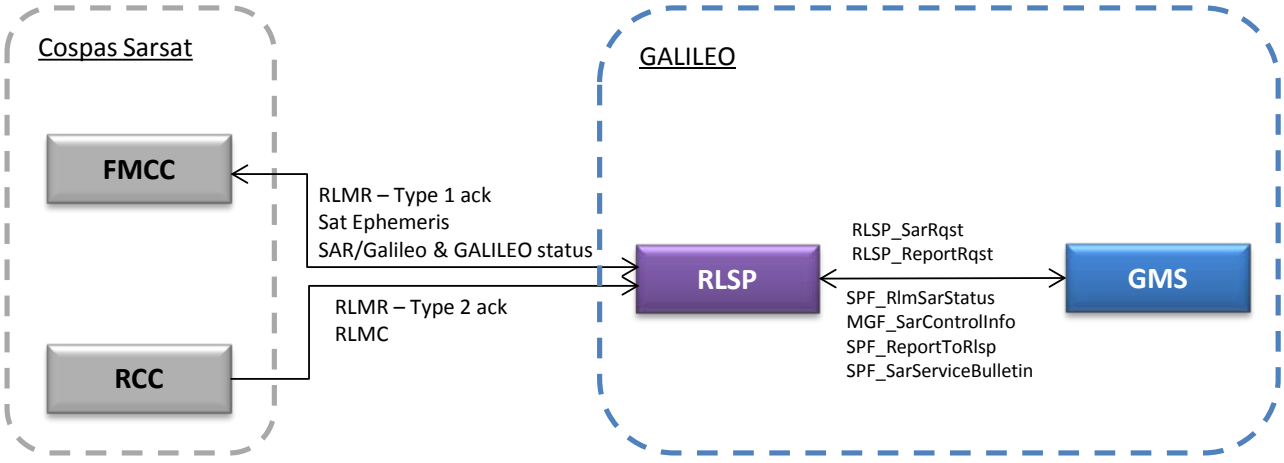


Figure 227: SFC8.2.3 Physical Static View

6.8.4.2 SFC8.2.1 SAR Return Link Resource Management

6.8.4.2.1 High Level Description

This SFC is responsible for the transmission and reception of various reports and bulletins regarding the status of the Galileo system, also the ephemeris data of the Galileo constellation is transmitted to the RLSP.

6.8.4.2.2 Physical Static Block Diagram

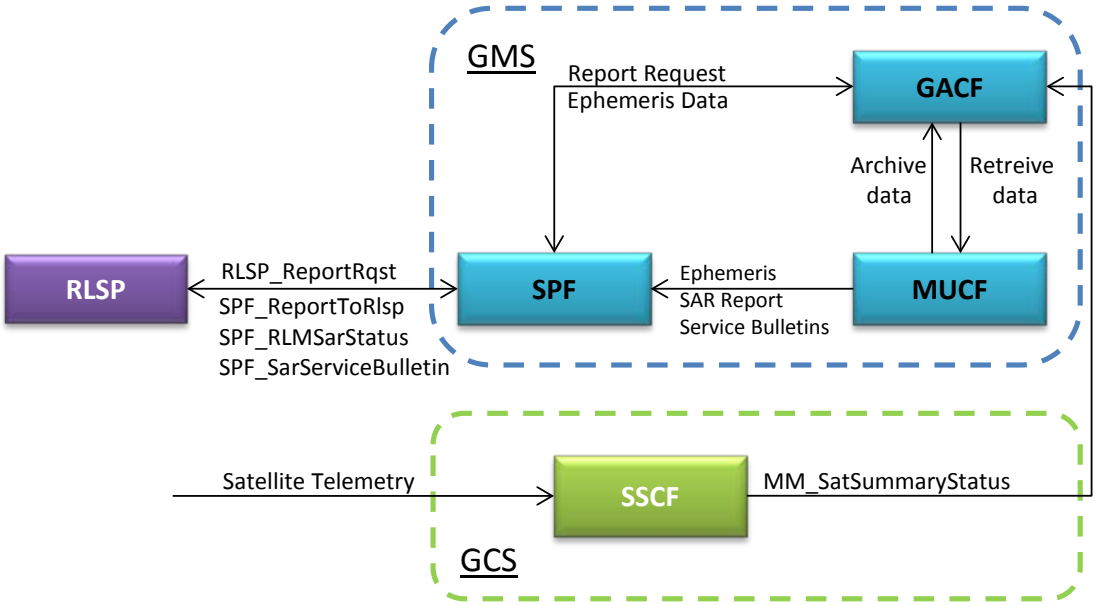


Figure 228: SFC8.2.1 Physical Static View

6.8.4.2.3 Overall Physical Description

The purpose of SFC SAR Return Link Resource Management is that RLSP receives periodic information from GMS about:

- Ephemeris parameters of Galileo Satellites, parameters for the GST UTC conversion every 10 min (NAV_EphemToRLSP) This allows the SAR service to define satellite allocations for RLM dissemination. This signal is created by the MUCF element in charge of supervising on-line the Galileo Mission.
- Information regarding the status of different SAR/Galileo components (MM_SARSystemStatus) every 15 minutes from the MUCF element in charge of monitoring the dissemination services, containing:
    - Current satellite status This will include the Spacecraft Id, the status of the SAR Forward Link payload, Centre Frequency, Polarisation, the gain control mode, the Frequency Inversion, Return Link Status and the bandwidth mode. Although this is not directly related to the return link service, the GMS-RLSP interface is the only means to give this information to a Cospas-Sarsat element.

- Current GMS SAR status (awaiting requests, unable to process requests, degraded modes, estimated RLM transmission delay based on average transmission delay of each new RLM processed within 15 min.)

GMS (MUCF) will receive information from GCS (SCCF) on the satellite status (MM_SatSummaryStatus) to forward to RLSP as part of SPF_RlmSarStatus.

SAR Service bulletin (PLN_SarServiceBulletin updated upon Mid Term Plan validation , once a week on average) is created by the MUCF element that manages the overall mission including:

- planned satellite Forward Link outages,
- planned satellite Return Link outages,
- GMS SAR Service Mode Status, and
- planned RLSP-GMS changes

This SFC support as well the report requests from RLSP to GMS asking for the ephemeris and GST-UTC offset data listed above when necessary. The request contains the report identification ID. (RLSP_ReportRqst).

6.8.4.2.3.1 Transmission of Report to RLSP: SPF

The Service Products Facility (SPF) sends to the RLSP periodic or upon request reports (every 15 minutes) about the status of SAR/Galileo and ephemeris data retrieved from the MUCF and the GST-UTC offset data.

6.8.4.2.3.2 Creation of RLSP Specific Report: MUCF

The Mission and Uplink Control Facility (MUCF) creates and sends the Ephemeris and SAR System Status reports and SAR Service Bulletins to the SPF for further distribution to RLSP. MUCF receives from GCS the Satellite Payload Status information as input to the SAR System Status reports.

6.8.4.2.3.3 SFC8.2.1 Related Constraints: SCCF

The SCCF receives the telemetry signal from Satellites and generates a Satellite status report to be sent to the MUCF as part of the GCS-GMS interface data transfer.

6.8.4.2.4 Dynamic Scenarios

There are 2 possible scenarios for sending non real time data to RLSP these are:

- Scenario 1: periodic delivery to the RLSP of updated status information regarding the status of different SAR/Galileo components (each 15 minutes) and Ephemeris data of Galileo satellites (after every update of each satellite navigation message every 100 minutes)

- Scenario 2: (#SFC8.2.1-1): Delivery of a report requested by RLSP at a non prescheduled instant (RLSP_ReportRqst). If the report is older than 7 days, the report will be retrieved from the central archive in the GACF.

A further operational scenario is also listed below to describe the start up of Galileo support to SAR Return link.

6.8.4.2.4.1 #SFC8.2.1-1: Delivery of a report requested by RLSP

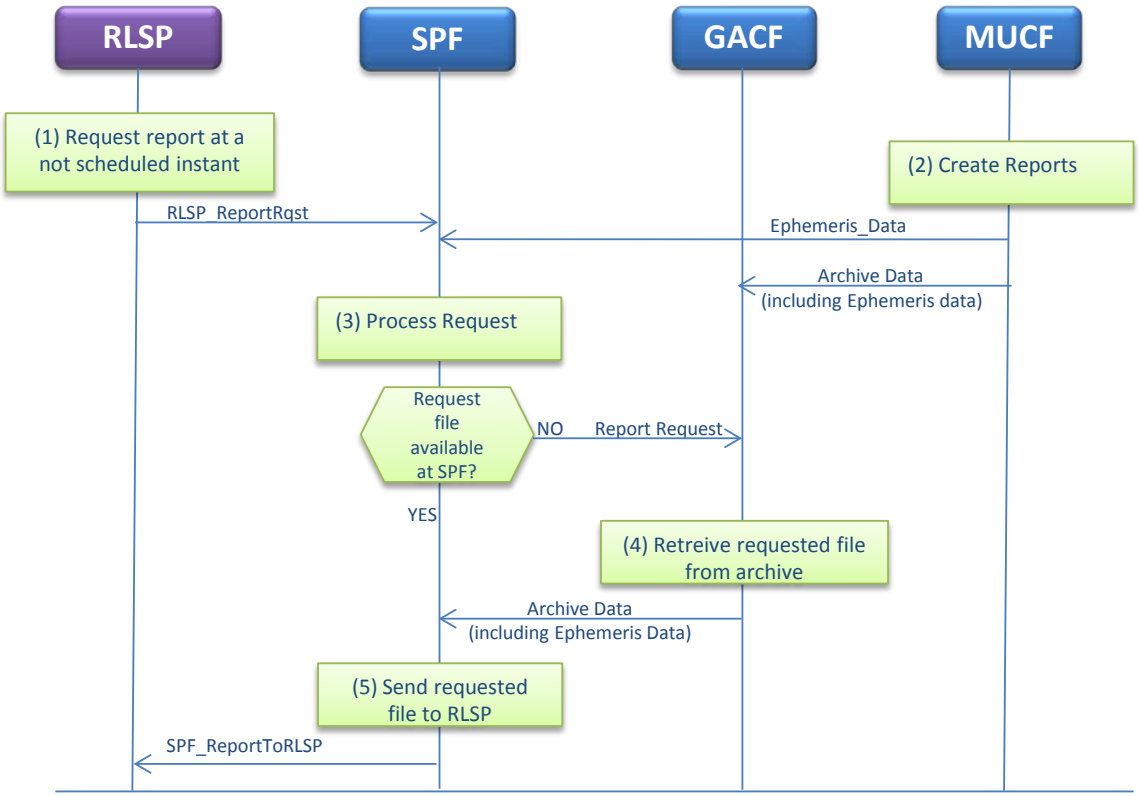


Figure 229 SFC8.2.1-1 Delivery of a report requested by RLSP

Scenario Textual Description:
0) General Description: Via the web interface, a user may download certain files or request an FTP delivery of a certain set of files. These files are historical versions of the products that the SPF provides during normal operation to the external entities under GMS initiative. The SPF offers this functionality to cover the cases where the external entity was not able to receive the files from the regular process or the cases where files on the external entity side have been lost (failure case, etc.)
Via the web interface, each of the external entities will be able to obtain files relating to certain products that represent a certain subset of the set of products that that particular entity receives from the GMS.
This scenario describes the process that take place to retrieve the requested file and send to the RLSP.
The sequential description is detailed below:

1. The RLSP operator fills an INF_ReportInfoRqst online via the SPF web server to request the file outside of the scheduled delivery. The RLSP can browse the SPF web server for files less than 7 days old (currently on the SPF), and those available in the archive. Note that a list of individual files available on the GACF archive is not provided as such a list would be unreasonably large and difficult to browse. The file available to the RLSP is NAV_EphemToRLSP
2. The MUCF is the original creator of the standard report and sends to SPF and GACF (for archiving purposes) at scheduled instances.
3. The SPF then processes the request to determine which files are to be retrieved and where from (either the SPF or GACF)

4. If the file is not stored at the SPF, the GACF is requested to retrieve the file from the archive.
5. Once the file is retrieved, the SPF makes an FTP put of the requested file/s to the address specified by the RLSP. The request is then considered closed with no further action.

6.8.4.2.4.2 #SFC8.2.1-2: Start up of GMS Support to SAR

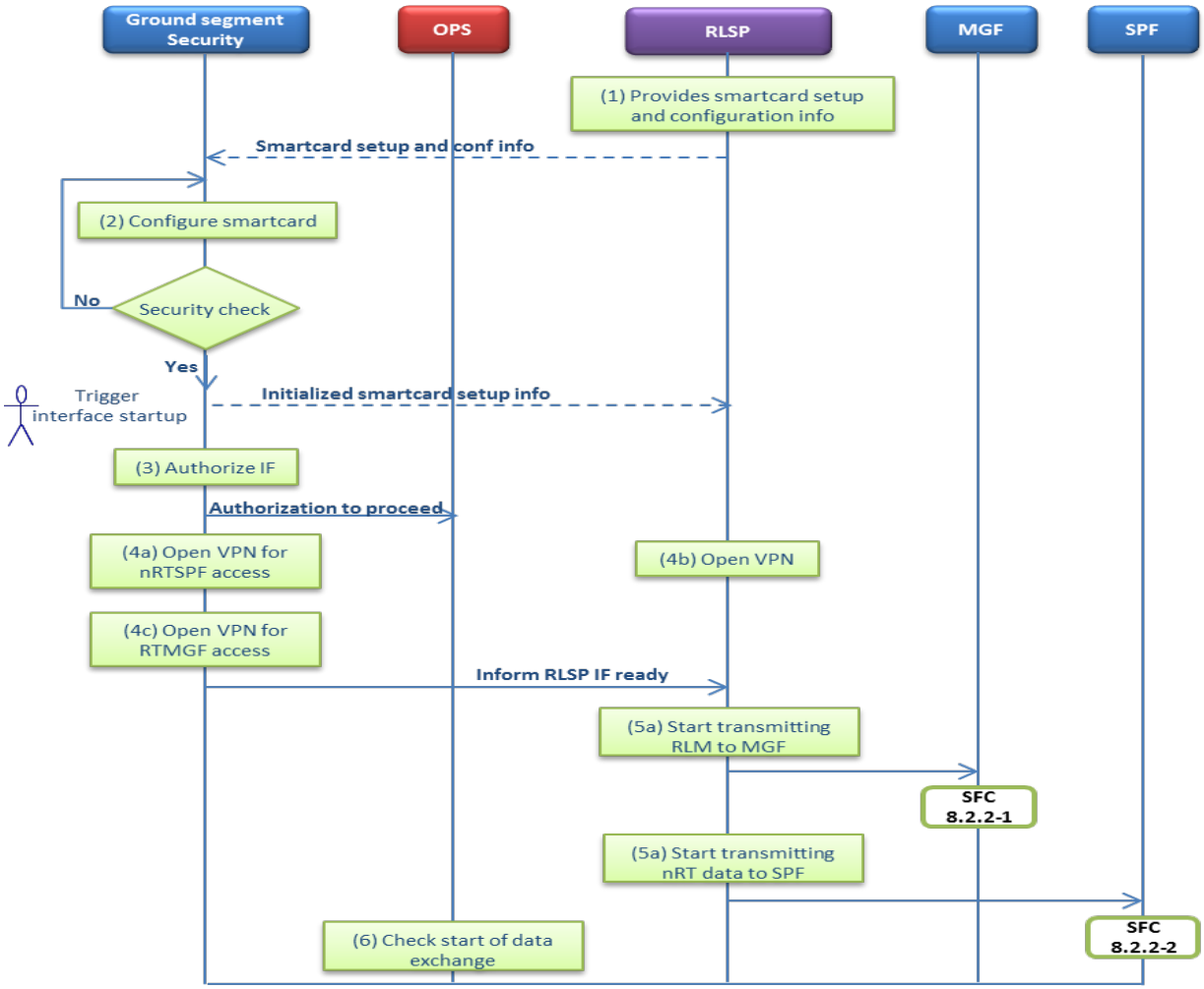


Figure 230: SFC8.2.1-2: Start up of GMS Support to SAR

Scenario Textual Description:
(0) General Description: This scenario describes the start up procedure for the support to SAR Return link.
The sequential description is detailed below:

1) RLSP provides smartcard, setup details and configuration information to the Galileo security team offline.
2) The Galileo security team then configure this smartcard ready for RLSP use. This is checked by security and sent to the RLSP offline.
3) The Galileo management trigger the request to start up the interface with the particular region and security authorise this interface for start up and informs Galileo operations.

4) The Galileo security team open the VPN from GNMF-s for non-real time access to the SPF and for real time access to the MGF. RLSP will also configure and open their VPN.
5) The RLSP then follows the steps in #SFC8.2.2-1 for RLM dissemination and the steps in #SFC8.2.1-1 for RLM resource management.
6) The Galileo Operations team then check the start of the data exchange

6.8.4.3 SFC8.2.2 SAR Return Link Dissemination

6.8.4.3.1 SFC8.2.2 SFC High Level Description

In support of the SAR Return Link Service, SAR/Galileo system will relay Return Link Messages from the COSPAS/SARSAT system, by means of a single Return Link Service Provider (RLSP), to future emergency beacons and devices equipped with a Galileo receiver using the navigation signals of the Galileo satellites

6.8.4.3.2 Physical Static Block Diagram

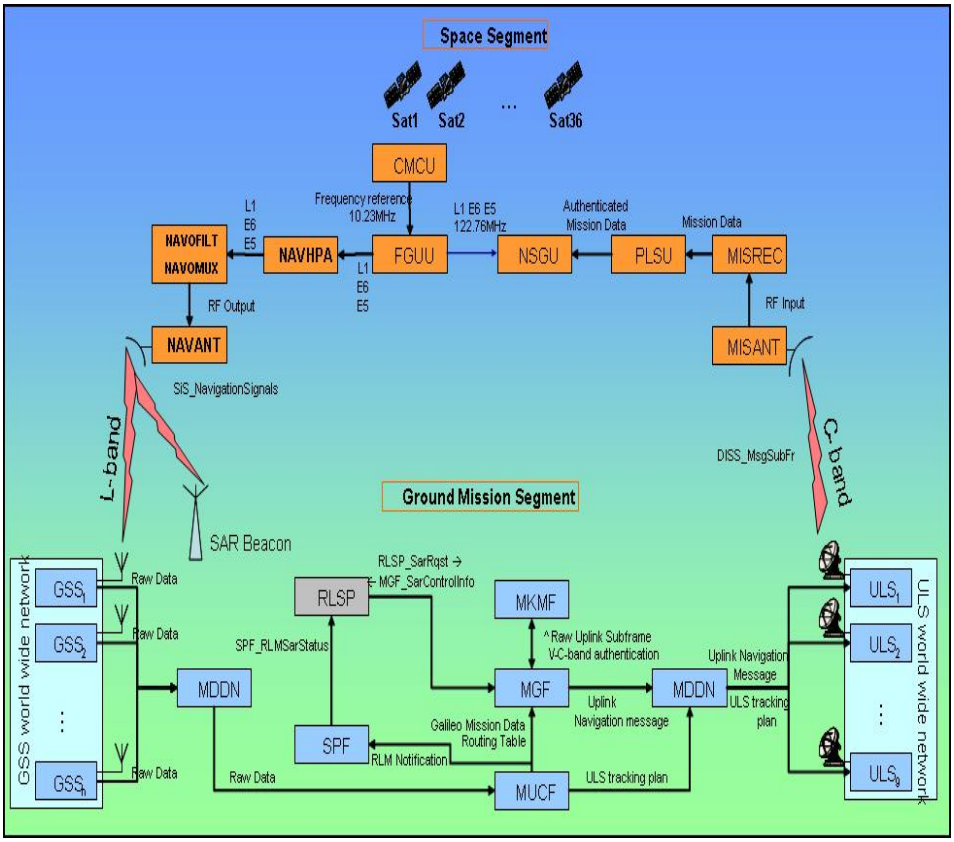


Figure 231: SFC8.2.2 Physical Static View

6.8.4.3.3 Overall Physical Description

6.8.4.3.3.1 RLM Dissemination

The RLM dissemination functional chain involves the following steps:

- RLSP sends to the MGF (GMS) a RLM dissemination request (RLSP_SarRqst), including one RLM and supporting information, twice with a delay of less than 1 minute between the two.
- The GMS (MGF) is able to receive up to 30 SAR RLM (short or long) requests every minute (RLSP_SarRqst).
- Within the 30 RLM requests per minute the GMS shall always prioritize new RLMs over repetitions of previous RLMs.
- The MGF selects the 2 satellites to be used for the dissemination of the RLM by using the list of satellites received from RLSP
- Whenever a request from the RLSP cannot be fulfilled by the GMS, the GMS will automatically inform the RLSP of this situation.
- The GMS performs the first transmission to the selected satellite 5 minutes after the reception of the RLM, GMS will then repeat the RLM transmission depending on the repetition strategy of that RLM (between 0 and 5 repetitions and 7 to 15 minutes between repetitions) or until it receives a cancellation notification.

(Default number of repetitions is currently defined as 5 To allow a different number of repetitions to be used this information would need to be included in the SAR_ReturnLinkAckNotif signal.)

- Each satellite has the capacity to disseminate 600bits/minute of SAR RLM data.
- The SAR RLM information is sent as part of the Galileo authenticated message sub-frame to the selected up-link stations of Galileo, (GMS_Uplink) and, from there to the selected satellites for broadcast to the end users as part of the L1-B signal.(SiS_NavigationSignals)
- For the case when a SAR Return Link cancellation request is received, the GMS shall stop the transmission of the RLM requested automatically assuming the RLM has been repeated less than the required number of times. The message ID is contained in the cancellation request.
- Each RLM cancellation request will be repeated once by the RLSP with no modification in the content of the message.
- Up to 30 SAR RLM cancellation requests every minute shall be processed by the GMS.

6.8.4.3.3.2 RLM Monitoring Reports

The Galileo system monitors the RLM dissemination in two ways:

☐ Uplink processing and saturation monitoring when the GMS fails to uplink the RLM within 5minutes of reception due to saturation of the SAR field in the uplink of the satellites in the satellite list sent by RLSP

☐ Monitoring of successful RLM transmission and reception by at least one GSS.

GMS(MGF) informs the RLSP (MGF_SarControlInfo) when triggered by the following events:

- Satellite list currently saturated
- Satellite list approaching saturation
- Satellite list saturation after RLM processing

Additionally every 5 minutes the GMS shall transmit monitoring data notifying the RLSP via SPF_RlmSarStatus the following information:

- For each RLM
  - List of satellites used of the dissemination of the RLM
  - Successful dissemination of the RLM, including reason for failure

- For each satellite in the Galileo constellation
  - Percentage of free SAR fields
  - Connection status

6.8.4.3.3.3 RLM Reception, Processing and Sub-frame Creation: MGF

As part of this SFC the MGF is responsible for:
- Receiving RLM requests and cancellations from RLSP
- Selecting 2 satellites for SAR RLM dissemination according to the list of satellites received from RLSP in the RLSP request
- RLM flow control determining and saturation management of the Galileo satellites
- Creating SAR packets to be included in the uplink message subframe per satellite.
- Transmission of authenticated uplink subframes to the ULS for uplink to Galileo satellites

6.8.4.3.3.3.1 RLM Reception: MGF

The MGF receives up to 30 RLM requests or cancellations per minute directly from the RLSP. The SAR RLSP is the single ground interface to the Galileo system.

The RLM to be sent to a Galileo equipped beacon request contains:
- message ID,
- message type,
- RLM content
- Support information with the estimated location of the beacon (no longer used by Galileo)
- A satellite list for dissemination including all S/C in view for the beacon and ordered by dissemination preference

6.8.4.3.3.3.2 Satellite Selection for RLMs: MGF

The MGF is in charge of determining the appropriate satellites to disseminate the RLM based on the list of satellites provided by the RLSP. All RLMs and corresponding repetitions must be disseminated through 2 satellites. The only exception to this rule is that if a new RLM arrives at GMS and there is only one satellite available within the next 5 minutes. In this case the GMS shall not reject the new RLM and shall manage the dissemination accordingly.

To select these satellites the MGF considers the list of preferred satellites provided by RLSP. The satellites shall be ordered in this list in the preferred broadcast order determined by the SAR MEOLUT.

To determine 2 ‘available’ satellites to use for the dissemination of the RLM MGF must determine the following by using the ULS Antenna to Satellite Allocation Plan
- A preferred satellite is connected to a Galileo ULS, this is carried out by consulting the satellite allocation plan from MUCF.
- The satellite is not to be used to transmit an integrity alert

The satellite has available free SAR fields in the uplink.

6.8.4.3.3.3.3 RLM Flow Control and Saturation Determination: MGF

Saturation management at GMS level is based on new RLM only and shall not reject the new RLM and shall manage the dissemination accordingly

The RLM multiplexing delay in the MGF will be estimated from the occupancy of the SAR message buffers, and the presence or not of integrity alarms messages (SAR messages parts are sent out at a maximum of 1Hz in the message sub-frames, but only when no alarms are being transmitted, therefore the presence of integrity alarms can add to the RLM multiplexing delay).

The monitoring of the RLM multiplexing delay will be performed for each satellite, and it will monitor the trends so as to detect when the multiplexing process is approaching saturation for any given “route”. In this case, as well as including the information in the mission monitoring data, it will also generate a “Warning” event in the Alert Log and will send a message to the RLSP.

The message elaboration and packet generation process detects when the message buffers are already full, and cannot store another message. That situation (route saturation, unable to process requests) is recorded in the Mission Monitoring data, and will also lead to a “Warning” event being generated in the Alert Log and to a warning message sent to the RLSP.

The status of the dissemination of SAR messages will also be included in the Mission Monitoring data provided to Monitoring & Control. This information will be sufficient to allow the MUCF to verify that the MGF is routing (or cancelling) the SAR messages correctly (as intended by the RLSP)
MGF_SarCtrlInfo_MM message is sent to RLSP at MGF initiative, whenever there are Return Link restrictions for the S/C list addressed by a new RLM request.

- *Selected S/C list currently saturated:*
  - MGF is not able to process the new RLM request (the RLM message cannot be sent
  - to at least one S/C in the list provided by RLSP within the next 5 min, because there
  - are not enough free slots in the satellites)
- *Selected S/C list saturated after RLM processing:*
  - Saturation reached for all the S/C in the S/C list (after scheduling of the new RLM
  - request received).This means that there is no capacity for neither long nor short new
  - RLMs in the next 5 min.
- *Selected S/C list approaching saturation:*
- Approaching saturation will be declared by the GMS when (after scheduling of the new RLM request received) all connected satellites in the S/C list have an used capacity of more than 80 % for the next 5 minutes. The percentage shall be calculated based on the slots used for transmitting new RLMs over the total slots available in the next 5 min

6.8.4.3.3.3.4 SAR Packet Generation: MGF

To route the messages, the MGF searches in the list of preferred satellites for free SAR slots in a pair of satellites, if it finds free slots the data is scheduled for be inserted in those slots. If there are not free slots to insert the SAR data, the MGF sends an alert on the mission monitoring info to RLSP.

MGF carries out the preparation of the Return Link Message packets to be sent for up-link at 1 Hz in the message sub-frames, using as source the RLM requests received every minute from the RLSP.

The MGF shall ensure the SAR messages are transmitted up to the following 5 minutes after the reception of the RLM request. Then it repeats this RLM up to 5 times with intervals of 7 to 15 min for repetition. These parameters (repetitions and intervals) are configurable through configuration file, which could be downloaded in all modes, but the contained configuration can be only uploaded in Loaded Mode.

6.8.4.3.3.3.5 Transmission to ULS: MGF

When the SAR packets are generated and included in the correct satellite uplink the MGF sends the complete authenticated sub-frames to the ULS for transmission to the Galileo satellites.

6.8.4.3.3.4 ULS Tracking Plan & RLM Monitoring: MUCF

6.8.4.3.3.4.1 Uplink Scheduling of RLM: MUCF

An element of the MUCF, is in charge of providing off-line the uplinks ULS-antenna / Satellite schedules taking into account the satellite visibility and all other constraints, such as the robustness to single SV/Antenna/ULS failure, availability of resources, etc. Uplink schedule contains mainly the ULS-Satellite contact plan, but also ULS tracking plans, and SAR, CS dissemination grids.

6.8.4.3.3.4.2 RLM Monitoring: MUCF

The MUCF is responsible for the monitoring of the RLM dissemination and the creation of a report to RLSP to notify them of the RLM dissemination status and Galileo capacity availability for SAR data.
At scheduled time (periodicity 5 min), MUCF builds a Return Link Service Status message (INF_RlmNotifMnt) containing the monitoring information regarding RLMdissemination/cancellation status for each currently active RLM.

- Number of active RLMs

More specifically, the RLSSM shall contain for each currently active RLM:

- RLM ID
- Confirmation that the RLM has been correctly scheduled for uplink,
- Confirmation that the RLM has been correctly disseminated (received by at least one GSS in less than 5 minutes),
- Confirmation that the dissemination of the RLM has been stopped as a result of a RLMCR,
- Additional information: number of satellites used for dissemination, satellite IDs, failure reasons.

6.8.4.3.3.5 Transmission of Monitoring Information to RLSP: SPF

The SPF element is in charge of forwarding files from the GMS to the External Entities and vice versa. It does not perform any scheduling of the delivery, but only forwards files that gets as

input to other entities based on certain configured rules. This includes the forwarding of the SAR report created by MUCF on the monitoring information regarding RLM status for each active RLM sent every 5 minutes to RLSP.

6.8.4.3.3.6 Uplink of Navigation Message: ULS

The ULS is responsible for the uplink of the multiplexed message to the selected satellites for RLM dissemination.

6.8.4.3.3.7 SFC8.2.2 Related Constraint: GSS

The GSS receives the Galileo downlink signal, including any present RLMs and delivers the raw data to the Galileo Control centre for processing.

6.8.4.3.3.8 Processing of SAR RLM Data On-Board: Navigation Payload

The Navigation payload receives through the mission receiver the Galileo uplink and creates the SiS Navigation Signal (including the SAR RLM (600bits/minute)) for broadcast to the user (receiver+beacon).

The payload includes the SAR RLM data in the downlink directly upon receiving it. It does not store the SAR RLM data or process it in any way. The payload places the SAR RLM data received on the I/NAV nominal page in the allocated SAR RLM field in the L1-B downlink signal.

The RLM is uplinked in 20bits/2 second and therefore for a long RLM of 160bits the whole RLM is disseminated by the satellite in 16 seconds.

6.8.4.3.4 Dynamic Scenarios

Two scenarios are envisaged for this SFC.

- the first corresponding to the RLM request nominal processing including the determination of saturation.
- the second one corresponding to scenario in which a RLM cancellation request is sent from RLSP.

The figures below show these two scenarios:

6.8.4.3.4.1 #SFC8.2.2-1 RLM Dissemination

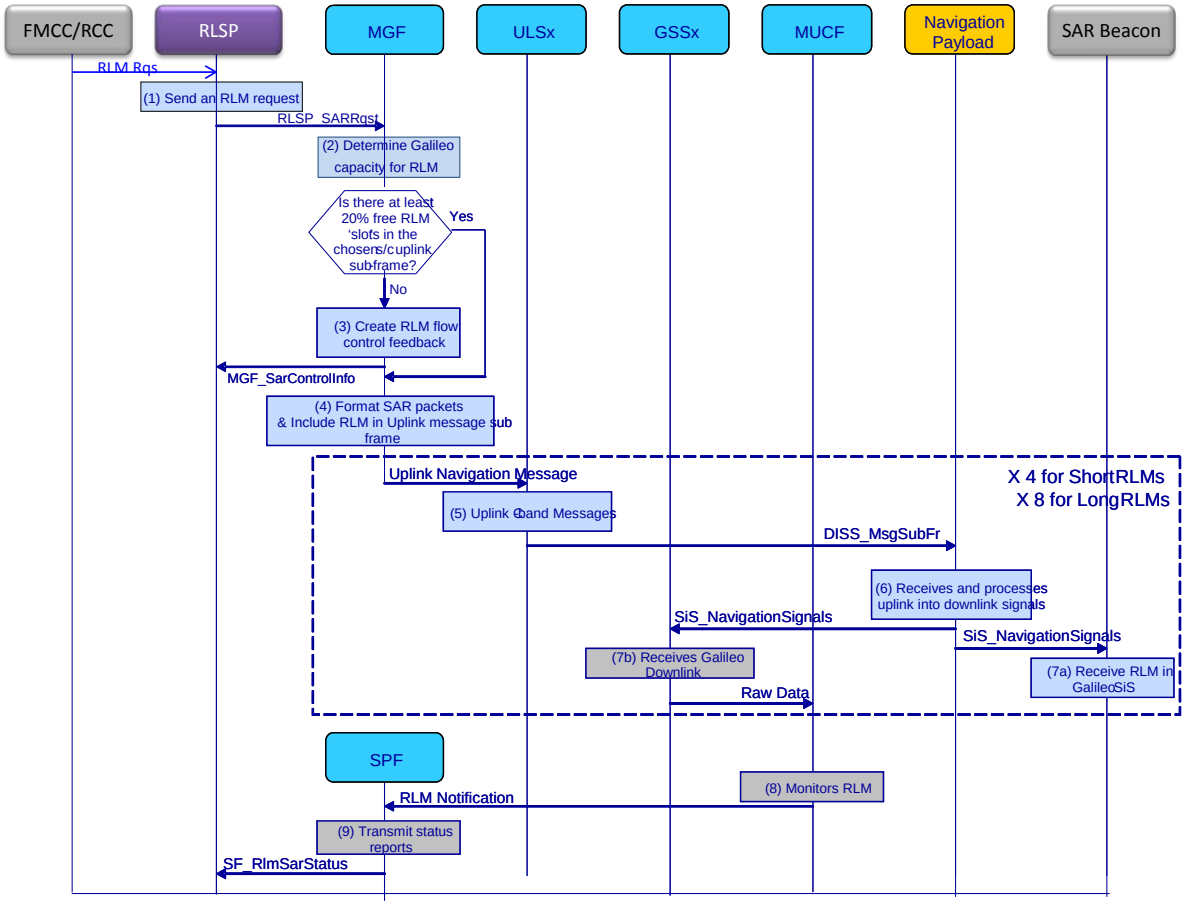


Figure 232: SFC8.2.2-1 RLM Dissemination

Scenario Textual Description:
0) General Description: When a distress is received by the Cospas-Sarsat ground system and the rescue operation is initiated a return link message can be sent to the SAR distress beacon in order to inform the person in distress that their distress is being acted on. The RLSP sends a Return Link Message (RLM) to the Galileo ground segment to disseminate to the beacon user. This dissemination is carried out with the process below.
The sequential description is detailed below:
(1) The RLSP sends the RLM as part of a request message to the MGF. This RLM request includes the RLM itself and a list of satellites the RLSP would like this message sent to. This list is determined by the MEOLUT and is based on the satellites used of the forward link dissemination of the corresponding distress message and the geometry of the satellites.
(2) The MGF checks each of the satellites listed in the RLM request to ensure that the satellite has a free SAR RLM space in the uplink subframe.

(3) If there is less than 20% free space in a satellite listed it checks each one in the list for spare capacity. If MGF cannot find at least 2 satellites in the list provided which have free SAR RLM space the MGF will flag saturation. The MGF takes the information regarding saturation status and creates a feedback message to RLSP to inform them of any possible saturation issues that they should take into account in their next choice of satellite lists and to be informed if the RLM did not get disseminated. This feedback notice can either be 1. satellite approaching saturation, 2. satellites reached saturation.

(4) If there is at least 20 % free SAR field slots in the satellite the MGF then formats the SAR packets into 20bit sections and includes them in the I/NAV uplink message sub-frame for the satellite. The MGF includes the RLM and 5 repetitions of the RLM in the uplink planning. As part of this step the MGF sends the uplink sub-frames to the MKMF to authenticate the sub-frame and add a security field to each sub-frame to allow transmission to the ULS. To transmit a whole RLM 4 message sub-frames are needed for short RLM and 8 for Long RLMs.
(5) The ULS then carries out the uplink of the Galileo uplink sub-frames to each satellite
(6) When the satellite receives the Galileo uplink it processes the data and creates the individual signals for the downlink
(7) The Galileo downlink is received by:
a)The SAR Beacon equipped with a Galileo receiver to be informed that his distress signal was received.
b)The Galileo Sensor Stations in the region, this raw data created by the Galileo sensor stations is transmitted to the MUCF.
(8) The MUCF receives the raw data from the GSS's and monitors this data for the RLM dissemination. The MUCF then creates the RLM Notification Monitoring report including information on all RLMs transmitted over the last 5 minutes and their status, also status of the Galileo satellites availability for further RLM to aid SAR in the planning of their list of satellite creation.

(9) The SPF sends the RLM Notification monitoring report to RLSP when received by MUCF every 5 minutes.

6.8.4.3.4.2 #SFC8.2.2-2 RLM Cancellation Dissemination

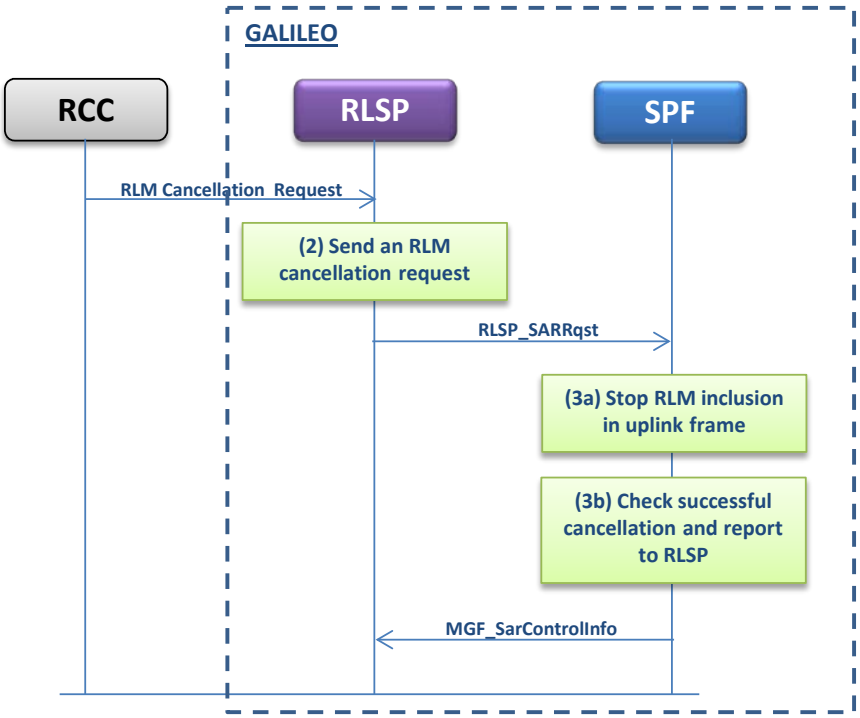


Figure 233: SFC8.2.2-2 RLM Cancellation Dissemination

Scenario Textual Description:

0) General Description: When a SAR distress is received by the Cospas-Sarsat ground system and the rescue operation is initiated a return link message can be sent to the SAR distress beacon in order to inform the person in distress that their distress is being acted on. There may be situations when it is requested to stop the transmission of this RLM . Possible reasons could involve: The RLM is found to be associated with a false alert, the RLM has been acknowledged to be received by the recipient, the rescue operation is complete, and therefore the Galileo system stops the required repetitions of the RLM. This cancellation is carried out with the process below.

The sequential description is detailed below:

1. The RLM cancellation request can be generated either:
    a. By RCC which transmits the request to the RLSP (ref Figure **233**)
    b. Or by the RLSP operator
2. The RLSP sends the RLM cancellation notice as part of a request message to the MGF. This RLM cancellation request includes the RLM ID to be cancelled.
3. (a) The MGF identifies the RLM to be cancelled and stops its inclusion in the Uplink frame to the Galileo
    (b) The MGF monitors the successful cancellation of the RLM
4. The information on the successful cancellation of the RLM is transmitted to RLSP.

6.9 SFC9 Support EDBS

6.9.1 Introduction

The objective of the Support to EDBS SFC is to provide the Galileo user with Open Service Authentication
This includes the reception of EDBS at the GCC and further dissemination via the Galileo constellation.

The following section describes the physical implementation of the SFC. It is based on the following documentation:

- ☐ System documents:
  - System Logical Architecture (SYS-RD.1)

- ☐ SSeg documents:
  - o Payload DDJF (IOVSS-RD.3 and FOCSS-RD.2)
- ☐ GMS documents:
  - GMS DDF1 (GMS-RD.1)
  - GMS DDF2 (GMS-RD.2)
  - o MUCF ICD (GMS-RD.11)
  - o SPF ICD (GMS-RD.16)
  - o GACF ICD (GMS-RD.12)
  - GMS to Ops concept (GMS-RD.41)

Note related to ERIS direct uplink removal:. In the following described section for ERIS support the direct dissemination path as part of the architecture remains incorporated but its functionality as such is to be trimmed by pertinent blocking / barrier mechanisms.
A note in each descriptive text will indicate a potential blocking method and / or indicate that the description has to be understood to be adopted by the direct ERIS data dissemination blocking mechanism or functionality.

6.9.2 SFC9 High Level Description

Galileo provides support to the Galileo Service Centre in terms of the possibility to disseminate External Data Broadcast Service (EDBS) data to Galileo users.
EDBS is an E1b mono frequency authentication service disseminated to Galileo users through E1b old ERIS bit allocation.

6.9.3 SFC9.1 EDBS Management

6.9.3.1 SFC9.1 High Level Description

Galileo Service Centre receives reports on the EDBS dissemination. This SFC describes the process taken to achieve the distribution of these reports across the GMS to GSC

6.9.3.2 Physical Static Block Diagram

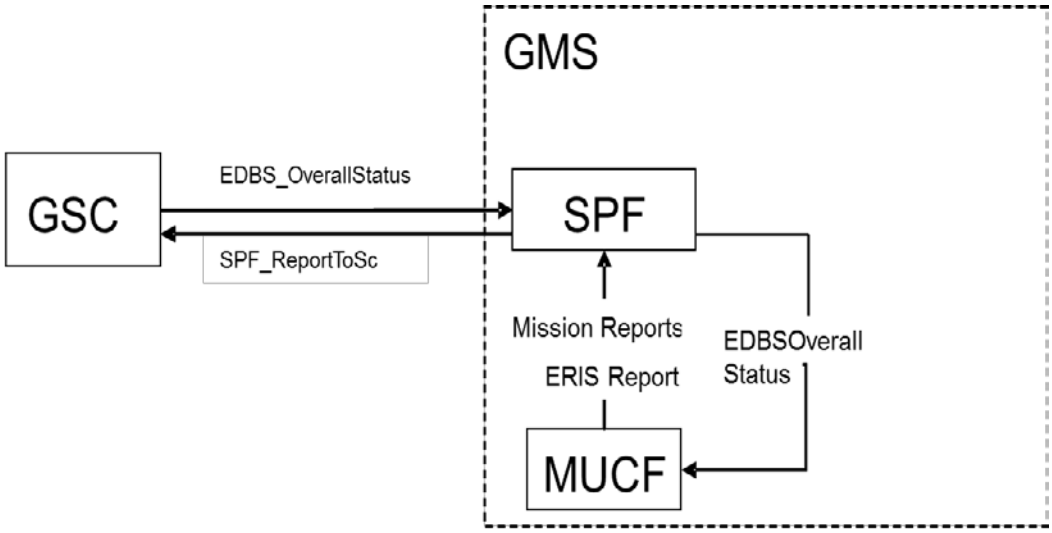


Figure 234 : SFC9.1 Physical Static View

6.9.3.3 Overall Physical Description

The purpose of this SFC is to allow information exchange (or reports) between GMS and the Galileo Service Centre in terms of EDBS

- EDBS_DisseminationReport
- CNF_SvMask

Other reports for GSC, namely for Commercial Service Support are defined into Commercial Service part.

The GSC reports to GMS include:
ERIS Status (MNT_EDBSStatus)

The EDBS dissemination reports from GMS to GSC are provided periodically (every 30 min)
These reports are produced by MUCF based on MGF mission monitoring and GSS data and sent to the SPF for transfer to the GSC
The EDBS reports from GSC to GMS are provided periodically (every 30 mins).

Note: The EDBS reports are not planned to be exchanged on GMS/GSC request.

6.9.3.3.1 Interface to GSC: SPF

The **Service Products Facility (SPF**) sends EDBS reports to Galileo Service Centre regularly (every 30 min). It also receives the EDBS status report from Galileo Service Centre.

SPF also provides EDBS with CNF_SvMask, including a correspondence table between: Identification of the Galileo satellites SVID and CCSDS Identifier

6.9.3.3.2 Report Creation Applicable to EDBS: MUCF

The **MUCF** creates the EDBS reports and sends to the SPF for further distribution to Galileo Service Centre.
The reports are created from top level reports from the mission monitoring functions of the MUCF.
The MUCF also accepts the EDBS status to aid the EDBS monitoring function described in SFC9.2.

6.9.3.3.3 Archive Storage: GACF

The **GACF** has 1 function in this functional chain:

- It archives the standard reports for the external entities

6.9.3.4 Dynamic Scenarios

Two main nominal scenarios have been identified:

- **Scenario 1:** GMS send all reports to Galileo Service Centre, and equally Galileo Service Centre sends its EDBS status report periodically.
- **Scenario 2:** This scenario is an operation scenario describing the start-up of the Galileo ground segment support to EDBS.Described in #SFC.9.1-1

6.9.3.4.1 #SFC9.1-1 Start up of GMS Support to EDBS

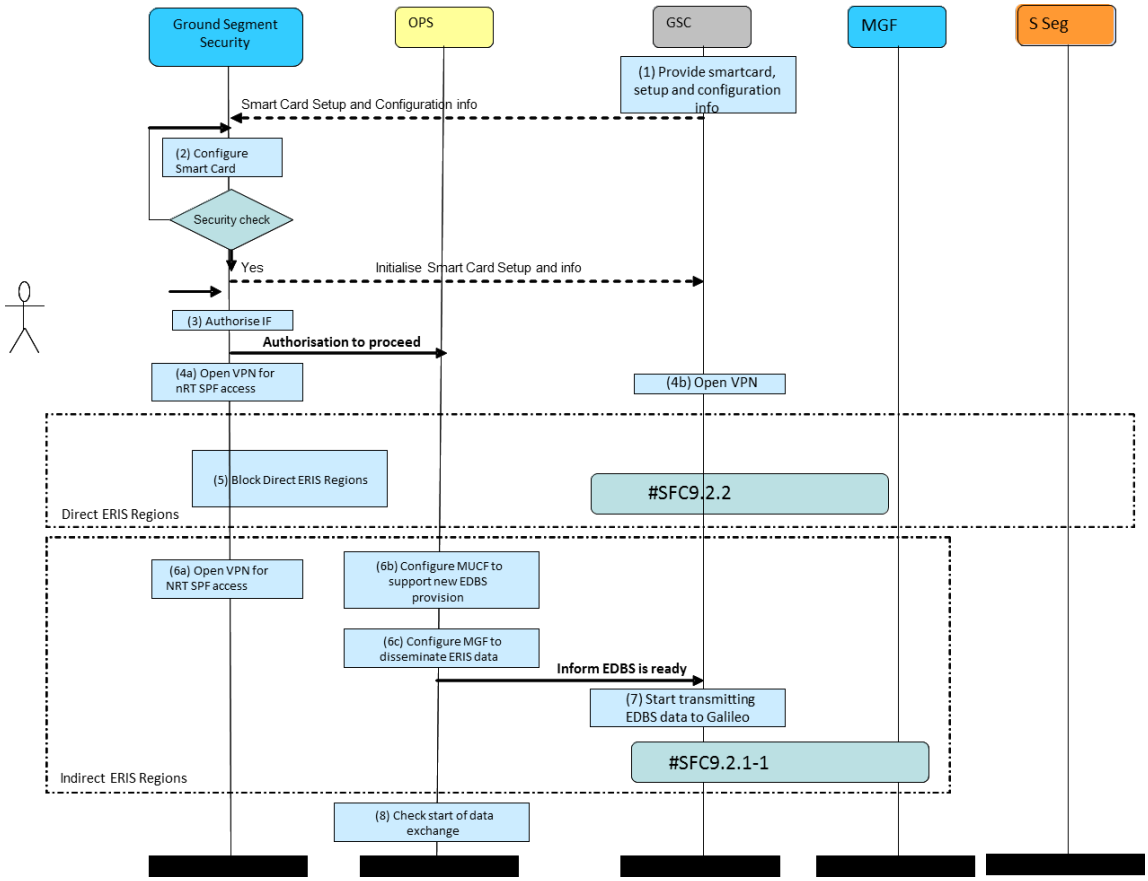


Figure 235 : #SFC9.1-1: Start up of GMS Support to EDBS

Scenario Textual Description:
0) General Description: This scenario describes the start-up procedure for the support to EDBS
The sequential description is detailed below:

1) GSC provides smartcard, setup details and configuration information to the Galileo security team offline.

2) The Galileo security team then configure this smartcard ready for EDBS use. This is checked by security and sent to GSC offline.

3) The Galileo management trigger the request to start up the interface with the GSC and security authorise this interface for start up and informs Galileo operations.

4) The Galileo security team opens the VPN from GNMF-s for non-real time access to the SPF. GSC will also configure and open their VPN.

5) ERIS Direct uplink is blocked following **#SFC9.2.2**
6) the real time VPN also needs to be opened and elements configured:
   a) The Galileo security team opens the VPN from GNMF-s for real time data flows.
   b) The operations team configure the MUCF to support EDBS provision
   c) The operations team configure the MGF to disseminate EDBS data

7) The GSC is informed that the interface is operational and can follow the steps in #SFC9.2.1-1 for EDBS dissemination

6.9.4 SFC9.2 EDBS Dissemination

6.9.4.1 High Level Description

Galileo provides support to Galileo Service Centre in terms of the possibility to disseminate External Data Broadcast Service to Galileo users.

GSC disseminates EDBS in an indirect way through Galileo Ground Mission Segment using previous ERIS Integrity subframe allocation (SFC 9.2.1)

The Galileo system supports EDBS indirect dissemination for Satellite in contact with ULS

6.9.4.2 Physical Static Block Diagram

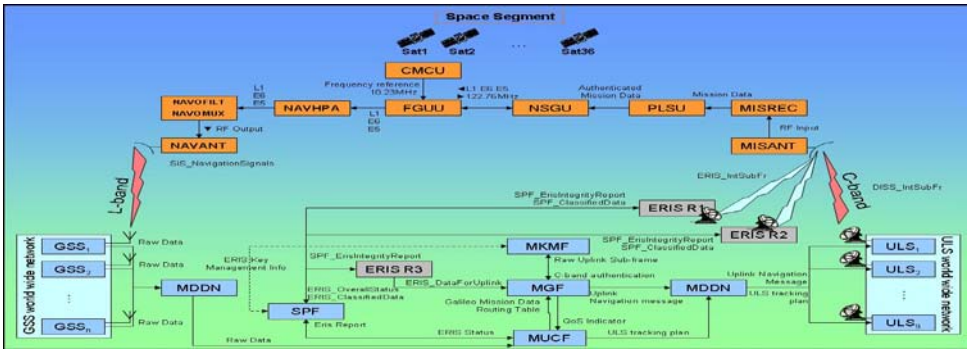


Figure 236 : SFC9.2 Physical Static Block Diagram

6.9.4.3 Overall Physical Description

Before further explaining EDBS scenario, it shall be noted that ERIS direct uplink has been removed from baseline. This means direct ERIS data transmission must not occur. EDBS dissemination will use previous indirect ERIS dissemination capabilities. During start up process of GMS Support to EDBS, ERIS direct uplink is blocked following #SFC9.2.2-1.

6.9.4.3.1 EDBS Data Handling on Board the Satellites

EDBS data is processed much in the same way as other integrity data within the satellite.
EDBS is received from GMS within the Integrity subframe and it is managed as previous ERIS indirect integrity. The 5 direct ERIS integrity are blocked.
The satellite takes the input (EDBS included in previous indirect ERIS bit allocation) and creates the downlink message and transmits this to the final users.

6.9.4.3.1.1 Reception of EDBS and GMS Uplinks: MISREC

For what relates to EDBS, the main functions of the **MISREC** are:

- Accept a single RF input from the C-band antenna comprising the CDMA signals from the GMS (including possibly EDBS data in previous indirect ERIS data
- Generate Mission uplink PRN codes, to despread and demultiplex the received CDMA signals
- Recover and track carrier
- Demodulates channel
- Soft decode the convolutional encoded symbol stream and resolves the node-symbol ambiguity
- Synchronise and de-scramble the frame
- Separates the integrity messages from mission data sub-frames, passing them to the output data stream with higher priority
- Processes the packets to provide one output data stream comprising integrity and navigation message data on a single serial interface.

6.9.4.3.1.2 Authentication of ERIS Channel Data: PLSU

The Payload xSU acts as a firewall for the C-band mission uplink during nominal operations, to authorize all mission data received on this link whilst in nominal mode, before passing the data to the NSGU for dissemination. The Payload xSU takes the MISREC output as its input (over a serial RS-422 interface), and outputs the authenticated mission data via a serial RS-422 interface to the NSGU.

note related to ERIS direct uplink removal: Barrier can be deployed for denying the ERIS direct uplink data to pass PLSU firewall.

6.9.4.3.1.3 Processing and Prioritisation of EDBS Data: NSGU

At the heart of the Galileo satellite payload, is the Navigation Signal Generation section, responsible for taking the 10.23MHz reference signal from the Reference Clock Section and the mission data signals from the C-band Receive Section as inputs, to generate the L-band downlink navigation broadcast signals.

The downlink allocation for EDBS data is 40 bits per second each two seconds on the I/NAV service. This consists of a 40 bit field with the same bit allocation and position as on the uplink field. The NSGU will therefore format the EDBS downlink using the EDBS within the Integrity Subframe from GMS. If no data is available there, the 40 bits for EDBS of the downlink will be filled with zeros.
EDBS will be only disseminated in E1b ERIS bit allocation.E5b ERIS bits are reserved for future uses

6.9.4.3.1.4 Dynamic Scenarios: #SFC9.2-1: EDBS Data Handling On-Board

Scenario Textual Description:
0) General Description: The Galileo Satellites receive EDBS data via Integrity Subframe from the Galileo C-Band uplink, the satellite must then process this data and include it in the correct place in the down link message
The sequential description is detailed below:

(1) The mission receiver in the Galileo satellite navigation payload receives the Galileo uplink from the Galileo core system Uplink Stations (including EDBS inside the integrity subframe)-
(2) The Payload security module then takes the uplink message from the mission receiver and authorises the messages

(3) The Navigation Signal Generation Unit (NSGU) receives the message stream and creates the downlink stream, composing the Navigation message, including the EDBS data from GMS source in the correct position in the L-Band downlink.

(4) The Frequency Generation Up converter Unit (FGUU) prepares the signal frequencies for each downlink carrier(5)The Navigation antenna transmits to earth the Galileo signals to the users and Sensor Stations.

6.9.4.3.2 EDBS Monitoring

The EDBS dissemination through the Galileo system is monitored by the Galileo ground segment. This allows feedback to the GSC of its dissemination performance and also informs the Galileo operators of the status of the support to EDBS service.
Galileo System monitors the uplink of the data from the MGF output and the downlink of the navigation message via the GSS.

6.9.4.3.2.1 Overall Physical Description

6.9.4.3.2.1.1 Monitoring of EDBS Data: MUCF

The MUCF exploits the GSS observables to check the correct dissemination of the EDBS data. MUCF checks also the successful uplink of the EDBS data sent to the GCC.

The GMS is able to generate EDBS reports. These reports tell GSC if the EDBS field was active, at what time and for what satellites The EDBS uplink report will show for each spacecraft the times at which activity started and stopped
MUCF confirm that the channel status will be monitored via the MMI. There will be a display showing for EDBS the monitored activity on each satellite.

6.9.4.3.2.1.2 Reception EDBS Data in Downlink: GSS

The GSS receives the SIS including the EDBS field and distributes the data to the GCC for monitoring.

6.9.4.3.2.1.3 EDBS Report Transmission: SPF

The SPF sends to GSC the EDBS report and also sends to the MUCF the EDBS status received from the GSC to the MUCF for monitoring purposes.

6.9.4.3.2.1.4 Integration of EDBS Data in Galileo Uplink: MGF

The MGF integrates the EDBS data into the uplink message and therefore sends a quality of service report to the MUCF to inform the MUCF of the correct uplink of the EDBS data received.

6.9.4.3.2.2 Dynamic Scenarios: #SFC9.2-2: EDBS Dissemination Monitoring

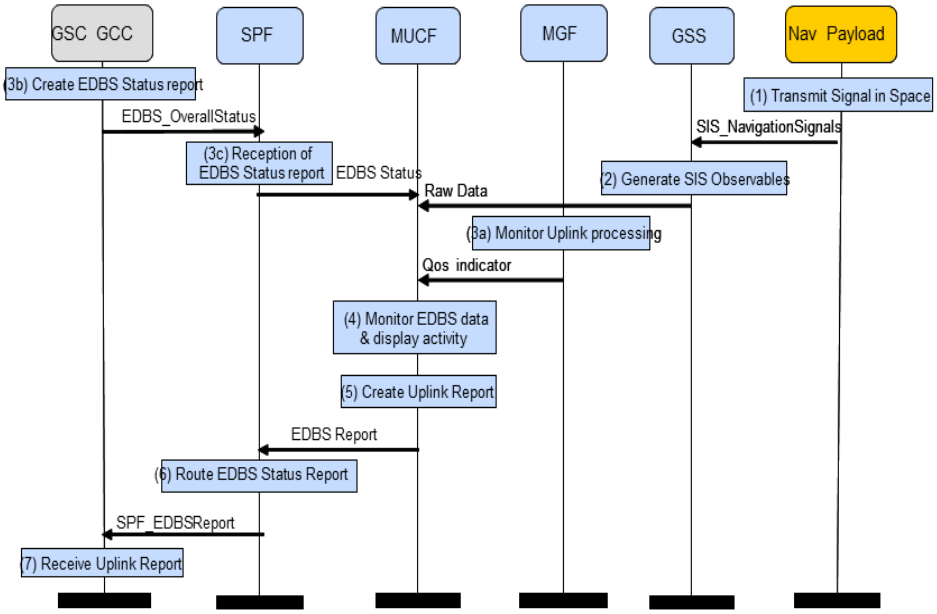


Figure 237 : #SFC9.2-2 : EDBS Dissemination Monitoring

Scenario Textual Description:
0) General Description: The Galileo Core system is responsible for the monitoring of the EDBS service. This entails checking the correct uplink of EDBS data received and of correct expected activities in the Galileo signal.
The sequential description is detailed below:

(1) The Galileo satellites transmit the Galileo signal

(2) The GSS receives the Galileo signals and produces raw data to be transmitted to the Galileo Control Centre

(3) Various information is sent to MUCF to aid monitoring:
- ☐ The MGF verifies that EDBS data is not corrupted with respect to received data
- ☐ The GSC sends information on its status to the MUCF.

(4) The MUCF collects all the data to monitor the EDBS service and displays this information on an MMI.

(5) The MUCF also creates a report on the monitoring for the EDBS service information

(6) The SPF distributes the EDBS reports to GSC. This report will include:
- ☐ Number of EDBS data received for each satellite

☐ The list of messages Time Tags received by MGF during the period of the report together with MGF reception status
☐ Evolution per satellite(status change events) of EDBS uplink of the integrity sub-frame performance parameter during the period of the report
☐ Evolution per satellite of EDBD uplink availability performance parameter during the period of the report
☐ Satellites used for EDBS data dissemination according to MGF

(7) The GSC receive the EDBS report for their use

6.9.4.4 SFC9.2.1 EDBS Uplink

6.9.4.4.1 SFC9.2.1 High Level Description

the GSC sends EDBS data to the Galileo ground centre for further dissemination to the Galileo satellites. The GSC must send this data at the correct time and with the correct format

6.9.4.4.2 Physical Static Block Diagram

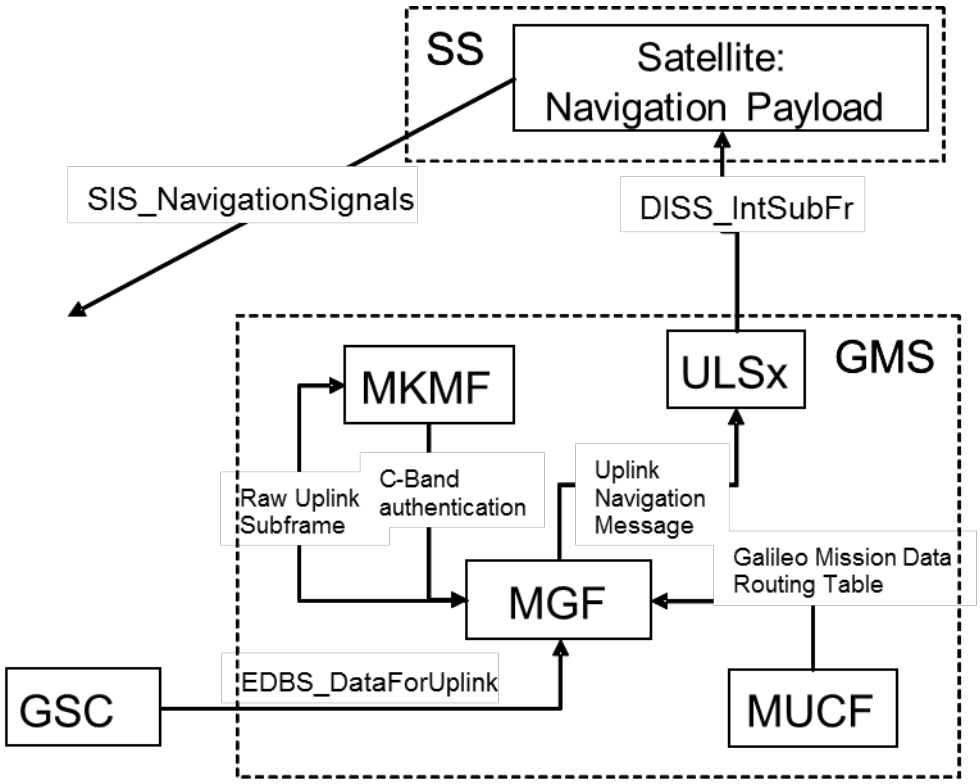


Figure 238 : SFC9.2.1 Physical Static View

6.9.4.4.3 Overall Physical Description

The elements responsible for the processing of the EDBS data are the MGF and MUCF for monitoring purposes, as mentioned above, and uplink scheduling. The Galileo satellite, to be specific, the Galileo satellite navigation payload, is also involved in this SFC in the processing of the EDBS data encapsulated within the Galileo uplink signal.

6.9.4.4.3.1 Reception of Indirect Data and Integration into Uplink Message: MGF

The Message Generation Facility (MGF) is the meeting point for all the data streams that make up the C-Band Uplink Messages. Its mission is to elaborate messages as specified in the GMS to Space ICD by multiplexing those data that allow providing the OS, CS, I/Nav PRS, SAR, and EDBS services.
The multiplexed streams are then forwarded to the ULS/antennas according to pre-defined routing table constrained by the ULS antennas tracking plan and by needs specific to each service.
MGF receives EDBS data from GSC.
The Galileo Service Centre is expected to send their data to the MGF in time for the required epoch for uplink to the satellites.

6.9.4.4.3.2 Processing of EDBS Data On-Board: Navigation Payload

The space segment element involved is the navigation payload. This payload receives through the mission receiver the Galileo uplink and creates the message for downlink to the user including the EDBS data.

6.9.4.4.3.3 Authentication of Uplink Subframes: MKMF

The MKMF is responsible for applying the protection on the whole integrity (including EDBS data) and message sub-frames, to allow access to the Galileo satellites. It is not a unique function for the EDBS service but to the whole message.

6.9.4.4.4 Dynamic Scenarios: #SFC9.2.1-1 EDBS Dissemination

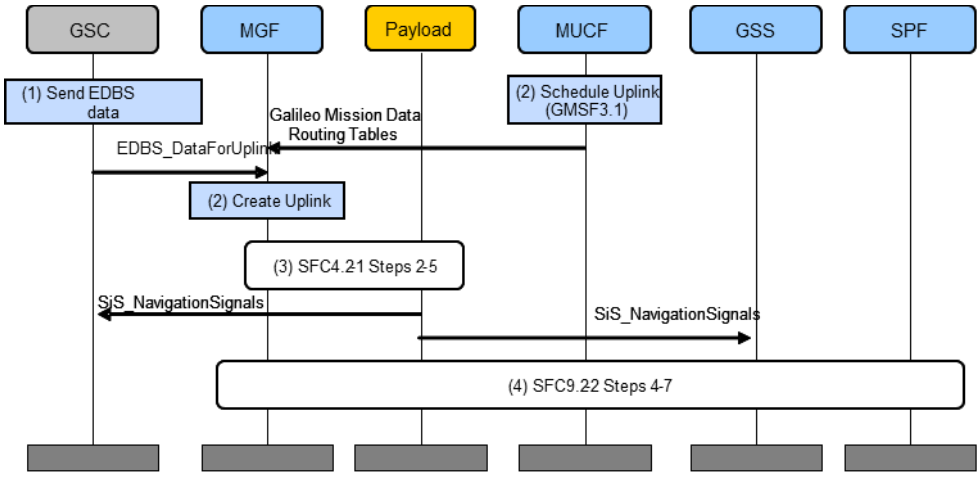


Figure 239 : #SFC9.2.1-1 : EDBS Dissemination

Scenario Textual Description:
0) General Description: The Galileo Service Centre can send the EDBS data to the Galileo ground segment for inclusion in the navigation uplink signal from the GMS ULS', to be transmitted to the EDBS users. This process is also monitored by the Galileo ground system for their own use and to inform the GSC on the dissemination success of the data.

GSC sends EDBS data only in epochs for which the related downlink corresponds for and Odd Page in E1b. (one each two seconds). For the other epochs (Even Page in E1b) no data is provided to MGF. Dummy values will be uplinked.
The sequential description is detailed below:

(1) The GSC sends to the Galileo ground centre MGF element EDBS data (40bits per satellite) at the required time to ensure the require epoch for uplink is achieved.

(2) The MGF receives the EDBS data and the uplink schedule for all satellites to be able to create the uplink sub-frames for all satellite.

(3) See dynamic scenario SFC4.2-1. The MGF sends the sub-frames to the MKMF and receives back protected sub-frames, and further sends the protected uplink sub-frames to the ULS for uplink to the satellite. The satellite then transfers the EDBS data directly into the downlink message to transmission to the users.

(4) The GMS also monitors the dissemination of the EDBS data as described in SFC9.2-2

6.9.4.5 SFC9.2.2 ERIS Direct Uplink Inhibition

6.9.4.5.1 High Level Description

Following GAL-DCN-TASI-SYST-X-0797, ERIS Direct Uplink will be blocked for each satellite by telecomanding the MISREC and NSGU to inhibit ERIS Direct reception. Once ERIS Direct Uplink is blocked, the indirect uplink will prevail and used for EDBS dissemination.

Note related to ERIS direct uplink removal: In the frame of a direct ERIS data barrier concept, Contact plan, blocking request, C-Band uplink keys are suggested not to be disseminated externally to prevent direct ERIS data uplinks.

7. DUAL GCC MANAGEMENT

In the FOC configuration two GCCs will be fully integrated with GCS and GMS equipment, such that the entire operations can be supported from a single GCC in case of need. The two GCC are identified as GCC-D (Oberpfaffenhofen, Germany) and GCC-I (Fucino, Italy).
In nominal conditions however, the system operations will be conducted from GCC-I for what concerns Mission services (GMS) operations and from GCC-D for the spacecraft constellation and GCS operations, with the other GCC capable of providing the complementary back up operations in case of need, but a specified conditions to reduce the overall man power nominally employed in each GCC [SYS-RD.6].

The possibility to hand-over the entire system operations to a single GCC is foreseen against the very remote possibility of a catastrophic failure affecting one GCC, while all availability requirements specified at system level apply independently within each single GCC.
For this reason, local redundancy is provided where necessary and it is expected that hand over of operations across GCCs (e.g. loss of one GCC) might happen only in exceptional conditions.

With this premises, it is logic that backup capability provided in each GCC are subject to certain assumptions and limitations, e.g. [SYS-RD.76]:

- all GMS elements at GCC-D are kept On, and where applicable synchronised with the data generated at GCC-I. The GMS elements of the mission chain (OSPF, MGF, x-KMFs) are remotely controlled from the MUCF at GCC-I (very limited GMS operators support foreseen at GCC-D in nominal conditions). This configuration at GCC-D will also support planned Mission service operations handover from GCC-I (for up to 4 weeks) when software upgrades are required in the prime GCC.
- All GCS elements at GCC-I are kept ON and, where applicable, kept synchronised with the data generated at GCC-D to ready to take the Master role in case of need and support a sufficiently quick ground response (despite limited GCS and Spacecraft operator support is foreseen at GCC-I in nominal conditions).

Refer to dedicated TN on GCC split concept (SYS-RD.76) for further details. The dynamic scenarios associated to GCC handover are provided in SFC5.1 (sections 6.5.1.2.2.4 and 6.5.1.2.2.5).

- #SFC5.1.1-4 GCS planned and unplanned handover
- #SFC5.1.1-5 GMS planned and unplanned handover

END OF DOCUMENT